

# R&S®CMW270/290/500

## WLAN Firmware Applications

### User Manual



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Version 35

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This user manual describes the following R&S®CMW options:

- R&S®CMW-KM650 (WLAN IEEE 802.11a/b/g, TX measurement)
- R&S®CMW-KM651 (WLAN IEEE 802.11n SISO, TX measurement)
- R&S®CMW-KM652 (WLAN IEEE 802.11, composite MIMO, TX measurement)
- R&S®CMW-KM654 (WLAN IEEE 802.11 true MIMO, TX measurement)
- R&S®CMW-KM655 (WLAN IEEE 802.11p, TX measurement)
- R&S®CMW-KM656 (WLAN IEEE 802.11ac SISO, TX measurement)
- R&S®CMW-KM657 (WLAN IEEE 802.11ax SISO, TX measurement)
- R&S®CMW-KM012 (TX measurement, multi-evaluation list mode)
- R&S®CMW-KS650 (WLAN IEEE 802.11a/b/g, basic signaling)
- R&S®CMW-KS651 (WLAN IEEE 802.11n, basic signaling)
- R&S®CMW-KS656 (WLAN IEEE 802.11ac, basic signaling)
- R&S®CMW-KS657 (WLAN IEEE 802.11ax, AP signaling)
- R&S®CMW-KS658 (WLAN IEEE 802.11ax, station signaling)
- R&S®CMW-KS660 (WLAN IEEE 802.11, advanced signaling)
- R&S®CMW-KS667 (WLAN IEEE 802.11ax signaling, multi-station testing)
- R&S®CMW-KS670 (WLAN IEEE 802.11, SU-MIMO 2x2 generic signaling)
- R&S®CMW-KS671 (WLAN IEEE 802.11ax, MU-MIMO signaling AP)
- R&S®CMW-KE100 (basic fading support: AWGN generator)
- R&S®CMW-KE670 (WLAN fading profiles)

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The following abbreviations are used throughout this manual: R&S®CMW is abbreviated as R&S CMW.

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# 1 Preface

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## 1.1 Documentation overview

This chapter provides an overview of the R&S CMW user documentation. Most documents are available in PDF format on the customer web.

The customer web is a section of GLORIS, the global Rohde & Schwarz information system, see <https://gloris.rohde-schwarz.com>. A registration is required.

### 1.1.1 Getting started manual / quick start guide

Depending on your instrument model, there is either a getting started manual or a quick start guide.

The document introduces the R&S CMW and describes how to set up and start working with the product.

A PDF file is available on the customer web.

### 1.1.2 User manual and help

Separate user manuals are provided for the base unit and the firmware applications:

- The base unit manual describes the base software and common features of the firmware applications. It also describes basic principles for manual operation and remote control. The manual includes the contents of the getting started manual or quick start guide.
- Additional documents, like this one, describe the firmware applications, including the graphical user interface of each application and the remote control commands.

All user manuals are available on the customer web.

The contents of the user manuals are also available as help in the R&S CMW. The help offers quick, context-sensitive access to the information needed for operation and programming. If you perform a software update, the help is also updated.

### 1.1.3 Features & functions

The features & functions documents list the available software options and the related features, settings and measurement results. The documents are available on the customer web.

### 1.1.4 Basic safety instructions

Contains safety instructions in many languages.

The printed document is delivered with the instrument.

### 1.1.5 Data sheet and brochure

The data sheet contains the technical specifications of the R&S CMW. It also lists the options and their order numbers and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

The documents are available on the customer web.

### 1.1.6 Release notes and open-source acknowledgment (OSA)

The release notes list new features, improvements and known issues per software version and installation package.

The open source acknowledgment provides verbatim license texts of the used open-source software.

The documents are available on the customer web.

### 1.1.7 Application notes, application cards, white papers, etc.

These documents deal with special applications or background information on particular topics.

The documents are available on the customer web.

## 1.2 How to read firmware application chapters

Each firmware application is described in a separate chapter. These chapters can be read independently of each other. However, they are all organized as follows:

1. General description
2. Application sheets (optional)
3. GUI reference
4. Programming examples
5. Command reference

### 1.2.1 General description

This chapter provides a general description of the firmware application, independent of a specific operation mode (manual or remote control). Background information related to the network standard is given as far as it is directly related to configurable parameters.

### 1.2.2 Application sheets

This optional chapter provides short application examples for selected issues and related background information.

### 1.2.3 GUI reference

The GUI reference describes the manual operation of the firmware application via the graphical user interface (GUI).

The description of a configuration dialog usually starts with a screenshot presenting the preset values of the parameters. Sometimes, preset values are modified to enable hidden parts of a dialog. Below the screenshot, all shown parameters are described. For each single parameter, a link to the corresponding command description in the command reference is provided. Ranges for numeric parameters and reset values are given there.

For measurement results, links to the corresponding command descriptions are provided (commands to retrieve the results).

### 1.2.4 Programming examples

The programming examples show how to control and configure the firmware application via a remote-control program and how to retrieve measurement results. The examples consist of comprehensive command sequences. You can check just a single command of a sequence to get an example for the syntax of this single command. But you can also consider an entire sequence showing the commands in the context of a command script, under consideration of dependencies and required orders of the commands.

The command sequences are written with the intention to list most commands of the firmware application. They do not show the fastest way for a given configuration task. The fastest way would use many reset values and omit the corresponding commands.

The examples are referenced by the command descriptions of the command reference.

### 1.2.5 Command reference

The command reference provides information on the remote commands of the firmware application. The commands are grouped according to their function.

Each command description indicates the syntax of the command header and of the parameters. For input parameters, the allowed ranges, reset values and default units are listed. Most commands have a command form and a query form. Exceptions are marked by "Setting only", "Query only" or "Event". Furthermore, a link to the programming examples is provided and the first software version supporting the command is indicated.



## 2 WLAN signaling

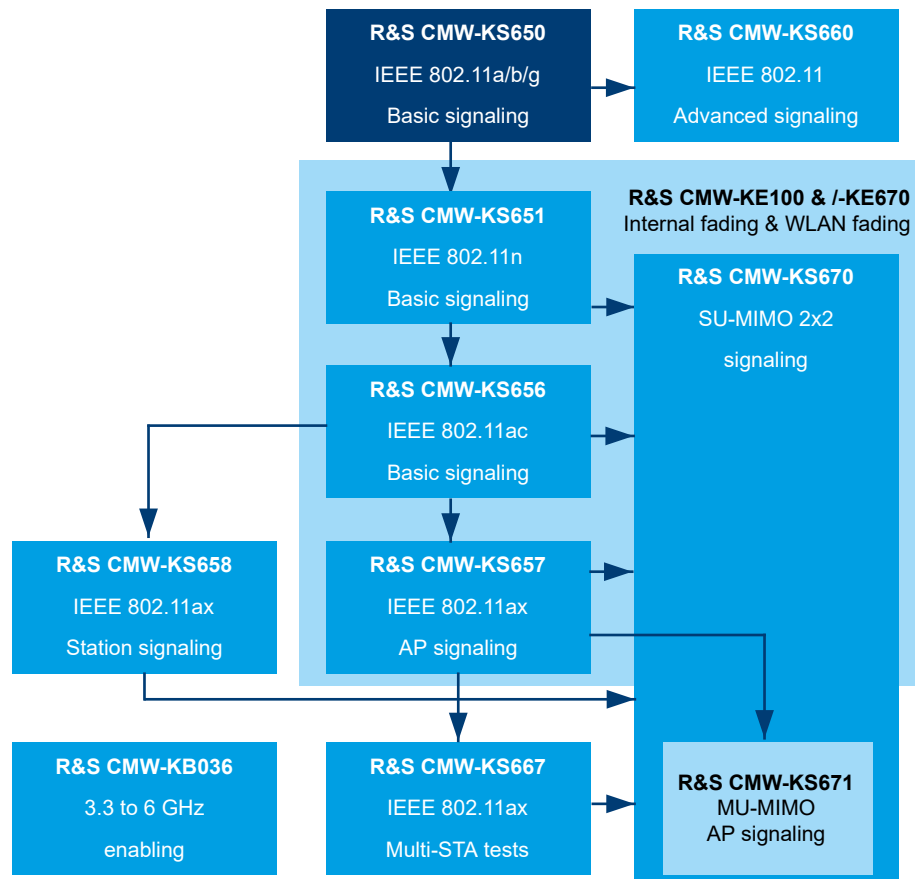
The WLAN signaling application emulates a WLAN device that communicates over WLAN with another WLAN device, the device under test (DUT).

The signaling application can assume the following roles:

- Access point (AP) in infrastructure mode (DUT is a WLAN station), SISO and MIMO 2x2
- Hotspot 2.0 access point (DUT is a WLAN station), SISO connections
- WLAN station (DUT is an access point), SISO and MIMO 2x2

The supported roles and WLAN IEEE 802.11 standard depend on the available options.

The following figure shows the option dependencies. Basic option R&S CMW-KS650 is always required.



The following software options are offered for WLAN signaling:

- R&S CMW-KS650 enables basic signaling functionality for 802.11a/b/g
- R&S CMW-KS651 adds basic functionality for 802.11n, SISO
- R&S CMW-KS656 adds basic functionality for 802.11ac, SISO
- R&S CMW-KS657 adds basic functionality for 802.11ax, SISO, AP mode

Both SUA partitions required (e.g. SUA1&2).

- R&S CMW-KS658 adds 802.11ax, SISO, station mode  
Both SUA partitions required (e.g. SUA1&2).
- R&S CMW-KS660 adds advanced functionality, such as Hotspot 2.0 and WPA/WPA2 enterprise for all supported IEEE 802.11 standards.
- R&S CMW-KS667 enables multi-STA testing for 802.11ax  
Both SUA partitions required (e.g. SUA1&2).
- R&S CMW-KS670 adds MIMO 2x2 in AP and station mode. One option R&S CMW-KB036 per TX/RX module is required for RF frequencies above 3.3 GHz.
- R&S CMW-KS671 extends MIMO 2x2 by adding MU-MIMO in AP mode for 802.11ax  
(This option requires R&S CMW-KS670 and R&S CMW-KS657)
- R&S CMW-KE100 enables basic internal fading
- R&S CMW-KE670 is required for WLAN fading, AP mode, SISO and SU-MIMO, and standards 802.11n, ac and ax

The generic option R&S CMW-KB036 "3.3 GHz to 6 GHz enabling" is required for RF frequencies above 3.3 GHz, for example for the WLAN 5-GHz or 6-GHz bands.



#### Hardware requirements

- WLAN signaling requires an R&S CMW500/2xx with TRX160.
- Both partitions of an SUA (e.g. SUA1&2) are required for:
  - 802.11ax
  - MIMO scenarios
  - 160 MHz channel bandwidth  
Note that only contiguous 160 MHz channel bandwidth is supported, 80+80 MHz bandwidth (non-contiguous) is not supported.
  - Internal fading (e.g. Fader1&2)
- The signaling options offer the following test functionality:
  - Operation as access point, Hotspot 2.0 or station
  - WPA/WPA2 personal
  - Basic configuration (channel, TX burst power, SSID, MAC address, supported rates etc.)
  - Frame rate control for management and data frames
  - SISO and MIMO 2x2 (if supported by WLAN standard)
  - Test of connection setup
  - Readout of MAC layer-related information received from the DUT
  - RX burst power measurement
  - IP packet generation (ICMP, UDP)
  - Packet error rate (PER) tests

- The signaling options R&S CMW-KS656/ -KS657 offer a subset of this test functionality for 802.11ac/ ax.

You can measure WLAN signals with the "WLAN measurements" firmware application (option R&S CMW-KM650 plus enhancements).

For basic information about WLAN signals, refer to IEEE standard 802.11.

## 2.1 What's new

This documentation describes version 4.0.20 and higher of the WLAN signaling application. Compared to version 3.8.20, it provides the following changes:

- WPA3 and OWE security modes added, see [Mode](#) and [SAE H2E Support](#)
- New reset values:
  - For coding of HE TB PPDU, see [Coding Type](#)
  - For spatial streams configuration of RX frame trigger, see [RX Space Time Streams](#)

## 2.2 General description

The following sections describe how to use the R&S CMW for WLAN signaling tests and how to interpret the results.

### 2.2.1 Introduction to WLAN signaling tests

The following sections help you to get started with the WLAN signaling application.

#### 2.2.1.1 Test setups

Connect the device under test (DUT) to one or two of the bidirectional RF COM connectors at the front panel of the R&S CMW. No additional cabling and no external trigger are needed. The input level ranges of all RF COM connectors are identical.

The following sections refer to the individual operation modes.

##### **AP mode**

The R&S CMW simulates a WLAN access point in infrastructure mode. The DUT is a WLAN station.

The standard cell scenario (with or without fading) uses a single RF COM connector for SISO connection.

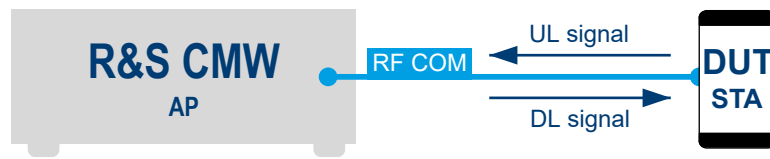


Figure 2-1: AP mode, SISO

For 802.11ax connections in 6 GHz band, a test setup with R&S CMW-Z800A is required. For details, refer to R&S CMW-Z800A product information published at customer web.

The following figure is one example of several possible test setups.



The MIMO scenario (with or without fading) typically uses two RF connectors and RF cables.

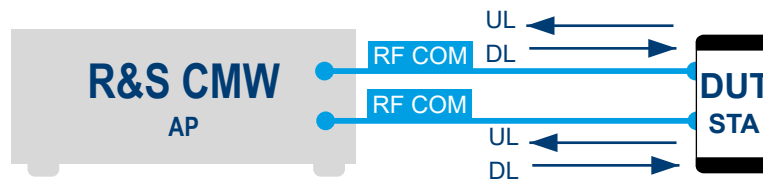


Figure 2-2: AP mode, MIMO 2x2

For test setups with multiple stations, use one combiner per stream to connect the RF port of the R&S CMW with several UEs.

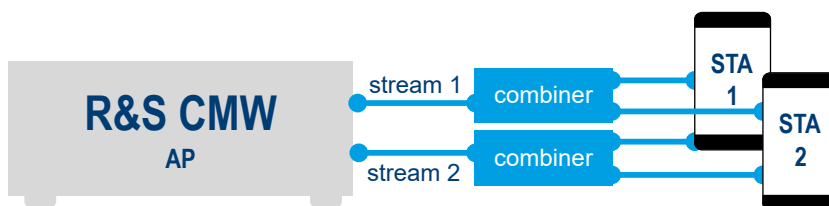


Figure 2-3: AP mode, MIMO 2x2, multiple stations

### Station mode

The R&S CMW simulates a WLAN station. The DUT is a WLAN access point.

The standard cell scenario (without fading) uses a single RF COM connector for SISO connection.

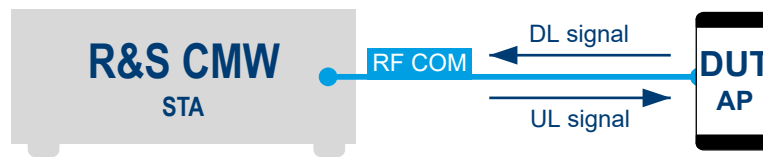


Figure 2-4: Station mode, SISO

The MIMO scenario typically uses two RF connectors and RF cables.

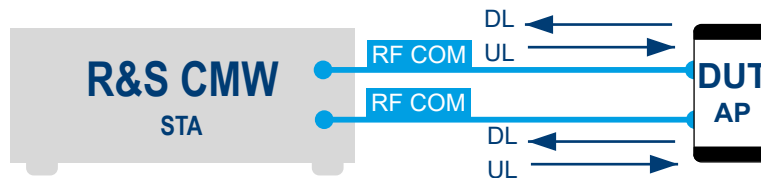


Figure 2-5: Station mode, MIMO 2x2

### Hotspot 2.0 mode

The R&S CMW simulates a Wi-Fi Hotspot 2.0 access point. The DUT is a WLAN station as Hotspot 2.0 client. Option R&S CMW-KS660 is required.



Figure 2-6: Hotspot 2.0 mode

#### 2.2.1.2 Establishing a WLAN association

An established WLAN association is a prerequisite for any WLAN tests.

Proceed as follows:

1. Connect your DUT to the R&S CMW (see [Chapter 2.2.1.1, "Test setups"](#), on page 15).
2. Configure the signaling application compatible to your DUT and the test case. You can modify the most important settings directly in the main view. The "Config" hot-key provides access to the configuration dialog box.

Typical settings:

- a) WLAN standard: 802.11a, g, n, ...
- b) Scenario selection: standard cell (SISO) or MIMO
- c) Operation mode selection: AP, station, ... (see [Table 2-6](#)).
- d) Used RF connectors
- e) For 802.11ax connections, select [Band](#)
- f) RF frequency / channel
- g) Power of the transmitted signal ("RF TX Burst Power")

h) Expected power of the DUT signal ("RF Exp. Peak Envelope Power")

**Note:** For the operation mode "Station", see also ["Special requirements in station mode"](#) on page 23.

3. To turn on the signal generator:
  - a) Press ON | OFF, or
  - b) Right-click the softkey "WLAN Signaling" > "ON".
4. Wait until the "WLAN Signaling" softkey indicates the "Run" state. The hour glass symbol must disappear.
5. Switch on the DUT and wait for association establishment. Note the connection status displayed in the main view.
 

**Note:** Depending on the DUT, it takes a few seconds until the association procedures have finished.
6. Perform an additional action if you have configured the "Station" operation mode with "Manual" connection mode.  
In that case, press the "Connect" hotkey.
 

**Note:** Depending on the DUT, it takes a few seconds until the association procedures have finished. With authentication, it takes up to two minutes.

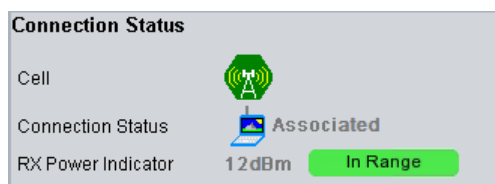
#### Checks if the association fails

- The "RF TX Burst Power" must be sufficient so that the DUT can receive the transmitted signal.
- The "RF Exp. Peak Envelope Power" must be in accordance with the received signal power.
- Check the "External Attenuation" for in and out paths.  
Typical external attenuation values:
  - 1 dB to 2 dB for a conducted connection, due to cables and connectors
  - >2 dB for splitters etc. in path
  - 25 dB to 30 dB for a shielded box
- The selected WLAN channel and WLAN standard must be supported by the DUT.
- The selected encryption algorithm must be supported by the DUT.

#### 2.2.1.3 Connection status

WLAN association involves a sequence of steps. The signaling application monitors the individual stages and displays the connection status in the main view.

If [Multi STA](#) is enabled, connection status is displayed for each station.



**Figure 2-7: Connection status area**

The "Connection Status" area also displays the "Cell" status: Signal switched on (green cell icon) or off (gray cell icon). The connection states are described in the following table.

**Table 2-1: Connection status overview**

| Cell | Connection status    | Description                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OFF  | Idle (default state) | WLAN signaling OFF                                                                                                                                                                                                                                                                                                                                                                                                 |
| ON   | Idle (default state) | The AP broadcasts beacon frames regularly. If the STA receives a beacon frame, it has sufficient knowledge about the AP to start authentication and association.                                                                                                                                                                                                                                                   |
|      | Authenticated        | The STA has sent an authentication request with authentication algorithm "open system" and the R&S CMW has sent a positive response.<br><br>The STA is now ready to start the association procedure.<br><br>Note that the R&S CMW does not support shared key authentication (WEP).                                                                                                                                |
|      | Associated           | The STA has sent an association request and the AP has sent a positive association response (including the STA's association ID).<br><br>If the AP does not require WLAN security, the STA can now exchange data packets with the AP.<br><br>For any WPA security mechanism, association is followed by a secondary authentication step. If successful, the DUT remains associated, otherwise it is disassociated. |
|      | Disassociated        | An association was established and:<br><ul style="list-style-type: none"> <li>• Either the AP or STA has sent a disassociation request (or a deauthentication request)</li> <li>• Or WPA authentication was not successful.</li> </ul> The STA is still "open system" authenticated, but data transfer is no longer possible.                                                                                      |
|      | Deauthenticated      | A STA was authenticated and either the AP or STA has sent a deauthentication request.<br><br>This action voids the existing authentication (and association) relationship.                                                                                                                                                                                                                                         |
|      | Connection timeout   | In station mode, the attempt to connect to the AP has timed out without response.                                                                                                                                                                                                                                                                                                                                  |

First the STA discovers the AP, optionally by sending a probe request displayed in event log. Afterward it attempts to establish a connection, the state changes from "Idle" via "Authenticated" to "Associated".

During disconnection, the state changes from "Associated" via "Disassociated" to "Deauthenticated".

In general, signaling measurements must be performed in the "Associated" state.

#### 2.2.1.4 Getting monitoring and measurement results

The WLAN signaling application provides the following types of results:

- Information in the main view, especially connection status, event log and DUT capabilities, see [Chapter 2.3.1, "Signaling view"](#), on page 32
- Results of a packet error rate (PER) test, see [Chapter 2.3.18.2, "Measurement results"](#), on page 103.

To obtain these results, proceed as follows:

1. Establish a connection as described in [Chapter 2.2.1.2, "Establishing a WLAN association"](#), on page 17
2. Check the information in the main view.
3. To access the PER measurement, press the "WLAN PER" softkey. Configure and start the measurement.

### 2.2.2 MIMO scenario

Option R&S CMW-KS670 is required for MIMO tests. It extends options R&S CMW-KS651, R&S CMW-KS656, R&S CMW-KS657, and R&S CMW-KS658 for standards 802.11n, ac, ax, supporting operating modes AP and station.



One option R&S CMW-KB036 per TX/RX module is required for RF frequencies above 3.3 GHz.

The R&S CMW simulates an access point supporting MIMO.

The following transmission modes are supported:

- **Spatial multiplexing**  
Data streams are transmitted on multiple spatial channels, with multiple antennas at the transmitter and receiver.
- **Space time block coding (STBC)**  
The number of spatial streams is less than the number of space-time streams.  
 $N_{SS} < N_{STS}$

The two DL signals are routed to different RF output connectors.

The supported connection standards are "802.11a/n", "802.11g(OFDM)/n", "802.11ac", and "802.11ax".

The following table shows the supported downlink MCS rates depending on the connection standard.



Table 2-2: Modulation and coding

| Standard 802.11... | Supported MCS values |
|--------------------|----------------------|
| a/n, g(OFDM)/n     | MCS 8 to 15          |
| ac                 | MCS 0 to 9           |
| ax                 | MCS 0 to 11          |

The R&S CMW supports the following channel bandwidths with MIMO.

- **For 802.11ac and ax:** 20 MHz, 40 MHz, 80 MHz
- **For 802.11n:** 20 MHz, 40 MHz

### 2.2.3 WLAN security

WLAN security comprises (mutual) authentication between AP and STA, packet encryption and message integrity verification. The beacon frames include a WPA parameter indicating the authentication and encryption algorithms to be used. The WLAN signaling application supports Wi-Fi protected access (WPA), Wi-Fi protected access version 2 (WPA2), and WPA3 - personal and enterprise mode.

#### WPA/ WPA2/ WPA3 personal security mode

In personal security mode, the authentication is based on a passphrase that generates a 256-bit pre-shared key (PSK). For the configuration, see ["Common Settings > Passphrase"](#) on page 81.

- In **WPA** personal security mode, packets are encrypted using the temporal key integrity protocol (TKIP). An additional message integrity check algorithm ("Michael") is used. WPA is only supported for 2.4 GHz and 5 GHz bands.
- **WPA2** personal security mode mandates CCMP-128 with PSK, an AES-based encryption mode. WPA2 is only supported for 2.4 GHz and 5 GHz bands. For operation mode station and WPA2 personal security mode, the encryption supports all combinations of TKIP and CCMP/AES. The dynamic selection algorithm always prioritizes CCMP/AES (if available) over TKIP.
- **WPA3** personal security mode mandates encryption CCMP-128 with SAE transition mode and protected management frames (PMF). WPA3 security mode supports all bands.

#### WPA/ WPA2/ WPA3 enterprise security mode

The enterprise mode requires the advanced signaling option R&S CMW-KS660.

- **WPA** enterprise security mode, packets are encrypted using TKIP with IEEE 802.1X with SHA-1. WPA is only supported for 2.4 GHz and 5 GHz bands.
- **WPA2** enterprise security mode mandates CCMP-128 with IEEE 802.1X with SHA-1. WPA2 is only supported for 2.4 GHz and 5 GHz bands. For operation mode station and WPA2 personal security mode, the encryption supports all combinations of TKIP and CCMP/AES. The dynamic selection algorithm always prioritizes CCMP/AES (if available) over TKIP.

- **WPA3** enterprise security mode mandates encryption CCMP-128 with IEEE 802.1X with SHA-256 and PMF. WPA3 security mode supports all bands.

The enterprise mode requires a RADIUS server and uses an extensible authentication protocol (EAP) for authentication. An internal or external RADIUS server can be set up for EAP authentication. When using an external RADIUS server, it must be reachable via LAN SWITCH connector no 2 of the Ethernet switch board (option R&S CMW-H661x). The complete test setup is shown below.

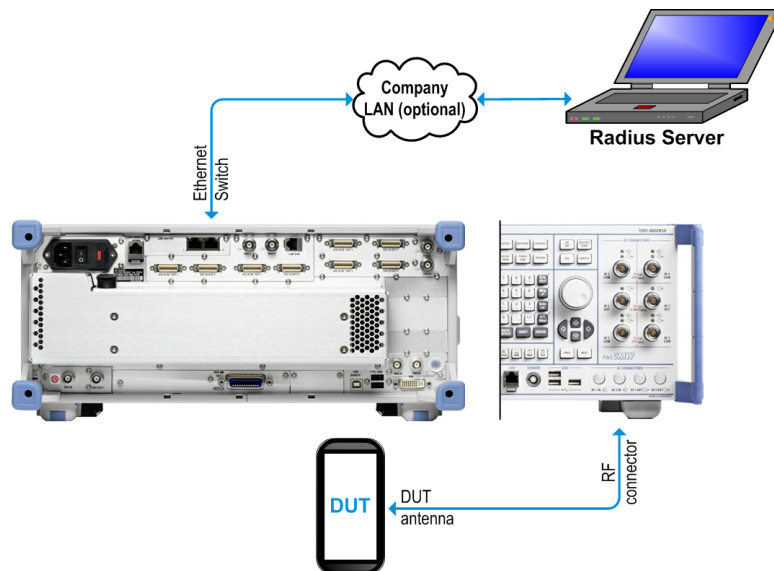


Figure 2-8: Test setup with external RADIUS server

### Opportunistic wireless encryption (OWE)

OWE with protected management frames (PMF) is used for 6 GHz band connections without encryption. This setting replaces encryption type "Disabled", that is relevant for the 2.4 GHz and 5 GHz bands. OWE is specified in RFC 8110.

We recommend that you use R&S CMWmars to obtain detailed information about the MAC messages exchanged during authentication.

Which of the described security mechanisms are supported, depends on the selected operation modes, as shown in the following table.

Table 2-3: Supported security mechanisms

|                 | AP | Station | Hotspot 2.0 |
|-----------------|----|---------|-------------|
| WPA personal    | X  | X       | -           |
| WPA2 personal   | X  | X       | -           |
| WPA3 personal   | X  | X       | -           |
| WPA enterprise  | X  | -       | -           |
| WPA2 enterprise | X  | -       | X           |
| WPA3 enterprise | X  | -       | -           |

|           | AP | Station | Hotspot 2.0 |
|-----------|----|---------|-------------|
| OWE       | X  | -       | -           |
| Automatic | -  | X       | -           |

For settings related to WLAN security, see [Chapter 2.3.8.11, "Security settings"](#), on page 79.

## 2.2.4 Station mode

In station mode, the R&S CMW simulates a WLAN station (STA). The DUT is a WLAN access point (AP).

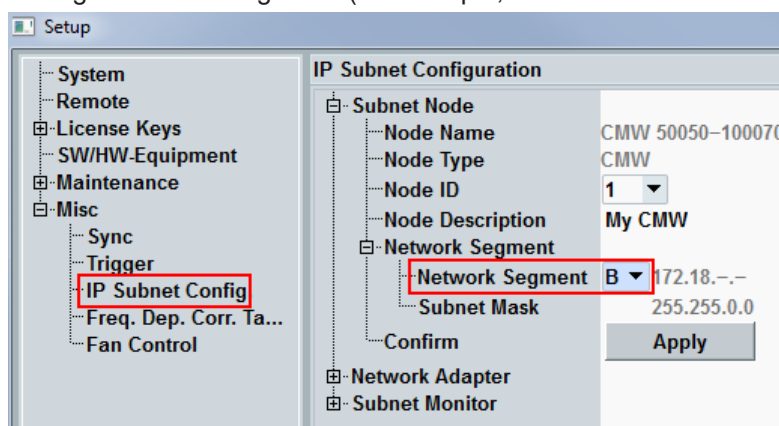
Station mode is supported with PER measurements, end-to-end data connections, RX statistics, and packet generators.

### Special requirements in station mode

- The configured WLAN channel must be the same as the one selected by the tested AP. The R&S CMW does not scan the supported frequency range.
- The configured standard must be compatible with the beacon format of the tested AP.
- The DUT - R&S CMW connection within the same network segment (for example: 172.22.x.y) , can lead to end-to-end connection failure. The following warning appears in the event log: "Subnet conflict: Check System IP subnet config".

The failure occurs, if the AP is within the same subnet as the R&S CMW's internal network segment. In this case, proceed as follows:

- Enter the "Setup" of the R&S CMW base system. To open the dialog box, press the [SETUP] key on the (soft-) front panel, or the  button.
- Select "Misc" > "IP Subnet Config"
- Change "Network Segment" (for example, from A to B -> 172.18.x.y)



**Limitations with 802.11ax**

Station mode for 802.11ax is supported for SISO and MIMO connections, HE\_SU, HE\_TB, and HE\_MU frame formats.

802.11ax features not applicable to station mode:

- Trigger frame generation (can only be initiated by an AP)
- NDP sounding (HE sounding protocol can only be initiated by an AP)
- Broadcast TWT not supported

Refer to:

- [Station mode](#) for the test setup
- [WPA/ WPA2/ WPA3 personal security mode](#) for the WLAN security mechanism
- [Station settings](#) for the configuration

### 2.2.5 Hotspot 2.0

Hotspot 2.0 is a specification from Wi-Fi Alliance. It was created to improve the end-user hotspot experience and to address the market/operator need for cellular offloading. The goal is to make the hotspot network access as easy as cellular network access.

In the corresponding operation mode, the R&S CMW simulates a Hotspot 2.0 access point to allow mobile devices to join a WiFi subscriber service automatically when entering a Hotspot 2.0 area.

To ensure that mobile devices always have seamless connection with WLAN and cellular network, the current release supports SIM/USIM authentication using EAP-SIM, EAP-AKA and EAP-AKA' (see [Chapter 2.2.3, "WLAN security"](#), on page 21).



Option R&S CMW-KS660 is required.

### 2.2.6 Signal properties

The IEEE 802.11 standard and its amendments specify different physical layer implementations. The following table gives an overview of the supported parts of the WLAN standards.



- 802.11n requires option R&S CMW-KS651
- 802.11ac requires option R&S CMW-KS651 and -KS656
- 802.11ax in AP mode, requires option R&S CMW-KS651, -KS656, and -KS657  
In addition, option R&S CMW-KS667 is required for multi-station testing.
- 802.11ax in station mode, requires option R&S CMW-KS651, -KS656, and -KS658
- Operating in the 5-GHz band (above 3.3 GHz in general) requires the 6-GHz option R&S CMW-KB036.

**Table 2-4: Supported IEEE 802.11 physical layer properties**

|                              | 802.11..                  | b <sup>5)</sup> | a <sup>5)</sup> | g <sup>5)</sup> | g (OFDM) <sup>5)</sup> | a/n | g/n <sup>5)</sup> | g (OFDM)/n | ac | ax <sup>1)</sup> |
|------------------------------|---------------------------|-----------------|-----------------|-----------------|------------------------|-----|-------------------|------------|----|------------------|
| Channel bandwidth            | 20 MHz                    | X               | X               | X               | X                      | X   | X                 | X          | X  | X                |
|                              | 40 MHz                    | -               | -               | -               | -                      | X   | X                 | X          | X  | X                |
|                              | 80 MHz                    | -               | -               | -               | -                      | -   | -                 | -          | X  | X                |
|                              | 160 MHz <sup>1)</sup>     | -               | -               | -               | -                      | -   | -                 | -          | X  | X                |
| Frequency band <sup>2)</sup> | 2.4 GHz                   | X               | -               | X               | X                      | -   | X                 | X          | -  | X                |
|                              | 5 GHz <sup>7)</sup>       | -               | X               | -               | -                      | X   | -                 | -          | X  | X                |
|                              | 6 GHz <sup>7,8)</sup>     | -               | -               | -               | -                      | -   | -                 | -          | -  | X                |
| Frame format                 | non-HT                    | X               | X               | X               | X                      | X   | X                 | X          | X  | X                |
|                              | HT MF                     | -               | -               | -               | -                      | X   | X                 | X          | X  | X                |
|                              | VHT                       | -               | -               | -               | -                      | -   | -                 | -          | X  | X                |
|                              | HE <sup>6)</sup>          | -               | -               | -               | -                      | -   | -                 | -          | -  | X                |
| Transmission scheme          | DSSS / CCK <sup>3)</sup>  | X               | -               | X               | -                      | -   | X                 | -          | X  | X                |
|                              | OFDM Non-HT <sup>4)</sup> | -               | X               | X               | X                      | X   | X                 | X          | X  | X                |
|                              | OFDM HT <sup>4)</sup>     | -               | -               | -               | -                      | X   | X                 | X          | X  | X                |
|                              | OFDM VHT <sup>4)</sup>    | -               | -               | -               | -                      | -   | -                 | -          | X  | X                |
|                              | OFDMA HE <sup>4)</sup>    | -               | -               | -               | -                      | -   | -                 | -          | -  | X                |

<sup>1)</sup> Both SUA partitions required (e.g. SUA1&2)

<sup>2)</sup> See [Table 2-7](#). Currently not limited - any channel frequency setting is possible.

<sup>3)</sup> DSSS 1 Mbit/s, DSSS 2 Mbit/s, CCK 5.5 Mbit/s, CCK 11 Mbit/s

<sup>4)</sup> See [Table 2-5](#)

<sup>5)</sup> SISO only

<sup>6)</sup> HE SU, HE MU, HE TB only

<sup>7)</sup> R&S CMW-KB036 is required

<sup>8)</sup> R&S CMW-Z800A is required

Table 2-5: OFDM transmission schemes: coding rate (MCS value)

|         | BPSK        | QPSK              | 16-QAM             | 64-QAM                       | 256-QAM          | 1024-QAM           |
|---------|-------------|-------------------|--------------------|------------------------------|------------------|--------------------|
| Non-HT  | 1/2, 3/4    | 1/2, 3/4          | 1/2, 3/4           | 2/3, 3/4                     | -                | -                  |
| HT SISO | 1/2 (MCS 0) | 1/2 (1), 3/4 (2)  | 1/2 (3), 3/4 (4)   | 2/3 (5), 3/4 (6), 5/6 (7)    | -                | -                  |
| HT MIMO | 1/2 (MCS 8) | 1/2 (9), 3/4 (10) | 1/2 (11), 3/4 (12) | 2/3 (13), 3/4 (14), 5/6 (15) | -                | -                  |
| VHT     | 1/2 (MCS 0) | 1/2 (1), 3/4 (2)  | 1/2 (3), 3/4 (4)   | 2/3 (5), 3/4 (6), 5/6 (7)    | 3/4 (8), 5/6 (9) | -                  |
| HE      | 1/2 (MCS 0) | 1/2 (1), 3/4 (2)  | 1/2 (3), 3/4 (4)   | 2/3 (5), 3/4 (6), 5/6 (7)    | 3/4 (8), 5/6 (9) | 3/4 (10), 5/6 (11) |

The available standards depend on the selected operation mode as shown in the following table.

Table 2-6: Standards supported per operation mode

| 802.11..               | b | a | g | g(OFDM) | a/n | g/n | g(OFDM)/n | ac | ax |
|------------------------|---|---|---|---------|-----|-----|-----------|----|----|
| AP                     | X | X | X | X       | X   | X   | X         | X  | X  |
| AP & multi-STA testing |   |   |   |         |     |     |           |    | X  |
| Station                | X | X | X | X       | X   | X   | X         | X  | X  |
| Hotspot 2.0            | - | - | - | -       | X   | X   | X         | X  | X  |

The frequency bands are divided into 20-MHz channels.

Table 2-7: Frequency bands

| Frequency band                          | Channel numbers | Carrier frequencies  |
|-----------------------------------------|-----------------|----------------------|
| 2.4 GHz                                 | 1 to 14         | 2412 MHz to 2484 MHz |
| 5 GHz <sup>1)</sup>                     | 34 to 196       | 5170 MHz to 6000 MHz |
| 6 GHz band <sup>1,2)</sup>              | 1 to 253        | 5955 MHz to 7125 MHz |
| <sup>1)</sup> R&S CMW-KB036 is required |                 |                      |
| <sup>2)</sup> R&S CMW-Z800A is required |                 |                      |



Country-specific regulations apply to both the allowed channels and maximum power levels within these channels. See [http://en.wikipedia.org/wiki/List\\_of\\_WLAN\\_channels](http://en.wikipedia.org/wiki/List_of_WLAN_channels) for a list of legally allowed WLAN channels.

The WLAN signaling application does not adhere to those regulations, e.g. it is possible to operate the AP on a channel that is "illegal" for the selected country code.

## 2.2.7 Trigger signals

The WLAN signaling application provides trigger signals that can be used by other R&S CMW applications to synchronize to the WLAN signal. These signals are especially useful to trigger WLAN TX measurements (options R&S CMW-KM65x). The signals can also be routed to the BNC connectors at the rear of the instrument.

The available trigger signals are described below.

To address the trigger signals in remote commands, use the following strings, with <i> replaced by the instance number of the signaling application:

- "WLAN Sig<i>: FrameTrigger"
- "WLAN Sig<i>: RXFrameTrigger"
- "WLAN Sig<i>: HE\_TB Trigger"

#### Frame trigger

This trigger signal is aligned to the start of the downlink MAC frames. It generates a pulse for the full duration of the TX frame. The trigger parameters are configurable, see ["TX Frame Trigger"](#) on page 89.

#### RX frame trigger

This trigger signal is aligned to the start of the uplink MAC frames. The trigger parameters are configurable, see ["RX Frame Trigger"](#) on page 90.

#### HE\_TB trigger

Trigger event after the transmission of trigger frame for the calculation of a time difference between the expected and actual arrival of an RX frame with HE TB PPDU.

### 2.2.8 Message logging

During a WLAN signaling test session, the R&S CMW can record the message flow between the tester and the DUT. You can view the message sequences with the universal message analyzer R&S CMWmars.

R&S CMWmars is the message analyzer for all R&S CMW signaling applications. It presents the logfile in various synchronized views that visualize the data from different perspectives, helping you to postprocess complex message logs in an intuitive and easy way.

The R&S CMWmars user interface combined with various tools and views helps you to narrow down the root cause of signaling protocol and lower layer problems, including the IP layer. It is well established as the standard analysis tool for customers such as chipset manufacturers, handset manufacturers and mobile network operators.



#### Pre-selecting message types

The message types that the R&S CMW replicates for logging can be selected in the "Debug" settings. A restricted subset of replicated message types saves processing power in the R&S CMW and can increase the maximum data throughput.

### 2.2.9 Internal fading

Testing under realistic air interface conditions is important in order to verify the receiver performance and the correct operation of the protocol stack implementation. For example, packet error rates can be evaluated.

The fading is only supported in AP mode using a whole SUA. The SUA comes with a fading simulator and AWGN generator that can be selectively enabled. It manipulates the generated downlink I/Q data stream to emulate typical signal conditions at the receiver.

The following instrument models and options are required to use the internal fading simulator:

- Both SUA partitions are required (e.g. Fader1&2)
- Option R&S CMW-KE100 for basic fading support and AWGN generator.
- Option R&S CMW-KE670 for fading profiles for WLAN 802.11n, ac, and ax. Refer to [Chapter A, "Predefined fading settings"](#), on page 633.



Internal fading is only supported for WLAN 802.11n, ac, and ax standards, in AP mode and channel bandwidth up to 40 MHz.

For MIMO fading, the R&S CMW transmits one or two streams via antennas Tx1 and Tx2 to the DUT antennas.

- At DUT antenna 1 (Rx1), you measure two fading channels: (Tx1->Rx1) + (Tx2->Rx1)
- At DUT antenna 2 (Rx2), you measure two fading channels: (Tx1->Rx2) + (Tx2->Rx2)

#### 2.2.9.1 Fading simulator

Multi-path fading is an effect which occurs in real world situations. A signal sent from the base station follows the direct line of sight and/or takes routes with reflections. The phase-shifted signals sum up at the receiving antenna. If the receiver is moving, there are also frequency-shifted signals.

The internal fading simulator supports multipath propagation conditions as described in [Chapter A, "Predefined fading settings"](#), on page 633.

A faded signal has a higher crest factor than an unfaded one. To avoid distortion, the baseband signal must be attenuated before entering the fading module. The necessary attenuation (insertion loss) depends on the selected fading profile.

In WLAN signaling, the insertion loss at the baseband level can be calculated automatically or set manually. It is automatically compensated on the HF level. The compensation implies a shift of the allowed DL power range to the same extent, but in opposite direction.



### 2.2.9.2 AWGN generator

Additional white Gaussian noise (AWGN) is typically modeled in receiver tests, because it can lead to a decrease of throughput. The quality of the received signal is affected by the ratio of the signal power to the surrounding traffic noise level (signal to noise ratio). The modulated signals from neighbor cells simply appear as noise. This effect is simulated by adding AWGN to the signal.

The internal fading module supports AWGN insertion with configurable bandwidth and signal to noise ratio. Insertion loss at the baseband level is calculated and compensated automatically at the HF.

### 2.2.10 IP communication

The IP configuration of the DUT and the instrument is used by the [Packet generator](#).

In station mode, WLAN signaling responds ARP requests for the assigned CMW IPv4 address and the CMW IPv4 address + 1. For instance, if the CMW is assigned the address 192.168.1.5, then it responds to ARP request for that address and 192.168.1.6. This feature can be used by single box tethering, to be able to ping in loop from DAU -> WLAN IP client -> RAT (e.g. LTE) -> DAU.

- For an R&S CMW without data application unit (DAU), the signaling application runs its own IPv4/IPv6 stack. DHCP and manual IP configuration is supported.
- If a DAU is installed, the WLAN signaling application relays incoming DHCP requests to the DAU. Refer to [Chapter 2.2.11, "Data end-to-end tests"](#), on page 29.

### 2.2.11 Data end-to-end tests

Use the data application unit (DAU) of the R&S CMW for data end-to-end tests.

You need the following options in addition to the WLAN signaling application:

- Data application unit (DAU, R&S CMW-B450x, mandatory)
- IPv4 enabler (R&S CMW-KA100, mandatory)  
IPv6 enabler (R&S CMW-KA150, optional)
- DAU measurements (R&S CMW-KM050, optional)
- Installed DAU software package 3.7.50.1 or later

If there is no space left for installation of a DAU in your instrument, you can install all these options in another R&S CMW and access the external DAU via LAN. For such a setup, you need R&S CMW-KA120 on both instruments. For more details, see section "Test setup with external DAU" in the DAU documentation.

### Setup procedure

To set up an end-to-end packet data connection, perform the following steps. The signaling application is initially switched off.

1. Set the IP version support required from the DAU. See [Chapter 2.3.10, "IP support"](#), on page 93.
2. If desired, configure the data end-to-end settings. See [Chapter 2.3.12, "Data end-to-end settings"](#), on page 96
  - a) For the operation mode "Station", you can configure additional destination IP addresses. The DAU routes traffic to these addresses to the DUT.
  - b) In AP mode, you can define the DUT IP address pool for the DHCP server of the DAU.
  - c) You can configure the access to an external DAU.
3. Prepare the other WLAN signaling settings as for a connection without the DAU.
4. Turn on the WLAN signaling application.
5. Establish a WLAN connection.
6. In the main view (or the configuration dialog under [IP support](#)) of the signaling application, note the IPv4 address and/or IPv6 prefix assigned to the DUT. You need this information for ping tests and iperf throughput tests.
7. Check the IP connectivity of the DUT, for example by pinging the DUT from the DAU or vice versa.

If you have connected your DUT to a PC with Windows operating system, you can open a command prompt at the PC and type `IPconfig` to display the active IP configuration.
8. Open the "Data Application Measurements" application.
9. At the top of the GUI, select the WLAN signaling application (parameter "Select RAN").
10. Initiate an end-to-end data transfer as described in the DAU user manual.

### 2.2.12 Packet generator

The WLAN signaling application includes packet generators that allow you to send IP traffic towards the DUT. The R&S CMW can run up to three packet generators.

The R&S CMW supports the generation of ICMP echo request ("Ping") packets and UDP packets including TID assignment. QoS TID control applies scheduling according to the TID priority or round robin.

The ICMP echo request packets are targeted at the DUT's IP stack. The DUT is expected to answer with a well-defined echo reply packet whose payload is identical to the payload of the corresponding request. "Pinging" triggers the DUT to transmit a burst of a certain length which is useful for combined signal path measurements.

UDP packets are used to put background traffic on the air interface, for example to prevent the DUT from entering the power save mode.

### 2.2.13 EDCA

Enhanced distributed channel access (EDCA) defines channel access mechanisms including the backoff time concept. The backoff time ensures channel access fairness between multiple STAs. EDCA also prioritizes different flows of TX traffic identified by TID and categorized by access class independently within a STA. Thus, a higher priority access class (AC) gains channel access before a lower priority AC.

#### EDCA Parameter Set

EDCA uses signaling information provided in the EDCA Parameter Set element.

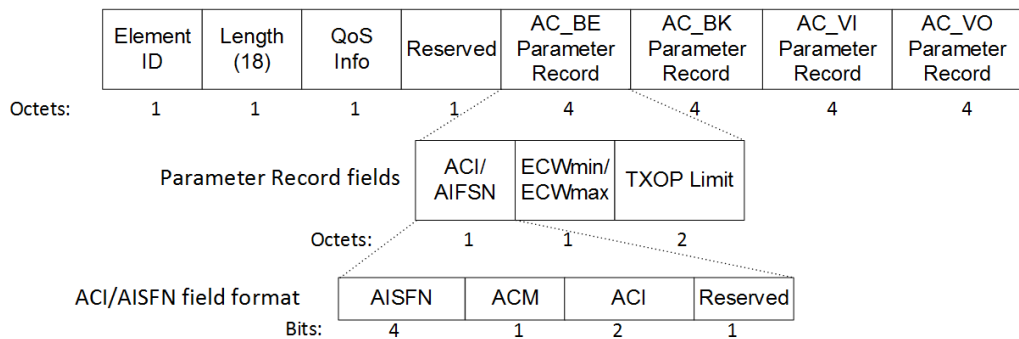


Figure 2-9: EDCA parameter set

The EDCA Parameter Set element uses AISFN values 2 to 15.

#### MU EDCA Parameter Set

802.11ax introduces the MU EDCA Parameter Set to provide the AP additional control over the scheduling of uplink MU channel access.

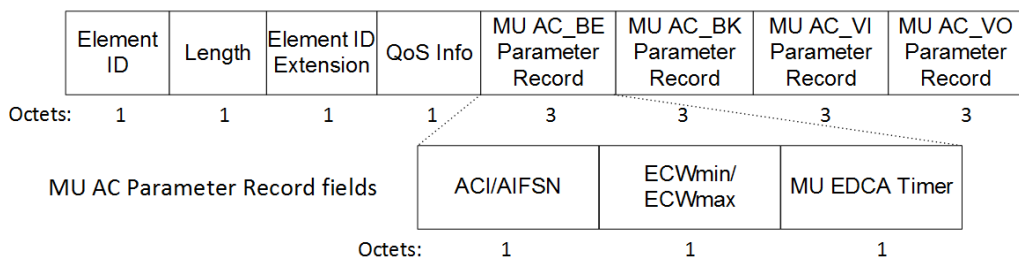


Figure 2-10: MU EDCA parameter set

The MU EDCA Parameter Set uses by default AISFN values 0 to disable channel access. This setting is only active for a finite period of time controlled by the MU EDCA timer. When the timer expires, the channel access is re-enabled and controlled by EDCA parameter set.

Using the transmission basic trigger frame (AP) -> HE TB response (STA) -> basic trigger frame (AP)... the AP forces STA to reset the MU EDCA timer to the signaled value. During the transmission of basic trigger frames at an interval less than the MU EDCA timer, the STA never attempts channel access; except as a result of a trigger frame. This way, the AP minimizes channel access collisions and maximizes the efficiency of traffic.

## 2.3 GUI reference

The following chapters provide detailed reference information on the parameters of the WLAN signaling application.

Most parameters are configured via a single configuration dialog. The packet error rate measurement ("WLAN PER") is configured and controlled via a separate view.

The GUI reference is structured as follows.

|                                                          |     |
|----------------------------------------------------------|-----|
| • <a href="#">Signaling view</a> .....                   | 32  |
| • <a href="#">Signaling and connection control</a> ..... | 44  |
| • <a href="#">Using the shortcut softkeys</a> .....      | 45  |
| • <a href="#">Global settings</a> .....                  | 46  |
| • <a href="#">MIMO settings</a> .....                    | 49  |
| • <a href="#">RF settings</a> .....                      | 50  |
| • <a href="#">Internal fading</a> .....                  | 55  |
| • <a href="#">Connection settings</a> .....              | 57  |
| • <a href="#">Trigger settings</a> .....                 | 89  |
| • <a href="#">IP support</a> .....                       | 93  |
| • <a href="#">Packet generator settings</a> .....        | 95  |
| • <a href="#">Data end-to-end settings</a> .....         | 96  |
| • <a href="#">Shortcut configuration</a> .....           | 98  |
| • <a href="#">Message monitoring settings</a> .....      | 98  |
| • <a href="#">Debug configuration</a> .....              | 99  |
| • <a href="#">Statistics settings</a> .....              | 100 |
| • <a href="#">Station configuration tabs</a> .....       | 101 |
| • <a href="#">PER measurement</a> .....                  | 102 |

### 2.3.1 Signaling view

In its left part, the signaling view displays status information. The most important settings are configurable in the right part.

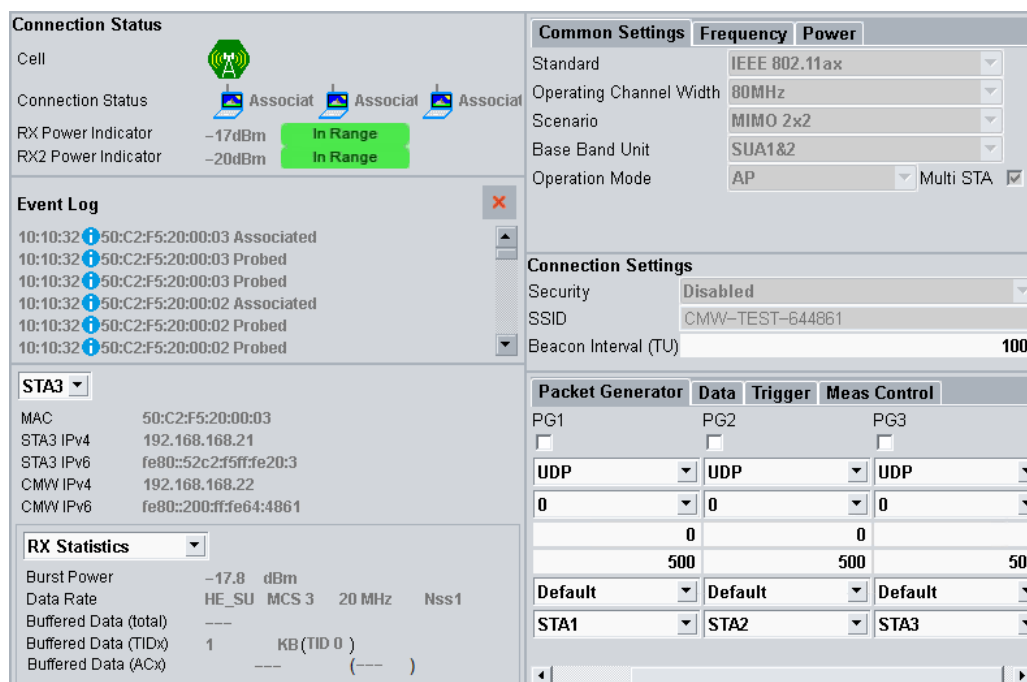


Figure 2-11: WLAN signaling view

For descriptions of the individual panes of the view, refer to the subchapters.

|                                                      |    |
|------------------------------------------------------|----|
| • <a href="#">Connection status</a> .....            | 33 |
| • <a href="#">Event log</a> .....                    | 35 |
| • <a href="#">DUT / UE info / STAx</a> .....         | 35 |
| • <a href="#">RX statistics</a> .....                | 36 |
| • <a href="#">HE capabilities</a> .....              | 37 |
| • <a href="#">HE TB info</a> .....                   | 38 |
| • <a href="#">RX trigger frames statistics</a> ..... | 38 |
| • <a href="#">Settings</a> .....                     | 39 |
| • <a href="#">Statistics</a> .....                   | 42 |

### 2.3.1.1 Connection status

Access: WLAN signaling main view > "Connection Status" pane.

The "Connection Status" pane displays status information for the simulated WLAN entity and the connection to the DUT.

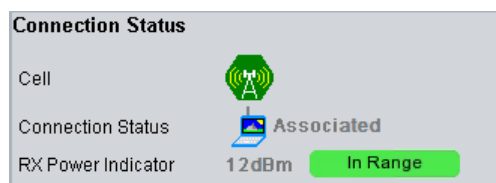


Figure 2-12: Connection status pane of the main view

The displayed information is described in the following.

|                            |    |
|----------------------------|----|
| Cell.....                  | 34 |
| Connection Status.....     | 34 |
| Station SSID.....          | 34 |
| EAP Connection Status..... | 34 |
| RX Power Indicator.....    | 34 |

### Cell

The cell icon indicates the overall signal state (green = on, gray = off, additional  = pending).

For control of the states, see [Chapter 2.3.2, "Signaling and connection control"](#), on page 44.

Remote command:

`SOURce:WLAN:SIGN<i>:STATe`

`SOURce:WLAN:SIGN<i>:STATe:ALL?`

### Connection Status

Displays the current state (or state transition) for all operation modes. If [Multi STA](#) is enabled, the connection status is displayed for each station in order from left to right: STA1, STA2, STA3.

For background information, see [Chapter 2.2.1.3, "Connection status"](#), on page 18.

Remote command:

`SOURce:WLAN:SIGN<i>:STATe:ALL?`

`FETCh:WLAN:SIGN<i>:PSWitched:STATe?`

`FETCh:WLAN:SIGN<i>:SECond:STATe?`

`FETCh:WLAN:SIGN<i>:THIRd:STATe?`

### Station SSID

The SSID of the associated AP is displayed in "Station" mode.

Remote command:

`SENSe:WLAN:SIGN<i>:UESinfo:APSSid?`

### EAP Connection Status

The current state of the EAP connection is displayed for the security mode "WPA / WPA2 Enterprise".

Remote command:

`SENSe:WLAN:SIGN<i>:SINfo:EAPStat?`

### RX Power Indicator

Indicates the RX power, received at the R&S CMW. This information helps you to set the expected power, see ["Exp. Peak Envelope Power"](#) on page 55.

The information is displayed for all operation modes. For MIMO scenarios, RX power indicator per DUT's antenna is displayed.

In addition, the R&S CMW also evaluates the quality of RX power:

- - - -: no signal from the DUT detected
- **In range:** DUT's power is in the expected range, according to the expected power.  
The highest MCS for the current WLAN mode can be decoded without IQ clipping.

- **Overdriven:** the DUT's power is above the expected range, without IQ clipping occurs. The receiver cannot decode a given burst. An inappropriate external attenuation is set, or the DUT does not transmit at the correct power.
- **Underdriven:** decoding of a received burst is almost impossible. An inappropriate external attenuation is set, the connecting cable is defective, or the DUT does not transmit at the correct power.

Remote command:

```
SENSe:WLAN:SIGN<i>:SINFo[:ANTenna<n>]:RXPindicator?
```

### 2.3.1.2 Event log

Access: WLAN signaling main view > "Event Log" pane.

The event log pane reports events and errors such as startup events and connection status changes.

The button to the right clears the displayed event entries.

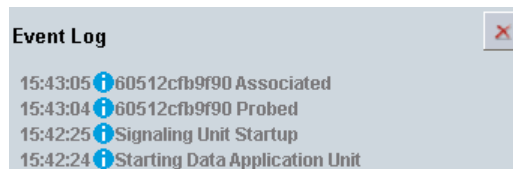


Figure 2-13: Event log pane of the main view

#### Event log entries

Each entry consists of a timestamp, an icon indicating the category of the event and a short text describing the event.

Meaning of the category icons: = information, warning, error

To clear the displayed event entries, press the button to the right.

Remote command:

```
SENSe:WLAN:SIGN<i>:ELOGging:ALL?
```

```
CLEAn:WLAN:SIGN<i>:ELOGging
```

### 2.3.1.3 DUT / UE info / STAx

Access: WLAN signaling main view > "DUT / UE Info" pane; or

Access for enabled **Multi STA**: WLAN signaling main view > select "STAx" in the field below the "Event Log" pane

The "DUT / UE Info" / "STAx" pane presents information about the associated device under test and the IP address of the R&S CMW.

| DUT / UE Info |                        |
|---------------|------------------------|
| MAC           | 50:C2:F5:20:00:03      |
| UE IPv4       | 192.168.168.21         |
| CMW IPv4      | 192.168.168.22         |
| UE IPv6       | fe80::52c2:f5ff:fe20:3 |
| CMW IPv6      | fe80::200:ff:fe64:4861 |

Figure 2-14: DUT / UE capabilities in main view

### MAC

The MAC address of the associated DUT is a 48-bit number, displayed as hexadecimal number. It uniquely identifies an IEEE 802.11 device. An AP registers a station's MAC address during association.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:UECapability:MAC:ADDRess?`

### IPv4 / IPv6 address

Displays the detected IPv4 and IPv6 addresses of the instrument and the connected DUT.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:UEAddress:IPV<n>?`

`SENSe:WLAN:SIGN<i>:UESinfo:CMWaddress:IPV<n>?`

## 2.3.1.4 RX statistics

Access: WLAN signaling main view > select "RX Statistics" in the field below the "DUT / UE Info" or "STAx" pane.

Displays statistical information about the signal received from the device under test.

| RX Statistics         |       |            |        |      |
|-----------------------|-------|------------|--------|------|
| Burst Power           | -17.8 | dBm        |        |      |
| Data Rate             | HE_SU | MCS 3      | 20 MHz | Nss1 |
| Buffered Data (total) | ---   |            |        |      |
| Buffered Data (TIDx)  | 1     | KB (TID 0) |        |      |
| Buffered Data (ACx)   | ---   |            | (---   | )    |

Figure 2-15: RX statistics in main view

### Burst Power

Average power of the last burst received from the DUT, indicating the quality of the physical link.

The power is measured by the R&S CMW over the preamble. A value is only available in associated state.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:RXBPower?`



**Data Rate**

Displays information related to the data rate of the DUT signal, for example the frame format, the modulation and coding scheme, the channel bandwidth and the number of spatial streams.

The displayed values represent the most recent and most prevalent data frame received. Ack frames are not included. Data received during each user-configured reporting interval is determined (see [Stats Settings \(hotkey\)](#)). Of the frames accumulated during that time, the value that appears most often is selected. The display is only updated if this value is different from the previous value.

Remote command:

```
SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:DRATe?
```

**Buffered Data**

Indicates asynchronous buffer status report (A-BSR) and QoS control buffer size report. An STA can asynchronously indicate the amount of buffered data to be transmit to assist in the scheduling decisions of the AP.

Option R&S CMW-KS657 is required.

Remote command:

```
SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:ABSReport?
```

**2.3.1.5 HE capabilities**

Access: WLAN signaling main view > select "HE Capabilities" in the field below the "DUT / UE Info" or "STAx" pane.

Displays high efficiency (HE) capabilities of the DUT. Refer to IEEE Std 802.11ax-2021, chapter 9.4.2.248 HE Capabilities Element.

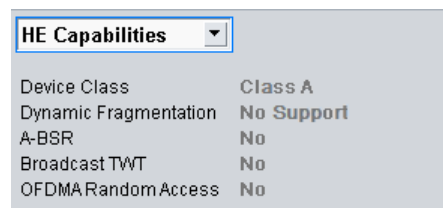


Figure 2-16: HE capabilities in main view

**Device Class**

Indicates device class for a non-AP STA based on its TX power and RSSI measurement accuracy capabilities.

Remote command:

```
SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?
```

**Dynamic Fragmentation**

Indicates fragmentation support including supported level.

Remote command:

```
SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?
```

**A-BSR**

Indicates support of an asynchronous buffer status report (BSR) control field.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?`

**Broadcast TWT**

Indicates support of broadcast target wake time (TWT) operation.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?`

**OFDMA Random Access**

Indicates support of OFDMA random access procedure.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?`

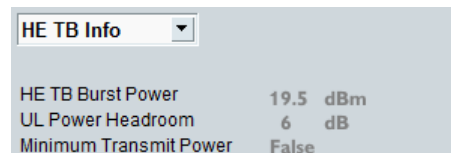
**2.3.1.6 HE TB info**

Access: WLAN signaling main view > select "HE TB Info" in the field below the "DUT / UE Info" or "STAx" pane.

Displays the main power connection parameters of HE trigger based bursts.

This section is relevant for 802.11ax standard, operating mode AP.

Option R&S CMW-KS657 is required.



*Figure 2-17: HE TB info in main view*

**HE TB Info**

Displays information related to uplink power headroom (UPH) control.

UL power headroom indicates the difference between current HE TB TX power and maximum TX power.

Minimum transmit power flag is set when the HE TB is sent at the STA's minimum TX power.

Remote command:

`SENSe:WLAN:SIGN<i>:STA<s>:HETBinfo:UPHinfo?`

**2.3.1.7 RX trigger frames statistics**

Access: WLAN signaling main view > select "RX Trigger Frames" in the field below the "DUT / UE Info" pane.

Displays the frames received by the instrument in station mode.

Option R&S CMW-KS658 is required.

| RX Trigger Frames   |      |
|---------------------|------|
| Total TF Received   | ---- |
| TF addressed to STA | ---- |

Figure 2-18: RX trigger frames statistics in main view

### RX Trigger Frames

Queries the total trigger frames received and the No. of trigger frames directed to the station (R&S CMW).

Remote command:

`SENSe:WLAN:SIGN<i>:UESInfo:RXTRigframe?`

### 2.3.1.8 Settings

Access: WLAN signaling main view > right panes.

The main view provides the most important settings for fast access.

| Common Settings         |               | Frequency | Power                                         |
|-------------------------|---------------|-----------|-----------------------------------------------|
| Standard                | IEEE 802.11ax |           |                                               |
| Operating Channel Width | 160MHz        |           |                                               |
| Scenario                | Standard Cell |           |                                               |
| Base Band Unit          | SUA1&2        |           |                                               |
| Operation Mode          | AP            |           | Multi STA <input checked="" type="checkbox"/> |

| Connection Settings  |                 |
|----------------------|-----------------|
| Security             | WPA2 Enterprise |
| SSID                 | CMW-AP          |
| Beacon Interval (TU) | 100             |

| Packet Generator |                          | Data                     | Trigger | Meas Control |
|------------------|--------------------------|--------------------------|---------|--------------|
|                  | PG1                      | PG2                      |         |              |
| Enable           | <input type="checkbox"/> | <input type="checkbox"/> |         |              |
| Protocol         | ICMP                     | ICMP                     |         |              |
| TID              | 0                        | 1                        |         |              |
| Interval (TU)    | 100                      | 100                      |         |              |
| Size (Byte)      | 500                      | 500                      |         |              |
| Payload Type     | Default                  | Default                  |         |              |
| Destination      | STA1                     | STA1                     |         |              |

Figure 2-19: Settings in the main view

All settings are also configurable via the main configuration dialog. Refer to the following chapters.

- "Common Settings" tab:

| Common Settings         | Frequency     | Power                              |
|-------------------------|---------------|------------------------------------|
| Standard                | IEEE 802.11ax |                                    |
| Operating Channel Width | 80MHz         |                                    |
| Scenario                | Standard Cell |                                    |
| Base Band Unit          | SUA1&2        |                                    |
| Operation Mode          | AP            | Multi STA <input type="checkbox"/> |

- "Standard" on page 47
- "Operating Channel Width" on page 47
- "Scenario" on page 47
- "Base Band Unit" on page 48
- "Operation Mode" on page 48
- "Fading" on page 49
- "Multi STA" on page 48

- "Frequency" tab:

| Common Settings            | Frequency    | Power |
|----------------------------|--------------|-------|
| Center Frequency / Channel | 5250.000 MHz | 50    |
| Np20 Index                 | 0            |       |
| Np20 Frequency / Channel   | 5180.000 MHz | 36    |

- "Center Frequency / Channel" on page 53
- "Np20 Index" on page 54
- "Np20 Frequency / Channel" on page 54

- "Power" tab:

| Common Settings             | Frequency | Power |
|-----------------------------|-----------|-------|
| TX Burst Power              | -20.0 dBm |       |
| RX Expected PEP             | 0.0 dBm   |       |
| Approximate RX Burst Power  | -13.0 dBm |       |
| RX2 Expected PEP            | 0.0 dBm   |       |
| Approximate RX2 Burst Power | -13.0 dBm |       |

- "TX Burst Power" on page 55
- "Exp. Peak Envelope Power" on page 55
- "Approximate Burst Power" on page 55

- "Connection Settings" pane:

| Connection Settings  |          |
|----------------------|----------|
| Security             | Disabled |
| SSID                 | CMW-AP   |
| Beacon Interval (TU) | 100      |

- "Mode" on page 80
- "SSID" on page 62

- ["Beacon Interval"](#) on page 61

- **"Packet Generator" tab:**

| Packet Generator | Data                     | Trigger                  | Meas Control             |
|------------------|--------------------------|--------------------------|--------------------------|
| Enable           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Protocol         | ICMP                     | ICMP                     | ICMP                     |
| TID              | 0                        | 0                        | 0                        |
| Interval (TU)    | 100                      | 100                      | 100                      |
| Size (Byte)      | 500                      | 500                      | 500                      |
| Payload Type     | Default                  | Default                  | Default                  |
| Destination      | STA1                     | STA1                     | STA1                     |

See [Chapter 2.3.11, "Packet generator settings"](#), on page 95

- **"Data" tab:**

| Packet Generator      | Data                                                                          | Trigger | Meas Control |
|-----------------------|-------------------------------------------------------------------------------|---------|--------------|
| Configuration         | <input checked="" type="radio"/> Automatic <input type="radio"/> User Defined |         |              |
| Frame Format          | HE_MU                                                                         |         |              |
| Channel Bandwidth     | 40MHz                                                                         |         |              |
| User1 Spatial Streams | 2                                                                             |         |              |
| User1 Rate            | MCS Index 7                                                                   |         |              |
| Guard Interval        | 1.6 $\mu$ s                                                                   |         |              |
| LTF Type              | 2x (6.4 $\mu$ s)                                                              |         |              |
| Coding Type           | LDPC                                                                          |         |              |

- ["Configuration"](#) on page 72
- ["Frame Format"](#) on page 73
- ["Non-HT, HT, VHT, HE Frame"](#) on page 73
- ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

- **"Meas Control" tab:**

| Packet Generator   | Data                                | Meas Control |
|--------------------|-------------------------------------|--------------|
| RX Format          | VHT Bursts                          |              |
| RX Bandwidth       | 20MHz                               |              |
| RX Rate            | MCS Index 5                         |              |
| Rate Restriction   | <input checked="" type="checkbox"/> |              |
| RX Spatial Streams | 1                                   |              |

See ["RX Frame Trigger"](#) on page 90

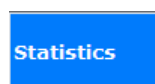
- **"Trigger" tab (only for 802.11ax in AP mode):**

See [Chapter 2.3.8.10, "Trigger frame settings"](#), on page 75

### 2.3.1.9 Statistics

Access: WLAN signaling main view > "Statistics" button to the right > one of the related hotkeys at the bottom

The related dialogs provide information concerning user data traffic.



| PG Data | RX PSDU Info | Stats Settings | Reset Stats |
|---------|--------------|----------------|-------------|
|---------|--------------|----------------|-------------|

#### PG Data (hotkey)

Opens a dialog with packet generator data. Table shows the statistics on generated ICMP and UDP packets and received ICMP packets. Information per packet generation is provided.

In three columns, the table displays the No. of frames, bytes and percentage of *ICMP Resp Received* versus *ICMP Req Sent*.

| Packet Generator Data |        |       |       |
|-----------------------|--------|-------|-------|
| Packet Generator 1    |        |       |       |
|                       | Frames | Bytes | %     |
| ICMP Req Sent         | 2      | 2944  |       |
| ICMP Resp Received    | 2      | 2944  | 100.0 |
| UDP Sent              | 0      | 0     |       |
| Packet Generator 2    |        |       |       |
|                       | Frames | Bytes | %     |
| ICMP Req Sent         | 4      | 1200  |       |
| ICMP Resp Received    | 4      | 1200  | 100.0 |
| UDP Sent              | 0      | 0     |       |
| Packet Generator 3    |        |       |       |
|                       | Frames | Bytes | %     |
| ICMP Req Sent         | 8      | 8000  |       |
| ICMP Resp Received    | 8      | 8000  | 100.0 |
| UDP Sent              | 0      | 0     |       |

**RX PSDU Info (hotkey)**

Opens a dialog with statistics on received (RX) traffic data. Statistics are grouped according to a frame format, data rate / MCS index, channel bandwidth, and spatial stream, typically one row per group.

For each group, the following current (within the latest statistical interval) and expected total values are displayed:

- No. of received frames
- Data volume in MB
- Data rate in Mbit/s
- RX power level in dBm

RX PSDU Info

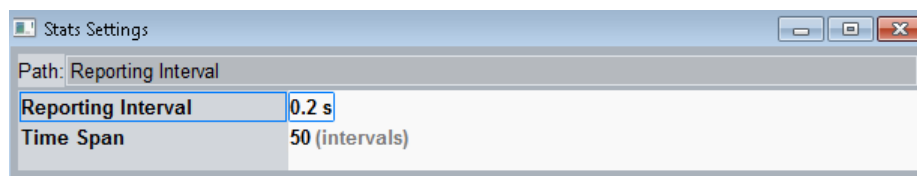
|         | BW      | Nss  | Frames: Curr. | Total | Megabytes: Curr. | Total  | Mbits/s: Curr. | Total | dBm: Curr. | Total |
|---------|---------|------|---------------|-------|------------------|--------|----------------|-------|------------|-------|
| Non-HT  |         |      |               |       |                  |        |                |       |            |       |
| 6 Mbps  | 20 MHz  | Nss1 | 0             | 98    | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | -7.1  |
| 12 Mbps | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| 9 Mbps  | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| 24 Mbps | 20 MHz  | Nss1 | 70            | 96252 | 0.00             | 4.92   | 0.1            | 0.1   | 11.8       | 12.1  |
| 48 Mbps | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| HT      |         |      |               |       |                  |        |                |       |            |       |
| VHT     |         |      |               |       |                  |        |                |       |            |       |
| HE_SU   |         |      |               |       |                  |        |                |       |            |       |
| MCS 0   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 1   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 2   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 3   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 4   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 5   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 6   | 160 MHz | Nss1 | 0             | 1     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | -19.1 |
| MCS 7   | 160 MHz | Nss1 | 9             | 12959 | 0.15             | 238.26 | 6.1            | 6.4   | 12.3       | 12.1  |
| MCS 8   | 160 MHz | Nss1 | 0             | 46    | 0.00             | 0.74   | 0.0            | 0.0   | 0.0        | -3.6  |
| MCS 9   | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 8   | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 10  | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 11  | 160 MHz | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 9   | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 10  | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| MCS 6   | 20 MHz  | Nss1 | 0             | 0     | 0.00             | 0.00   | 0.0            | 0.0   | 0.0        | 0.0   |
| HE_TB   |         |      |               |       |                  |        |                |       |            |       |

Remote command:

```
SENSe:WLAN:SIGN<i>:UESinfo:RXPSdu:HEMU?
SENSe:WLAN:SIGN<i>:UESinfo:RXPSdu:HEMU?
SENSe:WLAN:SIGN<i>:UESinfo:RXPSdu:HEMU?
SENSe:WLAN:SIGN<i>:UESinfo:RXPSdu:HEMU?
SENSe:WLAN:SIGN<i>:UESinfo:RXPSdu:HEMU?
SENSe:WLAN:SIGN<i>:UESinfo:RXPSdu:HEMU?
```

**Stats Settings (hotkey)**

Opens a dialog to configure the reporting interval and time span.



Stats Settings

Path: Reporting Interval

Reporting Interval: 0.2 s

Time Span: 50 (intervals)

For detailed description, refer to [Chapter 2.3.16, "Statistics settings"](#), on page 100.

Remote command:

```
CONFigure:WLAN:SIGN<i>:UESinfo:SETTings
```

**Reset Stats (hotkey)**

Clears entries in all statistic tables.

Remote command:

`CONFigure:WLAN:SIGN<i>:UESinfo:RESet`

**2.3.2 Signaling and connection control**

The individual connection states are controlled via the ON | OFF key, via hotkeys and via the DUT.

The available hotkeys depend on configured settings and the current connection state. All possible hotkeys are described in the following.

For background information, refer to [Chapter 2.2.1.3, "Connection status"](#), on page 18.



|                                                                 |    |
|-----------------------------------------------------------------|----|
| <a href="#">ON   OFF (key) / WLAN Signaling (softkey)</a> ..... | 44 |
| <a href="#">Connect (hotkey)</a> .....                          | 44 |
| <a href="#">Disconnect / Disconnect ST (hotkey)</a> .....       | 44 |
| <a href="#">Reconnect (hotkey)</a> .....                        | 45 |

**ON | OFF (key) / WLAN Signaling (softkey)**

To turn the signal transmission on or off, use the ON | OFF key or right-click the softkey. The current state is shown by the softkey. The signal transmission can be switched off anytime, independent of the current connection state. A yellow hour glass symbol indicates that the signaling generator is turning on or off.

In many modes, turning on the signal transmission and the DUT is sufficient to initiate a connection between the tester and the DUT.

Remote command:

`SOURce:WLAN:SIGN<i>:STATe`

`SOURce:WLAN:SIGN<i>:STATe:ALL?`

**Connect (hotkey)**

This hotkey requests to establish an association to the AP under test.

The hotkey is only visible if all the following apply:

- The operation mode equals "Station", see ["Operation Mode"](#) on page 48.
- The connection mode equals "Manual", see ["Connection Mode"](#) on page 63.
- The signaling state is ON and no association is established.

Remote command:

`CALL:WLAN:SIGN<i>:ACTion:STATION:CONNECT`

**Disconnect / Disconnect ST (hotkey)**

This hotkey is only visible in associated state. It requests the existing association to be terminated (disassociated and deauthenticated).



If [Multi STA](#) is enabled, the first related button from left is "Disconnect" controlling STA1. The second and third buttons are "Disconnect ST" buttons related to STA2 and STA3.

The order of buttons is fixed. For a station that is not connected, the related button is blank.

Remote command:

```
CALL:WLAN:SIGN<i>:STA<s>:ACTion:DISConnect
```

### Reconnect (hotkey)

This hotkey requests to re-establish the existing association. It has the same effect as a disconnect, followed immediately by a connect.

The hotkey is only visible if all the following apply:

- The operation mode equals "Station", see ["Operation Mode"](#) on page 48.
- The connection mode equals "Manual", see ["Connection Mode"](#) on page 63.
- An association is established.

Remote command:

```
CALL:WLAN:SIGN<i>:ACTion:STATion:REConnect
```

## 2.3.3 Using the shortcut softkeys

When using the WLAN signaling application and a WLAN measurement in parallel, use a shortcut softkey to switch to the measurement.



Using one of the softkeys ensures that the measurement is configured compatible to the settings of the signaling application. When you use the softkey to switch to the multi-evaluation measurement, the combined signal path scenario is activated automatically in the measurement.

Consequences:

- The measurement and the signaling application can be used in parallel, i.e. both DL signal transmission and measurement can be switched on.
- The signaling RF settings are also used for the measurement.
- Some input signal settings are configured compatible with the signaling settings.
- A suitable trigger signal provided by the signaling application is selected as trigger source for the measurement.

If the softkey label equals "Go to...", the softkey opens a dialog box with a list of all available measurements. If the softkey label indicates a measurement name instead of "Go to...", this measurement has been assigned to the softkey as fixed target, see [Chapter 2.3.13, "Shortcut configuration"](#), on page 98.

Three shortcut softkeys are available and can be set to different fixed targets.

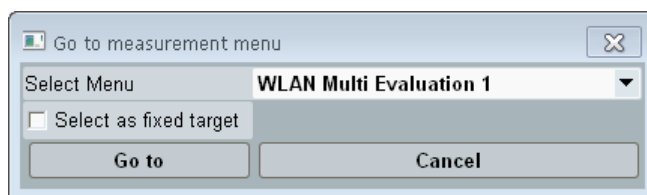


Figure 2-20: Dialog box opened by "Go to..." softkey

### Select Menu

Selects the target measurement you want to switch to.

### Select as fixed target

Sets the selected measurement as fixed target of the softkey. The softkey label indicates the measurement name and switches directly to the selected target without opening the dialog box.

When the dialog box has been disabled, you can still change the target measurement or re-enable the dialog box using the configuration menu, see [Chapter 2.3.13, "Short-cut configuration"](#), on page 98.

### Go to / Cancel

Press the "Go to" button to switch to the selected measurement or the "Cancel" button to abort.

## 2.3.4 Global settings

Access: "WLAN Signaling" softkey > "Config" hotkey > top of the configuration dialog.

The global settings are at the top of the configuration dialog. Configure them according to your test scenario, before configuring any other settings.

If [Multi STA](#) is enabled, the main configuration tree appears in a tab "CMW" and additional tabs "Station 1", "Station 2" and "Station 3" are displayed. See also [Chapter 2.3.17, "Station configuration tabs"](#), on page 101.

Depending on the global settings, only a particular set of actions and parameters is available and the GUI changes accordingly.

Parameters can even have different meanings depending on the global settings. For an example, see ["SSID"](#) on page 62.

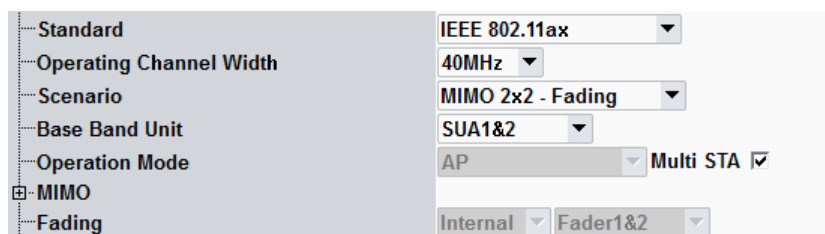


Figure 2-21: Global settings for scenario with fading

|                              |    |
|------------------------------|----|
| Standard.....                | 47 |
| Operating Channel Width..... | 47 |
| Scenario.....                | 47 |
| Base Band Unit.....          | 48 |
| Operation Mode.....          | 48 |
| Multi STA.....               | 48 |
| MIMO.....                    | 49 |
| Fading.....                  | 49 |

### Standard

Selects the IEEE 802.11 standard to be used. The available values depend on the selected operation mode.

For background information, see [Chapter 2.2.6, "Signal properties"](#), on page 24.

For the required options, see [Chapter 2, "WLAN signaling"](#), on page 13.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:STANdard`

### Operating Channel Width

Selects the channel bandwidth.

Option R&S CMW-KS651 is required for the bandwidth = 40 MHz.

Option R&S CMW-KS656 is required for the bandwidth  $\geq$  80 MHz.

Note that only contiguous 160 MHz channel bandwidth is supported, 80+80 MHz bandwidth (non-contiguous) is not supported.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:OCWidth`

### Scenario

Different test scenarios require different sets of parameters. Selecting a scenario hides/shows parts of the GUI as required by the scenario. The active scenario determines, for example, whether MIMO and fading are used.

"Standard Cell" SISO operation. Select the role of the R&S CMW and the DUT via the operation mode.

"MIMO 2x2" MIMO operation in downlink and uplink. MIMO connections are supported for standards 802.11n, ac, and ax.  
For further configuration, see [Chapter 2.3.5, "MIMO settings"](#), on page 49.  
Option R&S CMW-KS670 is required.

"Fading" scenarios "Standard Cell" or "MIMO 2x2" plus fading and/or AWGN insertion. Fading is for standards 802.11n, ac, ax and for AP operation mode. An additional parameter "Fading" is displayed, refer to ["Fading"](#) on page 49.  
Options R&S CMW-KE100 and R&S CMW-KE670 are required.

Remote command:

`ROUTE:WLAN:SIGN<i>?`

`ROUTE:WLAN:SIGN<i>:SCENario?`

`ROUTE:WLAN:SIGN<i>:SCENario:SCEl:l:FLEXible`

```
ROUTE:WLAN:SIGN<i>:SCENario:MIMO:FLEXible
ROUTE:WLAN:SIGN<i>:SCENario:SCFading:FLEXible
ROUTE:WLAN:SIGN<i>:SCENario:MIMFading:FLEXible
```

### Base Band Unit

Selects the signaling unit to be used. This setting is especially useful for multi-CMW setups, to select the instrument.

Each SUA module has two SUA partitions numbered as follows:

- Partition 1, 2 for SUA No. 1
- Partition 3, 4 for SUA No. 2, and so on.

See also [Chapter 2.5.1.2, "Values for signal path selection"](#), on page 124.

A prefix "1." to "4." selects the CMW in a multi-CMW setup.

Both partitions of an SUA are required for:

- 802.11ax
- MIMO scenarios
- 160 MHz channel bandwidth (contiguous)

Some examples:

- "SUA2" selects SUA partition 2 in a single-CMW setup.
- "SUA1&2" selects SUA partitions 1 and 2 in a single-CMW setup.
- "3.SUA1" selects SUA partition 1 in CMW number 3 in a multi-CMW setup.
- "2.SUA3&4" selects SUA partitions 3 and 4 in CMW number 2 in a multi-CMW setup.

Remote command:

```
ROUTE:WLAN:SIGN<i>:SCENario:SCEl1:FLEXible
ROUTE:WLAN:SIGN<i>:SCENario:MIMO:FLEXible
ROUTE:WLAN:SIGN<i>:SCENario:SCFading:FLEXible
ROUTE:WLAN:SIGN<i>:SCENario:MIMFading:FLEXible
```

### Operation Mode

The operation mode is configurable for scenario "Standard Cell".

For the supported IEEE standards, refer to [Table 2-6](#).

|               |                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "AP"          | The R&S CMW simulates an access point in infrastructure mode. The DUT is a WLAN station.<br>For 802.11ax, option R&S CMW-KS657 is required.                                              |
| "Station"     | The R&S CMW simulates a WLAN station. The DUT is an access point.<br>For 802.11ax, option R&S CMW-KS658 is required.                                                                     |
| "Hotspot 2.0" | The R&S CMW simulates a Wi-Fi Hotspot 2.0 access point. The DUT is a WLAN station. Option R&S CMW-KS660 is required. See also <a href="#">Chapter 2.2.5, "Hotspot 2.0"</a> , on page 24. |

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:OMODE
```

### Multi STA

Enables associations from multiple non-AP STAs allowing simultaneous control and support for multi-user (MU) operations.

Note, that:

- Option R&S CMW-KS667 is required.
- Maximum number of STAs is 3
- Fading scenarios are supported.
- Channel bandwidth is supported up to 160 MHz for SISO and up to 80MHz for MIMO.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:MSTation`

## MIMO

Refer to [Chapter 2.3.5, "MIMO settings"](#), on page 49.

## Fading

Selects the SUA partitions to be used for internal fading. Both partitions of an SUA are required. If two signaling applications use internal fading at the same time, use diverse internal faders to avoid conflict.

This parameter is displayed for the fading scenarios. Only internal fading is supported.

The fader setting selects both SUA partitions. The numbering is reversed, so that the lowest fader number selects the highest SUA partition. This reverse numbering allows you to select the same number for the fader and the baseband unit, because they address different SUA partitions.

Example: Baseband unit "2.SUA4" selects SUA partition 4 in CMW 2. Fader "2.Fader3&4" selects SUA partition 1&2 in CMW 2. So they can be combined. See also ["Base Band Unit"](#) on page 48.

The fader selection offers only values compatible with the baseband setting. If there is only one possible value, the selection is grayed out, displaying this value.

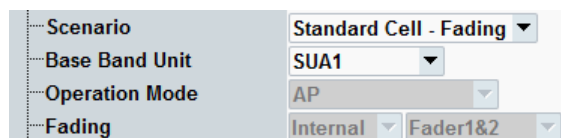


Figure 2-22: Scenario with fading

Remote command:

`ROUTe:WLAN:SIGN<i>:SCENario:SCFading:FLEXible`

`ROUTe:WLAN:SIGN<i>:SCENario:MIMFading:FLEXible`

## 2.3.5 MIMO settings

Access: "Config" hotkey > "MIMO" section.

If a MIMO scenario is selected, additional MIMO branch is displayed.

MIMO is supported for 802.11n, ac and ax standards, AP and station mode, up to 80 MHz channel bandwidth.

For background information, see [Chapter 2.2.2, "MIMO scenario"](#), on page 20.



Figure 2-23: MIMO settings

**TX MIMO Mode**

Selects the MIMO transmission mode: spatial multiplexing or space time block coding (STBC).

Automatic data frame [Configuration](#) applies the STBC.

For data frame [Configuration](#) = "User-defined", the STBC must be enabled within the data frame configuration for the particular frame format, see:

- [Chapter 2.3.8.9, "Data frame control settings"](#), on page 71 or
- ["Data Frame Control"](#) on page 106

Remote command:

`CONFigure:WLAN:SIGN<i>:MIMO:TMode`

**2.3.6 RF settings**

Access: "Config" hotkey > "RF Settings" section.

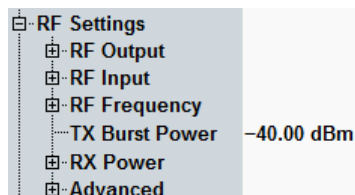


Figure 2-24: RF settings

The RF settings are described in the following chapters.

- [RF signal routing](#).....50
- [RF frequency](#).....52
- [RF power](#).....54

**2.3.6.1 RF signal routing**

Access: "Config" hotkey > "RF Settings" > "RF Output" / "RF Input" sections.

The two sections configure the RF input path and the RF output paths. Most parameters are configurable per path.

The number of configurable input and output paths depends on the selected scenario.

Depending on your hardware configuration, there are dependencies between some parameters. Configure the settings in the following order to access all possible values:

1. Select the signaling unit. See "[Base Band Unit](#)" on page 48.  
In a multi-CMW setup, this action selects also the instrument.
2. Select the RF connector.
3. Select the RF converter.

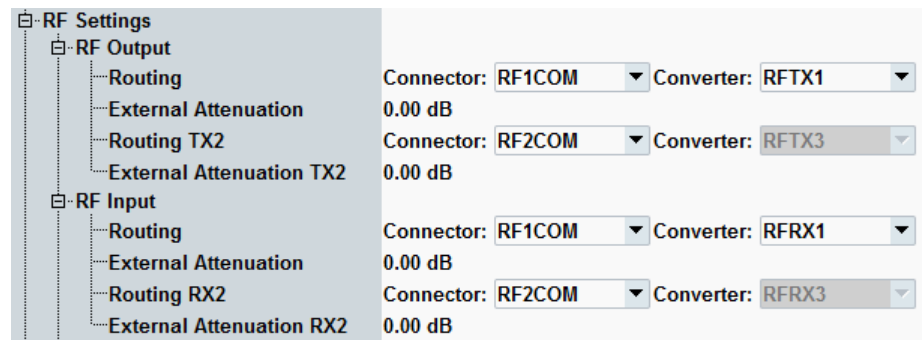


Figure 2-25: RF routing for MIMO 2x2

|                             |    |
|-----------------------------|----|
| RF Output.....              | 51 |
| L Routing.....              | 51 |
| L External Attenuation..... | 51 |
| RF Input.....               | 52 |
| L Routing.....              | 52 |
| L External Attenuation..... | 52 |

## RF Output

The following parameters configure the RF output paths of the R&S CMW.

### Routing ← RF Output

Selects the output paths for the generated RF signals, that means the output connector and the TX module for each output path.

Depending on your hardware configuration, there are dependencies between both parameters.

Remote command:

```
ROUTe:WLAN:SIGN<i>?
ROUTe:WLAN:SIGN<i>:SCENario:SCell:FLEXible
ROUTe:WLAN:SIGN<i>:SCENario:MIMO:FLEXible
ROUTe:WLAN:SIGN<i>:SCENario:SCFading:FLEXible
ROUTe:WLAN:SIGN<i>:SCENario:MIMFading:FLEXible
```

### External Attenuation ← RF Output

Defines the value of an external attenuation (or gain, if the value is negative) in the output path. With an external attenuation of x dB, the power of the generated signal is increased by x dB. The actual generated levels are equal to the displayed values plus the external attenuation.

If a correction table for frequency-dependent attenuation is active for the chosen connector, then the table name and a button are displayed. Press the button to display the table entries.

You can configure the external attenuation per output path. Antenna 2 is only available in MIMO scenarios.

Remote command:

```
CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:OUTPut
```

### RF Input

The following parameters configure the RF input path of the R&S CMW.

#### Routing ← RF Input

Selects the input path for the measured RF signal, i.e. the input connector and the RX module to be used.

You can configure the parameters per input path.

Remote command:

```
ROUTe:WLAN:SIGN<i>?
```

```
ROUTe:WLAN:SIGN<i>:SCENario:SCELl:FLEXible
```

```
ROUTe:WLAN:SIGN<i>:SCENario:MIMO:FLEXible
```

```
ROUTe:WLAN:SIGN<i>:SCENario:SCFading:FLEXible
```

```
ROUTe:WLAN:SIGN<i>:SCENario:MIMFading:FLEXible
```

#### External Attenuation ← RF Input

Defines the value of an external attenuation (or gain, if the value is negative) in the input path. The power readings are corrected by the external attenuation value.

The external attenuation value is also used in the calculation of the maximum configurable input power value.

If a correction table for frequency-dependent attenuation is active for the chosen connector, then the table name and a button are displayed. Press the button to display the table entries.

Antenna 2 is only available in MIMO scenarios with two input paths.

Remote command:

```
CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:INPut
```

### 2.3.6.2 RF frequency

Access: "Config" hotkey > "RF Settings" > "RF Frequency" section.

This section configures the channel bandwidth and the channel/frequency for uplink and downlink.

The settings for TX direction configure the center frequency of the RF analyzer. The settings for RX direction are defined via the offset to the specified center frequency of the generated WLAN signal.



|                         |                             |
|-------------------------|-----------------------------|
| RF Frequency            |                             |
| Operating Channel Width | 160MHz                      |
| Band                    | sub 6 GHz                   |
| Center Frequency        | 5630.000 MHz / Channel: 126 |
| Frequency Offset        | 0 Hz                        |
| Np20 Index              | 0                           |
| Np20 Frequency          | 5560.000 MHz / Channel: 112 |

Figure 2-26: Frequency settings for 80 MHz channel BW

|                                 |    |
|---------------------------------|----|
| Operating Channel Width.....    | 53 |
| Band.....                       | 53 |
| Center Frequency / Channel..... | 53 |
| Frequency Offset.....           | 53 |
| Np20 Index.....                 | 54 |
| Np20 Frequency / Channel.....   | 54 |

### Operating Channel Width

Same setting as "Operating Channel Width" on page 47.

### Band

Selects the operating band sub 6 GHz or 6 GHz band.

The R&S CMW itself supports frequencies up to 6 GHz (sub 6 GHz). To be able to use the 6 GHz band (5.925 GHz to 7.125 GHz), the external up-down-converter R&S CMW-Z800A is required. 6 GHz band is only supported for IEEE 802.11ax.

Option R&S CMW-KS657 or R&S CMW-KS658 is required.

For details on R&S CMW-Z800A, refer to its product information published at customer web.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:BAND`

### Center Frequency / Channel

Determines the center frequency of the generated WLAN signal and of the RF analyzer.

You can set the frequency either directly or via a valid channel number. For a list of WLAN channels, see [http://en.wikipedia.org/wiki/List\\_of\\_WLAN\\_channels](http://en.wikipedia.org/wiki/List_of_WLAN_channels).

Operating in the 5-GHz band (above 3.3 GHz in general) requires the 6-GHz option R&S CMW-KB036.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:FREQuency`

`CONFigure:WLAN:SIGN<i>:RFSettings:CHANnel`

### Frequency Offset

Positive or negative frequency offset to be added to the specified channel center frequency in receive direction.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:FOFFset`

**Np20 Index**

This setting is displayed for signals with more than 20 MHz bandwidth. The total bandwidth consists of several adjacent 20-MHz channels, half of them above the center frequency of the signal and the other half below the center frequency.

The index selects which of the 20-MHz channels is the primary channel. The lowest index value refers to the 20-MHz channel with the lowest frequency.

Remote command:

```
CONFigure:WLAN:SIGN<i>:RFSettings:NPIndex
```

**Np20 Frequency / Channel**

These settings are displayed for signals with more than 20 MHz bandwidth. They determine the center frequency of the primary 20-MHz channel.

The settings are coupled to the "Center Frequency / Channel" setting (center frequency of total bandwidth) and to the "Np20 Index".

To configure the frequencies of the 20-MHz channels and of the total bandwidth, set the "Np20 Index" and one of the other settings ("Center Frequency / Channel" or "Np20 Frequency / Channel").

Remote command:

```
CONFigure:WLAN:SIGN<i>:RFSettings:NPFrequency
CONFigure:WLAN:SIGN<i>:RFSettings:NPChannel
```

**2.3.6.3 RF power**

Access: "Config" hotkey > "RF Settings" > TX power parameters and "RX Power"/"Advanced" sections.

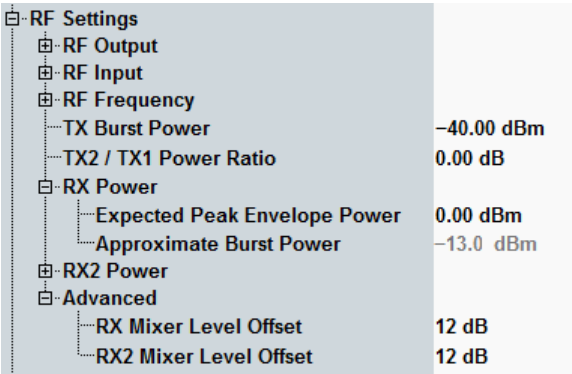


Figure 2-27: RF power settings for MIMO

TX Burst Power..... 55

TX2 / TX1 Power Ratio..... 55

Exp. Peak Envelope Power..... 55

Approximate Burst Power..... 55

RX Mixer Level Offset..... 55

**TX Burst Power**

Sets the power of the transmitted bursts. With MIMO scenarios, it specifies the total power of both signals.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:BOPower`

**TX2 / TX1 Power Ratio**

This parameter is only available in MIMO scenarios. It defines the power of the second signal relative to the power of the first signal.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:TSRatio`

**Exp. Peak Envelope Power**

Specifies the expected peak envelope power of the measured RF signal per antenna. It must account for the transmitted (average) uplink power plus the crest factor.

An expected PEP setting below the actual RF input power causes overflow in the input path. A setting which is much higher than the RF input power can reduce the measurement accuracy.

An optimum expected PEP setting results in an [Approximate Burst Power](#) value that is slightly higher than the [RX Power Indicator](#) value. See [Chapter 2.3.1.1, "Connection status"](#), on page 33.

The actual input power at the connectors must be within the level range of the selected RF input connector. Refer to the data sheet for details.

Antenna 2 is only available in MIMO scenarios with two input paths.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EPEPower`

**Approximate Burst Power**

The displayed power per Rx antenna is calculated from the specified expected PEP and the selected standard. Antenna 2 is only available in MIMO scenarios with two input paths.

Remote command:

`SENSe:WLAN:SIGN<i>:UESinfo[:ANTenna<n>]:ARXBpower?`

**RX Mixer Level Offset**

Varies the input level of the mixer in the analyzer path per antenna. Antenna 2 is only available in MIMO scenarios with two input paths.

Remote command:

`CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:MLOffset`

## 2.3.7 Internal fading

Access: "Config" hotkey > "Internal Fading" section.

This section is only relevant, if an internal fading scenario is selected.

For general prerequisites/required options and background information, see [Chapter 2.2.9, "Internal fading"](#), on page 28.

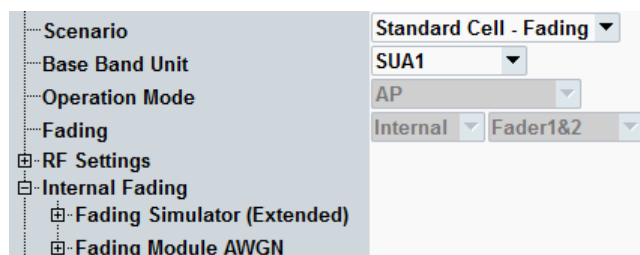


Figure 2-28: Internal fading settings

### 2.3.7.1 Fading simulator

Access: "Config" hotkey > "Internal Fading" > "Fading Simulator (Extended)".

The configuration enables and sets up the fading simulator. For background information, see [Chapter 2.2.9.1, "Fading simulator"](#), on page 28.

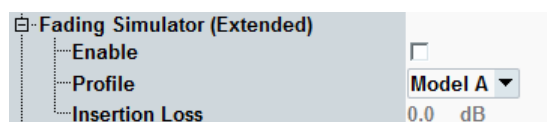


Figure 2-29: Fading simulator settings

|                                      |    |
|--------------------------------------|----|
| <a href="#">Enable</a> .....         | 56 |
| <a href="#">Profile</a> .....        | 56 |
| <a href="#">Insertion Loss</a> ..... | 56 |

#### Enable

Enables/disables the fading simulator.

Remote command:

`CONFfigure:WLAN:SIGN<i>:FADing:FSIMulator:ENABLe`

#### Profile

Selects a propagation condition profile according to IEEE 802.11-03/940r4.

The SISO and MIMO channel models A to F for 802.11n, ac, and ax, for 20 MHz and 40 MHz channel bandwidth are supported. (802.11ac and ax use identical profiles).

Refer to [Chapter A, "Predefined fading settings"](#), on page 633.

Remote command:

`CONFfigure:WLAN:SIGN<i>:FADing:FSIMulator:STANDard`

#### Insertion Loss

The insertion loss is the attenuation at the fader input. It is calculated automatically, based on the currently selected [Profile](#).

Remote command:

`CONFfigure:WLAN:SIGN<i>:FADing:FSIMulator:ILOSS:LOSS?`

### 2.3.7.2 Fading module AWGN

Access: "Config" hotkey > "Internal Fading" > "Fading Module AWGN".

The section enables and configures the AWGN insertion via the fading module. For background information, see [Chapter 2.2.9.2, "AWGN generator"](#), on page 29.

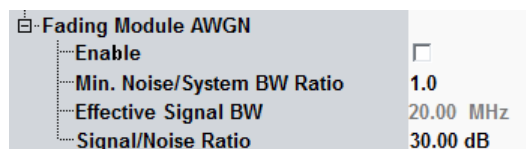


Figure 2-30: AWGN settings for MIMO

|                                 |    |
|---------------------------------|----|
| Enable.....                     | 57 |
| Min. Noise/System BW Ratio..... | 57 |
| Effective Signal BW.....        | 57 |
| Signal/Noise Ratio.....         | 57 |

#### Enable

Enables/disables AWGN insertion via the fading module.

Remote command:

`CONFigure:WLAN:SIGN<i>:FADing:AWGN:ENABle`

#### Min. Noise/System BW Ratio

Configures the minimum ratio between the noise bandwidth and the channel bandwidth.

Remote command:

`CONFigure:WLAN:SIGN<i>:FADing:AWGN:BWIDth:RATio`

#### Effective Signal BW

Displays the channel bandwidth used for the downlink.

#### Signal/Noise Ratio

Specifies the signal to noise ratio.

Remote command:

`CONFigure:WLAN:SIGN<i>:FADing:AWGN:SNRatIo`

## 2.3.8 Connection settings

Access: "Config" hotkey > "Connection" section.

The available connection settings depend on the operation mode as shown in the following figures.

|                          |                     |
|--------------------------|---------------------|
| <b>Connection</b>        |                     |
| <b>Association</b>       |                     |
| Beacon Interval (TU)     | 100                 |
| Target Wake Time (TWT)   |                     |
| DTIM Period              | 1                   |
| MAC-Address (BSSID)      | 000102030405        |
| SSID                     | CMW-AP              |
| BSS Color                | 1                   |
| Association ID Range     | Start: 1 Stop: 2007 |
| HT Smoothing             | Not Recommended ▼   |
| Country Code             | Disabled            |
| Supported Rates          | Automatic           |
| Management Frame Control | Automatic           |
| Data Frame Control       | Automatic           |
| Trigger Frame            |                     |
| Security                 | Disabled            |
| QoS Data TID             |                     |
| Block Ack                |                     |
| A-MPDU                   |                     |
| EDCA                     |                     |
| HE MAC Capabilities      |                     |
| NDP Sounding             |                     |

Figure 2-31: Connection settings (AP mode, 802.11ax)

|                          |                          |
|--------------------------|--------------------------|
| <b>Connection</b>        |                          |
| <b>Association</b>       |                          |
| Support of DSSS          | <input type="checkbox"/> |
| MAC-Address (BSSID)      | 000102030405             |
| HT Smoothing             | Not Recommended ▼        |
| <b>Station</b>           |                          |
| Connection Mode          | Auto Connect ▼           |
| SSID Connection          | Any ▼                    |
| Supported Rates          | Automatic                |
| Management Frame Control | Automatic                |
| Data Frame Control       | Automatic                |
| Security                 | Disabled                 |
| QoS Data TID             |                          |
| Block Ack                |                          |
| A-MPDU                   |                          |
| EDCA                     |                          |
| HE MAC Capabilities      |                          |

Figure 2-32: Connection settings (station mode, 802.11ax)

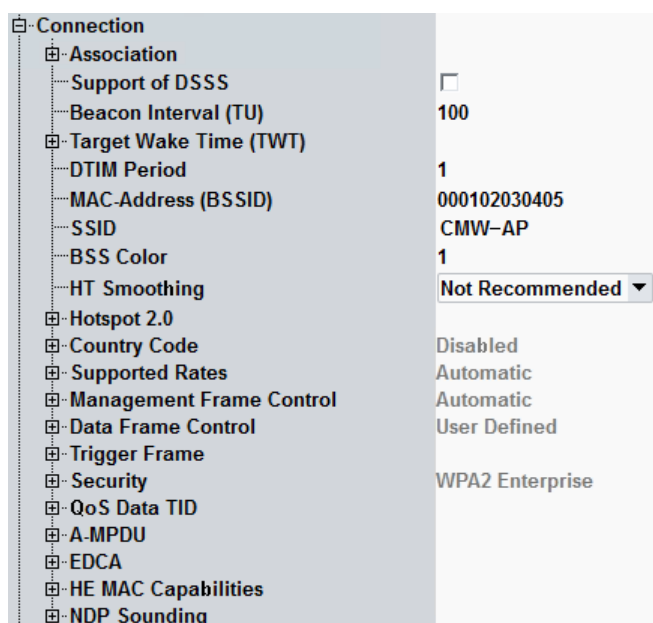


Figure 2-33: Connection settings (Hotspot 2.0 mode, 802.11ax)

The connection settings are described in the following chapters.

|                                          |    |
|------------------------------------------|----|
| • Association settings.....              | 59 |
| • General connection settings.....       | 61 |
| • Station settings.....                  | 63 |
| • Hotspot 2.0 settings.....              | 63 |
| • Target wake time settings.....         | 66 |
| • Country code settings.....             | 68 |
| • Supported rates settings.....          | 69 |
| • Management frame control settings..... | 70 |
| • Data frame control settings.....       | 71 |
| • Trigger frame settings.....            | 75 |
| • Security settings.....                 | 79 |
| • QoS data TID settings.....             | 82 |
| • Block Ack settings.....                | 83 |
| • A-MPDU settings.....                   | 84 |
| • EDCA settings.....                     | 85 |
| • HE MAC capabilities.....               | 86 |
| • NDP sounding settings.....             | 86 |

### 2.3.8.1 Association settings

Access: "Config" hotkey > "Connection" > "Association" section.

This section is relevant for AP and Hotspot 2.0 modes.

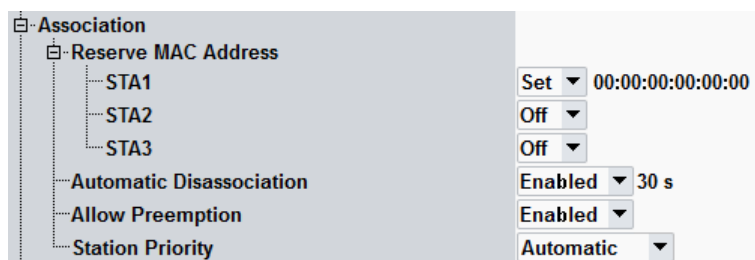


Figure 2-34: Association settings for enabled multi-STA

This chapter describes the following association settings.

|                               |    |
|-------------------------------|----|
| Reserve MAC Address.....      | 60 |
| Automatic Disassociation..... | 60 |
| Allow Preemption.....         | 60 |
| STA Priority.....             | 60 |

### Reserve MAC Address

If **Multi STA** is enabled, these parameters configure three slots available for STAs.

Option R&S CMW-KS667 is required.

- **Off** - the slot is disabled
- **Set** - reserves the slot for a particular MAC address
- **Any** - the slot is available to a STA of any MAC address

When a STA associates, the first attempt uses slot one. If the STA is not compatible with the slot settings, then slot two and finally slot three is attempted. If no slot is available, the association is rejected and an event log record is added.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:STA<s>:MACReserve`

### Automatic Disassociation

Enables or disables automatic STA disassociation, when a STA is no longer present. If enabled, the R&S CMW detects that a STA is absent, it automatically removes its association after some user-specified period of time.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:DISass`

### Allow Preemption

If enabled and all available slots are occupied, an incoming association compatible with the slot causes an existing STA to be disassociated. The prioritization is according to the slots STA1 to STA3.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:PREemption`

### STA Priority

Specifies how the stack prioritizes one STA over another.

- **Round Robin** - schedules equal number of TX opportunities to each station number



- **Automatic** - depends on TX buffer volume using TID priority sequence: 7, 6, 5, 4, 3, 0, 2, 1

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:STAPriority`

### 2.3.8.2 General connection settings

Access: "Config" hotkey > parameters in upper level of "Connection" section.

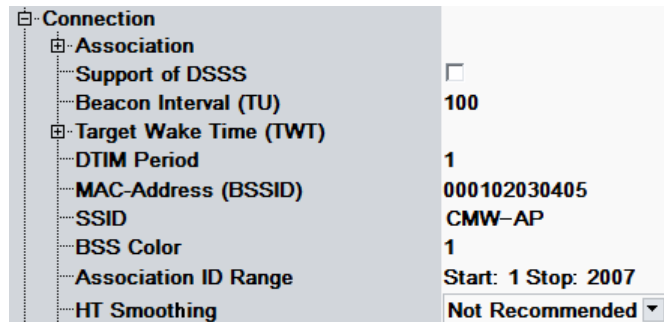


Figure 2-35: General connection settings

This chapter describes the following connection settings.

|                             |    |
|-----------------------------|----|
| Support of DSSS.....        | 61 |
| Beacon Interval.....        | 61 |
| Target Wake Time (TWT)..... | 62 |
| DTIM Period.....            | 62 |
| MAC Address (BSSID).....    | 62 |
| SSID.....                   | 62 |
| BSS Color.....              | 62 |
| Association ID Range.....   | 62 |
| HT Smoothing.....           | 62 |

#### Support of DSSS

Enables you to associate an 802.11b device using 802.11ac or ax standard. Also, it enables you to use non-HT management frames within 802.11ac and ax, see [Frame Format, Channel Bandwidth, Rate](#).

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:DSSS`

#### Beacon Interval

Defines the interval between two beacon frame transmissions. The interval is set in time units (1 TU = 1024  $\mu$ s).

The R&S CMW regularly broadcasts beacon frames to inform WLAN devices about the simulated infrastructure/ad-hoc network. Decreasing/increasing the interval makes the association procedure more/less responsive but decreases/increases the data throughput.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:BEACon`

**Target Wake Time (TWT)**

Refer to [Chapter 2.3.8.5, "Target wake time settings"](#), on page 66.

**DTIM Period**

Sets the number of beacon intervals between successive delivery traffic indication messages (DTIM).

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:DPERiod`

**MAC Address (BSSID)**

The 48-bit MAC address of the simulated WLAN entity as hexadecimal number. In AP mode, it is also known as basic service set ID (BSSID).

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:BSSid`

**SSID**

The service set identifier (SSID) is a textual identifier, with up to 32 characters (7-bit ASCII only).

The meaning depends on the operation mode:

- "Station": SSID of the access point to which the station is allowed to connect (if restricted at all, see ["SSID Connection"](#) on page 63)
- Other modes: SSID of the simulated entity

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:SSID`

**BSS Color**

Specifies the color code of the basic service set (BSS).

Option R&S CMW-KS657 is required.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:BSSColor`

**Association ID Range**

Specifies the range of IDs to be assigned by the access point to the connected DUTs. The setting is relevant in AP mode.

Option R&S CMW-KS657 is required.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:AID`

**HT Smoothing**

Indicates to the receiver whether the frequency-domain smoothing is recommended for channel estimation. The value "Non Recommended" instructs the DUT to perform only per-carrier independent channel (unsmoothed) estimation.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:SMOothing`

### 2.3.8.3 Station settings

Access: "Config" hotkey > "Connection" > "Station" section.

The parameters are visible in the operation mode "Station".

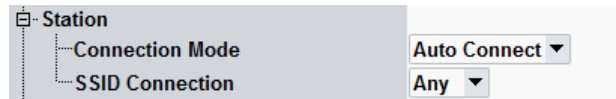


Figure 2-36: Station section

#### Connection Mode

This setting determines the connection behavior of the simulated station.

"Auto Connect" Associations are initiated automatically.

"Manual" Associations are initiated manually via hotkey, see [Chapter 2.3.2, "Signaling and connection control"](#), on page 44.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNECTION:STATION:CMODE`

#### SSID Connection

This setting specifies to which access points the simulated station is allowed to connect.

"Any" The station is allowed to connect to any available AP.

"SSID" The station is only allowed to connect to the AP configured via "SSID" on page 62.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNECTION:STATION:SCONNECTION`

### 2.3.8.4 Hotspot 2.0 settings

Access: "Config" hotkey > "Connection" > "Hotspot 2.0" section.

The parameters are visible in the operation mode "Hotspot 2.0".

The section specifies information to be sent to the DUT. The information is included in beacons, probe responses and GAS query responses as listed in the following table.

| Beacon / probe response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | GAS query response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interworking element <ul style="list-style-type: none"> <li>Access network options field               <ul style="list-style-type: none"> <li>Access network type</li> <li>Internet access</li> </ul> </li> <li>Venue info field               <ul style="list-style-type: none"> <li>Venue group</li> <li>Venue type</li> </ul> </li> <li>HESSID</li> </ul> BSS load element <ul style="list-style-type: none"> <li>Station count</li> <li>Channel utilization</li> <li>Available admission capacity</li> </ul> | <ul style="list-style-type: none"> <li>Venue name</li> <li>3GPP cellular network element               <ul style="list-style-type: none"> <li>PLMN</li> </ul> </li> <li>Domain name</li> <li>Operator-friendly name</li> <li>WAN metrics element               <ul style="list-style-type: none"> <li>Downlink speed</li> <li>Uplink speed</li> </ul> </li> <li>IP address type availability               <ul style="list-style-type: none"> <li>IPv4 address field</li> <li>IPv6 address field</li> </ul> </li> <li>Network authentication type</li> <li>NAI realm</li> </ul> |

The "Hotspot 2.0" section is described in the following.

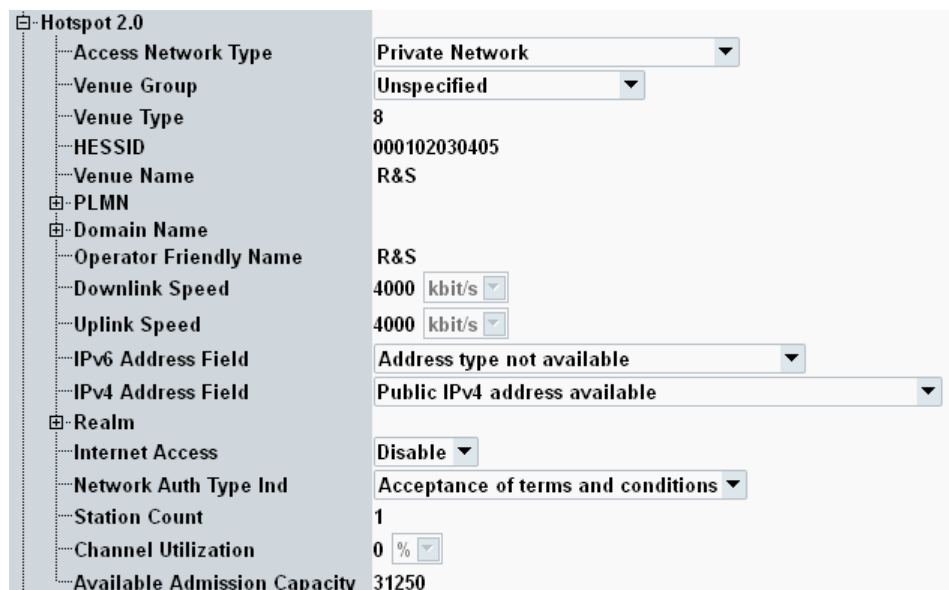


Figure 2-37: Hotspot 2.0 section

|                                                                       |    |
|-----------------------------------------------------------------------|----|
| Access Network Type to Venue Name.....                                | 64 |
| PLMN.....                                                             | 64 |
| Domain Name.....                                                      | 65 |
| Operator Friendly Name.....                                           | 65 |
| Downlink Speed to IPv4 Address Field.....                             | 65 |
| Realm.....                                                            | 66 |
| Internet Access, Network Auth Type Ind.....                           | 66 |
| Station Count, Channel Utilization, Available Admission Capacity..... | 66 |

### Access Network Type to Venue Name

The settings at the top of the section have the following meaning:

- "Access Network Type": 802.11u access network type advertised to the client  
See standard IEEE Std 802.11-2016, chapter 9.4.2.92 "Interworking element".
- "Venue Group" and "Venue Type": Optional venue information  
See standard IEEE Std 802.11-2016, chapter 9.4.1.35 "Venue Info field".
- "HESSID": Homogeneous extended service set identifier of the access point (12 hexadecimal digits)  
See standard IEEE Std 802.11-2016, chapter 9.4.2.92 "Interworking element".
- "Venue Name": Venue name of the IEEE 802.11 access network  
See standard IEEE Std 802.11-2016, chapter 9.4.5.4 "Venue Name ANQP-element".

Remote command:

CONFigure:WLAN:SIGN<i>:CONNection:HOTSspot:HSPar

### PLMN

Defines a list of 3GPP networks that the IEEE 802.11 access network provides service for.

| PLMN   | Enable                              | MCC | MNC | MNC Digits   |
|--------|-------------------------------------|-----|-----|--------------|
| PLMN 1 | <input checked="" type="checkbox"/> | 001 | 01  | Two Digits ▼ |
| PLMN 2 | <input type="checkbox"/>            | 001 | 01  | Two Digits ▼ |
| PLMN 3 | <input type="checkbox"/>            | 001 | 01  | Two Digits ▼ |
| PLMN 4 | <input type="checkbox"/>            | 001 | 01  | Two Digits ▼ |
| PLMN 5 | <input type="checkbox"/>            | 001 | 01  | Two Digits ▼ |

Each network is specified via its mobile country code (MCC, three digits) and its mobile network code (MNC, two- or three-digit format).

You can define up to five networks.

See standard IEEE Std 802.11-2016, chapter 9.4.5.11 "3GPP Cellular Network ANQP-element".

Remote command:

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar
```

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:PLMN<no>
```

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:MNDigits
```

### Domain Name

Defines a list of domain names of the entity operating the IEEE 802.11 access network.

| Domain Name   | Enable                              | Name         |
|---------------|-------------------------------------|--------------|
| Domain Name 1 | <input checked="" type="checkbox"/> | www.R&S.com  |
| Domain Name 2 | <input type="checkbox"/>            | www.R&S2.com |
| Domain Name 3 | <input type="checkbox"/>            | www.R&S3.com |
| Domain Name 4 | <input type="checkbox"/>            | www.R&S4.com |
| Domain Name 5 | <input type="checkbox"/>            | www.R&S5.com |

You can define up to five domain names.

See standard IEEE Std 802.11-2016, chapter 9.4.5.15 "Domain Name ANQP-element".

Remote command:

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar
```

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:DNAME<no>
```

### Operator Friendly Name

Name of the operator of the IEEE 802.11 access network

See Hotspot 2.0 technical specification, chapter 4.3 "Operator Friendly Name".

Remote command:

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar
```

### Downlink Speed to IPv4 Address Field

- "Downlink Speed, Uplink Speed": Speed of the link in kbit/s  
See Hotspot 2.0 technical specification, chapter 4.4 "WAN Metrics".
- "IPv6 Address Field", "IPv4 Address Field": Indicates which types of IP addresses can be allocated to the station.  
See standard IEEE Std 802.11-2016, chapter 9.4.5.9 "IP Address Type Availability ANQP-element".

Remote command:

```
CONFIGure:WLAN:SIGN<i>:CONNection:HOTSpot:HSSPar
```

### Realm

Defines a list of network access identifier (NAI) realms that are reachable via the IEEE 802.11 access network.

| Realm   | Enable                              | Name         | SIM                                 | TLS                      | TTLS                     | AKA                      | AKA'                     |
|---------|-------------------------------------|--------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Realm 1 | <input checked="" type="checkbox"/> | example.org  | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Realm 2 | <input type="checkbox"/>            | example2.org | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Realm 3 | <input type="checkbox"/>            | example3.org | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Realm 4 | <input type="checkbox"/>            | example4.org | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| Realm 5 | <input type="checkbox"/>            | example5.org | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

For each realm, the list indicates the realm name and the supported extensible authentication protocol (EAP) methods (SIM, TLS, TTLS, AKA, AKA').

You can define up to five realms.

See standard IEEE Std 802.11-2016, chapter 9.4.5.10 "NAI Realm ANQP-element".

Remote command:

```
CONFfigure:WLAN:SIGN<i>:CONNection:HOTSspot:HSSPar
CONFfigure:WLAN:SIGN<i>:CONNection:HOTSspot:REALm<no>
```

### Internet Access, Network Auth Type Ind

- "Internet Access": Specifies whether the hotspot provides internet access  
See standard IEEE Std 802.11-2016, chapter 9.4.2.92 "Interworking element".
- "Network Auth Type Ind": Indicates the network authentication type  
See standard IEEE Std 802.11-2016, chapter 9.4.5.6 "Network Authentication Type ANQP-element".

Remote command:

```
CONFfigure:WLAN:SIGN<i>:CONNection:HOTSspot:HSSPar
```

### Station Count, Channel Utilization, Available Admission Capacity

Configure the contents of the BSS load element, see standard IEEE Std 802.11-2016, chapter 9.4.2.28.

- "Station Count": Number of stations that are currently associated with the BSS
- "Channel Utilization": Percentage of time, that the access point sensed the primary channel was busy
- "Available Admission Capacity": Remaining time available via explicit admission control, in units of 32  $\mu$ s/s

Remote command:

```
CONFfigure:WLAN:SIGN<i>:CONNection:HOTSspot:CUTil
```

#### 2.3.8.5 Target wake time settings

Access: "Config" hotkey > "Connection" > "Target Wake Time (TWT)" section.

This section is relevant for 802.11ax standard, AP and Hotspot 2.0 operation mode.

The section configures broadcast target wake time (TWT) operation to define a specific time for a STA to access the AP.

Option R&S CMW-KS657 is required.

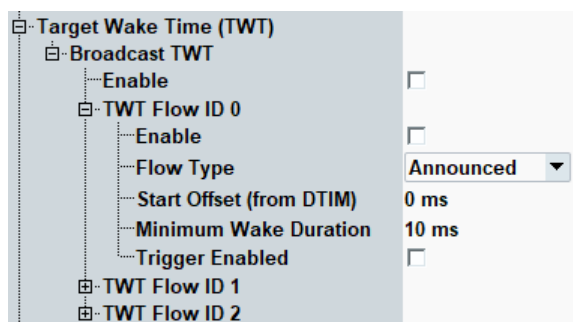


Figure 2-38: Broadcast TWT configuration

|                        |    |
|------------------------|----|
| Enable < Broadcast TWT | 67 |
| Enable < TWT Flow ID x | 67 |
| Flow Type              | 67 |
| Start Offset           | 68 |
| Minimum Wake Duration  | 68 |
| Trigger Enabled        | 68 |

**Enable < Broadcast TWT**

Enables/ disables broadcast target wake time (TWT) operation.

Remote command:

`CONFfigure:WLAN:SIGN<i>:CONNECTION:BTWT:ENABLe`

**Enable < TWT Flow ID x**

Enables/ disables particular schedule period.

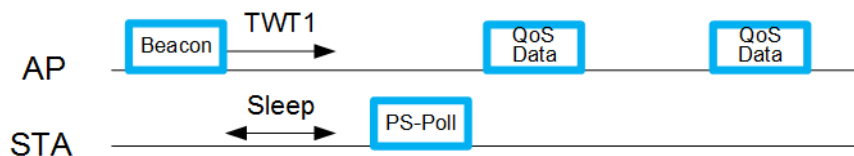
Remote command:

`CONFfigure:WLAN:SIGN<i>:CONNECTION:BTWT:SCHedule:ENABLe`

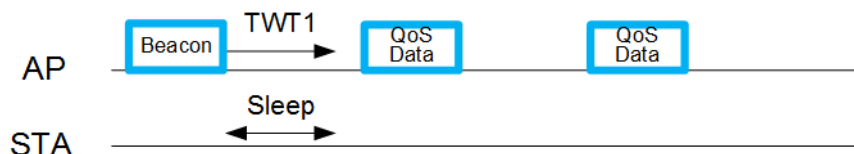
**Flow Type**

Specifies the flow type:

- Announced: requesting STA signals first its awake state, for example via a PS-poll.



- Unannounced: the AP sends a frame to the STA, without waiting for a PS-poll or APSD trigger frame.



Remote command:

`CONFfigure:WLAN:SIGN<i>:CONNECTION:BTWT:SCHedule:FTYPE`

Start Offset

Specifies the offset of a schedule period from beacon.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeodule:STIMe
```

Minimum Wake Duration

Sets the minimum wake time during the sleep period. If the STA does not receive a frame by this time, then it returns to sleep. If it receives a frame, then it waits for end of service period (EOSP) from the AP.

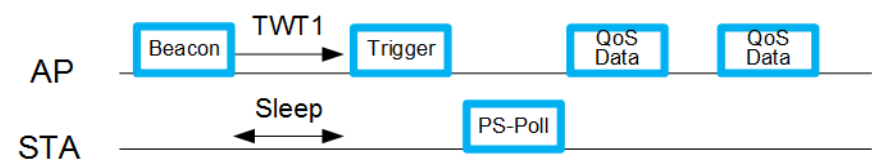
Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeodule:MWDuration
```

Trigger Enabled

Specify, whether the broadcast TWT operation is triggered.

In a trigger-enabled TWT, the AP always initiates frame exchange with a trigger frame. The following figure shows a trigger-based announced broadcast TWT message flow.



Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeodule:TENable
```

2.3.8.6 Country code settings

Access: "Config" hotkey > "Connection" > "Country Code" section.

These parameters define information transported to the DUT via the "Country" element. The information allows the station to identify the regulatory domain in which it is located.

The parameters are invisible in the operation mode "Station".

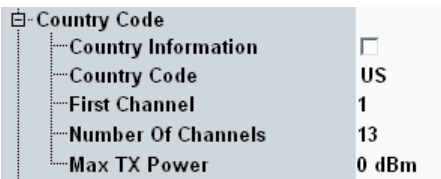


Figure 2-39: Country code settings

|                                        |    |
|----------------------------------------|----|
| Country Information.....               | 69 |
| Country Code.....                      | 69 |
| First Channel, Number of Channels..... | 69 |
| Max TX Power.....                      | 69 |



**Country Information**

Enables or disables the transport of the information.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCState
```

**Country Code**

Identifies the country or non-country entity in which the AP is operating. The non-country code is "XX". The country codes are defined in ISO/IEC 3166-1.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCConf
```

**First Channel, Number of Channels**

Define the allowed range of channels to be used.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCConf
```

**Max TX Power**

Defines the maximum allowed transmit power in dBm.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCConf
```

**2.3.8.7 Supported rates settings**

Access: "Config" hotkey > "Connection" > "Supported Rates" section.

This section defines which modulation/ coding/ data rates are supported by the simulated entity and which rates must be supported by the tested entity.

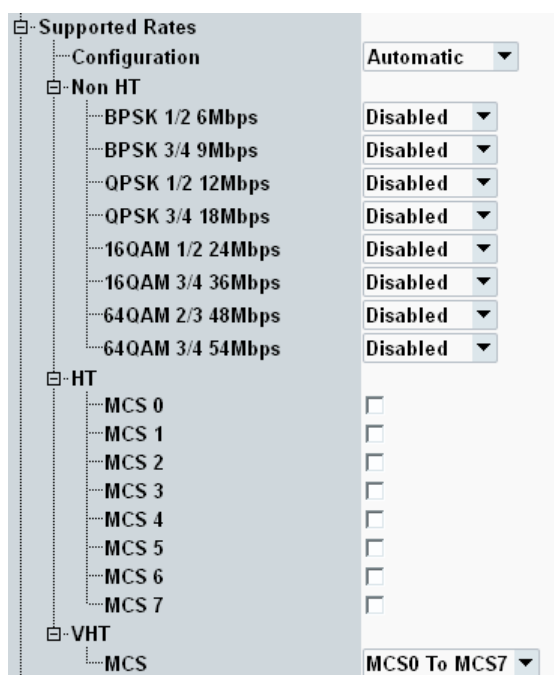


Figure 2-40: Rates settings (802.11ac)

### Configuration

To define the rates, set "Configuration" to "User Defined" and configure the entries. Which entries are available, depends on the selected standard.

For "Configuration" = "Automatic", the remaining settings have no effect.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:SRATes
```

### Non HT, HT, VHT Supported Rates

The available parameters depend on the selected standard.

"Disabled" means that the rate is not supported by the simulated entity. "Mandatory" and "Optional" mean that the rate is supported by the simulated entity and must / can be supported by the tested entity.

According to the 802.11 standard, a station must not associate with an AP if it cannot receive and transmit at all mandatory rates. See also [Table 2-5](#).

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:SRATes:DSSSconf
CONFigure:WLAN:SIGN<i>:CONNection:SRATes:OFDMconf
CONFigure:WLAN:SIGN<i>:CONNection:SRATes:OMCSconf
CONFigure:WLAN:SIGN<i>:CONNection:SRATes:VHTConf
```

#### 2.3.8.8 Management frame control settings

Access: "Config" hotkey > "Connection" > "Management Frame Control" section.

These parameters configure the modulation/ coding rate of the transmitted management frames.



Figure 2-41: Management frame control settings

### Configuration

Selects automatic settings or user-defined frame rate control.

- With automatic configuration, the following default frame rates are used:
  - 1 Mbit/s for 802.11b, g and g/n
  - 6 Mbit/s for 802.11a and g(OFDM)
  - 6 Mbit/s for 802.11a/n, g(OFDM)/n, ac, and ax
- For user-defined settings, the manual configuration of "Rate" applies, see [Frame Format, Channel Bandwidth, Rate](#).

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:MFDef`

### Frame Format, Channel Bandwidth, Rate

Select the modulation/ coding rate of the transmitted management frames (beacon, authentication, association response and probe response). The available values depend on the selected [Standard](#).

To use the non-HT management frames within standards 802.11ac or ax, the [Support of DSSS](#) must be enabled.

Configure the parameters top-down.

For "Channel Bandwidth", refer to ["Operating Channel Width"](#) on page 47.

The setting of rate is only relevant if [Configuration](#) = "User Defined".

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:MFDef`

### 2.3.8.9 Data frame control settings

Access: "Config" hotkey > "Connection" > "Data Frame Control" section.

If [Multi STA](#) is enabled, this section is not visible in the main configuration tree. The related settings appear in separate tabs, one tab per station, see [Chapter 2.3.17, "Station configuration tabs"](#), on page 101.

These parameters configure the modulation/ coding rate of the transmitted data frames.

Figure 2-42: Data frame control settings (MIMO)

|                                                   |    |
|---------------------------------------------------|----|
| Configuration.....                                | 72 |
| Frame Format.....                                 | 73 |
| Non-HT, HT, VHT, HE Frame.....                    | 73 |
| L 20 MHz Channel Settings (HE_MU frame only)..... | 74 |
| L Users and Dummies (HE_MU frame only).....       | 74 |

### Configuration

Selects automatic settings or user-defined frame rate control per station.

If **Multi STA** is enabled, the user can control one station while the stack controls another station automatically.

With automatic configuration, the data frame rate in transmit direction is dynamic. It is determined by several parameters:

- Operating bandwidth
- Standard
- DUT capabilities
- Channel throughput

Note that in the current version, HE\_MU rates are not supported with the automatic configuration. To generate an HE\_MU frame format, use the user-defined frame rate control.

Remote command:

CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:DFDef

### Frame Format

Select the modulation/ coding rate of the transmitted data frames. The available values depend on the selected standard.

Configure the parameters top-down.

Remote command:

CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:DFDef

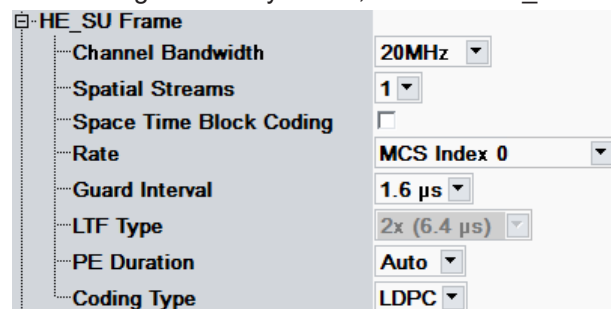
### Non-HT, HT, VHT, HE Frame

Selects the modulation/coding rate of the transmitted data frames. The available parameters depend on the selected standard. The settings are configurable per frame format.

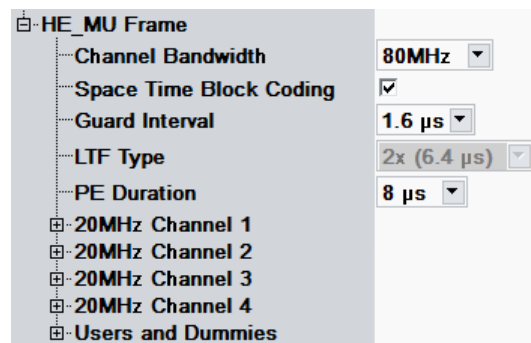
If you want to set a user-defined value, configure the parameters top-down.

- For **"Channel Bandwidth"**, refer to ["Operating Channel Width"](#) on page 47.
- **"Spatial streams"**:

The number of spatial streams is only visible for MIMO connections. This parameter is configurable only for HT, VHT and HE\_SU frames:



- **"Space Time Block Coding"**:  
For data frame [Configuration](#) = "User-defined", you have to enable STBC within the data frame configuration for the particular frame format manually.  
Automatic data frame [Configuration](#) applies the STBC automatically.
- **"Rate"**: With "User Defined" configuration, the selected rate is used for all data frame transmissions - except for packet error rate (PER) tests.  
With automatic configuration, the frame rates are set according to standard and also depend on channel bandwidth.
- **"Guard Interval"**:
  - Short or long guard interval for legacy (up to 802.11ac)
  - 0.8 µs, 1.6 µs, and 3.2 µs guard interval durations for 802.11ax
- **"LTF Type"** (HE frames only): 1x HE-LTF, 2x HE-LTF or 4x HE-LTF
- **"PE Duration"** (HE frames only): value of packet extension (PE) field to provide additional receive processing time at the end of the HE PPDU. Automatic setting is based on the reported DUTs' capabilities.
- **"Coding Type"** (for VHT and HE\_SU frames only): forward error correction coding low density parity check (LDPC) or binary convolution code (BCC).
- For **"20 MHz Channel"** configuration (HE\_MU frame only), refer to ["20 MHz Channel Settings \(HE\\_MU frame only\)"](#) on page 74.



- For "Users and Dummies" configuration, refer to "Users and Dummies (HE\_MU frame only)" on page 74.

Option R&S CMW-KS657 is required for 802.11ax.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:DFDef
```

### 20 MHz Channel Settings (HE\_MU frame only) ← Non-HT, HT, VHT, HE Frame

Configures allocation arrangement and sizes for the corresponding channel.

- **"RU Allocation Subfield"**: sets 8-bit indices for resource unit (RU) allocation to specify the number of RUs, their position and size. Refer to IEEE Std 802.11ax-2021, table 27-26, RU allocation subfield.
- **"RUs State"**: disables traffic data for an RU allocation or enables and selects the traffic data configuration to be transmitted in a particular RU. For an RU allocation, you can select one of the following traffic data configurations:
  - "User 1": real user data transmission. The traffic data configuration of the type "User" must be allocated to only one RU.
  - "Dummy": transmission of pseudo-random data. The same traffic data configuration of the type dummy (e.g. "Dummy2") can be applied up to all RU allocations, except for the one occupied by "User 1".
  - "Off": no traffic data are transmitted in a particular RU
- **"RUs Size"**: indicates the number of RU tones (pilot and data subcarriers). RUs sizes are selected via "RU Allocation Subfield"

Option R&S CMW-KS657 is required.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:ALSField
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
BLALlocation?
```

### Users and Dummies (HE\_MU frame only) ← Non-HT, HT, VHT, HE Frame

Configures traffic data. One "User" must be mapped to one RU within a channel to specify the traffic to the DUT. The remaining RUs can be disabled or optionally you can allocate pseudo-random data (dummies).

- For the traffic data configuration of the type "User", the configurable parameters are:  
Modulation, coding scheme, coding type and the number of spatial streams for MIMO
- For the traffic data configuration of the type "Dummy", the configurable parameters are:

### Modulation and coding scheme

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
RUALlocation
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
USER<index>:STReams
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
USER<index>:ALLocation
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
USER<index>:MCS
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
DUMMy<index>:MCS
CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:
USER<index>:CTYPe
```

#### 2.3.8.10 Trigger frame settings

Access: "Config" hotkey > "Connection" > "Trigger Frame " section.

The parameters are visible for 802.11ax.

This section configures HE trigger frame. The parameters of "Common Info" and "User Info" fields are specified in IEEE Std 802.11ax-2021, chapter 9.3.1.22 Trigger frame format.

If [Multi STA](#) is enabled, "User Info" settings are not visible in the main configuration tree. The related settings appear in separate tabs, one tab per station, see [Chapter 2.3.17, "Station configuration tabs"](#), on page 101.

Option R&S CMW-KS657 is required.

Figure 2-43: Trigger frame settings

|                                 |    |
|---------------------------------|----|
| Trigger Type.....               | 76 |
| Number of Symbols.....          | 77 |
| CS Required.....                | 77 |
| Bandwidth.....                  | 77 |
| GI and LTF.....                 | 77 |
| MU-MIMO LTF.....                | 77 |
| LDPC Extra Symbol.....          | 77 |
| AP Tx Power.....                | 78 |
| RU Allocation.....              | 78 |
| Coding Type.....                | 78 |
| MCS.....                        | 78 |
| DCM.....                        | 78 |
| Starting Spatial Stream.....    | 78 |
| Number of Spatial Streams.....  | 78 |
| Target RSSI.....                | 79 |
| Trigger Frame Transmission..... | 79 |

### Trigger Type

Specifies the trigger type as specified in the Common Info field.

The following trigger variants are supported by the R&S CMW:

- **"Basic Trigger"** carries any content.
- **"Beamforming Report Poll"** (BRP) solicits a beamforming report.
- **"MU-RTS"** solicits MU-CTS from multiple STAs.
- **"Buffer Status Report Poll"** (BSRP) solicits a buffer status report.



- **"Bandwidth Query Report Poll"** (BQRP) solicits a bandwidth query report.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:HETF:TTYP`

### Number of Symbols

Specifies the length of the HE trigger-based (TB) PPDU. HE TB PPDU is the HE PPDU that is the response to the trigger frame.

The value range is limited by the maximum PPDU duration (5484 µs) and dependent on the GI and LTF setting as follows:

- **1x HE-LTF + 1.6 µs GI:** 1 to 373 symbols
- **2x HE-LTF + 1.6 µs GI:** 1 to 372 symbols
- **4x HE-LTF + 3.2 µs GI:** 1 to 330 symbols

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:HETF:NOFSymbols`

### CS Required

If enabled, the STA must check the medium indication from CS mechanism before responding.

If disabled, specifies that the STA can respond without regard to the busy/idle state of the medium.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:HETF:CSR`

### Bandwidth

Specifies the channel bandwidth of the HE TB PPDU.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:HETF:CHBW`

### GI and LTF

Specifies the guard interval and LTF type of the HE TB PPDU.

Note that for **MU-MIMO LTF** set to `MASK`, the value "1x HE-LTF + 1.6 µs GI" is not supported.

See also **"Non-HT, HT, VHT, HE Frame"** on page 73.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:HETF:GILT`

### MU-MIMO LTF

Selects one of the following MU-MIMO LTF modes:

- Single stream pilots
- Mask LTF sequence of each spatial stream by a distinct orthogonal code

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:HETF:MLTF`

### LDPC Extra Symbol

Specifies the support of LDPC extra symbol segment.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:HETF:LDPC
```

### AP Tx Power

Specifies the value of an "AP TX Power" field the R&S CMW signals via trigger frames.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:HETF:APTpower?
```

### RU Allocation

Specifies the RU used by the HE TB PPDU.

In SISO scenario, up to three STAs can be allocated. For MIMO connections, only two STAs can be assigned an RU allocation.

The setting **x-tone RU-y** means the number of data and pilot subcarriers for x and RU index for y. Refer to IEEE Std 802.11ax-2021, tables 27-7 to 27-9.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:RUAL
```

### Coding Type

Specifies the coding used by the HE TB PPDU.

Instrument supports only one LDPC response, so the other STAs are using BCC.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:CTYP
```

### MCS

Specifies the modulation and coding scheme (MCS) used by the HE TB PPDU.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:MCS
```

### DCM

Specifies whether the HE TB response uses dual carrier modulation (DCM).

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:DCM
```

### Starting Spatial Stream

Sets the starting spatial stream for the HE TB PPDU.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:SSS?
```

### Number of Spatial Streams

Sets the number of HE TB PPDU spatial streams.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:NSS
```

Target RSSI

The "Target RSSI" indication displays the value that is carried in the trigger frame's "UL Target RSSI" field. The STA combines this value with its channel loss estimate to set the TX power for the HE TB response. When the "Target RSSI Mode" is set to "Auto", theR&S CMW automatically calculates this value. "Manual" mode allows direct control of the HE TB response power level.

Remote command:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:TRSSi?
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:TRSMODE
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:TSRControl
```

Trigger Frame Transmission

Enables you to transmit the trigger frame once or periodically in a specified interval. Periodic trigger can be switched on and off.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:HETF:SSTX
CONFigure:WLAN:SIGN<i>:CONNection:HETF:TXEN
CONFigure:WLAN:SIGN<i>:CONNection:HETF:TXP
```

2.3.8.11 Security settings

Access: "Config" hotkey > "Connection" > "Security" section.

The section contains the settings for all security modes.

For background information, see [Chapter 2.2.3, "WLAN security"](#), on page 21.

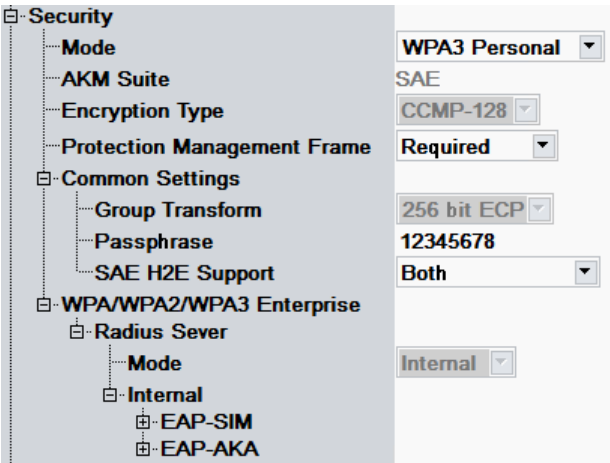


Figure 2-44: Security section

|                                        |    |
|----------------------------------------|----|
| Mode.....                              | 80 |
| AKM Suite.....                         | 80 |
| Encryption Type.....                   | 80 |
| Protection Management Frame.....       | 80 |
| SAE H2E Support.....                   | 80 |
| Common Settings > Group Transform..... | 81 |
| Common Settings > Passphrase.....      | 81 |

|                                               |    |
|-----------------------------------------------|----|
| WPA/WPA2/WPA3 Enterprise > Radius Server..... | 81 |
| L Mode.....                                   | 81 |
| L EAP-SIM.....                                | 81 |
| L EAP-AKA.....                                | 82 |
| L IP Address / Port Number / Shared Key.....  | 82 |

### Mode

Selects the security mechanism to be used.

The WLAN signaling application supports Wi-Fi protected access (WPA, WPA2, and WPA3), with personal mode and enterprise mode and opportunistic wireless encryption (OWE). The enterprise mode requires option R&S CMW-KS660.

Depending on the selected operation mode, only a subset of the security modes is supported, see [Table 2-3](#).

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:SECurity:TYPE`

### AKM Suite

Displays the used authentication key management (AKM) suite as defined in standard IEEE Std 802.11-2016, table 9-133.

WPA personal: TKIP with pre-shared key (**PSK**)

WPA enterprise: TKIP with IEEE Std **802.1X with SHA-1**

WPA2 personal: CCMP-128 with **PSK**

WPA2 enterprise CCMP-128 with IEEE Std **802.1X with SHA-1**

WPA3 personal: CCMP-128 with **SAE** and PMF

WPA3 enterprise CCMP-128 with IEEE Std **802.1X with SHA-256** and PMF

### Encryption Type

Selects the encryption type to be used in AP operation mode.

For AP operation mode, encryption for WPA is limited to TKIP and encryption for WPA2 and WPA3 is limited to CCMP-128.

For station operation mode, refer to [Chapter 2.2.3, "WLAN security"](#), on page 21.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ENCRyption`

### Protection Management Frame

Selects, whether the protection management frames are unsupported, supported or required.

This parameter applies to WPA2, WPA3 in AP operation mode.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNection:SECurity:PMF`

### SAE H2E Support

Selects authentication and key management mechanism supported by the R&S CMW:

- Hunting-and-pecking negotiated in SAE exchange

- Simultaneous authentication of equals (SAE) hash-to-element (H2E) mandatory for WPA3 and for SAE in 6 GHz band
- Both: the R&S CMW supports both authentication and key management mechanisms

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNECTION:SECurity:HASHtoelem`

### Common Settings > Group Transform

Specifies the group transform for WPA3 personal security mode.

This setting applies to WPA3 personal in AP operation mode.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNECTION:SECurity:GTRansform`

### Common Settings > Passphrase

This setting applies to WPA/WPA2/WPA3 personal in AP operation mode.

The used passphrase consists of 1 to 63 characters.

In station mode, the selection of ciphers is automatic, refer to [Chapter 2.2.3, "WLAN security"](#), on page 21.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNECTION:SECurity:PKEY`

`CONFigure:WLAN:SIGN<i>:CONNECTION:SECurity:PASSphrase`

### WPA/WPA2/WPA3 Enterprise > Radius Server

The settings in this section are related to the RADIUS server for the WPA / WPA2 enterprise mode.

#### Mode ← WPA/WPA2/WPA3 Enterprise > Radius Server

You can select between the internal RADIUS server and an external RADIUS server.

Remote command:

`CONFigure:WLAN:SIGN<i>:CONNECTION:SECurity:RSERver:MODE`

#### EAP-SIM ← WPA/WPA2/WPA3 Enterprise > Radius Server

This section configures EAP-SIM on the internal RADIUS server.

|         |                                  |
|---------|----------------------------------|
| EAP-SIM |                                  |
| RAND 1  | 55555555555555555555555555555555 |
| SRES 1  | 55545756                         |
| KC 1    | 0000000000000000                 |
| RAND 2  | 44444444444444444444444444444444 |
| SRES 2  | 44454647                         |
| KC 2    | 0000000000000000                 |
| RAND 3  | 33333333333333333333333333333333 |
| SRES 3  | 33323130                         |
| KC 3    | 0000000000000000                 |

Define three triplets (RAND, SRES, KC) to be used for authentication:

- RAND: The random challenge as 32-digit hexadecimal number.
- SRES: The signed response as 8-digit hexadecimal number.
- KC: The ciphering key as 16-digit hexadecimal number.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ESIM:KTOne
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ESIM:KTTwo
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ESIM:KTThree
```

### EAP-AKA ← WPA/WPA2/WPA3 Enterprise > Radius Server

This section configures EAP-AKA on the internal RADIUS server.

Define the keys and algorithm to be used:

- "Ki": The secret key is used for the authentication procedure (including a possible integrity check). The value is entered as 32-digit hexadecimal number.
- "OPC": The operator variant key is used for authentication and integrity check procedures with the MILENAGE algorithm set. The value is entered as 32-digit hexadecimal number.
- "Rand": The random number is used for the authentication procedure (including a possible integrity check). It is entered as 32-digit hexadecimal number.
- "Algorithm": The authentication algorithm to be used, which can be XOR or MILENAGE.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:EAKA:KALGo
```

### IP Address / Port Number / Shared Key ← WPA/WPA2/WPA3 Enterprise > Radius Server

These settings configure the address of an external RADIUS server and the shared secret that the RADIUS server uses to obfuscate passwords.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:ICONf
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:PNUMBER
CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:SKEY
```

#### 2.3.8.12 QoS data TID settings

Access: "Config" hotkey > "Connection" > "QoS Data TID" section.

This section configures traffic identifier (TID) transmission for different connection types.

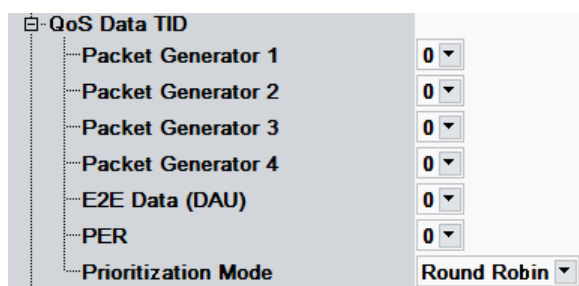


Figure 2-45: TID setting

**QoS Data TID**

Selects the TIDs for packet generators, end-to-end data connections including DAU, and PER measurements.

For HE bursts, TID controls allow you to set up generation of frames from multiple TIDs.

Remote command:

```
CONFfigure:WLAN:SIGN<i>:PGEN<index>:CONFig
CONFfigure:WLAN:SIGN<i>:CONNection:QOS:ETOE
CONFfigure:WLAN:SIGN<i>:PER:TIDentifier
```

**Prioritization Mode**

Selects the aggregation method of TIDs within an A-MPDU:

- Round-robin schedules equal transmission time to each TID
- TID priority selection prioritizes the transmission of highest TID values.

Remote command:

```
CONFfigure:WLAN:SIGN<i>:CONNection:QOS:PRIoritiz
```

**2.3.8.13 Block Ack settings**

Access: "Config" hotkey > "Connection" > "Block Ack" section.

This section configures a BlockAck session.

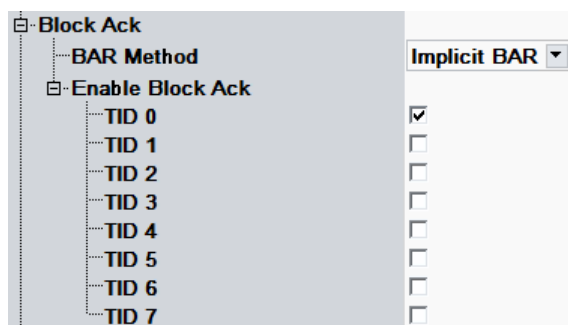


Figure 2-46: Block ack setting

**BAR Method**

Specifies the method used to request a BlockAck frame from the DUT.

The following methods are supported:

- **Explicit:** BlockAck frame transmitted in response to a block ack request
- **Implicit:** BlockAck frame transmitted in response to a frame containing an implicit block ack request
- **MU-BAR:** BlockAck frame transmitted in response to a multi-user block ack request (MU-BAR) trigger frame. With this method, BA or ACK frames from multiple STAs are multiplexed in uplink.

Remote command:

`CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:QOS:BARMethod`

### Enable Block Ack

Enables/ disables a Block Ack session for the specified TID.

Remote command:

`CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:QOS:BLAck`

## 2.3.8.14 A-MPDU settings

Access: "Config" hotkey > "Connection" > "A-MPDU" section.

These settings configure aggregate MPDUs (A-MPDU) with or without multiple TIDs for HT, VHT, and HE bursts.

|                                |                                     |
|--------------------------------|-------------------------------------|
| <b>A-MPDU</b>                  |                                     |
| TX Enable                      | <input checked="" type="checkbox"/> |
| Allow Multiple TID Aggregation | <input type="checkbox"/>            |
| Max A-MPDU Length (bytes)      | 131071                              |

### TX Enable

Enables or disables A-MPDU with multiple MPDUs.

A-MPDUs are sent at the rate specified by the data frame control, see ["Non-HT, HT, VHT, HE Frame"](#) on page 73.

Remote command:

`CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:AMPDu`

### Allow Multiple TID Aggregation

Enables or disables different TIDs within an A-MPDU. This setting is only relevant for HE frame formats, only when the DUT also supports reception and when data packets from more than one TID are configured in the R&S CMW.

See also [Chapter 2.3.11, "Packet generator settings"](#), on page 95.

Option R&S CMW-KS657 is required.

Remote command:

`CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:AMPDu`

### Max A-MPDU Length

Sets the maximal length of the entire A-MPDU. this setting effectively limits the number of subframes that can be aggregated.

Remote command:

`CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:AMPDu`



### 2.3.8.15 EDCA settings

Access: "Config" hotkey > "Connection" > "EDCA" section.

Configures enhanced distributed channel access (EDCA) for access categories (ACs) best effort (AC\_BE), background (AC\_BK), video (AC\_VI), and voice (AC\_VO). Settings for EDCA parameter set and MU EDCA parameter set are supported, to allow AP control over scheduling and channel access for UL MU transmission.

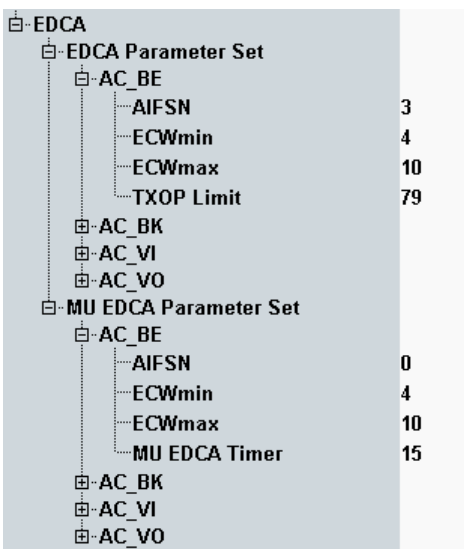


Figure 2-47: EDCA settings

|                            |    |
|----------------------------|----|
| EDCA Parameter Set.....    | 85 |
| MU EDCA Parameter Set..... | 86 |

#### EDCA Parameter Set

Defines EDCA prioritization parameters for all access categories.

- **"AIFSN"**: arbitration inter-frame space number
- **"ECWmin, ECWmax"**: the contention window (CW) represented by its minimum and maximum limits
- **"TXOP Limit"**: The transmission opportunity limit defines the time duration for the transmission of STA.

EDCA assigns smaller CW bounds, smaller AIFS and greater TXOPLimit to ACs with higher priorities and vice versa. For details, refer to [Chapter 2.2.13, "EDCA"](#), on page 31.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:EDCA:ACBE
CONFigure:WLAN:SIGN<i>:CONNection:EDCA:ACBK
CONFigure:WLAN:SIGN<i>:CONNection:EDCA:ACVI
CONFigure:WLAN:SIGN<i>:CONNection:EDCA:ACVO
```

**MU EDCA Parameter Set**

Configures the EDCA scheduling parameters for multi-user access in uplink. MU EDCA parameter set only applies for a finite period of time controlled by the MU EDCA timer. For details, refer to [EDCA](#)

Remote command:

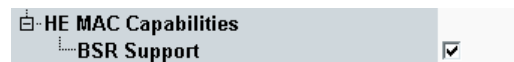
```
CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACBE
CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACBK
CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACVI
CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACVO
```

**2.3.8.16 HE MAC capabilities**

Access: "Config" hotkey > "Connection" > "HE MAC Capabilities" section.

This section sets the flags signaled within HE MAC capabilities.

Option R&S CMW-KS657 is required.



*Figure 2-48: HE MAC capabilities settings*

**BSR Support**

Indicates, whether the R&S CMW supports the buffer status report (BSR).

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:HEMac:BSRSupport
```

**2.3.8.17 NDP sounding settings**

Access: "Config" hotkey > "Connection" > "NDP Sounding" section.

This section is relevant for 802.11ax standard, AP and Hotspot 2.0 operation mode.

The following section describes measurements on HE null data packet (NDP) sounding protocol to facilitate beamforming and DL MU-MIMO transmission.

The measurement involves the following steps:

1. AP sends a VHT/HE NDP announcement frame, that signals the start of the procedure.
2. AP sends an HE sounding NDP exactly SIFS after the announcement .
3. STA uses the HE sounding NDP to measure the channel and returns to the AP an HE compressed beamforming/CQI frame containing the channel feedback.

Option R&S CMW-KS671 is required.

Figure 2-49: Configuration and control of NDP sounding

|                             |    |
|-----------------------------|----|
| Feedback Method.....        | 87 |
| STA.....                    | 87 |
| Bandwidth.....              | 87 |
| Spatial Streams.....        | 88 |
| GI and LTF.....             | 88 |
| Report Type.....            | 88 |
| RU Start Index.....         | 88 |
| RU End Index.....           | 88 |
| Codebook Size.....          | 88 |
| Number of Columns (Nc)..... | 88 |
| Subcarrier Grouping.....    | 88 |
| Transmission.....           | 88 |

### Feedback Method

Sets the feedback method for NDP sounding procedure. Trigger-based and non-trigger-based methods are supported.

Remote command:

CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:METHod

### STA

Specifies the STA to which the NDP sounding applies. This parameter is visible, if [Multi STA](#) is enabled.

Remote command:

CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TARGet

### Bandwidth

Selects the channel bandwidth for NDP sounding procedure.

Remote command:

CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:BW

**Spatial Streams**

Selects the number of spatial streams for NDP sounding procedure.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SPSTreams
```

**GI and LTF**

Selects the GI / LTF combination for NDP sounding procedure.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:LTFGi
```

**Report Type**

Selects the report type for NDP sounding procedure. All types of feedback are returned via the HE compressed Beamforming/CQI frame.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TYPE
```

**RU Start Index**

Specifies the last 26-tone RU to be measured (RU end index) during NDP sounding procedure.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:RUStart
```

**RU End Index**

Specifies the last 26-tone RU to be measured (RU end index) during NDP sounding procedure.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:RUEnd
```

**Codebook Size**

Sets the codebook size for HE trigger-based sounding: 0 or 1. Refer to standard IEEE Std 802.11ax-2021, table 9-29a.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:CBOok
```

**Number of Columns (Nc)**

Sets the number of columns value  $N_c$  for HE TB sounding.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:NUMColumns
```

**Subcarrier Grouping**

Sets the subcarrier grouping value  $N_g$  for HE TB sounding.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SUBGrouping
```

**Transmission**

For NDP sounding procedure, you can trigger the single-shot transmission or switches on the periodic transmission.

For the periodic transmission, set the transmission interval.

Remote command:

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SSTX
```

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TXEN
```

```
CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TXP
```

### 2.3.9 Trigger settings

Access: "Config" hotkey > "Trigger" section.

The "Trigger" parameters configure the trigger signals generated by the WLAN signaling application. The trigger signals can be used "internally" to trigger a measurement application or "externally" to synchronize external devices to the WLAN signal.

See also [Chapter 2.2, "General description"](#), on page 15.

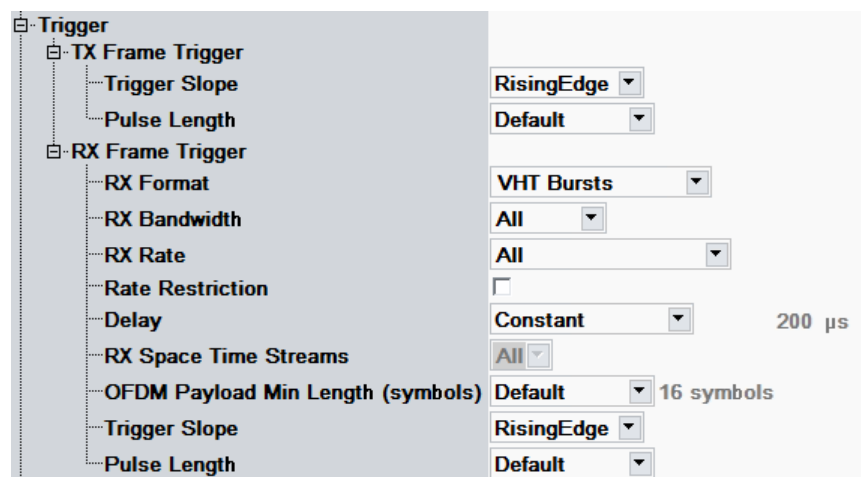


Figure 2-50: Trigger settings

|                                         |    |
|-----------------------------------------|----|
| TX Frame Trigger.....                   | 89 |
| L Trigger Slope.....                    | 90 |
| L Pulse Length.....                     | 90 |
| RX Frame Trigger.....                   | 90 |
| L RX Format.....                        | 90 |
| L RX Bandwidth.....                     | 91 |
| L RX Rate.....                          | 91 |
| L Rate Restriction.....                 | 91 |
| L Delay.....                            | 91 |
| L RX Space Time Streams.....            | 91 |
| L Min Length: DSSS or OFDM Payload..... | 91 |
| L Trigger Slope.....                    | 93 |
| L Pulse Length.....                     | 93 |

#### TX Frame Trigger

This section configures the TX frame trigger signal. It is aligned to the start of the downlink MAC frames.

To address this trigger signal in remote commands of other applications, use the string "WLAN Sig<i>: FrameTrigger", with <i> replaced by the instance number of the signaling application.

#### Trigger Slope ← TX Frame Trigger

Aligns either the rising edge or the falling edge of the trigger pulses with the start of the TX frames.

Changing the "Trigger Slope" changes the polarity of the trigger signal.

Remote command:

```
TRIGger:WLAN:SIGN<i>:TX:MACFrame:SLOPe
```

#### Pulse Length ← TX Frame Trigger

Configures the length of the trigger pulses.

"Default" The pulse duration is 1  $\mu$ s.

"Burst Length" The pulse length equals the burst length.

"User Defined" You can specify a pulse length.

Remote command:

```
TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgt h:MODE
TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgt h:VALue
```

#### RX Frame Trigger

This section configures the RX frame trigger signal. It is aligned to the start of the uplink MAC frames.

To address this trigger signal in remote commands of other applications, use the string "WLAN Sig<i>: RXFrameTrigger", with <i> replaced by the instance number of the signaling application.

#### RX Format ← RX Frame Trigger

Defines for which bursts a trigger pulse is generated.

Note that the trigger pulse is generated only for bursts matching the specified [RX Bandwidth](#) and [RX Rate](#).

"All Bursts" All received bursts result in an RX frame trigger pulse. There is no minimum length. Even ACK bursts result in a trigger pulse.

"OFDM Bursts" Only OFDM bursts with the configured minimum length result in an RX frame trigger pulse. This mode remains for backward compatibility. Use rather modes with the burst type specification (e.g. "VHT Bursts").

"DSSS/CCK Bursts" Only DSSS/CCK bursts with the configured minimum length result in an RX frame trigger pulse.

"Non-HT/HT/VHT/HE Bursts" Only particular bursts with the configured minimum length result in an RX frame trigger pulse.

Remote command:

```
TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe
```

**RX Bandwidth ← RX Frame Trigger**

Defines for which bandwidth of received bursts a trigger pulse is generated for the RX frame trigger signal.

Remote command:

`TRIGger:WLAN:SIGN<i>:RX:MACFrame:BW`

**RX Rate ← RX Frame Trigger**

Defines for which rate of received bursts a trigger pulse is generated for the RX frame trigger signal.

Remote command:

`TRIGger:WLAN:SIGN<i>:RX:MACFrame:RATE`

**Rate Restriction ← RX Frame Trigger**

When enabled, this feature selectively restricts (ignores) certain RX PPDU configurations so that the R&S CMW cannot acknowledge those frames. So the DUT gradually moves towards the specific target PPDU configuration that is to be measured. The effectiveness of the rate restriction technique depends on the particular device.

The rate restriction can control the DUT's TX bandwidth, so that it is even less than the operating channel width. It is also possible to control TX of only one spatial stream, even when configured for the MIMO scenario.

Rate restriction is provided for all supported standards except 802.11b.

Remote command:

`TRIGger:WLAN:SIGN<i>:RX:MACFrame:RREstriction`

**Delay ← RX Frame Trigger**

- "Constant": sets the delay of RX trigger to 200 µs (for OFDM signals only)  
The value is only approximated, as the processing delay and sample bus delay also contributes to this offset.  
Constant trigger delay is especially useful to measure several different IEEE 802.11 OFDM standards.
- "Burst Dependent": the trigger delay set automatically according to the generated signal

Remote command:

`TRIGger:WLAN:SIGN<i>:RX:MACFrame:CTDelay`

**RX Space Time Streams ← RX Frame Trigger**

Sets the number of antennas used by the RX frame trigger.

Remote command:

`TRIGger:WLAN:SIGN<i>:RX:MACFrame:STReams`

**Min Length: DSSS or OFDM Payload ← RX Frame Trigger**

This setting is relevant for all RX trigger formats except "All Bursts". It defines the number of symbols/bytes that need to be received before the trigger event is generated.

"Default" minimal length is usually appropriate for DSSS and OFDM standards 802.11a, g, n, and ac, 20 MHz. This mode uses a suitable minimum length, so that ACK bursts do not result in a trigger pulse.

With "User Defined", you can set the minimum number of symbols/bytes manually.

If you want to set a user-defined minimum length for OFDM, you can use the following formula to calculate the number of symbols from signal properties.

$$\text{Number of Data Symbols} = \left\lceil \frac{(\text{MAC overhead} + \text{PSDU Length}) * 8 + 22}{\text{Data bits per symbol}} \right\rceil$$

Where "MAC overhead" is the header in bytes and "PSDU Length" is the payload length in bytes. The value of "Data bits per symbol" ( $N_{\text{DBPS}}$ ) depends on the active transmission scheme, as given in the following table.

**Table 2-8: OFDM data bits per symbol depending on transmission scheme,  $N_{\text{SS}}=1$**

| Data rate in<br>Mbit/s | N <sub>DBPS</sub><br>(20 MHz)<br>a/g | MCS                          | N <sub>DBPS</sub><br>(20 MHz)<br>n, ac | N <sub>DBPS</sub><br>(40 MHz)<br>n, ac | N <sub>DBPS</sub><br>(80 MHz)<br>ac | N <sub>DBPS</sub><br>(160 MHz)<br>ac |
|------------------------|--------------------------------------|------------------------------|----------------------------------------|----------------------------------------|-------------------------------------|--------------------------------------|
| 6                      | 24                                   | MCS 0                        | 26                                     | 54                                     | 117                                 | 234                                  |
| 9                      | 36                                   | MCS 1                        | 52                                     | 108                                    | 234                                 | 468                                  |
| 12                     | 48                                   | MCS 2                        | 78                                     | 162                                    | 351                                 | 702                                  |
| 18                     | 72                                   | MCS 3                        | 104                                    | 216                                    | 468                                 | 936                                  |
| 24                     | 96                                   | MCS 4                        | 156                                    | 324                                    | 702                                 | 1404                                 |
| 36                     | 144                                  | MCS 5                        | 208                                    | 432                                    | 936                                 | 1872                                 |
| 48                     | 192                                  | MCS 6                        | 234                                    | 486                                    | 1053                                | 2106                                 |
| 54                     | 216                                  | MCS 7                        | 260                                    | 540                                    | 1170                                | 2340                                 |
|                        |                                      | MCS 8*                       | 312                                    | 648                                    | 1404                                | 2808                                 |
|                        |                                      | MCS 9*                       | -                                      | 720                                    | 1560                                | 3120                                 |
|                        |                                      | * only relevant for 802.11ac |                                        |                                        |                                     |                                      |

For further  $N_{\text{DBPS}}$  values, refer to IEEE Std 802.11-2016, chapters:

- 19.5 Parameters for HT MCSs
- 21.5 Parameters for VHT-MCSs

For the  $N_{\text{DBPS}}$  values for HE, refer to IEEE Std 802.11ax-2021, chapter 27.5 Parameters for HE-MCSs.

#### Example:

For 20 MHz channel with a header of 26 bytes and a payload length of 500 bytes, in 802.11n MCS 7 the number of data symbols are:

$$\lceil [(26+500)*8 + 22]/260 \rceil = 16$$

**Note:** The recommended "Min Length" of RX frame trigger for bandwidths  $\geq 80$  MHz is 6.

Remote command:

TRIGger:WLAN:SIGN<i>:RX:MACFrame:DSMLength

TRIGger:WLAN:SIGN<i>:RX:MACFrame:OFMLength



**Trigger Slope ← RX Frame Trigger**

Aligns either the rising edge or the falling edge of the trigger pulses with the start of the MAC frames.

Changing the "Trigger Slope" changes the polarity of the trigger signal.

Remote command:

```
TRIGger:WLAN:SIGN<i>:RX:MACFrame:SLOPe
```

**Pulse Length ← RX Frame Trigger**

Configures the length of the trigger pulses.

"Default" The pulse duration is 1 µs.

"User Defined" You can specify a pulse length.

Remote command:

```
TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:MODE
```

```
TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:VALue
```

## 2.3.10 IP support

Access: "Config" hotkey > "IP Support" section.

This section defines the required IP configuration. The IPv4 and IPv6 sections are only displayed if the support of the corresponding IP version is enabled.

- **When a DAU is present**, IP connectivity and end-to-end data functionality are provided by the data application unit (DAU), see [Chapter 2.2.11, "Data end-to-end tests"](#), on page 29.  
In AP mode, the "IP Support" section serves as a display of the IP settings configured in the DAU. When the WLAN signal is enabled, the values displayed here represent the network configuration that is advertised by the DHCP server in the DAU. In particular, the STA address field shows the IP address that is assigned to the DUT by DHCP.
- **When no DAU is present:**  
In AP mode, these fields are used to enter the IPv4 and IPv6 network configuration. The DHCP server in the WLAN signaling stack uses these values when queried by a DHCP client in the DUT



Figure 2-51: IP support settings (AP mode)

|                         |    |
|-------------------------|----|
| IP Version Support..... | 94 |
| IPv4 Interface.....     | 94 |
| IPv6 Interface.....     | 94 |

### IP Version Support

Defines the required IP version support. For background information, see [Chapter 2.2.10, "IP communication"](#), on page 29.

The IPv6 prefix entry is only available in the modes AP and Hotspot 2.0.

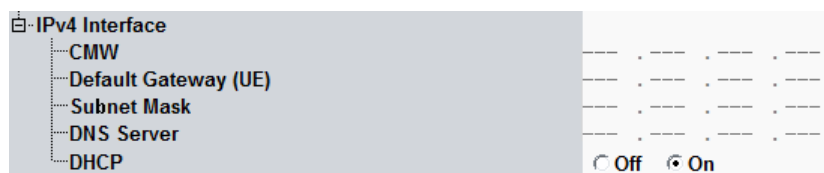
Remote command:

```
CONFfigure:WLAN:SIGN<i>:CONNection:IVSupport
```

### IPv4 Interface

Configures the internal IP addressing for IP V4. Configures the internal IPv4 stack and sets the destination address. These settings appear in station mode, with or without a DAU present.

These IP settings are used by the packet generator, see [Chapter 2.3.11, "Packet generator settings"](#), on page 95.



The manual and DHCP configuration is supported:

- To configure all parameters manually, set DHCP to "Off". Enter the static IP network configuration that is to be used by the R&S CMW.  
Mixing the static IPv4 and dynamic IPv6 addresses at the same time is supported.
- To use DHCP assignment, set DHCP to "On". The dialog becomes a read-only display of the network configuration obtained by the R&S CMW DHCP client from the DUT DHCP server.

Remote command:

```
CONFfigure:WLAN:SIGN<i>:IPVFour:STATic:IPADdress:CMW
CONFfigure:WLAN:SIGN<i>:IPVFour:STATic:IPADdress:STA<s>
CONFfigure:WLAN:SIGN<i>:IPVFour:STATic:IPADdress:GATeway
CONFfigure:WLAN:SIGN<i>:IPVFour:STATic:SMASK
CONFfigure:WLAN:SIGN<i>:IPVFour:STATic:IPADdress:DNS
CONFfigure:WLAN:SIGN<i>:IPVFour:DHCP
```

### IPv6 Interface

Defines the IPv6 prefix of the built-in IPv6 stack.

The setting is relevant in an R&S CMW without DAU, refer to [Chapter 2.2.10, "IP communication"](#), on page 29.

When a DAU is present, the parameter displays IPv6 prefix that is configured in the DAU.

Remote command:

```
CONFfigure:WLAN:SIGN<i>:IPVSix:PREFix
```

### 2.3.11 Packet generator settings

Access: "Config" hotkey > "Packet Generator" section.

This section configures up to three packet generators. For background information, see [Chapter 2.2.12, "Packet generator"](#), on page 30.

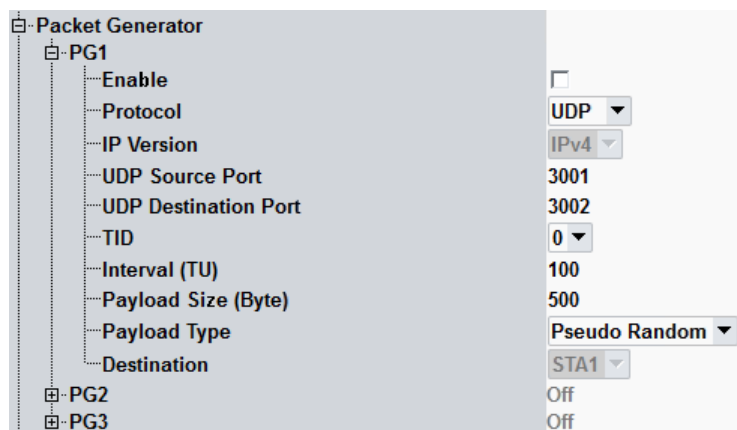


Figure 2-52: Packet generator settings

|                                  |    |
|----------------------------------|----|
| Enable.....                      | 95 |
| Protocol.....                    | 95 |
| IP Version.....                  | 95 |
| TID.....                         | 96 |
| UDP Source/Destination Port..... | 96 |
| Interval.....                    | 96 |
| Payload Size.....                | 96 |
| Payload Type.....                | 96 |
| Destination.....                 | 96 |

#### Enable

Enables or disables the packet generator. If enabled, the R&S CMW repeatedly transmits packets of the configured protocol type to the DUT.

Remote command:

`CONFigure:WLAN:SIGN<i>:PGEN<index>:CONFig`

#### Protocol

The protocol underlying the generated packets: ICMP ("Ping") or UDP.

Remote command:

`CONFigure:WLAN:SIGN<i>:PGEN<index>:PROToCol`

#### IP Version

Select the IP version to be used for generating packets.

The IP version selected for packet generators must be enabled by the [IP support](#) configuration.

Remote command:

`CONFigure:WLAN:SIGN<i>:PGEN<index>:IPVerSion`

**TID**

Selects the TID signaled by the packet generator.

Three instances of packet generator provide for HE bursts also the generation of multiple TIDs, see [Chapter 2.3.8.12, "QoS data TID settings"](#), on page 82.

Remote command:

```
CONFigure:WLAN:SIGN<i>:PGEN<index>:CONFig
```

**UDP Source/Destination Port**

The source/destination port of the UDP packet generator.

Remote command:

```
CONFigure:WLAN:SIGN<i>:PGEN<index>:UPORts
```

**Interval**

Time between packet transmissions in units of 1024  $\mu$ s.

Configure it long enough to transmit a packet with the given protocol and payload size at the current rate. If the interval is too short to transmit a packet, the scheduled packet transmissions are silently canceled.

Remote command:

```
CONFigure:WLAN:SIGN<i>:PGEN<index>:CONFig
```

**Payload Size**

Size of a packet's payload in bytes.

Remote command:

```
CONFigure:WLAN:SIGN<i>:PGEN<index>:CONFig
```

**Payload Type**

Bit sequence to be transmitted as payload.

Remote command:

```
CONFigure:WLAN:SIGN<i>:PGEN<index>:CONFig
```

**Destination**

Specifies the STA to which the packets are addressed to. This parameter is visible, if [Multi STA](#) is enabled.

Remote command:

```
CONFigure:WLAN:SIGN<i>:PGEN<index>:DESTination
```

## 2.3.12 Data end-to-end settings

Access: "Config" hotkey > "Data End to End" section.

This section provides settings related to an internal or external data application unit (DAU). For background information, see [Chapter 2.2.11, "Data end-to-end tests"](#), on page 29.

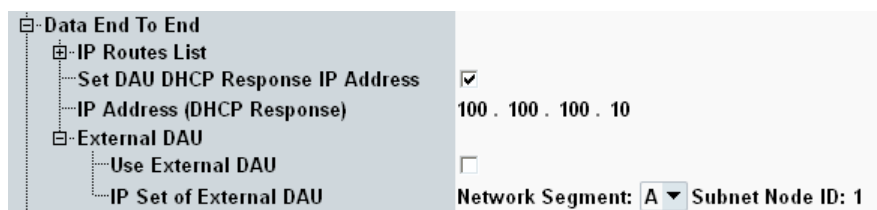


Figure 2-53: Data end-to-end settings (station)

|                                                                   |    |
|-------------------------------------------------------------------|----|
| IP Routes List.....                                               | 97 |
| Set DAU DHCP Response IP Address, IP Address (DHCP Response)..... | 97 |
| External DAU.....                                                 | 97 |
| L Use External DAU.....                                           | 98 |
| L IP Set of External DAU.....                                     | 98 |

### IP Routes List

Configures additional destination addresses for which the DAU routes packets to the DUT. Only for operation mode "Station".

The addresses of the DUT are known by the DAU. You do not need to enter them here. Enter, for example, addresses of a PC or server behind the DUT, for which traffic is to be routed via the DUT.

The default IP address of the second route can be used for single box tethering.

Depending on the configured IP version support, only the IPv4 part or the IPv6 part of the routes list is displayed (see [Chapter 2.3.10, "IP support"](#), on page 93).

| IP Routes List  | Enable                              | IPv4 Route / Mask (bit) |
|-----------------|-------------------------------------|-------------------------|
| Route Address 1 | <input checked="" type="checkbox"/> | 192 . 168 . 20 . 0 / 24 |
| Route Address 2 | <input checked="" type="checkbox"/> | 172 . 22 . 1 . 250 / 32 |
| Route Address 3 | <input type="checkbox"/>            | 0 . 0 . 0 . 0 / 0       |
| Route Address 4 | <input type="checkbox"/>            | 0 . 0 . 0 . 0 / 0       |
| Route Address 5 | <input type="checkbox"/>            | 0 . 0 . 0 . 0 / 0       |

To specify a destination, enable a line, enter the address and specify the number of bits identifying the subnet.

Example: "Mask" = 24 bit means that the first 24 bits identify the subnet. For IPv4, 24 bit corresponds to the subnet mask 255.255.255.0.

Remote command:

`CONFigure:WLAN:SIGN<i>:ETOE:IRList:IPAddress<n>`

### Set DAU DHCP Response IP Address, IP Address (DHCP Response)

If the checkbox is enabled, you can specify the IPv4 address that the DAU assigns to the DUT via DHCP.

Remote command:

`CONFigure:WLAN:SIGN<i>:ETOE:DUIP`

### External DAU

Configures and enables the usage of an external DAU, see also [Chapter 2.2.11, "Data end-to-end tests"](#), on page 29.

**Use External DAU ← External DAU**

Enable the checkbox if you want to use an external DAU.

The two instruments must be connected via LAN. For details, refer to the DAU documentation.

Remote command:

`CONFigure:WLAN:SIGN<i>:EDAU:ENABLE`

**IP Set of External DAU ← External DAU**

Select the "Network Segment" and "Subnet Node ID" of the R&S CMW where the external DAU is installed. You can check the values in the remote instrument's "Setup" dialog > "Misc" > "IP Subnet Config".

The two instruments must use the same network segment and must have different node IDs.

Remote command:

`CONFigure:WLAN:SIGN<i>:EDAU:NSEGment`

`CONFigure:WLAN:SIGN<i>:EDAU:NID`

### 2.3.13 Shortcut configuration

Access: "Config" hotkey > "Shortcut Softkey" section.

This section configures the three shortcut softkeys that provide a fast way to switch to selectable measurements.

See also [Chapter 2.3.3, "Using the shortcut softkeys"](#), on page 45

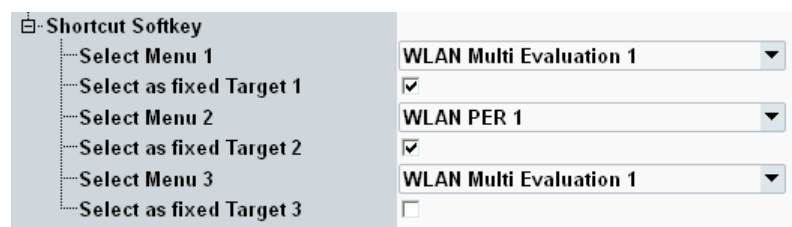


Figure 2-54: Shortcut configuration

**Select Menu**

Selects a measurement. The corresponding shortcut softkey opens a dialog presenting this measurement as default target or uses the measurement as fixed target.

**Select as fixed Target**

Configures and renames the corresponding shortcut softkey.

- **Enabled:** The softkey directly opens the measurement selected via [Select Menu](#).
- **Disabled:** The softkey opens a dialog box for selection of the target measurement.

### 2.3.14 Message monitoring settings

Access: "Config" hotkey > "Message Monitoring" section.

The "Message Monitoring" section allows you to configure the IPv4 connection to an R&S CMWmars instance and to enable the message replication.

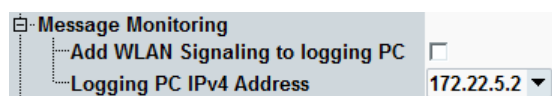


Figure 2-55: Message monitoring settings

For background information, see [Chapter 2.2.8, "Message logging"](#), on page 27.

### Add WLAN Signaling to Logging PC

Enables or disables R&S CMWmars message monitoring for the WLAN signaling application.

Remote command:

`CONFigure:WLAN:SIGN<i>:MMONitor:ENABLE`

### Logging PC IPv4 Address

Selects the IP address to which the messages are sent for monitoring.

The address pool is configured globally, see "Setup" dialog, section "Logging".

Remote command:

`CONFigure:WLAN:SIGN<i>:MMONitor:IPAddress`

## 2.3.15 Debug configuration

Access: "Config" hotkey > "Debug" section.

The "Debug" section configures the log file generation and selects which message types are replicated for analysis with R&S CMWmars.



Do not enable the debug mode during regular operation. It impacts the test performance, timing, hard disk usage, and so on.

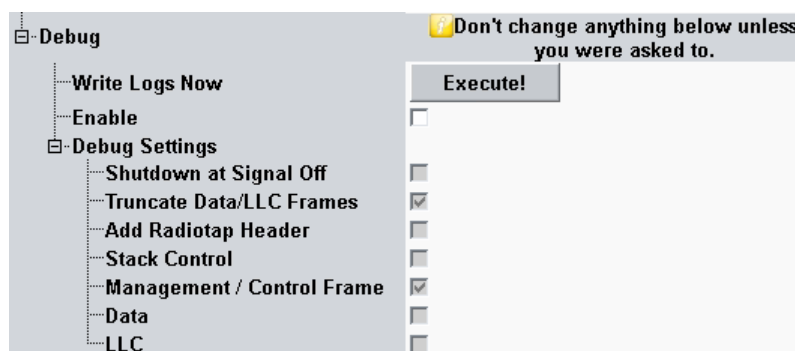


Figure 2-56: Debug configuration

The settings are not affected by a reset or preset. They are not stored in save files. A reboot restores the default settings.

|                     |     |
|---------------------|-----|
| Write Logs Now..... | 100 |
| Enable.....         | 100 |
| Debug Settings..... | 100 |

### Write Logs Now

Writes the WLAN log files to the system drive.

### Enable

Enables the configuration of the "Debug Settings".

### Debug Settings

Selects the types of control messages to be logged.

For enabled "Shutdown at Signal Off", the signaling unit shuts down after the signal is switched off. All selected logs are stored. It minimizes the performance impact during end-to-end data connections.

For enabled "Truncate Data/LLC Frames", MAC data and LLC data frames logged with the payload truncated include only the protocol headers (e.g. IP/UDP/DHCP).

If the "Add Radiotap Header" is enabled, radiotap headers provide additional information to supplement the raw frame capture data that can be derived by analyzing the 802.11 frames (e.g. via Wireshark).

For disabled "Data" logging, user plane control (UPC) logging is also disabled automatically.

The settings affect log files and message monitoring with R&S CMWmars.

## 2.3.16 Statistics settings

Access: "Config" hotkey > "Statistics" section.

Configures the collecting of statistics displayed in the WLAN signaling main view > "Statistics" button to the right, see [Chapter 2.3.1.9, "Statistics"](#), on page 42.



Figure 2-57: Statistics settings

### Reporting Interval

Sets the time interval for statistics reports for the calculation of current values

Remote command:

`CONFigure:WLAN:SIGN<i>:UESinfo:SETTings`

### Time Span

Determines how many reporting intervals are accumulated for the "Total" and "Average" columns of the statistics display. If a statistics report is older than "Time Span" allows, that report is excluded. If for instance, the newer reports do not contain newer statistics for a particular format and rate, then the values in total and average columns decrease.



Remote command:

CONFigure:WLAN:SIGN<i>:UESinfo:SETTings

### 2.3.17 Station configuration tabs

If [Multi STA](#) is enabled, the main configuration tree appears in a tab "CMW" (see [Chapter 2.3.4, "Global settings"](#), on page 46) and additional tabs "Station 1", "Station 2" and "Station 3" are displayed.

Option R&S CMW-KS667 is required.

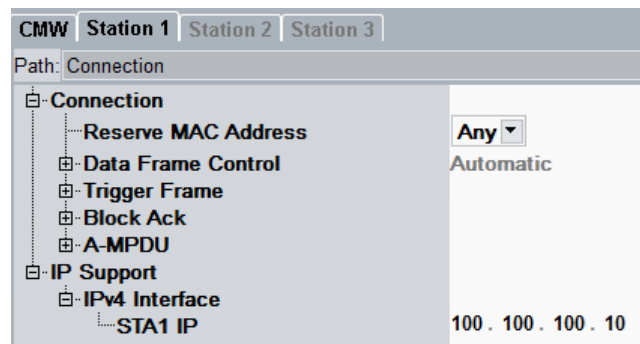


Figure 2-58: Settings per station

Many parameters per station are same as in main configuration tree. For the description, refer to:

- ["Reserve MAC Address"](#) on page 60
- [Chapter 2.3.8.9, "Data frame control settings"](#), on page 71
- [Chapter 2.3.8.10, "Trigger frame settings"](#), on page 75
  - [RU Allocation](#)
  - [Coding Type](#)
  - [MCS](#)
  - [DCM](#)
  - [Starting Spatial Stream](#)
  - [Number of Spatial Streams](#)
  - [Target RSSI](#)
- [Chapter 2.3.8.13, "Block Ack settings"](#), on page 83
- [Chapter 2.3.8.14, "A-MPDU settings"](#), on page 84

#### STAx IP

Sets the fixed IPv4 address of particular station.

Remote command:

CONFigure:WLAN:SIGN<i>:IPVFour:STATIC:IPAddress:STA<s>

### 2.3.18 PER measurement

The PER measurement transmits user data to the DUT and calculates the PER as the ratio of unacknowledged packets to transmitted packets. Unacknowledged packets are not retransmitted.

The number of MAC test packets to be transmitted is configurable. Each of the test packets carries the same amount of user data to the DUT. The test packets can be separated by unused packets.

A PER measurement requires an active association between AP and station, see [Chapter 2.2.1.2, "Establishing a WLAN association"](#), on page 17.

A packet error rate (PER) measurement is installed together with the WLAN signaling application. To access the measurement, press the softkey "WLAN PER" in the WLAN signaling main view.

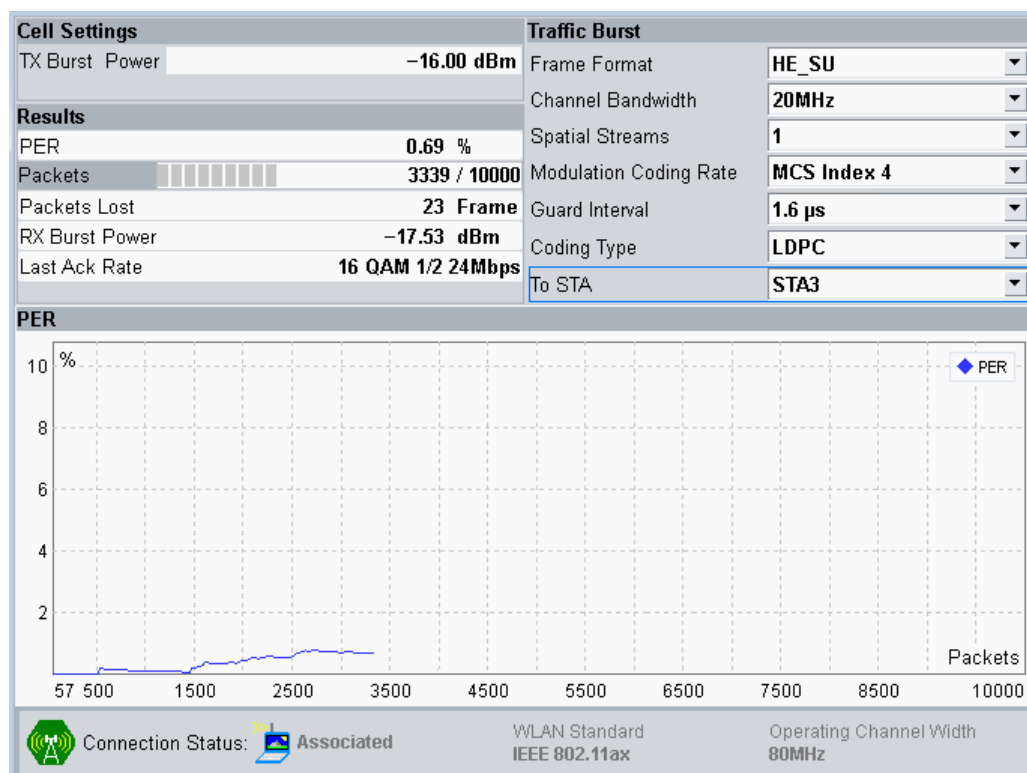


Figure 2-59: PER view

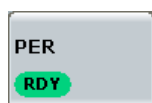
The "PER" view shows all measurement results, the most important settings and status information of the signaling application:

- Settings are the same as in the signaling configuration dialog, refer to [Measurement settings](#)
  - "TX Burst Power": see ["TX Burst Power"](#) on page 55.
  - "To STA" setting appears when [Multi STA](#) is enabled. Specify the station No. for which the traffic is measured.
  - For the description of remaining parameters, refer to [Chapter 2.3.8.9, "Data frame control settings"](#), on page 71.

- For status information, refer to [Chapter 2.3.1.1, "Connection status"](#), on page 33
- For results, refer to [Chapter 2.3.18.2, "Measurement results"](#), on page 103
- [Measurement control](#)..... 103
- [Measurement results](#)..... 103
- [Measurement settings](#)..... 104

### 2.3.18.1 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



#### PER (Softkey)

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:WLAN:SIGN<i>:PER
STOP:WLAN:SIGN<i>:PER
ABORt:WLAN:SIGN<i>:PER
FETCh:WLAN:SIGN<i>:PER:STATe?
FETCh:WLAN:SIGN<i>:PER:STATe:ALL?
```

### 2.3.18.2 Measurement results

Access: "WLAN Signaling" main view > "WLAN PER" softkey.

The "PER" results include scalar results and PER vs time diagram.

The results are described in the following.

#### Scalar results

| Results        |                                     |
|----------------|-------------------------------------|
| PER            | 0.20 %                              |
| Packets        | <div><div></div></div> 1 000 / 1000 |
| Packets Lost   | 2 Frame                             |
| RX Burst Power | -28.56 dBm                          |
| Last Ack Rate  | QPSK 1/2 12Mbps                     |

|                |                                                                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| "PER"          | Packet error rate calculated as ratio of unacknowledged packets to transmitted packets.                                                                   |
| "Packets"      | Progress indicator of the measurement, displaying the number of already transmitted packets and the configured total number of packets to be transmitted. |
| "Packets Lost" | Number of unacknowledged transmitted packets. Lost packets are not retransmitted.                                                                         |

"RX Burst Power" The average burst power of the received ACK frames, measured over the preamble.

"Last Ack Rate" The modulation and coding rate/scheme of the last received ACK frame.

Remote command:

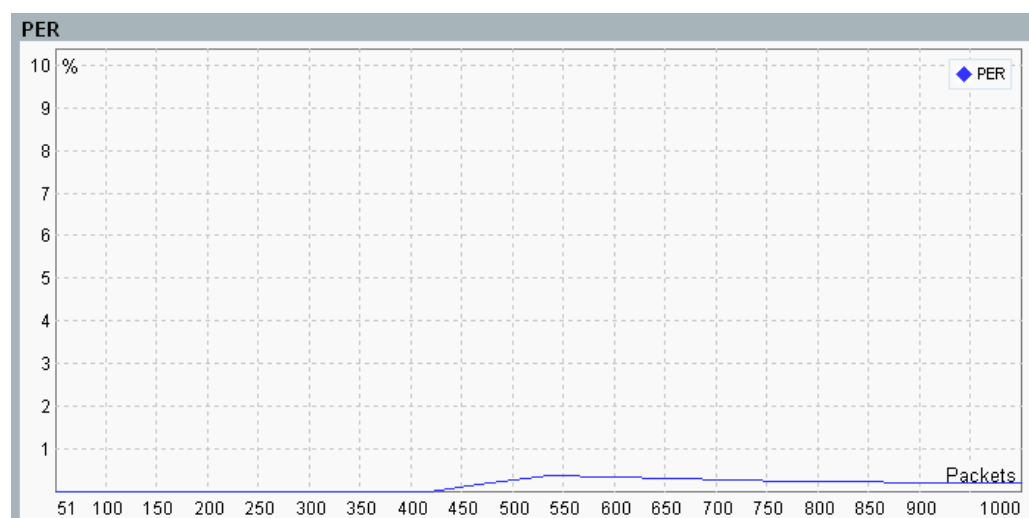
`FETCH:WLAN:SIGN<i>:PER?`

`READ:WLAN:SIGN<i>:PER?`

`FETCH:WLAN:SIGN<i>:PACKrate?`

### PER diagram

The diagram in the lower part of the dialog shows the PER vs. time/transmitted packets.



### 2.3.18.3 Measurement settings

Access: PER view > "PER" softkey > "Config" softkey.

The "Config" hotkey opens either the PER configuration dialog or the configuration dialog of the signaling application, depending on which softkey is active.

This chapter describes the settings in the PER configuration dialog.

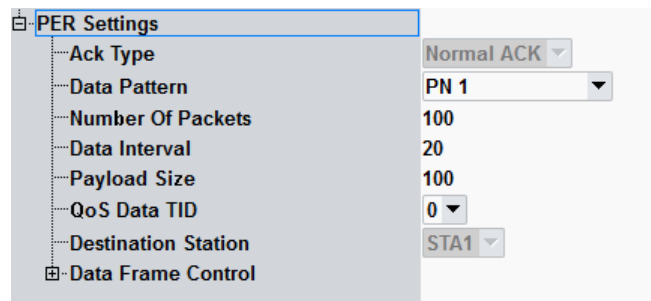


Figure 2-60: PER configuration dialog

|                                               |     |
|-----------------------------------------------|-----|
| Ack Type.....                                 | 105 |
| Data Pattern.....                             | 105 |
| Number of Packets.....                        | 105 |
| Data Interval.....                            | 105 |
| Payload Size.....                             | 105 |
| Qos Data TID.....                             | 106 |
| Destination Station.....                      | 106 |
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| L Link to SIG Frame Settings.....             | 106 |
| L Frame Format.....                           | 107 |
| L Non-HT, HT, VHT, HE Data Frame Control..... | 107 |
| L 20 MHz Channel Settings.....                | 108 |
| L Users and Dummies.....                      | 108 |

### Ack Type

Selects an evaluation scheme for the PER measurement. Currently, only the standard frame acknowledgment is supported.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:AType

### Data Pattern

Selects the data that the R&S CMW transfers to the DUT. The following bit patterns are supported:

- PN<no>: Pseudo-noise sequences of different length
- Pseudo-random bit sequence
- All zero: 00000...
- All one: 11111...
- Pattern 01: 010101...
- Pattern 10: 101010...

Remote command:

CONFigure:WLAN:SIGN<i>:PER:DPATtern

### Number of Packets

Specifies the number of user data MAC packets to be transmitted to the DUT. The PER view displays a progress bar, indicating this setting and the number of already transmitted packets. The PER measurement is stopped when the specified number of packets has been transmitted.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:PACKets

### Data Interval

Specifies the time interval (in units of 1024 µs) between data packet transmissions for the PER measurement.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:DINTerval

### Payload Size

Defines the payload size (in bytes) of the transmitted MAC data packets.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:PAYLoad:SIZE

### Qos Data TID

Sets the TID value for PER measurements.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:TIDentifier

### Destination Station

Specify the station for which the traffic is measured. This parameter is visible, if [Multi STA](#) is enabled.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:DESTination

### Data Frame Control

**Data Frame Control**

- Link to SIG Frame Settings ☐
- Frame Format: HE\_MU
- Non-HT Frame
- HT Frame
- VHT Frame
- HE\_SU Frame
- HE\_MU Frame
  - Channel Bandwidth: 20MHz
  - Space Time Block Coding ☐
  - Guard Interval: 1.6 µs
  - LTF Type: 2x (6.4 µs)
  - PE Duration: Auto
  - 20MHz Channel 1
    - RU Allocation Subfield: 0 (00000000)
    - RUs State: U1, Off, Off, Off, Off, Off, Off, Off
    - RUs Size: 26, 26, 26, 26, 26, 26, 26, 26
  - Users and Dummies
    - User1 (U1)
      - RU Allocated: 20MHz Channel 1
      - Spatial Streams: 1
      - MCS: MCS Index 9
      - Coding Type: LDPC
    - Dummy1 (D1)
    - Dummy2 (D2)
    - Dummy3 (D3)

### Link to SIG Frame Settings ← Data Frame Control

If enabled, the PER measurements use identical settings as configured in the signaling application.

All settings are also configurable via the main configuration dialog of the signaling application. Refer to the [Chapter 2.3.8.9, "Data frame control settings"](#), on page 71.

Remote command:

CONFigure:WLAN:SIGN<i>:CONNEction:SYNC

**Frame Format ← Data Frame Control**

Configures the downlink data frames. This setting is a critical parameter for reaching the maximum downlink throughput. The supported values depend on the selected standard.

Configure the parameters top-down. The "Modulation Coding Rate" parameter offers only values that are compatible to the other settings.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:FDEF

**Non-HT, HT, VHT, HE Data Frame Control ← Data Frame Control**

Selects the modulation/coding rate of the transmitted data frames. The available values depend on the selected standard. The settings are configurable per frame format.

If you want to set a user-defined value, configure the parameters top-down.

- For "Channel Bandwidth", refer to ["Operating Channel Width"](#) on page 47.
- "Spatial streams":  
The number of spatial streams is only visible for MIMO connections. This parameter is configurable only for HT, VHT and HE\_SU frames:

| Parameter               | Value                    |
|-------------------------|--------------------------|
| Channel Bandwidth       | 20MHz                    |
| Spatial Streams         | 1                        |
| Space Time Block Coding | <input type="checkbox"/> |
| Rate                    | MCS Index 0              |
| Guard Interval          | 1.6 µs                   |
| LTF Type                | 2x (6.4 µs)              |
| PE Duration             | Auto                     |
| Coding Type             | LDPC                     |

- "Space Time Block Coding":  
For data frame [Configuration](#) = "User-defined", you have to enable STBC within the data frame configuration for the particular frame format manually.  
Automatic data frame [Configuration](#) applies the STBC automatically.
- "Rate": The selected rate is only used for packet error rate (PER) tests.
- "Guard Interval":
  - Short or long guard interval for legacy (standards up to 802.11ac)
  - 0.8 µs, 1.6 µs, and 3.2 µs guard interval durations for 802.11ax
- "LTF Type" (HE frames only): 1x HE-LTF, 2x HE-LTF or 4x HE-LTF
- "PE Duration" (HE frames only): value of packet extension (PE) field to provide additional receive processing time at the end of the HE PPDU. Automatic setting is based on the reported DUTs' capabilities.
- "Coding Type" (for VHT and HE\_SU frames only): forward error correction coding low density parity check (LDPC) or binary convolution code (BCC).
- For "20 MHz Channel" configuration, refer to ["20 MHz Channel Settings"](#) on page 108.
- For "Users and Dummies" configuration, refer to ["Users and Dummies"](#) on page 108.

Remote command:

CONFigure:WLAN:SIGN<i>:PER:FDEF

### 20 MHz Channel Settings ← Non-HT, HT, VHT, HE Data Frame Control ← Data Frame Control

Configures arrangement and number of allocations.

- "RU Allocation Subfield": sets 8-bit indices for resource unit (RU) allocation to specify the number of RUs and their size. Refer to IEEE Std 802.11ax-2021, table 27-26, RU allocation subfield.
- "RUs State": disables or enables and selects user data to be transmitted in particular RU
- "RUs Size": specifies the number of RU tones; 26 means 26-tones RU. RUs sizes are selected via "RU Allocation Subfield"

Remote command:

```
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:ALSField
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:BLALlocation?
```

### Users and Dummies ← Non-HT, HT, VHT, HE Data Frame Control ← Data Frame Control

Configures payload. One "User" must be mapped to one RU within a channel. The remaining RUs can be disabled or you can allocate a "Dummy" as user data.

Modulation and coding scheme can be assigned to each user data type ("User" and "Dummy").

Coding type and the number of spatial streams for MIMO can be assigned only to the user data type "User".

Remote command:

```
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:RUAllocation
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:USER<index>:ALLocation
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:USER<index>:MCS
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:DUMMy<index>:MCS
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:USER<index>:CTYPe
CONFfigure:WLAN:SIGN<i>:PER:DFrame:HEMU:USER<index>:STReams
```

## 2.4 Programming

The following sections provide programming examples for the WLAN signaling application.

The examples contain SCPI commands supported by the R&S CMW and the following symbolic scripting commands:

- `// <comment>:`
- `WHILE <query> <> <value>:`  
Waits until the <query> returns a certain <value>, e.g. a specific state is reached.
- `WAITKEY <message>:`  
Displays a dialog box with a <message> and waits until you close the box.

- [General configuration](#)..... 109
- [Initialization](#)..... 109
- [Specifying general settings](#)..... 109



|                                         |     |
|-----------------------------------------|-----|
| • Configuring channel frequencies.....  | 111 |
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### 2.4.1 General configuration

The WLAN signaling application is programmed as follows:

- The application is controlled by SCPI commands with the following syntax: . . . :WLAN:SIGN: . . . .
- After a \*RST, the application is switched off.  
To activate the signal transmission, use `SOURCE:WLAN:SIGN:STATE ON`. \*OPC? ensures that the signal is available at the selected RF output before the next command line is executed.

### 2.4.2 Initialization

```
// *****
// Initial system-reset
// *****
*RST; *OPC?
*CLS; *OPC?
```

### 2.4.3 Specifying general settings

Each scenario is activated via a different `ROUTE` command. Use the "standalone scenario" section or the "MIMO scenario" section.

#### Standalone scenario

```
// *****
// Select the standalone scenario, the signaling unit and the RF path. Specify
// the external attenuation. Set the operation mode and multi-STA support.
// *****
ROUTE:WLAN:SIGN:SCENARIO:SCELL:FLEXible SUA012,RF1C,RX1,RF1C,TX1
CONFIGure:WLAN:SIGN:RFSettings:ANTennal:EATTenuation:OUTPut 2
CONFIGure:WLAN:SIGN:RFSettings:ANTennal:EATTenuation:INPut 2
CONFIGure:WLAN:SIGN:CONNECTION:OMODE AP
```

```
CONFigure:WLAN:SIGN:CONNECTION:MSTation ON
```

```
// *****
// Configure the burst power for the generated signal.
// *****
CONFigure:WLAN:SIGN:RFSettings:BOPower -30
```

### MIMO scenario

```
// *****
// Select the MIMO 2x2 scenario, the signaling unit and the RF paths.
// Specify the external attenuation.
// *****
ROUTE:WLAN:SIGN:SCENario:MIMO:FLEXible SUA012, RF1C, RX1, RF1C, TX1, RF1C, TX3
CONFigure:WLAN:SIGN:RFSettings:ANTenna1:EATTenuation:OUTPut 2
CONFigure:WLAN:SIGN:RFSettings:ANTenna2:EATTenuation:OUTPut 2
CONFigure:WLAN:SIGN:RFSettings:ANTenna1:EATTenuation:INPut 2
CONFigure:WLAN:SIGN:RFSettings:ANTenna2:EATTenuation:INPut 2

// *****
// Select the MIMO mode and configure the cyclic shift diversity.
// *****
CONFigure:WLAN:SIGN:MIMO:TMMode SMULtiplexin
CONFigure:WLAN:SIGN:MIMO:TCSD 0, 0

// *****
// Configure the total burst power of both generated signals and the ratio
// of the two signal powers.
// *****
CONFigure:WLAN:SIGN:RFSettings:BOPower -30
CONFigure:WLAN:SIGN:RFSettings:TSRatio 0
```

### Both scenarios

```
// *****
// Select the standard.
// *****
CONFigure:WLAN:SIGN:CONNECTION:STANDARD GONStd

// *****
// Configure the expected PEP of the measured signal and the mixer level offset.
// *****
CONFigure:WLAN:SIGN:RFSettings:ANTenna1:EPEPower -20
CONFigure:WLAN:SIGN:RFSettings:ANTenna1:MLOffset 1
```

## 2.4.4 Configuring channel frequencies

### 20-MHz channel

```
// *****
// Select the operating band sub 6 GHz.
// *****
CONFIGure:WLAN:SIGN:RFSettings:BAND BS6Ghz

// *****
// Select the channel bandwidth.
// *****
CONFIGure:WLAN:SIGN:RFSettings:OCWidth BW20

// *****
// Select the channel and query the channel center frequency.
// Configure the frequency offset.
// *****
CONFIGure:WLAN:SIGN:RFSettings:CHANnel 10
CONFIGure:WLAN:SIGN:RFSettings:FREQuency?
CONFIGure:WLAN:SIGN:RFSettings:FOFFset 0
```

### 40-MHz channel

```
// *****
// Select the channel bandwidth.
// *****
CONFIGure:WLAN:SIGN:RFSettings:OCWidth BW40

// *****
// Select the 40-MHz channel and query its center frequency.
// Configure the frequency offset.
// *****
CONFIGure:WLAN:SIGN:RFSettings:CHANnel 10
CONFIGure:WLAN:SIGN:RFSettings:FREQuency?
CONFIGure:WLAN:SIGN:RFSettings:FOFFset 0

// *****
// Configure which of the two 20-MHz channels is the primary channel.
// Query the center frequency and channel number of the primary 20-MHz channel.
// *****
CONFIGure:WLAN:SIGN:RFSettings:NPIndex 1
CONFIGure:WLAN:SIGN:RFSettings:NPFREquency?
CONFIGure:WLAN:SIGN:RFSettings:NPCHannel?
```

## 2.4.5 Configuring internal fading

```
// *****
// Select the standard and operation mode.
```

```
// Select the fading scenario, the signaling unit and the RF path.
// Specify the external attenuation.
// *****
Configure:WLAN:SIGN:CONNECTION:STANDARD ACSTd
Configure:WLAN:SIGN:CONNECTION:OMODE AP
ROUTE:WLAN:SIGN:SCENARIO:SCFading:FLEXible SUA012,RF1C,RX1,RF1C,TX1, FAD012
Configure:WLAN:SIGN:RFSettings:ANTENNA1:EATTenuation:OUTPUT 2
Configure:WLAN:SIGN:RFSettings:ANTENNA1:EATTenuation:INPUT 2

// *****
// Configure fading: enable fading simulator, select a fading profile.
// Query the insertion loss.
// *****
Configure:WLAN:SIGN:FADING:FSIMulator:ENABLE ON
Configure:WLAN:SIGN:FADING:FSIMulator:STANDARD MODA
Configure:WLAN:SIGN:FADING:FSIMulator:ILOSs:LOSS?

// *****
// Configure AWGN insertion: Enable AWGN, set min noise/system BW ratio,
// signal to noise ratio.
// *****
Configure:WLAN:SIGN:FADING:AWGN:ENABLE ON
Configure:WLAN:SIGN:FADING:AWGN:BWIDth:RATIo 1
Configure:WLAN:SIGN:FADING:AWGN:SNRatio 1
```

## 2.4.6 Configuring connection settings

```
// *****
// Configure associations: MAC address of STA1, enable automatic disassociation,
// allow preemption and set priority mechanism.
// *****
Configure:WLAN:SIGN:CONNECTION:ASSOCIATION:STA1:MACReserve SET 4A:55:FF:01:54:22
Configure:WLAN:SIGN:CONNECTION:ASSOCIATION:DISASS ON
Configure:WLAN:SIGN:CONNECTION:ASSOCIATION:PREEMPTION ON
Configure:WLAN:SIGN:CONNECTION:ASSOCIATION:STAPriority ROUR

// *****
// Enable associations of 802.11b device within 802.11ac or ax standard.
// Configure beacon interval, DTIM period, MAC address, SSID, country code,
// association ID range. Set HT smoothing bit.
// *****
Configure:WLAN:SIGN:CONNECTION:DSSS ON
Configure:WLAN:SIGN:CONNECTION:BEACON 110
Configure:WLAN:SIGN:CONNECTION:DPERIOD 2
Configure:WLAN:SIGN:CONNECTION:BSSID #H1230123
Configure:WLAN:SIGN:CONNECTION:SSID "myCMW"
Configure:WLAN:SIGN:CONNECTION:CCODE:CCState ENABLE
Configure:WLAN:SIGN:CONNECTION:CCODE:CCConf "US", 0, 50, 3
Configure:WLAN:SIGN:CONNECTION:AID 1,15
```

```

CONFigure:WLAN:SIGN:CONNection:SMOothing NREC

// *****
// Configure supported rates.
// *****
CONFigure:WLAN:SIGN:CONNection:SRATes ENABLE
CONF:WLAN:SIGN:CONN:SRATes:OFDMconf MAND,MAND,MAND,MAND,OPT,OPT,OPT,OPT
CONF:WLAN:SIGN:CONN:SRATes:OMCSconf SUPP,NOTS,SUPP,NOTS,SUPP,NOTS,SUPP,NOTS
CONF:WLAN:SIGN:CONN:SRATes:VHTConf MC08
// *****
// Configure frame rate control for management frames and data frames.
// *****
CONFigure:WLAN:SIGN:CONNection:MFDef ENABLE, HTM, MCS1
CONFigure:WLAN:SIGN:CONNection:DFDef ENABLE,HES,BW20,MCS10,GI08,X1,PE4,BCC,STR1

// *****
// Configure security settings for WPA personal:
// mechanism, encryption type
// *****
CONFigure:WLAN:SIGN:CONNection:SECurity:PASSphrase WPERsonal, "12345678"
CONFigure:WLAN:SIGN:CONNection:SECurity:ENCryption TKIP

// *****
// Configure EDCA parameter set for best effort, background, video and voice
// connection class, configure also parameter set for multi-user connections.
// *****
CONFigure:WLAN:SIGN:CONNection:EDCA:ACBE
CONFigure:WLAN:SIGN:CONNection:EDCA:ACBK
CONFigure:WLAN:SIGN:CONNection:EDCA:ACVI
CONFigure:WLAN:SIGN:CONNection:EDCA:ACVO
CONFigure:WLAN:SIGN:CONNection:MUEDca:ACBE
CONFigure:WLAN:SIGN:CONNection:MUEDca:ACBK
CONFigure:WLAN:SIGN:CONNection:MUEDca:ACVI
CONFigure:WLAN:SIGN:CONNection:MUEDca:ACVO

// *****
// Enable BSR support.
// *****
CONFigure:WLAN:SIGN:CONNection:HEMac:BSRSupport ON

```

### Other connection settings

The following connection settings are a preparation for other configurations. They are not relevant for the selected configuration (operation mode AP, standard 802.11n, WPA personal).

```

// *****
// DSSS: supported rates
// *****
CONFigure:WLAN:SIGN:CONNection:SRATes:DSSSconf DIS, DIS, DIS, DIS

```

```
// *****
// Hotspot 2.0: basic and supplementary parameters, BSS load element
// *****
Configure:WLAN:SIGN:CONNECTION:HOTSpot:MNDigits TWDigits
CONF:WLAN:SIGN:CONN:HOTS:HSPar PNET,UNSP,0,#H1234,"R&S",262,30,"op1","operator"
Configure:WLAN:SIGN:CONNECTION:HOTSpot:PLMN2 ON, 262, 31, TWDigits
Configure:WLAN:SIGN:CONNECTION:HOTSpot:DNAMe2 ON, "op2"
CONF:WLAN:SIGN:CONN:HOTSpot:HSSPar 5000, 5000, ATAAvailable, PIAavailable,
    "exmp.org", APRime, ON, ATConditions
Configure:WLAN:SIGN:CONNECTION:HOTSpot:REALm2 ON, "ex1.org", ON, ON, OFF, ON, ON
Configure:WLAN:SIGN:CONNECTION:HOTSpot:CUTil 10, 20, 30000

// *****
// Station: allowed AP and connection mode.
//
// For established connection, query SSID of the connected AP and trigger
// frames recieved.
// *****
Configure:WLAN:SIGN:CONNECTION:STATION:SCONNECTION ANY
Configure:WLAN:SIGN:CONNECTION:STATION:CMODE MANual

SENSe:WLAN:SIGN:UESinfo:APSSid?
SENSe:WLAN:SIGN:UESinfo:RXTRigframe?

// *****
// Security settings for WPA enterprise: radius server mode, internal radius
// server configuration, external radius server settings
// *****
Configure:WLAN:SIGN:CONNECTION:SECurity:RSErver:MODE Internal
Configure:WLAN:SIGN:CONNECTION:SECurity:ESIM:KTONE
    "15555555555555555555555555555551", "15545751", "1011111111111101"
Configure:WLAN:SIGN:CONNECTION:SECurity:ESIM:KTTWO
    "25555555555555555555555555555552", "25545752", "2011111111111102"
Configure:WLAN:SIGN:CONNECTION:SECurity:ESIM:KTTHree
    "35555555555555555555555555555553", "35545753", "3011111111111103"
Configure:WLAN:SIGN:CONNECTION:SECurity:EAKA:KALGo
    "000102030405060708090a0b0c0d0e0f", "12555555555555555555555555555551",
    "13555555555555555555555555555551", MILEnAge
Configure:WLAN:SIGN:CONNECTION:SECurity:RSErver:ICONf 172, 22, 1, 111
Configure:WLAN:SIGN:CONNECTION:SECurity:RSErver:PNUMber 1812
Configure:WLAN:SIGN:CONNECTION:SECurity:RSErver:SKEY "mykey"
```

## 802.11ax connection settings

The following connection settings are specific for high efficiency (HE) connections.

```
// *****
// Set BSS color, enable A-MPDU, multiple TIDs and set the length of A-MPDU.
// *****
Configure:WLAN:SIGN:CONNECTION:BSSColor 2
```

```

Configure:WLAN:SIGN:CONNECTION:AMPDu ENAB, ENAB, 150

// *****
// Configure broadcast TWT operation: enable it, enable schedule
// period, set flow type, minimal wake duration, start time, and enable
// triggering.
// *****
Configure:WLAN:SIGN:CONNECTION:BTWT:ENABLE ON
Configure:WLAN:SIGN:CONNECTION:BTWT:SCHEDULE:ENABLE ON
Configure:WLAN:SIGN:CONNECTION:BTWT:SCHEDULE:FTYPE ANN
Configure:WLAN:SIGN:CONNECTION:BTWT:SCHEDULE:STIME 10
Configure:WLAN:SIGN:CONNECTION:BTWT:SCHEDULE:MWDURATION 2
Configure:WLAN:SIGN:CONNECTION:BTWT:SCHEDULE:TENABLE ON

/// *****
// Configure frame rate control for data frames: enable data frames, set
// frame format, channel bandwidth, MCS, guard interval, LTF length, PE duration,
// coding, number of streams. Set RU allocation subfield, query RU allocations.
// Configure payload: map user 1 and dummy users to RU. Set MCS and coding
// type for corresponding users.
//
// For MIMO set the number of streams.
// *****
Configure:WLAN:SIGN:CONNECTION:DFDEF ENABLE,HES,BW20,MCS10,GI08,X1,PE4,BCC,STR1
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:ALSFIELD A200 CHA1
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:BLALLOCATION?
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:RUALLOCATION USR1,CHA1,RU1
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:RUALLOCATION DMY1,CHA1,RU2
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:RUALLOCATION DMY2,CHA1,RU3
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:USER1:ALLOCATION CHA1,RU1
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:USER1:MCS MCS5
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:USER1:CTYPE BCC
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:DUMMY1:MCS MCS10
Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:DUMMY2:MCS MCS10

Configure:WLAN:SIGN:CONNECTION:DFRAME:HEMU:USER1:STREAMS 2

/// *****
// Configure trigger frame - common info: select the trigger type, set length,
// require CS check. Set bandwidth, LTF type, MU-MIMO LTF, LDPC extra symbol.
// Query power level for trigger frames.
// *****
Configure:WLAN:SIGN:CONNECTION:HETF:TTYP BSRP
Configure:WLAN:SIGN:CONNECTION:HETF:NOFSYMBOLS 330
Configure:WLAN:SIGN:CONNECTION:HETF:CSR ON
Configure:WLAN:SIGN:CONNECTION:HETF:CHBW BW80
Configure:WLAN:SIGN:CONNECTION:HETF:GILT L432
Configure:WLAN:SIGN:CONNECTION:HETF:MLTF SING
Configure:WLAN:SIGN:CONNECTION:HETF:LDPC OFF

```

```

Configure:WLAN:SIGN:CONNECTION:HETF:APTXpower?

/// *****
// Configure trigger frame - user info: select RU size, coding type, MCS,
// DCM, query starting spatial stream and number of spatial streams.
// *****
Configure:WLAN:SIGN:CONNECTION:HETF:RUAL RU52
Configure:WLAN:SIGN:CONNECTION:HETF:CTYP LDPC
Configure:WLAN:SIGN:CONNECTION:HETF:MCS MCS11
Configure:WLAN:SIGN:CONNECTION:HETF:DCM ON
Configure:WLAN:SIGN:CONNECTION:HETF:SSS?
Configure:WLAN:SIGN:CONNECTION:HETF:NSS?

/// *****
// Configure trigger frame transmission: send the trigger once.
//
// Alternatively send the trigger periodically in specified intervals.
// *****
Configure:WLAN:SIGN:CONNECTION:HETF:SSTX

Configure:WLAN:SIGN:CONNECTION:HETF:TXP 100
Configure:WLAN:SIGN:CONNECTION:HETF:TXEN ON

/// *****
// Configure trigger frame power control: in manual mode set the target RSSI.
// Query expected Rx power of HE TB PPDU transmission.
// *****
Configure:WLAN:SIGN:CONNECTION:HETF:TRSMODE MAN
Configure:WLAN:SIGN:CONNECTION:HETF:TSRControl -20
Configure:WLAN:SIGN:CONNECTION:HETF:TRSSi?

```

## 2.4.7 Configuring trigger signals

```

// *****
// Configure the generated trigger signal (TX): slope, pulse length
// *****
TRIGGER:WLAN:SIGN:TX:MACFrame:SLOPE REDGE
TRIGGER:WLAN:SIGN:TX:MACFrame:PLENGTH:MODE UDEFIned
TRIGGER:WLAN:SIGN:TX:MACFrame:PLENGTH:VALue 0.001

// *****
// Configure the generated trigger signal (RX): burst types, bandwidth,
// rate. Set minimum length, slope, pulse length,
// spatial streams, delay type.
// *****
TRIGGER:WLAN:SIGN:RX:MACFrame:BTYPe HESBursts
TRIGGER:WLAN:SIGN:RX:MACFrame:BW BW80
TRIGGER:WLAN:SIGN:RX:MACFrame:RATE MCS10
TRIGGER:WLAN:SIGN:RX:MACFrame:OFMLength UDEFIned,1

```



```

TRIGger:WLAN:SIGN:RX:MACFrame:SLOPe REDGe
TRIGger:WLAN:SIGN:RX:MACFrame:PLENgtH:MODE UDEfined
TRIGger:WLAN:SIGN:RX:MACFrame:PLENgtH:VALue 0.001
TRIGger:WLAN:SIGN:RX:MACFrame:STReams ALL
TRIGger:WLAN:SIGN:RX:MACFrame:CTDelay CONSTant

// *****
// Alternatively select DSSS burst trigger and the minimal payload length.
// *****
TRIGger:WLAN:SIGN:RX:MACFrame:BTYPe DCBursts
TRIGger:WLAN:SIGN:RX:MACFrame:DSMLength UDEfined,16

```

### Configuring rate restriction

```

// *****
// Configure scenario, standard and OCW. Most DUTs attempt to use the most
// advanced format, highest bandwidth and most spatial streams available.
// *****
ROUTE:WLAN:SIGN:SCENario:SCELL:FLEXible SUA012,RF1C,RX1,RF1C,TX1
CONFIgure:WLAN:SIGN:CONNection:STANdard ACSTd
CONFIgure:WLAN:SIGN:RFSettings:OCWidth BW40

```

Turn on signal and let the DUT associate, see [Chapter 2.4.12, "Associating the DUT"](#), on page 119.

```

// *****
// Enable ICMP packet generator.
// *****
CONFIgure:WLAN:SIGN:PGEN1:PROTOcol ICMP
CONFIgure:WLAN:SIGN:PGEN1:CONFIg ON, 110, 500, BP01, TID0

// *****
// With rate restriction disabled, enable user-defined supported rates,
// set matching format, BW, Nss...
// *****
TRIGger:WLAN:SIGN:RX:MACFrame:RREStriction OFF
CONFIgure:WLAN:SIGN:CONNection:DFDef ENAB,VHT,BW40,MCS9,SHOR,KEEP,AUTO,
    BCC,STR1,OFF

// *****
// Enable rate restriction.
// *****
TRIGger:WLAN:SIGN:RX:MACFrame:RREStriction ON

```

Perform corresponding TX measurement.

Decrease MCS.

Reconfigure rate and repeat TX measurements with next lower MCS.

## 2.4.8 Configuring IP settings

```
// *****
// Configure the IP version support.
// *****
Configure:WLAN:SIGN:CONNECTION:IVSupport IV4V6

// *****
// Configure IP settings relevant if you have no DAU:
// subnet mask, IP address of instrument, DUT, gateway and DNS,
// DHCP, IPv6 prefix of stack
// *****
Configure:WLAN:SIGN:IPVFour:STATIC:SMASK 255, 255, 255, 0
Configure:WLAN:SIGN:IPVFour:STATIC:IPAddress:CMW 100, 100, 100, 162
Configure:WLAN:SIGN:IPVFour:STATIC:IPAddress:UE 100, 100, 100, 11
Configure:WLAN:SIGN:IPVFour:STATIC:IPAddress:GATeway 100, 100, 100, 1
Configure:WLAN:SIGN:IPVFour:STATIC:IPAddress:DNS 10, 0, 2, 166
Configure:WLAN:SIGN:IPVFour:DHCP OFF
Configure:WLAN:SIGN:IPVSix:PREFIX "fc01:abab:cafe:efe0::"
```

## 2.4.9 Configuring the packet generator

```
// *****
// Enable the generator one and configure it. Set interval, packet size,
// bit pattern, TID, protocol, ports, IP version, station and scheduling. Select
// BAR method and enable block acknowledgments for all TIDs.
// *****
Configure:WLAN:SIGN:PGEN1:CONFig ON, 110, 500, BP01, TID0
Configure:WLAN:SIGN:PGEN1:PROToCol UDP
Configure:WLAN:SIGN:PGEN1:UPOrts 3001, 3005
Configure:WLAN:SIGN:PGEN1:IPVersion IV4
Configure:WLAN:SIGN:PGEN1:DESTination STA1
Configure:WLAN:SIGN:CONNECTION:QOS:PRIoritiz ROUR
Configure:WLAN:SIGN:CONNECTION:QOS:BARMethod MUB
Configure:WLAN:SIGN:CONNECTION:QOS:BLACK ON, ON, ON, ON, ON, ON, ON, ON
```

## 2.4.10 Configuring DAU parameters

```
// *****
// Configure the IP address to be assigned to the DUT. Use an internal DAU.
// Set TID for end-to-end connections.
// *****
Configure:WLAN:SIGN:ETOE:DUIP ON, 100, 100, 100, 10
Configure:WLAN:SIGN:EDAU:ENABLE OFF
Configure:WLAN:SIGN:CONNECTION:QOS:ETOE TID1

// *****
// To use an external DAU: enable external DAU usage, define network segment
```

```
// and subnet node ID of the instrument with DAU.
// *****
CONFIGure:WLAN:SIGN:EDAU:ENABle ON
CONFIGure:WLAN:SIGN:EDAU:NSEgment A
CONFIGure:WLAN:SIGN:EDAU:NID 5

// *****
// Configure routes for the station mode.
// *****
CONF:WLAN:SIGN:ETOE:IRList:IPR1 ON,192,168,20,0,24,"fc01:abab:cafe:efe0::",64
CONF:WLAN:SIGN:ETOE:IRList:IPR1 ON,192,168,20,0,28,"fc01:abab:cafe:efe5::",64
CONF:WLAN:SIGN:ETOE:IRList:IPR1 ON,192,168,20,0,32,"fc01:abab:cafe:efe8::",64
```

### 2.4.11 Configuring message monitoring

```
// *****
// Enable message monitoring for WLAN, select address number 2 from the global
// logging PC address pool and query the corresponding IP address string.
// *****
CONFIGure:WLAN:SIGN:MMONitor:ENABle ON
CONFIGure:WLAN:SIGN:MMONitor:IPAddress IP2
CONFIGure:WLAN:SIGN:MMONitor:IPAddress?
```

### 2.4.12 Associating the DUT

```
// *****
// Clear old entries in event log.
// *****
CLEan:WLAN:SIGN:ELOGging

// *****
// Connect the DUT (switched off). Switch on the signaling application. Query the
// state until it equals ON,ADJ (signal available at RF output connector).
// *****
WAITKEY >Ensure that the DUT is connected to the instrument and switched off<
SOURce:WLAN:SIGN:STATe ON
WHILE SOURce:WLAN:SIGN:STATe:ALL? <> "ON,ADJ"

// *****
// Switch on the DUT. Wait until the DUT is associated.
// *****
WAITKEY >Switch on the DUT<
WHILE FETCh:WLAN:SIGN:PSWitched:STATe? <> "ASS"

// *****
// Query event log entries.
```

```
// *****
SENSe:WLAN:SIGN:ELOGging:ALL?
```

### 2.4.13 Querying UE information

```
// *****
// Query DUT information/capabilities.
// *****
SENSe:WLAN:SIGN:UECapability:MAC:ADdResS?
SENSe:WLAN:SIGN:UESinfo:RXBPower?
SENSe:WLAN:SIGN:UESinfo:ANTennal:RXPindicator?
SENSe:WLAN:SIGN:UESinfo:ANTennal:ARXBpower?
SENSe:WLAN:SIGN:UESinfo:UEADdress:IPV4?
SENSe:WLAN:SIGN:UESinfo:UEADdress:IPV6?
SENSe:WLAN:SIGN:UESinfo:DRATe?
SENSe:WLAN:SIGN:UECapability:HE?
SENSe:WLAN:SIGN:UESinfo:ABSReport?
SENSe:WLAN:SIGN:UESinfo:CMWaddress:IPV4?
SENSe:WLAN:SIGN:UESinfo:CMWaddress:IPV6?
```

### 2.4.14 HE TB list mode measurements

```
// *****
// Associate 802.11ax DUT.
// Enable TF transmission with target RSSI, query HE TB list mode state (off).
// *****
CONFIgure:WLAN:SIGN:CONNection:HETF:TRSMoDe MAN
CONFIgure:WLAN:SIGN:CONNection:HETF:TSRControl -10
FETCh:WLAN:SIGN:HETBased:STATe?

// *****
// Set HE TB measurement control parameters: number of frames.
// *****
CONFIgure:WLAN:SIGN:HETBased:FRAMes 2000

// *****
// Start HE TB list mode measurements. Query measurement state (run).
// Query UPH information and results (2000 records returned).
// *****
INITiate:WLAN:SIGN:HETBased
FETCh:WLAN:SIGN:HETBased:STATe?
SENSe:WLAN:SIGN:HETBinfo:UPHinfo?
FETCh:WLAN:SIGN:HETBased:UPHinfo?
```

### 2.4.15 HE NDP sounding procedure

Associate 802.11ax DUT, see ["802.11ax connection settings"](#) on page 114.

```

// *****
// For HE NDP sounding set: method, station feedback type, channel bandwidth, GI/LTF
// combination, RU index for start and end, the number of spatial streams
// Enable TF transmission with target RSSI, query HE TB list mode state (off).
// For HE TB sounding, set codebook, No. of columns and subcarrier grouping.
// *****

CONFIGure:WLAN:SIGN:CONNection:NDPSounding:METHod TBASed
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:TARGet STA1
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:TYPE SU
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:BW BW20
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:LTFGi L208
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:RUStart 0
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:RUEnd 8
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:SPSTreams STR2
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:CBOOk SIZE1
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:NUMColumns COL1
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:SUBGrouping GRP4

/// *****
// Configure the NDP sounding frame transmission: send the signal once.
//
// Alternatively send the signal periodically in specified intervals.
// *****
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:SSTX

CONFIGure:WLAN:SIGN:CONNection:NDPSounding:TXP 100
CONFIGure:WLAN:SIGN:CONNection:NDPSounding:TXEN ON

```

## 2.4.16 PER measurements

The PER measurement provided by the WLAN signaling application is programmed as follows:

- The application is controlled by SCPI commands with the following syntax: `...:WLAN:SIGN:PER:...`
- After a `*RST`, the measurement is switched off.  
You can start the measurement using `INIT:WLAN:SIGN:PER` and retrieve the results using `FETCh` commands.

The examples in this section focus on commands directly related to the PER measurement. For general configuration of the signaling application and setting up the connection, refer to [Chapter 2.4.1, "General configuration"](#), on page 109.

### 2.4.16.1 Configuring a PER measurement

```

// *****
// Configure the generated signal.
// *****

```

```

CONFigure:WLAN:SIGN:PER:FDEF HTM, BW20, MCS6
CONFigure:WLAN:SIGN:PER:DPATtern PN5
CONFigure:WLAN:SIGN:PER:PACKets 200
CONFigure:WLAN:SIGN:PER:DINterval 25
CONFigure:WLAN:SIGN:PER:PAYLoad:SIZE 90
CONFigure:WLAN:SIGN:PER:DESTination STA1

/// *****
// Use different configuration for PER Measurement and WLAN signaling.
// Configure data frame rate control for PER: set RU allocation subfield,
// query RU allocations.
// Configure payload: map user 1 and dummy users to RU. Set MCS and coding
// type and spatial streams for corresponding users. Set TID for PER.
// *****
CONFigure:WLAN:SIGN:CONNECTION:SYNC OFF
CONFigure:WLAN:SIGN:PER:FDEF HES, BW20, MCS10, GI08, X1, PE4, BCC
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:ALSField A200 CHA1
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:BLAllocation?
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:RUALlocation USR1,CHA1,RU1
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:RUALlocation DMY1,CHA1,RU2
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:RUALlocation DMY2,CHA1,RU3
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:USER1:ALlocation CHA1,RU1
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:USER1:MCS MCS5
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:USER1:CTYPe BCC
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:USER1:STReams STR1
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:DUMMy1:MCS MCS10
CONFigure:WLAN:SIGN:PER:DFrame:HEMU:DUMMy2:MCS MCS10
CONFigure:WLAN:SIGN:PER:TIDentifier TID6

```

#### 2.4.16.2 Running a PER measurement

```

// *****
// Enhance user data statistics: Clear statistic entries and set reporting
// interval and time span to collect statistics.
// *****
CONFigure:WLAN:SIGN:UESinfo:RESet
CONFigure:WLAN:SIGN:UESinfo:SETTings 2,1000

// *****
// Start a PER measurement.
// Wait until the measurement is finished.
// *****
INIT:WLAN:SIGN:PER
WHILE FETCh:WLAN:SIGN:PER:STATE? <> "RDY"

// *****
// Query the measurement results.
// *****
FETCh:WLAN:SIGN:PER?

```

```
// *****
// Query information about last ACK frame.
// *****
FETCh:WLAN:SIGN:PACKrate?

// *****
// Query user data statistics for all frame formats.
// *****
SENSe:WLAN:SIGN:UESinfo:RXPSdu:HEMU?
SENSe:WLAN:SIGN:UESinfo:RXPSdu:HESU?
SENSe:WLAN:SIGN:UESinfo:RXPSdu:HETB?
SENSe:WLAN:SIGN:UESinfo:RXPSdu:HT?
SENSe:WLAN:SIGN:UESinfo:RXPSdu:NONHt?
SENSe:WLAN:SIGN:UESinfo:RXPSdu:VHT?
```

## 2.5 Command reference

The following sections provide detailed reference information on the remote control commands of the WLAN signaling application.

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### 2.5.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

### 2.5.1.1 SIGN<i>

SIGN<i> is used as abbreviation of SIGNaling<instance>. For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same firmware application. It can be omitted if the instrument supports only one instance, or to address the first instance.

### 2.5.1.2 Values for signal path selection

Which connectors and modules can be specified in a command, depends on the installed hardware and the active subinstrument or instance <i>.

This chapter lists all values available for path selection. Depending on your configuration, only a subset is relevant for you. Virtual connector names are only relevant for setting commands. Queries return the physical connector names.

Additional information is available in the base software documentation. It describes typical instrument configurations with the allowed RF connector - TX/RX module combinations and the mapping of virtual connector names to physical connectors.

#### Single-CMW setup

Selection values:

- SUA as base band unit:  
SUA1 | SUA2 | SUA3 | SUA4 | SUA5 | SUA6  
SUA012 | SUA034 | SUA056  
SUA partition 1, 2, 3, 4, 5, 6, 1&2, 3&4, 5&6
- RX module:  
RX1 | RX2 | RX3 | RX4
- TX module:  
TX1 | TX2 | TX3 | TX4
- RX connector:  
RF1C | RF2C | RF3C | RF4C | RFAC | RFBC  
RF 1 COM to RF 4 COM plus virtual connector names
- TX connector:  
RF1C | RF2C | RF3C | RF4C | RFAC | RFBC  
RF1O | RF3O | RFAO  
RF 1 COM to RF 4 COM plus virtual connector names  
RF 1 OUT and RF 3 OUT plus virtual connector name
- SUA as fader:  
FAD012 | FAD034 | FAD056  
SUA fader 1&2, 3&4, 5&6  
The SUA fader numbering is reversed. The highest SUA fader uses the lowest SUA partition.



**CMWflexx setup with R&S CMWC and several R&S CMW500**

Selection values:

- **SUA as base band unit:**  
 SUA11 | SUA12 | SUA13 | SUA14 | SUA15 | SUA16  
 SUA21 | SUA22 | SUA23 | SUA24 | SUA25 | SUA26  
 SUA31 | SUA32 | SUA33 | SUA34 | SUA35 | SUA36  
 SUA41 | SUA42 | SUA43 | SUA44 | SUA45 | SUA46  
 SUA112 | SUA134 | SUA156 | SUA212 | SUA234 | SUA256  
 SUA312 | SUA334 | SUA356 | SUA412 | SUA434 | SUA456  
**SUA<a><b>: CMW <a>, SUA partition <b>**  
 Example SUA21: CMW 2, SUA partition 1  
**SUA<a><b><c>: CMW <a>, SUA partitions <b>&<c>**  
 Example SUA312: CMW 3, SUA partitions 1&2
- **RX module:**  
 RX11 | RX12 | RX13 | RX14 | RX21 | RX22 | RX23 | RX24  
 RX31 | RX32 | RX33 | RX34 | RX41 | RX42 | RX43 | RX44  
**RX<a><b>: CMW <a>, RX <b>**  
 Example RX24: CMW 2, RX module 4
- **TX module:**  
 TX11 | TX12 | TX13 | TX14 | TX21 | TX22 | TX23 | TX24  
 TX31 | TX32 | TX33 | TX34 | TX41 | TX42 | TX43 | TX44  
**TX<a><b>: CMW <a>, TX <b>**  
 Example TX13: CMW 1, TX module 3
- **RX connector:**  
 R11C | R12C | R13C | R14C | R21C | R22C | R23C | R24C  
 R31C | R32C | R33C | R34C | R41C | R42C | R43C | R44C  
**R<a><b>C: CMW <a>, connector RF <b> COM**  
 Example R12C: CMW 1, RF 2 COM
- **TX connector:**  
 R11C | R12C | R13C | R14C | R21C | R22C | R23C | R24C  
 R31C | R32C | R33C | R34C | R41C | R42C | R43C | R44C  
 R11O | R13O | R21O | R23O | R31O | R33O | R41O | R43O  
**R<a><b>C: CMW <a>, connector RF <b> COM**  
**R<a><b>O: CMW <a>, connector RF <b> OUT**  
 Example R21O: CMW 2, RF 1 OUT
- **SUA as fader:**  
 FAD112 | FAD134 | FAD156 | FAD212 | FAD234 | FAD256  
 FAD312 | FAD334 | FAD356 | FAD412 | FAD434 | FAD456  
**FAD<a><b><c>: CMW <a>, SUA fader <b>&<c>**  
 Example FAD312: CMW 3, SUA fader 1&2  
 The SUA fader numbering is reversed. The highest SUA fader uses the lowest SUA partition.

For CMW 1, you can alternatively use the single-CMW selection values.

### 2.5.1.3 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**

If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:

- "Setting only": The command can only be used to set parameters.
- "Query only": The command can only be used to query parameters.
- "Event": The command initiates an event.

- **Parameter usage**

The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query.
- "Setting parameters" are sent with a setting command.
- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.

- **Parameter type**

The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:

- "string": characters enclosed in quotation marks (" or ')
- "numeric", "decimal": decimal number
- "integer": integer number
- "float": floating point number
- "hex": hexadecimal number starting with #H
- "binary": binary number starting with #B
- "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).

- **Firmware/Software:**

Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

### 2.5.1.4 Reliability indicator

The reliability indicator has one of the following values:

- **0 ("No Error"):**

Measurement values available, no error detected.

- **1 ("Measurement Timeout"):**

The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INValid` results or has not completed the full statistic count.

- **2 ("Capture Buffer Overflow"):**

The measurement configuration results in a capture length that exceeds the available memory.

- **3 ("Input Overdriven") / 4 ("Input Underdriven"):**  
The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 ("Trigger Timeout"):**  
The measurement could not be started or continued because no trigger event was detected.
- **7 ("Acquisition Error"):**  
The R&S CMW could not properly decode the RF input signal.
- **8 ("Sync Error"):**  
The R&S CMW could not synchronize to the RF input signal.
- **9 ("Uncal"):**  
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the specified data sheet limits.
- **15 ("Reference Frequency Error"):**  
The instrument has been configured to use an external reference signal. But the reference oscillator could not be phase-locked to the external signal (for example signal level too low, frequency out of range or reference signal not available at all).
- **16 ("RF Not Available"):**  
The measurement could not be started because the configured RF input path was not active.  
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 ("RF Level Not Settled") / 18 ("RF Frequency Not Settled"):**  
The measurement could not be started because the R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 ("Call Not Established"):**  
For measurements: The measurement could not be started because no signaling connection to the DUT was established.  
For DAU IMS service: Establishing a voice over IMS call failed.
- **20 ("Call Type Not Usable"):**  
For measurements: The measurement could not be started because the established signaling connection had wrong properties.  
For DAU IMS service: The voice over IMS settings could not be applied.
- **21 ("Call Lost"):**  
For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.  
For DAU IMS service: The voice over IMS call was lost.
- **23 ("Missing Option"):**  
An action cannot be executed due to a missing option.  
For GPRF generator: The ARB file cannot be played due to a missing option.
- **24 ("Invalid RF Setting"):**  
The desired RF TX level or RF RX reference level could not be applied.
- **25 ("Level Overrange"):**  
The RF TX level is in overrange. The signal quality can be degraded.
- **26 ("Resource Conflict"):**

The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application. Stop the application that has allocated the conflicting resources and try again.

- **28 ("Unexpected Parameter Change"):**  
One or more measurement configuration parameters were changed while the measurement completed. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.
- **29 ("Invalid RF Frequency Setting"):**  
The desired RF TX frequency or RF RX frequency could not be applied.
- **30 ("File Not Found"):**  
The specified file could not be found.
- **40 ("ARB File CRC Error"):**  
The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.
- **42 ("ARB Header Tag Invalid"):**  
The ARB file selected in the GPRF generator contains an invalid header tag.
- **43 ("ARB Segment Overflow"):**  
The number of segments in the multi-segment ARB file is higher than the allowed maximum.
- **44 ("ARB File Not Found"):**  
The selected ARB file could not be found.
- **45 ("ARB Memory Overflow"):**  
The ARB file length is greater than the available memory.
- **46 ("ARB Sample Rate Out of Range"):**  
The clock rate of the ARB file is either too high or too low.
- **47 ("ARB Cycles Out of Range"):**  
The repetition mode equals "Single Shot" and the playback length is too long. Reduce the playback length or set the repetition mode to "Continuous".  
$$\text{<Length>} = (\text{<Cycles>} * \text{<Samples>} + \text{<Additional Samples>}) / \text{<Clock Rate>}$$
- **50 ("Startup Error"):**  
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 ("No Reply"):**  
The DAU has received no response, for example for a ping request.
- **52 ("Connection Error")**  
A connection setup failed or a connection was lost.  
For DAU: The DAU could not establish a connection to internal components. Restart the instrument.
- **53 ("Configuration Error"):**  
The current DAU configuration is incomplete or wrong and could not be applied. Check especially the IP address configuration.
- **54 ("Filesystem Error"):**  
The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **70 ("Wrong Standard")**  
The standard of the measured signal does not match the configured standard.

- **71 ("Wrong Bandwidth")**  
The bandwidth of the measured signal does not match the configured bandwidth.
- **72 ("Wrong Burst Type")**  
The burst type of the measured signal does not match the configured burst type.
- **73 ("MIMO Signal Detected")**  
The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.
- **74 ("More Streams than Antennas")**  
The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 ("Matrix Inversion Failed")**  
The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.
- **76 ("SIG CRC Failed")**  
The cyclic redundancy check of a SIGNAL field failed.
- **77 ("Parity Check Failed")**  
The parity check of a SIGNAL field failed.
- **78 ("Bursts Not Identical")**  
In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 ("Wrong Modulation")**  
The modulation type of the measured signal does not match the configured modulation type.
- **80 ("SISO Signal Detected")**  
Received signal contains only one stream.
- **81 ("Less Streams than Antennas")**  
Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.
- **82 ("Signal Fields of Different Antennas Are Not Identical")**  
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 ("Bit Errors in Some of the Received Symbols")**  
Adjust input signal power and repeat training.
- **84 ("First and Last Symbols Need to Be Skipped")**  
Due to a sequence counter in the MAC header the first and the last symbol (due to the FCS) change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 ("Received and Saved Bursts Are Not Identical")**  
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 ("No Symbols for Measurement")**  
Too many head and tail symbols have been skipped.

- **93 ("OCXO Oven Temperature Too Low"):**  
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **101 ("Firmware Error"):**  
Indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.  
If the error occurs again, consider the following hints:
  - Firmware errors can often be repaired by restoring the factory default settings.
  - If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog, section "SW/HW-Equipment > Installed Software".
  - Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.
- **102 ("Unidentified Error"):**  
Indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 (firmware error)".
- **103 ("Parameter Error"):**  
Indicates that the measurement could not be performed due to internal conflicting parameter settings.  
A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.  
If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <http://www.service.rohde-schwarz.com>).
- **104 ("Not Functional"):**  
The application could not be started with the configured parameter set.

## 2.5.2 Signaling control and states

The following commands control the connection to the DUT and retrieve connection-related information.

|                                                                              |     |
|------------------------------------------------------------------------------|-----|
| <a href="#">SOURce:WLAN:SIGN&lt;i&gt;:STATe</a> .....                        | 131 |
| <a href="#">SOURce:WLAN:SIGN&lt;i&gt;:STATe:ALL?</a> .....                   | 131 |
| <a href="#">CALL:WLAN:SIGN&lt;i&gt;:ACTioN:STATioN:CONNeCT</a> .....         | 131 |
| <a href="#">CALL:WLAN:SIGN&lt;i&gt;:STA&lt;s&gt;:ACTioN:DISConnect</a> ..... | 132 |
| <a href="#">CALL:WLAN:SIGN&lt;i&gt;:ACTioN:STATioN:REConnect</a> .....       | 132 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:SEConD:STATe?</a> .....                 | 132 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:THIRd:STATe?</a> .....                  | 132 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:PSWitched:STATe?</a> .....              | 132 |
| <a href="#">SENSe:WLAN:SIGN&lt;i&gt;:UESinfo:APSSid?</a> .....               | 133 |
| <a href="#">SENSe:WLAN:SIGN&lt;i&gt;:SINFo:EAPStat?</a> .....                | 133 |

|                                                          |     |
|----------------------------------------------------------|-----|
| SENSe:WLAN:SIGN<i>:SINFo[:ANTenna<n>]:RXPindicator?..... | 133 |
| SENSe:WLAN:SIGN<i>:UESinfo[:ANTenna<n>]:ARXBpower?.....  | 134 |
| SENSe:WLAN:SIGN<i>:UESinfo:CMWaddress:IPV<n>?.....       | 134 |

---

### SOURce:WLAN:SIGN<i>:STATe <MainState>

Turns the generator (the cell) on or off.

#### Parameters:

<MainState>            ON | OFF | 1 | 0  
                              Switch generator **ON** or **OFF**  
                              \*RST:        OFF

**Example:**            See [Associating the DUT](#)

**Firmware/software:** V2.1.30

**Manual operation:** See "Cell" on page 34

---

### SOURce:WLAN:SIGN<i>:STATe:ALL?

Returns detailed information about the WLAN signaling generator state and the connection state of station one to three. See also [Chapter 2.2.1.3, "Connection status"](#), on page 18.

#### Return values:

<MainState>            OFF | ON

<SyncState>            PENDING | ADJUSTED

**PENDING:** The generator has been turned on (off) but the signal is not yet (still) available.

**ADJUSTED:** The physical output signal corresponds to the main generator state.

<DUT1>                IDLE | AUTHENTICATED | ASSOCIATED | DEAUTHENTICATED | DISASSOCIATED | CTIMEOUT

<DUT2>                IDLE | AUTHENTICATED | ASSOCIATED | DEAUTHENTICATED | DISASSOCIATED | CTIMEOUT

<DUT3>                IDLE | AUTHENTICATED | ASSOCIATED | DEAUTHENTICATED | DISASSOCIATED | CTIMEOUT

**Example:**            See [Associating the DUT](#)

**Usage:**                Query only

**Firmware/software:** V2.1.30, V3.8.20: added <DUTx> for multi STA support

**Manual operation:** See "Cell" on page 34

---

### CALL:WLAN:SIGN<i>:ACTion:STATION:CONNECT

Initiates an association to the AP under test. The command is only relevant in the operation mode "Station" with connection mode "Manual".

**Usage:** Event

**Firmware/software:** V3.2.10

**Manual operation:** See ["Connect \(hotkey\)"](#) on page 44

#### **CALL:WLAN:SIGN<i>:STA<s>:ACTion:DISConnect**

Disassociates and deauthenticates the DUT by sending a deauthentication frame.

**Suffix:**

<s> 1..3  
Station No.

**Usage:** Event

**Firmware/software:** V2.1.30, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Disconnect / Disconnect ST \(hotkey\)"](#) on page 44

#### **CALL:WLAN:SIGN<i>:ACTion:STATION:REConnect**

Re-establishes the existing association to the AP under test. The command has the same effect as a disconnect, followed immediately by a connect.

The command is only relevant in the operation mode "Station" with connection mode "Manual".

**Usage:** Event

**Firmware/software:** V3.2.10

**Manual operation:** See ["Reconnect \(hotkey\)"](#) on page 45

#### **FETCh:WLAN:SIGN<i>:SECOnd:STATe?**

#### **FETCh:WLAN:SIGN<i>:THIRd:STATe?**

#### **FETCh:WLAN:SIGN<i>:PSWitched:STATe?**

Gets the connection status, refer to [Chapter 2.2.1.3, "Connection status"](#), on page 18. Commands for station one, two and three are available.

**Return values:**

<PS State> IDLE | PROBed | AUTHenticated | ASSociated |  
DEAuthenticating | DISassociated | CTIMEout

**Example:** See [Associating the DUT](#)

**Usage:** Query only

**Firmware/software:** V2.1.30, V3.2.10: added CTIMEout  
V3.8.20: added commands for multi-STA

**Manual operation:** See ["Connection Status"](#) on page 34



**SENSe:WLAN:SIGN<i>:UESInfo:APSSid?**

Returns the SSID of the associated access point. The command is only relevant in operation mode "Station".

**Return values:**

<SSID> string  
Service set identifier as string

**Example:** See [Other connection settings](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Manual operation:** See ["Station SSID"](#) on page 34

**SENSe:WLAN:SIGN<i>:SINFo:EAPStat?**

Queries the current state of the EAP connection for the security mode "WPA / WPA2 Enterprise".

**Return values:**

<State> IDLE | SUCCess | FAILure  
EAP connection state

<Type> IDENTITY | NOTification | NAK | MD5 | OTP | GTC | TLS | CLEap |  
SIM | TTLS | AKA | AKAPrime  
Authentication stage

**Usage:** Query only

**Firmware/software:** V3.5.20

**Options:** R&S CMW-KS660

**Manual operation:** See ["EAP Connection Status"](#) on page 34

**SENSe:WLAN:SIGN<i>:SINFo[:ANTenna<n>]:RXPindicator?**

Queries the Rx burst power per individual antenna and evaluates the quality of the RX signal from the connected DUT. Antenna 2 is only available in a MIMO scenario.

**Suffix:**

<n> 1..2  
Antenna

**Return values:**

<PowerValue> float  
Default unit: dBm

<PowerIndicator> UNDerdriven | RANGE | OVERdriven

**Example:** See [Querying UE information](#)

**Usage:** Query only

**Firmware/software:** V3.7.30, V3.7.40: removed FAIRlygood, V3.7.60: added suffix :ANTenna<n>

**Manual operation:** See ["RX Power Indicator"](#) on page 34

#### SENSe:WLAN:SIGN<i>:UESinfo[:ANTenna<n>]:ARXBpower?

Queries the approximate RX burst power per individual antenna, calculated from the expected PEP and the configured standard. Antenna 2 is only available in a MIMO scenario.

**Suffix:**

<n> 1..2  
Antenna

**Return values:**

<ApproxPower> float  
Default unit: dBm

**Example:** See [Querying UE information](#)

**Usage:** Query only

**Firmware/software:** V3.5.120, V3.7.60: added suffix :ANTenna<n>

**Manual operation:** See ["Approximate Burst Power"](#) on page 55

#### SENSe:WLAN:SIGN<i>:UESinfo:CMWaddress:IPV<n>?

Queries the IP address of the R&S CMW.

**Suffix:**

<n> 4, 6  
IP version

**Return values:**

<IPAddress> string  
The assigned IPv4 or IPv6 addresses.

**Example:** See [Querying UE information](#)

**Usage:** Query only

**Firmware/software:** V3.7.50

**Manual operation:** See ["IPv4 / IPv6 address"](#) on page 36

### 2.5.3 Event log

The following commands clear and retrieve event log entries.

#### CLEan:WLAN:SIGN<i>:ELOGging

Clears the event log.

**Example:** See [Associating the DUT](#)

**Usage:** Event

**Firmware/software:** V3.8.20

**Manual operation:** See ["Event log entries"](#) on page 35

### SENSe:WLAN:SIGN<i>:ELOGging:ALL?

Queries all entries of the event log.

For each entry, three parameters are returned, from oldest to latest entry: {<Category>, <Timestamp>, <Description>}<sub>entry 1</sub>, {<Category>, <Timestamp>, <Description>}<sub>entry 2</sub>, ...

#### Return values:

<Category> INFO | WARNing | ERRor | EMPTy  
Category of the entry, as indicated in the main view by an icon  
EMPTy means that there are no entries.

<Timestamp> string  
Timestamp of the entry as string in the format "hh:mm:ss"

<Description> string  
Text string describing the event

**Example:** See [Associating the DUT](#)

**Usage:** Query only

**Firmware/software:** V3.5.20

**Manual operation:** See ["Event log entries"](#) on page 35

## 2.5.4 UE capabilities and statistics

The following commands return the information that the UE reports to the R&S CMW.

|                                                     |     |
|-----------------------------------------------------|-----|
| SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?          | 135 |
| SENSe:WLAN:SIGN<i>:STA<s>:UECapability:MAC:ADDRESS? | 136 |
| SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:ABSReport?        | 136 |
| SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:DRATe?            | 137 |
| SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:RXBPower?         | 138 |
| SENSe:WLAN:SIGN<i>:UESinfo:RXTRigframe?             | 139 |
| SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:UEAddress:IPV<n>? | 139 |

### SENSe:WLAN:SIGN<i>:STA<s>:UECapability:HE?

Indicates the reported UE HE capabilities.

#### Suffix:

<s> 1..3  
Station No.

**Return values:****<DeviceClass>** A | B**<DynFragment>** NO | L1 | L2 | L3

Dynamic fragmentation not supported, or dynamic fragmentation supported with level 1 to 3.

**<ABSR>** NO | YES

Indicates support of a buffer status report (BSR) control field.

**<BroadcastTWT>** NO | YES

Indicates support of broadcast target wake time (TWT) operation.

**<OFDMARandAcc>** NO | YES

Indicates support of OFDMA random access procedure.

**Example:** See [Querying UE information](#).**Usage:** Query only**Firmware/software:** V3.7.20, V3.8.20: added <s>**Options:** R&S CMW-KS667 for :STA2: and :STA3:**Manual operation:** See ["Device Class"](#) on page 37**SENSe:WLAN:SIGN<i>:STA<s>:UECapability:MAC:ADDRess?**

Gets the MAC address of an associated DUT.

**Suffix:****<s>** 1..3  
Station No.**Return values:****<MACAddress>** string  
Hexadecimal MAC address as string**Example:** See [Querying UE information](#)**Usage:** Query only**Firmware/software:** V2.1.30, V3.8.20: added <s>**Options:** R&S CMW-KS667 for :STA2: and :STA3:**Manual operation:** See ["MAC"](#) on page 36**SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:ABSRReport?**

Indicates reported buffered data for a UE supporting a HE buffer status report (BSR) control field.

**Suffix:**

<s> 1..3  
Station No.

**Return values:**

<Total> decimal  
Maximum of all reports received in preceding interval  
Range: 0 KB to 4.145152E+6 KB  
Default unit: kB

<BufferedData\_TID> decimal  
Maximum of all QoS control reports  
Range: 0 KB to 4.145152E+6 KB  
Default unit: kB

<TIDx> TID0 | TID1 | TID2 | TID3 | TID4 | TID5 | TID6 | TID7  
Indication of TID, for which the buffer status  
<BufferedData\_TID> is reported

<BufferedData\_AC> decimal  
Maximum AC-specific queue size of all AC control reports  
Range: 0 KB to 4.145152E+6 KB  
Default unit: kB

<ACx> ACBE | ACBK | ACVI | ACVO  
Indication of access category (ACI bitmap subfield) for which the  
buffer status <BufferedData\_AC> is reported  
**ACBE:** AC\_BE (best effort)  
**ACBK:** AC\_BK (background)  
**ACVI:** AC\_VI (video)  
**ACVO:** AC\_VO (voice)

**Example:** See [Querying UE information](#).

**Usage:** Query only

**Firmware/software:** V3.7.30, V3.8.20: added <s>, unit bit changed to kilobyte

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Buffered Data"](#) on page 37

**SENSE:WLAN:SIGN<i>:STA<s>:UESinfo:DRATe?**

Queries information related to the data rate of the DUT signal.

**Suffix:**

<s> 1..3  
Station No.

**Return values:**

<Format> NHT | HT | VHT | HESU | HEMU | HETB

|                           |                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Frame format                                                                                                                                                                                                                                                                                                                                                    |
|                           | <b>NHT</b> : non-high throughput format (non-HT)                                                                                                                                                                                                                                                                                                                |
|                           | <b>HT</b> : high throughput format                                                                                                                                                                                                                                                                                                                              |
|                           | <b>VHT</b> : very high throughput format                                                                                                                                                                                                                                                                                                                        |
|                           | <b>HESU</b> : high efficiency format, single user MIMO                                                                                                                                                                                                                                                                                                          |
|                           | <b>HEMU</b> : high efficiency format, multi user MIMO                                                                                                                                                                                                                                                                                                           |
|                           | <b>HETB</b> : high efficiency format, trigger based uplink single user MIMO                                                                                                                                                                                                                                                                                     |
| <Rate>                    | MB1   MB2   MB5   MB6   MB9   MB11   MB12   MB18   MB24   MB36   MB48   MB54   MCS0   MCS1   MCS2   MCS3   MCS4   MCS5   MCS6   MCS7   MCS8   MCS9   MCS10   MCS11   MCS12   MCS13   MCS14   MCS15<br><b>MBx</b> : data rate for NHT in Mbit/s {1, 2, 5.5, 6, 9, 11, 12, 18, 24, 36, 48, 54}<br><b>MCSx</b> : modulation and coding scheme x for HT, VHT and HE |
| <CBW>                     | BW20   BW40   BW80   BW88   BW16<br>Channel bandwidth in MHz: 20, 40, 80, 80+80, 160                                                                                                                                                                                                                                                                            |
| <NSS>                     | NSS1   NSS2   NSS3   NSS4   NSS5   NSS6   NSS7   NSS8<br>Number of spatial streams                                                                                                                                                                                                                                                                              |
| <b>Example:</b>           | See <a href="#">Querying UE information</a>                                                                                                                                                                                                                                                                                                                     |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                                                                                                                                                                      |
| <b>Firmware/software:</b> | V3.7.10, V3.7.20: added <CBW> BW80, V3.7.30: added <Format> HE, <CBW> BW88, BW16, V3.7.60: added <Rate> MCS12 to MCS15, V3.7.70: added <Rate> MB11<br>V3.8.20: added <s> and for <Format> removed HE, added HESU, HEMU, HETB                                                                                                                                    |
| <b>Options:</b>           | R&S CMW-KS657 for 802.11ax<br>R&S CMW-KS667 for :STA2: and :STA3:                                                                                                                                                                                                                                                                                               |
| <b>Manual operation:</b>  | See <a href="#">"Data Rate"</a> on page 37                                                                                                                                                                                                                                                                                                                      |

---

**SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:RXBPower?**

Queries the average power of the last burst received from the DUT.

**Suffix:**

<s> 1..3  
Station No.

**Return values:**

<Power> float  
Default unit: dBm

**Example:** See [Querying UE information](#)

**Usage:** Query only

**Firmware/software:** V3.5.120, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Burst Power"](#) on page 36

---

#### **SENSe:WLAN:SIGN<i>:UESinfo:RXTrigframe?**

Queries the total trigger frames received and the number of trigger frames directed to the station, i.e. R&S CMW.

**Return values:**

<Total\_TF> decimal

<Station\_TF> decimal

**Example:** See [Other connection settings](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KS658

**Manual operation:** See ["RX Trigger Frames"](#) on page 39

---

#### **SENSe:WLAN:SIGN<i>:STA<s>:UESinfo:UEAddress:IPV<n>?**

Queries the detected IPv4 / IPv6 addresses of the connected DUT.

**Suffix:**

<n> 4, 6  
Selects the IP version

<s> 1..3  
Station No.

**Return values:**

<IPAddress> string  
IPv4 / IPv6 addresses as string

**Example:** See [Querying UE information](#)

**Usage:** Query only

**Firmware/software:** V3.0.20, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["IPv4 / IPv6 address"](#) on page 36

## **2.5.5 User data statistics**

The following commands query statistics on traffic data and configure retrieval of statistics.

|                                               |     |
|-----------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:UESInfo:RESet.....     | 140 |
| CONFigure:WLAN:SIGN<i>:UESInfo:SETTings.....  | 140 |
| SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HEMU?.....  | 140 |
| SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HESU?.....  | 140 |
| SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HETB?.....  | 140 |
| SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HT?.....    | 140 |
| SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:NONHt?..... | 140 |
| SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:VHT?.....   | 140 |

---

### CONFigure:WLAN:SIGN<i>:UESInfo:RESet

Clears entries in all statistic tables concerning user data traffic.

**Example:** See [Running a PER measurement](#)

**Usage:** Event

**Firmware/software:** V3.8.20

**Manual operation:** See ["Reset Stats \(hotkey\)"](#) on page 44

---

### CONFigure:WLAN:SIGN<i>:UESInfo:SETTings <reporting\_interval>, <time\_span>

Sets reporting interval and time span used for enhanced statistics of user data traffic.

#### Parameters:

|                      |              |
|----------------------|--------------|
| <reporting_interval> | float        |
| Range:               | 0.2 s to 5 s |
| *RST:                | 0.2 s        |
| <time_span>          | integer      |
| Range:               | 1 to 1500    |
| *RST:                | 50           |

**Example:** See [Running a PER measurement](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["Stats Settings \(hotkey\)"](#) on page 43

---

SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HEMU?  
 SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HESU?  
 SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HETB?  
 SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:HT?  
 SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:NONHt?  
 SENSe:WLAN:SIGN<i>:UESInfo:RXPSdu:VHT?

Queries statistics on RX traffic data for the following IEEE 802.11 frame formats: high efficiency format multi user (HEMU), high efficiency format single user (HESU), high efficiency format trigger based (HETB), high throughput format (HT), non-high throughput format (NHT), and very high throughput format (VHT).



**Return values:**

|                 |                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <1_Rate>        | MB1   MB2   MB5   MB6   MB9   MB11   MB12   MB18   MB24   MB36   MB48   MB54   MCS0   MCS1   MCS2   MCS3   MCS4   MCS5   MCS6   MCS7   MCS8   MCS9   MCS10   MCS11   MCS12   MCS13   MCS14   MCS15<br><b>MBx</b> : data rate for NHT in Mbit/s {1, 2, 5.5, 6, 9, 11, 12, 18, 24, 36, 48, 54}<br><b>MCSx</b> : modulation and coding scheme x for HT, VHT and HE |
| <2_BW>          | BW20   BW40   BW80   BW88   BW16<br>Channel bandwidth in MHz: 20, 40, 80, 80+80, 160                                                                                                                                                                                                                                                                            |
| <3_NSS>         | NSS1   NSS2   NSS3   NSS4   NSS5   NSS6   NSS7   NSS8<br>Number of spatial streams                                                                                                                                                                                                                                                                              |
| <4_Frames_curr> | decimal                                                                                                                                                                                                                                                                                                                                                         |
| <5_Frames_tot>  | No. of recieved frames: current number and expectetd total number of frames                                                                                                                                                                                                                                                                                     |
| <6_Bytes_curr>  | float                                                                                                                                                                                                                                                                                                                                                           |
| <7_Bytes_tot>   | Volume of recieved user data: current value and expectetd total number of bytes                                                                                                                                                                                                                                                                                 |
| <8_Mbps_curr>   | float                                                                                                                                                                                                                                                                                                                                                           |
| <9_Mbps_tot>    | Data rate: current and average (total) value over the time span                                                                                                                                                                                                                                                                                                 |
| <10_Power_curr> | float                                                                                                                                                                                                                                                                                                                                                           |
| <11_Power_tot>  | RX power: current and average (total) value over the time span                                                                                                                                                                                                                                                                                                  |

**Example:** See [Running a PER measurement](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Manual operation:** See ["RX PSDU Info \(hotkey\)"](#) on page 43

## 2.5.6 Scenario selection and operation mode

The following commands configure the scenario, select the RF paths for the generated signal and the analyzed signal, set the operation mode and multi-STA support.

|                                                                            |     |
|----------------------------------------------------------------------------|-----|
| <a href="#">ROUTE:WLAN:SIGN&lt;i&gt;:SCENario?</a> .....                   | 142 |
| <a href="#">ROUTE:WLAN:SIGN&lt;i&gt;?</a> .....                            | 142 |
| <a href="#">ROUTE:WLAN:SIGN&lt;i&gt;:SCENario:SCELI:FLEXible</a> .....     | 143 |
| <a href="#">ROUTE:WLAN:SIGN&lt;i&gt;:SCENario:SCFading:FLEXible</a> .....  | 143 |
| <a href="#">ROUTE:WLAN:SIGN&lt;i&gt;:SCENario:MIMO:FLEXible</a> .....      | 144 |
| <a href="#">ROUTE:WLAN:SIGN&lt;i&gt;:SCENario:MIMFading:FLEXible</a> ..... | 144 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:OMODE</a> .....        | 145 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:MSTation</a> .....     | 145 |

**ROUTe:WLAN:SIGN<i>:SCENario?**

Returns the active scenario.

**Return values:**

<Scenario>                      STANdard | MIMO2 | SCFading | MIMFading

**STANdard**

Standard SISO scenario

**MIMO2**

MIMO 2x2 (DL and UL)

**SCFading**

Standard SISO scenario with fading

**MIMFading**

MIMO 2x2 scenario with fading

**Usage:**                              Query only

**Firmware/software:** V2.1.30, V3.2.11 added MIMO, V3.7.50: added SCFading, V3.7.60: removed MIMO, added MIMO2, MIMFading

**Manual operation:**    See "[Scenario](#)" on page 47

**ROUTe:WLAN:SIGN<i>?**

Queries the active test scenario, the used TRX modules and the used RF connectors.

For the STANdard and SCFading scenarios, the first six parameters are returned. For the MIMO scenario, all eight parameters are returned.

For possible connector and converter values, see [Chapter 2.5.1.2, "Values for signal path selection"](#), on page 124.

**Return values:**

<Scenario>                      STANdard cell | MIMO2 | SCFading | MIMFading

**STANdard**

Standard SISO scenario

**MIMO2**

MIMO 2x2 (DL and UL)

**SCFading**

Standard SISO scenario with fading

**MIMFading**

MIMO 2x2 scenario with fading

<Controller>                      string

For future use - returned value not relevant

<RXConnector>                      RF connector for the input path 1

<RXConverter>                      RX module for the input path 1

<TXConnector>                      RF connector for output path 1

<TXConverter>                      TX module for output path 1

<TXConnector2> RF connector for output path 2

<TXConverter2> TX module for output path 2

<RXConnector2> RF connector for the input path 2

<RXConverter2> RX module for the input path 2

**Usage:** Query only

**Firmware/software:** V2.1.30, V3.7.50: added <RXConnector2>. <RXConverter2> and <Scenario> SCFading

**Manual operation:** See "[Scenario](#)" on page 47

V3.7.60: removed <Scenario> MIMO, added MIMO2, MIMFading

**ROUTE:WLAN:SIGN<i>:SCENario:SCELI:FLEXible** <PCCBBBoard>, <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>

Defines the standard RX and TX routing (no MIMO).

For possible connector and converter values, see [Chapter 2.5.1.2, "Values for signal path selection"](#), on page 124.

**Parameters:**

<PCCBBBoard> Signaling unit

<RXConnector> RF connector for the input path

<RXConverter> RX module for the input path

<TXConnector> RF connector for the output path

<TXConverter> TX module for the output path

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.5.120

**Manual operation:** See "[Scenario](#)" on page 47

**ROUTE:WLAN:SIGN<i>:SCENario:SCFading:FLEXible** <PCCBBBoard>, <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>, <PCCFadingBoard>

Activates the "Standard Cell Fading" scenario and selects the signal paths.

For possible parameter values, see [Chapter 2.5.1.2, "Values for signal path selection"](#), on page 124.

**Parameters:**

<PCCBBBoard> Signaling unit

<RXConnector> RF connector for the input path

<RXConverter> RX module for the input path

<TXConnector> RF connector for the output path

<TXConverter> TX module for the output path

<PCCFadingBoard> Internal fader

**Example:** See [Configuring internal fading](#)

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Scenario"](#) on page 47

**ROUTe:WLAN:SIGN<i>:SCENario:MIMO:FLEXible** <PCCBBBoard>,  
 <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>,  
 <TX2Connector>, <TX2Converter>[, <RX2Connector>, <RX2Converter>]

Defines the RX and TX routing for the MIMO scenarios.

For possible connector and converter values, see [Chapter 2.5.1.2, "Values for signal path selection"](#), on page 124.

**Parameters:**

<PCCBBBoard> Signaling unit

<RXConnector> RF connector for the input path 1

<RXConverter> RX module for the input path 1

<TXConnector> RF connector for output path 1

<TXConverter> TX module for output path 1

<TX2Connector> RF connector for output path 2

<TX2Converter> TX module for output path 2. Select different modules for the two paths.

<RX2Connector> RF connector for the input path 2

<RX2Converter> RX module for the input path 2. Select different modules for the two paths.

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.5.120, V3.7.50: added <RX2Connector>, <RX2Converter>

**Options:** R&S CMW-KS670

**Manual operation:** See ["Scenario"](#) on page 47

**ROUTe:WLAN:SIGN<i>:SCENario:MIMFading:FLEXible** <PCCBBBoard>,  
 <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>,  
 <TX2Connector>, <TX2Converter>, <RX2Connector>, <RX2Converter>,  
 <PCCFadingBoard>

Activates the "MIMO 2x2 Fading" scenario and selects the signal paths.

For possible parameter values, see [Chapter 2.5.1.2, "Values for signal path selection"](#), on page 124.

**Parameters:**

|                  |                                    |
|------------------|------------------------------------|
| <PCCBBBoard>     | Signaling unit                     |
| <RXConnector>    | RF connector for the input path 1  |
| <RXConverter>    | RX module for the input path 1     |
| <TXConnector>    | RF connector for the output path 1 |
| <TXConverter>    | TX module for the output path 1    |
| <TX2Connector>   | RF connector for the output path 2 |
| <TX2Converter>   | TX module for the output path 2    |
| <RX2Connector>   | RF connector for the input path 2  |
| <RX2Converter>   | RX module for the input path 2     |
| <PCCFadingBoard> | Internal fader                     |

**Example:** See [Configuring internal fading](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Scenario"](#) on page 47

**CONFigure:WLAN:SIGN<i>:CONNection:OMODE <mode>**

Selects the operation mode, that is the type of WLAN entity simulated by the WLAN signaling application.

**Parameters:**

|                 |                                     |
|-----------------|-------------------------------------|
| <mode>          | AP   STATION   HSPot2               |
| <b>AP:</b>      | access point in infrastructure mode |
| <b>STATION:</b> | WLAN station                        |
| <b>HSPot2:</b>  | WiFi Hotspot 2.0 access point       |
| <b>*RST:</b>    | AP                                  |

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.2.10, V3.2.20: added IBSS, V3.2.60: added WDIRect and HSPOT2  
V3.7.60: removed IBSS and WDIRect

**Options:** R&S CMW-KS660 for HSPot2

**Manual operation:** See ["Operation Mode"](#) on page 48

**CONFigure:WLAN:SIGN<i>:CONNection:MSTation <enable>**

Enables or disables the support of multi-station connection.

**Parameters:**

|              |          |
|--------------|----------|
| <Enable>     | OFF   ON |
| <b>*RST:</b> | OFF      |

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KS667

**Manual operation:** See ["Multi STA"](#) on page 48

## 2.5.7 MIMO settings

The following command provides additional settings for the MIMO scenario.

[CONFigure:WLAN:SIGN<i>:MIMO:TMMode](#)..... 146

---

**CONFigure:WLAN:SIGN<i>:MIMO:TMMode <mode>**

Selects the transmission mode for MIMO connections.

This command supports only spatial multiplexing and space time block coding (STBC).

In addition, to enable STBC in a particular data frame, use the commands:

[CONFigure:WLAN:SIGN<i>:STA<s>:CONNECTION:DFDef](#) or

[CONFigure:WLAN:SIGN<i>:PER:FDEF](#)

**Parameters:**

<mode> STBC | SMULtiplexin  
\*RST: SMUL

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.2.11, V3.7.60: removed STBC, TXDiversity, V3.8.10: added STBC

**Options:** R&S CMW-KS670

**Manual operation:** See ["TX MIMO Mode"](#) on page 50

## 2.5.8 RF path properties

The following commands define path properties to be compensated (external attenuation).

[CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:OUTPut](#)..... 146

[CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:INPut](#)..... 147

---

**CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:OUTPut  
<ExtAttenuation>**

Defines an external attenuation (or gain, if the value is negative), to be applied to the individual antennas. Antenna 2 is only available in a MIMO scenario.

**Suffix:**

<n> 1..2  
Antenna

**Parameters:**

<ExtAttenuation> numeric  
Range: -50 dB to 90 dB  
\*RST: 0 dB  
Default unit: dB

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.2.11, V3.7.60: changed suffix to :ANTenna<n>

**Options:** R&S CMW-KS670

**Manual operation:** See ["External Attenuation"](#) on page 51

---

**CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:INPut  
<ExtAttenuation>**

Specifies the external attenuation for the specified antenna in the analyzer path.  
Antenna 2 is only available in a MIMO scenario with two input paths.

**Suffix:**

<n> 1..2  
Antenna

**Parameters:**

<ExtAttenuation> numeric  
Range: -50 dB to 55 dB  
\*RST: 0 dB  
Default unit: dB

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.7.60

**Manual operation:** See ["External Attenuation"](#) on page 52

## 2.5.9 Signal settings

The following commands provide settings for the generated WLAN signals.

|                                                     |     |
|-----------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:RFSettings:OCWidth.....      | 148 |
| CONFigure:WLAN:SIGN<i>:RFSettings:BAND.....         | 148 |
| CONFigure:WLAN:SIGN<i>:RFSettings:FREQUENCY.....    | 148 |
| CONFigure:WLAN:SIGN<i>:RFSettings:CHANnel.....      | 149 |
| CONFigure:WLAN:SIGN<i>:RFSettings:FOFFset.....      | 149 |
| CONFigure:WLAN:SIGN<i>:RFSettings:NPIndex.....      | 149 |
| CONFigure:WLAN:SIGN<i>:RFSettings:NPFRrequency..... | 149 |
| CONFigure:WLAN:SIGN<i>:RFSettings:NPCHannel.....    | 150 |
| CONFigure:WLAN:SIGN<i>:RFSettings:BOPower.....      | 150 |

|                                                            |     |
|------------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:RFSettings:TSRatio.....             | 151 |
| CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EPEPower..... | 151 |
| CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:MLOffset..... | 151 |

---

### CONFigure:WLAN:SIGN<i>:RFSettings:OCWidth <Value>

Sets the operating channel bandwidth.

#### Parameters:

<Value> BW20 | BW40 | BW80 | BW160  
**BW20:** 20 MHz  
**BW40:** 40 MHz  
**BW80:** 80 MHz  
**BW160:** 160 MHz  
 \*RST: BW20

**Example:** See [Configuring channel frequencies](#)

**Firmware/software:** V3.5.120, V3.7.20: added BW80, V3.7.30: added BW160

**Options:** R&S CMW-KS656 for BW80 and BW160

**Manual operation:** See ["Operating Channel Width"](#) on page 47

---

### CONFigure:WLAN:SIGN<i>:RFSettings:BAND <freq\_band>

Selects the operating band sub 6 GHz or 6 GHz band.

#### Parameters:

<freq\_band> BS6Ghz | B6GHz  
 \*RST: BS6G

**Example:** See [Configuring channel frequencies](#).

**Firmware/software:** V3.8.20

**Manual operation:** See ["Band"](#) on page 53

---

### CONFigure:WLAN:SIGN<i>:RFSettings:FREQuency <Frequency>

Sets the center frequency of the generated WLAN signal and of the RF analyzer.

#### Parameters:

<Frequency> numeric  
 Range: 70E+6 Hz to 6E+9 Hz  
 \*RST: 2412E+6 Hz  
 Default unit: Hz

**Example:** See [Configuring channel frequencies](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["Center Frequency / Channel"](#) on page 53



**CONFigure:WLAN:SIGN<i>:RFSettings:CHANnel** <Channel>

Sets the RF channel number.

**Parameters:**

<Channel> integer  
 Range: 1 to 196  
 \*RST: 1

**Example:** See [Configuring channel frequencies](#)

**Firmware/software:** V3.0.20

**Manual operation:** See ["Center Frequency / Channel"](#) on page 53

**CONFigure:WLAN:SIGN<i>:RFSettings:FOFFset** <Offset>

Specifies a positive or negative frequency offset to be added to the configured center frequency.

**Parameters:**

<Offset> integer  
 Range: -100E+3 Hz to 100E+3 Hz  
 \*RST: 0 Hz  
 Default unit: Hz

**Firmware/software:** V3.5.20

**Manual operation:** See ["Frequency Offset"](#) on page 53

**CONFigure:WLAN:SIGN<i>:RFSettings:NPIndex** <Np20>

Selects the position of the primary 20-MHz channel, for a signal with more than 20 MHz channel bandwidth.

**Parameters:**

<Np20> integer  
 Index of the 20-MHz channel configured as primary channel  
 Range: 0 to <no of channels> - 1  
 Increment: 1  
 \*RST: 0

**Example:** See [Configuring channel frequencies](#)

**Firmware/software:** V3.5.120

**Manual operation:** See ["Np20 Index"](#) on page 54

**CONFigure:WLAN:SIGN<i>:RFSettings:NPFrequency** <Np20Freq>

Sets the center frequency of the primary 20-MHz channel, for a signal with more than 20 MHz bandwidth.

**Parameters:**

<Np20Freq>            numeric  
                          Range:        70E+6 Hz to 6E+9 Hz  
                          \*RST:        2.412E+9 Hz  
                          Default unit: Hz

**Example:**            See [Configuring channel frequencies](#)

**Firmware/software:** V3.5.120

**Manual operation:** See ["Np20 Frequency / Channel"](#) on page 54

**CONFigure:WLAN:SIGN<i>:RFSettings:NPChannel** <Np20Channel>

Sets the channel number of the primary 20-MHz channel, for a signal with more than 20 MHz bandwidth.

**Parameters:**

<Np20Channel>        integer  
                          Range:        1 to 196  
                          \*RST:        1

**Example:**            See [Configuring channel frequencies](#)

**Firmware/software:** V3.5.120

**Manual operation:** See ["Np20 Frequency / Channel"](#) on page 54

**CONFigure:WLAN:SIGN<i>:RFSettings:BOPower** <BurstOutputPow>

Sets the burst power of the transmitted signal.

The allowed value range depends on the used connector and the external attenuation in the output path.

$$\text{Minimum} = \text{level}_{\text{connector, min}} - \text{ext. att.}_{\text{out}}$$

$$\text{Maximum} = \text{level}_{\text{connector, max}} - \text{ext. att.}_{\text{out}}$$

With  $\text{level}_{\text{connector, min}} = -145.98 \text{ dBm}$  ( $-137.98 \text{ dBm}$ ),  $\text{level}_{\text{connector, max}} = -15.98 \text{ dBm}$  ( $-2.98 \text{ dBm}$ ) for RF COM (RF OUT); please also notice the ranges quoted in the data sheet.

**Parameters:**

<BurstOutputPow>    numeric  
                          Range:        see above  
                          \*RST:        -40 dBm  
                          Default unit: dBm

**Example:**            See [Specifying general settings](#)

**Firmware/software:** V2.1.30

V2.1.40: range shifted

V3.2.11: reset changed from -80 dBm to -40 dBm

**Manual operation:** See "TX Burst Power" on page 55

**CONFigure:WLAN:SIGN*i*:RFSettings:TSRatio** <Ratio>

Sets the power ratio of TX2 to TX1 for the MIMO scenario.

### Parameters:

```
<Ratio>          numeric
                  TX2/TX1
                  *RST:      0 dB
                  Default unit: dB
```

**Example:** See [Specifying general settings](#)

Firmware/software: V3.2.11

Options: R&amp;S CMW-KS670

**Manual operation:** See "TX2 / TX1 Power Ratio" on page 55

**CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EPEPower** <Expected Peak Envelop Power>

Specifies the expected peak envelope power of the specified antenna at the I/Q input. Antenna 2 is only available in MIMO scenarios with two input paths.

The correct DUT range to be set is  $(-20 \text{ dBm} + \text{external attenuation})$  to  $(55 \text{ dBm} + \text{external attenuation})$ .

**Suffix:**

<n> 1..2  
Antenna

### Parameters:

|               |                                                                       |
|---------------|-----------------------------------------------------------------------|
| <ExpectedPEP> | numeric                                                               |
| Range:        | -20 dBm (CMW limit) to (55 dBm + external<br>attenuation = DUT limit) |
| *RST:         | 0 dBm                                                                 |
| Default unit: | dBm                                                                   |

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.7.60

**Manual operation:** See "Exp. Peak Envelope Power" on page 55

**CONFigure:WLAN:SIGN*i*:RFSettings:ANTenna<n>:MLOffset** <value>

Varies the input level of the mixer for the specified antenna in the analyzer path. Antenna 2 is only available in MIMO scenarios with two input paths.

**Suffix:**

<n> 1..2  
Antenna

**Parameters:**

<value> integer  
 Range: -10 dB to 12dB  
 Default unit: dB

**Example:** See [Specifying general settings](#)

**Firmware/software:** V3.7.60

**Manual operation:** See ["RX Mixer Level Offset"](#) on page 55

## 2.5.10 Internal fading

The following commands configure the internal fader of the R&S CMW.

- [Fading simulator](#).....152
- [Fading module AWGN](#).....153

### 2.5.10.1 Fading simulator

The following commands configure the fading simulator of the internal fader.

- [CONFigure:WLAN:SIGN<i>:FADing:FSIMulator:ENABLE](#).....152
- [CONFigure:WLAN:SIGN<i>:FADing:FSIMulator:ILOSs:LOSS?](#).....152
- [CONFigure:WLAN:SIGN<i>:FADing:FSIMulator:STANdard](#).....153

---

#### **CONFigure:WLAN:SIGN<i>:FADing:FSIMulator:ENABLE <Enable>**

Enables/disables the fading simulator.

**Parameters:**

<Enable> OFF | ON  
 \*RST: OFF

**Example:** See [Configuring internal fading](#)

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Enable"](#) on page 56

---

#### **CONFigure:WLAN:SIGN<i>:FADing:FSIMulator:ILOSs:LOSS?**

Sets the insertion loss for the fading simulator.

**Return values:**

<InsertionLoss> float  
 Range: -3.02 dB to 30 dB  
 Default unit: dB

**Example:** See [Configuring internal fading](#)

**Usage:** Query only

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Insertion Loss"](#) on page 56

---

#### **CONFigure:WLAN:SIGN<i>:FADing:FSIMulator:STANdard** <profile>

Selects a propagation condition profile for fading, see [Chapter A, "Predefined fading settings"](#), on page 633.

##### **Parameters:**

<Profile>                    MODA | MODB | MODC | MODD | MODE | MODF  
                                 Mode A to F  
                                 \*RST:        MODA

**Example:**                See [Configuring internal fading](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Profile"](#) on page 56

### **2.5.10.2 Fading module AWGN**

The following commands configure the AWGN generator of the internal fader.

|                                                                             |     |
|-----------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:FADing:AWGN:BWIDth:RATio</a> ..... | 153 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:FADing:AWGN:ENABLe</a> .....       | 153 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:FADing:AWGN:SNRatiO</a> .....      | 154 |

---

#### **CONFigure:WLAN:SIGN<i>:FADing:AWGN:BWIDth:RATio** <Ratio>

Specifies the minimum ratio between the noise bandwidth and the channel bandwidth.

##### **Parameters:**

<Ratio>                    numeric  
                                 Range:        1 to 1000  
                                 \*RST:        1

**Example:**                See [Configuring internal fading](#)

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Min. Noise/System BW Ratio"](#) on page 57

---

#### **CONFigure:WLAN:SIGN<i>:FADing:AWGN:ENABLe** <Enable>

Enables or disables AWGN insertion via the fading module.

**Parameters:**

<Enable> OFF | ON  
 \*RST: OFF

**Example:** See [Configuring internal fading](#)

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Enable"](#) on page 57

---

**CONFigure:WLAN:SIGN<i>:FADing:AWGN:SNRatio <Ratio>**

Specifies the signal to noise ratio for the AWGN inserted on the internal fading module.

**Parameters:**

<Ratio> numeric  
 Range: 0 dB to 40 dB  
 \*RST: 30 dB  
 Default unit: dB

**Example:** See [Configuring internal fading](#)

**Firmware/software:** V3.7.50, V3.7.60: changed range and reset value

**Options:** R&S CMW-KE100 and R&S CMW-KE670

**Manual operation:** See ["Signal/Noise Ratio"](#) on page 57

## 2.5.11 Connection settings

The following commands define the upper layer BSS properties.

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| • <a href="#">General connection settings</a> .....                | 154 |
| • <a href="#">Association settings</a> .....                       | 163 |
| • <a href="#">Hotspot 2.0 settings</a> .....                       | 165 |
| • <a href="#">Target wake time settings</a> .....                  | 171 |
| • <a href="#">Management and data frame control settings</a> ..... | 173 |
| • <a href="#">Trigger frame control settings</a> .....             | 181 |
| • <a href="#">Security settings</a> .....                          | 190 |
| • <a href="#">QoS data settings</a> .....                          | 196 |
| • <a href="#">A-MPDU settings</a> .....                            | 198 |
| • <a href="#">EDCA settings</a> .....                              | 199 |

### 2.5.11.1 General connection settings

This section describes the following connection commands.

|                                                                        |     |
|------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:DSSS</a> .....     | 155 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:STANdard</a> ..... | 155 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BEACon</a> .....   | 156 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:DPERiod</a> .....  | 156 |

|                                                            |     |
|------------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:BSSid.....               | 156 |
| CONFigure:WLAN:SIGN<i>:CONNection:SSID.....                | 157 |
| CONFigure:WLAN:SIGN<i>:CONNection:BSSColor.....            | 157 |
| CONFigure:WLAN:SIGN<i>:CONNection:AID.....                 | 157 |
| CONFigure:WLAN:SIGN<i>:CONNection:SMOothing.....           | 158 |
| CONFigure:WLAN:SIGN<i>:CONNection:STATION:SCONnection..... | 158 |
| CONFigure:WLAN:SIGN<i>:CONNection:STATION:CMODE.....       | 158 |
| CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCState.....       | 159 |
| CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCConf.....        | 159 |
| CONFigure:WLAN:SIGN<i>:CONNection:SRATes.....              | 160 |
| CONFigure:WLAN:SIGN<i>:CONNection:SRATes:DSSSconf.....     | 160 |
| CONFigure:WLAN:SIGN<i>:CONNection:SRATes:OFDMconf.....     | 161 |
| CONFigure:WLAN:SIGN<i>:CONNection:SRATes:OMCSconf.....     | 162 |
| CONFigure:WLAN:SIGN<i>:CONNection:SRATes:VHTConf.....      | 162 |
| CONFigure:WLAN:SIGN<i>:CONNection:HEMac:BSRSupport.....    | 163 |

---

### CONFigure:WLAN:SIGN<i>:CONNection:DSSS <Support\_of\_DSSS>

Enables you to associate an 802.11b device using 802.11ac or ax standard. Also, it enables you to use non-HT management frames within 802.11ac and ax.

#### Parameters:

<Support\_of\_DSSS> OFF | ON

\*RST: OFF

**Example:** See [Configuring connection settings](#).

**Firmware/software:** V3.7.70

**Manual operation:** See ["Support of DSSS"](#) on page 61

---

### CONFigure:WLAN:SIGN<i>:CONNection:STANdard <Typ>

Selects the IEEE 802.11 WLAN standard to be used.

#### Parameters:

<Typ> ASTD | GOSTd | ANSTd | GONStd | BSTD | GSTD | GNSTd |  
ACSTd | AXSTd

**BSTD:** 802.11b

**ASTD:** 802.11a

**GSTD:** 802.11g

**GOSTd:** 802.11g (OFDM)

**ANSTd:** 802.11a/n

**GNSTd:** 802.11g/n

**GONStd:** 802.11g (OFDM)/n

**ACSTd:** 802.11ac

**AXSTd:** 802.11ax

\*RST: ASTD

**Example:** See [Specifying general settings](#)

**Firmware/software:** V2.1.40, V3.0.20: added BSTD, GSTD, GNSTd, V3.5.120: added ACSTd, V3.7.30: added AXSTd, V3.7.60: removed NGFStd

**Options:** R&S CMW-KS651 for ANSTd, GNSTd, GONSTd, ACSTd  
R&S CMW-KS656 for ACSTd  
R&S CMW-KS657 for AXSTd

**Manual operation:** See ["Standard"](#) on page 47

#### **CONFigure:WLAN:SIGN<i>:CONNection:BEACon** <BeaconInterval>

Sets the interval between two beacon frame transmissions for a simulated infrastructure/ ad-hoc network.

##### **Parameters:**

<BeaconInterval> integer  
Interval in time units (1 TU = 1024  $\mu$ s)  
Range: 20 to 16000  
\*RST: 100

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V2.1.40  
V3.0.20: minimum increased to 20, maximum decreased to 1000  
V3.2.11: maximum increased to 16000

**Manual operation:** See ["Beacon Interval"](#) on page 61

#### **CONFigure:WLAN:SIGN<i>:CONNection:DPERiod** <Period>

Sets the number of beacon intervals between successive delivery traffic indication messages (DTIM).

##### **Parameters:**

<Period> integer  
Number of beacon intervals  
Range: 1 to 10  
\*RST: 1

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.60

**Manual operation:** See ["DTIM Period"](#) on page 62

#### **CONFigure:WLAN:SIGN<i>:CONNection:BSSId** <BSSID>

Sets the 48-bit MAC address of the WLAN interface.



**Parameters:**

<BSSID> hex  
 Hexadecimal number with 12 digits  
 Leading zeros can be omitted.  
 Range: #H0 to #HFFFFFFFFFFFF  
 \*RST: #H102030405

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["MAC Address \(BSSID\)"](#) on page 62

**CONFigure:WLAN:SIGN<i>:CONNection:SSID <SSID>**

Sets the service set identifier (SSID).

**Parameters:**

<SSID> string  
 String with up to 32 characters (7-bit ASCII only)  
 \*RST: "CMW-AP"

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["SSID"](#) on page 62

**CONFigure:WLAN:SIGN<i>:CONNection:BSSColor <Value>**

Specifies the color code of basic service set (BSS).

**Parameters:**

<Value> numeric  
 Range: 1 to 63  
 \*RST: 1

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.7.50: changed range and reset value

**Options:** R&S CMW-KS657

**Manual operation:** See ["BSS Color"](#) on page 62

**CONFigure:WLAN:SIGN<i>:CONNection:AID <start>, <stop>**

Specifies the range of IDs to be assigned by the access point to the connected DUTs.

**Parameters:**

<start> numeric  
 Range: 1 to 2007  
 \*RST: 1

<stop> numeric  
 Range: 1 to 2007  
 \*RST: 2007

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KS657

**Manual operation:** See ["Association ID Range"](#) on page 62

#### **CONFigure:WLAN:SIGN<i>:CONNection:SMOothing <bit>**

Indicates to the receiver whether the frequency-domain smoothing is recommended for channel estimation.

##### **Parameters:**

<bit> NRECommended | RECommended  
 Not recommended or recommended  
 \*RST: NREC

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.7.30

**Manual operation:** See ["HT Smoothing"](#) on page 62

#### **CONFigure:WLAN:SIGN<i>:CONNection:STAtion:SCONnection <mode>**

Specifies to which access points the simulated station is allowed to connect (operation mode "Station").

##### **Parameters:**

<mode> ANY | SSID  
**ANY:** Any AP is allowed.  
**SSID:** Only a specific AP is allowed, identified by the configured SSID, see [CONFigure:WLAN:SIGN<i>:CONNection:SSID](#) on page 157.  
 \*RST: ANY

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.10

**Manual operation:** See ["SSID Connection"](#) on page 63

#### **CONFigure:WLAN:SIGN<i>:CONNection:STAtion:CMODE <mode>**

Selects the connection behavior of the simulated station (operation mode "Station").

**Parameters:**

<mode> ACONnect | MANual  
**ACONnect:** automatic connection  
**MANual:** manual connection via [CALL:WLAN:SIGN<i>:ACTION:STATION:CONNeCT](#) or [CALL:WLAN:SIGN<i>:ACTION:STATION:REConnect](#)  
 \*RST: ACON

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.10

**Manual operation:** See ["Connection Mode"](#) on page 63

**CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCSTate <State>**

Enables/disables the broadcast of regulatory domain information in beacon frames.

**Parameters:**

<State> DISable | ENABle  
 \*RST: DIS

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.10  
 V3.0.20: "OFF | ON" changed to "DISable | ENABle"

**Manual operation:** See ["Country Information"](#) on page 69

**CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCConf <CodeDigit>, <FirstChannel>, <NbOfChannels>, <MaxTXPower>**

Sets the regulatory domain information to be transmitted in beacon frames.

To enable the transmission, see [CONFigure:WLAN:SIGN<i>:CONNection:CCODE:CCSTate](#) on page 159.

**Parameters:**

<CodeDigit> string  
 Country code as string  
 \*RST: US

<FirstChannel> integer  
 First in the range of allowed channels  
 Range: 0 to 255  
 \*RST: 1

<NbOfChannels> integer  
 Number of allowed channels  
 Range: 0 to 255  
 \*RST: 13

<MaxTXPower> integer  
 Maximum transmit power  
 Range: -40 dBm to 40 dBm  
 \*RST: 0 dBm  
 Default unit: dBm

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.10

**Manual operation:** See ["Country Code"](#) on page 69

### **CONFigure:WLAN:SIGN<i>:CONNection:SRATes <State>**

Enables/disables user-defined supported rates.

#### **Parameters:**

<State> ENABLE | DISable  
 \*RST: DIS

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.20

**Manual operation:** See ["Configuration"](#) on page 70

### **CONFigure:WLAN:SIGN<i>:CONNection:SRATes:DSSSconf <D1MB>, <D2MB>, <C55M>, <C11M>**

Definition of DSSS/CCK supported rates.

These settings apply only if user-defined supported rates are enabled, see [CONFigure:WLAN:SIGN<i>:CONNection:SRATes](#) on page 160.

#### **Parameters:**

<D1MB> DISabled | MANDatory | OPTional  
 Support for DSSS, 1 Mbit/s  
 \*RST: DIS

<D2MB> DISabled | MANDatory | OPTional  
 Support for DSSS, 2 Mbit/s  
 \*RST: DIS

<C55M> DISabled | MANDatory | OPTional  
 Support for CCK, 5.5 Mbit/s  
 \*RST: DIS

<C11M> DISabled | MANDatory | OPTional  
 Support for CCK, 11 Mbit/s  
 \*RST: DIS

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.20

**Manual operation:** See ["Non HT, HT, VHT Supported Rates"](#) on page 70

---

**CONFigure:WLAN:SIGN<i>:CONNection:SRATes:OFDMconf** <BR12>, <BR34>, <QR12>, <QR34>, <Q1M12>, <Q1M34>, <Q6M23>, <Q6M34>

Definition of OFDM non-HT supported rates (modulation, coding rate, data rate).

These settings apply only if user-defined supported rates are enabled, see [CONFigure:WLAN:SIGN<i>:CONNection:SRATes](#) on page 160.

**Parameters:**

|         |                                                                                    |
|---------|------------------------------------------------------------------------------------|
| <BR12>  | DISabled   MANDatory   OPTional<br>Support for BPSK, 1/2, 6 Mbit/s<br>*RST: DIS    |
| <BR34>  | DISabled   MANDatory   OPTional<br>Support for BPSK, 3/4, 9 Mbit/s<br>*RST: DIS    |
| <QR12>  | DISabled   MANDatory   OPTional<br>Support for QPSK, 1/2, 12 Mbit/s<br>*RST: DIS   |
| <QR34>  | DISabled   MANDatory   OPTional<br>Support for QPSK, 3/4, 18 Mbit/s<br>*RST: DIS   |
| <Q1M12> | DISabled   MANDatory   OPTional<br>Support for 16-QAM, 1/2, 24 Mbit/s<br>*RST: DIS |
| <Q1M34> | DISabled   MANDatory   OPTional<br>Support for 16-QAM, 3/4, 36 Mbit/s<br>*RST: DIS |
| <Q6M23> | DISabled   MANDatory   OPTional<br>Support for 64-QAM, 2/3, 48 Mbit/s<br>*RST: DIS |
| <Q6M34> | DISabled   MANDatory   OPTional<br>Support for 64-QAM, 3/4, 54 Mbit/s<br>*RST: DIS |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.20

**Manual operation:** See ["Non HT, HT, VHT Supported Rates"](#) on page 70

---

**CONFigure:WLAN:SIGN<i>:CONNection:SRATes:OMCSconf** <MCS0>, <MCS1>, <MCS2>, <MCS3>, <MCS4>, <MCS5>, <MCS6>, <MCS7>

Definition of supported OFDM HT modulation and coding schemes (MCS).

These settings apply only if user-defined supported rates are enabled, see [CONFigure:WLAN:SIGN<i>:CONNection:SRATes](#) on page 160.

**Parameters:**

|        |                                        |
|--------|----------------------------------------|
| <MCS0> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS1> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS2> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS3> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS4> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS5> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS6> | NOTSupported   SUPPorted<br>*RST: NOTS |
| <MCS7> | NOTSupported   SUPPorted<br>*RST: NOTS |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.20

**Manual operation:** See ["Non HT, HT, VHT Supported Rates"](#) on page 70

---

**CONFigure:WLAN:SIGN<i>:CONNection:SRATes:VHTConf** <VHT\_rates>

Definition of supported OFDM VHT modulation and coding schemes (MCS).

These settings apply only if user-defined supported rates are enabled, see [CONFigure:WLAN:SIGN<i>:CONNection:SRATes](#) on page 160.

**Parameters:**

|             |                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| <VHT_rates> | MC07   MC08   MC09<br><b>MC07:</b> MCS 0 to MCS 7<br><b>MC08:</b> MCS 0 to MCS 8<br><b>MC09:</b> MCS 0 to MCS 9<br>*RST: MC07 |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.7.10

**Manual operation:** See ["Non HT, HT, VHT Supported Rates"](#) on page 70

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:HEMac:BSRSupport** <Supported>

Indicates, whether the R&S CMW supports the buffer status report (BSR).

##### **Parameters:**

<Supported>            OFF | ON  
                              \*RST:        ON

**Example:**            See [Configuring connection settings](#)

**Firmware/software:** V3.7.60

**Options:**            R&S CMW-KS657

**Manual operation:** See ["BSR Support"](#) on page 86

### 2.5.11.2 Association settings

The following commands configure connection settings related to association.

|                                                                                                   |     |
|---------------------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:ASSociation:DISass</a> .....                  | 163 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:ASSociation:PREemption</a> .....              | 163 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:ASSociation:STA&lt;s&gt;:MACReserve</a> ..... | 164 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:ASSociation:STAPriority</a> .....             | 164 |

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:DISass** <enable>[, <timeout>]

Enables or disables automatic STA disassociation, when a STA is no longer present. If enabled, the R&S CMW detects that a STA is absent, it automatically removes its association after some user-specified period of time.

##### **Parameters:**

<Enable>            OFF | ON  
                              \*RST:        OFF  
  
 <Timeout>            numeric  
                              Range:        1 s to 3600 s  
                              \*RST:        30 s

**Example:**            See [Configuring connection settings](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["Automatic Disassociation"](#) on page 60

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:PREemption** <enable>

If enabled, then the existing association possible with any MAC addresses is replaced by a new incoming one.

**Parameters:**

<Enable> OFF | ON  
 \*RST: ON

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["Allow Preemption"](#) on page 60

**CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:STA<s>:MACReserve**  
 <reservation>[, <address>]

Configures three slots available for STAs, if

[CONFigure:WLAN:SIGN<i>:CONNection:MSTation](#) is set to ON

**Suffix:**

<s> 1..3  
 Station No.

**Parameters:**

<Reservation> ANY | SET | OFF  
**ANY** - the slot is available to a STA of any MAC address  
**SET** - reserves the slot for a particular MAC address  
**OFF** - the slot is disabled  
 \*RST: ANY

<Address> string  
 MAC address of the DUT for <Reservation> = SET  
 \*RST: 00:00:00:00:00:00

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KS667  
 R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Reserve MAC Address"](#) on page 60

**CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:STAPriority** <mode>

Specifies how the stack prioritizes one STA over another in multi-STA connections.

**Parameters:**

<Mode> AUTO | ROURobin  
 Automatic or round robin  
 \*RST: AUTO

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["STA Priority"](#) on page 60



### 2.5.11.3 Hotspot 2.0 settings

The following commands configure connection settings for Hotspot 2.0.

|                                                          |     |
|----------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar.....     | 165 |
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSSPar.....    | 166 |
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:PLMN<no>.....  | 168 |
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:MNDigits.....  | 169 |
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:DNAMe<no>..... | 169 |
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:REALm<no>..... | 169 |
| CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:CUTil.....     | 170 |

---

**CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar** <AccessNetType>, <VenueGroup>, <VenueType>, <HESSID>, <VenueName>, <MCC>, <MNC>, <DomainName>, <OpFrieName>

Defines basic parameters of the Hotspot 2.0 operation mode.

**Parameters:**

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <AccessNetType> | PNETwork   PNWGaccess   CPNetwork   FPNetwork   PDNetwork   ESONetwork   TOEXperiment   WILDcard<br><b>PNETwork</b> : private network<br><b>PNWGaccess</b> : private network with guest access<br><b>CPNetwork</b> : chargeable public network<br><b>FPNetwork</b> : free public network<br><b>PDNetwork</b> : personal device network<br><b>ESONetwork</b> : emergency services only network<br><b>TOEXperiment</b> : test or experimental<br><b>WILDcard</b> : wildcard<br>*RST: PNET |
| <VenueGroup>    | numeric   UNSpecified   ASSEMBly   BUSiness   EDUCational   FAINDustrial   INSTititutional   MERCantile   RESidential   STORage   UAMisc   VEHicular   OUTDoor<br><b>FAINDustrial</b> : factory and industrial<br><b>UAMisc</b> : utility and miscellaneous<br>*RST: UNSP                                                                                                                                                                                                               |
| <VenueType>     | numeric<br>Range: 0 to 255<br>*RST: 8                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <HESSID>        | numeric<br>Homogeneous extended service set identifier<br>Range: #H0 to #HFFFFFFFFFFFF<br>*RST: #H102030405                                                                                                                                                                                                                                                                                                                                                                             |
| <VenueName>     | string<br>String with up to 252 ASCII characters<br>*RST: R&S                                                                                                                                                                                                                                                                                                                                                                                                                           |

|              |                                                                                                                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MCC>        | numeric<br>Mobile country code of 3GPP network reachable via the hotspot<br>To configure more than one PLMN, use <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:HOTSpot:PLMN&lt;no&gt;</a> on page 168.<br>Range: 1 to 999<br>*RST: 1               |
| <MNC>        | numeric<br>Mobile network code of 3GPP network reachable via the hotspot<br>To configure more than one PLMN, use <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:HOTSpot:PLMN&lt;no&gt;</a> on page 168.<br>Range: 1 to 999 <sup>1)</sup><br>*RST: 1 |
| <DomainName> | string<br>Domain name of the network operator as string<br>To configure more than one domain name, use <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:HOTSpot:DNAME&lt;no&gt;</a> on page 169.                                                      |
| <OpFrieName> | string<br>Friendly name of the network operator as string<br>*RST: R&S                                                                                                                                                                                   |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.60

**Options:** R&S CMW-KS660

**Manual operation:** See ["Access Network Type to Venue Name"](#) on page 64

<sup>1)</sup> Depending on the MNC length setting, the maximum is 999 or 99, see [CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:MNDigits](#) on page 169.

---

**CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSSPar** <DownlinkSpeed>, <UplinkSpeed>, <IPV6AddField>, <IPV4AddField>, <RealmName>, <EAPType>, <InternetAccess>, <NetAuthTypeInd>

Defines supplementary parameters of the Hotspot 2.0 operation mode.

**Parameters:**

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| <DownlinkSpeed> | numeric<br>Range: 0 kbit/s to 300000 kbit/s<br>*RST: 4000 kbit/s<br>Default unit: kbit/s |
|-----------------|------------------------------------------------------------------------------------------|

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <UplinkSpeed>    | <p>numeric</p> <p>Range: 0 kbit/s to 300000 kbit/s</p> <p>*RST: 4000 kbit/s</p> <p>Default unit: kbit/s</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <IPv6AddField>   | <p>ATNavailable   ATAVailable   AATNknown</p> <p>Indicates whether an IPv6 address can be allocated to the station</p> <p><b>ATNavailable:</b> address type not available</p> <p><b>ATAVailable:</b> address type available</p> <p><b>AATNknown:</b> availability of the address type is not known</p> <p>*RST: ATN</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <IPv4AddField>   | <p>ATNavailable   PIAavailable   PRIaavailabl   SNPiaavailab   DNPiaavailab   PSNiaavailab   PDNiaavailab   AATNknown</p> <p>Indicates whether an IPv4 address can be allocated to the station</p> <p><b>ATNavailable:</b> address type not available</p> <p><b>PIAavailable:</b> public IPv4 address available</p> <p><b>PRIaavailabl:</b> port-restricted IPv4 address available</p> <p><b>SNPiaavailab:</b> single-NATed private IPv4 address available</p> <p><b>DNPiaavailab:</b> double-NATed private IPv4 address available</p> <p><b>PSNiaavailab:</b> port-restricted and single-NATed IPv4 address available</p> <p><b>PDNiaavailab:</b> port-restricted and double-NATed IPv4 address available</p> <p><b>AATNknown:</b> availability of the address type not known</p> <p>*RST: PIA</p> |
| <RealmName>      | <p>string</p> <p>Name of reachable NAI realm as string</p> <p>To configure more than one realm, use <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:HOTSpot:REALm&lt;no&gt;</a> on page 169.</p> <p>*RST: example.org</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <EAPType>        | <p>SIM   TTLS   AKA   APRime   TLS</p> <p>Supported extensible authorization protocol type</p> <p>EAP-SIM, EAP-TTLS, EAP-AKA, EAP-AKA' or EAP-TLS</p> <p>To enable multiple EAP types, use <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:HOTSpot:REALm&lt;no&gt;</a> on page 169.</p> <p>*RST: SIM</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <InternetAccess> | <p>OFF   ON</p> <p>Specifies whether the hotspot provides internet access</p> <p>*RST: OFF</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <NetAuthTypeInd> | <p>ATConditions   OESupported   HREDirection   DREDirection</p> <p>Network authentication type</p> <p><b>ATConditions:</b> acceptance of terms and conditions</p> <p><b>OESupported:</b> on-line enrollment supported</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**HREDirection:** http/https redirection

**DREDirection:** DNS redirection

\*RST: ATC

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.70, V3.5.20: EAP type TLS added

**Options:** R&S CMW-KS660

**Manual operation:** See ["Downlink Speed to IPv4 Address Field"](#) on page 65

**CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:PLMN<no>** <State>, <MCC>, <MNC>, <NumOfDigits>

Defines a list of 3GPP networks that the hotspot provides service for.

The MCC and MNC of the first PLMN can also be defined via [CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar](#) on page 165.

**Suffix:**

<no> 1..5  
List item no

**Parameters:**

<State> OFF | ON  
Disables/enables the list entry  
\*RST: Entry 1 ON, other entries OFF

<MCC> integer  
Mobile country code  
Range: 1 to 999  
\*RST: 1

<MNC> integer  
Mobile network code  
Range: Depends on <NumOfDigits>  
\*RST: 1

<NumOfDigits> TWDigits | THDigits  
Length of the MNC  
**TWDigits:** two digits (1 to 99)  
**THDigits:** three digits (1 to 999)  
\*RST: TWD

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.5.20

**Options:** R&S CMW-KS660

**Manual operation:** See ["PLMN"](#) on page 64

**CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:MNDigits <NumOfDigits>**

Defines the length of the MNC of the first PLMN in Hotspot 2.0 operation mode.

**Parameters:**

<NumOfDigits>      TWDigits | THDigits  
                          **TWDigits:** two digits  
                          **THDigits:** three digits  
                          \*RST:        TWD

**Example:**            See [Configuring connection settings](#)

**Firmware/software:** V3.2.70

**Options:**            R&S CMW-KS660

**Manual operation:** See ["PLMN"](#) on page 64

**CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:DNAME<no> <State>, <Name>**

Defines a list of domain names of the entity operating the IEEE 802.11 access network.

The first domain name can also be defined via [CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSPar](#) on page 165.

**Suffix:**

<no>                    1..5  
                          List item no

**Parameters:**

<State>                OFF | ON  
                          Disables/enables the list entry  
                          \*RST:        Entry 1 ON, other entries OFF

<Name>                string  
                          Domain name as string

**Example:**            See [Configuring connection settings](#)

**Firmware/software:** V3.5.20

**Options:**            R&S CMW-KS660

**Manual operation:** See ["Domain Name"](#) on page 65

**CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:REALm<no> <State>, <Name>, <Sim>, <TLS>, <TTLS>, <AKA>, <AKAprime>**

Defines a list of NAI realms that are reachable via the hotspot.

The first realm can also be defined via [CONFigure:WLAN:SIGN<i>:CONNection:HOTSpot:HSSPar](#) on page 166.

**Suffix:**

<no> 1..5  
List item no

**Parameters:**

<State> OFF | ON  
Disables/enables the list entry  
\*RST: Entry 1 ON, other entries OFF

<Name> string  
Realm name as string

<Sim> OFF | ON  
Realm supports EAP-SIM  
\*RST: ON for realm 1, OFF for others

<TLS> OFF | ON  
Realm supports EAP-TLS  
\*RST: OFF

<TTLS> OFF | ON  
Realm supports EAP-TTLS  
\*RST: OFF

<AKA> OFF | ON  
Realm supports EAP-AKA  
\*RST: OFF

<AKAprime> OFF | ON  
Realm supports EAP-AKA'  
\*RST: OFF

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.5.20

**Options:** R&S CMW-KS660

**Manual operation:** See ["Realm"](#) on page 66

---

**CONFigure:WLAN:SIGN<i>:CONNECTION:HOTSpot:CUTil** <StationCount>,  
<ChannelUtilization>, <AvailableAdmissionCapacity>

Configures the contents of the BSS load element.

**Parameters:**

<StationCount> numeric  
Number of stations that are currently associated with the BSS  
Range: 0 to 65535  
\*RST: 1

<ChannelUtilization> numeric

Percentage of time, that the access point sensed the primary channel was busy

Range: 0 % to 100 %

\*RST: 0 %

Default unit: %

<AvailAdmCapacity> numeric

Remaining time available via explicit admission control, in units of 32 µs/s

Range: 0 to 31250

\*RST: 31250

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.5.40

**Manual operation:** See "[Station Count, Channel Utilization, Available Admission Capacity](#)" on page 66

#### 2.5.11.4 Target wake time settings

The following commands configure 802.11ax-specific broadcast target wake time (TWT) operation.

|                                                                                           |     |
|-------------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BTWT:ENABLE</a> .....                 | 171 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BTWT:SCHechedule:ENABLE</a> .....     | 171 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BTWT:SCHechedule:FTYPE</a> .....      | 172 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BTWT:SCHechedule:MWDuration</a> ..... | 172 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BTWT:SCHechedule:STIME</a> .....      | 173 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:BTWT:SCHechedule:TENable</a> .....    | 173 |

---

##### **CONFigure:WLAN:SIGN<i>:CONNection:BTWT:ENABLE <Enable>**

Enables/ disables broadcast target wake time (TWT) operation.

##### **Parameters:**

<Enable> OFF | ON

\*RST: OFF

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KS657

**Manual operation:** See "[Enable < Broadcast TWT](#)" on page 67

---

##### **CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHechedule:ENABLE <FlowID>, <Enable>**

##### **CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHechedule:ENABLE? <FlowID>**

Enables/ disables particular schedule period.

**Parameters:**

<Enable> OFF | ON  
 \*RST: OFF

**Parameters for setting and query:**

<FlowID> integer

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KS657

**Manual operation:** See ["Enable < TWT Flow ID x" on page 67](#)

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHedule:FTYPE** <FlowID>,  
 <FlowType>

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHedule:FTYPE?** <FlowID>

Specifies the broadcast TWT flow type for the specified schedule period.

**Parameters:**

<FlowType> ANNounced | UNANNounced  
 \*RST: ANN

**Parameters for setting and query:**

<FlowID> integer

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KS657

**Manual operation:** See ["Flow Type" on page 67](#)

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHedule:MWDuration** <FlowID>,  
 <MinWakeDuration>

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHedule:MWDuration?**  
 <FlowID>

Specifies the minimum wake duration for the specified scheduled period.

**Parameters:**

<MinWakeDuration> numeric  
 Range: 0 ms to 100 ms  
 \*RST: 10 ms

**Parameters for setting and query:**

<FlowID> integer

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.70

**Options:** R&S CMW-KS657



**Manual operation:** See ["Minimum Wake Duration"](#) on page 68

---

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeDule:STIME** <FlowID>, <StartTime>

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeDule:STIME?** <FlowID>

Specifies the offset of the specified schedule period from beacon.

**Parameters:**

<StartTime>                      numeric  
                                          Range:        0 ms to 100 ms  
                                          \*RST:        0 ms

**Parameters for setting and query:**

<FlowID>                      integer

**Example:**                      See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40

**Options:**                      R&S CMW-KS657

**Manual operation:** See ["Start Offset"](#) on page 68

---

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeDule:TENable** <FlowID>, <Enable>

**CONFigure:WLAN:SIGN<i>:CONNection:BTWT:SCHeDule:TENable?** <FlowID>

Enables/disables the broadcast TWT trigger for the specified schedule period.

**Parameters:**

<Enable>                      OFF | ON  
                                          \*RST:        OFF

**Parameters for setting and query:**

<FlowID>                      integer

**Example:**                      See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40

**Options:**                      R&S CMW-KS657

**Manual operation:** See ["Trigger Enabled"](#) on page 68

### 2.5.11.5 Management and data frame control settings

The following commands configure the management and data frame control of all supported standards.

|                                                                                                      |     |
|------------------------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:MFDDef</a> .....                                 | 174 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:STA&lt;s&gt;:CONNection:DFDef</a> .....                     | 175 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:STA&lt;s&gt;:DFRame:HEMU:ALSField</a> .....      | 176 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:STA&lt;s&gt;:DFRame:HEMU:BLALlocation?</a> ..... | 177 |

|                                                                                      |     |
|--------------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:DUMMy<index>:MCS....            | 178 |
| CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:RUAllocation.....               | 178 |
| CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:<br>AlLocation..... | 179 |
| CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:CTYPe....           | 179 |
| CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:MCS.....            | 180 |
| CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:STReams             | 180 |

---

### CONFigure:WLAN:SIGN<i>:CONNection:MFDef <State>, <Format>, <Rate>

Enables and configures the user-defined frame rate control for management frames.

#### Parameters:

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <State>  | DISable   ENABle<br>Disables/enables the user-defined frame rate control<br>*RST:       DIS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <Format> | NHT   HTM   VHT   HES   HEM<br>Selects the frame format<br><b>NHT</b> : non-high throughput format (non-HT)<br><b>HTM</b> : HT mixed format (HT MF)<br><b>VHT</b> : very high throughput format<br><b>HES</b> : high efficiency single-user format (HE SU)<br><b>HEM</b> : high efficiency multi-user format (HE MU)                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <Rate>   | D1MBit   D2Mbits   C55Mbits   C11Mbits   BR12   BR34   QR12  <br>QR34   Q1M12   Q1M34   Q6M23   Q6M34   MCS   MCS1  <br>MCS2   MCS3   MCS4   MCS5   MCS6   MCS7   MCS8   MCS9  <br>MCS10   MCS11   MCS12   MCS13   MCS14   MCS15<br>Sets the rate<br><b>D1MBit</b> : DSSS, 1 Mbit/s<br><b>D2Mbits</b> : DSSS, 2 Mbit/s<br><b>C55Mbits</b> : CCK, 5.5 Mbit/s<br><b>C11Mbits</b> : CCK, 11 Mbit/s<br><b>BR12</b> : BPSK, 1/2, 6 Mbit/s<br><b>BR34</b> : BPSK, 3/4, 9 Mbit/s<br><b>QR12</b> : QPSK, 1/2, 12 Mbit/s<br><b>QR34</b> : QPSK, 3/4, 18 Mbit/s<br><b>Q1M12</b> : 16-QAM, 1/2, 24 Mbit/s<br><b>Q1M34</b> : 16-QAM, 3/4, 36 Mbit/s<br><b>Q6M23</b> : 64-QAM, 2/3, 48 Mbit/s<br><b>Q6M34</b> : 64-QAM, 3/4, 54 Mbit/s<br><b>MCS, MCS1,...,MCS15</b> : MCS 0 to MCS 15<br>*RST:       BR12 |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.5.120, V3.7.30: added HE,  
V3.7.60: removed <Format> HTG

**Options:** R&S CMW-KS651 for 802.11n (HT)  
R&S CMW-KS656 for 802.11ac (VHT)

R&S CMW-KS657 for 802.11ax (HE)

**Manual operation:** See ["Configuration"](#) on page 71

---

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:DFDef** <State>, <Format>, <ChanBW>, <Rate>[, <GuardInterval>, <LTFTType>, <PEDuration>, <CTYPE>, <Streams>, <STBC>]

Enables and configures the user-defined frame rate control for data frames.

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<State> DISable | ENABle  
Disables/enables the user-defined frame rate control  
\*RST: DIS

<Format> NHT | HTM | VHT | HES | HEM  
Selects the frame format  
**NHT:** non-high throughput format (non-HT)  
**HTM:** HT mixed format (HT MF)  
**VHT:** very high throughput format  
**HES:** high efficiency single-user format (HE SU)  
**HEM:** high efficiency multi-user format (HE MU)

<ChanBW> BW20 | BW40 | BW80 | BW160  
Channel bandwidth  
The value must not exceed the operating channel bandwidth, see [CONFigure:WLAN:SIGN<i>:RFSettings:OCWidth](#) on page 148.  
\*RST: BW20

<Rate> D1MBit | D2Mbits | C55Mbits | C11Mbits | BR12 | BR34 | QR12 | QR34 | Q1M12 | Q1M34 | Q6M23 | Q6M34 | MCS | MCS1 | MCS2 | MCS3 | MCS4 | MCS5 | MCS6 | MCS7 | MCS8 | MCS9 | MCS10 | MCS11 | MCS12 | MCS13 | MCS14 | MCS15  
See rate list in [CONFigure:WLAN:SIGN<i>:CONNection:MFDef](#)  
\*RST: BR12

<GuardInterval> LONG | SHORT | GI08 | GI16 | GI32  
**SHORT, LONG:** short or long guard interval (up to 802.11ac)  
**GI08, GI16, GI32:** 0.8 µs, 1.6 µs, and 3.2 µs guard interval durations (for 802.11ax)

<LTFTType> X1 | X2 | X4  
1x HE-LTF, 2x HE-LTF, 4x HE-LTF for 802.11ax

|                           |                                                                                                                                                                                                                                                                                               |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <PEDuration>              | PE0   PE4   PE8   PE12   PE16   AUTO<br><b>PEx</b> : additional receive processing time of x $\mu$ s signaled in packet extension (PE) field (only for 802.11ax)<br><b>AUTO</b> : automatic setting based on the reported DUTs capabilities                                                   |
| <CTYPE>                   | LDPC   BCC<br>Coding type (for 802.11ax - VHT, HE_SU, HE_MU frames only): low density parity check or binary convolution code<br>*RST: LDPC                                                                                                                                                   |
| <Streams>                 | STR1   STR2<br>Number of streams<br>*RST: STR1                                                                                                                                                                                                                                                |
| <STBC>                    | OFF   ON<br>Enables / disables space time block coding (STBC). If disabled, spatial multiplexing is used.<br>*RST: OFF                                                                                                                                                                        |
| <b>Example:</b>           | See <a href="#">Configuring connection settings</a>                                                                                                                                                                                                                                           |
| <b>Firmware/software:</b> | V3.5.120, V3.7.30: added <Format> for HE, <ChanBW> BW160, <GuardInterval> GI08, GI16, GI32, <LTFTType>, <PEDuration>, <CTYPE>, V3.7.40: added <Streams>, V3.7.60: removed <Format> HTG, added <STBC> V3.7.70: added support of LDPC <CTYPE> also for VHT and HE_MU frames, V3.8.20: added <s> |
| <b>Options:</b>           | R&S CMW-KS651 for 802.11n (HT)<br>R&S CMW-KS656 for 802.11ac (VHT)<br>R&S CMW-KS657 for 802.11ax (HE)<br>R&S CMW-KS667 for :STA2: and :STA3:                                                                                                                                                  |
| <b>Manual operation:</b>  | See <a href="#">"Configuration"</a> on page 72                                                                                                                                                                                                                                                |

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:ALSField**  
<CH20\_Index>, <Subfield>

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:ALSField?**  
<CH20\_Index>

Sets 8-bit indices for resource unit (RU) allocation for the selected channel. The <Subfield> parameter specifies the number of RUs, their position and size. Refer to IEEE Std 802.11ax-2021, table 27-26, RU allocation subfield.

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<Subfield> A000 | A001 | A002 | A003 | A004 | A005 | A006 | A007 | A008 |  
 A009 | A010 | A011 | A012 | A013 | A014 | A015 | A016 | A024 |  
 A032 | A040 | A048 | A056 | A064 | A072 | A080 | A088 | A096 |  
 A112 | A113 | A114 | A115 | A116 | A120 | A128 | A192 | A200 |  
 A208 | A216 | A224  
 \*RST: A192

**Parameters for setting and query:**

<CH20\_Index> CHA1 | CHA2 | CHA3 | CHA4

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
 R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["20 MHz Channel Settings \(HE\\_MU frame only\)"](#)  
 on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:BLALlocation?**  
 <CH20\_Index>

Queries the allocation state and size of an entire 20MHz block for the specified channel.

**Suffix:**

<s> 1..3  
 Station No.

**Query parameters:**

<CH20\_Index> CHA1 | CHA2 | CHA3 | CHA4

**Return values:**

{<1\_RU1> OFF | USR1 | DMY1 | DMY2 | DMY3  
 User mapping for to the corresponding RU (no user, user 1 or dummy user)  
 <2\_Size1>}. T26 | T52 | T106 | T242 | T484 | T996 | T2X9  
 RU size: 26-, 52-, 106-, 242-, 484, 996-tone RU, 2x996-tone RU  
 ..{<17\_RU9> OFF | USR1 | DMY1 | DMY2 | DMY3  
 <18\_Size9>}. T26 | T52 | T106 | T242 | T484 | T996 | T2X9

**Example:** See [802.11ax connection settings](#)

**Usage:** Query only

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
 R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["20 MHz Channel Settings \(HE\\_MU frame only\)"](#)  
 on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:  
DUMMy<index>:MCS <MCS\_Index>**

Sets the modulation and coding scheme for the corresponding dummy user.

**Suffix:**

|         |                              |
|---------|------------------------------|
| <index> | 1..3<br>Dummy user selection |
| <s>     | 1..3<br>Station No.          |

**Parameters:**

|             |                                                                                                                                                       |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MCS_Index> | MCS   MCS1   MCS2   MCS3   MCS4   MCS5   MCS6   MCS7   MCS8   MCS9   MCS10   MCS11<br><b>MCS, MCS1,...,MCS11:</b> MCS 0 to MCS 11<br><b>*RST:</b> MCS |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:RUAllocation  
<CH20\_Index>, <RU\_Index>, <AllocState>**
**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:RUAllocation?  
<CH20\_Index>, <RU\_Index>**

Configures allocations for specified channel and resource unit (RU). Maps a user to the RU, sets the size of allocation.

**Suffix:**

|     |                     |
|-----|---------------------|
| <s> | 1..3<br>Station No. |
|-----|---------------------|

**Parameters:**

|              |                                                                        |
|--------------|------------------------------------------------------------------------|
| <AllocState> | OFF   USR1   DMY1   DMY2   DMY3<br>User mapping for to the selected RU |
|--------------|------------------------------------------------------------------------|

**Parameters for setting and query:**

|              |                                                                                |
|--------------|--------------------------------------------------------------------------------|
| <CH20_Index> | CHA1   CHA2   CHA3   CHA4                                                      |
| <RU_Index>   | RU1   RU2   RU3   RU4   RU5   RU6   RU7   RU8   RU9<br>Resource unit selection |

**Return values:**

|        |                                                                                                                                     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------|
| <Size> | T26   T52   T106   T242   T484   T996   T2X9<br>RU size: 26-, 52-, 106-, 242-, 484, 996-tone RU, 2x996-tone RU<br><b>*RST:</b> T242 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------|

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.7.50: changed <Size> default value  
V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:  
ALLocation** <CH20\_Index>, <RU\_Index>

Configures allocations for the user in HE MU PPDU. Maps the user to a resource unit (RU).

**Suffix:**

|         |                     |
|---------|---------------------|
| <index> | 1<br>User           |
| <s>     | 1..3<br>Station No. |

**Parameters:**

|              |                                                                  |
|--------------|------------------------------------------------------------------|
| <CH20_Index> | CHA1   CHA2   CHA3   CHA4<br>*RST: CHA1                          |
| <RU_Index>   | RU1   RU2   RU3   RU4   RU5   RU6   RU7   RU8   RU9<br>*RST: RU1 |

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:  
CTYPe** <CodingType>

Selects the coding type related to a user for HE MU PPDU.

**Suffix:**

|         |                     |
|---------|---------------------|
| <index> | 1<br>User           |
| <s>     | 1..3<br>Station No. |

**Parameters:**

|              |                          |
|--------------|--------------------------|
| <CodingType> | LDPC   BCC<br>*RST: LDPC |
|--------------|--------------------------|

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:  
MCS <MCS\_Index>**

Sets the modulation and coding scheme for user 1 (user data assigned to the DUT).

**Suffix:**

|         |                     |
|---------|---------------------|
| <index> | 1<br>User           |
| <s>     | 1..3<br>Station No. |

**Parameters:**

|             |                                                                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MCS_Index> | MCS   MCS1   MCS2   MCS3   MCS4   MCS5   MCS6   MCS7  <br>MCS8   MCS9   MCS10   MCS11<br><b>MCS, MCS1,...,MCS11:</b> MCS 0, MCS 1 to MCS 11<br>*RST: MCS |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

---

**CONFigure:WLAN:SIGN<i>:CONNection:STA<s>:DFRame:HEMU:USER<index>:  
STReams <streams>**

Sets the number of streams used by the user for MIMO connections.

**Suffix:**

|         |                     |
|---------|---------------------|
| <index> | 1<br>User           |
| <s>     | 1..3<br>Station No. |

**Parameters:**

|           |                                                 |
|-----------|-------------------------------------------------|
| <streams> | STR1   STR2<br>One or two streams<br>*RST: STR1 |
|-----------|-------------------------------------------------|

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40, V3.8.20: added <s>



**Options:** R&S CMW-KS657  
 R&S CMW-KS670  
 R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Users and Dummies \(HE\\_MU frame only\)"](#) on page 74

### 2.5.11.6 Trigger frame control settings

The following commands configure the trigger frame control of all supported standards.

|                                                          |     |
|----------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:HETF:APTpower?         | 181 |
| CONFigure:WLAN:SIGN<i>:CONNection:HETF:CHBW              | 182 |
| CONFigure:WLAN:SIGN<i>:CONNection:HETF:CSR               | 182 |
| CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:CTYP       | 182 |
| CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:DCM        | 183 |
| CONFigure:WLAN:SIGN<i>:CONNection:HETF:GILT              | 183 |
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---

#### CONFigure:WLAN:SIGN<i>:CONNection:HETF:APTpower?

Specifies the value of "AP TX Power" the R&S CMW signals via a trigger frame.

##### Return values:

<int\_value> decimal  
 Range: 0 to 60

<dBm\_value> decimal  
 Range: -20 dBm to 40 dBm  
 Default unit: dBm

**Example:** See [802.11ax connection settings](#)

**Usage:** Query only

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KS657

**Manual operation:** See ["AP Tx Power"](#) on page 78

---

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:CHBW <Bandwidth>**

Specifies the channel bandwidth of the HE TB PPDU.

**Parameters:**

<Bandwidth> BW20 | BW40 | BW80 | BW160  
\*RST: BW20

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KS657

**Manual operation:** See ["Bandwidth"](#) on page 77

---

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:CSR <Required>**

Specifies, whether the check of medium status is required before responding.

**Parameters:**

<Required> OFF | ON  
\*RST: OFF

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.7.50: reset value changed

**Options:** R&S CMW-KS657

**Manual operation:** See ["CS Required"](#) on page 77

---

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:CTYP <Type>**

Specifies the coding used by the HE TB PPDU.

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<Type> BCC | LDPC  
\*RST: LDPC

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>,  
V4.0.20: changed reset value

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Coding Type"](#) on page 78

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:DCM <DCM>**

Specifies whether the HE TB response uses dual carrier modulation (DCM).

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<DCM> OFF | ON  
\*RST: OFF

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["DCM"](#) on page 78

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:GILT <GI\_LTF>**

Specifies the guard interval and LTF type of the HE TB PPDU.

For [CONFigure:WLAN:SIGN<i>:CONNection:HETF:MLTF MASK](#), the value "1x HE-LTF + 1.6 µs GI" is not supported.

**Parameters:**

<GI\_LTF> L116 | L216 | L432  
LTF type and corresponding GI:  
**L116:** 1x HE-LTF + 1.6 µs GI  
**L216:** 2x HE-LTF + 1.6 µs GI  
**L432:** 4x HE-LTF + 3.2 µs GI  
\*RST: L432

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["GI and LTF"](#) on page 77

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:LDPC <Extra\_symbol>**

Specifies the support of LDPC extra symbol segment.

**Parameters:**

<Extra\_symbol> OFF | ON  
\*RST: OFF

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["LDPC Extra Symbol"](#) on page 77

---

#### **CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:MCS <MCS>**

Specifies the modulation and coding scheme (MCS) used by the HE TB PPDU.

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<MCS> MCS | MCS1 | MCS2 | MCS3 | MCS4 | MCS5 | MCS6 | MCS7 |  
MCS8 | MCS9 | MCS10 | MCS11  
**MCS, MCS1,...,MCS11:** MCS 0 to MCS 11  
**\*RST:** MCS

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["MCS"](#) on page 78

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:HETF:MLTF <MU\_MIMO\_LTF>**

Sets MU-MIMO long training fields (LTF).

**Parameters:**

<MU\_MIMO\_LTF> SING | MASK  
**SING:** single stream pilots  
**MASK:** mask LTF sequence of each spatial stream by a distinct  
orthogonal code  
**\*RST:** SING

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["MU-MIMO LTF"](#) on page 77

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:HETF:NOFSymbols <NumOfSymbols>**

Specifies the length of the HE TB PPDU.

**Parameters:**

<NumOfSymbols> integer  
 Range: 1 to 330  
 \*RST: 100  
 Default unit: symbol

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40, V3.7.50: reset value changed

**Options:** R&S CMW-KS657

**Manual operation:** See ["Number of Symbols"](#) on page 77

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:NSS <Number\_SS>**

Sets the number of HE TB PPDU spatial streams.

**Suffix:**

<s> 1..3  
 Station No.

**Parameters:**

<Number\_SS> integer  
 Range: 1 to 8  
 \*RST: 1

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
 R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Number of Spatial Streams"](#) on page 78

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:RUAL <RU\_allocation>[, <channel\_80MHz>]**

Specifies the RU used by the HE TB PPDU. Refer to IEEE Std 802.11ax-2021, table 9-29i B7–B1 of the RU Allocation subfield.

**Suffix:**

<s> 1..3  
 Station No.

**Parameters:**

<RU\_allocation> RU0 | RU1 | RU2 | RU3 | RU4 | RU5 | RU6 | RU7 | RU8 | RU9 |  
 RU10 | RU11 | RU12 | RU13 | RU14 | RU15 | RU16 | RU17 |  
 RU18 | RU19 | RU20 | RU21 | RU22 | RU23 | RU24 | RU25 |  
 RU26 | RU27 | RU28 | RU29 | RU30 | RU31 | RU32 | RU33 |  
 RU34 | RU35 | RU36 | RU37 | RU38 | RU39 | RU40 | RU41 |  
 RU42 | RU43 | RU44 | RU45 | RU46 | RU47 | RU48 | RU49 |  
 RU50 | RU51 | RU52 | RU53 | RU54 | RU55 | RU56 | RU57 |  
 RU58 | RU59 | RU60 | RU61 | RU62 | RU63 | RU64 | RU65 |  
 RU66 | RU67 | RU68 | OFF

**"OFF"**: No resource unit for the specified station is allocated

**"RUx"**: Bits 7 to 1 of the RU allocation subfield, see table below.

<Channel\_80MHz> PRIMary | SECondary

For RU67 and RU68 applying 160 MHz channel, sets the bit 0 of the RU allocation subfield that indicates primary 80 MHz or secondary 80 MHz channel.

**Example:**

See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

V3.8.20: added <s>, <Channel\_80MHz> and

<RU\_allocation> OFF

**Options:**

R&S CMW-KS657

R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["RU Allocation"](#) on page 78

**Table 2-9: The encoding of B7–B1 of the RU allocation subfield**

| RU size     | 20 MHz |   |    |   |   |   |    |   | 20 MHz |   |    |   |    |   |   |   |    | 20 MHz |    |   |    |    |   |    |   | 20 MHz |   |    |   |    |   |    |   |    |   |   |   |    |  |    |  |  |  |  |  |    |  |  |  |
|-------------|--------|---|----|---|---|---|----|---|--------|---|----|---|----|---|---|---|----|--------|----|---|----|----|---|----|---|--------|---|----|---|----|---|----|---|----|---|---|---|----|--|----|--|--|--|--|--|----|--|--|--|
| 26-tone     | 0      | 1 | 2  | 3 | 4 | 5 | 6  | 7 | 8      | 9 | 1  | 1 | 1  | 1 | 1 | 1 | 1  | 1      | 1  | 2 | 2  | 2  | 2 | 2  | 2 | 2      | 2 | 2  | 3 | 3  | 3 | 3  | 3 | 3  |   |   |   |    |  |    |  |  |  |  |  |    |  |  |  |
|             |        |   |    |   |   |   |    |   |        |   | 0  | 1 | 2  | 3 | 4 | 5 | 6  | 7      | 8  |   | 9  | 0  | 1 | 2  | 3 | 4      | 5 | 6  | 7 | 8  | 9 | 0  | 1 | 2  | 3 | 4 | 5 | 6  |  |    |  |  |  |  |  |    |  |  |  |
| 52-tone     | 37     |   | 38 |   |   |   | 39 |   | 40     |   | 41 |   | 42 |   |   |   | 43 |        | 44 |   |    | 45 |   | 46 |   |        |   | 47 |   | 48 |   | 49 |   | 50 |   |   |   | 51 |  | 52 |  |  |  |  |  |    |  |  |  |
| 106-tone    | 53     |   |    |   |   |   |    |   | 54     |   |    |   | 55 |   |   |   |    |        |    |   | 56 |    |   |    |   | 57     |   |    |   |    |   |    |   | 58 |   |   |   | 59 |  |    |  |  |  |  |  | 60 |  |  |  |
| 242-tone    | 61     |   |    |   |   |   |    |   | 62     |   |    |   |    |   |   |   |    | 63     |    |   |    |    |   |    |   | 64     |   |    |   |    |   |    |   |    |   |   |   |    |  |    |  |  |  |  |  |    |  |  |  |
| 484-tone    | 65     |   |    |   |   |   |    |   |        |   |    |   |    |   |   |   |    | 66     |    |   |    |    |   |    |   |        |   |    |   |    |   |    |   |    |   |   |   |    |  |    |  |  |  |  |  |    |  |  |  |
| 996-tone    | 67     |   |    |   |   |   |    |   |        |   |    |   |    |   |   |   |    |        |    |   |    |    |   |    |   |        |   |    |   |    |   |    |   |    |   |   |   |    |  |    |  |  |  |  |  |    |  |  |  |
| 2x 996-tone | 68     |   |    |   |   |   |    |   |        |   |    |   |    |   |   |   |    |        |    |   |    |    |   |    |   |        |   |    |   |    |   |    |   |    |   |   |   |    |  |    |  |  |  |  |  |    |  |  |  |

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNEction:HETF:SSS?**

Sets the starting spatial stream for the HE TB PPDU.

**Suffix:**

<s> 1..3  
 Station No.

**Return values:**

<Starting\_SS> decimal  
Range: 1 to 8

**Example:** See [802.11ax connection settings](#)

**Usage:** Query only

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Starting Spatial Stream"](#) on page 78

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:SSTX**

Transmits the trigger frame once.

**Example:** See [802.11ax connection settings](#)

**Usage:** Event

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["Trigger Frame Transmission"](#) on page 79

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:TRSMoDe <Mode>**

Specifies the trigger frame power control mode.

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<Mode> AUTO | MANual | MAXPower  
**AUTO:** AP\_TX\_Power and Target\_RSSI calculated automatically  
**MAN:** The value Target RSSI Control defines adjustment to the Target\_RSSI calculation  
**MAXP:** Sets the Target\_RSSI to 127, the UE transmits the HE TB PPDU at maximum Tx power  
 \*RST: AUTO

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Target RSSI"](#) on page 79

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:TRSSi?**

Specifies the expected Rx power of HE TB PPDU transmission as a response to trigger frame.

**Suffix:**

<s> 1..3  
Station No.

**Return values:**

<int\_value> decimal  
Target\_RSSI index  
**0 to 90:** map to -110 dBm to -20 dBm  
**91-126:** reserved  
**127:** station is commanded to transmit at maximum power for the assigned MCS  
Range: 0 to 127

<dBm\_value> decimal  
Target\_RSSI value  
Range: -110 dBm to -20 dBm

**Example:** See [802.11ax connection settings](#)

**Usage:** Query only

**Firmware/software:** V3.7.40, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Target RSSI"](#) on page 79

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:TSRControl <pwr\_dB>**

Specifies the value Target RSSI Control for adjustment to the Target\_RSSI. This parameter is only relevant in manual mode for target RSSI calculation.

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<pwr\_dB> integer  
Range: -40 dB to 0 dB  
\*RST: 0 dB

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.40, V3.7.50: range extended, V3.8.20: added <s>

**Options:** R&S CMW-KS657  
R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Target RSSI"](#) on page 79



---

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:TTYP <Type>**

Specifies the trigger type as specified in the Common Info field.

**Parameters:**

<Type>                    BTR | BRP | MRTS | BSRP | BQRP  
**BTR:** Basic Trigger  
**BRP:** Beamforming Report Poll  
**MRTS:** MU-RTS  
**BSRP:** Buffer Status Report Poll  
**BQRP:** Bandwidth Query Report Poll  
 \*RST:            BTR

**Example:**            See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

**Options:**            R&S CMW-KS657

**Manual operation:** See ["Trigger Type"](#) on page 76

---

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:TXEN <State>**

Enables/ disables the periodical trigger frame.

**Parameters:**

<State>                    OFF | ON  
 \*RST:            OFF

**Example:**            See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

**Options:**            R&S CMW-KS657

**Manual operation:** See ["Trigger Frame Transmission"](#) on page 79

---

**CONFigure:WLAN:SIGN<i>:CONNection:HETF:TXP <Interval>**

Sets the interval for periodical trigger frame.

**Parameters:**

<Interval>                integer  
                               Range:        1 to 10E+3  
                               \*RST:        100  
                               Default unit: ms

**Example:**            See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30

**Options:**            R&S CMW-KS657

**Manual operation:** See ["Trigger Frame Transmission"](#) on page 79

### 2.5.11.7 Security settings

The following commands configure connection settings for WLAN security.

|                                                                 |     |
|-----------------------------------------------------------------|-----|
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---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:EAKA:KALGo** <Ki>, <Opc>, <Rand>, <Algorithm>

Configures EAP-AKA on the internal RADIUS server.

**Parameters:**

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| <Ki>        | string<br>Secret key as string with 32 hexadecimal digits<br>*RST: 000102030405060708090a0b0c0d0e0f           |
| <Opc>       | string<br>Operator variant key as string with 32 hexadecimal digits<br>*RST: 00000000000000000000000000000000 |
| <Rand>      | string<br>Random number as string with 32 hexadecimal digits<br>*RST: 00000000000000000000000000000000        |
| <Algorithm> | MILenage   XOR<br>Authentication algorithm to be used<br>*RST: XOR                                            |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.60

**Options:** R&S CMW-KS660

**Manual operation:** See "EAP-AKA" on page 82

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ENCryption** <Encryption type>

Sets the encryption type for AP operation mode, if WPA, WPA2, or WPA3 personal security mode is selected.

**Parameters:**

<Encryption type> AES | TKIP | DISabled | GCMP  
**AES:** AES basd CCMP-128 with PSK (for WPA2)  
**TKIP:** TKIP with PSK (for WPA)  
**DISabled:** encryption not used  
**GCMP:** CCMP-128 with SAE and PMF (WPA3)

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.10  
 V4.0.20: added GCMP

**Manual operation:** See ["Encryption Type"](#) on page 80

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ESIM:KTONE** <Rand>, <Sres>, <Kc>

Defines the first triplet for EAP-SIM authentication (internal RADIUS server).

**Parameters:**

<Rand> string  
 Random challenge as string with 32 hexadecimal digits  
 \*RST: 55555555555555555555555555555555  
 <Sres> string  
 Signed response as string with 8 hexadecimal digits  
 \*RST: 55545756  
 <Kc> string  
 Cipherring key as string with 16 hexadecimal digits  
 \*RST: 0000000000000000

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.60

**Options:** R&S CMW-KS660

**Manual operation:** See ["EAP-SIM"](#) on page 81

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ESIM:KTTHree** <Rand>, <Sres>, <Kc>

Defines the third triplet for EAP-SIM authentication (internal RADIUS server).

**Parameters:**

|        |                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------|
| <Rand> | string<br>Random challenge as string with 32 hexadecimal digits<br>*RST: 33333333333333333333333333333333 |
| <Sres> | string<br>Signed response as string with 8 hexadecimal digits<br>*RST: 33323130                           |
| <Kc>   | string<br>Cipherring key as string with 16 hexadecimal digits<br>*RST: 0000000000000000                   |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.60

**Options:** R&S CMW-KS660

**Manual operation:** See ["EAP-SIM"](#) on page 81

---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:ESIM:KTTWo** <Rand>, <Sres>, <Kc>

Defines the second triplet for EAP-SIM authentication (internal RADIUS server).

**Parameters:**

|        |                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------|
| <Rand> | string<br>Random challenge as string with 32 hexadecimal digits<br>*RST: 44444444444444444444444444444444 |
| <Sres> | string<br>Signed response as string with 8 hexadecimal digits<br>*RST: 44454647                           |
| <Kc>   | string<br>Cipherring key as string with 16 hexadecimal digits<br>*RST: 0000000000000000                   |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.60

**Options:** R&S CMW-KS660

**Manual operation:** See ["EAP-SIM"](#) on page 81

---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:GTRansform**  
<group\_transform>

Specifies the group transform for WPA3 personal security mode.

**Parameters:**

<group\_transform> ECP256 | ECP384  
256-bit ECP or 384-bit ECP

**Firmware/software:** V4.0.20

**Manual operation:** See ["Common Settings > Group Transform"](#) on page 81

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:HASHtoelem <mode>**

Selects authentication and key management mechanism supported by the R&S CMW:

**Parameters:**

<mode> HUNT | H2E | BOTH  
**HUNT:** hunting-and-pecking negotiated in SAE exchange  
**H2E:** SAE hash-to-element is mandatory for WPA3 and for SAE in 6 GHz band  
**BOTH:** the R&S CMW supports both techniques  
\*RST: BOTH

**Manual operation:** See ["SAE H2E Support"](#) on page 80

V4.0.20

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:PASSphrase <SecurityType>[, <Passphrase>]**

Selects the WLAN security mechanism to be used and defines the passphrase for WPA/WPA2/WPA3 personal.

For supported values depending on operation mode, see [Table 2-3](#).

**Parameters:**

<SecurityType> DISabled | AUTO | WPERsonal | WENTERprise | W2Personal | W2ENTERprise | OWE | W3Personal | W3ENTERprise  
**DISabled:** no security (only for the 2.4 and 5 GHz bands)  
**AUTO:** automatic selection of any supported security type  
**WPERsonal:** WPA personal  
**WENTERprise:** WPA enterprise  
**W2Personal:** WPA2 personal  
**W2ENTERprise:** WPA2 enterprise  
**OWE:** opportunistic wireless encryption with protected management frames (PMF)  
**W3Personal:** WPA3 personal  
**W3ENTERprise:** WPA3 enterprise  
\*RST: DIS  
  
<Passphrase> string  
Passphrase for AP operation mode as a string, 1 to 63 characters  
\*RST: "12345678"

- Example:** See [Configuring connection settings](#)
- Firmware/software:** V3.7.10, V3.7.20: <Passphrase> length extended,  
V4.0.20: added <SecurityType> OWE, W3Personal, W3Enterprise
- Manual operation:** See "[Common Settings > Passphrase](#)" on page 81

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:PKEY** <key\_mode>[,  
<private\_key>]

Sets the private security key for WPA3 personal as a string, 1 to 63 characters.

**Parameters:**

<key\_mode>               RANDOM | FIXEd  
**RAND:** private key assigned automatically  
**FIX:** private key assigned manually via <private\_key>  
 \*RST:               RAND

<private\_key>           string  
 \*RST:               123.456789012346E+60

**Manual operation:** See "[Common Settings > Passphrase](#)" on page 81

V4.0.20

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:PMF** <protection>

Selects, whether the protection management frames are unsupported, supported or required.

This parameter applies to WPA2, WPA3 in AP operation mode.

**Parameters:**

<protection>           UNSupported | SUPPorted | REQuired  
 \*RST:               UNS

**Firmware/software:** V4.0.20

**Manual operation:** See "[Protection Management Frame](#)" on page 80

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:ICONf** <IPfirstPart>,  
<IPsecondPart>, <IPthirdPart>, <IPfourthPart>

Sets the IPv4 address of an external RADIUS server.

**Parameters:**

<B1>,...,<B4>           integer  
 Parts of the address <B1> . <B2> . <B3> . <B4>  
 Range:           0 to 255 (each octet)  
 \*RST:           172, 22, 1, 222

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.10

**Options:** R&S CMW-KS660

**Manual operation:** See ["IP Address / Port Number / Shared Key"](#) on page 82

---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:MODE <Mode>**

Selects the RADIUS server mode for WPA/WPA2 enterprise.

**Parameters:**

<Mode> INTernal | EXTernal

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.70

**Options:** R&S CMW-KS660

**Manual operation:** See ["Mode"](#) on page 81

---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:PNUMber <number>**

Sets the UDP port number of an external RADIUS server.

**Parameters:**

<number> integer

Range: 1 to 65535

\*RST: 1812

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.10

**Manual operation:** See ["IP Address / Port Number / Shared Key"](#) on page 82

---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:RSERver:SKEY <string>**

Sets the shared key of an external RADIUS server.

**Parameters:**

<string> string

Shared key as string

\*RST: test12345

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.2.10

**Manual operation:** See ["IP Address / Port Number / Shared Key"](#) on page 82

---

**CONFigure:WLAN:SIGN<i>:CONNection:SECurity:TYPE** <SecurityType>,  
<endPart>

Selects the WLAN security mechanism to be used and defines the last character of the passphrase for WPA/WPA2/WPA3 personal.

For supported values depending on operation mode, see [Table 2-3](#).

**Parameters:**

|                             |                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------|
| <b>&lt;SecurityType&gt;</b> | DISabled   AUTO   WPERsonal   WENTerprise   W2Personal   W2ENTERprise   OWE   W3Personal   W3ENTERprise |
|                             | <b>DISabled:</b> no security (only for the 2.4 GHz and 5 GHz bands)                                     |
|                             | <b>AUTO:</b> automatic selection of any supported security type (station mode only)                     |
|                             | <b>WPERsonal:</b> WPA personal (only for the 2.4 GHz and 5 GHz bands)                                   |
|                             | <b>WENTerprise:</b> WPA enterprise (only for the 2.4 GHz and 5 GHz bands)                               |
|                             | <b>W2Personal:</b> WPA2 personal (only for the 2.4 GHz and 5 GHz bands)                                 |
|                             | <b>W2ENTERprise:</b> WPA2 enterprise (only for the 2.4 GHz and 5 GHz bands)                             |
|                             | <b>OWE:</b> opportunistic wireless encryption (only for 6 GHz band)                                     |
|                             | <b>W3Personal:</b> WPA3 personal (all bands)                                                            |
|                             | <b>W3ENTERprise:</b> WPA3 enterprise (all bands)                                                        |
|                             | *RST: DIS                                                                                               |
| <b>&lt;endPart&gt;</b>      | string                                                                                                  |
|                             | Last passphrase character as string                                                                     |

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.0.20, V3.2.10: WENTerprise, W2Personal, W2ENTERprise, V4.0.20: added OWE, W3Personal, W3ENTERprise

**Manual operation:** See ["Mode"](#) on page 80

### 2.5.11.8 QoS data settings

The following commands configure quality of service (QoS) settings.

|                                                                                          |     |
|------------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:STA&lt;s&gt;:CONNection:QOS:BLACK</a> .....     | 196 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:QOS:ETOE</a> .....                   | 197 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:QOS:PRloritiz</a> .....              | 197 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:STA&lt;s&gt;:CONNection:QOS:BARMethod</a> ..... | 197 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:PER:TIDentifier</a> .....                       | 198 |

---

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:QOS:BLACK** <TID0>, <TID1>,  
<TID2>, <TID3>, <TID4>, <TID5>, <TID6>, <TID7>

Enables/ disables a block ack session per TID (8 values).



**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<TID0> ... <TID7> OFF | ON  
\*RST: ON

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["Enable Block Ack"](#) on page 84

**CONFigure:WLAN:SIGN<i>:CONNection:QOS:ETOE <TID>**

Sets the TID value to be used for the end-to-end connection using DAU.

**Parameters:**

<TID> TID0 | TID1 | TID2 | TID3 | TID4 | TID5 | TID6 | TID7  
\*RST: TID0

**Example:** See [Configuring DAU parameters](#)

**Firmware/software:** V3.7.30

**Manual operation:** See ["QoS Data TID"](#) on page 83

**CONFigure:WLAN:SIGN<i>:CONNection:QOS:PRioritiz <mode>**

Prioritization mode selects the transmission sequence.

- Round-robin schedules equal transmission time to each TID
- TID priority selection prioritizes the transmission of highest TID values
- Automatic selection

**Parameters:**

<mode> ROURobin | TIDPriority | AUTO  
\*RST: ROUR

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.7.30, V3.8.20: added AUTO

**Manual operation:** See ["Prioritization Mode"](#) on page 83

**CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:QOS:BARMethod <method>**

Specifies the method used to request a BlockAck frame from the DUT

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<method> IMPBar | EXPBar | MUBar  
Implicit, explicit or multi-user block acknowledgment request

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.7.40, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["BAR Method"](#) on page 83

**CONFIGure:WLAN:SIGN<i>:PER:TIDentifier <TID>**

Sets the TID value to be used for PER measurements.

**Parameters:**

<TID> TID0 | TID1 | TID2 | TID3 | TID4 | TID5 | TID6 | TID7  
\*RST: TID0

**Example:** See [Configuring a PER measurement](#)

**Firmware/software:** V3.7.30

**Manual operation:** See ["QoS Data TID"](#) on page 83

**2.5.11.9 A-MPDU settings**

The following command configures aggregate MPDUs (A-MPDU).

**CONFIGure:WLAN:SIGN<i>:STA<s>:CONNEction:AMPDu <Enable>, <Multi\_TID>, <MaxLength>**

Configures aggregate MPDUs (A-MPDU).

**Suffix:**

<s> 1..3  
Station No.

**Parameters:**

<Enable> DISable | ENABLE  
Enables/ disables the A-MPDUs  
\*RST: ENAB

<Multi\_TID> DISable | ENABLE  
Enables/ disables multi-TID A-MPDU  
\*RST: DIS

<MaxLength> integer  
The maximal length of entire A-MPDU  
Range: 50 to 131.071E+3  
\*RST: 131.071E+3  
Default unit: byte

**Example:** See [802.11ax connection settings](#)

**Firmware/software:** V3.7.30, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["TX Enable"](#) on page 84

### 2.5.11.10 EDCA settings

The following command configures enhanced distributed channel access (EDCA) for aggregate MPDUs.

|                                                                      |     |
|----------------------------------------------------------------------|-----|
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:EDCA:ACBE</a>   | 199 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:EDCA:ACBK</a>   | 199 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:EDCA:ACVI</a>   | 199 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:EDCA:ACVO</a>   | 199 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:MUEdca:ACBE</a> | 200 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:MUEdca:ACBK</a> | 200 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:MUEdca:ACVI</a> | 200 |
| <a href="#">CONFfigure:WLAN:SIGN&lt;i&gt;:CONNection:MUEdca:ACVO</a> | 200 |

---

**CONFfigure:WLAN:SIGN<i>:CONNection:EDCA:ACBE** <AIFSN>, <ECWmin>, <ECWmax>, <TXOPlim>

**CONFfigure:WLAN:SIGN<i>:CONNection:EDCA:ACBK** <AIFSN>, <ECWmin>, <ECWmax>, <TXOPlim>

**CONFfigure:WLAN:SIGN<i>:CONNection:EDCA:ACVI** <AIFSN>, <ECWmin>, <ECWmax>, <TXOPlim>

**CONFfigure:WLAN:SIGN<i>:CONNection:EDCA:ACVO** <AIFSN>, <ECWmin>, <ECWmax>, <TXOPlim>

Configures the record fields of EDCA parameter set.

**Parameters:**

|          |                                      |
|----------|--------------------------------------|
| <AIFSN>  | integer                              |
|          | Arbitration inter-frame space number |
|          | Range: 2 to 15                       |
|          | *RST: See table below                |
| <ECWmin> | integer                              |
|          | Minimal contention window            |
|          | Range: 0 to 15                       |
|          | *RST: See table below                |
| <ECWmax> | integer                              |
|          | Maximal contention window            |
|          | Range: 0 to 15                       |
|          | *RST: See table below                |

<TXOPlim> integer  
 Transmission opportunity limit  
 Range: 0 to 255  
 \*RST: See table below

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.7.40

**Manual operation:** See ["EDCA Parameter Set"](#) on page 85

*Table 2-10: Default values*

| Access class | AIFSN | ECWmin | ECWmax | TXOP limit |
|--------------|-------|--------|--------|------------|
| AC_BK        | 7     | 4      | 10     | 79         |
| AC_BE        | 3     | 4      | 10     | 79         |
| AC_VI        | 2     | 1      | 4      | 128        |
| AC_VO        | 2     | 0      | 1      | 65         |

**CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACBE** <AIFSN>, <ECWmin>, <ECWmax>, <timer>

**CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACBK** <AIFSN>, <ECWmin>, <ECWmax>, <timer>

**CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACVI** <AIFSN>, <ECWmin>, <ECWmax>, <timer>

**CONFigure:WLAN:SIGN<i>:CONNection:MUEDca:ACVO** <AIFSN>, <ECWmin>, <ECWmax>, <timer>

Configures the record fields of MU EDCA parameter set.

**Parameters:**

<AIFSN> integer  
 Arbitration inter-frame space number. Zero disables channel access.  
 Range: 0, 2 to 15  
 \*RST: See table below

<ECWmin> integer  
 Minimal contention window  
 Range: 0 to 15  
 \*RST: See table below

<ECWmax> integer  
 Maximal contention window  
 Range: 0 to 15  
 \*RST: See table below

<timer> integer  
 MU EDCA timer  
 Range: 1 to 255  
 \*RST: See table below  
 Default unit: 8x TUs (8x 1024 µs)

**Example:** See [Configuring connection settings](#)

**Firmware/software:** V3.7.40

**Manual operation:** See ["MU EDCA Parameter Set"](#) on page 86

*Table 2-11: Default values*

| Access class | AISFN | ECWmin | ECWmax | MU EDCA timer |
|--------------|-------|--------|--------|---------------|
| AC_BK        | 0     | 4      | 10     | 15            |
| AC_BE        | 0     | 4      | 10     | 15            |
| AC_VI        | 0     | 1      | 4      | 15            |
| AC_VO        | 0     | 0      | 1      | 15            |

## 2.5.12 Trigger settings

The following commands define the generated trigger signal.

|                                                     |     |
|-----------------------------------------------------|-----|
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe.....         | 201 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:BW.....            | 202 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:DSMLength.....     | 203 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:OFMLength.....     | 203 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:MODE.....  | 204 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:VALue..... | 204 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:RATE.....          | 204 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:RREStriction.....  | 205 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:CTDelay.....       | 205 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:STReams.....       | 205 |
| TRIGger:WLAN:SIGN<i>:RX:MACFrame:SLOPe.....         | 206 |
| TRIGger:WLAN:SIGN<i>:TX:MACFrame:SLOPe.....         | 206 |
| TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgtH:MODE.....  | 206 |
| TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgtH:VALue..... | 206 |

---

### TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe <Type>

Defines for which bursts a trigger pulse is generated for the RX frame trigger signal.

Note that the trigger pulse is generated only for bursts matching the specified trigger bandwidth and trigger rate.

Refer to:

[TRIGger:WLAN:SIGN<i>:RX:MACFrame:BW](#)

TRIGger:WLAN:SIGN<i>:RX:MACFrame:RATE

**Parameters:**

<Type>

ABURsts | OBUrsts | DCBURsts | NHTBURsts | HTBURsts |  
VHTBURsts | HESBURsts

**ABURsts**

All received bursts result in an RX frame trigger pulse.

**OBUrsts**

Only OFDM bursts with the configured minimum length result in an RX frame trigger pulse.

**DCBURsts**

Only DSSS/CCK bursts with the configured minimum length result in an RX frame trigger pulse.

**NHTBURsts**

Only non-HT bursts with the configured minimum length result in an RX frame trigger pulse.

**HTBURsts**

Only HT bursts with the configured minimum length result in an RX frame trigger pulse.

**VHTBURsts**

Only VHT bursts with the configured minimum length result in an RX frame trigger pulse.

**HESBURsts**

Only HE SU bursts with the configured minimum length result in an RX frame trigger pulse.

\*RST: ABUR

**Example:**

See [Configuring trigger signals](#)

**Firmware/software:**

V3.2.11, V3.7.30: added NHTBURsts, HTBURsts, VHTBURsts, HESBURsts

**Options:**

An SUA required for NHTBURsts, HTBURsts, VHTBURsts, HESBURsts

**Manual operation:**

See ["RX Format"](#) on page 90

---

TRIGger:WLAN:SIGN<i>:RX:MACFrame:BW <TriggerBandwidth>

Defines for which bandwidth of received bursts a trigger pulse is generated for the RX frame trigger signal.

**Parameters:**

<TriggerBandwidth>

BW20 | BW40 | BW80 | BW160 | ALL | ON | OFF

**BWx:** RX frame trigger signal generated for the received bursts with the bandwidth of x MHz

**ALL:** RX frame trigger signal generated for all bandwidths

**ON:** RX frame trigger signal switched on

**OFF:** RX frame trigger signal switched off

\*RST: ALL

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.30

**Manual operation:** See ["RX Bandwidth"](#) on page 91

---

### TRIGger:WLAN:SIGN<i>:RX:MACFrame:DSMLength <Mode>[, <Length>]

Defines the minimum length for the RX frame trigger mode DSSS/CCK Bursts, see [TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe](#).

#### Parameters:

**<Mode>**                      DEFault | UDEFined  
**DEFault:** automatically calculated value  
**UDEFined:** configured <Length>  
**\*RST:**                      DEF

**<Length>**                      numeric  
 Minimum number of bytes for UDEFined mode  
**Range:**                      16 to 1500  
**\*RST:**                      299  
 Default unit: byte

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.40

**Manual operation:** See [" Min Length: DSSS or OFDM Payload"](#) on page 91

---

### TRIGger:WLAN:SIGN<i>:RX:MACFrame:OFMLength <Mode>[, <Length>]

Defines the minimum length for the all OFDM RX frame trigger modes (all modes except All Bursts and DSSS/CCK Bursts), see [TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe](#).

#### Parameters:

**<Mode>**                      DEFault | UDEFined  
**DEFault:** automatically calculated value  
**UDEFined:** configured <Length>  
**\*RST:**                      DEF

**<Length>**                      numeric  
**Range:**                      1 to 1500  
**\*RST:**                      16

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.40

**Manual operation:** See [" Min Length: DSSS or OFDM Payload"](#) on page 91

**TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:MODE** <PulseLengthMode>

Configures the length of generated RX MAC frame trigger pulses.

**Parameters:**

<PulseLengthMode> DEFault | UDEfined

**DEFault**

The pulse length is 1 µs.

**UDEfined**

The pulse length is specified via [TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:VALue](#) on page 204.

\*RST: DEFault

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.0.10

**Manual operation:** See "[Pulse Length](#)" on page 93

**TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:VALue** <PulseLengthVal>

Sets the pulse length for the mode UDEfined of the RX frame trigger, see [TRIGger:WLAN:SIGN<i>:RX:MACFrame:PLENgtH:MODE](#) on page 204.

**Parameters:**

<PulseLengthVal> numeric | ON | OFF

Range: 1E-6 s to 0.01 s

\*RST: 1E-6 s

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.0.10

**Manual operation:** See "[Pulse Length](#)" on page 93

**TRIGger:WLAN:SIGN<i>:RX:MACFrame:RATE** <TriggerRate>

Defines for which rate of received bursts a trigger pulse is generated for the RX frame trigger signal.

**Parameters:**

<TriggerRate> BR12 | QR12 | QR34 | Q1M12 | Q1M34 | Q6M23 | Q6M34 | BR34 | MCS0 | MCS1 | MCS2 | MCS3 | MCS4 | MCS5 | MCS6 | MCS7 | D1MBit | D2Mbits | C55Mbits | C11Mbits | MCS8 | MCS9 | MCS10 | MCS11 | MCS12 | MCS13 | MCS14 | MCS15 | ALL | ON | OFF

**D1MBit:** DSSS, 1 Mbit/s

**D2Mbits:** DSSS, 2 Mbit/s

**C55Mbits:** CCK, 5.5 Mbit/s

**C11Mbits:** CCK, 11 Mbit/s

**BR12:** BPSK, 1/2, 6 Mbit/s

**BR34:** BPSK, 3/4, 9 Mbit/s



**QR12:** QPSK, 1/2, 12 Mbit/s

**QR34:** QPSK, 3/4, 18 Mbit/s

**Q1M12:** 16-QAM, 1/2, 24 Mbit/s

**Q1M34:** 16-QAM, 3/4, 36 Mbit/s

**Q6M23:** 64-QAM, 2/3, 48 Mbit/s

**Q6M34:** 64-QAM, 3/4, 54 Mbit/s

**MCS, MCS1,...,MCS15:** MCS 0 to MCS 15

**ALL:** RX frame trigger signal generated for all rates

**ON:** RX frame trigger signal switched on

**OFF:** RX frame trigger signal switched off

\*RST: ALL

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.30

**Manual operation:** See ["RX Rate"](#) on page 91

#### **TRIGger:WLAN:SIGN<i>:RX:MACFrame:RREstriction <enable>**

Enables or disables the rate control of the DUT.

##### **Parameters:**

<Enable> OFF | ON

\*RST: OFF

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.70

**Manual operation:** See ["Rate Restriction"](#) on page 91

#### **TRIGger:WLAN:SIGN<i>:RX:MACFrame:CTDelay <DelayType>**

Sets the trigger delay type burst (automatic configuration) or constant delay of 200 µs for RX frame trigger, OFDM.

##### **Parameters:**

<DelayType> BURSt | CONStant

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.60

**Manual operation:** See ["Delay"](#) on page 91

#### **TRIGger:WLAN:SIGN<i>:RX:MACFrame:STReams <SpatialStreams>**

Sets the spatial streams for RX frame trigger for MIMO connections.

**Parameters:**

<SpatialStreams> ALL | STR1 | STR2 | ON | OFF  
 Both streams, stream 1, or stream 2 used for the trigger signal  
 \*RST: ALL

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V3.7.60, V4.0.20: changed reset value

**Manual operation:** See ["RX Space Time Streams"](#) on page 91

**TRIGger:WLAN:SIGN<i>:RX:MACFrame:SLOPe** <TrigSlope>

**TRIGger:WLAN:SIGN<i>:TX:MACFrame:SLOPe** <TrigSlope>

Aligns either the rising edge or the falling edge of the trigger pulses with the start of the MAC frames.

**Parameters:**

<TrigSlope> REDGe | FEDGe  
 \*RST: REDGe

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V2.1.30, V3.0.10 RX MAC frame trigger

**Manual operation:** See ["Trigger Slope"](#) on page 90

**TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgtH:MODE** <PulseLengthMode>

Configures the length of generated TX MAC frame trigger pulses.

**Parameters:**

<PulseLengthMode> DEFault | BLENgtH | UDEFined  
**DEFault**  
 The pulse length is 1 µs.  
**BLENgtH**  
 The pulse length equals the burst length.  
**UDEFined**  
 The pulse length is specified via [TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgtH:VALue](#) on page 206.  
 \*RST: DEFault

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V2.1.30, V3.0.20: BLENgtH added

**Manual operation:** See ["Pulse Length"](#) on page 90

**TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgtH:VALue** <PulseLengthVal>

Sets the pulse length for the mode UDEFined of the TX frame trigger, see [TRIGger:WLAN:SIGN<i>:TX:MACFrame:PLENgtH:MODE](#) on page 206.

**Parameters:**

<PulseLengthVal> numeric | ON | OFF  
 Range: 1E-6 s to 0.01 s  
 \*RST: 1E-6 s

**Example:** See [Configuring trigger signals](#)

**Firmware/software:** V2.1.30

**Manual operation:** See "[Pulse Length](#)" on page 90

## 2.5.13 Packet generator

The following commands configure the IP packet generator.

|                                                                                  |     |
|----------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:PGEN&lt;index&gt;:CONFig</a> .....      | 207 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:PGEN&lt;index&gt;:DESTination</a> ..... | 208 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:PGEN&lt;index&gt;:PROTocol</a> .....    | 208 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:PGEN&lt;index&gt;:IPVersion</a> .....   | 208 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:PGEN&lt;index&gt;:UPORTs</a> .....      | 209 |

---

**CONFigure:WLAN:SIGN<i>:PGEN<index>:CONFig** <State>, <Interval>,  
 <PayloadSize>, <PayloadType>[, <TID>]

Configures the packet generator.

**Suffix:**

<index> 1..3

**Parameters:**

<State> OFF | ON  
 Disables/enables the packet generator  
 \*RST: OFF

<Interval> integer  
 Time interval between packet transmissions in units of 1024 µs  
 Range: 0 to 10E+3  
 \*RST: 100

<PayloadSize> integer  
 Payload size of generated packets in bytes  
 Range: 0 to 1472  
 \*RST: 500

<PayloadType> DEFault | AZERoes | AONes | BP01 | BP10 | PRANdom  
 Bit sequence to be transmitted as payload  
**DEFault:** an implementation-specific default pattern  
**AZERoes:** all zeroes  
**AONes:** all ones  
**BP01:** bit pattern 010101...  
**BP10:** bit pattern 101010...

**PRANdom:** a pseudo-random bit sequence

\*RST: DEF

<TID> TID0 | TID1 | TID2 | TID3 | TID4 | TID5 | TID6 | TID7

TID signaled by the packet generator

\*RST: TID0

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.0.10, V3.2.40: added AONes, BP01, BP10, PRANdom  
V3.7.20: interval extended, V3.7.30: added <index>, <TID>

**Manual operation:** See ["QoS Data TID"](#) on page 83

#### **CONFigure:WLAN:SIGN<i>:PGEN<index>:DESTination <station>**

Specifies the STA to which the packets are addressed to. This parameter is visible, if [Multi STA](#) is enabled.

##### **Suffix:**

<index> 1..3

##### **Parameters:**

<station> STA1 | STA2 | STA3

\*RST: STA1

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["Destination"](#) on page 96

#### **CONFigure:WLAN:SIGN<i>:PGEN<index>:PROTOcol <Type>**

Sets the protocol of the packet generator.

##### **Suffix:**

<index> 1..3

##### **Parameters:**

<Type> ICMP | UDP

\*RST: ICMP

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.0.20, V3.7.30: added <index>

**Manual operation:** See ["Protocol"](#) on page 95

#### **CONFigure:WLAN:SIGN<i>:PGEN<index>:IPVersion <Version>**

Sets the IP version to be used for generating packets.

##### **Suffix:**

<index> 1..3

**Parameters:**

<Version> IV4 | IV6  
 \*RST: IV4

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.2.60, V3.7.30: added <index>

**Manual operation:** See ["IP Version"](#) on page 95

**CONFigure:WLAN:SIGN<i>:PGEN<index>:UPOrts** <SourcePort>,  
 <DestinationPort>

Sets the source and destination ports for the UDP protocol.

**Suffix:**

<index> 1..3

**Parameters:**

<SourcePort> integer  
 Range: 0 to 65535  
 \*RST: 3001

<DestinationPort> integer  
 Range: 0 to 65535  
 \*RST: 3002

**Example:** See [Configuring the packet generator](#)

**Firmware/software:** V3.0.20, V3.7.30: added <index>

**Manual operation:** See ["UDP Source/Destination Port"](#) on page 96

## 2.5.14 IP interface parameters

The following commands configure the IPv4/v6 stacks built into the WLAN signaling application.

The settings are only relevant for instruments without DAU.

|                                                                                          |     |
|------------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:IVSupport</a> .....                  | 209 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVFour:STAtic:IPADdress:CMW</a> .....          | 210 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVFour:STAtic:IPADdress:STA&lt;s&gt;</a> ..... | 210 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVFour:STAtic:IPADdress:GAteWay</a> .....      | 211 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVFour:STAtic:SMASk</a> .....                  | 211 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVFour:STAtic:IPADdress:DNS</a> .....          | 211 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVFour:DHCP</a> .....                          | 212 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:IPVSix:PREFix</a> .....                         | 212 |

**CONFigure:WLAN:SIGN<i>:CONNection:IVSupport** <version>

Defines the required IP version support.

**Parameters:**

<version> IV4 | IV6 | IV4V6  
 IPv4 only, IPv6 only, or both  
 \*RST: IV4V6

**Example:** See [Configuring IP settings](#)

**Firmware/software:** V3.2.60

**Options:** R&S CMW-KA100

**Manual operation:** See ["IP Version Support"](#) on page 94

**CONFigure:WLAN:SIGN<i>:IPVFour:STATIC:IPADdress:CMW** <First>, <Sec>,  
 <Third>, <Fourth>

Defines the static IP V4 address of the R&S CMW. The setting is relevant for instruments without a DAU.

**Parameters:**

<Octet1>, <Octet2>, integer  
 <Octet3>, <Octet4> Address <Octet1>.<Octet2>.<Octet3>.<Octet4>  
 Range: 0 to 255  
 \*RST: 100, 100, 100, 50

**Example:** See [Configuring IP settings](#)

**Firmware/software:** V3.7.50

**Manual operation:** See ["IPv4 Interface"](#) on page 94

**CONFigure:WLAN:SIGN<i>:IPVFour:STATIC:IPADdress:STA<s>** <FirstNumber>,  
 <SecNumber>, <ThirdNumber>, <FourthNumber>

Defines the static IP V4 address of the DUT. The setting is only relevant for access point and instruments without a DAU.

**Suffix:**

<s> 1..4  
 Station No.

**Parameters:**

<Octet1>, <Octet2>, integer  
 <Octet3>, <Octet4> Address <Octet1>.<Octet2>.<Octet3>.<Octet4>  
 Range: 0 to 255  
 \*RST: 100, 100, 100, 10

**Example:** See [Configuring IP settings](#)

**Firmware/software:** V3.7.50, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["IPv4 Interface"](#) on page 94

---

**CONFigure:WLAN:SIGN<i>:IPVFour:STATIC:IPADdress:GATeway** <First>, <Sec>, <Third>, <Fourth>

Provides the IPv4 address of the default gateway.

The setting is relevant for instruments without DAU.

**Parameters:**

<Octet1>, <Octet2>, integer  
 <Octet3>, <Octet4> Address <Octet1>.<Octet2>.<Octet3>.<Octet4>  
 Range: 0 to 255  
 \*RST: 100, 100, 100, 1

**Example:** See [Configuring IP settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["IPv4 Interface"](#) on page 94

---

**CONFigure:WLAN:SIGN<i>:IPVFour:STATIC:SMASK** <FirstOctet>, <SecondOctet>, <ThirdOctet>, <FourthOctet>

Specifies the subnet mask of the built-in IPv4 stack.

The setting is relevant for instruments without DAU.

**Parameters:**

<Octet1>, <Octet2>, integer  
 <Octet3>, <Octet4> Mask <Octet1>.<Octet2>.<Octet3>.<Octet4>  
 Range: 0 to 255  
 \*RST: 255, 255, 255, 0

**Example:** See [Configuring IP settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["IPv4 Interface"](#) on page 94

---

**CONFigure:WLAN:SIGN<i>:IPVFour:STATIC:IPADdress:DNS** <First>, <Sec>, <Third>, <Fourth>

Provides the IPv4 address of a DNS server to the built-in IPv4 stack.

The setting is relevant for instruments without DAU.

**Parameters:**

<Octet1>, <Octet2>, integer  
 <Octet3>, <Octet4> Address <Octet1>.<Octet2>.<Octet3>.<Octet4>  
 Range: 0 to 255  
 \*RST: 10, 0, 2, 166

**Example:** See [Configuring IP settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["IPv4 Interface"](#) on page 94

---

#### **CONFigure:WLAN:SIGN<i>:IPVFour:DHCP <Activate>**

Enables or disables the built-in DHCP server.

The setting is relevant for station mode, with or without a DAU.

This setting is not relevant for AP mode.

##### **Parameters:**

<Activate>                      OFF | ON  
                                      \*RST:            ON

**Example:**                      See [Configuring IP settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["IPv4 Interface"](#) on page 94

---

#### **CONFigure:WLAN:SIGN<i>:IPVSix:PREFIX <Prefix>**

Defines the IPv6 prefix of the built-in IPv6 stack.

##### **Parameters:**

<Prefix>                        string  
                                      IPv6 prefix as string  
                                      \*RST:            fc01:abab:cdcd:efe0::

**Example:**                      See [Configuring IP settings](#)

**Firmware/software:** V3.2.60

**Manual operation:** See ["IPv6 Interface"](#) on page 94

## **2.5.15 Data end-to-end**

The following commands are related to an internal or external data application unit (DAU).

|                                                                                    |     |
|------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:ETOE:IRList:IPRaddress&lt;n&gt;</a> ..... | 212 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:ETOE:DUIP</a> .....                       | 213 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:EDAU:ENABLE</a> .....                     | 214 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:EDAU:NSEgment</a> .....                   | 214 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:EDAU:NID</a> .....                        | 214 |

---

#### **CONFigure:WLAN:SIGN<i>:ETOE:IRList:IPRaddress<n> <State>, <Ip4\_1>, <Ip4\_2>, <Ip4\_3>, <Ip4\_4>, <Ip4Netmask>, <Ip6Prefix>, <Ip6Netmask>**

Configures an entry of the routes list. The routes list defines destination addresses for which the DAU routes packets to the DUT.



**Suffix:**

<n> 1..5  
Entry number

**Parameters:**

<State> OFF | ON  
Entry disabled or enabled  
\*RST: OFF

<Ip4\_1> integer  
First octet of the IPv4 destination address  
Range: 0 to 255

<Ip4\_2> integer  
Second octet  
Range: 0 to 255

<Ip4\_3> integer  
Third octet  
Range: 0 to 255

<Ip4\_4> integer  
Fourth octet  
Range: 0 to 255

<Ip4Netmask> integer  
Number of subnet bits for the IPv4 address  
Range: 1 to 32

<Ip6Prefix> string  
IPv6 prefix as string, for example "fc01:abab:cdcd:efe0::"

<Ip6Netmask> integer  
Number of subnet bits for the IPv6 address  
Range: 1 to 128

**Example:** See [Configuring DAU parameters](#)

**Firmware/software:** V3.5.40

**Manual operation:** See ["IP Routes List"](#) on page 97

---

**CONFigure:WLAN:SIGN<i>:ETOE:DUIP** <State>, <FirstNumber>, <SecNumber>, <ThirdNumber>, <FourthNumber>

Allows you to specify the IPv4 address that the DAU assigns to the DUT via DHCP.

**Parameters:**

<State> OFF | ON  
Disables/enables the IP address configuration

<Octet1>,<Octet2>, integer  
 <Octet3>,<Octet4> Address <Octet1>.<Octet2>.<Octet3>.<Octet4>  
 Range: 0 to 255  
 \*RST: 100, 100, 100, 10

**Example:** See [Configuring DAU parameters](#)

**Firmware/software:** V3.5.20

**Manual operation:** See ["Set DAU DHCP Response IP Address, IP Address \(DHCP Response\)"](#) on page 97

### CONFigure:WLAN:SIGN<i>:EDAU:ENABLE <ExternalDau>

Enables usage of an external DAU.

#### Parameters:

<ExternalDau> OFF | ON  
 \*RST: OFF

**Example:** See [Configuring DAU parameters](#)

**Firmware/software:** V3.5.20

**Options:** R&S CMW-KA120

**Manual operation:** See ["Use External DAU"](#) on page 98

### CONFigure:WLAN:SIGN<i>:EDAU:NSEGment <SegmentNumber>

Specifies the network segment of the instrument where the external DAU is installed.

#### Parameters:

<SegmentNumber> A | B | C  
 \*RST: A

**Example:** See [Configuring DAU parameters](#)

**Firmware/software:** V3.5.20

**Options:** R&S CMW-KA120

**Manual operation:** See ["IP Set of External DAU"](#) on page 98

### CONFigure:WLAN:SIGN<i>:EDAU:NID <NodeId>

Specifies the subnet node ID of the instrument where the external DAU is installed.

#### Parameters:

<NodeId> integer  
 Range: 1 to 254  
 \*RST: 1

**Example:** See [Configuring DAU parameters](#)

**Firmware/software:** V3.5.20

**Options:** R&S CMW-KA120

**Manual operation:** See ["IP Set of External DAU"](#) on page 98

## 2.5.16 Message monitoring

The following commands are related to message logging with R&S CMWmars.

|                                                                       |     |
|-----------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:MMONitor:ENABLE</a> .....    | 215 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:MMONitor:IPADdress</a> ..... | 215 |

---

### **CONFigure:WLAN:SIGN<i>:MMONitor:ENABLE** <Enable>

Enables or disables message monitoring for the WLAN signaling application.

**Parameters:**

<Enable> OFF | ON  
\*RST: OFF

**Example:** See [Configuring message monitoring](#)

**Firmware/software:** V3.5.20

**Manual operation:** See ["Add WLAN Signaling to Logging PC"](#) on page 99

---

### **CONFigure:WLAN:SIGN<i>:MMONitor:IPADdress** <Index>

Selects the IP address to which signaling messages are sent for message monitoring. The address pool is configured globally via

CONFigure:BASE:MMONitor:IPADdress<n>.

A query returns both the current index and the resulting IP address.

**Parameters:**

<Index> IP1 | IP2 | IP3  
Address pool index

**Return values:**

<IPAddress> string  
Used IP address as string

**Example:** See [Configuring message monitoring](#)

**Firmware/software:** V3.5.20

**Manual operation:** See ["Logging PC IPv4 Address"](#) on page 99

## 2.5.17 HE TB list mode measurements

The following commands provide HE TB - related measurements on UL power head-room (UPH) in several sequential frames.

The measurements are only supported per remote control. They have no corresponding GUI parameters.

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| <a href="#">INITiate:WLAN:SIGN&lt;i&gt;:HETBased</a> .....                    | 216 |
| <a href="#">STOP:WLAN:SIGN&lt;i&gt;:HETBased</a> .....                        | 216 |
| <a href="#">ABORT:WLAN:SIGN&lt;i&gt;:HETBased</a> .....                       | 216 |
| <a href="#">CONFigure:WLAN:SIGN&lt;i&gt;:HETBased:FRAMes</a> .....            | 216 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:HETBased:STATe?</a> .....                | 217 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:HETBased:UPHinfo?</a> .....              | 217 |
| <a href="#">SENSe:WLAN:SIGN&lt;i&gt;:STA&lt;s&gt;:HETBinfo:UPHinfo?</a> ..... | 218 |

---

#### **INITiate:WLAN:SIGN<i>:HETBased**

#### **STOP:WLAN:SIGN<i>:HETBased**

#### **ABORT:WLAN:SIGN<i>:HETBased**

Starts, stops, or aborts the measurement:

- **INITiate** . . . starts or restarts the measurement. The measurement enters the "RUN" state.
- **STOP** . . . halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- **ABORT** . . . halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use **FETCh** . . . **STATe?** to query the current measurement state.

**Example:** See [HE TB list mode measurements](#)

**Usage:** Event

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS657

---

#### **CONFigure:WLAN:SIGN<i>:HETBased:FRAMes <no\_of\_frames>**

Sets the number of frames for HE TB list mode measurements. This setting determines the statistic count of the measurement.

#### **Parameters:**

|              |           |
|--------------|-----------|
| <NoOfFrames> | integer   |
| Range:       | 1 to 2000 |
| *RST:        | 1         |

**Example:** See [HE TB list mode measurements](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS657

**FETCh:WLAN:SIGN<i>:HETBased:STATe?**

Queries the main measurement state. Use `FETCh: . . . :STATe:ALL?` to query the measurement state including the substates. Use `INITiate...`, `STOP...`, `ABORt . . .` to change the measurement state.

**Return values:**

<State>                      OFF | RUN | RDY

**OFF:** measurement switched off, no resources allocated, no results available (when entered after `ABORt . . .`)

**RDY:** measurement has been terminated, valid results can be available

**RUN:** measurement running (after `INITiate...`, `READ...`), synchronization pending or adjusted, resources active or queued

\*RST:                      OFF

**Example:**                      See [HE TB list mode measurements](#)

**Usage:**                      Query only

**Firmware/software:** V3.7.60

**Options:**                      R&S CMW-KS657

**FETCh:WLAN:SIGN<i>:HETBased:UPHinfo?**

Queries the results of the signaling HE TB list mode measurements on UL power headroom (UPH).

The result groups are listed in a sequence as  $\{<BurstPower>, <UPH>, <MinTXPowerFlag>, <LST>\}_{frame\_1}, \dots, \{<BurstPower>, <UPH>, <MinTXPowerFlag>, <LST>\}_{frame\_n}$ .

The number of results n is set via

[CONFigure:WLAN:SIGN<i>:HETBased:FRAMES](#)

**Return values:**

<BurstPower>                      float

Measured burst power in uplink

Range:                      -999 dBm to 999 dBm

Default unit: dBm

<UPH>                      decimal

UL power headroom

Range:                      0 dB to 31 dB

Default unit: dB

<MinTXPowerFlag>                      OFF | ON

Indication whether the HE TB bursts are sent at the minimum transmit power of the station

<LST>                      float

Rx timestamp from physical layer

**Example:** See [HE TB list mode measurements](#)

**Usage:** Query only

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS657

---

#### SENSe:WLAN:SIGN<i>:STA<s>:HETBinfo:UPHinfo?

Queries actual information related to uplink power headroom (UPH) control.

##### Suffix:

<s> 1..3  
Station No.

##### Return values:

<BurstPower> float  
Indication of HE TB burst power.  
Range: -999 dBm to 999 dBm

<UPH> decimal  
Indication of UL power headroom.  
Range: 0 dB to 31 dB

<MinTXPowerFlag> OFF | ON  
Indication whether the HE TB bursts are sent at the minimum transmit power of the station.

**Example:** See [HE TB list mode measurements](#)

**Usage:** Query only

**Firmware/software:** V3.7.60, V3.8.20: added <s>

**Options:** R&S CMW-KS667 for :STA2: and :STA3:

**Manual operation:** See ["HE TB Info"](#) on page 38

## 2.5.18 HE sounding emulation

The following commands provide measurements on HE sounding protocol to facilitate beamforming and DL MU-MIMO transmission.

|                                                           |     |
|-----------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:BW          | 219 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:CBOok       | 219 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:LTFGi       | 219 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:METHod      | 220 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:NUMColumns  | 220 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:RUENd       | 220 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:RUSart      | 220 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SUBGrouping | 221 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SPSTreams   | 221 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SSTX        | 221 |

|                                                           |     |
|-----------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TARGet..... | 222 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TXEN.....   | 222 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TXP.....    | 222 |
| CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TYPE.....   | 223 |

---

### CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:BW <band>

Selects the channel bandwidth for NDP sounding procedure.

#### Parameters:

<band> BW20 | BW40 | BW80 | BW160

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["Bandwidth"](#) on page 87

---

### CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:CBOok <size>

Sets the codebook size for HE TB sounding: 0 or 1.

#### Parameters:

<size> SIZE0 | SIZE1  
\*RST: SIZE1

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KS671

**Manual operation:** See ["Codebook Size"](#) on page 88

---

### CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:LTFGi <LTF\_GI>

Selects the GI / LTF combination for NDP sounding procedure.

#### Parameters:

<LTF\_GI> L208 | L216 | L432  
**"L208"**: 2x HE-LTF + 0.8  $\mu$ s GI  
**"L216"**: 2x HE-LTF + 1.6  $\mu$ s GI  
**"L432"**: 4x HE-LTF + 3.2  $\mu$ s GI (optional)  
 \*RST: L208

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["GI and LTF"](#) on page 88

---

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:METhod** <method>

Sets the feedback method for NDP sounding procedure.

**Parameters:**

<method>                   NONTrigger | TBASed  
                              **"NONTrigger"**: non-trigger-based  
                              **"TBASed"**: trigger-based

**Example:**                   See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60, V3.8.20: added TBASed

**Options:**                   R&S CMW-KS671

**Manual operation:**   See ["Feedback Method"](#) on page 87

---

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:NUMColumns** <num\_col>

Sets the number of columns value  $N_c$  for HE TB sounding.

**Parameters:**

<num\_col>                   COL1 | COL2  
                              \*RST:       COL1

**Example:**                   See [HE NDP sounding procedure](#)

**Firmware/software:** V3.8.20

**Options:**                   R&S CMW-KS671

**Manual operation:**   See ["Number of Columns \( \$N\_c\$ \)"](#) on page 88

---

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:RUEnd** <RU\_idx>

Specifies the last 26-tone RU to be measured (RU end index) during NDP sounding procedure.

**Parameters:**

<RU\_idx>                   integer  
                              Range:       8 to 8  
                              \*RST:       8

**Example:**                   See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:**                   R&S CMW-KS671

**Manual operation:**   See ["RU End Index"](#) on page 88

---

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:RUStart** <RU\_idx>

Specifies the 26-tone RU marking the beginning of the measured bandwidth (RU start index) for NDP sounding procedure.



**Parameters:**

<RU\_idx> integer  
 Range: 0 to 0  
 \*RST: 0

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["RU Start Index"](#) on page 88

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SUBGrouping <Ng>**

Sets the subcarrier grouping value Ng for HE TB sounding.

**Parameters:**

<Ng> GRP4 | GRP16  
 \*RST: GRP4

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KS671

**Manual operation:** See ["Subcarrier Grouping"](#) on page 88

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SPSTreams  
<num\_streams>**

Selects the number of spatial streams for NDP sounding procedure.

**Parameters:**

<num\_streams> STR1 | STR2

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["Spatial Streams"](#) on page 88

**CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:SSTX**

Triggers the single-shot transmission for NDP sounding procedure.

**Example:** See [HE NDP sounding procedure](#)

**Usage:** Event

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["Transmission"](#) on page 88

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TARGet <station>**

Selects the STA to which the NDP sounding applies. This parameter is visible, if [Multi STA](#) is enabled.

##### **Parameters:**

<station> STA1 | STA2 | STA3  
\*RST: STA1

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KS671

**Manual operation:** See ["STA"](#) on page 87

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TXEN <State>**

Switches on or off the periodic transmission for NDP sounding procedure.

##### **Parameters:**

<State> OFF | ON  
\*RST: OFF

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["Transmission"](#) on page 88

---

#### **CONFigure:WLAN:SIGN<i>:CONNection:NDPSounding:TXP <Interval>**

Selects the periodic transmission interval for NDP sounding procedure.

##### **Parameters:**

<Interval> integer  
Range: 1 to 10E+3  
\*RST: 100  
Default unit: ms

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["Transmission"](#) on page 88

**CONFigure:WLAN:SIGN<i>:CONNction:NDPSounding:TYPE <type>**

Selects the report type for NDP sounding procedure. All types of feedback are returned via the HE Compressed Beamforming/CQI Frame.

**Parameters:**

<type> SU | MU | CQI  
**"SU"**: single-user feedback  
**"MU"**: multi-user feedback (only for trigger-based sounding procedure)  
**"CQI"**: channel quality index feedback

**Example:** See [HE NDP sounding procedure](#)

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KS671

**Manual operation:** See ["Report Type"](#) on page 88

**2.5.19 PER measurement**

The following sections describe the commands related to the packet error rate (PER) measurement.

- [Measurement control and states](#).....223
- [Measurement settings](#).....225
- [Measurement results](#).....233

**2.5.19.1 Measurement control and states**

The following commands control the PER measurement and return the current measurement state.

|                                                               |     |
|---------------------------------------------------------------|-----|
| <a href="#">INITiate:WLAN:SIGN&lt;i&gt;:PER</a> .....         | 223 |
| <a href="#">STOP:WLAN:SIGN&lt;i&gt;:PER</a> .....             | 223 |
| <a href="#">ABORT:WLAN:SIGN&lt;i&gt;:PER</a> .....            | 223 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:PER:STATe?</a> .....     | 224 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:PER:STATe:ALL?</a> ..... | 224 |

**INITiate:WLAN:SIGN<i>:PER****STOP:WLAN:SIGN<i>:PER****ABORT:WLAN:SIGN<i>:PER**

Starts, stops, or aborts the measurement:

- **INITiate**... starts or restarts the measurement. The measurement enters the "RUN" state.
- **STOP**... halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.

- `ABORT...` halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use `FETCH...STATE?` to query the current measurement state.

**Example:** See [Running a PER measurement](#)

**Usage:** Event

**Firmware/software:** V2.1.30

**Manual operation:** See ["PER \(Softkey\)"](#) on page 103

#### **FETCH:WLAN:SIGN<i>:PER:STATE?**

Queries the main measurement state. Use `FETCH:...:STATE:ALL?` to query the measurement state including the substates. Use `INITiate...`, `STOP...`, `ABORT...` to change the measurement state.

##### **Return values:**

<State> OFF | RDY | RUN

**OFF:** measurement switched off, no resources allocated, no results available (when entered after `ABORT...`)

**RDY:** measurement has been terminated, valid results can be available

**RUN:** measurement running (after `INITiate...`, `READ...`), synchronization pending or adjusted, resources active or queued

\*RST: OFF

**Example:** See [Running a PER measurement](#)

**Usage:** Query only

**Firmware/software:** V2.1.30

**Manual operation:** See ["PER \(Softkey\)"](#) on page 103

#### **FETCH:WLAN:SIGN<i>:PER:STATE:ALL?**

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use `FETCH:...:STATE?` to query the main measurement state only. Use `INITiate...`, `STOP...`, `ABORT...` to change the measurement state.

**Return values:**

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MainState>               | OFF   RDY   RUN<br><b>OFF:</b> measurement switched off, no resources allocated, no results available (when entered after <code>STOP...</code> )<br><b>RDY:</b> measurement has been terminated, valid results can be available<br><b>RUN:</b> measurement running (after <code>INITiate...</code> , <code>READ...</code> ), synchronization pending or adjusted, resources active or queued<br><b>*RST:</b> OFF |
| <SyncState>               | PEND   ADJ   INV<br><b>PEND:</b> waiting for resource allocation, adjustment, hardware switching ("pending")<br><b>ADJ:</b> all necessary adjustments finished, measurement running ("adjusted")<br><b>INV:</b> not applicable because <main_state>: OFF or RDY ("invalid")                                                                                                                                      |
| <RessourceState>          | QUE   ACT   INV<br><b>QUE:</b> measurement without resources, no results available ("queued")<br><b>ACT:</b> resources allocated, acquisition of results in progress but not complete ("active")<br><b>INV:</b> not applicable because <main_state>: OFF or RDY ("invalid")                                                                                                                                      |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Firmware/software:</b> | V2.1.30                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Manual operation:</b>  | See <a href="#">"PER (Softkey)"</a> on page 103                                                                                                                                                                                                                                                                                                                                                                  |

**2.5.19.2 Measurement settings**

The following commands configure the PER measurement.

|                                                                                              |     |
|----------------------------------------------------------------------------------------------|-----|
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:ATYPe</code> .....                                    | 226 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DPATtern</code> .....                                 | 226 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:PACKets</code> .....                                  | 226 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DINTerval</code> .....                                | 227 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:PAYLoad:SIZE</code> .....                             | 227 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DESTination</code> .....                              | 227 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:CONNECTION:SYNC</code> .....                              | 228 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:FDEF</code> .....                                     | 228 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DFRame:HEMU:ALSField</code> .....                     | 229 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DFRame:HEMU:BLALlocation?</code> .....                | 230 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DFRame:HEMU:DUMMy&lt;index&gt;:MCS</code> .....       | 230 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DFRame:HEMU:RUALlocation</code> .....                 | 231 |
| <code>CONFigure:WLAN:SIGN&lt;i&gt;:PER:DFRame:HEMU:USER&lt;index&gt;:ALLocation</code> ..... | 231 |

|                                                                 |     |
|-----------------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:MCS.....     | 232 |
| CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:CTYPe.....   | 232 |
| CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:STReams..... | 233 |

---

#### CONFigure:WLAN:SIGN<i>:PER:ATYPE <AckType>

Selects an evaluation scheme for the PER measurement. Currently, only the standard frame acknowledgment is available.

##### Parameters:

<AckType>           ACK

**Firmware/software:** V2.1.30

**Manual operation:** See ["Ack Type"](#) on page 105

---

#### CONFigure:WLAN:SIGN<i>:PER:DPATtern <Pattern>

Selects the data that the R&S CMW transfers to the DUT.

##### Parameters:

<Pattern>           PN1 | PN2 | PN3 | PN4 | PN5 | PN6 | PN7 | PN8 | PN9 | PN10 |  
 PN11 | PN12 | PN13 | PN14 | PN15 | PN16 | PN17 | PN18 |  
 PN19 | PN20 | PN21 | PN22 | PN23 | PN24 | PN25 | PN26 |  
 PN27 | PN28 | PN29 | PN30 | PN31 | PN32 | PRANdom |  
 AZERo | AONE | PT01 | PT10

**PN1,...,PN32:** pseudo-noise bit sequences of different lengths

**PRANdom:** random bit sequence

**AZERo:** all zero pattern "000..."

**AONE:** all one pattern "111..."

**PT01:** alternating sequence starting with zero "010101..."

**PT10:** alternating sequence starting with one "101010..."

\*RST:           PN1

**Example:** See [Configuring a PER measurement](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["Data Pattern"](#) on page 105

---

#### CONFigure:WLAN:SIGN<i>:PER:PACKets <NumberOfPackets>

Sets the number of user data MAC packets to be transmitted to the DUT.

##### Parameters:

<NumberOfPackets> numeric

Range:           1 to 1E+6

\*RST:           100

**Example:** See [Configuring a PER measurement](#)

**Firmware/software:** V2.1.30

**Manual operation:** See ["Number of Packets"](#) on page 105

**CONFigure:WLAN:SIGN<i>:PER:DINTerval <Interval>**

Specifies the time interval (in units of 1024  $\mu$ s) between data packet transmissions for the PER measurement.

**Parameters:**

<Interval> integer  
 Range: 0 to 100  
 \*RST: 20

**Example:** See [Configuring a PER measurement](#)

**Firmware/software:** V2.1.30, V3.7.20: interval extended

**Manual operation:** See ["Data Interval"](#) on page 105

**CONFigure:WLAN:SIGN<i>:PER:PAYLoad:SIZE <Size>**

Specifies the payload size (in bytes) for the PER measurement.

**Parameters:**

<Size> integer  
 Range: see table below  
 Default unit: byte

**Example:** See [Configuring a PER measurement](#)

**Firmware/software:** V2.1.30, V3.7.50: range extended

**Manual operation:** See ["Payload Size"](#) on page 105

| Frame types  | Supported size in bytes |
|--------------|-------------------------|
| DSSS, non-HT | 0 to 2000               |
| HT, VHT, HE  | 0 to 4096               |

**CONFigure:WLAN:SIGN<i>:PER:DESTination <sta>**

Specify the station for which the traffic is measured. This parameter is visible, if [Multi STA](#) is enabled.

**Parameters:**

<sta> STA1 | STA2 | STA3  
 \*RST: STA1

**Example:** See [Configuring a PER measurement](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["Destination Station"](#) on page 106

**CONFigure:WLAN:SIGN<i>:CONNection:SYNC <sync>**

If enabled, the PER measurements use identical settings as configured in the signaling application.

Refer to the [Chapter 2.3.8.9, "Data frame control settings"](#), on page 71.

**Parameters:**

<sync> OFF | ON  
\*RST: OFF

**Example:** See [Configuring a PER measurement](#).

**Firmware/software:** V3.7.40

**Manual operation:** See ["Link to SIG Frame Settings"](#) on page 106

**CONFigure:WLAN:SIGN<i>:PER:FDEF <Format>, <Bandwidth>, <CodeRate>[, <GuardInterval>, <LTFTType>, <PEDuration>, <CTYPE>, <Streams>, <STBC>]**

Configures the downlink data frames for PER measurements.

**Parameters:**

<Format> NHT | HTM | VHT | HES | HEM  
Selects the frame format  
**NHT:** non-high throughput format (non-HT)  
**HTM:** HT mixed format (HT MF)  
**VHT:** very high throughput format  
**HES:** high efficiency single-user format (HE SU)  
**HEM:** high efficiency multi-user format (HE MU)

<Bandwidth> BW20 | BW40 | BW80 | BW160  
Channel bandwidth  
The value must not exceed the operating channel bandwidth, see [CONFigure:WLAN:SIGN<i>:RFSettings:OCWidth](#) on page 148.  
\*RST: BW20

<CodeRate> BR12 | QR12 | QR34 | Q1M12 | Q1M34 | Q6M23 | Q6M34 | BR34 | MCS | MCS1 | MCS2 | MCS3 | MCS4 | MCS5 | MCS6 | MCS7 | D1MBit | D2Mbits | C55Mbits | C11Mbits | MCS8 | MCS9 | MCS10 | MCS11 | MCS12 | MCS13 | MCS14 | MCS15  
See rate list in [CONFigure:WLAN:SIGN<i>:CONNection:MFDef](#)  
\*RST: BR12

<GuardInterval> LONG | SHORT | GI08 | GI16 | GI32  
**SHORT, LONG:** short or long guard interval (up to 802.11ac)  
**GI08, GI16, GI32:** 0.8 µs, 1.6 µs, and 3.2 µs guard interval durations (for 802.11ax)



|                           |                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <LTFTType>                | X1   X2   X4<br>1x HE-LTF, 2x HE-LTF, 4x HE-LTF for 802.11ax                                                                                                                                                                                                              |
| <PEDuration>              | PE0   PE4   PE8   PE12   PE16   AUTO<br><b>PEx</b> : additional receive processing time of x $\mu$ s signaled in packet extension (PE) field (only for 802.11ax)<br><b>AUTO</b> : automatic setting based on the reported DUTs capabilities                               |
| <CTYPE>                   | LDPC   BCC<br>Coding type (for 802.11ax - VHT, HE_SU, HE_MU frames only): low density parity check or binary convolution code<br>*RST: LDPC                                                                                                                               |
| <Streams>                 | STR1   STR2<br>Number of streams<br>*RST: STR1                                                                                                                                                                                                                            |
| <STBC>                    | OFF   ON<br>Enables / disables space time block coding (STBC). If disabled, spatial multiplexing is used.<br>*RST: OFF                                                                                                                                                    |
| <b>Example:</b>           | See <a href="#">Configuring a PER measurement</a>                                                                                                                                                                                                                         |
| <b>Firmware/software:</b> | V3.5.120, V3.7.30: added <Format> for HE, <ChanBW> BW160, <GuardInterval> GI08, GI16, GI32, <LTFTType>, <PEDuration>, <CTYPE>, V3.7.50: added <Streams>, V3.7.60: removed <Format> HTG, added <STBC> V3.7.70: added support of LDPC <CTYPE> also for VHT and HE_MU frames |
| <b>Options:</b>           | R&S CMW-KS651 for 802.11n (HT)<br>R&S CMW-KS656 for 802.11ac (VHT)<br>R&S CMW-KS657 for 802.11ax (HE)                                                                                                                                                                     |
| <b>Manual operation:</b>  | See <a href="#">"Frame Format"</a> on page 107                                                                                                                                                                                                                            |

---

**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:ALSField** <CH20\_Index>, <Subfield>

**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:ALSField?** <CH20\_Index>

Sets 8-bit indices for resource unit (RU) allocation for the selected channel. The <Subfield> parameter specifies the number of RUs, their position and size. Refer to IEEE Std 802.11ax-2021, table 27-26, RU allocation subfield.

**Parameters:**

<Subfield> A000 | A001 | A002 | A003 | A004 | A005 | A006 | A007 | A008 |  
 A009 | A010 | A011 | A012 | A013 | A014 | A015 | A016 | A024 |  
 A032 | A040 | A048 | A056 | A064 | A072 | A080 | A088 | A096 |  
 A112 | A113 | A114 | A115 | A116 | A120 | A128 | A192 | A200 |  
 A208 | A216 | A224  
 \*RST: A000

**Parameters for setting and query:**

<CH20\_Index> CHA1 | CHA2 | CHA3 | CHA4

**Example:** See [Configuring a PER measurement](#).

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["20 MHz Channel Settings"](#) on page 108

**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:BLALlocation? <CH20\_Index>**

Queries the allocation state and size of an entire 20MHz block for the specified channel.

**Query parameters:**

<CH20\_Index> CHA1 | CHA2 | CHA3 | CHA4

**Return values:**

{<1\_RU1> OFF | USR1 | DMY1 | DMY2 | DMY3  
 User mapping for to the corresponding RU (no user, user 1 or dummy user)  
 <2\_Size1>}.. T26 | T52 | T106 | T242 | T484 | T996 | T2X9  
 RU size: 26-, 52-, 106-, 242-, 484, 996-tone RU, 2x996-tone RU  
 ..{<17\_RU9> OFF | USR1 | DMY1 | DMY2 | DMY3  
 <18\_Size9>} T26 | T52 | T106 | T242 | T484 | T996 | T2X9

**Example:** See [Configuring a PER measurement](#).

**Usage:** Query only

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["20 MHz Channel Settings"](#) on page 108

**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:DUMMy<index>:MCS**

<MCS\_Index>

Sets the modulation and coding scheme for the corresponding dummy user.

**Suffix:**

<index> 1..3  
Dummy user selection

**Parameters:**

<MCS\_Index> MCS | MCS1 | MCS2 | MCS3 | MCS4 | MCS5 | MCS6 | MCS7 | MCS8 | MCS9 | MCS10 | MCS11  
**MCS, MCS1,...,MCS11:** MCS 0 to MCS 11  
\*RST: MCS

**Example:** See [Configuring a PER measurement](#).

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["Users and Dummies"](#) on page 108

**CONFIGure:WLAN:SIGN<i>:PER:DFRame:HEMU:RUAllocation** <CH20\_Index>, <RU\_Index>, <AllocState>

**CONFIGure:WLAN:SIGN<i>:PER:DFRame:HEMU:RUAllocation?** <CH20\_Index>, <RU\_Index>

Configures allocations for specified channel and resource unit (RU). Maps a user to the RU, sets the size of allocation.

**Parameters:**

<AllocState> OFF | USR1 | DMY1 | DMY2 | DMY3  
User mapping for to the selected RU

**Parameters for setting and query:**

<CH20\_Index> CHA1 | CHA2 | CHA3 | CHA4  
<RU\_Index> RU1 | RU2 | RU3 | RU4 | RU5 | RU6 | RU7 | RU8 | RU9  
Resource unit selection

**Return values:**

<Size> T26 | T52 | T106 | T242 | T484 | T996 | T2X9  
RU size: 26-, 52-, 106-, 242-, 484, 996-tone RU, 2x996-tone RU  
\*RST: T242

**Example:** See [Configuring a PER measurement](#).

**Firmware/software:** V3.7.30, V3.7.50: changed <Size> default value

**Options:** R&S CMW-KS657

**Manual operation:** See ["Users and Dummies"](#) on page 108

**CONFIGure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:Allocation** <CH20\_Index>, <RU\_Index>

Configures allocations for the user in HE MU PPDU. Maps the user to a resource unit (RU).

**Suffix:**

&lt;index&gt; 1

**Parameters:**

&lt;CH20\_Index&gt; CHA1 | CHA2 | CHA3 | CHA4

\*RST: CHA1

&lt;RU\_Index&gt; RU1 | RU2 | RU3 | RU4 | RU5 | RU6 | RU7 | RU8 | RU9

\*RST: RU1

**Example:**See [Configuring a PER measurement](#).**Firmware/software:** V3.7.30**Options:** R&S CMW-KS657**Manual operation:** See ["Users and Dummies"](#) on page 108**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:MCS**  
<MCS\_Index>

Sets the modulation and coding scheme for user 1 (user data assigned to the DUT).

**Suffix:**

&lt;index&gt; 1

**Parameters:**

&lt;MCS\_Index&gt; MCS | MCS1 | MCS2 | MCS3 | MCS4 | MCS5 | MCS6 | MCS7 | MCS8 | MCS9 | MCS10 | MCS11

**MCS, MCS1,...,MCS11:** MCS 0 to MCS 11

\*RST: MCS

**Example:**See [Configuring a PER measurement](#).**Firmware/software:** V3.7.30**Options:** R&S CMW-KS657**Manual operation:** See ["Users and Dummies"](#) on page 108**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:CTYPe** <Coding Type>

Selects the coding type related to a user for HE MU PPDU.

**Suffix:**

&lt;index&gt; 1

**Parameters:**

&lt;Coding Type&gt; LDPC | BCC

\*RST: LDPC

**Example:**See [Configuring a PER measurement](#).**Firmware/software:** V3.7.30

**Options:** R&S CMW-KS657

**Manual operation:** See ["Users and Dummies"](#) on page 108

---

**CONFigure:WLAN:SIGN<i>:PER:DFRame:HEMU:USER<index>:STReams**  
 <streams>

Sets the number of streams for PER measurements used by the user for MIMO connections.

**Suffix:**

<index> 1  
 User

**Parameters:**

<streams> STR1 | STR2  
 One or two streams  
 \*RST: STR1

**Example:** See [Configuring a PER measurement](#).

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KS657

**Manual operation:** See ["Users and Dummies"](#) on page 108

### 2.5.19.3 Measurement results

The following commands return the results of the PER measurement.

|                                                    |     |
|----------------------------------------------------|-----|
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:PER?</a>      | 233 |
| <a href="#">READ:WLAN:SIGN&lt;i&gt;:PER?</a>       | 233 |
| <a href="#">FETCh:WLAN:SIGN&lt;i&gt;:PACKrate?</a> | 234 |

---

**FETCh:WLAN:SIGN<i>:PER?**

**READ:WLAN:SIGN<i>:PER?**

Returns all results of the PER measurement.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<PER> float  
 Range: 0 % to 100 %

<CurrentNoPackets> decimal  
 Number of MAC data packets that were sent since the start of the measurement.  
 Range: 0 to total number of packets (see [CONFigure:WLAN:SIGN<i>:PER:PACKets](#))

|                           |                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------|
| <PacketsLost>             | decimal<br>Number of packets lost (unacknowledged) since the start of the measurement. No retransmission. |
| <RXBurstPower>            | float<br>Average received burst power of uplink ACK frames<br>Default unit: dBm                           |
| <b>Example:</b>           | See <a href="#">Running a PER measurement</a>                                                             |
| <b>Usage:</b>             | Query only                                                                                                |
| <b>Firmware/software:</b> | V2.1.30<br>V3.0.10 RX burst power parameter                                                               |
| <b>Manual operation:</b>  | See <a href="#">"Scalar results"</a> on page 103                                                          |

---

#### FETCH:WLAN:SIGN<i>:PACKrate?

Returns the modulation and coding rate/scheme of the last received ACK frame.

##### Return values:

|                           |                                                                                                                                                                                                                                                                                                                    |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>             | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                                                                                   |
| <rate>                    | BR12   QR12   QR34   Q1M12   Q1M34   Q6M23   Q6M34  <br>BR34   MCS   MCS1   MCS2   MCS3   MCS4   MCS5   MCS6  <br>MCS7   D1MBit   D2Mbits   C55Mbits   C11Mbits   MCS8  <br>MCS9   MCS10   MCS11   MCS12   MCS13   MCS14   MCS15<br>See rate list in <a href="#">CONFIGure:WLAN:SIGN&lt;i&gt;:CONNection:MFDef</a> |
| <b>Example:</b>           | See <a href="#">Running a PER measurement</a>                                                                                                                                                                                                                                                                      |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                                                                                                                         |
| <b>Firmware/software:</b> | V3.2.40                                                                                                                                                                                                                                                                                                            |
| <b>Manual operation:</b>  | See <a href="#">"Scalar results"</a> on page 103                                                                                                                                                                                                                                                                   |

## 2.6 List of commands

|                                                                      |     |
|----------------------------------------------------------------------|-----|
| ABORT:WLAN:SIGN<i>:HETBased.....                                     | 216 |
| ABORT:WLAN:SIGN<i>:PER.....                                          | 223 |
| CALL:WLAN:SIGN<i>:ACTion:STATION:CONNection.....                     | 131 |
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| CALL:WLAN:SIGN<i>:STA<s>:ACTion:DISConnect.....                      | 132 |
| CLEAn:WLAN:SIGN<i>:ELOGging.....                                     | 134 |
| CONFigure:WLAN:SIGN<i>:CONNection:AID.....                           | 157 |
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|-----------------------------------------------------------------|-----|
| CONFigure:WLAN:SIGN<i>:CONNection:ASSociation:STAPriority.....  | 164 |
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## 3 WLAN measurements

The WLAN measurements firmware application provides TX tests on WLAN signals.

The TX tests cover the following DUT's transmitter properties:

- Power vs. time for received burst
- Rise and fall power ramp
- DC power
- Modulation accuracy per OFDM symbol (EVM)
- Modulation accuracy per carrier for OFDM signals (EVM)
- Modulation accuracy per chip for DSSS signals (EVM)
- Signal statistics: center frequency error, timing error, symbol clock error, 802.11ax: frequency offset error
- I/Q constellation diagram
- Power of unoccupied 802.11ax subcarriers (unused tone error measurement)
- DUT's power variation across the OFDM subcarriers (spectrum flatness)
- Out-of-band emissions (transmit spectrum mask)

Many of the tests and conformance requirements are specified in IEEE standard 802.11. The application supports WLAN signals according to:

- IEEE Std 802.11-2016
- IEEE Std 802.11ax-2021

To generate a WLAN signal in parallel to a TX measurement, use the GPRF generator and WLAN waveform file.

To process a WLAN waveform file, you need the following options:

- R&S CMW-KW650 for 802.11a/b/g/n/j/p,
- R&S CMW-KW656 for 802.11ac, and
- R&S CMW-KW657 for 802.11ax.

See [Chapter 3.2.6, "WLAN signal properties"](#), on page 247.

### 3.1 What's new

This documentation describes version 4.0.20 and higher of the WLAN measurement application. Compared to version 3.8.20, it provides the following features and changes:

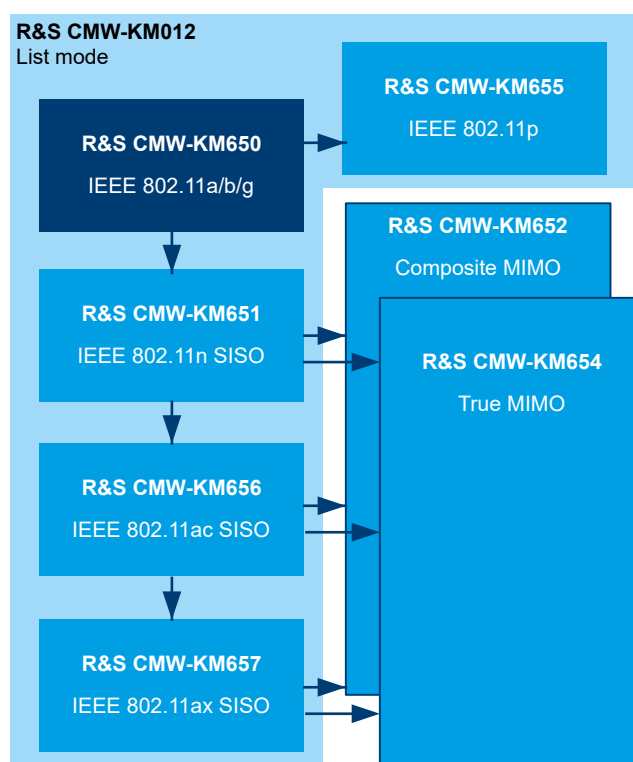
- Sections related to no longer supported hardware have been removed. Concerns:
  - R&S CMWS
  - R&S CMW100 (still supported on different software branch, with separate documentation)
- Added EVM measurement method according to standard IEEE Std 802.11-2016 for 802.11b signals, see [DSSS EVM Method](#).

## 3.2 General description

With basic option R&S CMW-KM650, it is possible to perform 802.11a, b and g TX measurements.



Measuring a signal in the 4 GHz or 5 GHz band requires a 6 GHz option. Refer to the base user manual of your instrument model.



**Figure 3-1: WLAN measurement options**

The following options unlock enhanced functionality:

- R&S CMW-KM651 adds support for 802.11n SISO signals
- R&S CMW-KM655 adds support for 802.11p measurements
- R&S CMW-KM656 adds support for 802.11ac SISO measurements (R&S CMW-KM651 also required)
- R&S CMW-KM657 adds support for 802.11ax SISO measurements (R&S CMW-KM651 and R&S CMW-KM656 also required)
- R&S CMW-KM652 enables composite MIMO measurements:
  - 802.11n MIMO signals require also option R&S CMW-KM651.
  - 802.11ac MIMO signals require also option R&S CMW-KM656.

- 802.11ax MIMO signals require also option R&S CMW-KM657.

See [Chapter 3.2.2.1, "Composite MIMO measurements"](#), on page 244

- R&S CMW-KM654 enables true MIMO measurements:
  - 802.11n true MIMO signals require also option R&S CMW-KM651.
  - 802.11ac true MIMO signals require also option R&S CMW-KM656.
  - 802.11ax true MIMO signals require also option R&S CMW-KM657.

One option R&S CMW-KM654 per R&S CMW is required.

See [Chapter 3.2.2.2, "True MIMO measurements"](#), on page 245.

- R&S CMW-KM012 activates the multi-evaluation list mode for all standards (see [Chapter 3.2.7, "Multi-evaluation list mode"](#), on page 252).

WLAN signaling application allows you to set up an access point or station and to perform "combined signaling path" measurements, see [Chapter 3.2.5, "Combined signaling and measurement"](#), on page 247.

The following sections describe how to perform and configure the WLAN multi-evaluation measurement.

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| • <a href="#">Test setup</a> .....                            | 242 |
| • <a href="#">How to measure a WLAN signal</a> .....          | 243 |
| • <a href="#">Defining the scope of the measurement</a> ..... | 245 |
| • <a href="#">Trigger signal</a> .....                        | 246 |
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| • <a href="#">WLAN signal properties</a> .....                | 247 |
| • <a href="#">Multi-evaluation list mode</a> .....            | 252 |

### 3.2.1 Test setup

Connect the external RF signal source (mobile station, signal generator etc.) to one of the bidirectional RF connectors on the front panel of the R&S CMW.

Connect your DUT with the R&S CMW according to the "Receive Mode" configuration:

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| • <a href="#">SISO</a> .....           | 242 |
| • <a href="#">Composite MIMO</a> ..... | 242 |
| • <a href="#">True MIMO</a> .....      | 243 |

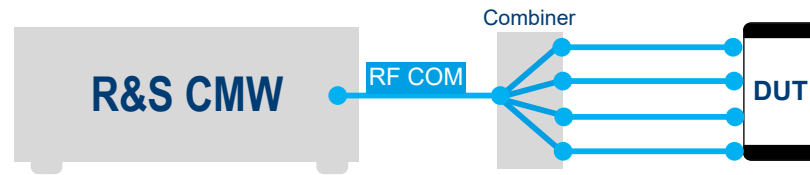
#### 3.2.1.1 SISO

Connect the external RF signal source (mobile station, signal generator etc.) to one of the bidirectional RF connectors on the front panel of the R&S CMW.

#### 3.2.1.2 Composite MIMO

For [Composite MIMO measurements](#), connect the RF signal source to a single bidirectional RF connector on the front panel.

To test a WLAN unit with several antennas, the antenna signals of the DUT must be combined externally. The R&S CMW uses only a single RF input port.

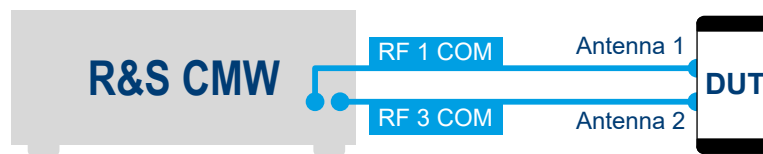


**Figure 3-2: Composite MIMO connector assignment**

To facilitate the evaluation of measurement results, ensure that the RF connections between the individual antennas and the external combiner have the same physical properties, e.g. the same attenuation.

### 3.2.1.3 True MIMO

Two transmitting antennas of the DUT are connected to two RF input ports. Different Rx modules for the two paths are used. R&S CMW500/2xx supports only true MIMO 2x2.



**Figure 3-3: True MIMO with R&S CMW500/2xx**

## 3.2.2 How to measure a WLAN signal

After connecting your WLAN unit to the R&S CMW, you have to adjust the following settings:

- The IEEE standard the signal complies to (802.11n, 802.11ac, 802.11p, ...)
- For 802.11n, ac, and ax signals, the receive mode (SISO, composite MIMO, or true MIMO)
- For 802.11n, 802.11p, 802.11ac, or 802.11ax signals, the signal bandwidth
- The measurements to be performed (enabling / disabling views)
- For true MIMO measurements, select a MIMO scenario
- The used RF connectors
- For 802.11n signals, the transmitted burst type (mixed or greenfield)
- For 802.11p signals, the power class
- The channels to be measured - either by channel number or center frequency
- The "Expected Nominal Power", "User Margin" and "External Attenuation". Recommended values: "Expected Nominal Power" = peak WLAN burst power; "User Margin" = 0 dB.

For the first listed settings, there are dependencies, so that the order is important: Configure first the standard, then the receive mode, then the bandwidth, then the assigned views.



The measurement of a DSSS signal with low data rate and large payload sizes can take up to 40 s. In remote control mode, set the measurement timeout to an adequate value, e.g. to 60 s.

With the settings described above, the R&S CMW is able to synchronize to the incoming bursts. For SISO signals and true MIMO measurements, it decodes the signal.

### 3.2.2.1 Composite MIMO measurements

During composite MIMO (CMIMO) measurements, the R&S CMW "compares" an externally combined MIMO signal to a previously recorded reference signal. Reference signals are recorded in a dedicated training mode.



- This feature requires option R&S CMW-KM652.
- In training mode, the reference DUT has to be configured to send repeatedly the same burst for all antenna measurements.
- In measurement mode, the DUT has to be configured to send the same burst that was used to generate the active training data. Possible heading and trailing symbols due to sequence numbering and checksums can be skipped by the measurement control logic.
- Composite MIMO measurements do not allow decoding of multiple spatial streams. They only enable "composite" modulation accuracy measurements.
- Composite MIMO is available for 802.11n, 802.11ac, and 802.11ax signals.
- Up to 8 streams are supported for 802.11ac and 802.11ax signals  
Up to 4 streams are supported for 802.11n signals

The reference signal is recorded "antenna per antenna". One by one, the individual antennas of a reference DUT ("golden device") have to be connected and recorded. When all antenna signals are successfully recorded, the R&S CMW decodes the burst and stores the acquired "training data" in internal memory. They are available as long as the measurement application is running and can also be stored to a file. In "measurement mode", either training data from internal memory ("internal data") can be used or a previously generated training data file can be loaded.



On an R&S CMW500/2xx, the WLAN measurement generates a file with extension `mif`.

The file formats are not compatible between the different platforms. Example: You cannot reuse a `xmif` file on an R&S CMW500/2xx.



### 3.2.2.2 True MIMO measurements



- True MIMO feature requires one option R&S CMW-KM654 per R&S CMW.
- True MIMO 2x2 is supported with an R&S CMW500/2xx.
- True MIMO is available for 802.11n, 802.11ac, and 802.11ax signals.

In a true MIMO scenario (TMIMO), the R&S CMW records two DUT's antennas, simultaneously connected to different Rx modules of the R&S CMW. The R&S CMW records the signals of all antennas in parallel, see [Chapter 3.2.1.3, "True MIMO"](#), on page 243.

## 3.2.3 Defining the scope of the measurement

### OFDM / OFDMA signals

For OFDM / OFDMA signals, you can configure the number of PSDU symbols to be measured. If the PSDU is shorter than the configured number of symbols, the available PSDU symbols are measured.

In general, the preamble/header is not measured, except for power measurements.

Power vs. time measure over the entire burst, including preamble and payload. Even LTF and data portion of the bursts can be evaluated separately. The following figure shows the PPDU formats with highlighted LTF portion and data portion of the bursts.

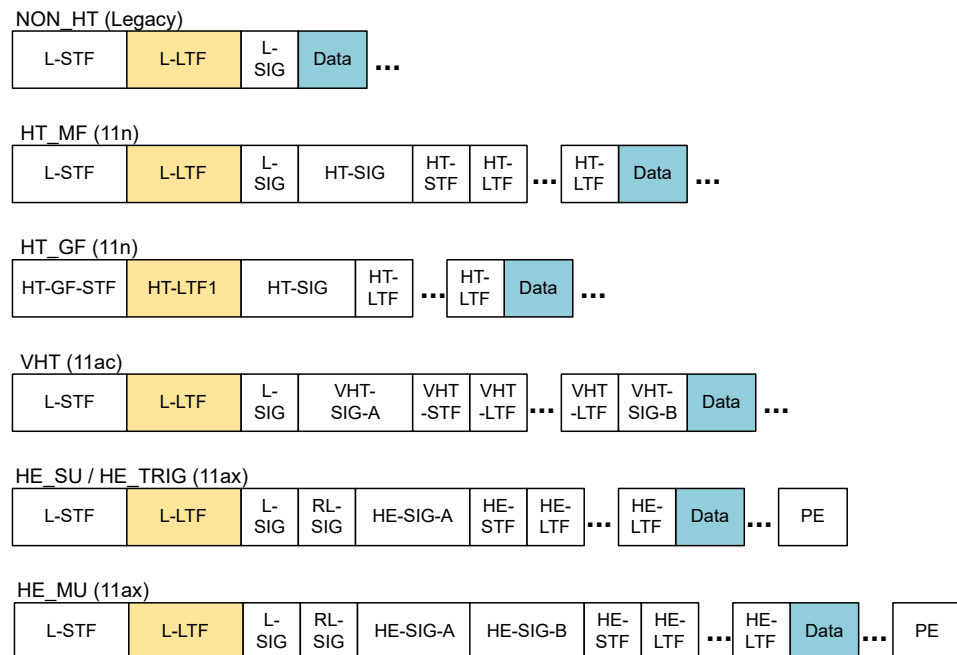


Figure 3-4: PPDU format

### DSSS signals

For DSSS signals, you can configure the number of PSDU chips to be measured. You can also measure the preamble and header, or skip them.

If the PSDU is shorter than the configured number of chips, the available PSDU chips are measured.

The first 26 chips of the burst and the last 26 chips of the burst are never measured.

## 3.2.4 Trigger signal

### Trigger modes for WLAN measurements

The WLAN multi-evaluation measurement starts according to the configured trigger source, see ["Trigger Source"](#) on page 303.

### Trigger signals provided by the WLAN measurement

The WLAN multi-evaluation measurement provides trigger signal that can be used for true MIMO measurements, options R&S CMW-KM654.

To address the trigger signals in remote commands, use "WLAN Meas<i>: Multi Evaluation", with <i> replaced by the instance number of the measurement application.

For configuration, see [Chapter 3.2.2.2, "True MIMO measurements"](#), on page 245.

### 3.2.5 Combined signaling and measurement

The multi-evaluation measurement can be used in combination with the WLAN signaling application (option R&S CMW-KS65x). A connection to the DUT can be set up by the signaling application and the resulting DUT signal can be measured using the multi-evaluation measurement.



#### Limitations

- Channel bandwidth 80+80 MHz is not supported in combined signal path.
- Transmit spectrum mask measurements in combined signal path are fully supported only up to 40 MHz bandwidth.  
For bandwidths > 40 MHz, the measurements only display the results in the frequency range from -80 MHz to 80 MHz.
- An R&S CMW500/2xx is required for CSP

To use both applications in parallel, the "Combined Signal Path" scenario must be activated (see "[Scenario = Combined Signal Path](#)" on page 289). The signal routing and analyzer settings are then configured by the signaling application. The multi-evaluation measurement displays the corresponding signaling settings instead of its own settings. These signaling settings can be configured both in the measurement GUI and in the GUI of the signaling application.

To configure these settings via remote commands, the commands of the signaling application must be used. For a command-mapping table, see [Chapter 3.5.7, "Combined signal path commands"](#), on page 609.



If the same TRX path is used by WLAN measurement application and GPRF generator or WLAN signaling (CSP), the power vs. time measurements partly show also R&S CMW Tx power. The difference in displayed results is approx.  $6\text{ dB} + \text{ExtAttIn} + \text{ExtAttOut}$ .

Whenever the combined signal path scenario is activated or the controlling application is changed, the RX frame trigger signal of the signaling application is selected automatically as a trigger source. Its default settings are suitable for standards up to 802.11ac, channel bandwidth 20 MHz. For other input signals, specify a suitable number of symbols that need to be received before the trigger event is generated ("Min Length"). This trigger source is mandatory for combined signal path measurements.



The recommended "Min Length" of RX frame trigger for bandwidths  $\geq 80\text{ MHz}$  is 6.

### 3.2.6 WLAN signal properties

- [Bands and channel numbers](#)..... 248
- [Physical layer](#)..... 248
- [MIMO signals](#)..... 251

### 3.2.6.1 Bands and channel numbers

The IEEE 802.11 standard and its amendments currently describe WLAN applications in the 2.4 GHz, 3.6 GHz, 4.9 GHz, and 5.0 GHz bands.

The R&S CMW covers all these bands (not all the applications) and implements the canonical channel numbering for the 2.4 GHz, 4 GHz and 5 GHz band. The channel center frequencies are defined at integer multiples of 5 MHz. The relationship between center frequency and channel number is given by the following formula:

$$\text{Channel center frequency} = \text{<lower band edge>} + 5 \times n_{\text{ch}} \text{ (MHz)}$$

Where

- <lower band edge> = 2407 and  $n_{\text{ch}} = 1, 2, \dots, 14$  for the 2.4 GHz band
- <lower band edge> = 4000 and  $n_{\text{ch}} = 0, 1, \dots, 200$  for the 4 GHz band
- <lower band edge> = 5000 and  $n_{\text{ch}} = 0, 1, \dots, 200$  for the 5 GHz band



The set of valid operating channel numbers by regulatory domain is defined in annex E of standard IEEE Std 802.11-2016.

IEEE Std 802.11ax-2021 adds 6 GHz band and defines the related channels in annex E.1 as shown in the following figure.

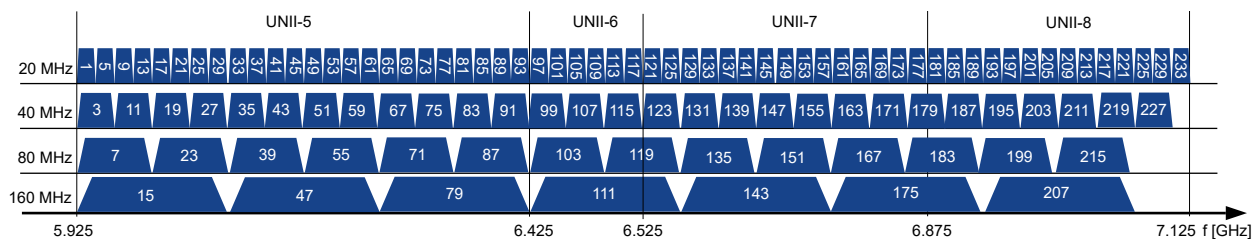


Figure 3-5: Channels in 6 GHz band

### 3.2.6.2 Physical layer

The IEEE 802.11 standard and its amendments describe different physical layer implementations. The following table gives an overview of the supported WLAN standards a, b, g, n, ac, ax, and p.

Table 3-1: 802.11 physical layer properties

| Standard            | 802.11a/g/p OFDM                                                             | 802.11b/g DSSS        | 802.11n                                           | 802.11ac                                                          |
|---------------------|------------------------------------------------------------------------------|-----------------------|---------------------------------------------------|-------------------------------------------------------------------|
| Frequency band      | 802.11a: 5 GHz<br>802.11p: 5.850 to 5.925 GHz<br>802.11g: 2.4 GHz            | 2.4 GHz               | 2.4 GHz, 5 GHz<br>4 GHz <sup>1)</sup>             | 5 GHz                                                             |
| Channel width       | 802.11a/g: 20 MHz<br>802.11p: 5 <sup>2)</sup> , 10 <sup>2)</sup> , or 20 MHz | 20 MHz                | 5 <sup>2)</sup> , 10 <sup>2)</sup> , 20 or 40 MHz | 5 <sup>2)</sup> , 10 <sup>2)</sup> , 20, 40, 80, 160 or 80+80 MHz |
| Transmission scheme | OFDM                                                                         | Single carrier / DSSS | OFDM                                              | OFDM                                                              |

| Standard                                                                                                                                              | 802.11a/g/p OFDM                                                                                                                                                                                                                                                                                                                                         | 802.11b/g DSSS                                                                                            | 802.11n                                                                                                                                               | 802.11ac                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Burst type / PLCP type                                                                                                                                | Direct link                                                                                                                                                                                                                                                                                                                                              | Long PLCP<br>Short PLCP                                                                                   | HT mixed<br>HT greenfield                                                                                                                             | VHT                                                                                                                                                                                                                     |
| Guard interval                                                                                                                                        | Long (0.8 µs)                                                                                                                                                                                                                                                                                                                                            | Long (0.8 µs)                                                                                             | Short (0.4 µs) or long (0.8 µs)                                                                                                                       | Short (0.4 µs) or long (0.8 µs)                                                                                                                                                                                         |
| MU-MIMO                                                                                                                                               | -                                                                                                                                                                                                                                                                                                                                                        | -                                                                                                         | -                                                                                                                                                     | Downlink                                                                                                                                                                                                                |
| Modulation                                                                                                                                            | <b>Modulation type, coding rate (data rates for quarter- half-  full-clocked operation, respectively)</b><br>BPSK, 1/2 (1.5 3 6 Mbps)<br>BPSK, 3/4 (2.25 4.5 9 Mbps)<br>QPSK, 1/2 (3 6 12 Mbps)<br>QPSK, 3/4 (4.5 9 18 Mbps)<br>16-QAM, 1/2 (6 12 24 Mbps)<br>16-QAM, 3/4 (9 18 36 Mbps)<br>64-QAM, 2/3 (12 24 48 Mbps)<br>64-QAM, 3/4 (13.5 27 54 Mbps) | <b>Modulation type (data rate)</b><br>DBPSK (1 Mbps)<br>DQPSK (2 Mbps)<br>CCK (5.5 Mbps)<br>CCK (11 Mbps) | <b>Modulation type, coding rate</b><br>BPSK, 1/2<br>QPSK, 1/2<br>QPSK, 3/4<br>16-QAM, 1/2<br>16-QAM, 3/4<br>64-QAM, 2/3<br>64-QAM, 3/4<br>64-QAM, 5/6 | <b>Modulation type, coding rate</b><br>BPSK, 1/2<br>QPSK, 1/2<br>QPSK, 3/4<br>16-QAM, 1/2<br>16-QAM, 3/4<br>64-QAM, 2/3<br>64-QAM, 3/4<br>64-QAM, 5/6<br>256-QAM, 3/4<br>256-QAM, 5/6<br>1024-QAM, 3/4<br>1024-QAM, 5/6 |
| Payload length                                                                                                                                        | 1 to 1366 symbols                                                                                                                                                                                                                                                                                                                                        | 1 to 4095 bytes                                                                                           | 16 to 1366 symbols                                                                                                                                    | 16 to 1024 symbols                                                                                                                                                                                                      |
| Subcarrier spacing                                                                                                                                    | 312.5 kHz                                                                                                                                                                                                                                                                                                                                                | -                                                                                                         | 312.5 kHz                                                                                                                                             | 312.5 kHz                                                                                                                                                                                                               |
| Symbol time                                                                                                                                           | 3.2 µs                                                                                                                                                                                                                                                                                                                                                   | 3.2 µs                                                                                                    | 3.2 µs                                                                                                                                                | 3.2 µs                                                                                                                                                                                                                  |
| <sup>1)</sup> band not defined by IEEE but supported by the measurement<br><sup>2)</sup> 10 MHz = "half-clocked", 5 MHz = "quarter-clocked" operation |                                                                                                                                                                                                                                                                                                                                                          |                                                                                                           |                                                                                                                                                       |                                                                                                                                                                                                                         |

| Standard            | 802.11ax                                                                                                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frequency band      | 2.4 GHz, 5 GHz                                                                                                                                                                |
| Channel width       | 20 MHz, 40 MHz, 80 MHz, 80+80 MHz, or 160 MHz                                                                                                                                 |
| Transmission scheme | OFDM, OFDMA                                                                                                                                                                   |
| Burst type          | High efficiency single user PSDU ( <b>HE SU PPDU</b> ), one or more PSDUs to one or more users ( <b>HE MU PPDU</b> ), HE trigger-based single user PSDU ( <b>HE TB PPDU</b> ) |
| Guard interval      | 0.8 µs, 1.6 µs, or 3.2 µs                                                                                                                                                     |
| LTF size            | 1x LTF (3.2 µs), 2x LTF (6.4 µs), 4x LTF (12.8 µs)                                                                                                                            |
| MU-MIMO             | Downlink and uplink                                                                                                                                                           |

|                           |                                                                                                                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Standard</b>           | <b>802.11ax</b>                                                                                                                                                                                                                                                                         |
| <b>Modulation</b>         | <b>Modulation type, coding rate</b><br>BPSK, 1/2, BPSK, 1/2 DCM<br>QPSK, 1/2, QPSK, 1/2 DCM<br>QPSK, 3/4<br>16-QAM, 1/2, 16-QAM, 1/2 DCM<br>16-QAM, 3/4, 16-QAM, 3/4 DCM<br>64-QAM, 2/3<br>64-QAM, 3/4<br>64-QAM, 5/6<br>256-QAM, 3/4<br>256-QAM, 5/6<br>1024-QAM, 3/4<br>1024-QAM, 5/6 |
| <b>Payload length</b>     | 16 to 1024 symbols                                                                                                                                                                                                                                                                      |
| <b>Subcarrier spacing</b> | 78.125 kHz                                                                                                                                                                                                                                                                              |
| <b>Symbol time</b>        | 12.8 µs                                                                                                                                                                                                                                                                                 |

The radio resources for OFDM signals are divided into subcarriers (SC) in the frequency domain and into OFDM symbols in the time domain. Within a channel, each subcarrier is identified by an index, with the carrier frequency located in SC number 0. The SC index range, the positions of pilot SC and unused SC indices depend on the WLAN standard. The following table provides an overview of the most important configurations.

**Table 3-2: OFDM subcarriers**

| Protocol<br>802.11... | Channel BW                                | Mode    | SC index                  | Unused SC<br>(Result INV)             | Pilot SC                                          |
|-----------------------|-------------------------------------------|---------|---------------------------|---------------------------------------|---------------------------------------------------|
| a/g                   | 20 MHz                                    | non-HT  | -26 to 26                 | 0                                     | ±7, ±21                                           |
| p                     | 5, 10 or 20 MHz                           |         |                           |                                       |                                                   |
| n, ac                 | 5, 10 or 20 MHz                           | HT, VHT | -28 to 28                 | 0                                     | ±7, ±21                                           |
| n, ac                 | 40 MHz                                    | HT, VHT | -58 to 58                 | -1 to 1                               | ±11, ±25, ±53                                     |
| ac                    | 80 MHz                                    | VHT     | -122 to 122               | -1 to 1                               | ±11, ±39, ±75, ±103                               |
| ac                    | 160 MHz<br>(contiguous)                   | VHT     | -250 to 250               | -129 to -127<br>-5 to 5<br>127 to 129 | ±25, ±53, ±89,<br>±117, ±139, ±167,<br>±203, ±231 |
| ac                    | 80+80 MHz<br>(=non-contiguous<br>160 MHz) | VHT     | -122 to 122 <sup>1)</sup> | -1 to 1 <sup>1)</sup>                 | ±11, ±39, ±75,<br>±103 <sup>1)</sup>              |

<sup>1)</sup> per 80 MHz segment

For 802.11ax in the OFDMA frequency allocation, the resource units (RU) can contain 26, 52, 106, 242, 484 or 996 tones (aka subcarriers) and are in fixed locations. The

tones/subcarriers in the resource units are adjacent and contiguous except in the middle of the channel where direct current (DC) carrier is present. For data subcarrier indices, refer to IEEE Std 802.11ax-2021, section 27.3.2 Subcarrier and resource allocation.

**Table 3-3: OFDM subcarriers for 802.11ax (mode HE)**

| Channel BW           | RU size               | Unused SC                                                                                                                                                                                     | Pilot SC                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 MHz               | 26, 52                | 0, $\pm 69$ , $\pm 122$                                                                                                                                                                       | $\pm 10$ , $\pm 22$ , $\pm 36$ , $\pm 48$ , $\pm 62$ , $\pm 76$ , $\pm 90$ , $\pm 102$ , $\pm 116$                                                                                                                                                                                                                                                                                                                                                   |
|                      | 106, 242              | 0                                                                                                                                                                                             | $\pm 22$ , $\pm 48$ , $\pm 90$ , $\pm 116$                                                                                                                                                                                                                                                                                                                                                                                                           |
| 40 MHz               | 26, 52                | 0, $\pm 3$ , $\pm 56$ , $\pm 57$ , $\pm 110$ , $\pm 137$ , $\pm 190$ , $\pm 191$ , $\pm 244$                                                                                                  | $\pm 10$ , $\pm 24$ , $\pm 36$ , $\pm 50$ , $\pm 64$ , $\pm 78$ , $\pm 90$ , $\pm 104$ , $\pm 116$ , $\pm 130$ , $\pm 144$ , $\pm 158$ , $\pm 170$ , $\pm 184$ , $\pm 198$ , $\pm 212$ , $\pm 224$ , $\pm 238$                                                                                                                                                                                                                                       |
|                      | 106                   | 0, $\pm 3$ , $\pm 110$ , $\pm 137$ , $\pm 244$                                                                                                                                                | $\pm 10$ , $\pm 36$ , $\pm 78$ , $\pm 104$ , $\pm 144$ , $\pm 170$ , $\pm 212$ , $\pm 238$                                                                                                                                                                                                                                                                                                                                                           |
|                      | 242, 484              | 0                                                                                                                                                                                             | $\pm 10$ , $\pm 36$ , $\pm 78$ , $\pm 104$ , $\pm 144$ , $\pm 170$ , $\pm 212$ , $\pm 238$                                                                                                                                                                                                                                                                                                                                                           |
| 80 MHz               | 26, 52                | 0, $\pm 17$ , $\pm 70$ , $\pm 71$ , $\pm 124$ , $\pm 151$ , $\pm 204$ , $\pm 205$ , $\pm 258$ , $\pm 259$ , $\pm 312$ , $\pm 313$ , $\pm 366$ , $\pm 393$ , $\pm 446$ , $\pm 447$ , $\pm 500$ | $\pm 10$ , $\pm 24$ , $\pm 38$ , $\pm 50$ , $\pm 64$ , $\pm 78$ , $\pm 92$ , $\pm 104$ , $\pm 118$ , $\pm 130$ , $\pm 144$ , $\pm 158$ , $\pm 172$ , $\pm 184$ , $\pm 198$ , $\pm 212$ , $\pm 226$ , $\pm 238$ , $\pm 252$ , $\pm 266$ , $\pm 280$ , $\pm 292$ , $\pm 306$ , $\pm 320$ , $\pm 426$ , $\pm 440$ , $\pm 454$ , $\pm 468$ , $\pm 480$ , $\pm 494$ , $\pm 334$ , $\pm 346$ , $\pm 360$ , $\pm 372$ , $\pm 386$ , $\pm 400$ , $\pm 414$ , |
|                      | 106                   | 0, $\pm 17$ , $\pm 124$ , $\pm 151$ , $\pm 258$ , $\pm 259$ , $\pm 366$ , $\pm 393$ , $\pm 500$                                                                                               | $\pm 24$ , $\pm 50$ , $\pm 92$ , $\pm 118$ , $\pm 158$ , $\pm 184$ , $\pm 226$ , $\pm 252$ , $\pm 266$ , $\pm 292$ , $\pm 334$ , $\pm 360$ , $\pm 400$ , $\pm 426$ , $\pm 468$ , $\pm 494$                                                                                                                                                                                                                                                           |
|                      | 242, 484              | 0                                                                                                                                                                                             | $\pm 24$ , $\pm 50$ , $\pm 92$ , $\pm 118$ , $\pm 158$ , $\pm 184$ , $\pm 226$ , $\pm 252$ , $\pm 266$ , $\pm 292$ , $\pm 334$ , $\pm 360$ , $\pm 400$ , $\pm 426$ , $\pm 468$ , $\pm 494$                                                                                                                                                                                                                                                           |
|                      | 996                   | 0                                                                                                                                                                                             | $\pm 24$ , $\pm 92$ , $\pm 158$ , $\pm 226$ , $\pm 266$ , $\pm 334$ , $\pm 400$ , $\pm 468$                                                                                                                                                                                                                                                                                                                                                          |
| 80+80 MHz<br>160 MHz | 26, 52, 106, 242, 484 | 0                                                                                                                                                                                             | {pilot subcarrier indices in 80 MHz - 512, pilot subcarrier indices in 80 MHz + 512}                                                                                                                                                                                                                                                                                                                                                                 |
|                      | 996                   | 0                                                                                                                                                                                             | {for the lower 80 MHz, pilot subcarrier indices in 80 MHz - 512, for the upper 80 MHz, pilot subcarrier indices in 80 MHz + 512}                                                                                                                                                                                                                                                                                                                     |



The multi-evaluation measurement has the following limitations:

- For 802.11n MIMO signals, only the high throughput (HT) mode is supported.

### 3.2.6.3 MIMO signals

The R&S CMW supports the measurement of MIMO signals.

- Option R&S CMW-KM652 enables [Composite MIMO measurements](#)
- Option R&S CMW-KM654 enables [True MIMO measurements](#)

Option R&S CMW-KM651 is a prerequisite for all MIMO options.

This section introduces some terms specific to MIMO signals and helpful for understanding the measurement results. The general properties of corresponding SISO signals apply also to MIMO (see [Table 3-1](#)).

The transmitter block diagrams for MIMO signals are complex and described in detail in IEEE Std 802.11-2016, section 19.3.3. In short, the transmission procedure for data fields consists of the following steps:

- The data passes a scrambler, an encoder parser and FEC encoders.

- A stream parser divides the data into  $N_{SS}$  **spatial streams**.
- Each spatial stream separately passes an interleaver and a constellation mapper.
- The  $N_{SS}$  spatial streams are mapped to  $N_{STS}$  **space-time streams**. Two cases must be distinguished:
  - $N_{SS} < N_{STS}$ : Space-time block coding (STBC) spreads the constellation points of the spatial streams into the space-time streams.
  - $N_{SS} = N_{STS}$ : The constellation points of each spatial stream are mapped unmodified to a space-time stream.
- A spatial mapper maps the  $N_{STS}$  space-time streams to  $N_{TX}$  transmit chains, with  $N_{STS} \leq N_{TX}$ . If constellation points from each space-time stream are mapped directly onto the transmit chains (one-to-one mapping), this method is called direct mapping.
- For each separate transmit chain, the following actions are performed: inverse discrete Fourier transform (IDFT), cyclic shift insertion, guard interval insertion and windowing. Then the RF signal of each chain is transmitted via a separate antenna.

Summary:  $N_{SS}$  spatial streams are mapped to  $N_{STS}$  space-time streams which are mapped to  $N_{TX}$  transmit chains and thus antennas, with  $1 \leq N_{SS} \leq N_{STS} \leq N_{TX} \leq 4$ . Usually, equal values are selected for  $N_{STS}$  and  $N_{TX}$ .

### 3.2.7 Multi-evaluation list mode

Option R&S CMW-KM012 provides multi-evaluation list mode (MELM). In this list mode, the measurement interval can be subdivided into multiple segments, each with different basic input signal properties. The list mode can be used to measure different signal types supported by a DUT, in a single measurement.

The MELM for WLAN measurements supports modulation accuracy and transmit spectrum mask measurements. Segments can differ in the underlying standard, segment time duration, level, frequency, channel bandwidth, etc.



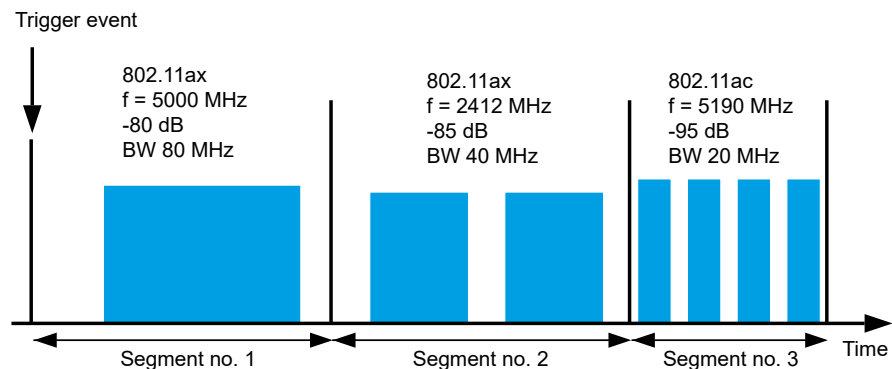
MELM supports only SISO signals with the channel bandwidth of 20-MHz, 40-MHz, 80-MHz or 160-MHz.

For 80+80 MHz signals, measure the two 80 MHz signal segments separately in two different list mode segments:

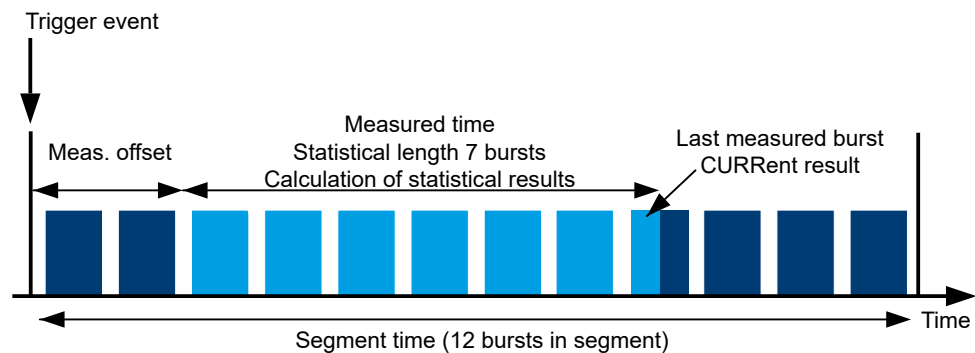
- For one list mode segment, set the central frequency to the value matching with the left 80 MHz segment.
- For another list mode segment, set the central frequency to the value matching with the right 80 MHz segment.

Each segment contains an integer number of bursts with the same underlying configuration as, e.g., standard, channel, level, channel bandwidth. The bursts can have a different duration and also the gaps between bursts can vary. For each segment, the segment duration, measurement offset and measurement time are configurable. The figure below shows a series of three segments. Blue rectangles depict WLAN bursts.





In list mode, the R&S CMW can measure modulation and transmit spectrum mask results. The measured quantities can be enabled or disabled individually for each segment.



#### Limitations:

- The R&S CMW supports up to 100 segments with up to 2000 bursts per segment for modulation measurements. Spectrum measurements support up to 100 segments with up to 1000 bursts per segment.
- The maximum segment time is 1 s. The configuration must fulfill the following criterion:  
 $\text{Segment time} \geq \text{Meas. offset} + \text{Meas. time}$
- The maximum total captured time for all segments is limited (approximately 4.5 s - depends on hardware). Capture buffer overflow indicates, that the volume of captured data exceeds available memory and the measurement time must be reduced.

It is possible to measure all bursts of a segment or to limit the measurement to some bursts at the beginning of the segment. For this limitation, you can define statistical length or measurement time.

The "current" result of a segment refers to the last measured burst of the statistical length. Additional statistical values (average, maximum and standard deviation) are

calculated for the entire statistical length. The example below shows a segment with 10 bursts and a statistical length of five measured bursts.

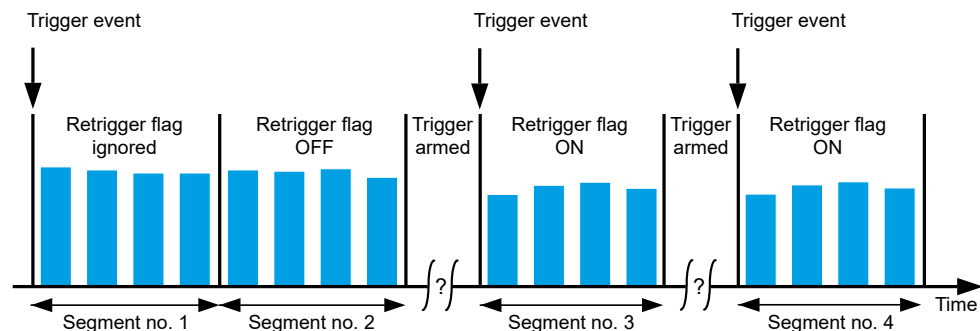
Results of bursts that cannot be measured accurately because of overflow, low signal or synchronization error can be discarded (parameter [Measure on Exception](#) = OFF). In that case, the measurement still tries to provide results for the specified statistical length. If not enough bursts of the segment can be measured, a shorter statistical length is used. In the example shown below, overflow occurs in the fourth burst. The samples of this burst are discarded and burst number 6 is measured to reach the specified statistical length. The reached statistical length, a reliability indicator for the measurement and a reliability indicator for the segment are included in the measurement results.

### Trigger modes

A list mode measurement can either be triggered only once, or it can be retriggered at the beginning of specified segments.

The retrigger flag of each segment is evaluated. It defines whether the measurement waits for a trigger event before measuring the segment, or not. Retriggering the measurement is recommended if the timing of the first timeslot of a segment is inaccurate, e.g. because of signal reconfiguration at the DUT. Furthermore retriggering from time to time can compensate for a possible time drift of the DUT. The retrigger flag of the first segment of the measurement is always ignored (implicitly set to ON).

In the example shown below, the "Segment" trigger mode is enabled. The retrigger flag is OFF for the second segment and ON for the third segment. Thus the measurement stops when the first and second segments have been captured and waits for a trigger event before capturing the third segment.



The list mode is essentially a single-shot remote control application. When a measurement is initiated in list mode, all defined segments are measured once. Afterwards, the results can be retrieved using `FETCH` commands. The following remote control commands are used for list mode settings and result retrieval:

| Parameters                                                                                                                          | SCPI commands                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Activate / deactivate list mode                                                                                                     | CONFigure:WLAN:MEAS<i>:MEValuation:LIST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Enabling results                                                                                                                    | CONFigure:WLAN:MEAS<i>:MEValuation:LIST:RESult:<br>MODulation<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:RESult:TSMask                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Number of segments                                                                                                                  | CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Statistical length (segment time, measurement time, measurement offset, number of measured bursts) per segment and for all segments | CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:<br>STIME<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:<br>MTIME<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:<br>MOFFset<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:<br>SCount<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:STIME<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:MTIME<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:MOFFset<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SCount:<br>MODulation<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SCount:TSMask |
| Signal configuration for all segments: standard, level, frequency, retriggering...                                                  | CONFigure:WLAN:MEAS<i>:MEValuation:LIST:STANdard<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:ENPower<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:FREQuency<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:RTRigger etc., see Chapter 3.5.3.4, "List mode settings", on page 386                                                                                                                                                                                                                                                                          |
| Signal configuration per segment: standard, level, frequency, retriggering...                                                       | CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:<br>SETup<br>CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:<br>RTRigger etc.                                                                                                                                                                                                                                                                                                                                                                                                             |
| Retrieve results                                                                                                                    | FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGment<no>:...<br>FETCh:WLAN:MEAS<i>:MEValuation:LIST:...<br>See Chapter 3.5.5.24, "List mode results (one segment)", on page 580 and Chapter 3.5.5.25, "List mode results (all segments)", on page 589                                                                                                                                                                                                                                                                                                       |

The list mode can be deactivated via command (see table above) and also via the GUI:

- Go to local using the corresponding hotkey.  
The active list mode is indicated in the upper right corner of the current view by the words "List Mode!".
- In the configuration dialog box, section "Measurement Control", disable the list mode.



### Global and list mode parameters

Some settings are available as special list mode settings and as multi-evaluation settings (e.g. WLAN standard). In list mode, the R&S CMW ignores these multi-evaluation parameters. All other settings not available as special list mode settings are taken from the multi-evaluation measurement, e.g.:

- External attenuation
- Measure on exception
- Trigger settings

## 3.3 GUI reference

The following sections provide detailed reference information on the graphical user interface (GUI) and the parameters of the WLAN measurement firmware application.

|                                                 |     |
|-------------------------------------------------|-----|
| • <a href="#">Measurement control</a> .....     | 256 |
| • <a href="#">Measurement results</a> .....     | 256 |
| • <a href="#">Parameters and settings</a> ..... | 286 |
| • <a href="#">CMIMO training mode</a> .....     | 327 |

### 3.3.1 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



#### Multi Evaluation (Softkey)

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:WLAN:MEAS<i>:MEvaluation
ABORt:WLAN:MEAS<i>:MEvaluation
STOP:WLAN:MEAS<i>:MEvaluation
FETCh:WLAN:MEAS<i>:MEvaluation:STATe?
FETCh:WLAN:MEAS<i>:MEvaluation:STATe:ALL?
```

### 3.3.2 Measurement results

The results of the WLAN multi-evaluation measurement are displayed in several different views. Use the "Display" parameters to select the views and to change the appearance and contents of the views.

Which views are available depends on the WLAN standard, receive mode (SISO/MIMO), and channel bandwidth as listed in the following table.

**Table 3-4: Supported measurements**

| Result view                                                                                                                             | SISO               |                          |                  |                   | MIMO (n, ac, ax)   |
|-----------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------------|------------------|-------------------|--------------------|
|                                                                                                                                         | 802.11b, g<br>DSSS | 802.11a, g, n, p<br>OFDM | 802.11ac<br>OFDM | 802.11ax<br>OFDMA | MIMO+8080 (ac, ax) |
| Power vs. time                                                                                                                          | X                  | X                        | X                | X                 | X <sup>1)</sup>    |
| EVM vs symbol                                                                                                                           |                    | X                        | X                | X                 |                    |
| EVM vs carrier                                                                                                                          |                    | X                        | X                | X                 | X <sup>1)</sup>    |
| EVM vs chip                                                                                                                             | X                  |                          |                  |                   |                    |
| I/Q constellation                                                                                                                       | X                  | X                        | X                | X                 |                    |
| Unused tone error                                                                                                                       |                    |                          |                  | X                 | X <sup>1)</sup>    |
| Spectrum flatness                                                                                                                       |                    | X                        | X                | X                 | X <sup>1,2)</sup>  |
| Transmit spectrum mask                                                                                                                  | X                  | X                        | X                | X                 | X <sup>1)</sup>    |
| TX measurement scalar                                                                                                                   | X                  | X                        | X                | X                 | X                  |
| <sup>1)</sup> Only supported for switched and true MIMO.                                                                                |                    |                          |                  |                   |                    |
| <sup>2)</sup> The view is not displayed at the GUI. But the measurement is supported and the results are available via remote commands. |                    |                          |                  |                   |                    |

For the required options, refer to [Chapter 3.2, "General description"](#), on page 241.

The views are described in the following sections.

|                                                                                         |     |
|-----------------------------------------------------------------------------------------|-----|
| ● <a href="#">Overview</a> .....                                                        | 257 |
| ● <a href="#">Views error vector magnitude (EVM)</a> .....                              | 258 |
| ● <a href="#">View IQ constellation</a> .....                                           | 260 |
| ● <a href="#">View unused tone error</a> .....                                          | 261 |
| ● <a href="#">View spectrum flatness</a> .....                                          | 263 |
| ● <a href="#">View transmit spectrum mask</a> .....                                     | 264 |
| ● <a href="#">View power vs. time</a> .....                                             | 268 |
| ● <a href="#">View TX measurement (scalar) for DSSS</a> .....                           | 271 |
| ● <a href="#">View TX measurement (scalar) for OFDM</a> .....                           | 272 |
| ● <a href="#">View TX measurement (scalar) for CMIMO (802.11n, ac, ax)</a> .....        | 275 |
| ● <a href="#">View TX measurement (scalar) for SMIMO, TMIMO (802.11n, ac, ax)</a> ..... | 276 |
| ● <a href="#">View signal fields info</a> .....                                         | 278 |
| ● <a href="#">View TX measurement (scalar) for OFDMA</a> .....                          | 282 |
| ● <a href="#">Selecting and modifying views</a> .....                                   | 284 |
| ● <a href="#">Using markers</a> .....                                                   | 285 |
| ● <a href="#">Common view elements</a> .....                                            | 285 |

### 3.3.2.1 Overview

The "Overview" dialog shows several or all the detailed views in a single diagram.

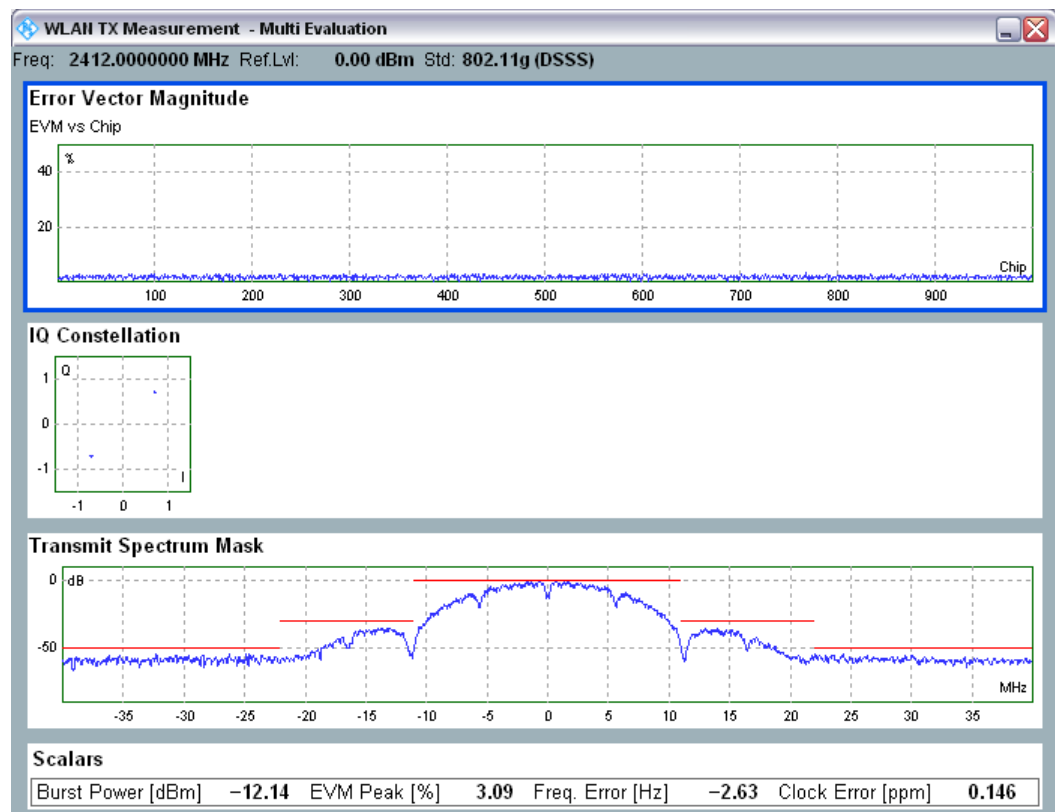


Figure 3-6: Overview

The results to be evaluated and displayed in the overview can be limited using the hotkey "Assign Views" (softkey "Multi Evaluation").

Via "Display" > "Select View" (or double-click), you can enlarge one of the diagrams in the overview and show a detailed view with additional measurement results, see [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284. The particular detailed views are described in the following sections.

### 3.3.2.2 Views error vector magnitude (EVM)

Access: "Display" softkey > "Select View" hotkey > "EVM".

This section applies to the following detailed views:

- "EVM vs Symbol"
- "EVM vs Carrier"
- "EVM vs Chip" (DSSS only)

Each of the detailed views shows a diagram and a statistical overview of results.

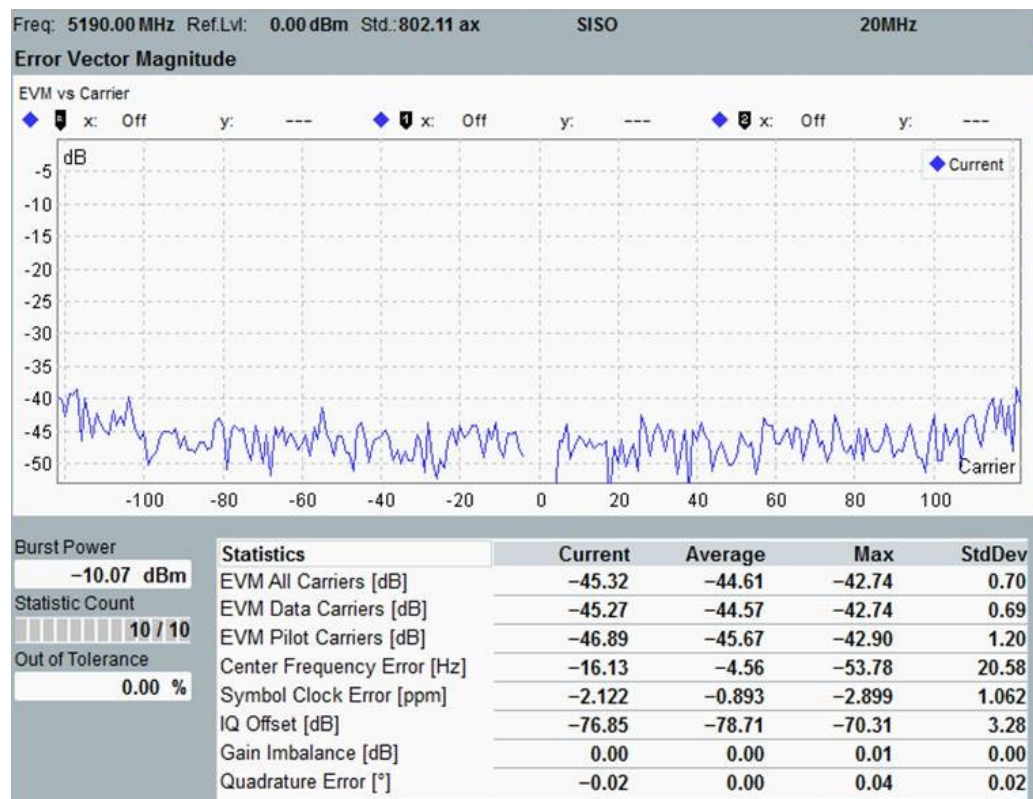


Figure 3-7: EVM vs carrier

Diagram contents:

- EVM vs symbol**  
 The diagram contains one measurement result per OFDM symbol.
- EVM vs carrier**  
 The diagram contains one measurement result per subcarrier. The subcarriers are labeled according to IEEE 802.11. Subcarrier 0 is associated with the center frequency. It is not measured. For additional unused subcarriers, refer to [Table 3-2](#). The subcarrier structure depends on the standard, see [Chapter 3.2.6.2, "Physical layer"](#), on page 248.  
 Except unused subcarriers, for 802.11ax, the R&S CMW automatically excludes from EVM measurements  $\pm 3$  subcarriers; i.e. subcarrier -3 to subcarriers +3 from the central frequency. For meaningful results, especially of I/Q offset and DC power, set the center frequency of the R&S CMW to the same frequency as used by the DUT's local oscillator (LO). It is usually the center frequency of the DUT.  
 I/Q offset is always measured in the center of the signal  
 For 80+80 MHz, two results for I/Q offset and DC power are displayed. The measurement is done in the center of both 80 MHz bands.  
 For MIMO, the results for all received streams or the results for a selected stream can be displayed.

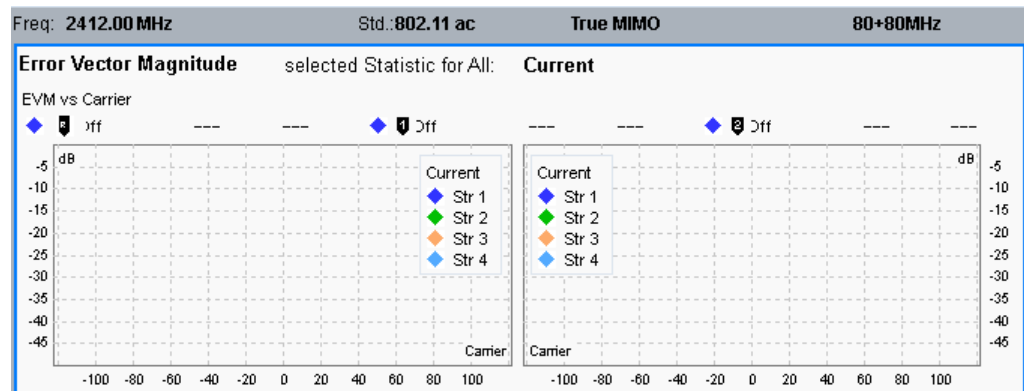


Figure 3-8: EVM vs. carrier (true MIMO 80+80 MHz)

- **EVM vs chip**

Each value is RMS averaged over the samples of a chip. The view is only relevant for DSSS signals.

For a description of the statistical overview, refer to:

- [Chapter 3.3.2.9, "View TX measurement \(scalar\) for OFDM"](#), on page 272
- [Chapter 3.3.2.8, "View TX measurement \(scalar\) for DSSS"](#), on page 271

For query of the diagram contents via remote control, see:

- [Chapter 3.5.5.2, "EVM versus chip traces"](#), on page 459
- [Chapter 3.5.5.3, "EVM versus symbol traces"](#), on page 460
- [Chapter 3.5.5.4, "EVM versus carrier traces"](#), on page 460
- [Chapter 3.5.5.8, "Carrier Frequency offset error traces"](#), on page 468

For query of statistical results via remote control, see:

- [Chapter 3.5.5.20, "Modulation single values, DSSS"](#), on page 522
- [Chapter 3.5.5.21, "Modulation single values, OFDM"](#), on page 524
- [Chapter 3.5.5.22, "Modulation single values, OFDMA"](#), on page 537

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

### 3.3.2.3 View IQ constellation

The I/Q constellation diagram displays the modulation symbols as points in the I/Q plane.



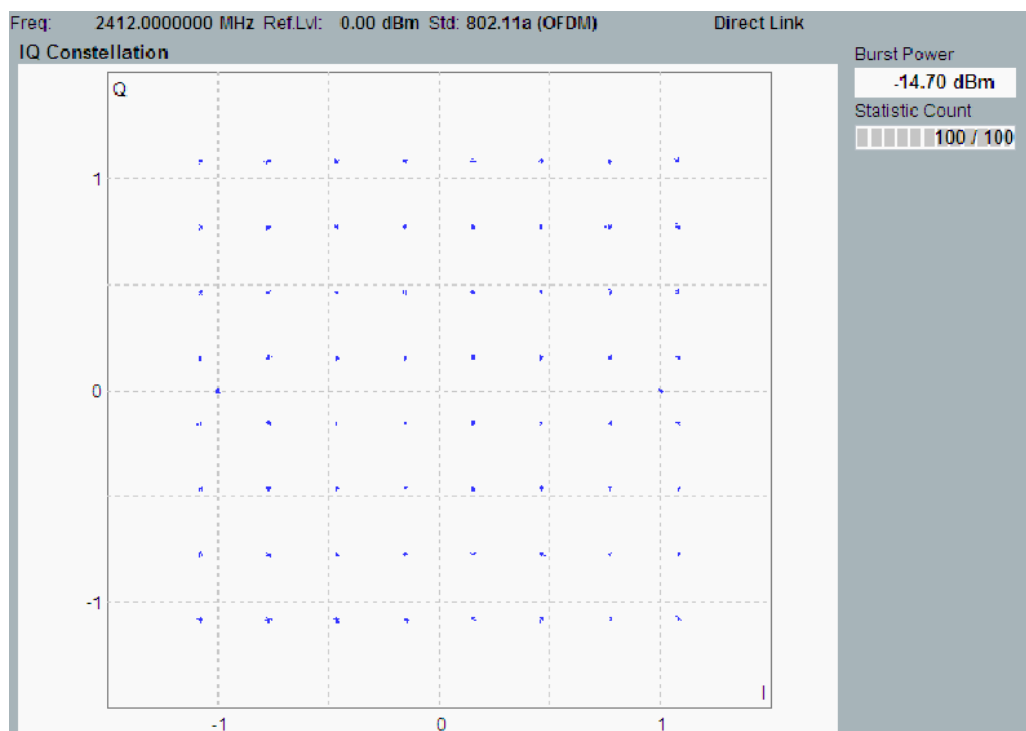


Figure 3-9: I/Q constellation diagram

The expected results depend on the modulation type. For an ideal QPSK signal, the constellation points are on a circle around the origin, with relative phase angles of 90 deg.

Constellation diagrams give a graphical representation of the signal quality and can help to reveal typical modulation errors causing signal distortions.



For correct measurements, set the center frequency of the R&S CMW to the same frequency as used by the DUT's local oscillator (LO). It is usually the center frequency of the DUT.

For 80+80 MHz, the R&S CMW measures in the center of both 80 MHz bands.

For query of the diagram contents via remote control, see:

- [Chapter 3.5.5.5, "I/Q constellation traces"](#), on page 465

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285.

#### 3.3.2.4 View unused tone error

The view comprises a diagram and a table of statistical results of unoccupied tones for an HE TB PPDU signal.

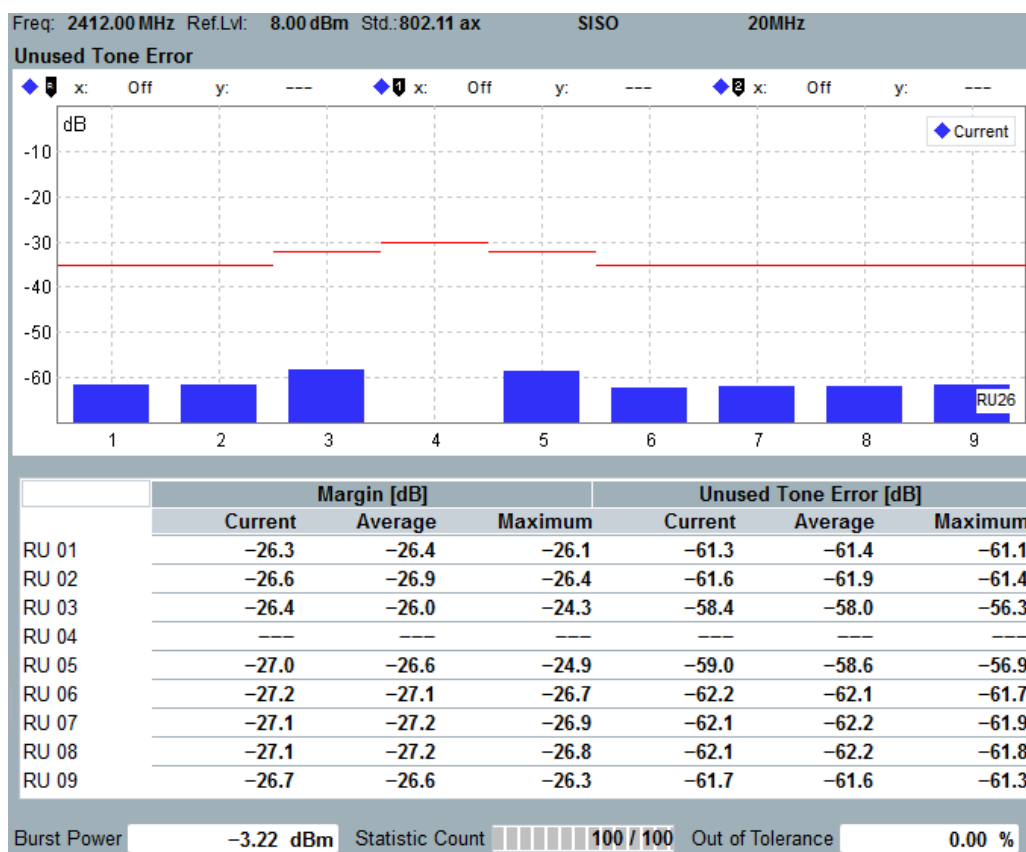


Figure 3-10: Unused tone error diagram

### Diagram

The unused tone error measures the EVM of unoccupied tones averaged over 26 subcarriers (2 MHz). The blue bars correspond to the current resource unit (RU) distribution. The occupied subcarrier in this example is the fourth RU within 26-tone RUs and it is excluded from this measurement.

The limit of the occupied carrier is set as "TB high power" if the "Transmit Power" for unused tone error is set to high. The limit of the occupied carrier is set as "TB low power" if the "Transmit Power" for unused tone error is set to low. See ["Modulation limits 802.11ax"](#) on page 311. The limit line for unoccupied subcarriers is calculated according to IEEE Std 802.11ax-2021, equations (27-131) and (27-132).

### Table

The "Margin" table below the diagram contains values which are relevant for the limit check. The "Margin" values indicate the vertical distance between an estimated limit line and the result trace.

A positive margin indicates that the limit is exceeded. Margin results are displayed for the current, average and maximum trace.

"Unused Tone Error" indicates an unoccupied subcarrier  $EVM_{RMS}$  for each unoccupied 26-tone RU across detected HE TB PPDU.

For query of the results via remote control, see:

- [Chapter 3.5.5.6, "Unused tone error traces"](#), on page 466
- [Chapter 3.5.5.7, "Unused tone error margin values"](#), on page 467

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285.

### 3.3.2.5 View spectrum flatness

The "Spectrum Flatness" view comprises a diagram and a table with margin results.

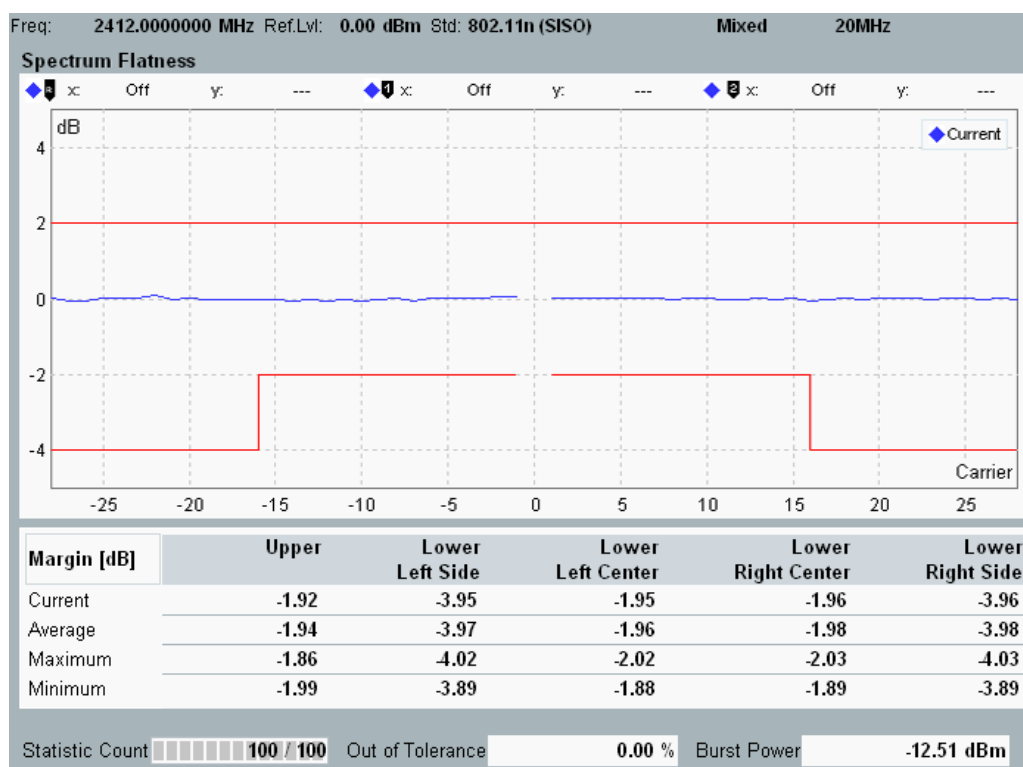


Figure 3-11: Spectrum flatness view (OFDM contiguous)

The diagram displays:

- Relative or absolute spectrum flatness result values. A hotkey "Display Mode" switches between the results, refer to [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.
- Average energy in all allocated subcarriers as a blue line.  
The subcarrier energies are normalized to the arithmetic mean of the average energies in the center carriers, hence the line is vertically centered at 0 dB.
- Configured upper and lower limits as red lines, if limit checking is enabled.  
For background information, see ["Spectrum Flatness Limits OFDM"](#) on page 314.

The margin table presents the margins to the upper and lower spectrum flatness limits, as described in ["Spectrum Flatness Limits OFDM"](#) on page 314.

For 80+80 MHz signals, all results are available per segment. The diagram is split into two diagrams. The margin table is also split into two tables.

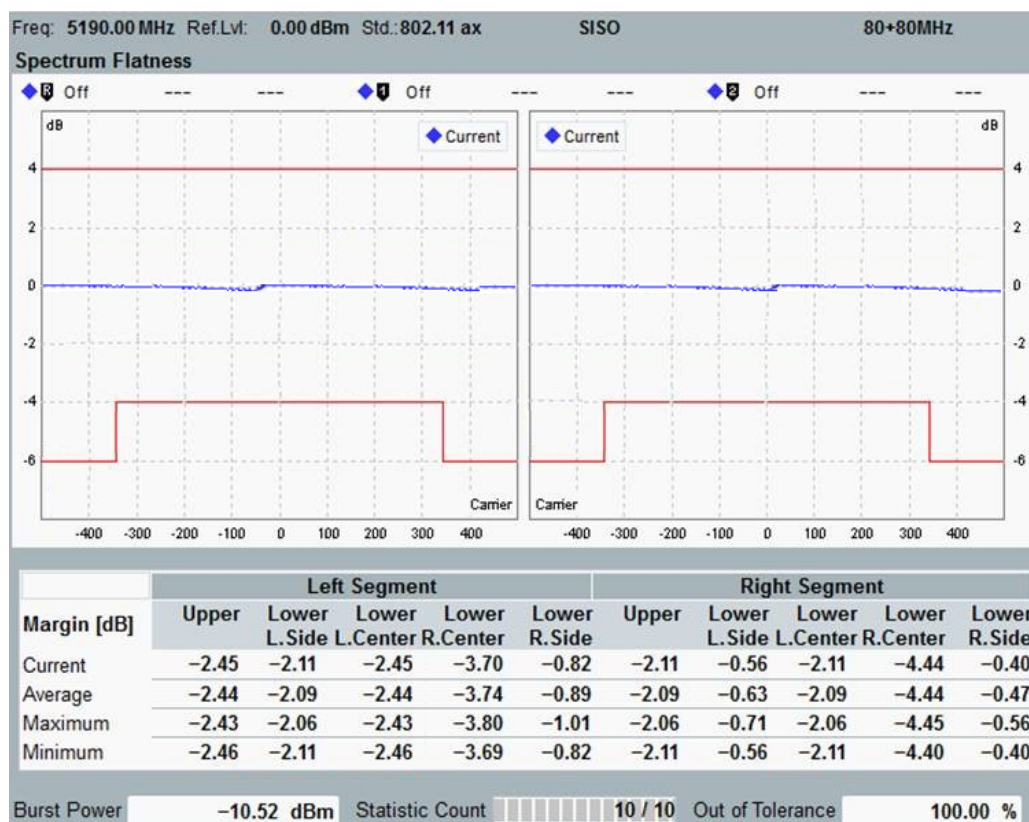


Figure 3-12: Spectrum flatness view (802.11ac, 80+80 MHz)

For query of the results via remote control, see:

- [Chapter 3.5.5.9, "Spectrum flatness traces"](#), on page 469
- [Chapter 3.5.5.10, "Spectrum flatness margin values"](#), on page 475
- [Chapter 3.5.5.11, "Spectrum flatness margin positions"](#), on page 477

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

### 3.3.2.6 View transmit spectrum mask

#### General description

The "Transmit Spectrum Mask" view comprises a diagram and a table of statistical results.

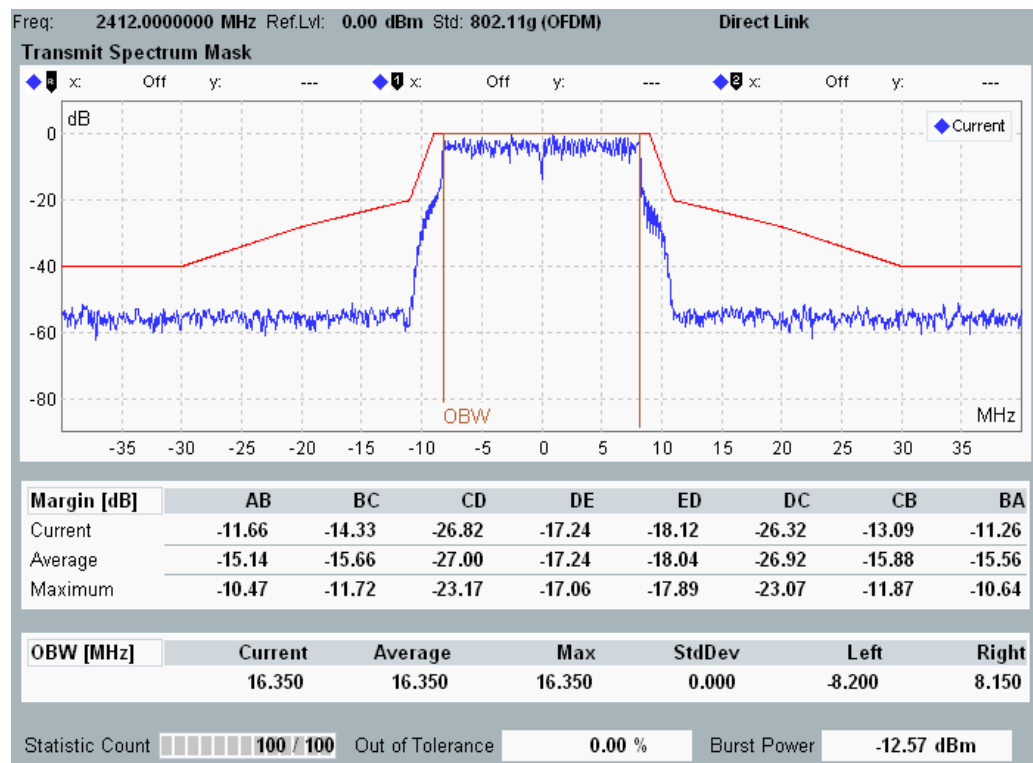


Figure 3-13: IEEE transmit spectrum mask results (OFDM; OBW enabled)

The spectrum measurement covers a symmetric frequency range that is four times the channel bandwidth wide. The 0-MHz point of the X-axis is at the center of the channel.

A value of 0 dB corresponds to the maximum power value (DSSS) or spectral density (OFDM) of the signal measured within the current burst.

The "Margin" table below the diagram contains values which are relevant for the limit check.

See ["Transmit spectrum mask DSSS"](#) on page 316 and subsequent sections.

The "Margin" values indicate the vertical distance between the spectrum mask and the result trace. For each section  $S$  of the spectrum mask (e.g. "AB" or "DC"),  $Margin(S)$  represents the "worst" trace value:

$$Margin(S) = \max \{ Trace(f) - Limit(f) \mid f \text{ in } S \}$$

A positive margin indicates that the limit is exceeded. Margin results are displayed for the current, average and maximum trace.

For 80+80 MHz signals, the margin results are available per segment. For small channel distances, both segments are covered by a single diagram, with the 0-MHz point of the X-axis at the center of the left segment. For large channel distances, there are two diagrams - one per segment.



Transmit spectrum mask measurements in combined signal path are fully supported only up to 40 MHz.

For bandwidths > 40 MHz, the measurements only display the results in the frequency range from -80 MHz to 80 MHz.

For switched and true MIMO measurements, use the hotkey "Display" > "Select Ant. Spectrum" to select the results per antenna or the results for all antennas.

For 80 + 80 MHz, use the hotkey "Display" > "Graphs & Margins" to select the trace and margins results for both segments or individual segment.

- The appearance of **results per antenna/ stream** is similar to SISO results, refer to [View transmit spectrum mask](#).
- The **results for all antennas/ streams** show only average values. They do not display limit lines or OBW results. With the hotkey "Display" > "Select Trace Spectrum", you can select the traces of individual antennas or all antennas including statistics.

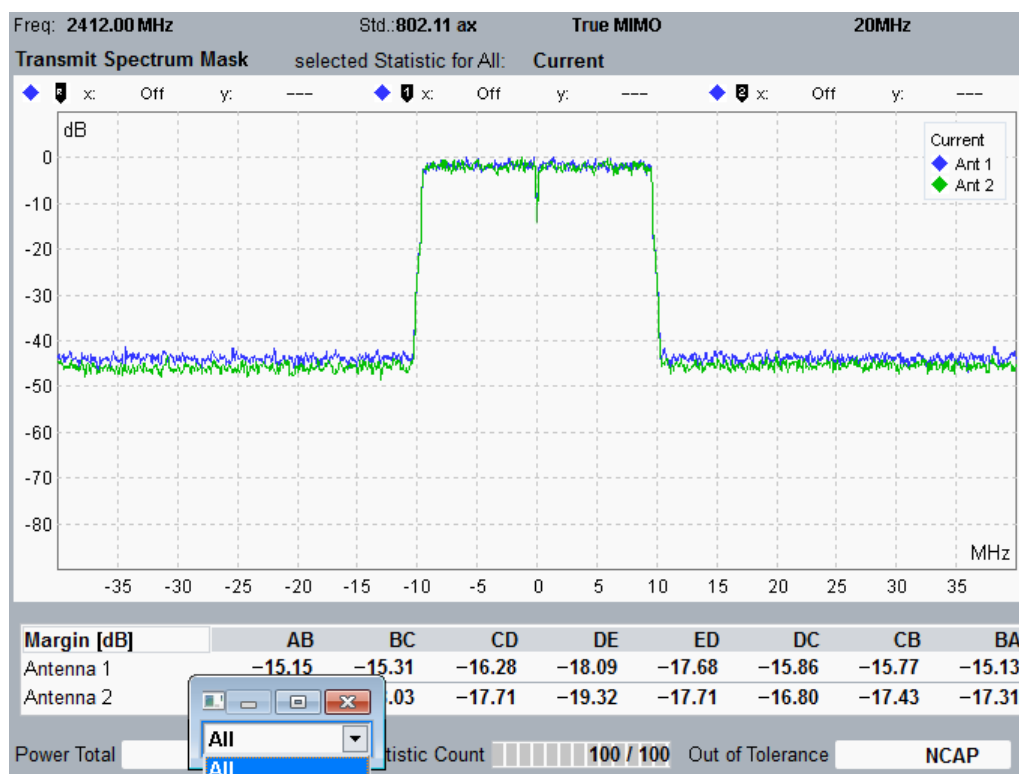


Figure 3-14: Transmit spectrum mask results (MIMO, all antennas)

For query of the results via remote control, see:

- [Chapter 3.5.5.12, "Transmit spectrum mask traces"](#), on page 479
- [Chapter 3.5.5.14, "Transmit spectrum mask margin values"](#), on page 487
- [Chapter 3.5.5.15, "Transmit spectrum mask margin positions"](#), on page 492

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

### Relative vs. absolute limits

For 802.11p signals, you can switch between limit sets with absolute or relative limits. The Y-axis of the diagram changes accordingly (dBm or dB). You can select a mask via the "Display" hotkey "Mask Selection" or via the measurement control settings, see "[Mask Selection](#)" on page 301.

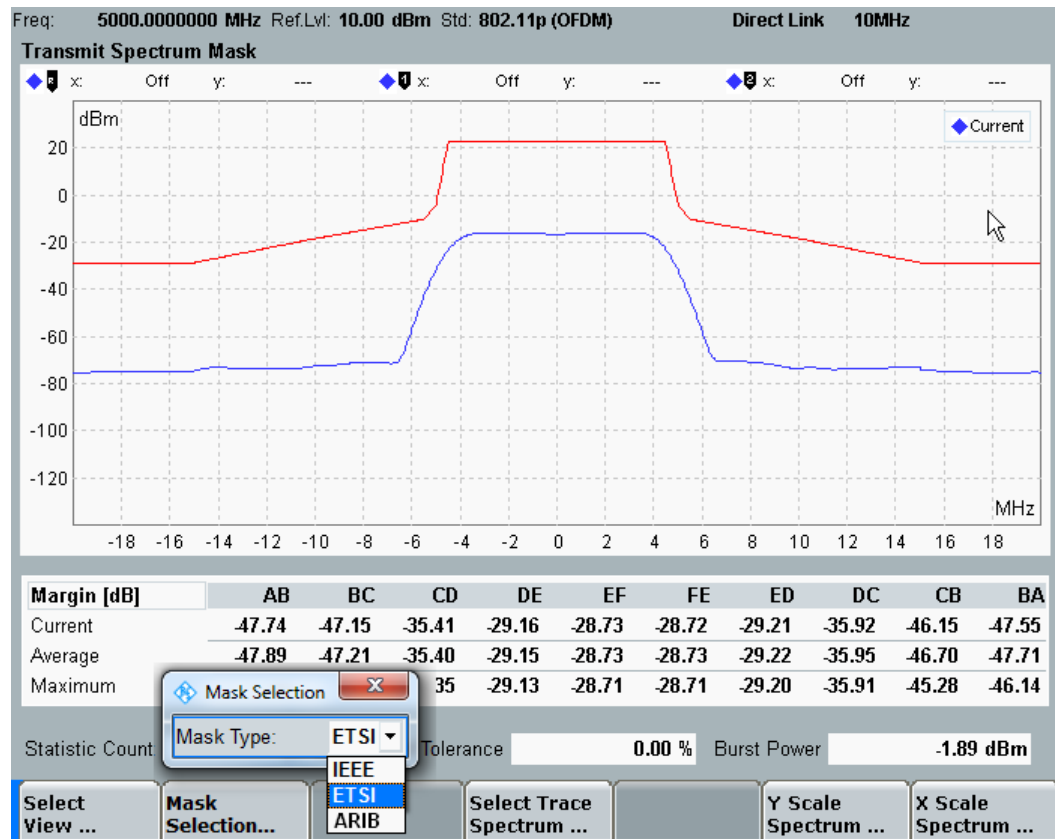


Figure 3-15: ETSI transmit spectrum mask results (802.11p)

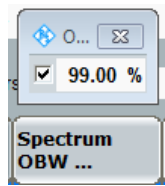
The resolution bandwidth used for the IEEE mask and the ARIB mask is 100 kHz. For the ETSI mask, the resolution bandwidth is 1 MHz.

### Occupied bandwidth

If enabled via the config tree or the "Spectrum OBW" hotkey, the OBW results are shown in the transmit spectrum mask view, see [Figure 3-13](#).

The OBW is indicated by its upper and lower edges in the transmit spectrum mask diagram (orange vertical lines). The numerical results are presented in a dedicated "OBW" table below the "Margin" table. For 80+80 MHz signals, the OBW results are available per segment.

To set the OBW percentage quickly, use the "Spectrum OBW" hotkey.



The OBW percentage refers to the power in the overall covered frequency range (four times the channel bandwidth).

In switched MIMO and true MIMO scenarios, the OBW results are only available in diagrams per antenna (softkey "Display" > "Select Ant. Spectrum"). OBW is not available in diagrams displaying all antennas.

For query of the results via remote control, see:

- [Chapter 3.5.5.16, "Transmit spectrum OBW results"](#), on page 498

### 3.3.2.7 View power vs. time

The "Power vs. Time" results are displayed in three diagrams that can be selectively enabled or disabled:

- The "Burst" diagram, showing the low-resolution burst traces, i.e. signal power in dBm vs. time
- The "Rise" and "Fall" diagrams, showing the high-resolution ramp traces

For DSSS signals, an additional table is displayed below the diagrams, presenting statistical "Rise Time" and "Fall Time" values and highlighting violations of the corresponding limits.

For OFDM signals, the table presents statistical timing error distribution results, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285.



The calculation of timing error is only supported for OFDM standards. It requires a suitable trigger to be specified as a trigger source.

Internal IF power trigger and MAC frame RX trigger cannot be used.



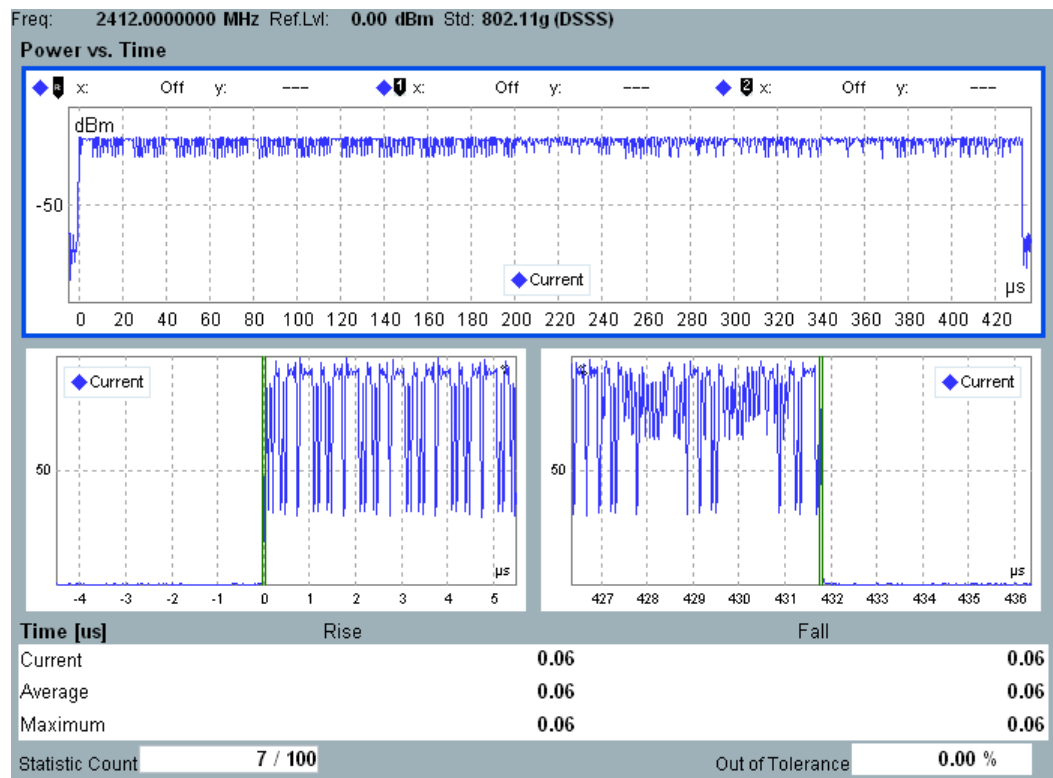


Figure 3-16: WLAN multi-evaluation: Power vs. time results

Configure the measurement always to cover whole bursts.

The content of the ramp diagrams depends on the selected "Standard":

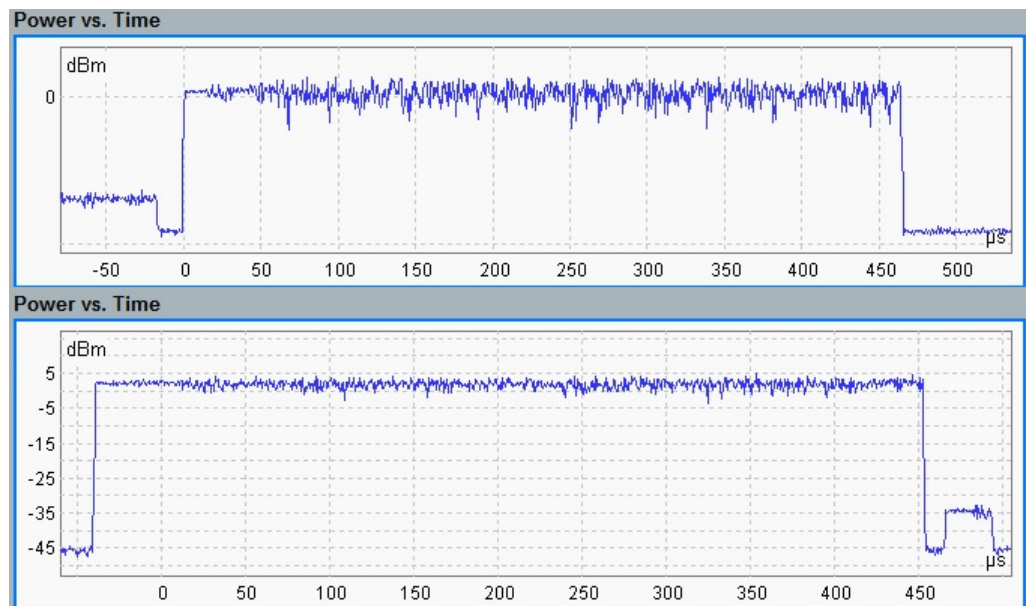
- For DSSS signals, the focus is on the associated limits (see ["Power vs. time limits"](#) on page 305). The signal power is presented as a percentage of the "Reference Power", which can be set to either the maximum or to the mean burst power. The green vertical lines indicate the 10% and 90% threshold crossings, the 0 μs position corresponds to the start of the measured frame.
- For OFDM signals, the diagrams show the absolute signal power in dBm.



For correct measurements, set the center frequency of the R&S CMW to the same frequency as used by the DUT's local oscillator (LO). It is usually the center frequency of the DUT.

### Deviation in combined signal path

If the same TRX path is used by WLAN measurement application and GPRF generator or WLAN signaling (CSP), the power vs. time measurements partly show also R&S CMW Tx power. The difference in displayed results is approx.  $6 \text{ dB} + \text{ExtAttIn} + \text{ExtAttOut}$ . The following figure shows measurements at HE SU and HE TB burst types. In upper graph for HE SU, the trigger frame before the Rx burst is visible, for HE\_TB an ACK after the Rx burst is visible.



**Figure 3-17: Power vs. time graphs in CSP**

Upper graph = HE TB burst measurement

Lower graph = HE SU burst measurement

## MIMO

For true MIMO measurements, the results for all RX antennas or the results for a selected antenna can be displayed.

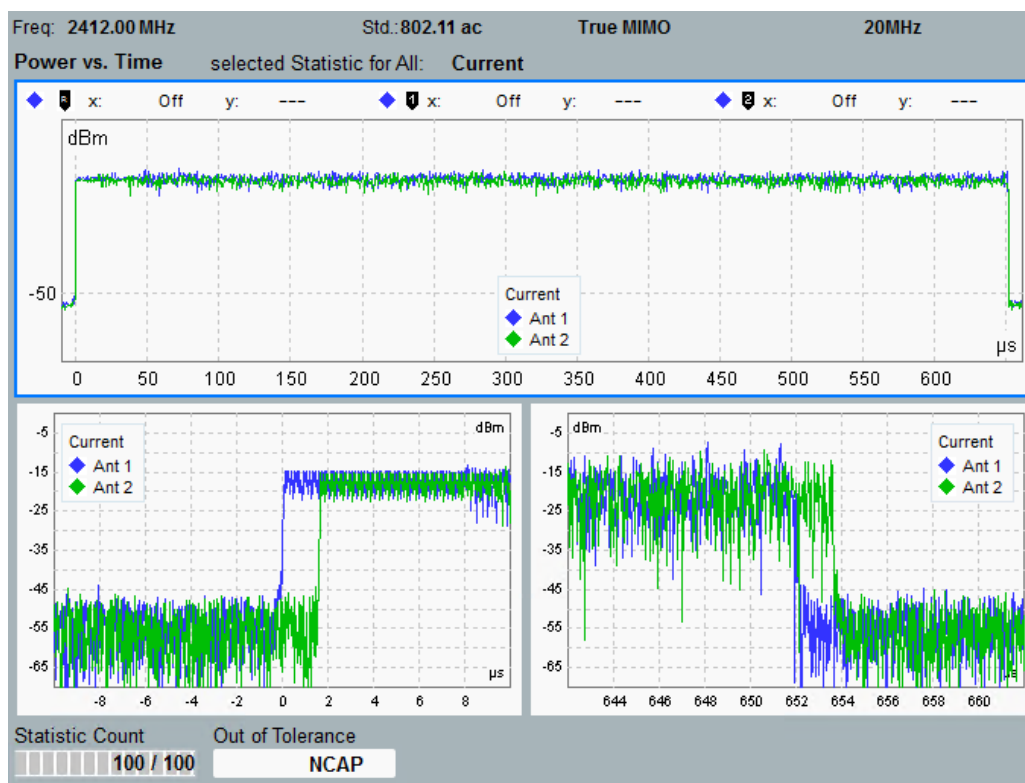


Figure 3-18: Power vs. time graphs (true MIMO)

### 80+80 MHz

For 80+80 MHz, the R&S CMW measures in the center of both 80 MHz bands.

The default view shows the "Burst" diagrams: the left segment in the upper part and the right segment in the lower part.

Double-click one segment diagram to display "Burst", "Rise" and "Fall" diagrams of the selected segment.

With escape key, you return from detailed results of one segment to "Burst" diagrams of the default appearance.

For query of the results via remote control, see:

- [Chapter 3.5.5.17, "Power vs. time burst traces"](#), on page 500
- [Chapter 3.5.5.18, "Power vs. time ramp traces"](#), on page 507
- [Chapter 3.5.5.19, "Power vs. time single values"](#), on page 517
- [Chapter 3.5.5.1, "Timing error traces"](#), on page 458

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

#### 3.3.2.8 View TX measurement (scalar) for DSSS

The view contains a table of statistical results.

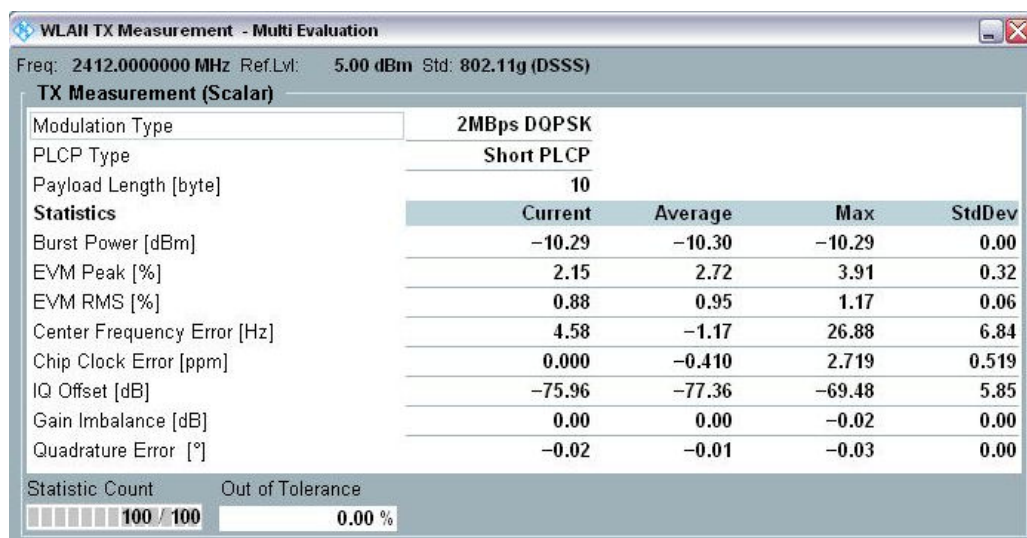


Figure 3-19: WLAN multi-evaluation: Results (DSSS signal)

The table provides the following results in the upper part:

- "Modulation Type" and "PLCP Type" (i.e. burst type): see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "Payload Length": number of bytes in the payload of the measured burst

In the lower part, the table provides results with statistical evaluation:

- "Burst Power": RMS value over the entire burst, including preamble and payload
- Error vector magnitude (EVM) peak and RMS values
- Center frequency error
- Chip clock error
- I/Q origin offset
- "Gain Imbalance": absolute value of (1 + I/Q imbalance)
- "Quadrature Error": angular component of (1 + I/Q imbalance)

For query of the results via remote control, see:

- [Chapter 3.5.5.20, "Modulation single values, DSSS"](#), on page 522

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285.

### 3.3.2.9 View TX measurement (scalar) for OFDM

The view contains a table of statistical results.

For the supported MIMO results, refer to [Table 3-4](#).

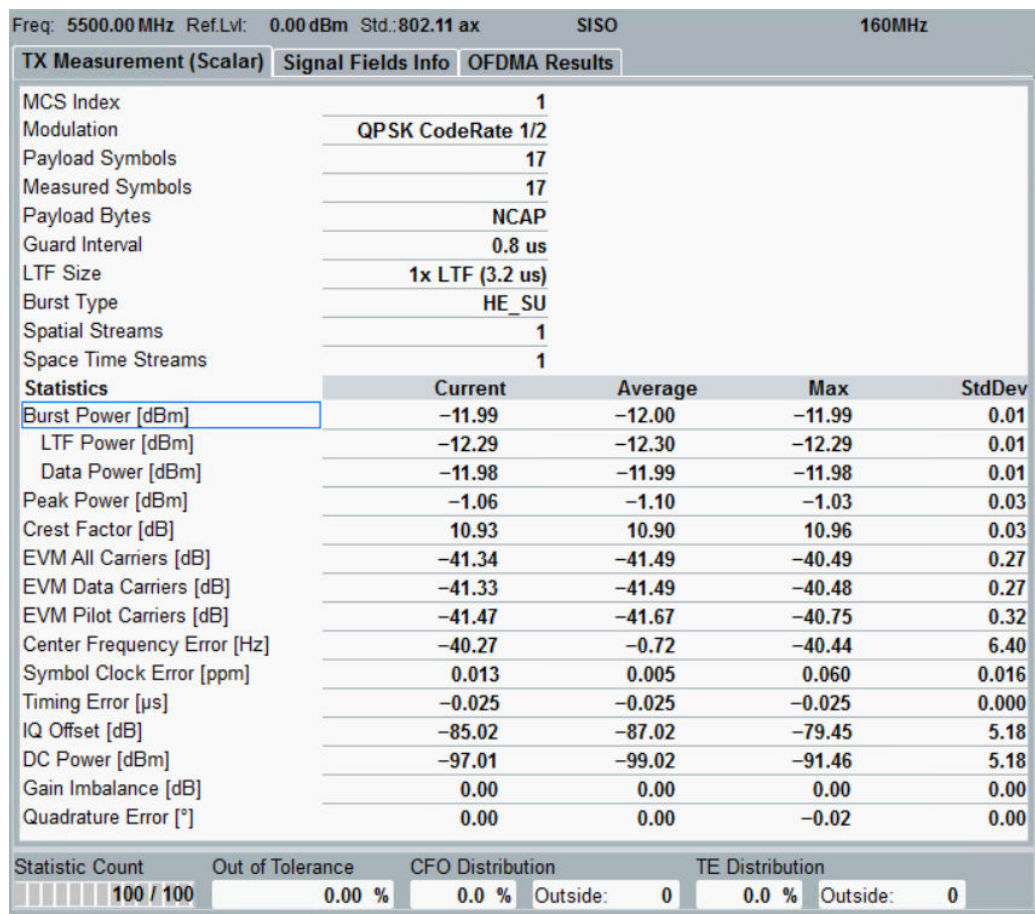


Figure 3-20: OFDM SISO scalar results

The table provides the following results in the upper part:

- "MCS Index": modulation and coding scheme index, determining, for example, the modulation type and coding rate  
Because the coding rate of HE TB PPDU is not signaled, the MCS index of HE TB PPDU cannot be displayed.
- "Modulation": see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "Payload Symbols": number of OFDM symbols in the payload of the measured burst
- "Measured Symbols": number of measured OFDM symbols  
The number of "Measured Symbols" can be smaller than "Payload Symbols" if the evaluation length is reduced manually (see [Evaluation Length, Skip Preamble and Header](#)) or automatically due to memory restrictions.
- "Payload Bytes": number of bytes in the payload of the measured burst
- "Guard Interval": see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "LTF Size": 802.11ax LTF type 1x LTF, 2x LTF or 4x LTF
- "Burst Type": see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "Spatial Streams", "Space Time Streams": number of spatial streams  $N_{SS}$  and space-time streams  $N_{STS}$

- "Power Backoff": minimum distance of signal power to reference level since the start of the measurement  
The backoff result helps you to configure optimum values for the expected power and the user margin.  
The result is not yet displayed on the GUI, but available via remote command.

In the lower part, the table provides results with statistical evaluation:

- "Burst Power": RMS value over the entire burst, including preamble and payload  
With the hotkey "LTF- & Data-Power..." you can show or hide the following burst power results:
  - The power of long training fields (LTF) portion of the burst
  - The power of data portion of the burst
 Refer to [Chapter 3.2.3, "Defining the scope of the measurement"](#), on page 245.
- "Peak Power": peak value over the entire burst (only up to 40 MHz channel bandwidth)
- "Crest Factor": crest factor of the burst
- "EVM": error vector magnitudes for all subcarriers (data and pilot), for data subcarriers only and for pilot subcarriers only
- "Center frequency error"
- "Symbol clock error"
- "Timing Error": The timing error measurement is only supported for OFDM standards. It requires a suitable trigger signal to derive the expected timing. Internal IF power trigger and MAC frame RX trigger cannot be used.
- "I/Q Offset": measured in the center of the signal
- "DC Power": power of the DC subcarriers (center frequency)
- "Gain Imbalance": absolute value of (1 + I/Q imbalance)  
Gain imbalance cannot be calculated if the spectrum is not symmetrical, e.g. for HT\_TB and HE\_MU.
- "Quadrature Error": angular component of (1 + I/Q imbalance)  
Quadrature error cannot be calculated if the spectrum is not symmetrical, e.g. for HT\_TB and HE\_MU.



According to the standard, the test must be performed over at least 20 PPDU's. The PPDU's under test must be at least 16 data OFDM symbols long. A smaller "Payload Length" is highlighted in red and indicated by a warning message.

|                         |                                                                     |
|-------------------------|---------------------------------------------------------------------|
| Payload Length [symbol] | <span style="color: red;">1</span> Standard requires >= 16 symbols! |
|-------------------------|---------------------------------------------------------------------|

For query of the results via remote control, see [Chapter 3.5.5.21, "Modulation single values, OFDM"](#), on page 524.

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

### 3.3.2.10 View TX measurement (scalar) for CMIMO (802.11n, ac, ax)

The view contains a table of statistical results.

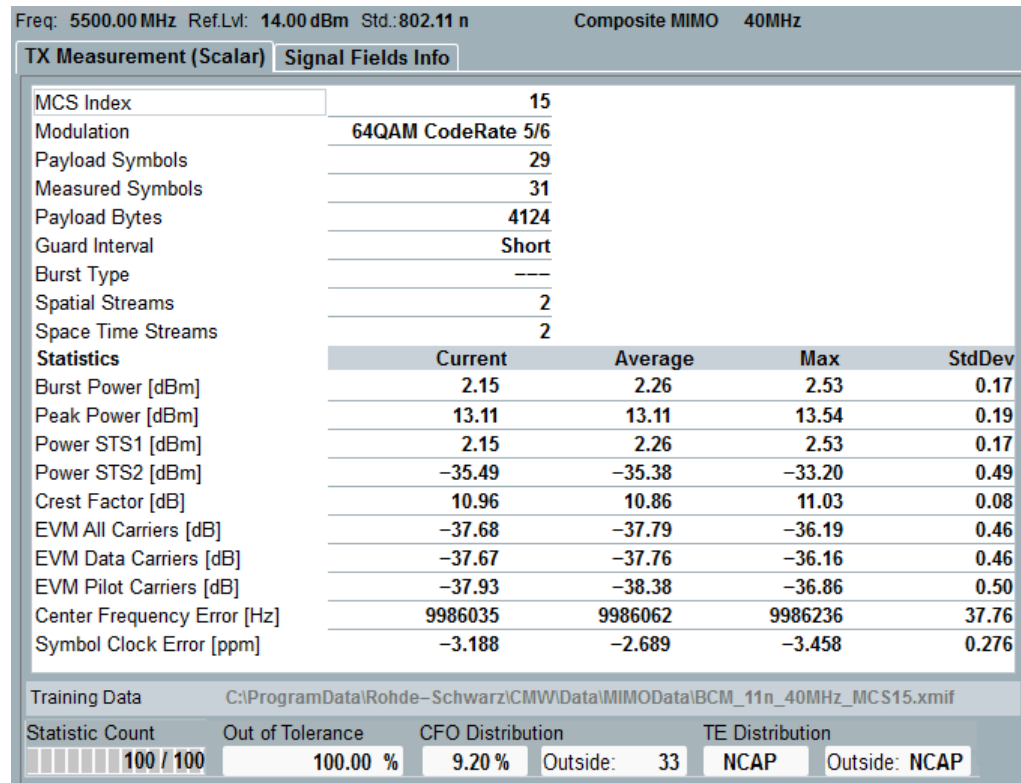


Figure 3-21: Composite MIMO scalar results

The table provides the following results in the upper part:

- "MCS Index": modulation and coding scheme index, determining, for example the modulation type, coding rate and number of spatial streams. For overview tables, see IEEE Std 802.11-2016, section 19.5.
- "Modulation": see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "Payload Symbols": number of OFDM symbols in the payload of the measured burst
- "Measured Symbols": number of measured OFDM symbols  
The number of "Measured Symbols" can be smaller than "Payload Symbols" if the evaluation length is reduced manually (see [Evaluation Length, Skip Preamble and Header](#)) or automatically due to memory restrictions.
- "Payload Bytes": number of bytes in the payload of the measured burst
- "Guard Interval": see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "LTF Size": 802.11ax LTF type 1x LTF, 2x LTF or 4x LTF
- "Burst Type": see [Chapter 3.2.6.2, "Physical layer"](#), on page 248
- "Spatial Streams", "Space Time Streams": number of spatial streams  $N_{SS}$  and space-time streams  $N_{STS}$



- "Spatial Streams", "Space Time Streams": Number of spatial streams  $N_{SS}$  and space-time streams  $N_{STS}$ , see also [Chapter 3.2.6.3, "MIMO signals"](#), on page 251

In the lower part, the table provides results with statistical evaluation:

- "Burst Power", "Peak Power": RMS and peak power of the measured burst  
With the hotkey "LTF- & Data-Power..." you can show or hide the following burst power results:
  - The power of long training fields (LTF) portion of the burst
  - The power of data portion of the burst

Refer to [Chapter 3.2.3, "Defining the scope of the measurement"](#), on page 245.

- "Power STS1" to "Power STS4": RMS power for the individual space-time streams
- EVM: Error vector magnitude of the composite signal. Ideal signals are calculated per space-time stream and combined to a sum signal. The error vector is determined by comparison of the ideal sum signal with the received sum signal. EVM values can always be determined for pilot subcarriers. If the signal contains more than one spatial stream (MCS index > 7), the EVM for the data carriers cannot be determined unless training data are used for the measurement. Without training data, the EVM values for "All Carriers" and "Data Carriers" are not available. See [Chapter 3.2.2.1, "Composite MIMO measurements"](#), on page 244.
- Center frequency error
- "Training Data": see ["Training Data"](#) on page 296

For query of the results via remote control, see:

- [Chapter 3.5.5.21, "Modulation single values, OFDM"](#), on page 524

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

### 3.3.2.11 View TX measurement (scalar) for SMIMO, TMIMO (802.11n, ac, ax)

The view contains a table of statistical results.

Results of switched MIMO and true MIMO measurements are similar.



|                             |  |                    |  |                  |  |          |  |        |  |
|-----------------------------|--|--------------------|--|------------------|--|----------|--|--------|--|
| Freq: 5000.00 MHz           |  | Std.: 802.11 ax    |  | True MIMO        |  | 20MHz    |  |        |  |
| TX Measurement (Scalar)     |  | Signal Fields Info |  | OFDMA Results    |  |          |  |        |  |
| Spatial Streams             |  | 2                  |  |                  |  |          |  |        |  |
| Space Time                  |  | 2                  |  |                  |  |          |  |        |  |
| Data Symbols                |  | 34                 |  |                  |  |          |  |        |  |
| Statistics                  |  | Current            |  | Average          |  | Max      |  | StdDev |  |
| Overview                    |  |                    |  |                  |  |          |  |        |  |
| EVM All [dB]                |  | -43.72             |  | -43.96           |  | -42.33   |  | 0.28   |  |
| EVM Data [dB]               |  | -43.73             |  | -43.96           |  | -42.30   |  | 0.28   |  |
| EVM Pilot [dB]              |  | -43.58             |  | -43.86           |  | -42.78   |  | 0.37   |  |
| Power Total [dBm]           |  | -7.13              |  | -7.13            |  | -7.12    |  | 0.00   |  |
| Symbol Clock Error [ppm]    |  | -0.022             |  | -0.004           |  | 0.158    |  | 0.041  |  |
| Center Frequency Error [Hz] |  | 1.52               |  | -0.49            |  | -16.25   |  | 2.30   |  |
| Streams                     |  |                    |  |                  |  |          |  |        |  |
| EVM All 1 [dB]              |  | -43.83             |  | -44.31           |  | -42.59   |  | 0.38   |  |
| EVM All 2 [dB]              |  | -43.62             |  | -43.63           |  | -41.56   |  | 0.34   |  |
| EVM Data 1 [dB]             |  | -43.85             |  | -44.32           |  | -42.53   |  | 0.38   |  |
| EVM Data 2 [dB]             |  | -43.61             |  | -43.63           |  | -41.51   |  | 0.34   |  |
| EVM Pilot 1 [dB]            |  | -43.30             |  | -44.19           |  | -42.52   |  | 0.54   |  |
| EVM Pilot 2 [dB]            |  | -43.87             |  | -43.55           |  | -41.74   |  | 0.44   |  |
| Antenna                     |  |                    |  |                  |  |          |  |        |  |
| Burst Power 1 [dBm]         |  | -9.86              |  | -9.85            |  | -9.84    |  | 0.00   |  |
| LTF Power 1 [dBm]           |  | -10.21             |  | -10.21           |  | -10.19   |  | 0.00   |  |
| Data Power 1 [dBm]          |  | -9.85              |  | -9.85            |  | -9.83    |  | 0.00   |  |
| Burst Power 2 [dBm]         |  | -10.45             |  | -10.45           |  | -10.43   |  | 0.00   |  |
| LTF Power 2 [dBm]           |  | -10.79             |  | -10.79           |  | -10.76   |  | 0.00   |  |
| Data Power 2 [dBm]          |  | -10.45             |  | -10.45           |  | -10.43   |  | 0.00   |  |
| Peak Power 1 [dBm]          |  | -0.68              |  | -0.67            |  | -0.57    |  | 0.04   |  |
| Peak Power 2 [dBm]          |  | -0.21              |  | -0.10            |  | 0.02     |  | 0.06   |  |
| Crest Factor 1 [dB]         |  | 9.18               |  | 9.18             |  | 9.28     |  | 0.04   |  |
| Crest Factor 2 [dB]         |  | 10.24              |  | 10.35            |  | 10.46    |  | 0.06   |  |
| IQ Offset 1 [dB]            |  | -88.73             |  | -84.65           |  | -75.83   |  | 6.05   |  |
| IQ Offset 2 [dB]            |  | -90.26             |  | -84.48           |  | -74.26   |  | 5.75   |  |
| DC Power 1 [dBm]            |  | -98.59             |  | -94.51           |  | -85.68   |  | 6.05   |  |
| DC Power 2 [dBm]            |  | -100.72            |  | -94.93           |  | -84.71   |  | 5.75   |  |
| Gain Imbalance 1 [dB]       |  | 0.00               |  | 0.00             |  | 0.00     |  | 0.00   |  |
| Gain Imbalance 2 [dB]       |  | 0.00               |  | 0.00             |  | 0.00     |  | 0.00   |  |
| Quad. Error 1 [°]           |  | 0.00               |  | 0.00             |  | 0.03     |  | 0.01   |  |
| Quad. Error 2 [°]           |  | -0.01              |  | 0.00             |  | -0.03    |  | 0.01   |  |
| Statistic Count             |  | Out of Tolerance   |  | CFO Distribution |  |          |  |        |  |
| 100 / 100                   |  | 0.00 %             |  | 0.0 %            |  | Outside: |  | 0      |  |

Figure 3-22: True MIMO scalar results

The table provides the following general and overall results:

- "Spatial Streams", "Space Time": number of spatial streams  $N_{SS}$  and space-time streams  $N_{STS}$ , see also [Chapter 3.2.6.3, "MIMO signals"](#), on page 251
- "Data Symbols": number of OFDM symbols in the payload of the measured burst
- EVM: error vector magnitudes for all subcarriers (data and pilot), for data subcarriers only and for pilot subcarriers only
- "Power Total": power of the overall signal
- "Symbol clock error"
- "Carrier frequency error"

Results per stream:

- "EVM All", only available if  $N_{SS}=N_{STS}$
- "EVM Data" measured per spatial stream
- "EVM Pilot" measured per space-time stream

Results per antenna:

- "Power Backoff": minimum distance of signal power to reference level since the start of the measurement  
The backoff result helps you to configure optimum values for the expected power and the user margin.  
The result is not yet displayed on the GUI, but available via remote command.
- "Burst Power": RMS value over the entire burst, including preamble and payload  
With the hotkey "LTF- & Data-Power..." you can show or hide the following burst power results:
  - The power of long training fields (LTF) portion of the burst
  - The power of data portion of the burst

Refer to [Chapter 3.2.3, "Defining the scope of the measurement"](#), on page 245.

- "Peak Power": peak value over the entire burst, only available up to 40 MHz bandwidth
- Crest factor, only available up to 40 MHz bandwidth
- I/Q origin offset
- "DC Power": power of the direct current (DC) subcarriers (center frequency)
- "Gain Imbalance": absolute value of (1 + I/Q imbalance)  
Gain imbalance cannot be calculated if the spectrum is not symmetrical, e.g. for HT\_TB and HE\_MU.
- "Quadrature Error": angular component of (1 + I/Q imbalance)  
Quadrature error cannot be calculated if the spectrum is not symmetrical, e.g. for HT\_TB and HE\_MU.

With the hotkey "Display" > "Antenna Filter", you can display only the results of an individual antenna.

For query of the results via remote control, see:

- [Chapter 3.5.5.21, "Modulation single values, OFDM"](#), on page 524

For additional information, refer to [Chapter 3.3.2.16, "Common view elements"](#), on page 285 and [Chapter 3.3.2.14, "Selecting and modifying views"](#), on page 284.

### 3.3.2.12 View signal fields info

The view contains a table with the reported signal fields of various PPDU formats. The column "Bits" lists bit numbers, where the parameter is signaled. The "Value" columns indicate the signaled values in binary and decimal formats.

For check bits, the instrument indicates passed (green) or failed (red) check verdict.

The displayed results depend on signal PPDU format:

|                         |     |
|-------------------------|-----|
| L- SIG.....             | 279 |
| HT-SIG.....             | 279 |
| VHT-SIG-A.....          | 279 |
| HE-SIG-A (HE_SU).....   | 280 |
| HE-SIG-A (HE_MU).....   | 281 |
| HE-SIG-A (HE_TRIG)..... | 281 |
| HE-SIG-B.....           | 282 |

## L- SIG

Legacy signal field for NON\_HT (SISO only).

See also IEEE Std 802.11-2016, section 19.3.9.3.5, L-SIG definition.

|                                                                 |         |                    |       |
|-----------------------------------------------------------------|---------|--------------------|-------|
| Freq: 2412.00 MHz Ref.Lvl: 0.00 dBm Std.: 802.11 a/g SISO 20MHz |         |                    |       |
| TX Measurement (Scalar)                                         |         | Signal Fields Info |       |
| L- SIG                                                          | Bits    | Value [Binary]     | Value |
| Rate                                                            | B0-B3   | 1100               | 12    |
| Reserved                                                        | B4      | 0                  | 0     |
| Length                                                          | B5-B16  | 001111101000       | 1000  |
| Parity                                                          | B17     | 0                  | 0     |
| Tail                                                            | B18-B23 | 000000             | 0     |

Figure 3-23: Signal fields info results

Remote command:

`FETCh:WLAN:MEAS<i>:MEValuation:SINfo:LSIG:LENGTH?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

## HT-SIG

High throughput signal field for 802.11n (single user MIMO only).

See also IEEE Std 802.11-2016, table 19-11, HT-SIG fields.

|              |         |                  |       |
|--------------|---------|------------------|-------|
| HT-SIG       |         |                  |       |
| HT-SIG1      | Bits    | Value [Binary]   | Value |
| MCS          | B0-B6   | 0000111          | 7     |
| CBW 20/40    | B7      | 1                | 1     |
| HT Length    | B8-B23  | 0100000000000010 | 16386 |
| HT-SIG2      | Bits    | Value [Binary]   | Value |
| Smoothing    | B0      | 1                | 1     |
| Not Sounding | B1      | 1                | 1     |
| Reserved     | B2      | 1                | 1     |
| Aggregation  | B3      | 0                | 0     |
| STBC         | B4-B5   | 00               | 0     |
| FEC Coding   | B6      | 0                | 0     |
| Short GI     | B7      | 0                | 0     |
| N of ESS     | B8-B9   | 00               | 0     |
| CRC          | B10-B17 | 10100110         | 166   |
| Tail         | B18-B23 | 000000           | 0     |

Remote command:

`FETCh:WLAN:MEAS<i>:MEValuation:SINfo:HTSig:AGGRegation?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

## VHT-SIG-A

Very high throughput signal field VHT-SIG-A for 802.11ac.

See also IEEE Std 802.11-2016, table 21-12, Fields in the VHT-SIG-A field.

| VHT-SIG-A                    |         |                |
|------------------------------|---------|----------------|
| VHT-SIG-A1                   |         |                |
| BW                           | Bits    | Value [Binary] |
| Reserved1                    | B0-B1   | 11             |
| STBC                         | B2      | 1              |
| Group ID                     | B3      | 0              |
| SU NSTS                      | B4-B9   | 111111         |
| Partial AID                  | B10-B12 | 000            |
| TXOP_PS_NOT_ALLOWED          | B13-B21 | 000000000      |
| Reserved2                    | B22     | 1              |
|                              | B23     | 1              |
| VHT-SIG-A2                   |         |                |
| Short GI                     | B0      | 0              |
| Short GI NSYM Disambiguation | B1      | 0              |
| FEC Coding                   | B2      | 0              |
| LDPC Extra Symbol            | B3      | 0              |
| SU VHT-MCS                   | B4-B7   | 1001           |
| Beamformed                   | B8      | 0              |
| Reserved3                    | B9      | 1              |
| CRC                          | B10-B17 | 10011010       |
| Tail                         | B18-B23 | 000000         |

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:SINfo:VHTSig:BEAMformed?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

### HE-SIG-A (HE\_SU)

High efficiency signal field HE-SIG-A for 802.11ax, single user MIMO.

See also IEEE Std 802.11ax-2021, 27.3.11.7 HE-SIG-A.

| HE-SIG-A (HE_SU)       |         |                |
|------------------------|---------|----------------|
| HE-SIG-A1              |         |                |
| Format                 | Bits    | Value [Binary] |
| Beam Change            | B0      | 1              |
| UL/DL                  | B1      | 0              |
| MCS                    | B2      | 0              |
| DCM                    | B3-B6   | 1011           |
| BSS Color              | B7      | 0              |
| Reserved1              | B8-B13  | 000101         |
| Spatial Reuse          | B14     | 1              |
| Bandwidth              | B15-B18 | 0000           |
| GI+LTF Size            | B19-B20 | 11             |
| Nsts                   | B21-B22 | 00             |
|                        | B23-B25 | 000            |
| HE-SIG-A2              |         |                |
| TXOP                   | Bits    | Value [Binary] |
| Coding                 | B0-B6   | 1111111        |
| LDPC Extra Symbol      | B7      | 1              |
| STBC                   | B8      | 0              |
| TxBF                   | B9      | 0              |
| Pre-FEC Padding Factor | B10     | 0              |
| PE Disambiguity        | B11-B12 | 01             |
| Reserved2              | B13     | 0              |
| Doppler                | B14     | 1              |
| CRC                    | B15     | 0              |
| Tail                   | B16-B19 | 1011           |
|                        | B20-B25 | 000000         |

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:HESU:BEAMchange?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

### HE-SIG-A (HE\_MU)

HE signal field HE-SIG-A for 802.11ax, multi user MIMO.

See also IEEE Std 802.11ax-2021, table 27-20, HE-SIG-A field of an HE MU PPDU.

| HE-SIG-A (HE_MU)       |         |                |       |
|------------------------|---------|----------------|-------|
| HE-SIG-A1              |         |                |       |
| UL/DL                  | Bits    | Value [Binary] | Value |
|                        | B0      | 0              | 0     |
| SIGB MCS               | B1-B3   | 101            | 5     |
| SIGB DCM               | B4      | 0              | 0     |
| BSS Color              | B5-B10  | 000101         | 5     |
| Spatial Reuse          | B11-B14 | 0000           | 0     |
| Bandwidth              | B15-B17 | 010            | 2     |
| N of HE-SIG-B Symbols  | B18-B21 | 0010           | 2     |
| SIGB Compression       | B22     | 0              | 0     |
| GI+LTF Size            | B23-B24 | 10             | 2     |
| Doppler                | B25     | 0              | 0     |
| HE-SIG-A2              |         |                |       |
| TXOP                   | Bits    | Value [Binary] | Value |
|                        | B0-B6   | 1111111        | 127   |
| Reserved               | B7      | 1              | 1     |
| N of HE-LTF Symbols    | B8-B10  | 000            | 0     |
| LDPC Extra Symbol      | B11     | 0              | 0     |
| STBC                   | B12     | 0              | 0     |
| Pre-FEC Padding Factor | B13-B14 | 10             | 2     |
| PE Disambiguity        | B15     | 0              | 0     |
| CRC                    | B16-B19 | 0111           | 7     |
| Tail                   | B20-B25 | 000000         | 0     |

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:HEMU:BDCM?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

### HE-SIG-A (HE\_TRIG)

HE signal field HE-SIG-A for 802.11ax, trigger based uplink single user MIMO.

See also IEEE Std 802.11ax-2021, table 27-21, HE-SIG-A field of an HE TB PPDU.

| HE-SIG-A (HE_TRIG) |         |                |       |
|--------------------|---------|----------------|-------|
| HE-SIG-A1          |         |                |       |
| Format             | Bits    | Value [Binary] | Value |
|                    | B0      | 0              | 0     |
| BSS Color          | B1-B6   | 000101         | 5     |
| Spatial Reuse1     | B7-B10  | 0000           | 0     |
| Spatial Reuse2     | B11-B14 | 0000           | 0     |
| Spatial Reuse3     | B15-B18 | 0000           | 0     |
| Spatial Reuse4     | B19-B22 | 0000           | 0     |
| Reserved1          | B23     | 1              | 1     |
| Bandwidth          | B24-B25 | 00             | 0     |
| HE-SIG-A2          |         |                |       |
| TXOP               | Bits    | Value [Binary] | Value |
|                    | B0-B6   | 1111111        | 127   |
| Reserved2          | B7-B15  | 11111111       | 511   |
| CRC                | B16-B19 | 1110           | 14    |
| Tail               | B20-B25 | 000000         | 0     |

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:HETB:BSSColor?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

### HE-SIG-B

HE signal field HE-SIG-B for 802.11ax, OFDMA and multi user MIMO.

See also IEEE Std 802.11ax-2021, section 27.3.11.8.3 common field and section 27.3.11.8.4 user-specific field.

| HE-SIG-B              |         |                |       |
|-----------------------|---------|----------------|-------|
| Content Channel 1     |         |                |       |
| Common Field          |         |                |       |
| RU Allocation         | 8 bits  | 01011000       | 88    |
| Center 26-tone RU     | 1 bit   | NCAP           | NCAP  |
| CRC                   | 4 bits  | 0100           | 4     |
| Tail                  | 6 bits  | 000000         | 0     |
| User Field 1          |         |                |       |
| STA-ID                | 11 bits | 00000001011    | 11    |
| NSTS                  | 3 bits  | 000            | 0     |
| Tx Beamforming        | 1 bit   | 0              | 0     |
| Spatial Configuration | 4 bits  | NCAP           | NCAP  |
| MCS                   | 4 bits  | 0101           | 5     |
| DCM                   | 1 bit   | 0              | 0     |
| Reserved              | 1 bit   | NCAP           | NCAP  |
| Coding                | 1 bit   | 0              | 0     |
| CRC                   | 4 bits  | NCAP           | NCAP  |
| Tail                  | 6 bits  | NCAP           | NCAP  |
| User Field 2          |         |                |       |
| STA-ID                | 11 bits | 00000010110    | 22    |
| NSTS                  | 3 bits  | 000            | 0     |
| Tx Beamforming        | 1 bit   | 0              | 0     |
| Spatial Configuration | 4 bits  | NCAP           | NCAP  |
| MCS                   | 4 bits  | 1001           | 9     |
| DCM                   | 1 bit   | 0              | 0     |
| Reserved              | 1 bit   | NCAP           | NCAP  |
| Coding                | 1 bit   | 0              | 0     |
| CRC                   | 4 bits  | 1000           | 8     |
| Tail                  | 6 bits  | 000000         | 0     |
| User Field 3          |         |                |       |
|                       | Bits    | Value [Binary] | Value |

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:HEB:CHANnel<ch_index>:`

`CFIeld:CRC?` etc.

Refer to [Chapter 3.5.5.23, "Signal fields info"](#), on page 541.

### 3.3.2.13 View TX measurement (scalar) for OFDMA

The view shows the results of OFDMA 802.11ax measurements.

|                         |  |                    |  |                |  |               |  |                               |  |
|-------------------------|--|--------------------|--|----------------|--|---------------|--|-------------------------------|--|
| Freq: 2412.00 MHz       |  | Ref.Lvl: 0.00 dBm  |  | Std.:802.11 ax |  | Switched MIMO |  | 20MHz                         |  |
| TX Measurement (Scalar) |  | Signal Fields Info |  | OFDMA Results  |  |               |  |                               |  |
| No of Users             |  | 4                  |  |                |  |               |  |                               |  |
| No of RUs               |  | 3                  |  |                |  |               |  |                               |  |
| Statistics              |  | Current            |  | Average        |  | Max           |  | StdDev                        |  |
| EVM All [dB]            |  | -65.33             |  | -65.33         |  | -65.33        |  | 0.00                          |  |
| EVM Data [dB]           |  | -65.27             |  | -65.27         |  | -65.27        |  | 0.00                          |  |
| EVM Pilot [dB]          |  | -66.65             |  | -66.65         |  | -66.65        |  | 0.00                          |  |
| RU 1                    |  | RU Size: 106       |  | RU Index: 1    |  | RU26 Index: 1 |  |                               |  |
| Power Ant1              |  | -12.47             |  | -12.47         |  | -12.47        |  | 0.00                          |  |
| Power Ant2              |  | -11.65             |  | -11.65         |  | -11.65        |  | 0.00                          |  |
| User 1                  |  | MCS: 0             |  | DCM: 0         |  | NSTS: 1       |  | STA-ID: 1 TxBf: N Coding: BCC |  |
| EVM All [dB]            |  | -66.57             |  | -66.57         |  | -66.57        |  | 0.00                          |  |
| EVM Data [dB]           |  | -66.56             |  | -66.56         |  | -66.56        |  | 0.00                          |  |
| EVM Pilot [dB]          |  | -66.86             |  | -66.86         |  | -66.86        |  | 0.00                          |  |
| User 2                  |  | MCS: 0             |  | DCM: 0         |  | NSTS: 1       |  | STA-ID: 1 TxBf: N Coding: BCC |  |
| RU 2                    |  | RU Size: 52        |  | RU Index: 3    |  | RU26 Index: 6 |  |                               |  |
| Power Ant1              |  | -15.56             |  | -15.56         |  | -15.56        |  | 0.00                          |  |
| Power Ant2              |  | -96.79             |  | -96.79         |  | -96.79        |  | 0.00                          |  |
| User 3                  |  | MCS: 2             |  | DCM: 0         |  | NSTS: 1       |  | STA-ID: 1 TxBf: 0 Coding: BCC |  |
| RU 3                    |  | RU Size: 52        |  | RU Index: 4    |  | RU26 Index: 8 |  |                               |  |
| Power Ant1              |  | -15.56             |  | -15.56         |  | -15.56        |  | 0.00                          |  |
| Power Ant2              |  | -14.77             |  | -14.77         |  | -14.77        |  | 0.00                          |  |
| User 4                  |  | MCS: 9             |  | DCM: 0         |  | NSTS: 2       |  | STA-ID: 1 TxBf: 0 Coding: BCC |  |
| EVM All [dB]            |  | -63.84             |  | -63.84         |  | -63.84        |  | 0.00                          |  |
| EVM Data [dB]           |  | -63.67             |  | -63.67         |  | -63.67        |  | 0.00                          |  |
| EVM Pilot [dB]          |  | -66.74             |  | -66.74         |  | -66.74        |  | 0.00                          |  |
| Stream 1                |  | -64.58             |  | -64.58         |  | -64.58        |  | 0.00                          |  |
| Stream 2                |  | -63.21             |  | -63.21         |  | -63.21        |  | 0.00                          |  |

Figure 3-24: OFDMA results

At the top, the view indicates OFDMA allocations. Power and EVM results show current, average, max and standard deviation values in individual columns. The view shows EVM overall results and values measured per particular user and stream. Power results are displayed per Rx antenna.

The view contains a table with the following results:

- **"No. of Users, No. of RUs"**: OFDMA common information with the actual allocations (number of all allocated users and resource units)
- **"EVM x"**: EVM results show current, average, max and standard deviation values in individual columns. The view shows EVM overall results and values measured per particular RU, user, and stream. Individual EVM results for all carriers, data carriers and pilot carriers are shown.
- **"Power x"**: Power results show current, average, max and standard deviation values in individual columns. The power results are displayed per RU and Rx antenna.
- **"RU x"**: OFDMA user information signaled in HE signal field is displayed in gray rows for each RU. In the row, RU size, its index and the index based on RU-26 are indicated.



- **"User x"**: OFDMA user information signaled in HE signal field is displayed in gray rows for each user. In the row, the reported MCS, DCM field, STA-ID field, NSTS field, TxBF field are indicated for each user in the RU.

For query of the results via remote control, see:

```
FETCh:WLAN:MEAS<i>:MEValuation:OFDMa:INFO?
```

```
FETCh:WLAN:MEAS<i>:MEValuation:OFDMa:UINfo<user>?.
```

```
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:CURRent?
```

```
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:
USER<user>:CURRent?
```

```
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:
USER<user>:STReam<str>:CURRent? etc.
```

### 3.3.2.14 Selecting and modifying views

Press the "Display" softkey to show hotkeys for view configuration. They change the appearance and contents of the active view. The following hotkeys are available at the bottom of the GUI:

| Hotkey                                  | Description                                                                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Select View ..."                       | Switches to a certain detailed view or overview.<br>Alternatively, select a diagram in the overview and press [ ENTER] or the rotary knob.                                     |
| "Select Ant..."                         | During switched MIMO and true MIMO measurements, displays graphs of all antennas or the selected antenna.                                                                      |
| "Graphs..." or "Graphs & Margins..."    | For 80+80 MHz bandwidth, selects trace (and margin) results for both segments or individual segment                                                                            |
| "Select Trace..." or "Trace Select ..." | Selects the trace types to be displayed in the view. For SMIMO and TMIMO, you can also display the results of one or more antennas/streams.                                    |
| "X Scale... / Y Scale..."               | Modifies the ranges of the X-axis and the Y-axis.                                                                                                                              |
| "LTF- & Data-Power..."                  | For "TX Measurement (Scalar)" view, shows or hides burst power results of the LTF and data portion of the measured burst.                                                      |
| "Antenna Filter..."                     | During switched MIMO and true MIMO measurements, for "TX Measurement (Scalar)" view, enables/disables a filter to display only results of one selected antenna                 |
| "Display Mode"                          | In the "Spectrum Flatness" or "Transmit Spectrum Mask" view, switches between relative and absolute result values (dB vs. dBm), see <a href="#">"Display Mode"</a> on page 326 |
| "EVMCar MIMO Str. Select"               | Selects the streams to be displayed in the EVM vs. carrier view for MIMO connections.                                                                                          |

Additional options are available in the "Measurement Control" section of the configuration dialog.



### 3.3.2.15 Using markers

Press the "Marker" softkey to show hotkeys for marker configuration. They activate markers and modify their position. The following hotkeys are available at the bottom of the GUI:

| Hotkey              | Description                                                                                                                                                                  |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Ref. Marker ..."   | Enable or disable the reference marker, select a trace and the marker position on that trace.                                                                                |
| "Marker 1 / 2 ..."  | Enable or disable marker 1 or 2 and define the marker position (absolute or relative to the reference marker).<br>Depending on the trace mode, a trace can also be selected. |
| "Trace Select Mode" | Define whether all markers are collectively set to the same trace or to individual traces.                                                                                   |

### 3.3.2.16 Common view elements

The following information is available in most result views.

#### Current settings

The most important settings of the WLAN multi-evaluation measurement are displayed in the measurement dialog.

|                       |                   |                     |             |             |     |
|-----------------------|-------------------|---------------------|-------------|-------------|-----|
| Freq: 2412.000000 MHz | Ref.Lvl: 0.00 dBm | Std: 802.11g (OFDM) | Direct Link | 12Mbps QPSK | 426 |
|-----------------------|-------------------|---------------------|-------------|-------------|-----|

For configuration, see [Chapter 3.3.3.2, "Signal routing and analyzer settings"](#), on page 288 and [Chapter 3.3.3.3, "Input signal settings"](#), on page 293.

#### Statistic Count

Progress bar for the measurement. During the first single shot after the start of the measurement, the bar shows the number of elapsed measurement intervals relative to the "Statistic Count". A filled progress bar indicates that the first shot is complete and the statistical depth has been reached.

#### Out of Tolerance

Percentage of measurement intervals (bursts) that were failed because at least one result exceeded the limits.

All results that use the same statistic count setting contribute to the same "Out of Tolerance" result.

#### CFO Distribution

Shows the percentage of tolerated carrier frequency offset (CFO) values and the number of CFO values exceeding the specified limit ("Outside"). For the specified limit, refer to [Limit setting for I/Q offset, frequency error, symbol clock error, CFO distribution](#).

The CFO distribution measurement is only supported for 802.11ax. Exceeding the CFO distribution limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:MODulation:CFDistrib? etc.`

### TE Distribution

Shows the percentage of tolerated timing error (TE) values and the number of TE values exceeding the specified limit ("Outside"). For the specified limit, refer to [Power vs. time limits](#).

The calculation of timing error is only supported for OFDM standards. It requires a suitable trigger to be specified as a trigger source. Internal IF power trigger and MAC frame RX trigger cannot be used.

Exceeding the timing error distribution limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

Remote command:

`FETCh:WLAN:MEAS<i>:MEvaluation:PVTime:TEDistrib? etc.`

### Statistics in the result tables

Most result diagrams and result tables provide statistical results.

The statistical values are calculated as follows:

- **"Current"**: Result obtained in the last measurement interval. For some modulation results, an RMS value and a peak value are available. They are calculated as the average and peak of all samples in the measured slot.
- **"Average"**: Weighted average of all "Current" values since the start of the measurement.
- **"Max", "Min"**: Largest or smallest "Current" value since the start of the measurement.
- **"StdDev"**: Standard deviation of all "Current" values since the start of the measurement.

## 3.3.3 Parameters and settings

All settings are defined via softkeys and hotkeys or using the configuration dialog. The configuration dialog is described in the following sections. To open the dialog, press the "Multi Evaluation" softkey and then the "Config" hotkey.

### 3.3.3.1 Basic input signal settings

Access: "Config" hotkey > top of the configuration dialog.

The parameters at the top of the configuration tree configure the most important input signal settings.

|            |           |               |        |
|------------|-----------|---------------|--------|
| Standard   | 802.11 ac | Receive Mode: | SISO   |
| Band Width | 80+80MHz  | Ch. Distance: | 80 MHz |

Figure 3-25: Signal routing and analyzer settings (802.11ac, 80+80 MHz)

|                   |     |
|-------------------|-----|
| Standard.....     | 287 |
| Receive Mode..... | 287 |
| Band Width.....   | 287 |
| Ch. Distance..... | 288 |

### Standard

Selects the IEEE 802.11 standard. The availability of certain signal properties depends on the selected standard.

See [Chapter 3.2.6.2, "Physical layer"](#), on page 248.

Required options:

- "802.11n" requires R&S CMW-KM651
- "802.11ac" requires R&S CMW-KM651 and R&S CMW-KM656
- "802.11ax" requires R&S CMW-KM651, R&S CMW-KM656, and R&S CMW-KM657
- "802.11p" requires R&S CMW-KM655

In the combined signal path (CSP), consider the dependency between this parameter and the burst type, set for the RX frame trigger in the signaling application.

See [TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe](#)

Selecting a standard that is not compatible with the current scenario restores the "Standalone" scenario.

Remote command:

[CONFigure:WLAN:MEAS<i>:ISIGnal:STANdard](#)

### Receive Mode

Allows you to select a receive mode.

The available receive modes depend on the selected IEEE 802.11 standard, as summarized in the following table:

| Receive mode                      | Standard 802.11... |     |   |    |    |   |
|-----------------------------------|--------------------|-----|---|----|----|---|
|                                   | b/g                | a/g | n | ac | ax | p |
| SISO                              | ✓                  | ✓   | ✓ | ✓  | ✓  | ✓ |
| Composite MIMO<br>(R&S CMW-KM652) | –                  | –   | ✓ | ✓  | ✓  | – |
| True MIMO<br>(R&S CMW-KM654)      | –                  | –   | ✓ | ✓  | ✓  | – |

Refer to [Chapter 3.2.2, "How to measure a WLAN signal"](#), on page 243.

Remote command:

[CONFigure:WLAN:MEAS<i>:ISIGnal:RMODe](#)

### Band Width

Defines the signal bandwidth. You can select between:

- 5, 10, 20, and 40 MHz for 802.11n
- 5, 10, 20, 40, 80, 80+80 MHz, and 160 MHz for 802.11ac
- 20, 40, 80, 80+80 MHz, and 160 MHz for 802.11ax

- 5, 10, and 20 MHz for 802.11p

The bandwidth for all other supported standards is fixed.

In the combined signal path (CSP), consider the dependency between this parameter and the trigger bandwidth, set for the RX frame trigger in the signaling application.

`TRIGger:WLAN:SIGN<i>:RX:MACFrame:BW`

For 802.11ax and trigger source set to "HE\_TB Trigger", the setting depends on the trigger configuration.

See `CONFigure:WLAN:SIGN<i>:CONNECTION:HETF:CHBW`

Remote command:

`CONFigure:WLAN:MEAS<i>:ISIGNAL:BWIDth`

### Ch. Distance

Distance between the center frequencies of the two 80-MHz segments for the bandwidth 80+80 MHz. A distance of 80 MHz means that the two segments are contiguous.

Remote command:

`CONFigure:WLAN:MEAS<i>:ISIGNAL:CDISTance`

### 3.3.3.2 Signal routing and analyzer settings

Access: "Config" hotkey > top of the configuration dialog.

The following parameters configure the RF input path.

|                              |                                   |
|------------------------------|-----------------------------------|
| Scenario                     | StandAlone (Non Signaling)        |
| RF Routing                   | Connector: 1.1 Converter: 1.RFRX1 |
| External Attenuation (Input) | 0.00 dB                           |
| Band / Center Channel        | 5 GHz 7 Ch                        |
| Side Channel(s)              | 7 Ch 9 Ch 13 Ch 17 Ch             |
| Center Frequency             | 5035.000000 MHz                   |
| Expected Nominal Power       | 0.00 dBm Ref.Level: 0.00 dBm      |
| User Margin                  | 0.00 dB                           |
| Mixer Level Offset           | 12.00 dB                          |
| Frequency Offset             | 0.00 Hz                           |

Figure 3-26: Signal routing and analyzer settings

|                                             |     |
|---------------------------------------------|-----|
| Scenario = StandAlone.....                  | 289 |
| Scenario = Combined Signal Path.....        | 289 |
| Scenario = True MIMO.....                   | 289 |
| Use Sep. Ant. for 80+80 MHz.....            | 290 |
| Number of Antennas.....                     | 290 |
| Training Mode.....                          | 290 |
| RF Routing.....                             | 290 |
| External Attenuation (Input) /Ext. Att..... | 291 |
| Band / Channels /Center Frequency.....      | 291 |
| Expected Nominal Power.....                 | 292 |
| User Margin.....                            | 293 |
| Mixer Level Offset.....                     | 293 |
| Frequency Offset.....                       | 293 |

**Scenario = StandAlone**

The multi-evaluation measurement is used standalone.

Non-signaling stand alone scenario is only available in the combination with the [Receive Mode](#) = "SISO" or "Composite MIMO".

See also [Chapter 3.2.2.1, "Composite MIMO measurements"](#), on page 244.

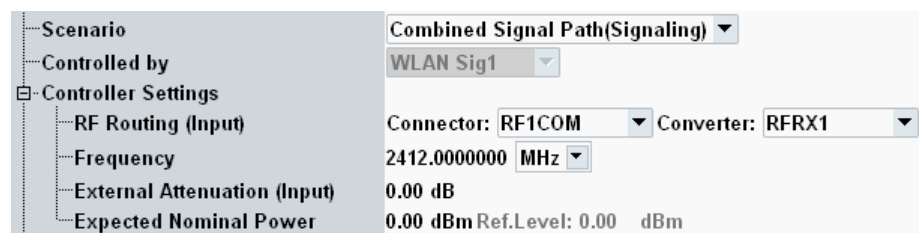
Remote command:

```
ROUTe:WLAN:MEAS<i>:SCENario:SALone
ROUTe:WLAN:MEAS<i>:SCENario?
ROUTe:WLAN:MEAS<i>?
ROUTe:WLAN:MEAS<i>:CATalog:SCENario?
```

**Scenario = Combined Signal Path**

The multi-evaluation measurement is combined with an instance of the WLAN signaling application (option R&S CMW-KS65x). The WLAN signaling application establishes a connection to the DUT. The additional parameter "Controlled by" selects the signaling application instance.

In this scenario, some parameters of the measurement are set or restricted by the signaling instance. The configuration tree changes accordingly:



The multi-evaluation measurement settings are remembered in the background and enabled again, when switching back to the standalone scenario.

Connection status information of the signaling application is displayed at the bottom of the measurement GUI. Softkeys and hotkeys provide access to the settings of the signaling application and allow you to switch the signal on or off, see [Chapter 3.3.3.8, "Additional softkeys and hotkeys"](#), on page 325.

For additional information, see [Chapter 3.2.5, "Combined signaling and measurement"](#), on page 247.

Remote command:

```
ROUTe:WLAN:MEAS<i>:SCENario:CSPath
ROUTe:WLAN:MEAS<i>:SCENario?
ROUTe:WLAN:MEAS<i>?
ROUTe:WLAN:MEAS<i>:CATalog:SCENario?
CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n>
```

**Scenario = True MIMO**

True MIMO scenarios are only available for standards "802.11n", "802.11ac", and "802.11ax" and receive mode "True MIMO".

R&S CMW500/2xx supports true MIMO 2x2 only.

One option R&S CMW-KM654 per R&S CMW is required.

External attenuation and expected nominal power are configurable per antenna. The configuration tree changes accordingly.

| Connector | Ext. Att. | ExpNomPwr. |
|-----------|-----------|------------|
| 1.1       | 0.00 dB   | 0.00 dBm   |
| 2.1       | 0.00 dB   | 0.00 dBm   |
| 3.1       | 0.00 dB   | 0.00 dBm   |

Remote command:

```
ROUTE:WLAN:MEAS<i>:SCENario:TMIMo<PathCount>
CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n>
ROUTE:WLAN:MEAS<i>:CATalog:SCENario?
```

### Use Sep. Ant. for 80+80 MHz

This setting is only displayed for 80+80 MHz signals.

Enable the checkbox if your DUT uses separate antennas for the two segments. You can then connect the antennas to different RF connectors and avoid using a combiner.

Remote command:

```
CONFigure:WLAN:MEAS<i>:RFSettings:SANTennas
```

### Number of Antennas

Specifies the number of TX paths of the DUT that are connected to the R&S CMW in MIMO scenarios.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MIMO:NOAntennas
```

### Training Mode

Activates the training mode and displays the corresponding view for **Receive Mode** = "Composite MIMO". Refer to [Chapter 3.3.4, "CMIMO training mode"](#), on page 327.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MODE
```

### RF Routing

Selects the input path for the measured RF signal, i.e. the input connector and the RX module to be used.

In the standalone (SA) scenario, these parameters are controlled by the measurement. In the combined signal path (CSP) scenario, they are controlled by the signaling application.

Remote command:

```
ROUTE:WLAN:MEAS<i>?
ROUTE:WLAN:MEAS<i>:SCENario:SALone (SA)
CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n> (SA)
```

```
ROUTE:WLAN:MEAS<i>:SCENario:CSPath (CSP)
ROUTE:WLAN:SIGN<i>:SCENario:... (CSP)
```

### External Attenuation (Input) /Ext. Att.

Defines the value of an external attenuation (or gain, if the value is negative) in the input path. The power readings are corrected by the external attenuation value.

The external attenuation value is also used in the calculation of the maximum configurable input power value.

If a correction table for frequency-dependent attenuation is active for the chosen connector, then the table name and a button are displayed. Press the button to display the table entries.

In the "MIMO" scenarios, you can configure a connector-specific attenuation value.

All configured attenuations are added: global external attenuation + connector-specific attenuation + active frequency-dependent attenuation table.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:WLAN:MEAS<i>:RFSettings:EATTenuation (SA)
CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:INPut (CSP)
CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n> (SA)
```

### Band / Channels /Center Frequency

Using the band and channel settings, you can specify the input signal in terms of standard WLAN channels in a supported band. The center frequency of the overall channel can also be set directly, which allows for non-standard WLAN channels. If a channel frequency corresponds to a channel index number, this number is displayed.

#### Note:

- The available channel index numbers and related frequencies depend on the selected band.
- The R&S CMW does not limit the set of available channels based on the selected standard, nor does it enforce any regulatory restrictions. Exception: The 4-GHz band can only be configured for 802.11n.
- Measuring signals above 3.3 GHz requires a 6 GHz option.

Depending on the signal bandwidth the input signal comprises one, two, four or eight RF side channels.

The available settings change accordingly:

- For 5 MHz, 10 MHz or 20 MHz signal bandwidth, there is only a single channel. "Frequency" sets the center frequency of the channel.



Figure 3-27: Frequency settings for 20 MHz 802.11n

- A 40 MHz 802.11n signal consists of a primary 20 MHz channel and a secondary 20 MHz channel that is one of the non-overlapping channels right above ("Secondary Channel": 1) or right below ("Secondary Channel": -1) the "Primary Channel".

The frequency offset between primary and secondary channel is  $\pm 20$  MHz. The secondary channel number is  $\pm 4$  (if it is in the allowed channel number range). "Center Frequency" sets the center frequency of the 40 MHz bandwidth.

|                       |                                                          |                                      |
|-----------------------|----------------------------------------------------------|--------------------------------------|
| Band / Center Channel | 5 GHz ▾ 1 Ch                                             | Primary Ch./Freq.: 3 Ch 5015.000 MHz |
| Secondary Channel     | -1 ▾ 1 = 20 MHz above, -1 = 20 MHz below primary channel |                                      |
| Center Frequency      | 5005.000000 MHz ▾                                        |                                      |

Figure 3-28: Frequency settings for 40 MHz

- An 802.11ac or ax signal with 40 MHz, 80 MHz or 160 MHz bandwidth consists of 2, 4 or 8 adjacent, non-overlapping 20 MHz channels, respectively. Setting one of the 20 MHz channel indices automatically adjusts the remaining ones. "Center Frequency" sets the center frequency of the total bandwidth. For 80+80 MHz signals, it sets the center frequency of the left segment.

|                       |                                                |
|-----------------------|------------------------------------------------|
| Band / Center Channel | 5 GHz ▾ 7 Ch                                   |
| Side Channel(s)       | 1 Ch 5 Ch 9 Ch 13 Ch   17 Ch 21 Ch 25 Ch 29 Ch |
| Center Frequency      | 5035.000000 MHz ▾                              |

Figure 3-29: Frequency settings for 80+80 MHz

- For IEEE 802.11ax measurements with an R&S CMW-Z800A, the RF frequency of the DUT is in the range 5.925 GHz to 7.125 GHz. The measurement application measures on down-converted signal from the R&S CMW-Z800A ( $f_{\text{DUT}} - 4800$  MHz). The measurement shows the actual measurement frequency and not the frequency of the DUT. The measurement is only used in combined signal path and configures the lower frequency automatically.

In the standalone (SA) scenario, these parameters are controlled by the measurement. In the combined signal path (CSP) scenario, they are controlled by the signaling application.

Remote command:

```

CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:BAND (SA)
CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency (SA)
CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:CHANnels (SA)
CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:CHANnels<Ch> (SA)
CONFigure:WLAN:SIGN<i>:RFSettings:FREQuency (CSP)
CONFigure:WLAN:SIGN<i>:RFSettings:CHANnel (CSP)

```

### Expected Nominal Power

Sets the analyzer in accordance with the nominal power of the signal to be measured. The nominal power is the average output power at the DUT during the measurement intervals where the transmitter is on.

The reference level is calculated as the expected peak power at the output of the DUT:

*Reference level = expected nominal power + user margin*

The actual input power at the connectors must be within the level range of the selected RF input connector; refer to the data sheet. With correctly configured power settings, the input power equals the "Reference Level" minus the "External Attenuation (Input)" value.



In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFigure:WLAN:MEAS<i>:RFSettings:ENPower<connector>` (SA)

`CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EPEPower` (CSP)

### User Margin

Margin that the measurement adds to the expected nominal power to determine its reference power (reference level). The margin is typically used to account for the known variations of the RF input signal power, for example, the variations due to a specific channel configuration.

The appropriate values depend on the configuration of the received WLAN signal, e.g. on the modulation scheme.

Remote command:

`CONFigure:WLAN:MEAS<i>:RFSettings:UMARgin<connector>`

### Mixer Level Offset

Varies the input level of the mixer in the analyzer path. A negative value reduces the mixer input level. A positive value increases it. Optimize the mixer input level according to the properties of the measured signal.

| Mixer level offset | Advantages                                                                              | Possible shortcomings                                      |
|--------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------|
| < 0 dB             | Suppression of distortion (e.g. of the intermodulation products generated in the mixer) | Lower dynamic range (due to smaller signal-to-noise ratio) |
| > 0 dB             | High signal-to-noise ratio, higher dynamic range                                        | Risk of intermodulation, smaller overdrive reserve         |

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFigure:WLAN:MEAS<i>:RFSettings:MLOffset` (SA)

`CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:MLOffset` (CSP)

### Frequency Offset

Positive or negative frequency offset to be added to the specified center frequency of the RF analyzer. This setting is useful to adjust the measurement frequency.

Remote command:

`CONFigure:WLAN:MEAS<i>:RFSettings:FOffset`

#### 3.3.3.3 Input signal settings

Access: "Config" hotkey > "Input Signal" section.

The WLAN "Input Signal" parameters define basic WLAN signal properties. The values in this section must be set in accordance with the measured signal.

See [Chapter 3.2.2, "How to measure a WLAN signal"](#), on page 243.

Figure 3-30: WLAN input signal settings

|                                                       |     |
|-------------------------------------------------------|-----|
| Standard, Receive Mode, Band Width, Ch. Distance..... | 294 |
| Burst Type.....                                       | 294 |
| Evaluation Length, Skip Preamble and Header.....      | 294 |
| Modulation Filter.....                                | 295 |
| Power Class.....                                      | 295 |
| Training Data.....                                    | 296 |
| IQ Swap.....                                          | 296 |
| Coding Rate.....                                      | 296 |

### Standard, Receive Mode, Band Width, Ch. Distance

These parameters are also listed at the top of the configuration tree.

See [Chapter 3.3.3.1, "Basic input signal settings"](#), on page 286

### Burst Type

Displays the burst type, depending on the selected standard, see [Chapter 3.2.6.2, "Physical layer"](#), on page 248.

A selection can only be made for standard 802.11n: Choose "Green Field" for signals with greenfield header format and "Mixed" for all other signals.

Remote command:

CONFigure:WLAN:MEAS<i>:ISIGnal:BTYPe

### Evaluation Length, Skip Preamble and Header

- For DSSS signals, you can configure the number of payload chips to measured. The checkbox selects whether the preamble and header are skipped or measured in addition to the configured number of payload chips. For spectrum emission measurements with 802.11b, set the maximum evaluation length to 270000 chips.
- For OFDM signals, you can configure the number of payload symbols to be measured. The preamble/header is not measured.

The evaluation length can be also reduced automatically due to memory restrictions.

For more details, see [Chapter 3.2.3, "Defining the scope of the measurement"](#), on page 245.

Remote command:

```
CONFigure:WLAN:MEAS<i>:ISIGnal:DSSS:ELENgth
CONFigure:WLAN:MEAS<i>:ISIGnal:OFDM:ELENgth
```

### Modulation Filter

Refines the basic burst filter of IEEE standard to limit the evaluation to bursts of a particular modulation type.

While the value "All" is selected and if the signal-to-noise ratio is low, the modulation estimation is not able to detect the correct modulation. If it fails, usually 1024 QAM is estimated because noise looks similar. In that case, select the correct modulation manually.

If the received burst has a different modulation, the reliability indicates "Wrong Modulation", see [Chapter 3.5.1.8, "Reliability indicator"](#), on page 351.

- In stand-alone for HE\_TRIG bursts (802.11ax) measurements, a wrong burst is ignored, but makes EVM worse.
- In stand-alone measurements for all other bursts and standards, a after 10 wrong bursts a "Wrong Modulation" is indicated. Wrong bursts are not included in EVM calculation.
- In combined signal path (CSP), a wrong burst is ignored also for EVM calculation.

In **CSP**, consider the dependency between this parameter and the data rate, set for the RX frame trigger in the signaling application.

```
TRIGger:WLAN:SIGN<i>:RX:MACFrame:RATE
```

For 802.11ax and trigger source set to "HE\_TB Trigger", the setting depends on the trigger configuration:

```
CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:MCS
```

When changing the trigger source from "HE\_TB Trigger" to any other trigger source, the modulation filter is set to "All".

Remote command:

```
CONFigure:WLAN:MEAS<i>:ISIGnal:MODFilter
```

### Power Class

This setting is only displayed for standard 802.11p. It selects the STA transmit power class as defined in annex D.2.2 of standard IEEE Std 802.11-2016.

The power class determines the transmit spectrum mask to be applied, see also ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

- |                |                                                                  |
|----------------|------------------------------------------------------------------|
| "Class A"      | The "Class A" transmit spectrum mask is used for limit checks.   |
| "Class B"      | The "Class B" transmit spectrum mask is used for limit checks.   |
| "Class C/D"    | There is no transmit spectrum limit check.                       |
| "User Defined" | The "User Def." transmit spectrum mask is used for limit checks. |

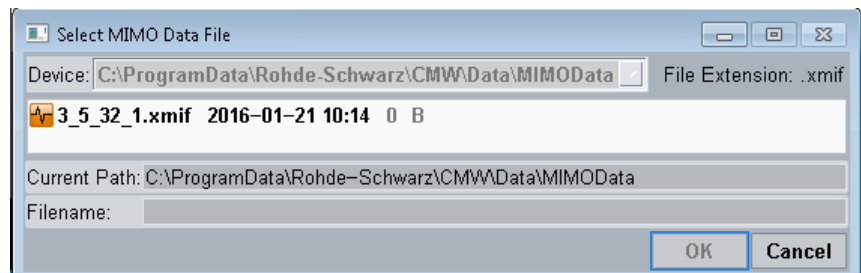
Remote command:

```
CONFigure:WLAN:MEAS<i>:ISIGnal:PClass
```

**Training Data**

This setting is only displayed for 802.11n, ac and ax composite MIMO measurements. It specifies the reference signal, see [Chapter 3.2.2.1, "Composite MIMO measurements"](#), on page 244.

|                   |                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "no data"         | Unknown signal                                                                                                                                                    |
| "Internal Data"   | The R&S CMW expects to (continuously) receive the same burst that was "recorded" during the preceding training mode session (still available in internal memory). |
| "File Data"       | The R&S CMW expects to (continuously) receive the burst that was "recorded" to the selected file.                                                                 |
| "Select New File" | The R&S CMW opens a dialog to select a file with training data.                                                                                                   |



Remote command:

`CONFigure:WLAN:MEAS<i>:ISIGnal:TData`

`CONFigure:WLAN:MEAS<i>:ISIGnal:TData:FILE:DATE?`

**IQ Swap**

With active I/Q swap, the I values are mapped to the Q channel and vice versa. As a result, the points in the I/Q constellation diagrams are reflected at the first bisector ( $Q = I$ ). In spectral representation, the spectrum is reflected at the RF carrier frequency.

Remote command:

`CONFigure:WLAN:MEAS<i>:ISIGnal:IQSWap`

**Coding Rate**

For 802.11ax measurements at HE TB PPDU, the coding rate of HE\_TRIG bursts is not signaled. Specify it within the limit settings, see ["Modulation limits 802.11ax"](#) on page 311.

Option R&S CMW-KM657 is required.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:  
EVMall:TBCoderate`

**3.3.3.4 Measurement control settings**

Access: "Config" hotkey > "Measurement Control" section.

The measurement control parameters configure the scope of the measurement.

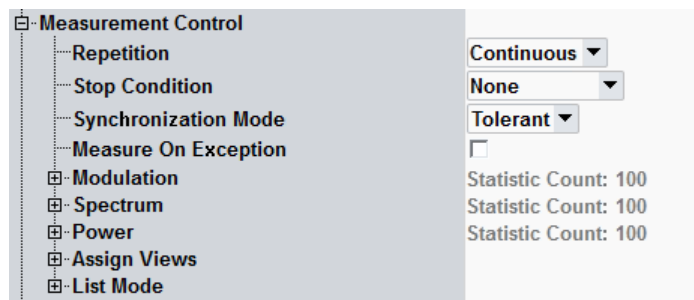


Figure 3-31: Measurement control settings

|                                  |     |
|----------------------------------|-----|
| Repetition.....                  | 297 |
| Stop Condition.....              | 298 |
| Measure on Exception.....        | 298 |
| Synchronization Mode.....        | 298 |
| Modulation.....                  | 298 |
| L Statistic Count.....           | 298 |
| L Tracking.....                  | 299 |
| L Channel Estimation.....        | 300 |
| L FFT Demodulation Offset.....   | 300 |
| L DSSS Tx Filter Estimation..... | 300 |
| L DSSS EVM Method.....           | 300 |
| L Skip OFDM Symbols.....         | 300 |
| Spectrum.....                    | 301 |
| L Statistic Count.....           | 301 |
| L Spectrum Trigger Time.....     | 301 |
| L Average FFT Num.....           | 301 |
| L Occupied Bandwidth (OBW).....  | 301 |
| L Mask Selection.....            | 301 |
| Power.....                       | 302 |
| L Statistic Count.....           | 302 |
| L Average Length.....            | 302 |
| L Reference Power.....           | 302 |
| Assign Views.....                | 302 |
| List Mode > Enable.....          | 303 |

### Repetition

Defines how often the measurement is repeated if it is not stopped explicitly or by a failed limit check.

- **Continuous:** The measurement is continued until it is explicitly terminated. The results are periodically updated.
- **Single-Shot:** The measurement is stopped after one measurement cycle.

Single-shot is preferable if only a single measurement result is required under fixed conditions, which is typical for remote-controlled measurements. The continuous mode is suitable to monitor the evolution of the measurement results and to observe how they depend on the measurement configuration, which is typically done in manual control.

Remote command:

CONFigure:WLAN:MEAS<i>:MEvaluation:REPetition

### Stop Condition

Specifies the conditions for an early termination of the measurement:

- **"None"**: The measurement is performed according to its "Repetition" mode and "Statistic Count", irrespective of the limit check results.
- **"On Limit Failure"**: The measurement is stopped when one of the limits is exceeded, irrespective of the repetition mode set. If no limit failure occurs, it is performed according to its "Repetition" mode and "Statistic Count". Use this setting for measurements that are intended for checking limits, e.g. production tests.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEValuation:SCONdition`

### Measure on Exception

Specifies whether results that the measurement identifies as faulty or inaccurate are rejected. A faulty result occurs, for example, when an overload is detected. In remote control, the cause of the error is indicated by the "reliability indicator".

- **Off**: Faulty results are rejected. The measurement is continued. The statistical counters are not reset. Use this mode to ensure that a single faulty result does not affect the entire measurement.
- **On**: Results are never rejected. Use this mode for development purposes, if you want to analyze the reason for occasional wrong transmissions.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEValuation:MOEXception`

### Synchronization Mode

Sets the synchronization mode:

- **Normal**: synchronization according to preamble detection
- **Tolerant**: synchronization with the second part of preamble when the first part cannot be detected

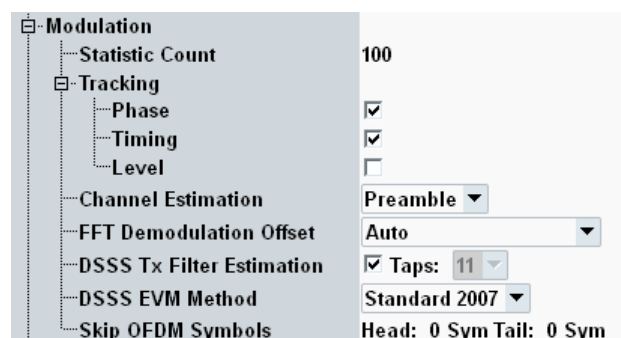
Remote command:

`CONFigure:WLAN:MEAS<i>:MEValuation:SMODE`

### Modulation

Access: "Config" hotkey > "Measurement Control" > "Modulation" section.

The section contains the following settings for modulation measurements.



### Statistic Count ← Modulation

Defines the number of measurement intervals per measurement cycle. A measurement cycle is the duration of a single-shot measurement.

In the WLAN multi-evaluation measurement, the measurement interval is completed when the R&S CMW has measured the specified evaluation length of a burst.

See also ["Evaluation Length, Skip Preamble and Header"](#) on page 294.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:MODulation`

#### **Tracking ← Modulation**

The WLAN burst signal is transformed to the frequency domain using an FFT operation. The FFT is performed symbol-wise for all symbols of the payload. The calculated FFTs include:

- The common phase drift of all subcarriers at symbol  $i$
- The phase shift of subcarrier  $k$  at symbol  $i$  caused by the timing drift
- The gain at symbol  $i$  in relation to the reference gain at the long symbol (LS)

Thus, the frequency estimate is not error-free in general. Therefore, the overall frequency deviation of the DUT must be compensated.

#### **"Phase":**

The common phase drift of all subcarriers at symbol  $i$  is made up of the frequency deviation and the phase jitter at symbol  $i$ . The phase jitter is composed of a constant part and a nonlinear part.

The constant part is caused by the uncompensated frequency deviation. The measurement of the phase starts at the first symbol of the payload, whereas the channel frequency response represents the channel at the long symbol of the preamble. The uncompensated frequency deviation produces a phase drift between the long symbol and the first symbol of the payload. It appears as a constant part in the common phase drift.

The nonlinear part can be caused by the phase noise of the DUT oscillator. Or it can be caused by the increase of the DUT amplifier temperature at the beginning of the burst.

Referring to the IEEE 802.11 standard, the common phase drift must be estimated and compensated from the pilots. Therefore, the phase tracking is normally activated as the default setting.

For composite MIMO and 802.11ac input signals, it is always enabled.

#### **"Timing":**

This phase is caused by the relative clock deviation of the reference oscillator. Timing drift tracking is activated by default. It depends on the standard, whether the tracking is requested by IEEE 802.11 or not.

#### **"Level":**

A change of the gain at symbol  $i$  (caused, for example, by an increase of the DUT amplifier temperature) can lead to symbol errors, especially for large symbol alphabets. In this case, the estimation and compensation of the gain can be useful. However, according to the IEEE 802.11 standard, the compensation of the gain is not a requirement, and therefore is not activated per default.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:PHASe
CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:TIMing
CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:LEVel
```

### Channel Estimation ← Modulation

Specifies whether channel estimation is done in the payload (and the preamble) or in the preamble only. Channel estimation in the payload improves the accuracy of EVM measurement results. IEEE 802.11 requests channel estimation in the preamble.

This setting is only relevant for OFDM signals.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:CESTimation
```

### FFT Demodulation Offset ← Modulation

Defines the mode how the start offset of the FFT demodulation is determined.

- "Guard Interval Center": The guard interval center is used as FFT start offset.
- "Peak": The peak of the fine-timing metric is used to determine the FFT start offset.
- "Auto" (default): The measurement application determines the optimal FFT start offset.

This setting is only relevant for OFDM signals.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEValuation:DEMod:FFT:OFFSet
```

### DSSS Tx Filter Estimation ← Modulation

This parameter is relevant for 802.11b/g DSSS signals only. It determines if and how accurate the transmit filter is estimated.

The transmitter is using some type of filtering to satisfy the spectral mask requirement of the 802.11b/g standard. As the standard does not specify what type of filtering to use, the modulation results are computed with incorrect knowledge of the transmitter filtering. The wrong filtering causes the EVM for a perfect 802.11b signal to have a nonzero EVM level. Accurate transmit filter estimation reduces the measured EVM.

For maximum measurement speed, disable "DSSS Tx Filter Estimation". With filter estimation enabled, the higher the number of equalizer filter "Taps", the more accurate the estimation, but also the lower the measurement speed.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:EFTaps
```

### DSSS EVM Method ← Modulation

This parameter is relevant for 802.11b signals only. It selects the EVM measurement method - according to IEEE Std 802.11-2007, IEEE Std 802.11b-1999, or according to standard IEEE Std 802.11-2016.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEValuation:EMETHod
```

### Skip OFDM Symbols ← Modulation

Allows you to exclude a configurable number of head and tail bits from modulation measurements.



This feature is, for example, useful for composite MIMO measurements, where non-repeating parts of the retransmitted burst (such as sequence number and checksum) would otherwise invalidate the modulation measurement.

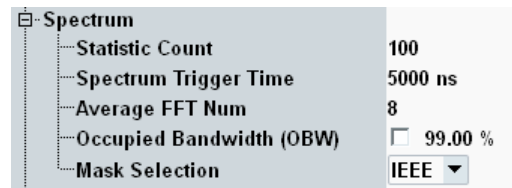
Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:COMPensation:SKIPsymbols`

### Spectrum

Access: "Config" hotkey > "Measurement Control" > "Spectrum" section.

The section contains the following settings for spectrum measurements.



#### Statistic Count ← Spectrum

Defines the number of measurement intervals per measurement cycle. A measurement cycle is the duration of a single-shot measurement.

In the WLAN multi-evaluation measurement, the measurement interval is completed when the R&S CMW has measured the specified evaluation length of a burst.

See also ["Evaluation Length, Skip Preamble and Header"](#) on page 294.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:TSMask`

#### Spectrum Trigger Time ← Spectrum

Specifies a trigger offset time. After the trigger event, the R&S CMW waits this time before it performs the first Fast Fourier Transform (FFT).

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:TROTime`

#### Average FFT Num ← Spectrum

Specifies the number of FFT operations per burst. The results of the individual FFT operations are averaged to calculate the burst results.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:AFFTnum`

#### Occupied Bandwidth (OBW) ← Spectrum

Enables/disables OBW measurement and sets the OBW percentage.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:OBWPercent`

#### Mask Selection ← Spectrum

This setting is only displayed for standard 802.11p. It determines the transmit spectrum mask to be applied to 802.11p channels.

See also ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

- IEEE: Relative spectral density limits as specified in IEEE 802.11
  - ETSI: Absolute emission limits as specified in ETSI EN 302 571 V1.1.1 (2008-09)
  - ARIB: Absolute emission limits as specified in ARIB STD-T109
- This selection is only available for 10 MHz bandwidth.

The setting switches also between relative and absolute result values (dB vs. dBm) in the "Transmit Spectrum Mask" view.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:MSElection`

### Power

Access: "Config" hotkey > "Measurement Control" > "Power" section.

The section contains the following settings for power measurements.



#### Statistic Count ← Power

Defines the number of measurement intervals per measurement cycle. A measurement cycle is the duration of a single-shot measurement.

In the WLAN multi-evaluation measurement, the measurement interval is completed when the R&S CMW has measured the specified evaluation length of a burst.

See also ["Evaluation Length, Skip Preamble and Header"](#) on page 294.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:PVTime`

#### Average Length ← Power

The "Average Length" is the length of the moving average filter, which smoothes the power trace and thus eliminates the modulation. While a long average length leads to more stable measurement results, it increases the rise/fall times compared to no averaging.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:PVTime:ALENgt`

#### Reference Power ← Power

In DSSS, the thresholds for rising and falling edge results are defined as percentage values of the reference power. The reference can be set to either the max or mean power of the burst.

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:PVTime:RPOwer`

### Assign Views

Selects the view types to be displayed in the overview dialog. The measurement does not evaluate the results for disabled views. Limiting the number of assigned views can speed up the measurement.

The available views depend on other settings. Configure first the standard, the receive mode and the bandwidth.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult[:ALL]
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:EVM
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:EVMCarrier
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:EVMSymbol
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:IQConst
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:MSCalar
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:PVTime
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:SFLatness
CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:TSMask
```

List Mode > Enable

Shows whether the list mode is enabled and disables an enabled list mode.



Enabling the list mode is only possible via the remote control command below. The list mode is essentially a remote control feature. For an introduction, see [Chapter 3.2.7, "Multi-evaluation list mode"](#), on page 252.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MEvaluation:LIST
```

3.3.3.5 Trigger settings

Access: "Config" hotkey > "Trigger" section.

The trigger settings configure the trigger system for the WLAN multi-evaluation measurement.

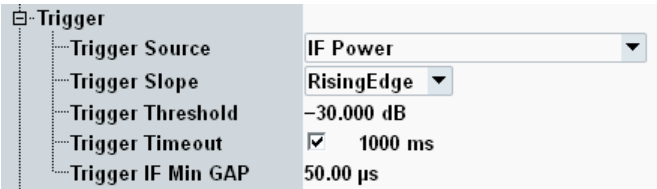


Figure 3-32: Trigger settings

|                         |     |
|-------------------------|-----|
| Trigger Source.....     | 303 |
| Trigger Slope.....      | 304 |
| Trigger Threshold.....  | 304 |
| Trigger Timeout.....    | 304 |
| Trigger IF Min GAP..... | 304 |

Trigger Source

Selects the source of the trigger event.

Only the measurement-specific trigger sources are listed here. Additional trigger sources are provided by the base software and by other applications like the GPRF generator. For a description of these trigger sources, refer to the documentation of the corresponding software part.

"IF Power" The measurement is triggered by the power of the received signal, converted into an IF signal. The trigger event coincides with the rising or falling edge of the measured WLAN burst.

Remote command:

```
TRIGger:WLAN:MEAS<i>:MEValuation:SOURce
```

```
TRIGger:WLAN:MEAS<i>:MEValuation:CATalog:SOURce?
```

### Trigger Slope

Selects whether the trigger event is generated at the rising edge or at the falling edge of the trigger pulse. This setting is relevant for power trigger sources.

Remote command:

```
TRIGger:WLAN:MEAS<i>:MEValuation:SLOPe
```

### Trigger Threshold

Defines the input signal power where a trigger event is generated for power trigger sources.

The threshold is a dB value, relative to the reference level minus the external attenuation. If the reference level equals the maximum output power of the DUT and the external attenuation settings are correct, the trigger threshold is relative to the maximum input power.

A low threshold ensures that the input signal is always detected. A higher threshold can prevent unintended trigger events.

Remote command:

```
TRIGger:WLAN:MEAS<i>:MEValuation:THReshold
```

### Trigger Timeout

Sets a time after which a started measurement must have received a trigger event. If no trigger event is received, a trigger timeout is indicated in manual operation mode. In remote control mode, the measurement is stopped. The parameter can be disabled so that no timeout occurs.

Remote command:

```
TRIGger:WLAN:MEAS<i>:MEValuation:TOUT
```

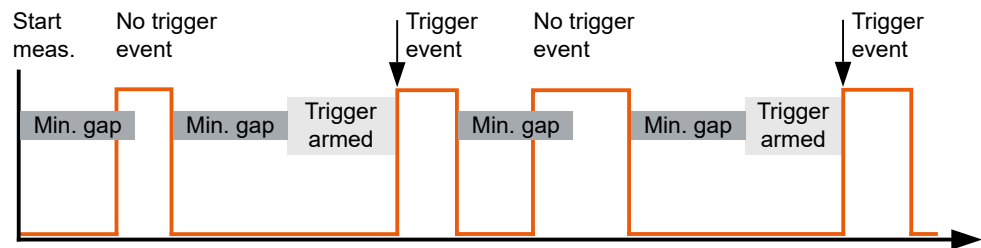
### Trigger IF Min Gap

Defines a minimum duration of the power-down periods (gaps) between two triggered power pulses. This setting is valid for "(IF) Power" trigger sources.

The trigger system is controlled by a timer which is reset to zero in the following instances:

- At the IF power-down ramp of each triggered or untriggered pulse, even if the previous counter has not yet elapsed. A power-down ramp is detected when the signal power falls below the trigger threshold.
- At the beginning of each measurement. The minimum gap defines the minimum time between the start of the measurement and the first trigger event.

The trigger system is rearmed when the timer has reached the specified minimum gap.



This parameter can be used to prevent unwanted trigger events due to fast power variations.

For a [Trigger Source](#) different from "IF Power", it is hidden.

Remote command:

`TRIGger:WLAN:MEAS<i>:MEvaluation:MGAP`

### 3.3.3.6 Limit settings

The following sections give an overview of the WLAN multi-evaluation limit settings and the related test requirements.

Conformance requirements for WLAN transmitter tests are specified in the IEEE 802.11 family of standards.

|                                                                      |     |
|----------------------------------------------------------------------|-----|
| • <a href="#">Power vs. time limits</a> .....                        | 305 |
| • <a href="#">Modulation limits DSSS</a> .....                       | 306 |
| • <a href="#">Modulation limits 802.11a/g/p OFDM</a> .....           | 307 |
| • <a href="#">Modulation limits 802.11n</a> .....                    | 308 |
| • <a href="#">Modulation limits 802.11ac</a> .....                   | 309 |
| • <a href="#">Modulation limits 802.11ax</a> .....                   | 311 |
| • <a href="#">Spectrum Flatness Limits OFDM</a> .....                | 314 |
| • <a href="#">Transmit spectrum mask DSSS</a> .....                  | 316 |
| • <a href="#">Transmit spectrum mask OFDM, default masks</a> .....   | 317 |
| • <a href="#">Transmit spectrum mask OFDM, by regulation</a> .....   | 320 |
| • <a href="#">Transmit spectrum mask OFDM, absolute limits</a> ..... | 322 |

#### Power vs. time limits

Access: "Config" hotkey > "Limits" > "Power vs. Time" section.

Standard IEEE Std 802.11ax-2021 defines requirements for HE PHY in section 27.3.15.3 "Pre-correction accuracy requirements". After the AP triggers the uplink transmissions, the STA transmits the HE TB PPDU at a specified time. The accuracy of this time must be  $\pm 0.4 \mu\text{s}$  relative to the frame sent by the AP to trigger the UL transmissions.



The calculation of timing error is supported for OFDM standards. It requires a suitable trigger to be specified as a trigger source.

Internal IF power trigger and MAC frame RX trigger cannot be used.

A limit for the distribution of timing error is not defined in standard. The limit check applies only for OFDM SISO measurements. Exceeding the timing error distribution limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

The rise time and fall time represent a basic pulse shaping criterion for DSSS signals.

IEEE Std 802.11-2016 defines related requirements in sections 15.4.5.8 and 16.3.7.7 "Transmit power-on and power-down ramp". The transmit power-on ramp (rise time) for 10% to 90% of maximum power must be no greater than 2  $\mu$ s. The transmit power-down ramp (fall time) for 90% to 10% of maximum power must also be no greater than 2  $\mu$ s.

All limits can be set in the configuration dialog.

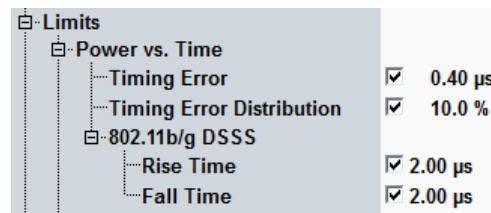


Figure 3-33: Power vs. time limit settings

As an alternative to the maximum power, it is also possible to select the average power as the reference level.

See also "[Reference Power](#)" on page 302.

The pulse shape of DSSS signals is complemented by their spectrum mask, see "[Transmit spectrum mask DSSS](#)" on page 316 for details.

For remote commands, refer to [Power vs. time limits](#).

### Modulation limits DSSS

Access: "Config" hotkey > "Limits" > "Modulation" > "802.11b/g DSSS" section.

A poor modulation accuracy of the transmitter increases the transmission errors. The error vector magnitude (EVM) is the critical quantity to assess the modulation accuracy of a transmitter.

According to IEEE, the peak EVM of a DSSS signal must not exceed 35 %. This limit must be checked for 1000 samples (chips) of a burst. Center frequency and chip clock tolerance are specified in ppm of the carrier frequency: Errors must not exceed  $\pm 25$  ppm. For channel 1, the limit is 25 ppm of 2.412 GHz = 60.3 kHz.

Table 3-5: Modulation limits: 802.11b/g DSSS

| Characteristics                        | Refer to IEEE Std 802.11-2016, section...             | Specified limit              |
|----------------------------------------|-------------------------------------------------------|------------------------------|
| Peak EVM                               | 15.4.5.10/16.3.7.9 Transmit modulation accuracy       | < 35%                        |
| Frequency error                        | 15.4.5.6/16.3.7.5 Transmit center frequency tolerance | < $\pm 25$ ppm <sup>1)</sup> |
| Chip clock error                       | 15.4.5.7/16.3.7.6 Chip clock frequency tolerance      | < $\pm 25$ ppm <sup>1)</sup> |
| <sup>1)</sup> of the carrier frequency |                                                       |                              |

All limits can be set in the configuration dialog:

|                        |                                     |                   |
|------------------------|-------------------------------------|-------------------|
| 802.11b/g DSSS         |                                     |                   |
| EVM Peak               | <input checked="" type="checkbox"/> | 35.00 %           |
| EVM RMS                | <input checked="" type="checkbox"/> | 35.00 %           |
| IQ Offset              | <input checked="" type="checkbox"/> | -15.00 dB         |
| Center Frequency Error | <input checked="" type="checkbox"/> | 60000.00 Hz (+/-) |
| Chip Clock Error       | <input checked="" type="checkbox"/> | 25.000 ppm (+/-)  |

Figure 3-34: Modulation limit settings for 802.11b/g DSSS

For remote commands, refer to [Chapter 3.5.4.2, "Modulation limits: 802.11b/g DSSS"](#), on page 404.

### Modulation limits 802.11a/g/p OFDM

Access: "Config" hotkey > "Limits" > "Modulation" > "802.11a/g" or "802.11p" section.

IEEE defines EVM limits from -25 dB to -5 dB, depending on the modulation type and coding rate. Center frequency and symbol clock tolerance are specified in ppm of the carrier frequency.

For 802.11p, they depend on the frequency band and the channel width.

Table 3-6: Modulation limits: 802.11a/g/p OFDM

| Characteristics    | Refer to IEEE Std 802.11-2016, section...                                   | Specified limit                                                                                                 |
|--------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| EVM (RMS)          | 17.3.9.7.4 Transmitter constellation error for 802.11a and p (5 GHz band)   | < -5 dB (BPSK, 1/2)                                                                                             |
|                    |                                                                             | < -8 dB (BPSK, 3/4)                                                                                             |
|                    |                                                                             | < -10 dB (QPSK, 1/2)                                                                                            |
|                    |                                                                             | < -13 dB (QPSK, 3/4)                                                                                            |
|                    |                                                                             | < -16 dB (16-QAM, 1/2)                                                                                          |
|                    |                                                                             | < -19 dB (16-QAM, 3/4)                                                                                          |
|                    |                                                                             | < -22 dB (64-QAM, 2/3)<br>< -25 dB (64-QAM, 3/4)                                                                |
| Frequency error    | 17.3.9.5 Transmit center frequency tolerance for 802.11a and p (5 GHz band) | < $\pm 20$ ppm <sup>1)</sup> for 20 MHz and 10 MHz channels,<br>< $\pm 10$ ppm <sup>1)</sup> for 5 MHz channels |
|                    | 18.4.7.4 Transmit center frequency tolerance for 802.11g (2.4 GHz band)     | < $\pm 25$ ppm                                                                                                  |
| Symbol clock error | 17.3.9.6 Symbol clock frequency tolerance for 802.11a and p (5 GHz band)    | < $\pm 20$ ppm for 20 MHz and 10 MHz channels,<br>< $\pm 10$ ppm for 5 MHz channels                             |
|                    | 18.4.7.5 Symbol clock frequency tolerance for 802.11g (2.4 GHz band)        | < $\pm 25$ ppm                                                                                                  |

The configuration dialog allows you to set these limits as absolute values:

|                        |                                     |                   |
|------------------------|-------------------------------------|-------------------|
| 802.11a/g              |                                     |                   |
| EVM BPSK 1/2           | <input checked="" type="checkbox"/> | -5.00 dB          |
| EVM BPSK 3/4           | <input checked="" type="checkbox"/> | -8.00 dB          |
| EVM QPSK 1/2           | <input checked="" type="checkbox"/> | -10.00 dB         |
| EVM QPSK 3/4           | <input checked="" type="checkbox"/> | -13.00 dB         |
| EVM 16QAM 1/2          | <input checked="" type="checkbox"/> | -16.00 dB         |
| EVM 16QAM 3/4          | <input checked="" type="checkbox"/> | -19.00 dB         |
| EVM 64QAM 2/3          | <input checked="" type="checkbox"/> | -22.00 dB         |
| EVM 64QAM 3/4          | <input checked="" type="checkbox"/> | -25.00 dB         |
| EVM Pilot              | <input checked="" type="checkbox"/> | -8.00 dB          |
| IQ Offset              | <input checked="" type="checkbox"/> | -15.00 dB         |
| Center Frequency Error | <input checked="" type="checkbox"/> | 60000.00 Hz (+/-) |
| Symbol Clock Error     | <input checked="" type="checkbox"/> | 25.000 ppm (+/-)  |
| 802.11p                |                                     |                   |
| EVM BPSK 1/2           | <input checked="" type="checkbox"/> | -5.00 dB          |
| EVM BPSK 3/4           | <input checked="" type="checkbox"/> | -8.00 dB          |
| EVM QPSK 1/2           | <input checked="" type="checkbox"/> | -10.00 dB         |
| EVM QPSK 3/4           | <input checked="" type="checkbox"/> | -13.00 dB         |
| EVM 16QAM 1/2          | <input checked="" type="checkbox"/> | -16.00 dB         |
| EVM 16QAM 3/4          | <input checked="" type="checkbox"/> | -19.00 dB         |
| EVM 64QAM 2/3          | <input checked="" type="checkbox"/> | -22.00 dB         |
| EVM 64QAM 3/4          | <input checked="" type="checkbox"/> | -25.00 dB         |
| EVM Pilot              | <input checked="" type="checkbox"/> | -8.00 dB          |
| IQ Offset              | <input checked="" type="checkbox"/> | -15.00 dB         |
| Center Frequency Error | <input checked="" type="checkbox"/> | 60000.00 Hz (+/-) |
| Symbol Clock Error     | <input checked="" type="checkbox"/> | 25.000 ppm (+/-)  |

Figure 3-35: Modulation limit settings for 802.11a/g/p OFDM

For remote commands, refer to:

- [Chapter 3.5.4.3, "Modulation limits: 802.11a/g OFDM"](#), on page 406
- [Chapter 3.5.4.4, "Modulation limits: 802.11p OFDM"](#), on page 409

### Modulation limits 802.11n

Access: "Config" hotkey > "Limits" > "Modulation" > "802.11n" section.

IEEE defines EVM limits from -28 dB to -5 dB, depending on the modulation type and coding rate.

For the frequency error and the symbol clock error, IEEE specifies limits in ppm of the carrier frequency and depending on the frequency band:  $\pm 20$  ppm for the 5 GHz band and  $\pm 25$  ppm for the 2.4 GHz band. Example: For channel 36 of the 5 GHz band, the limit is 20 ppm of 5180 MHz = 103.6 kHz.

For MIMO measurements, modulation limits do not apply. Limit check is deactivated.



Table 3-7: Modulation limits: 802.11n

| Characteristics    | Refer to IEEE Std 802.11-2016, section...     | Specified limit                                                                                                                                                                                       |
|--------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EVM (RMS)          | 19.3.18.7.3 Transmitter constellation error   | < -5 dB (BPSK, 1/2)<br>< -10 dB (QPSK, 1/2)<br>< -13 dB (QPSK, 3/4)<br>< -16 dB (16-QAM, 1/2)<br>< -19 dB (16-QAM, 3/4)<br>< -22 dB (64-QAM, 2/3)<br>< -25 dB (64-QAM, 3/4)<br>< -27 dB (64-QAM, 5/6) |
| I/Q offset         | 19.3.18.7.2 Transmit center frequency leakage | < -15 dB (20 MHz)<br>< -20 dB (40 MHz)                                                                                                                                                                |
| Frequency error    | 19.3.18.4 Transmit center frequency tolerance | < $\pm 20$ ppm (5 GHz band)<br>< $\pm 25$ ppm (2.4 GHz band)                                                                                                                                          |
| Symbol clock error | 19.3.18.6 Symbol clock frequency tolerance    | < $\pm 20$ ppm (5 GHz band)<br>< $\pm 25$ ppm (2.4 GHz band)                                                                                                                                          |

The SISO limits apply also for MIMO as a limit per user and stream.

The configuration dialog allows you to set these limits as absolute values:

|                        |                                     |                   |
|------------------------|-------------------------------------|-------------------|
| 802.11n                |                                     |                   |
| EVM BPSK 1/2           | <input checked="" type="checkbox"/> | -5.00 dB          |
| EVM QPSK 1/2           | <input checked="" type="checkbox"/> | -10.00 dB         |
| EVM QPSK 3/4           | <input checked="" type="checkbox"/> | -13.00 dB         |
| EVM 16QAM 1/2          | <input checked="" type="checkbox"/> | -16.00 dB         |
| EVM 16QAM 3/4          | <input checked="" type="checkbox"/> | -19.00 dB         |
| EVM 64QAM 2/3          | <input checked="" type="checkbox"/> | -22.00 dB         |
| EVM 64QAM 3/4          | <input checked="" type="checkbox"/> | -25.00 dB         |
| EVM 64QAM 5/6          | <input checked="" type="checkbox"/> | -27.00 dB         |
| EVM Pilot              | <input checked="" type="checkbox"/> | -8.00 dB          |
| IQ Offset              |                                     |                   |
| 5MHz                   | <input checked="" type="checkbox"/> | -15.00 dB         |
| 10MHz                  | <input checked="" type="checkbox"/> | -15.00 dB         |
| 20MHz                  | <input checked="" type="checkbox"/> | -15.00 dB         |
| 40MHz                  | <input checked="" type="checkbox"/> | -20.00 dB         |
| Center Frequency Error | <input checked="" type="checkbox"/> | 60000.00 Hz (+/-) |
| Symbol Clock Error     | <input checked="" type="checkbox"/> | 25.000 ppm (+/-)  |

Figure 3-36: Modulation limit settings for 802.11n

For remote commands, refer to [Chapter 3.5.4.5, "Modulation limits: 802.11n SISO"](#), on page 412.

### Modulation limits 802.11ac

Access: "Config" hotkey > "Limits" > "Modulation" > "802.11ac" section.

Modulation limits for 802.11ac signals are specified in the standard IEEE Std 802.11-2016. The R&S CMW supports limit checks for the following modulation characteristics:

**Table 3-8: Modulation limits: 802.11ac**

| Characteristics                        | Refer to IEEE Std 802.11-2016, section...                                | Specified limit                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EVM (RMS)                              | 21.3.17.4.3 Transmitter constellation error                              | < -5 dB (BPSK, 1/2)<br>< -10 dB (QPSK, 1/2)<br>< -13 dB (QPSK, 3/4)<br>< -16 dB (16-QAM, 1/2)<br>< -19 dB (16-QAM, 3/4)<br>< -22 dB (64-QAM, 2/3)<br>< -25 dB (64-QAM, 3/4)<br>< -27 dB (64-QAM, 5/6)<br>< -30 dB (256-QAM, 3/4)<br>< -32 dB (256-QAM, 5/6)<br>< -35 dB (1024-QAM, 3/4)<br>< -35 dB (1024-QAM, 5/6)<br>< -8 dB (pilot) |
| I/Q offset                             | 21.3.17.4.2 Transmit center frequency leakage                            | < -17.48 dB (5 to 20 MHz)<br>< -20.57 dB (40 MHz)<br>< -23.84 dB (80 MHz)<br>< -23.84 dB and -20 dBm (80+80 MHz)<br>< -26.85 dB (160 MHz)                                                                                                                                                                                              |
| Frequency error and symbol clock error | 21.3.17.3 Transmit center frequency and symbol clock frequency tolerance | < $\pm 20$ ppm                                                                                                                                                                                                                                                                                                                         |

For MIMO measurements, modulation limits do not apply. Limit check is deactivated.

All limits can be set in the configuration dialog:

|                        |                                     |                                               |
|------------------------|-------------------------------------|-----------------------------------------------|
| 802.11ac               |                                     |                                               |
| EVM BPSK 1/2           | <input checked="" type="checkbox"/> | -5.00 dB                                      |
| EVM QPSK 1/2           | <input checked="" type="checkbox"/> | -10.00 dB                                     |
| EVM QPSK 3/4           | <input checked="" type="checkbox"/> | -13.00 dB                                     |
| EVM 16QAM 1/2          | <input checked="" type="checkbox"/> | -16.00 dB                                     |
| EVM 16QAM 3/4          | <input checked="" type="checkbox"/> | -19.00 dB                                     |
| EVM 64QAM 2/3          | <input checked="" type="checkbox"/> | -22.00 dB                                     |
| EVM 64QAM 3/4          | <input checked="" type="checkbox"/> | -25.00 dB                                     |
| EVM 64QAM 5/6          | <input checked="" type="checkbox"/> | -27.00 dB                                     |
| EVM 256QAM 3/4         | <input checked="" type="checkbox"/> | -30.00 dB                                     |
| EVM 256QAM 5/6         | <input checked="" type="checkbox"/> | -32.00 dB                                     |
| EVM 1024QAM 3/4        | <input checked="" type="checkbox"/> | -35.00 dB                                     |
| EVM 1024QAM 5/6        | <input checked="" type="checkbox"/> | -35.00 dB                                     |
| EVM Pilot              | <input checked="" type="checkbox"/> | -8.00 dB                                      |
| IQ Offset              |                                     |                                               |
|                        |                                     | relative absolute                             |
| 5MHz                   | <input checked="" type="checkbox"/> | -17.48 dB                                     |
| 10MHz                  | <input checked="" type="checkbox"/> | -17.48 dB                                     |
| 20MHz                  | <input checked="" type="checkbox"/> | -17.48 dB                                     |
| 40MHz                  | <input checked="" type="checkbox"/> | -20.57 dB                                     |
| 80MHz                  | <input checked="" type="checkbox"/> | -23.84 dB                                     |
| 80+80MHz               | <input checked="" type="checkbox"/> | -23.84 dB <input type="checkbox"/> -20.00 dBm |
| 160MHz                 | <input checked="" type="checkbox"/> | -26.85 dB                                     |
| Center Frequency Error | <input checked="" type="checkbox"/> | 100000.00 Hz (+/-)                            |
| Symbol Clock Error     | <input checked="" type="checkbox"/> | 20.000 ppm (+/-)                              |

Figure 3-37: Modulation limit settings for 802.11ac

For remote commands, refer to [Chapter 3.5.4.6, "Modulation limits: 802.11ac"](#), on page 415.

### Modulation limits 802.11ax

Access: "Config" hotkey > "Limits" > "Modulation" > "802.11ax" section.

Modulation limits for 802.11ax signals are specified in IEEE Std 802.11ax-2021. The R&S CMW supports limit checks for the following modulation characteristics:

EVM (RMS) limits are defined by IEEE, refer to IEEE Std 802.11ax-2021, section 27.3.19.4.3 Transmitter constellation error.

Table 3-9: Modulation limits for 802.11ax: EVM

| Specified limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HE SU PPDU and HE MU PPDU:</b><br>< -5 dB (BPSK, 1/2, with DCM)<br>< -5 dB (BPSK, 1/2, without DCM)<br>< -5 dB (QPSK, 1/2, with DCM)<br>< -10 dB (QPSK, 1/2, without DCM)<br>< -13 dB (QPSK, 3/4, without DCM)<br>< -10 dB (16-QAM, 1/2, with DCM)<br>< -13 dB (16-QAM, 3/4, with DCM)<br>< -16 dB (16-QAM, 1/2, without DCM)<br>< -19 dB (16-QAM, 3/4, without DCM)<br>< -22 dB (64-QAM, 2/3, without DCM)<br>< -25 dB (64-QAM, 3/4, without DCM)<br>< -27 dB (64-QAM, 5/6, without DCM)<br>< -30 dB (256-QAM, 3/4, without DCM)<br>< -32 dB (256-QAM, 5/6, without DCM)<br>< -35 dB (1024-QAM, 3/4, without DCM)<br>< -35 dB (1024-QAM, 5/6, without DCM)                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>HE TB PPDU <math>P_{TX} &gt; \max P_{MCS7}</math></b><br>< -13 dB (BPSK, 1/2, with DCM)<br>< -13 dB (BPSK, 1/2, without DCM)<br>< -13 dB (QPSK, 1/2, with DCM)<br>< -13 dB (QPSK, 1/2, without DCM)<br>< -13 dB (QPSK, 3/4, without DCM)<br>< -13 dB (16-QAM, 1/2, with DCM)<br>< -13 dB (16-QAM, 3/4, with DCM)<br>< -16 dB (16-QAM, 1/2, without DCM)<br>< -19 dB (16-QAM, 3/4, without DCM)<br>< -22 dB (64-QAM, 2/3, without DCM)<br>< -25 dB (64-QAM, 3/4, without DCM)<br>< -27 dB (64-QAM, 5/6, without DCM)<br>< -30 dB (256-QAM, 3/4, without DCM)<br>< -32 dB (256-QAM, 5/6, without DCM)<br>< -35 dB (1024-QAM, 3/4, without DCM)<br>< -35 dB (1024-QAM, 5/6, without DCM) | <b>HE TB PPDU <math>P_{TX} \leq \max P_{MCS7}</math></b><br>< -27 dB (BPSK, 1/2, with DCM)<br>< -27 dB (BPSK, 1/2, without DCM)<br>< -27 dB (QPSK, 1/2, with DCM)<br>< -27 dB (QPSK, 1/2, without DCM)<br>< -27 dB (QPSK, 3/4, without DCM)<br>< -27 dB (16-QAM, 1/2, with DCM)<br>< -27 dB (16-QAM, 3/4, with DCM)<br>< -27 dB (16-QAM, 1/2, without DCM)<br>< -27 dB (16-QAM, 3/4, without DCM)<br>< -27 dB (64-QAM, 2/3, without DCM)<br>< -27 dB (64-QAM, 3/4, without DCM)<br>< -27 dB (64-QAM, 5/6, without DCM)<br>< -30 dB (256-QAM, 3/4, without DCM)<br>< -32 dB (256-QAM, 5/6, without DCM)<br>< -35 dB (1024-QAM, 3/4, without DCM)<br>< -35 dB (1024-QAM, 5/6, without DCM) |

The limit line for unused tone error is configured indirectly via specifying a limit for occupied subcarrier HE TB PPDU transmitted with high or low power. See also IEEE Std 802.11ax-2021, equations (27-131) and (27-132). The coding rate of HE TB PPDU is not signaled, specify it within the limit settings.

Table 3-10: Limit setting for I/Q offset, frequency error, symbol clock error, CFO distribution

| Characteristics                        | Refer to IEEE Std 802.11ax-2021 section...                               | Specified limit                                                                                                                                                       |
|----------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I/Q offset                             | 27.3.19.4.2 Transmit center frequency leakage                            | < -32 dB and -20 dBm (20 MHz)<br>< -32 dB and -20 dBm (40 MHz)<br>< -32 dB and -20 dBm (80 MHz)<br>< -32 dB and -20 dBm (80+80 MHz)<br>< -32 dB and -20 dBm (160 MHz) |
| Frequency error and symbol clock error | 27.3.19.3 Transmit center frequency and symbol clock frequency tolerance | < ±20 ppm                                                                                                                                                             |
| Carrier frequency offset error         | 27.3.15.3 Pre-correction accuracy requirements                           | < 350 Hz for data subcarriers<br>measured as the 10% point of the complementary cumulative distribution function (CCDF) <sup>*)</sup>                                 |

<sup>\*)</sup> Exceeding the CFO distribution limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

For MIMO measurements, modulation limits do not apply. Limit check is deactivated.

Limits can be set in the configuration dialog:

| 802.11ax          |                                     |
|-------------------|-------------------------------------|
| SU & MU           |                                     |
| EVM BPSK 1/4 DCM  | -5.00 dB                            |
| EVM BPSK 1/2      | -5.00 dB                            |
| EVM QPSK 1/4 DCM  | -5.00 dB                            |
| EVM QPSK 1/2      | -10.00 dB                           |
| EVM QPSK 3/4      | -13.00 dB                           |
| EVM 16QAM 1/4 DCM | -10.00 dB                           |
| EVM 16QAM 3/8 DCM | -13.00 dB                           |
| EVM 16QAM 1/2     | -16.00 dB                           |
| EVM 16QAM 3/4     | -19.00 dB                           |
| EVM 64QAM 2/3     | -22.00 dB                           |
| EVM 64QAM 3/4     | -25.00 dB                           |
| EVM 64QAM 5/6     | -27.00 dB                           |
| EVM 256QAM 3/4    | -30.00 dB                           |
| EVM 256QAM 5/6    | -32.00 dB                           |
| EVM 1024QAM 3/4   | -35.00 dB                           |
| EVM 1024QAM 5/6   | -35.00 dB                           |
| EVM Pilot         | -8.00 dB                            |
| Trigger Based     |                                     |
|                   | TB high power TB low power Coderate |
| EVM BPSK          | -13.00 dB -27.00 dB 1/2             |
| EVM QPSK          | -13.00 dB -27.00 dB 3/4             |
| EVM 16QAM         | -19.00 dB -27.00 dB 3/4             |
| EVM 64QAM         | -27.00 dB -27.00 dB 5/6             |
| EVM 256QAM        | -32.00 dB -32.00 dB 5/6             |
| EVM 1024QAM       | -35.00 dB -35.00 dB 5/6             |
| EVM Pilot         | -13.00 dB -27.00 dB                 |
| Unused Tone Error | -13.00 dB TB high power             |

Figure 3-38: Modulation limit settings for 802.11ax, part 1

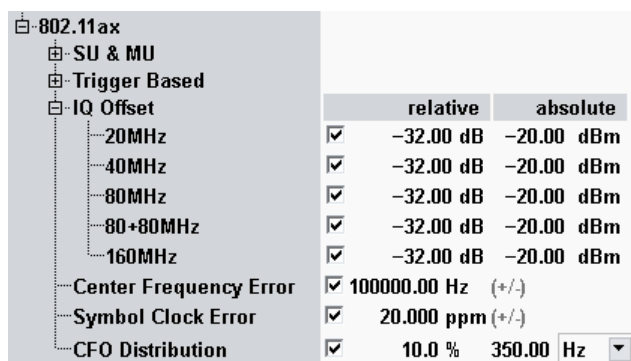


Figure 3-39: Modulation limit settings for 802.11ax, part 2

For remote commands, refer to [Chapter 3.5.4.7, "Modulation limits: 802.11ax"](#), on page 420.

### Spectrum Flatness Limits OFDM

Access: "Config" hotkey > "Limits" > "Spectrum Flatness" > any section.

The spectrum flatness limits are defined depending on the WLAN standard and the channel bandwidth.

The corresponding IEEE references are:

- For 802.11a/g/p: IEEE Std 802.11-2016, section 17.3.9.7.3 Transmitter spectral flatness
- For 802.11n (HT): IEEE Std 802.11-2016, section 19.3.18.2 Spectral flatness
- For 802.11ac (VHT): IEEE Std 802.11-2016, section 21.3.17.2 Spectral flatness
- For 802.11ax (HE): IEEE Std 802.11ax-2021, section 27.3.19.2 Spectral flatness

The subcarrier ranges and the related spectrum flatness limits are defined by IEEE as listed in the following table.

Table 3-11: Subcarrier ranges for spectrum flatness

| Standard             | Subcarriers                 | Lower limit | Upper limit |
|----------------------|-----------------------------|-------------|-------------|
| 802.11a/g/p          | -16 to -1, 1 to 16 (center) | -4 dB       | 4 dB        |
|                      | -26 to -17, 17 to 26 (side) | -6 dB       | 4 dB        |
| 802.11n (20 MHz)     | -16 to -1, 1 to 16 (center) | -4 dB       | 4 dB        |
|                      | -28 to -17, 17 to 28 (side) | -6 dB       | 4 dB        |
| 802.11n (40 MHz, HT) | -42 to -2, 2 to 42 (center) | -4 dB       | 4 dB        |
|                      | -58 to -43, 43 to 58 (side) | -6 dB       | 4 dB        |
| 802.11ac (20 MHz)    | -16 to -1, 1 to 16 (center) | -4 dB       | 4 dB        |
|                      | -28 to -17, 17 to 28 (side) | -6 dB       | 4 dB        |
| 802.11ac (40 MHz)    | -42 to -2, 2 to 42 (center) | -4 dB       | 4 dB        |
|                      | -58 to -43, 43 to 58 (side) | -6 dB       | 4 dB        |
| 802.11ac (80 MHz)    | -84 to -2, 2 to 84 (center) | -4 dB       | 4 dB        |

| Standard             | Subcarriers                                                                                                      | Lower limit | Upper limit |
|----------------------|------------------------------------------------------------------------------------------------------------------|-------------|-------------|
|                      | -122 to -85, 85 to 122 (side)                                                                                    | -6 dB       | 4 dB        |
| 802.11ac (80+80 MHz) | contiguous: limits as for 802.11ac (160 MHz)<br>non-contiguous: limits for each segment as for 802.11ac (80 MHz) |             |             |
| 802.11ac (160 MHz)   | -172 to -130, -126 to -44, 44 to 126, and 130 to 172 (center)                                                    | -4 dB       | 4 dB        |
|                      | -250 to -173, -43 to -6, 6 to 43, and 173 to 250 (side)                                                          | -6 dB       | 4 dB        |
| 802.11ax (20 MHz)    | -84 to -2, 2 to 84 (center)                                                                                      | -4 dB       | 4 dB        |
|                      | -122 to -85, 85 to 122 (side)                                                                                    | -6 dB       | 4 dB        |
| 802.11ax (40 MHz)    | -168 to -3, 3 to 168 (center)                                                                                    | -4 dB       | 4 dB        |
|                      | -244 to -169, 169 to 244 (side)                                                                                  | -6 dB       | 4 dB        |
| 802.11ax (80 MHz)    | -344 to -3, 3 to 344 (center)                                                                                    | -4 dB       | 4 dB        |
|                      | -500 to -345, 345 to 500 (side)                                                                                  | -6 dB       | 4 dB        |
| 802.11ax (80+80 MHz) | contiguous: limits as for 802.11ax (160 MHz)<br>non-contiguous: limits for each segment as for 802.11ax (80 MHz) |             |             |
| 802.11ax (160 MHz)   | -696 to -515, -509 to -166, 166 to 509, and 515 to 696 (center)                                                  | -4 dB       | 4 dB        |
|                      | -1012 to -697, -165 to -12, 12 to 165, and 697 to 1012 (side)                                                    | -6 dB       | 4 dB        |

At the R&S CMW, a limit set consists of an upper limit applicable to all subcarriers (except subcarrier 0) and lower limits for the "center" and "side" subcarrier ranges. All limits can be set in the configuration dialog.

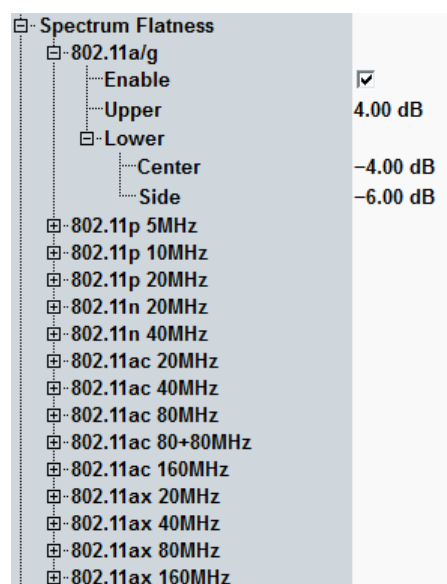


Figure 3-40: Spectrum flatness limit settings

For remote commands, refer to:

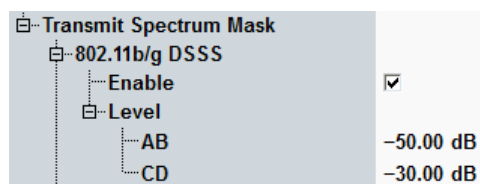
- [Chapter 3.5.4.8, "Spectrum flatness limits: 802.11a/g OFDM"](#), on page 429
- [Chapter 3.5.4.9, "Spectrum flatness limits: 802.11p OFDM"](#), on page 430
- [Chapter 3.5.4.10, "Spectrum flatness limits: 802.11n"](#), on page 432
- [Chapter 3.5.4.11, "Spectrum flatness limits: 802.11ac"](#), on page 433
- [Chapter 3.5.4.12, "Spectrum flatness limits: 802.11ax"](#), on page 435

### Transmit spectrum mask DSSS

Access: "Config" hotkey > "Limits" > "Transmit Spectrum Mask" > "802.11b/g DSSS" section.

The energy that spills outside the designated radio channel increases the interference with adjacent channels and decreases the system capacity. The amount of unwanted off-carrier energy is assessed by a spectrum measurement.

Limits are defined via a transmit spectrum mask divided into several frequency areas.



**Figure 3-41: Transmit spectrum mask settings for DSSS**

The transmit spectrum mask is defined via points connected by horizontal lines. The mask is symmetric relative to the center frequency of the RF analyzer (offset 0 MHz), see figure below. The borders of the areas are fixed as defined by IEEE, with the points A at the edges of the frequency range covered by the measurement.

You can configure a power level limit for the points A/B, and C/D. The limit for points E always equals 0 dB.



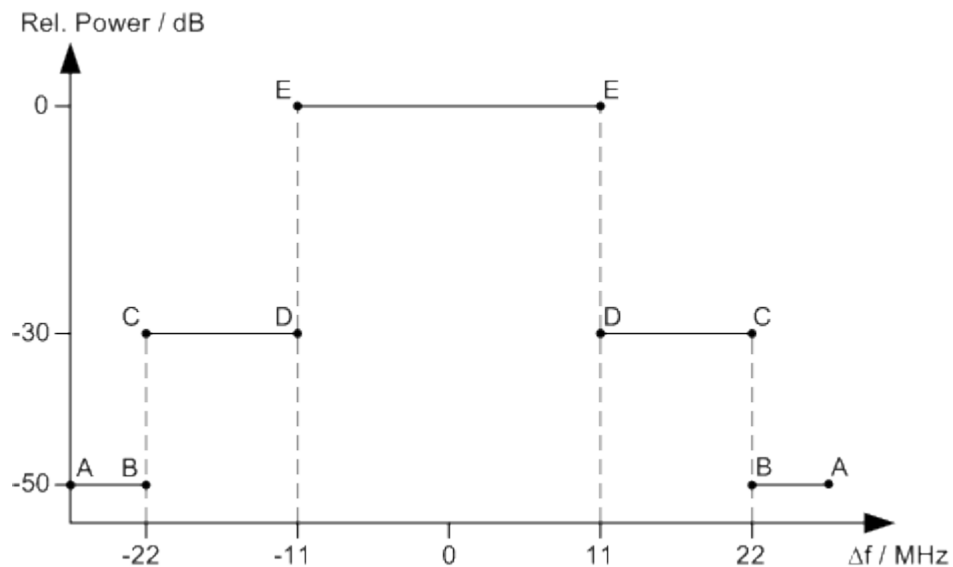


Figure 3-42: Transmit spectrum mask (DSSS)

The default settings shown in the figure above are specified in IEEE Std 802.11-2016, section 15.4.5.5/16.3.7.4 Transmit spectrum mask.

For remote commands, refer to [Chapter 3.5.4.13, "Transmit spectrum mask limits: 802.11b/g DSSS"](#), on page 437.

#### Transmit spectrum mask OFDM, default masks

Access: "Config" hotkey > "Limits" > "Transmit Spectrum Mask" > "802.11a/g", "802.11n", "802.11ac", or "802.11ax" section.

The default masks for OFDM signals are defined via points A,B,C,D,E connected by straight lines. The masks are symmetric relative to the center frequency of the RF analyzer (offset 0 MHz). The points A are at the edges of the frequency range covered by the measurement.

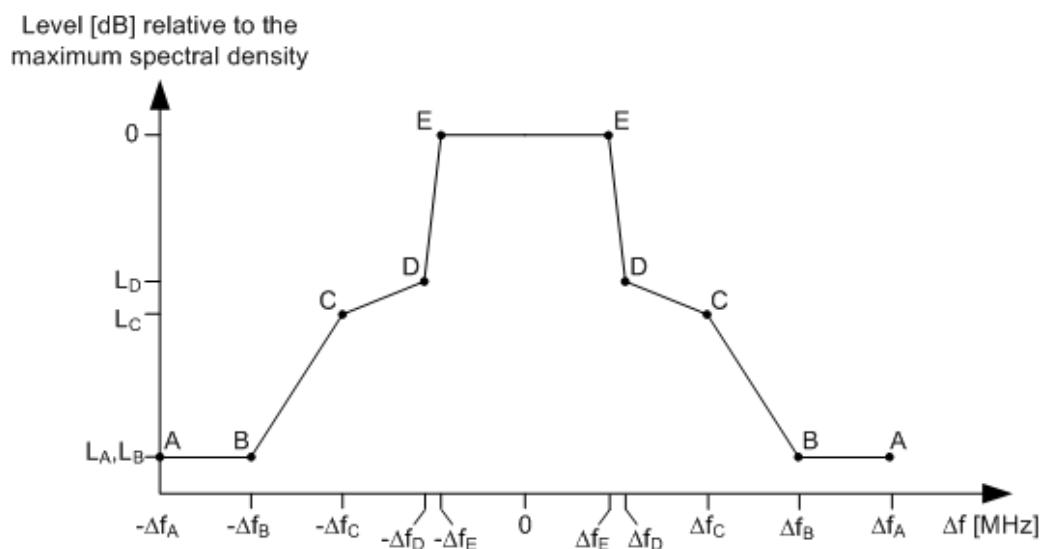


Figure 3-43: Default transmit spectrum mask (OFDM)

The frequency offsets  $\Delta f_A$ , ...,  $\Delta f_E$  and levels  $L_A$ , ...,  $L_E$  are specified in:

- IEEE Std 802.11-2016, section 17.3.9.3 for 802.11a/g/p
- IEEE Std 802.11-2016, section 19.3.18.1 for 802.11n (OFDM HT)
- IEEE Std 802.11-2016, section 21.3.17.1 for 802.11ac (OFDM VHT)
- IEEE Std 802.11ax-2021, section 27.3.19.1 for 802.11ax (OFDM HE)

Table 3-12: Default transmit spectrum masks for 802.11a/g/p, n, and ac

| Point | Frequency offset<br>(relative to channel BW)                            | Relative limit<br>• 802.11a/g/p<br>• 802.11n in 5 GHz<br>band | Relative limit<br>• 802.11n in 2.4 GHz<br>band |
|-------|-------------------------------------------------------------------------|---------------------------------------------------------------|------------------------------------------------|
| A     | $\Delta f_A = 2 \text{ BW}$                                             | $L_A = -40 \text{ dB}$                                        | $L_A = -45 \text{ dB}$                         |
| B     | $\Delta f_B = 1.5 \text{ BW}$                                           | $L_B = -40 \text{ dB}$                                        | $L_B = -45 \text{ dB}$                         |
| C     | $\Delta f_C = \text{BW}$                                                | $L_C = -28 \text{ dB}$                                        | $L_C = -28 \text{ dB}$                         |
| D     | $\Delta f_D = 0.5 \text{ BW} + \min \{0.05 \text{ BW}, 1 \text{ MHz}\}$ | $L_D = -20 \text{ dB}$                                        | $L_D = -20 \text{ dB}$                         |
| E     | $\Delta f_E = 0.5 \text{ BW} - \min \{0.05 \text{ BW}, 1 \text{ MHz}\}$ | $L_E = 0 \text{ dB}$                                          | $L_E = 0 \text{ dB}$                           |

Table 3-13: Exception: Default transmit spectrum masks for 802.11ax

| Point | Frequency offset in MHz according to channel BW |        |        |         | Relative limit         |
|-------|-------------------------------------------------|--------|--------|---------|------------------------|
|       | 20 MHz                                          | 40 MHz | 80 MHz | 160 MHz |                        |
| A     | $\Delta f_A > 30$                               | 60     | 120    | 240     | $L_A = -40 \text{ dB}$ |
| B     | $\Delta f_B = 30$                               | 60     | 120    | 240     | $L_B = -40 \text{ dB}$ |
| C     | $\Delta f_C = 20$                               | 40     | 80     | 160     | $L_C = -28 \text{ dB}$ |

| Point | Frequency offset in MHz according to channel BW |        |        |         | Relative limit         |
|-------|-------------------------------------------------|--------|--------|---------|------------------------|
|       | 20 MHz                                          | 40 MHz | 80 MHz | 160 MHz |                        |
| D     | $\Delta f_D = 10.5$                             | 20.5   | 40.5   | 80.5    | $L_D = -20 \text{ dB}$ |
| E     | $\Delta f_E = 9.75$                             | 19.5   | 39.5   | 79.5    | $L_E = 0 \text{ dB}$   |



In the presence of additional regulatory restrictions, the DUT needs to meet both the regulatory requirements and the mask defined here. Its emissions must be no higher at any frequency offset than the minimum of the values specified in the regulatory and default masks.

The transmit spectrum masks by regulatory domain for 802.11p are described in ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

The spectrum mask offsets at the R&S CMW are the default spectrum mask offsets and the default relative limits are used as preset values. You can configure non-standard power level limits  $L_A, \dots, L_D$  though:

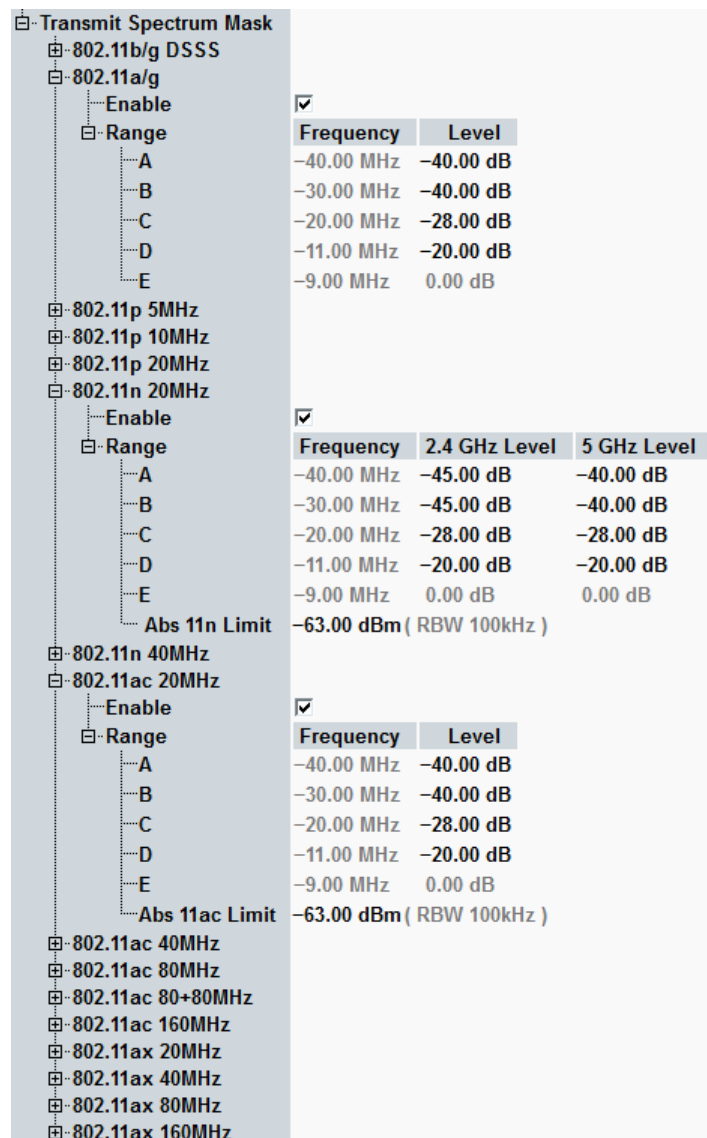


Figure 3-44: Default transmit spectrum masks for OFDM

$L_E$  always equals 0 dB.

For remote commands, refer to:

- [Chapter 3.5.4.14, "Transmit spectrum mask limits: 802.11a/g OFDM"](#), on page 438
- [Chapter 3.5.4.15, "Transmit spectrum mask limits: 802.11p OFDM"](#), on page 439
- [Chapter 3.5.4.16, "Transmit spectrum mask limits: 802.11n"](#), on page 449
- [Chapter 3.5.4.17, "Transmit spectrum mask limits: 802.11ac"](#), on page 452
- [Chapter 3.5.4.18, "Transmit spectrum mask limits: 802.11ax"](#), on page 455

### Transmit spectrum mask OFDM, by regulation

Access: "Config" hotkey > "Limits" > "Transmit Spectrum Mask" > "802.11p" section.

Regulatory masks are defined for 802.11p stations. The IEEE masks are described in this section. For the ETSI and ARIB masks, see ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322.

The IEEE masks are defined via points P1 to P5 connected by straight lines. The masks are symmetric relative to the center frequency of the RF analyzer (offset 0 MHz).

Level [dB] relative to the maximum spectral density

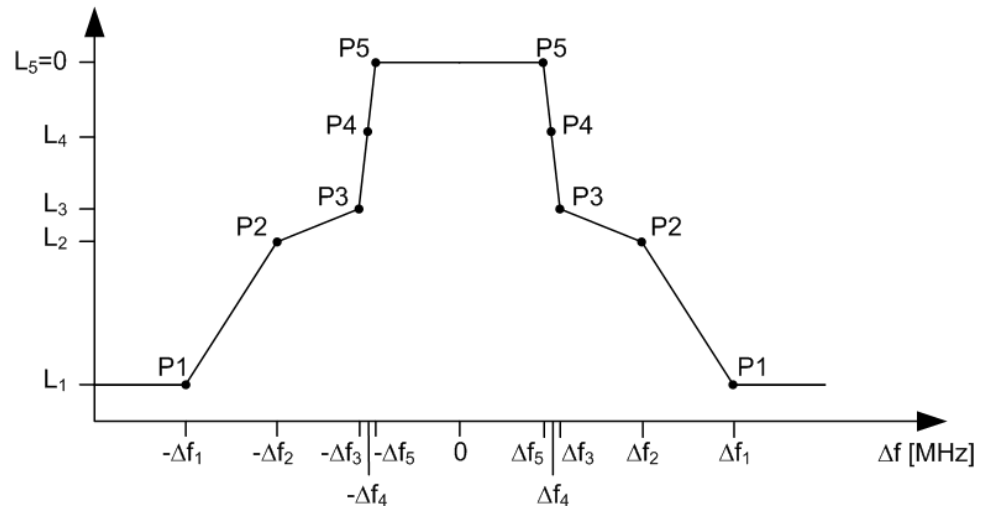


Figure 3-45: Regulatory IEEE transmit spectrum mask

The frequency offsets  $\Delta f_1, \dots, \Delta f_5$  and relative limits  $L_A, \dots, L_E$  are specified in annex D.2.3 of standard IEEE Std 802.11-2016. A distinction is made between different transmit power level classes A to D, as defined in annex D.2.2 of the same standard.

Table 3-14: Transmit spectrum mask by regulatory restrictions in the 5.85 GHz to 5.925 GHz band (802.11p)

| Point | Frequency offset<br>(relative to channel BW<br>5, 10, 20 MHz) | Relative limit (by power class) |                        |                        |                        |
|-------|---------------------------------------------------------------|---------------------------------|------------------------|------------------------|------------------------|
|       |                                                               | A                               | B                      | C                      | D                      |
| P1    | $\Delta f_1 = 1.5 \text{ BW}$                                 | $L_1 = -40 \text{ dB}$          | $L_1 = -40 \text{ dB}$ | $L_1 = -50 \text{ dB}$ | $L_1 = -65 \text{ dB}$ |
| P2    | $\Delta f_2 = \text{BW}$                                      | $L_2 = -28 \text{ dB}$          | $L_2 = -28 \text{ dB}$ | $L_2 = -40 \text{ dB}$ | $L_2 = -55 \text{ dB}$ |
| P3    | $\Delta f_3 = 0.55 \text{ BW}$                                | $L_3 = -20 \text{ dB}$          | $L_3 = -20 \text{ dB}$ | $L_3 = -32 \text{ dB}$ | $L_3 = -45 \text{ dB}$ |
| P4    | $\Delta f_4 = 0.5 \text{ BW}$                                 | $L_4 = -10 \text{ dB}$          | $L_4 = -16 \text{ dB}$ | $L_4 = -26 \text{ dB}$ | $L_4 = -35 \text{ dB}$ |
| P5    | $\Delta f_5 = 0.45 \text{ BW}$                                | $L_5 = 0 \text{ dB}$            | $L_5 = 0 \text{ dB}$   | $L_5 = 0 \text{ dB}$   | $L_5 = 0 \text{ dB}$   |

The limit check depends on the configured power class (see [Chapter 3.3.3.3, "Input signal settings"](#), on page 293) as follows:

- For the power classes A, B and user-defined, the R&S CMW takes the intersection of the default spectrum mask (see ["Transmit spectrum mask OFDM, default"](#)



- -59 dBm per MHz for 80, 80+80, and 160 MHz channels

See:

- IEEE Std 802.11-2016, section 19.3.18.1 for 802.11n,
- IEEE Std 802.11-2016, section 21.3.17.1 for 802.11ac, and
- IEEE Std 802.11ax-2021, section 27.3.19.1 for 802.11ax.

The absolute limits for standards 802.11n, ac, and ax are defined for a resolution bandwidth of 100 kHz. These values are used as the default "Abs 11n Limit", "Abs 11ac Limit", and "Abs 11ax Limit" values at the R&S CMW.

For configuration, refer to [Figure 3-44](#).

**Table 3-15: Absolute IEEE limits**

| Channel bandwidth | Absolute 802.11n, ac, ax limits |
|-------------------|---------------------------------|
| 5 MHz             | -57 dBm                         |
| 10 MHz            | -60 dBm                         |
| 20 MHz            | -63 dBm                         |
| 40 MHz            | -66 dBm                         |
| 80 MHz            | -69 dBm                         |
| 80+80 MHz         | -69 dBm                         |
| 160 MHz           | -69 dBm                         |

ETSI EN 302 571 V1.1.1 (2008-09) imposes absolute emission limits for intelligent transport systems (ITS), operating in the 5.85 GHz to 5.925 GHz band. The limits are specified in section 6.4.2.2 and are summarized in the following table.

Linear interpolation is performed according to [Figure 3-45](#).

These values are used as preset "Abs Level"s (see [Figure 3-46](#)).

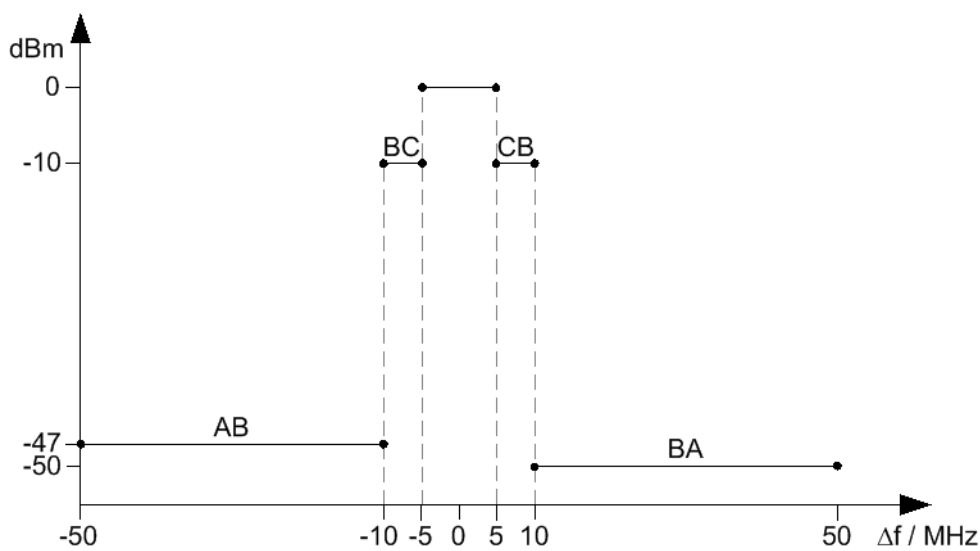
**Table 3-16: Absolute ETSI limits for the 5 GHz ITS band**

| Point | Frequency offset               | Limit in dBm (per 1 MHz) |                   |
|-------|--------------------------------|--------------------------|-------------------|
|       |                                | Channel BW 10 MHz        | Channel BW 20 MHz |
| P1    | $\Delta f_1 = 1.5 \text{ BW}$  | -27 dBm                  | ---               |
| P2    | $\Delta f_2 = \text{BW}$       | -17 dBm                  | ---               |
| P3    | $\Delta f_3 = 0.55 \text{ BW}$ | -9 dBm                   | -9 dBm            |
| P4    | $\Delta f_4 = 0.5 \text{ BW}$  | -3 dBm                   | -3 dBm            |
| P5    | $\Delta f_5 = 0.45 \text{ BW}$ | 23 dBm                   | 23 dBm            |

ARIB STD-T109 also defines absolute emission limits for 802.11p. They apply only for 10 MHz bandwidth. The limits are summarized in the following table. The ARIB mask is not configurable.

**Table 3-17: Absolute ARIB limits for 802.11p, 10 MHz**

| Range | Frequency offset   | Limit in dBm (per 100 kHz) |
|-------|--------------------|----------------------------|
| AB    | -50 MHz to -10 MHz | -47 dBm                    |
| BC    | -10 MHz to -5 MHz  | -10 dBm                    |
| CB    | 5 MHz to 10 MHz    | -10 dBm                    |
| BA    | 10 MHz to 50 MHz   | -50 dBm                    |

**Figure 3-47: ARIB transmit spectrum mask**

For ...:ABSolute:... or ...:ABSLimit remote commands, refer to:

- [Chapter 3.5.4.15, "Transmit spectrum mask limits: 802.11p OFDM"](#), on page 439
- [Chapter 3.5.4.16, "Transmit spectrum mask limits: 802.11n"](#), on page 449
- [Chapter 3.5.4.17, "Transmit spectrum mask limits: 802.11ac"](#), on page 452
- [Chapter 3.5.4.18, "Transmit spectrum mask limits: 802.11ax"](#), on page 455

### 3.3.3.7 Generator shortcut

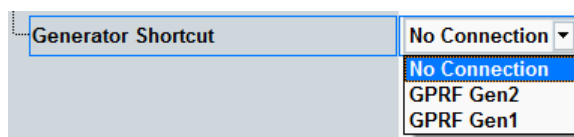
Access: "Config" hotkey > "Generator Shortcut" section.

A standard use case in "StandAlone (Non-Signaling)" mode is to use the GPRF generator to generate a signal the DUT replies to. In particular, one is interested in the DUT's reaction on varying generator signals.

To provide better support for this use case, the parameters of the GPRF generator can be set from the WLAN measurement GUI.

Select the appropriate GPRF generator instance.





When a connection to the GPRF generator instance is established, two additional softkeys (with corresponding hotkey bars) provide access to the generator configuration.

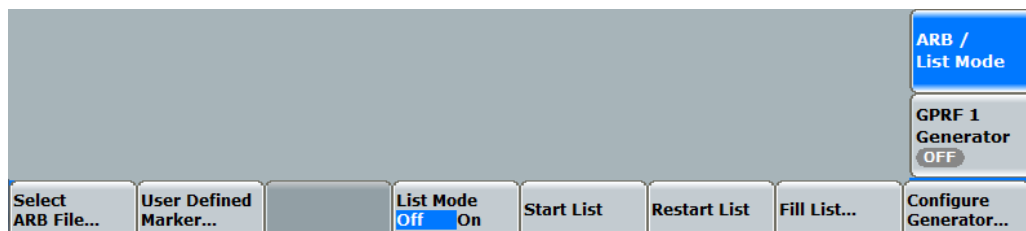


Figure 3-48: ARB: softkey and hotkey bar

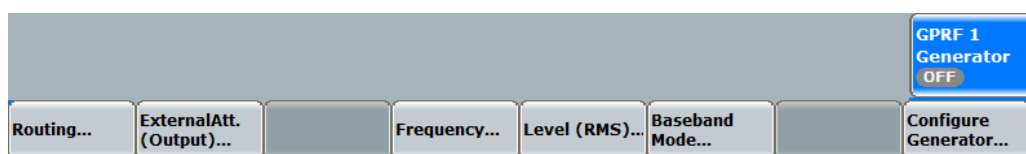


Figure 3-49: GPRF<instance> generator: softkey and hotkey bar

Use the appropriate softkey/hotkey combination to access the generator parameters.



- The "Configure Generator" softkey opens the configuration tree of the connected GPRF generator instance.
- For details on the available parameters, refer to the GPRF generator documentation.

### 3.3.3.8 Additional softkeys and hotkeys

The measurement provides some softkey/hotkey combinations which have no equivalent in the configuration dialog. Most of these hotkeys provide display configurations (like diagram scaling). They are self-explanatory and do not have any remote-control commands assigned.

The remaining softkeys > hotkeys are described below.

The softkey "WLAN Signaling" is displayed only if the combined signal path scenario is active and is provided by the "WLAN Signaling" application selected as controlling application. See also ["Scenario = Combined Signal Path"](#) on page 289.

The softkeys "ARB / List Mode" and "GPRF Generator" are displayed only if the stand-alone scenario is active and the generator shortcut is enabled, see [Chapter 3.3.3.7, "Generator shortcut"](#), on page 324.

While one of the signaling or generator softkeys is selected, the "Config" hotkey opens the configuration dialog of the generator or signaling application, not the configuration dialog of the measurement.

### WLAN Signaling

Select this softkey and press [ON | OFF] to turn the signal transmission on or off. Alternatively, right-click the softkey.

Press the softkey two times (select it and press it again) to switch to the signaling application.

Remote command:

Use the commands of the signaling application.

### ARB / List Mode > ...

Provides access to the most important ARB and list mode settings of the GPRF generator.

Remote command:

Use the commands of the GPRF generator.

### GPRF Generator

Select this softkey and press [ON | OFF] to turn the GPRF generator on or off. Alternatively, right-click the softkey.

Press the softkey two times (select it and press it again) to switch to the generator application.

The hotkeys assigned to this softkey provide access to the most important GPRF generator settings.

Remote command:

Use the commands of the GPRF generator.

### Display Mode

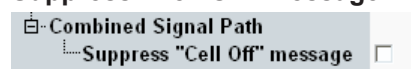
In the "Spectrum Flatness" or "Transmit Spectrum Mask" view, select "Display" and "Display Mode" to switch between relative and absolute result values (dB vs. dBm).

Remote command:

`CONFigure:WLAN:MEAS<i>:MEvaluation:SFlatness:DMODE`

`CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:DMODE`

### Suppress "Cell Off" Message



This setting is located in the configuration tree. It is only visible if the combined signal path scenario is active.

The checkbox suppresses a warning that is displayed if the "WLAN Signaling" softkey is selected and the signal is on and you press [ON | OFF]. The warning asks you whether you really want to switch off the signal.

Remote command:

n/a

### 3.3.4 CMIMO training mode

The training mode is relevant for composite MIMO (CMIMO) measurements.

See also:

- [Chapter 3.2.2.1, "Composite MIMO measurements"](#), on page 244
- [Chapter 3.3.2.10, "View TX measurement \(scalar\) for CMIMO \(802.11n, ac, ax\)"](#), on page 275

#### 3.3.4.1 Activation and deactivation

If the receive mode is set to "Composite MIMO", the following hotkeys allow you to toggle between training mode and measurement mode.

##### Training Mode

Activate the training mode and display the corresponding view via:

- "Training Mode (Button)" available in the configuration tree. Refer to ["Training Mode"](#) on page 290.
- "Training Mode (Hotkey)" available in the measurement mode. Press the softkey "Multi Evaluation" to display the hotkey.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MODE
```

##### Measure Mode (Hotkey)

This hotkey activates the measurement mode and displays the corresponding view. It is only available in training mode. Press the softkey "Antenna <i> Data Acqui." to display the hotkey.

Remote command:

```
CONFigure:WLAN:MEAS<i>:MODE
```

#### 3.3.4.2 Training data acquisition

Like any other measurement, the training data acquisition is controlled using the [ON | OFF] or [RESTART | STOP] keys. The context (for the selected TX antenna) is set via the following softkey:



##### Antenna <i> Data Acqui. (Softkey)

Sets the focus of the [ON | OFF] or [RESTART | STOP] keys to the training mode view and activates the training mode hotkeys.

Remote command:

```
INITiate:WLAN:MEAS<i>:TMODe:ANTenna<Antennas>
ABORt:WLAN:MEAS<i>:TMODe
```

### 3.3.4.3 Training mode view

This view displays the current training mode settings and the results of the preceding training measurements.

|                                                                               |           |               |                  |
|-------------------------------------------------------------------------------|-----------|---------------|------------------|
| Freq: 2412.000000 MHz Ref.Lvl: 0.00 dBm Std: 802.11n (MIMO) Green Field 40MHz |           |               |                  |
| TX Measurement (Scalar)                                                       |           |               |                  |
| Settings                                                                      |           |               |                  |
| Number of Antennas                                                            | 4         |               |                  |
| Selected Action                                                               | Antenna 4 |               |                  |
| Results                                                                       | MCS       | Power [ dBm ] | Pilot EVM [ dB ] |
| Antenna 1 Data                                                                | 42        | -12.85        | -43.59           |
| Antenna 2 Data                                                                | 42        | -12.85        | -42.24           |
| Antenna 3 Data                                                                | 42        | -12.85        | -42.94           |
| Antenna 4 Data                                                                | 42        | -12.83        | -43.04           |
| Decoding Status                                                               | OK        |               |                  |
| File Name                                                                     |           |               |                  |

Figure 3-50: MIMO training mode view

In the "Settings" part, the training mode view displays:

- The number of antennas contributing to the MIMO training signal
- The currently selected antenna

#### Results

The "Results" table displays the training mode results that were gathered during the current WLAN measurement session.

Remote command:

```
FETCh:WLAN:MEAS<i>:TMODe:ANTenna<Antennas>?
READ:WLAN:MEAS<i>:TMODe:ANTenna<Antennas>?
```

#### Save Data to File (Hotkey)

If the "Decoding Status" is OK, it is possible to save the acquired training data to a file for later reuse. This hotkey calls up a "Save File" dialog.

Remote command:

```
CONFigure:WLAN:MEAS<i>:TMODe:FILE:SAVE
CONFigure:WLAN:MEAS<i>:TMODe:FILE:DATE?
```

#### Clear Data (Hotkey)

This hotkey clears the training results and resets the internal data of the training mode.

Remote command:

```
CLEAr:WLAN:MEAS<i>:TMODe:DATA
```

### 3.3.4.4 Antenna configuration

Training mode and measurement mode share the basic WLAN input signal configuration (see [Chapter 3.3.3.3, "Input signal settings"](#), on page 293). The antenna configuration is done via hotkeys.



#### No of Antennas (Hotkey)

This hotkey opens a popup that allows you to set the number of TX antennas in the MIMO signal to be recorded.

Remote command:

`CONFfigure:WLAN:MEAS<i>:TMODe:NOAntennas`

#### Select Antenna (Hotkey)

This hotkey opens a popup that allows you to set the TX antenna signal to be recorded.

Note that there is no dedicated remote command for selecting the antenna without starting the training data. At the GUI, the antenna number is kept in memory until the training mode measurement is started.

## 3.4 Programming

The following sections provide programming examples for the WLAN multi-evaluation measurement.

The WLAN multi-evaluation measurement is programmed as follows:

- The measurement is controlled by SCPI commands with the following syntax: `...WLAN:MEAS:MEValuation...`
- Use general commands of the type `...:WLAN:MEAS...` (no `:MEValuation` mnemonic) to define the signal routing and perform RF and analyzer settings.
- Use general commands of the type `...:WLAN:MEAS:ISIGnal...` (no `:MEValuation` mnemonic) to inform the R&S CMW about the basic properties of the measured WLAN signal.
- After a `*RST`, the measurement is switched off. Use `READ:WLAN:MEAS:MEValuation...?` to initiate a single-shot measurement and retrieve the results. You can also start the measurement using `INIT:WLAN:MEAS:MEValuation` and retrieve the results using `FETCH:WLAN:MEAS:MEValuation...?`.
- [Specifying general measurement settings](#)..... 330
- [Specifying measurement-specific settings](#)..... 331
- [Configuring the trigger system](#)..... 332
- [Specifying limits \(DSSS\)](#)..... 332
- [Specifying power vs. time limits](#)..... 333
- [Specifying modulation limits \(OFDM\)](#)..... 333

|                                                                    |     |
|--------------------------------------------------------------------|-----|
| • Specifying spectrum flatness limits.....                         | 334 |
| • Specifying transmit spectrum masks (OFDM).....                   | 335 |
| • Performing single-shot measurements (OFDM, SISO).....            | 336 |
| • Performing single-shot measurements (OFDM, CMIMO).....           | 337 |
| • Performing single-shot measurements (OFDM, MIMO, 80+80 MHz)..... | 338 |
| • Performing single-shot measurements (DSSS).....                  | 339 |
| • Single-shot and continuous measurements.....                     | 340 |
| • Retrieving signal fields reports.....                            | 340 |
| • CMIMO training mode.....                                         | 343 |
| • Using WLAN list mode.....                                        | 344 |

### 3.4.1 Specifying general measurement settings

```
// *****
// Initial system-reset
// *****
*RST; *OPC?
*CLS; *OPC?

// *****
// Define signal routing, perform RF and analyzer settings for a WLAN
// signal with a carrier frequency of 2422 MHz, a peak power of 17 dBm,
// no user margin and 1 dB mixer level offset.
// *****
ROUTE:WLAN:MEAS:SCENario:SALone RF1C, RX1
CONFIGure:WLAN:MEAS:RFSettings:EATTenuation 2
CONFIGure:WLAN:MEAS:RFSettings:FREQuency:BAND B24Ghz
CONFIGure:WLAN:MEAS:RFSettings:FREQuency 2422E+6
CONFIGure:WLAN:MEAS:RFSettings:ENPower 17
CONFIGure:WLAN:MEAS:RFSettings:UMARgin 0
CONFIGure:WLAN:MEAS:RFSettings:MLOffset 1

// *****
// Define input signal settings for 802.11n:
// Select standard, receive mode, burst type, bandwidth and I/Q swap.
// *****
CONFIGure:WLAN:MEAS:ISIGnal:STANdard HTOFdm
CONFIGure:WLAN:MEAS:ISIGnal:RMODE SISO
CONFIGure:WLAN:MEAS:ISIGnal:BTYPe MIXed
CONFIGure:WLAN:MEAS:ISIGnal:BWIDth BW20
CONFIGure:WLAN:MEAS:ISIGnal:IQSWap OFF

// *****
// Alternatively define input signal settings for 802.11ac: Set
// standard, receive mode, evaluation length, bandwidth and channel
// distance. Use separate antennas for the two segments.
// *****
CONFIGure:WLAN:MEAS:ISIGnal:STANdard VHTofdm
CONFIGure:WLAN:MEAS:ISIGnal:RMODE SISO
```

```

CONFigure:WLAN:MEAS:ISIGnal:OFDM:ELENgth 1377
CONFigure:WLAN:MEAS:ISIGnal:BWIDth BW88mhz
CONFigure:WLAN:MEAS:ISIGnal:CDIStance 80
CONFigure:WLAN:MEAS:RFSettings:SANTennas ON

/// *****
// Alternatively configure the true MIMO connection 2x2 with R&S CMW500. Set
// external attenuation and nominal power per antenna
// *****
ROUTe:WLAN:MEAS:SCENario:TMIMo2
CONFigure:WLAN:MEAS:MIMO:NOANTennas 2
CONFigure:WLAN:MEAS:ISIGnal:RMODE TMIMo
CONFigure:WLAN:MEAS:RFSettings:Antenna1 RF1C,0,-20
CONFigure:WLAN:MEAS:RFSettings:Antenna2 RF3C,0,-20

```

### 3.4.2 Specifying measurement-specific settings

```

// *****
// Define synchronization mode, error handling.
// *****
CONFigure:WLAN:MEAS:MEValuation:SMODE TOLerant
CONFigure:WLAN:MEAS:MEValuation:SCONdition SLFail
CONFigure:WLAN:MEAS:MEValuation:MOEXception ON

// *****
// Define modulation measurement settings.
// *****
CONFigure:WLAN:MEAS:MEValuation:SCount:MODulation 20
CONFigure:WLAN:MEAS:MEValuation:COMPensation:TRACking:PHASe OFF
CONFigure:WLAN:MEAS:MEValuation:COMPensation:TRACking:TIMing OFF
CONFigure:WLAN:MEAS:MEValuation:COMPensation:TRACking:LEVel OFF

CONFigure:WLAN:MEAS:MEValuation:COMPensation:CESTimation PAYL
CONFigure:WLAN:MEAS:MEValuation:DEMod:FFT:OFFSet AUTO
CONFigure:WLAN:MEAS:MEValuation:COMPensation:EFTaps OFF
CONFigure:WLAN:MEAS:MEValuation:EMETHod ST2007
CONFigure:WLAN:MEAS:MEValuation:COMPensation:SKIPsymbols 1, 1

// *****
// Define spectrum measurement settings.
// *****
CONFigure:WLAN:MEAS:MEValuation:SCount:TSMask 20
CONFigure:WLAN:MEAS:MEValuation:TSMask:TROTime 0.0001
CONFigure:WLAN:MEAS:MEValuation:TSMask:AFFTnum 5
CONFigure:WLAN:MEAS:MEValuation:TSMask:OBWPercent ON

// *****
// Define power measurement settings.

```

```
// *****
CONFigure:WLAN:MEAS:MEvaluation:SCount:PVTime 20
CONFigure:WLAN:MEAS:MEvaluation:PVTime:ALENgtH 5
CONFigure:WLAN:MEAS:MEvaluation:PVTime:RPOWer MEAN

// *****
// Enable all measurements (DSSS and OFDM).
// *****
CONFigure:WLAN:MEAS:MEvaluation:RESult ON,ON,ON,ON,ON,ON,ON,ON,ON,ON
```

### 3.4.3 Configuring the trigger system

```
// *****
// Set trigger source, timeout, trigger level, slope and minimum trigger gap.
// *****
TRIGger:WLAN:MEAS:MEvaluation:SOURce 'IF Power'
TRIGger:WLAN:MEAS:MEvaluation:TOUT 1
TRIGger:WLAN:MEAS:MEvaluation:THReshold -25
TRIGger:WLAN:MEAS:MEvaluation:SLOPe REDGe
TRIGger:WLAN:MEAS:MEvaluation:MGAP 0.00002
```

### 3.4.4 Specifying limits (DSSS)

```
// *****
// Define Power vs. Time limits for DSSS signals.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:PVTime:REDGe:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:PVTime:REDGe 2.1E-6
CONFigure:WLAN:MEAS:MEvaluation:LIMit:PVTime:FEDGE:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:PVTime:FEDGE 2.1E-6

// *****
// Define modulation limits for DSSS signals.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:DSSS:EVMPeak 30
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:DSSS:EVMRms 25
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:DSSS:IQOffset -16
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:DSSS:CFERror 50000
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:DSSS:CCERror 21

// *****
// Define transmission spectrum mask for DSSS signals.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:DSSS:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:DSSS:Y:AB -49
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:DSSS:Y:CD -29
```



### 3.4.5 Specifying power vs. time limits

```
// *****
// Define timing error limit and its maximal distribution error.
// *****
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:PVTime:TERRor 0.4E-6,ON
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:PVTime:TEDistrib 10
```

### 3.4.6 Specifying modulation limits (OFDM)

```
// *****
// Define modulation limits for OFDM signals, standards 802.11a, g.
// *****
CONF:WLAN:MEAS:MEV:LIMit:MODulation:LOFDm:EVM -5,-8,-10,-13,OFF,OFF,-23,-27
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:LOFDm:EVMPilot -10
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:LOFDm:IQOffset -16
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:LOFDm:CFERror 45000
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:LOFDm:SCERror 21

// *****
// Define modulation limits for standard 802.11n.
// *****
CONF:WLAN:MEAS:MEV:LIMit:MODulation:HTOFdm:EVM -5,-8,-10,-13,OFF,OFF,-23,-27
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HTOFdm:EVMPilot -10
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HTOFdm:IQOffset:BW40 -16
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HTOFdm:CFERror 45000
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HTOFdm:SCERror 21

// *****
// Define modulation limits for standard 802.11ac.
// *****
CONF:WLAN:MEAS:MEV:LIM:MOD:VHT:EVM -5,-8,-13,-16,-19,-22,-25,-27,-30,-32,-35,-35
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:VHTofdm:EVMPilot -10
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:VHTofdm:IQOffset:BW40 -20

CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:VHTofdm:IQOffset:BW8080 -25,-20

CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:VHTofdm:CFERror 45000
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:VHTofdm:SCERror 21

// *****
// Define modulation limits for standard 802.11ax.
// *****
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:EVMall:TBCoderate CR12,
    CR34,CR34,CR56,CR56,CR34
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:EVMall:TBHigh -13,-13,
    -30,-32,-35,-35
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:EVMall:TBLow -27,-27,
    -30,-32,-35,-35
CONFIGure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:EVMPilo:TBHigh -13
```

```

CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:EVMPilot:TBLow -27

CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:IQOffset:BW20 -17,-20
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:IQOffset:BW40 -20,-20
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:IQOffset:BW80 -23,-20
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:IQOffset:BW160 -32,-20

CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:IQOffset:BW8080 -26,-20

CONFigure:WLAN:MEAS:MEvaluation:LIMit:UTEPower HIGH
CONFigure:WLAN:MEAS:MEvaluation:LIMit:UTERror ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:CFError 50000
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:HEOFdm:CFDistrib 10,50000

// *****
// Define modulation limits for standard 802.11p.
// *****
CONF:WLAN:MEAS:MEV:LIMit:MODulation:POFDm:EVM -6,-8,-10,-13,-16,-19,-22,-25
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:POFDm:EVMPilot -7
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:POFDm:IQOffset -16
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:POFDm:CFError 61000
CONFigure:WLAN:MEAS:MEvaluation:LIMit:MODulation:POFDm:SCError 24

```

### 3.4.7 Specifying spectrum flatness limits

```

// *****
// Define spectrum flatness limits for OFDM signals, standards 802.11a, g.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:LOFDm:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:LOFDm:UPPer 3
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:LOFDm:LOWer -3,-4

// *****
// Define spectrum flatness limits for standard 802.11n, 40 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:HTOFdm:BW40:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:HTOFdm:BW40:UPPer 3
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:HTOFdm:BW40:LOWer -3,-4

// *****
// Define spectrum flatness limits for standard 802.11ac, 40 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:VHTofdm:BW40:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:VHTofdm:BW40:UPPer 3
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:VHTofdm:BW40:LOWer -4,-7

// *****
// Define spectrum flatness limits for standard 802.11ax, 80 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:HEOfdm:BW80:ENABle ON

```

```
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:HEOfdm:BW80:UPPer 3
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:HEOfdm:BW80:LOWer -4,-7
```

```
// *****
// Define spectrum flatness limits for standard 802.11p, 20 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:POFDm:BW20:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:POFDm:BW20:UPPer 3
CONFigure:WLAN:MEAS:MEvaluation:LIMit:SFLatness:POFDm:BW20:LOWer -4,-7
```

### 3.4.8 Specifying transmit spectrum masks (OFDM)

```
// *****
// Define spectrum mask for OFDM signals, standards 802.11a, g.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:LOFDm:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:LOFDm:Y:A -39
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:LOFDm:Y:B -39
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:LOFDm:Y:C -27
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:LOFDm:Y:D -19

// *****
// Define spectrum mask for standard 802.11n, 40 MHz, 5-GHz band.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HTOFdm:BW40:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HTOFdm:BW40:Band5:Y:A -46
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HTOFdm:BW40:Band5:Y:B -46
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HTOFdm:BW40:Band5:Y:C -30
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HTOFdm:BW40:Band5:Y:D -20
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HTOFdm:BW40:ABSLimit -70

// *****
// Define spectrum mask for standard 802.11ac, 40 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:VHTofdm:BW40:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:VHTofdm:BW40:Y:A -44
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:VHTofdm:BW40:Y:B -44
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:VHTofdm:BW40:Y:C -27
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:VHTofdm:BW40:Y:D -19
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:VHTofdm:BW40:ABSLimit -65

// *****
// Define spectrum mask for standard 802.11ax, 80 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HEOfdm:BW80:ENABle ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HEOfdm:BW80:Y:A -44
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HEOfdm:BW80:Y:B -44
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HEOfdm:BW80:Y:C -27
```

```

CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HEOfdm:BW80:Y:D -19
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:HEOfdm:BW80:ABSLimit -65

// *****
// Define spectrum mask for standard 802.11p, 10 MHz.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ENABLE ON
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CA:Y:A -41
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CA:Y:B -41
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CA:Y:C -29
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CA:Y:D -21
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CA:Y:E -11
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CB:Y:A -41
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CB:Y:B -41
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CB:Y:C -29
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CB:Y:D -21
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:CB:Y:E -17
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:UDEFined:Y:A -51
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:UDEFined:Y:B -51
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:UDEFined:Y:C -41
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:UDEFined:Y:D -33
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:UDEFined:Y:E -27
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ABSolute:Y:A -28
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ABSolute:Y:B -28
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ABSolute:Y:C -18
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ABSolute:Y:D -10
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ABSolute:Y:E -4
CONFigure:WLAN:MEAS:MEvaluation:LIMit:TSMask:POFDm:BW10:ABSolute:Y:F 22

```

### 3.4.9 Performing single-shot measurements (OFDM, SISO)

```

// *****
// After configuring the measurement for an 802.11a OFDM signal:
// Start the measurement and return an EVM trace.
// Query the measurement state (should be "RDY").
// *****
READ:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:SYMBOL:CURRENT?
FETCh:WLAN:MEAS:MEvaluation:STATE?

// *****
// Configure external trigger, query timing error and its distribution.
// *****
TRIGger:WLAN:MEAS:MEvaluation:SOURce "Basel: External TRIG A"
READ:WLAN:MEAS:MEvaluation:PVTime:TERRor:CURRENT?
FETCh:WLAN:MEAS:MEvaluation:PVTime:TEDistrib?

// *****
// Query traces for EVM, I/Q constellation, spectrum, PvT (power and time
// results). The results are from the previous measurement.

```

```
// *****
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:SYMBol:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:IQConst:INPhase?
FETCh:WLAN:MEAS:MEvaluation:TRACe:IQConst:QUADrature?
FETCh:WLAN:MEAS:MEvaluation:TRACe:SFLatness:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:FREQuency?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:TIME?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TERRor?
FETCh:WLAN:MEAS:MEvaluation:TRACe:CFERror?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:REDGe:MINimum?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:FEDGe:MINimum?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:REDGe:TIME?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:FEDGe:TIME?

// *****
// Query statistical results from the previous measurement.
// *****
FETCh:WLAN:MEAS:MEvaluation:MODulation:CURRent?
FETCh:WLAN:MEAS:MEvaluation:MODulation:CFDistrib?
FETCh:WLAN:MEAS:MEvaluation:SFLatness:AVERage?
FETCh:WLAN:MEAS:MEvaluation:SFLatness:MINimum?
FETCh:WLAN:MEAS:MEvaluation:SFLatness:X:MINimum?
FETCh:WLAN:MEAS:MEvaluation:TSMask:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:FREQuency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:OBW?

// *****
// Alternatively configure a measurement for an 802.11ax signal. Afterwards
// start the unused tone error measurement and return an EVM
// trace for unoccupied HE TB PPDUs and margins.
// Query the measurement state (should be "RDY").
// *****
READ:WLAN:MEAS:MEvaluation:UTERror1:CURRent?
FETCh:WLAN:MEAS:MEvaluation:UTERror1:MARGin:CURRent?
FETCh:WLAN:MEAS:MEvaluation:STATE?
```

### 3.4.10 Performing single-shot measurements (OFDM, CMIMO)

```
// *****
// After configuring the measurement for 802.11n composite MIMO measurements:
// Start single-shot measurement and return the average results.
// Query the measurement state (should be "RDY").
// *****
READ:WLAN:MEAS:MEvaluation:MODulation:MIMO1:AVERage?
READ:WLAN:MEAS:MEvaluation:MODulation:MIMO2:AVERage?
```

```

READ:WLAN:MEAS:MEvaluation:MODulation:MIMO3:AVERage?
READ:WLAN:MEAS:MEvaluation:MODulation:MIMO4:AVERage?
FETCh:WLAN:MEAS:MEvaluation:STATe?

// *****
// Query spectrum traces for the first two antennas/streams.
// *****
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MIMO1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MIMO2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:FREQuency?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK:MIMO1?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK:MIMO2?

// *****
// Query statistical results for the first two antennas/streams.
// *****
FETCh:WLAN:MEAS:MEvaluation::MODulation:MIMO1:CURRent?
FETCh:WLAN:MEAS:MEvaluation::MODulation:MIMO2:CURRent?
FETCh:WLAN:MEAS:MEvaluation::MODulation:MIMO3:CURRent?
FETCh:WLAN:MEAS:MEvaluation::MODulation:MIMO4:CURRent?
FETCh:WLAN:MEAS:MEvaluation:MODulation:CMIMo:PSTS:CURRent?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO1:FREQuency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO2:FREQuency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:OBW:MIMO1?
FETCh:WLAN:MEAS:MEvaluation:TSMask:OBW:MIMO2?

```

### 3.4.11 Performing single-shot measurements (OFDM, MIMO, 80+80 MHz)

```

// *****
// After configuring the measurement for 802.11ac, switched MIMO2x2, 80+80 MHz:
// Start single-shot measurement and return first results.
// Query the measurement state (should be "RDY").
// *****
READ:WLAN:MEAS:MEvaluation:TRACe:TSMask:FREQuency?
FETCh:WLAN:MEAS:MEvaluation:STATe?

// *****
// Query EVM vs carrier traces for both antennas/streams and segments.
// *****
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:SEGMENT1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:SEGMENT2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO1:SEGMENT1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO2:SEGMENT1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO3:SEGMENT1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO4:SEGMENT1:AVERage?

```

```

FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO1:SEGment2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO2:SEGment2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO3:SEGment2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:CARRier:MIMO4:SEGment2:AVERage?

// *****
// Query spectrum traces for both antennas/streams and segments.
// *****
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MIMO1:SEGment1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MIMO1:SEGment2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MIMO2:SEGment1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MIMO2:SEGment2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:FREQuency?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK:MIMO1:SEGment1?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK:MIMO1:SEGment2?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK:MIMO2:SEGment1?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK:MIMO2:SEGment2?

// *****
// Query statistical results for both antennas/streams and segments.
// *****
FETCh:WLAN:MEAS:MEvaluation:MODulation:MIMO1:SEGments:AVERage?
FETCh:WLAN:MEAS:MEvaluation:MODulation:MIMO2:SEGments:AVERage?
FETCh:WLAN:MEAS:MEvaluation:MODulation:MIMO1:AVERage?
FETCh:WLAN:MEAS:MEvaluation:MODulation:MIMO2:AVERage?
FETCh:WLAN:MEAS:MEvaluation:MODulation:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO1:SEGments:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO2:SEGments:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO1:SEGments:FREQuency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:MIMO2:SEGments:FREQuency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:OBW:MIMO1:SEGments?
FETCh:WLAN:MEAS:MEvaluation:TSMask:OBW:MIMO2:SEGments?

```

### 3.4.12 Performing single-shot measurements (DSSS)

```

// *****
// After configuring the measurement for an 802.11b DSSS signal:
// Start single-shot measurement and return an EVM trace.
// Query the measurement state (should be "RDY").
// *****
READ:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:DSSS:CURREnt?
FETCh:WLAN:MEAS:MEvaluation:STATe?

// *****
// Read EVM, I/Q constellation, spectrum and PvT traces obtained in the
// last measurement without re-starting the measurement.
// *****
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:DSSS:CURREnt?
FETCh:WLAN:MEAS:MEvaluation:TRACe:IQConst:INPhase?

```

```

FETCh:WLAN:MEAS:MEvaluation:TRACe:IQConst:QUADrature?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:FREQuency?
FETCh:WLAN:MEAS:MEvaluation:TRACe:TSMask:MASK?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:MINimum?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:TIME?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:REDGe:MINimum?
FETCh:WLAN:MEAS:MEvaluation:TRACe:PVTime:FEDGe:MINimum?

// *****
// Read statistical results obtained in the last measurement
// without re-starting the measurement.
// *****
FETCh:WLAN:MEAS:MEvaluation:MODulation:DSSS:CURRent?
FETCh:WLAN:MEAS:MEvaluation:TSMask:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:FREQuency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:TSMask:OBW?
FETCh:WLAN:MEAS:MEvaluation:PVTime:REDGe:AVERage?
FETCh:WLAN:MEAS:MEvaluation:PVTime:FEDGe:AVERage?

```

### 3.4.13 Single-shot and continuous measurements

```

// *****
// Start single-shot measurement and return current statistical
// modulation results (without repeating the measurement).
// Query the measurement state (should be "RDY").
// *****
INIT:WLAN:MEAS:MEvaluation
FETCh:WLAN:MEAS:MEvaluation:MODulation:DSSS:CURRent?
FETCh:WLAN:MEAS:MEvaluation:STATE?

// *****
// Start continuous measurement; wait for 5 ms and return average
// EVM RMS trace results of the last measured cycle.
// Query measurement state and substates (should be "RUN,ADJ,ACT").
// *****
CONFigure:WLAN:MEAS:MEvaluation:REPetition CONTinuous
INIT:WLAN:MEAS:MEvaluation
Pause 5000
FETCh:WLAN:MEAS:MEvaluation:TRACe:EVMagnitude:DSSS:RMS:AVERage?
FETCh:WLAN:MEAS:MEvaluation:STATE:ALL?

```

### 3.4.14 Retrieving signal fields reports

```

// *****
// Retrieve the reported values of L-SIG field for legacy signal (NON_HT).
// *****
FETCh:WLAN:MEAS:MEvaluation:SINfo:LSIG:LENGth?

```



```

FETCh:WLAN:MEAS:MEvaluation:SINFo:LSIG:PARity?
FETCh:WLAN:MEAS:MEvaluation:SINFo:LSIG:RATE?
FETCh:WLAN:MEAS:MEvaluation:SINFo:LSIG:REServed?
FETCh:WLAN:MEAS:MEvaluation:SINFo:LSIG:TAIL?

// *****
// Retrieve the reported values of HT-SIG field for 802.11n signal.
// *****
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:AGGRegation?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:CBW?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:CRC?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:FECCoding?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:HTLength?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:MCS?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:NESS?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:NSounding?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:REServed?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:SHORTgi?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:SMOothing?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:STBCoding?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HTSig:TAIL?

// *****
// Retrieve the reported values of VHT-SIG field for 802.11ac signal.
// *****
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:BEAMformed?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:BW?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:CRC?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:FECCoding?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:GID?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:LDPC?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:PAID?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:REServed1?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:REServed2?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:SDISambig?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:SGI?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:SMCS?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:STBC?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:SUNSts?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:TAIL?
FETCh:WLAN:MEAS:MEvaluation:SINFo:VHTSig:TXOP?

// *****
// Retrieve the reported values of HE_SU-SIG field for 802.11ax single user MIMO.
// *****
FETCh:WLAN:MEAS:MEvaluation:SINFo:HESU:BEAMchange?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HESU:BSSColor?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HESU:BW?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HESU:CODing?
FETCh:WLAN:MEAS:MEvaluation:SINFo:HESU:CRC?

```

```

FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:DCM?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:DOPPler?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:FORMat?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:GILTfsize?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:LDPC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:MCS?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:NSTS?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:PEDisambig?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:PFECpadding?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:REServed1?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:REServed2?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:SPATialreuse?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:STBC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:TAIL?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:TXBF?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:TXOP?
FETCh:WLAN:MEAS:MEValuation:SINFo:HESU:ULDL?

// *****
// Retrieve the reported HE_MU-SIG field for 802.11ax multi user MIMO SIG-A.
// *****
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:BW?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:CRC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:DOPPler?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:GILTfsize?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:LDPC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:NLTfsymbols?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:NSBSymbols?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:PEDisambig?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:PFECpadding?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:REServed?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:SBCompress?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:SPATialreuse?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:STBC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:TAIL?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:TXOP?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEMU:ULDL?

// *****
// Retrieve the reported HE_MU-SIG field for 802.11ax multi user MIMO SIG-B
// for the channel index one and the user index one.
// *****
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:CFIeld:CRC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:CFIeld:CRU?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:CFIeld:RUAllocation?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:CFIeld:TAIL?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:UFIeld1:Coding?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:UFIeld1:CRC?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:UFIeld1:DCM?
FETCh:WLAN:MEAS:MEValuation:SINFo:HEB:CHANnel1:UFIeld1:MCS?

```

```

FETCh:WLAN:MEAS:MEvaluation:SINfo:HEB:CHANnel1:UFIeld1:NSTs?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HEB:CHANnel1:UFIeld1:REServed?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HEB:CHANnel1:UFIeld1:SPAConfig?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HEB:CHANnel1:UFIeld1:STAdid?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HEB:CHANnel1:UFIeld1:TAIL?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HEB:CHANnel1:UFIeld1:TXBeamform?

// *****
// Retrieve the reported values of HE_TB-SIG field for 802.11ax trigger
// based uplink single user MIMO PPDU.
// *****
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:BSSColor?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:BW?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:CRC?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:FORMat?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:REServed1?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:REServed2?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:SPATialreusel?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:SPATialreuse2?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:TAIL?
FETCh:WLAN:MEAS:MEvaluation:SINfo:HETB:TXOP?

```

### 3.4.15 CMIMO training mode

```

// *****
// Prerequisite: Adequate input signal settings
// *****
CONFigure:WLAN:MEAS:MODE TMODE
CLEar:WLAN:MEAS:TMODE:DATA
CONFigure:WLAN:MEAS:TMODE:NOAntennas 2

// *****
// Make sure antenna 1 is connected to the configured RX port.
// *****
READ:WLAN:MEAS:TMODE:ANTenna1?

// *****
// Disconnect antenna 1 and connect antenna 2.
// *****
READ:WLAN:MEAS:TMODE:ANTenna2?

// *****
// If the decode status is OK, save acquired training data to file.
// *****
CONFigure:WLAN:MEAS:TMODE:FILE:SAVE "my_training_data.mif"

// *****
// Switch back to measurement mode.
// *****

```

```

CONFigure:WLAN:MEAS:MODE MMODE

// *****
// Use training data still present in internal memory.
// *****
CONFigure:WLAN:MEAS:ISIGnal:TData "Internal"

// *****
// Alternatively, load training data from file.
// *****
CONFigure:WLAN:MEAS:ISIGnal:TData "my_training_data.mif"

// *****
// Make sure the composite signal is fed to the configured RX port.
// Then start measuring ...
// *****

```

### 3.4.16 Using WLAN list mode

The WLAN multi-evaluation list mode is programmed as follows:

- The measurement is controlled by SCPI commands with the following syntax: `...WLAN:MEAS:MEvaluation:LIST...`
- Use general commands of the type `...:WLAN:MEAS...` (no `:MEvaluation` mnemonic) to define the signal routing and perform RF and analyzer settings.
- After a `*RST`, the measurement is switched off and list mode is disabled. Use `CONFigure:WLAN:MEAS:MEvaluation:LIST ON` to enable the list mode and `INIT:WLAN:MEAS:MEvaluation` to initiate a single-shot measurement.
- Use `FETCh:WLAN:MEAS:MEvaluation:LIST:...?` commands to retrieve the results.

#### Speed considerations

The following measurement settings have an impact on the measurement speed:

- The number and size of the segments and the number of measured bursts in each segment
- The number and type of results that the R&S CMW needs to calculate
- [Specifying general measurement settings](#)..... 344
- [Specifying measurement-specific settings](#)..... 345
- [Performing single-shot measurements](#)..... 346

#### 3.4.16.1 Specifying general measurement settings

```

// *****
// System-Reset
// *****
*RST; *OPC?

```

```
*CLS; *OPC?
```

```
// *****
// Define signal routing.
// *****
ROUTE:WLAN:MEAS:SCENario:SALone RX1, RF1C

CONFigure:WLAN:MEAS:RFSettings:EATTenuation 2
CONFigure:WLAN:MEAS:RFSettings:UMARgin 0
CONFigure:WLAN:MEAS:RFSettings:FREQuency:BAND B24Ghz

// *****
// Use a power trigger to start the measurement.
// *****
TRIGger:WLAN:MEAS:MEValuation:SOURce 'IF POWER'
TRIGger:WLAN:MEAS:MEValuation:SLOPe REDGe
TRIGger:WLAN:MEAS:MEValuation:THReshold -25
TRIGger:WLAN:MEAS:MEValuation:TOUT 1
```

### 3.4.16.2 Specifying measurement-specific settings

```
// *****
// Define 2 segments with a segment duration of 1 s each. Set measurement
// offset and measurement time.
// *****
CONFigure:WLAN:MEAS:MEValuation:LIST:COUNT 2
CONFigure:WLAN:MEAS:MEValuation:LIST:STIME 1, 1
CONFigure:WLAN:MEAS:MEValuation:LIST:MOFFset 0.1,0.1
CONFigure:WLAN:MEAS:MEValuation:LIST:MTIME 0.6,0.6

// *****
// Enable modulation and transmit spectrum mask results for both segments.
// Select an averaging length of 20 (all measured bursts in the segment).
// *****
CONFigure:WLAN:MEAS:MEValuation:LIST:RESult:MODulation ON,ON
CONFigure:WLAN:MEAS:MEValuation:LIST:RESult:TSMask ON,OF
CONFigure:WLAN:MEAS:MEValuation:LIST:SCount:MODulation 20,20
CONFigure:WLAN:MEAS:MEValuation:LIST:SCount:TSMask 20,20

// *****
// Use retriggering for the measurements of both segments.
// *****
CONFigure:WLAN:MEAS:MEValuation:LIST:RTRigger ON,ON

// *****
// Specify input signal: standard, burst type, channel bandwidth, expected
// nominal power and frequency for both segments
// *****
```

```

CONFigure:WLAN:MEAS:MEvaluation:LIST:STANdard HTOF,HTOF
CONFigure:WLAN:MEAS:MEvaluation:LIST:BTYPe MIX,GRE
CONFigure:WLAN:MEAS:MEvaluation:LIST:BWIDth BW40,BW20
CONFigure:WLAN:MEAS:MEvaluation:LIST:ENPower -35,-40
CONFigure:WLAN:MEAS:MEvaluation:LIST:FREQuency 2412,2452

// *****
// Change settings for segment 1
// *****
CONF:WLAN:MEAS:MEV:LIST:SEGment1:SETup 1,0.7,0.1,-35,5185,HEOF,BW40

// *****
// Change settings for segment 2 per parameter: enable modulation and spectrum
// measurements, disable triggering per segment, specify statistics count for
// modulation and spectrum measurements. Set segment time, standard, burst type,
// measurement offset and time, channel bandwidth. Change level and frequency.
// Change connector used for segment 2.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:RESult ON,ON
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:RTRigger OFF
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:SCount 20,10
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment1:STIME 1
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment1:STANdard HTOF
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:BTYPe MIX
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:MOFFset 0.3
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:MTIME 0.4
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:BWIDth BW40
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:ENPower -35
CONFigure:WLAN:MEAS:MEvaluation:LIST:SEGment2:FREQuency 2412

```

### 3.4.16.3 Performing single-shot measurements

```

// *****
// Enable the list mode.
// Start single-shot measurement.
// *****
CONFigure:WLAN:MEAS:MEvaluation:LIST ON
INIT:WLAN:MEAS:MEvaluation

// *****
// For DSSSS signals, return average results for segment 1 and segment 2 and
// segment-independent results.
// Query the measurement state (should be "RDY").
// *****
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment1:MODulation:DSSS:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment2:MODulation:DSSS:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:BPOWer:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:EVMRms:AVERage?

```

```

FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:CFERror:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:CCERror:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:GIMBalance:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:IQOffset:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DSSS:QERror:AVERage?
FETCh:WLAN:MEAS:MEvaluation:STATE?

// *****
// Alternatively return OFDM average results for segment 1 and segment 2.
// Query the measurement state (should be "RDY").
// *****
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment1:MODulation:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment2:MODulation:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment1:TSMask:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment2:TSMask:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment1:TSMask:FREquency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:SEGment2:TSMask:FREquency:AVERage?
FETCh:WLAN:MEAS:MEvaluation:STATE?

// *****
// Alternatively, use segment-independent commands for OFDM signals
// to retrieve the results for all segments.
// *****
FETCh:WLAN:MEAS:MEvaluation:LIST:SRELiability?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:BPOWer:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DCPower:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:DPOWer:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:LTFPower:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:PPOWer:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:PBACKoff?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:CFACTOR:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:CFERror:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:EVMall:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:EVMPilot:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:GIMBalance:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:IQOffset:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:QERror:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:SCERror:AVERage?
FETCh:WLAN:MEAS:MEvaluation:LIST:MODulation:SCount?
FETCh:WLAN:MEAS:MEvaluation:LIST:TSMask:SCount?

// *****
// Disable the list mode.
// *****
CONFIgure:WLAN:MEAS:MEvaluation:LIST OFF

```

## 3.5 Command reference

The following sections provide detailed reference information on the remote control commands of the "WLAN Measurements" application.

|                                                                       |     |
|-----------------------------------------------------------------------|-----|
| • <a href="#">Conventions and general information</a> .....           | 348 |
| • <a href="#">General measurement settings</a> .....                  | 355 |
| • <a href="#">Multi-evaluation measurement control commands</a> ..... | 370 |
| • <a href="#">Measurement limit settings</a> .....                    | 401 |
| • <a href="#">Measurement result commands</a> .....                   | 458 |
| • <a href="#">CMIMO training mode</a> .....                           | 606 |
| • <a href="#">Combined signal path commands</a> .....                 | 609 |

### 3.5.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

#### 3.5.1.1 MEAS<i>

MEAS<i> is used as abbreviation of "MEASurement<instance>". For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same measurement application. It can be omitted if the instrument supports only one instance, or to address the first instance.

#### 3.5.1.2 FETCh, READ and CALCulate commands

All commands are used to retrieve measurement results:

- **FETCh...** returns the results of the measurement cycle after the cycle is complete. Before sending a **FETCh...**, start the measurement via an **INITiate...**
- **READ...** starts a new single-shot measurement and returns the results.
- **CALCulate...** returns one limit check result per **FETCh** result:
  - **OK:** The **FETCh** result is located within the limits or no limit has been defined/enabled for this result.
  - **ULEU** ("User limit exceeded upper"): An upper limit is violated. The **FETCh** result is located above the limit.
  - **ULEL** ("User limit exceeded lower"): A lower limit is violated. The **FETCh** result is located below the limit.



### 3.5.1.3 Current and statistical results

The measurement is repeated according to the selected statistic count and repetition mode. Consecutive measurement values are stored and used to calculate statistical results.

Commands for result query often contain a mnemonic that indicates which results are returned: **CURRent**, **AVERage**, **EXTReme**, **MINimum**, **MAXimum** or **SDEViation**.

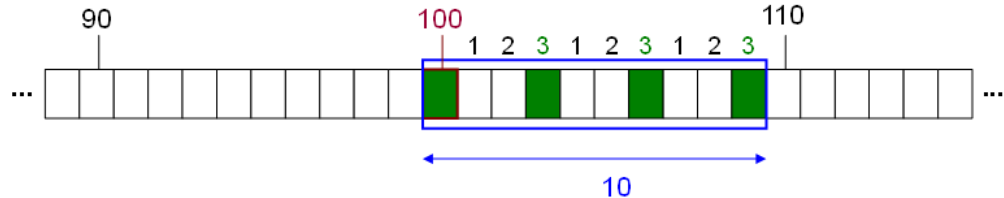
### 3.5.1.4 Trace Sub-Arrays

The optional query parameters **<Start>**, **<Count>** and **<Decimation>** can be used to limit the value array that is returned by a "TRACe" command:

1. Limit the returned array to the segment beginning with position **<Start>**.
2. If necessary, further limit the segment to **<Count>** values.
3. If necessary, further decimate the segment **<Start> ... <Start>+<Count>-1** by returning only every **n**th value (**<Decimation> = n**)

#### Example:

**FETCh:<FWA>:MEAS:MEV:TRACe:xx? 100, 10, 3** returns every 3rd value of the 10 values beginning with the 100th one.



### 3.5.1.5 Values for RF path selection

To select an RF input path, you must specify an RF connector and an RX module (converter).

Which connectors and modules can be specified in a command, depends on the installed hardware, the test setup and the active subinstrument or instance **<i>**.

This chapter lists all values available for path selection. Depending on your configuration, only a subset is relevant for you. Virtual connector names are only relevant for setting commands. Queries return the physical connector names.

Additional information is available in the base unit documentation. It describes typical instrument configurations with the allowed RF connector - TX/RX module combinations and the mapping of virtual connector names to physical connectors.

**Single R&S CMW500 / 270 / 290**

RF path selection values:

- **RX module:**  
RX1 | RX2 | RX3 | RX4
- **RX connector:**  
RF1C | RF2C | RF3C | RF4C | RFAC | RFBC  
RF 1 COM to RF 4 COM plus virtual connector names

**CMWflexx setup with R&S CMWC and several R&S CMW500**

RF path selection values:

- **RX module:**  
RX11 | RX12 | RX13 | RX14 | RX21 | RX22 | RX23 | RX24  
RX31 | RX32 | RX33 | RX34 | RX41 | RX42 | RX43 | RX44  
RX<a><b>: CMW <a>, RX <b>  
Example RX34: RX module 4 of CMW 3
- **RX connector:**  
R11C | R12C | R13C | R14C | R21C | R22C | R23C | R24C  
R31C | R32C | R33C | R34C | R41C | R42C | R43C | R44C  
R<a><b>C: CMW <a>, connector RF <b> COM  
Example R34C: RF 4 COM of CMW 3

For CMW 1, you can alternatively use the single-CMW RF path selection values.

**3.5.1.6 Frequency ranges**

The following frequency ranges are supported:

- Basic frequency range RX: 150 MHz to 3.3 GHz
- Basic frequency range TX: 70 MHz to 3.3 GHz
- Frequency range extension up to 6 GHz: R&S CMW-KB036

**3.5.1.7 Keywords**

Selected keywords used in the command description are described in the following.

- **Command usage**  
If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:
  - "Setting only": The command can only be used to set parameters.
  - "Query only": The command can only be used to query parameters.
  - "Event": The command initiates an event.
- **Parameter usage**  
The parameter usage is indicated by the keyword preceding the parameters:
  - "Parameters" are sent with a setting or query command and are returned as the result of a query.
  - "Setting parameters" are sent with a setting command.

- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.
- **Parameter type**  
The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:
  - "string": characters enclosed in quotation marks (" or ')
  - "numeric", "decimal": decimal number
  - "integer": integer number
  - "float": floating point number
  - "hex": hexadecimal number starting with #H
  - "binary": binary number starting with #B
  - "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).
- **Firmware/Software:**  
Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

### 3.5.1.8 Reliability indicator

The first value in the output arrays of `FETCh...?`, `READ...?` and `CALCulate...?` queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

The reliability indicator has one of the following values:

- **0 ("No Error"):**  
Measurement values available, no error detected.
- **1 ("Measurement Timeout"):**  
The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INValid` results or has not completed the full statistic count.
- **2 ("Capture Buffer Overflow"):**  
The measurement configuration results in a capture length that exceeds the available memory.
- **3 ("Input Overdriven") / 4 ("Input Underdriven"):**  
The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 ("Trigger Timeout"):**  
The measurement could not be started or continued because no trigger event was detected.

- **7 ("Acquisition Error"):**  
The R&S CMW could not properly decode the RF input signal.
- **8 ("Sync Error"):**  
The R&S CMW could not synchronize to the RF input signal.
- **9 ("Uncal"):**  
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the specified data sheet limits.
- **15 ("Reference Frequency Error"):**  
The instrument has been configured to use an external reference signal. But the reference oscillator could not be phase-locked to the external signal (for example signal level too low, frequency out of range or reference signal not available at all).
- **16 ("RF Not Available"):**  
The measurement could not be started because the configured RF input path was not active.  
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 ("RF Level Not Settled") / 18 ("RF Frequency Not Settled"):**  
The measurement could not be started because the R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 ("Call Not Established"):**  
For measurements: The measurement could not be started because no signaling connection to the DUT was established.  
For DAU IMS service: Establishing a voice over IMS call failed.
- **20 ("Call Type Not Usable"):**  
For measurements: The measurement could not be started because the established signaling connection had wrong properties.  
For DAU IMS service: The voice over IMS settings could not be applied.
- **21 ("Call Lost"):**  
For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.  
For DAU IMS service: The voice over IMS call was lost.
- **23 ("Missing Option"):**  
An action cannot be executed due to a missing option.  
For GPRF generator: The ARB file cannot be played due to a missing option.
- **24 ("Invalid RF Setting"):**  
The desired RF TX level or RF RX reference level could not be applied.
- **25 ("Level Overrange"):**  
The RF TX level is in overrange. The signal quality can be degraded.
- **26 ("Resource Conflict"):**  
The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application.  
Stop the application that has allocated the conflicting resources and try again.
- **28 ("Unexpected Parameter Change"):**  
One or more measurement configuration parameters were changed while the measurement completed. The results were not obtained with these new parameter val-

ues. Repeat the measurement. This situation can only occur in remote single-shot mode.

- **29 ("Invalid RF Frequency Setting"):**  
The desired RF TX frequency or RF RX frequency could not be applied.
- **30 ("File Not Found"):**  
The specified file could not be found.
- **40 ("ARB File CRC Error"):**  
The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.
- **42 ("ARB Header Tag Invalid"):**  
The ARB file selected in the GPRF generator contains an invalid header tag.
- **43 ("ARB Segment Overflow"):**  
The number of segments in the multi-segment ARB file is higher than the allowed maximum.
- **44 ("ARB File Not Found"):**  
The selected ARB file could not be found.
- **45 ("ARB Memory Overflow"):**  
The ARB file length is greater than the available memory.
- **46 ("ARB Sample Rate Out of Range"):**  
The clock rate of the ARB file is either too high or too low.
- **47 ("ARB Cycles Out of Range"):**  
The repetition mode equals "Single Shot" and the playback length is too long. Reduce the playback length or set the repetition mode to "Continuous".  
$$\langle \text{Length} \rangle = (\langle \text{Cycles} \rangle * \langle \text{Samples} \rangle + \langle \text{Additional Samples} \rangle) / \langle \text{Clock Rate} \rangle$$
- **50 ("Startup Error"):**  
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 ("No Reply"):**  
The DAU has received no response, for example for a ping request.
- **52 ("Connection Error")**  
A connection setup failed or a connection was lost.  
For DAU: The DAU could not establish a connection to internal components. Restart the instrument.
- **53 ("Configuration Error"):**  
The current DAU configuration is incomplete or wrong and could not be applied. Check especially the IP address configuration.
- **54 ("Filesystem Error"):**  
The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **70 ("Wrong Standard")**  
The standard of the measured signal does not match the configured standard.
- **71 ("Wrong Bandwidth")**  
The bandwidth of the measured signal does not match the configured bandwidth.
- **72 ("Wrong Burst Type")**  
The burst type of the measured signal does not match the configured burst type.
- **73 ("MIMO Signal Detected")**

The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.

- **74 ("More Streams than Antennas")**  
The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 ("Matrix Inversion Failed")**  
The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.
- **76 ("SIG CRC Failed")**  
The cyclic redundancy check of a SIGNAL field failed.
- **77 ("Parity Check Failed")**  
The parity check of a SIGNAL field failed.
- **78 ("Bursts Not Identical")**  
In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 ("Wrong Modulation")**  
The modulation type of the measured signal does not match the configured modulation type.
- **80 ("SISO Signal Detected")**  
Received signal contains only one stream.
- **81 ("Less Streams than Antennas")**  
Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.
- **82 ("Signal Fields of Different Antennas Are Not Identical")**  
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 ("Bit Errors in Some of the Received Symbols")**  
Adjust input signal power and repeat training.
- **84 ("First and Last Symbols Need to Be Skipped")**  
Due to a sequence counter in the MAC header the first and the last symbol (due to the FCS) change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 ("Received and Saved Bursts Are Not Identical")**  
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 ("No Symbols for Measurement")**  
Too many head and tail symbols have been skipped.
- **93 ("OCXO Oven Temperature Too Low"):**  
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **101 ("Firmware Error"):**

Indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.

If the error occurs again, consider the following hints:

- Firmware errors can often be repaired by restoring the factory default settings.
- If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog, section "SW/HW-Equipment > Installed Software".
- Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.

- **102 ("Unidentified Error"):**  
Indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 (firmware error)".
- **103 ("Parameter Error"):**  
Indicates that the measurement could not be performed due to internal conflicting parameter settings.  
A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.  
If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <http://www.service.rohde-schwarz.com>).
- **104 ("Not Functional"):**  
The application could not be started with the configured parameter set.

### 3.5.2 General measurement settings

The commands valid for all WLAN measurements are divided into the groups listed below.

|                                               |     |
|-----------------------------------------------|-----|
| • <a href="#">Signal routing</a> .....        | 355 |
| • <a href="#">Analyzer settings</a> .....     | 360 |
| • <a href="#">Input signal settings</a> ..... | 364 |

#### 3.5.2.1 Signal routing

The following commands configure the scenario, select the input path for the measured signal and define an external attenuation value.

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| <a href="#">ROUTe:WLAN:MEAS&lt;i&gt;?.....</a>                                | 356 |
| <a href="#">ROUTe:WLAN:MEAS&lt;i&gt;:CATalog:SCENario?.....</a>               | 356 |
| <a href="#">ROUTe:WLAN:MEAS&lt;i&gt;:SCENario?.....</a>                       | 356 |
| <a href="#">ROUTe:WLAN:MEAS&lt;i&gt;:SCENario:CSPath.....</a>                 | 357 |
| <a href="#">ROUTe:WLAN:MEAS&lt;i&gt;:SCENario:SALone.....</a>                 | 357 |
| <a href="#">ROUTe:WLAN:MEAS&lt;i&gt;:SCENario:TMIMo&lt;PathCount&gt;.....</a> | 357 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MIMO:NOANtennas.....</a>             | 358 |

|                                                     |     |
|-----------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n>.....   | 358 |
| CONFigure:WLAN:MEAS<i>:RFSettings:SANTennas.....    | 359 |
| CONFigure:WLAN:MEAS<i>:RFSettings:EATTenuation..... | 359 |

---

### ROUTe:WLAN:MEAS<i>?

Returns the configured routing settings for SISO scenarios.

For possible connector and converter values, see [Chapter 3.5.1.5, "Values for RF path selection"](#), on page 349.

#### Return values:

|                           |                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <Scenario>                | SALone   SALone   CSPath<br><b>SALone:</b> Standalone (non-signaling)<br><b>CSPath:</b> Combined signal path (with WLAN signaling) |
| <Controller>              | string<br>Controlling application for scenario CSPath                                                                              |
| <RXConnector1>            | RF connector for the input path                                                                                                    |
| <RXConverter1>            | RX module for the input path                                                                                                       |
| <b>Usage:</b>             | Query only                                                                                                                         |
| <b>Firmware/software:</b> | V2.0.10<br>V2.1.30: added CSPath                                                                                                   |
| <b>Manual operation:</b>  | See " <a href="#">Scenario = StandAlone</a> " on page 289                                                                          |

---

### ROUTe:WLAN:MEAS<i>:CATalog:SCENario?

Returns all scenarios possible for the instrument.

#### Return values:

|                           |                                                                                                                                                                                                                                         |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Valid_Scenarios>         | UNDefined   SALone   CSPath   SMI4   MIMO2x2   MIMO4x4   MIMO8x8   TMIMo   SALone   CSPath   TMIMo<br><b>SALone:</b> Standalone (non-signaling)<br><b>CSPath:</b> Combined signal path (with WLAN signaling)<br><b>TMIMo:</b> True MIMO |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                                              |
| <b>Firmware/software:</b> | V3.7.60                                                                                                                                                                                                                                 |
| <b>Manual operation:</b>  | See " <a href="#">Scenario = StandAlone</a> " on page 289                                                                                                                                                                               |

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### ROUTe:WLAN:MEAS<i>:SCENario?

Returns the active scenario.



**Return values:**

<GUI\_Scenario> SALone | TMIMo | CSPath  
**SALone:** Standalone (non-signaling)  
**CSPath:** Combined signal path (with WLAN signaling)  
**TMIMo:** True MIMO

**Usage:** Query only

**Firmware/software:** V2.0.10, V2.1.30: added CSPath, V3.7.31: added TMIMo

**Manual operation:** See "[Scenario = StandAlone](#)" on page 289

**ROUTe:WLAN:MEAS<i>:SCENario:CSPath <Controller>**

Activates the combined signal path scenario and selects the controlling application. The selected application controls the signal routing and analyzer settings while the combined signal path scenario is active.

**Parameters:**

<Controller> string  
 String parameter selecting the controlling application, e.g.,  
 'WLAN Sig1' or 'WLAN Sig2'

**Firmware/software:** V2.1.30

**Manual operation:** See "[Scenario = Combined Signal Path](#)" on page 289

**ROUTe:WLAN:MEAS<i>:SCENario:SALone <RXConnector>, <RXConverter>**

Activates the standalone scenario and selects the RF input path for the measured RF signal.

For possible connector and converter values, see [Chapter 3.5.1.5, "Values for RF path selection"](#), on page 349.

**Parameters:**

<RXConnector> RF connector for the input path  
 <RXConverter> RX module for the input path

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V1.0.15.20

**Manual operation:** See "[Scenario = StandAlone](#)" on page 289

**ROUTe:WLAN:MEAS<i>:SCENario:TMIMo<PathCount>**

Selects the number of paths for the true MIMO scenario via <PathCount>. The connectors are configurable via [CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n>](#).

**Suffix:**

<PathCount> 2, 3, 4  
 Selects true MIMO 2x2 (R&S CMW500/2xx supports true MIMO 2x2 only).  
 A query ignores the suffix.

**Return values:**

<GUI\_Scenario> TMIMo  
 Returns the active scenario (true MIMO).

**Example:** See [Specifying general measurement settings](#).

**Firmware/software:** V3.7.60

**Options:** R&S CMW-KM654 per R&S CMW

**Manual operation:** See "[Scenario = True MIMO](#)" on page 289

**CONFigure:WLAN:MEAS<i>:MIMO:NOAntennas <NumOfAntennas>**

Sets the number of connected antennas for true MIMO measurements.

**Parameters:**

<NumOfAntennas> decimal  
 Number of antennas 1 to 2

**Firmware/software:** V3.7.31

**Options:** R&S CMW-KM654 per R&S CMW for true MIMO

**Manual operation:** See "[Number of Antennas](#)" on page 290

**CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n> <ConnectorAll>[, <ExtAtt>[, <ENP>]]**

Assigns an RF input connector and optionally an external attenuation and expected nominal power to antenna <n>.

The command is relevant for true MIMO and for 80+80 MHz signals with a separate antenna per segment.

For 80+80 MHz signals, the antenna numbering continues for the second segment. Example: Two antennas per segment. Segment 0 uses antenna 1 and 2. Segment 1 uses antenna 3 and 4.

**Suffix:**

<n> 1..8  
 Antenna  
 For true MIMO with R&S CMW500/2xx, limited to maximum 2 antennas

**Parameters:**

|          |                                                                                                                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ExtAtt> | numeric<br>External attenuation component for antenna <n><br>Range: 50 dB to 90 dB<br>*RST: 0 dB<br>Default unit: dB                                                                                                                                                                      |
| <ENP>    | numeric<br>Expected nominal power for antenna <n>. It can be calculated as follows:<br>$\text{Range (Expected Nominal Power)} = \text{Range (Input Power)} + \text{External Attenuation} - \text{User Margin}$<br>The input power range is stated in the data sheet.<br>Default unit: dBm |

**Setting parameters:**

**Connector** For connector values, see [Chapter 3.5.1.5, "Values for RF path selection"](#), on page 349.

**Firmware/software:** V3.5.121, V3.7.10: added <ExtAtt>, V3.7.31: added <ENP>

**Manual operation:** See ["Scenario = Combined Signal Path"](#) on page 289

**CONFigure:WLAN:MEAS<i>:RFSettings:SANTennas <SepAnt>**

It specifies whether the DUT uses separate antennas for the two segments of an 80+80 MHz signal.

To assign RF input connectors to the two antennas, see [CONFigure:WLAN:MEAS<i>:RFSettings:ANTenna<n>](#) on page 358.

**Parameters:**

|          |                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------|
| <SepAnt> | ON   OFF<br><b>ON:</b> separate antennas for each segment<br><b>OFF:</b> same antennas for both segments<br>*RST: OFF |
|----------|-----------------------------------------------------------------------------------------------------------------------|

**Firmware/software:** V3.5.121  
V3.7.50: R&S CMW500/2xx also supported

**Manual operation:** See ["Use Sep. Ant. for 80+80 MHz"](#) on page 290

**CONFigure:WLAN:MEAS<i>:RFSettings:EATTenuation <RFInputExtAtt>**

Defines an external attenuation (or gain, if the value is negative), to be applied to all RF input connectors (global external attenuation).

For the combined signal path scenario, use [CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:EATTenuation:INPut](#).

**Parameters:**

<RFInputExtAtt>      Range:        -50 dB to 90 dB  
                          \*RST:        0 dB  
                          Default unit: dB

**Example:**            See [Specifying general measurement settings](#)

**Firmware/software:** V1.0.10.1

**Manual operation:** See ["External Attenuation \(Input\) /Ext. Att."](#) on page 291

**3.5.2.2 Analyzer settings**

The following commands configure the RF input path.

|                                                                                      |     |
|--------------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FREQUENCY:BAND</a>               | 360 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FREQUENCY</a>                    | 360 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FREQUENCY:SCHannel</a>           | 361 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FREQUENCY:CHANnels</a>           | 361 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FREQUENCY:CHANnels&lt;Ch&gt;</a> | 362 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:ENPower&lt;connector&gt;</a>     | 363 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:UMARgin&lt;connector&gt;</a>     | 364 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:MLOffset</a>                     | 364 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FOFFset</a>                      | 364 |

---

**CONFigure:WLAN:MEAS<i>:RFSettings:FREQUENCY:BAND <FreqBand>**

Selects the frequency band.

**Parameters:**

<FreqBand>            B24Ghz | B5GHz | B4GHz  
                          **B24Ghz:** 2.4 GHz band  
                          **B4GHz:** 4 GHz band  
                          **B5GHz:** 5 GHz band  
                          \*RST:        B24G

**Example:**            See [Specifying general measurement settings](#)

**Firmware/software:** V1.0.10.50  
                          V3.2.71: added 4 GHz band

**Options:**            R&S CMW-KM651

**Manual operation:** See ["Band / Channels /Center Frequency"](#) on page 291

---

**CONFigure:WLAN:MEAS<i>:RFSettings:FREQUENCY <Frequency>**

Configures the center frequency of the RF analyzer. Set it to the center frequency of the received 20-MHz, 40-MHz, 80-MHz or 160-MHz WLAN channel.

For 80+80 MHz signals, set the center frequency of the left segment. The frequency of the right segment is calculated using channel distance, see [CONFigure:WLAN:MEAS<i>:ISIGnal:CDistance](#).

For the combined signal path scenario, use:

- `CONFigure:WLAN:SIGN<i>:RFSettings:FREQuency`
- `CONFigure:WLAN:SIGN<i>:RFSettings:CHANnel`

The supported frequency range depends on the instrument model and the available options. The supported range can be smaller than stated here. Refer to the preface of your model-specific base unit manual.

**Parameters:**

<Frequency>                      numeric  
                                          Range:        70E+6 Hz to 6E+9 Hz  
                                          \*RST:        2.412E+9 Hz

**Example:**                      See [Specifying general measurement settings](#)

**Firmware/software:** V1.0.10.1  
                                  V3.2.11: Minimum frequency decreased to 70 MHz

**Manual operation:**        See ["Band / Channels /Center Frequency"](#) on page 291

**CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:SCHannel <SecondChannel>**

Sets the position of the secondary channel relative to the primary channel for 40 MHz 802.11n signals.

**Parameters:**

<SecondChannel>        POSitive | NEGative  
                                  **POSitive:** Secondary channel right above the primary channel  
                                  **NEGative:** Secondary channel right below the primary channel  
                                  \*RST:        POS

**Firmware/software:** V2.1.30

**Options:**                      R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:CHANnels <Channel>**

The command logic depends on the standard. This description applies to the standards 802.11 a/b/g/n/p. For other standards, see `CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:CHANnels<Ch>` on page 362.

The command configures the center frequency of the RF analyzer. It sets the frequency to the center of the channel with the index <Channel>.

Before using this command, configure the standard, the bandwidth and the band, see:

- `CONFigure:WLAN:MEAS<i>:ISIGnal:STANdard` on page 365
- `CONFigure:WLAN:MEAS<i>:ISIGnal:BWIDth` on page 366
- `CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:BAND` on page 360

**Parameters:**

<Channel>                      numeric  
                                          Channel index

The range depends on the band:

2.4-GHz band: 1 to 14

4-GHz and 5-GHz bands: 0 to 200

Range: 0 to 200

\*RST: 1

**Firmware/software:** V3.2.71

**Manual operation:** See "[Band / Channels /Center Frequency](#)" on page 291

---

### **CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:CHANnels<Ch> <Channel>**

The command logic depends on the standard. This description applies to the standards 802.11ac and ax. For other standards, see [CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:CHANnels](#) on page 361.

A setting command sets channel number <Ch> to the channel index <Channel>. The other 20-MHz channels of the bandwidth are configured automatically, resulting in a sequence of channel indices with the increment 4, see examples.

A query returns the channel indices of all 20-MHz channels as comma-separated list.

Before using this command, configure the standard, the bandwidth and the band, see:

- [CONFigure:WLAN:MEAS<i>:ISignal:STANdard](#) on page 365
- [CONFigure:WLAN:MEAS<i>:ISignal:BWIDth](#) on page 366
- [CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency:BAND](#) on page 360

#### **Suffix:**

<Ch> 1..8

Selects the 20-MHz channel to be configured. For a query, the suffix is ignored.

The range depends on the signal bandwidth, see table below.

#### **Setting parameters:**

<Channel> numeric

Channel index for the 20-MHz channel number <Ch>

For a valid configuration, all 20-MHz channels must fit into the band. So the effective ranges depend on <Ch>, see table below.

Range: 0 to 200

\*RST: 1

#### **Return values:**

<Channels> decimal

Comma-separated list of channel indices

1, 2, 4, or 8 values, see table below

Range: 0 to 200

#### **Example:**

Bandwidth 80 MHz, valid configuration:

CONF:WLAN:MEAS:RFS:FREQuency:CHANnels3 10

Resulting channels: 2, 6, 10, 14

**Example:** Bandwidth 80 MHz, invalid configuration:  
 CONF:WLAN:MEAS:RFS:FREQuency:CHANnels4 10  
 Resulting channels: INV, 2, 6, 10

**Firmware/software:** V3.2.10

**Options:** R&S CMW-KM656

**Manual operation:** See ["Band / Channels /Center Frequency"](#) on page 291

| Band-width / MHz | <Channel> ranges for valid configuration |          |          |           |           |           |           |           |
|------------------|------------------------------------------|----------|----------|-----------|-----------|-----------|-----------|-----------|
|                  | <Ch>=1                                   | <Ch>=2   | <Ch>=3   | <Ch>=4    | <Ch>=5    | <Ch>=6    | <Ch>=7    | <Ch>=8    |
| 20               | 0 to 200                                 | -        | -        | -         | -         | -         | -         | -         |
| 40               | 0 to 196                                 | 4 to 200 | -        | -         | -         | -         | -         | -         |
| 80               | 0 to 188                                 | 4 to 192 | 8 to 196 | 12 to 200 | -         | -         | -         | -         |
| 80+80<br>160     | 0 to 172                                 | 4 to 176 | 8 to 180 | 12 to 184 | 16 to 188 | 20 to 192 | 24 to 196 | 28 to 200 |

**CONFigure:WLAN:MEAS<i>:RFS**Settings:ENPower<connector> <ExpNomPwr>

Sets the expected nominal power of the measured RF signal.

For the combined signal path scenario, use [CONFigure:WLAN:SIGN<i>:RFS](#)Settings:ANTenna<n>:EPEPower.

**Suffix:**

<connector> 1  
 For future use, currently the values cannot be set independently.

**Parameters:**

<ExpNomPwr> numeric  
 The range of the expected nominal power can be calculated as follows:  

$$\text{Range (Expected Nominal Power)} = \text{Range (Input Power)} + \text{External Attenuation} - \text{User Margin}$$
  
 Range: -47 dBm to 34 dBm for the input power at RF 1 COM and RF 2 COM (please notice also the ranges quoted in the data sheet).  
 \*RST: 0 dBm  
 Default unit: dBm

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V3.0.30

**Manual operation:** See ["Expected Nominal Power"](#) on page 292

**CONFigure:WLAN:MEAS<i>:RFSettings:UMARgin<connector> <UserMargin>**

Sets the margin that the measurement adds to the expected nominal power to determine the reference power. The reference power minus the external input attenuation must be within the power range of the selected input connector. Refer to the data sheet.

**Suffix:**

<connector> 1  
For future use, currently the values cannot be set independently.

**Parameters:**

<UserMargin> numeric  
Range: 0 dB to (34 dB + external attenuation - expected nominal power)  
\*RST: dB

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V3.0.30

**Manual operation:** See ["User Margin"](#) on page 293

**CONFigure:WLAN:MEAS<i>:RFSettings:MLOffset <MLOffset>**

Varies the input level of the mixer in the analyzer path.

For the combined signal path scenario, use [CONFigure:WLAN:SIGN<i>:RFSettings:ANTenna<n>:MLOffset](#).

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V1.0.15.20

**Manual operation:** See ["Mixer Level Offset"](#) on page 293

**CONFigure:WLAN:MEAS<i>:RFSettings:FOFFset <FreqOffset>**

Sets a positive or negative frequency offset to be added to the center frequency ([CONFigure:WLAN:MEAS<i>:RFSettings:FREQuency](#)).

**Parameters:**

<FreqOffset> numeric  
Range: -80E+3 Hz to 80E+3 Hz  
\*RST: 0 Hz

**Firmware/software:** V3.2.10

**Manual operation:** See ["Frequency Offset"](#) on page 293

**3.5.2.3 Input signal settings**

The following commands define basic WLAN signal properties.



|                                                      |     |
|------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:ISIGnal:STANdard.....         | 365 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:RMODE.....            | 365 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:BWIDth.....           | 366 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:CDIStance.....        | 367 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:BTYPe.....            | 367 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:DSSS:ELENgth.....     | 367 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:OFDM:ELENgth.....     | 368 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:PCLass.....           | 368 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:TDAta.....            | 369 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:TDAta:FILE:DATE?..... | 369 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:MODFilter.....        | 369 |
| CONFigure:WLAN:MEAS<i>:ISIGnal:IQSWap.....           | 370 |

### CONFigure:WLAN:MEAS<i>:ISIGnal:STANdard <Standard>

Selects the IEEE 802.11 standard. Several WLAN signal properties depend on the selected standard, see [Chapter 3.2.6.2, "Physical layer"](#), on page 248.

In the combined signal path (CSP), consider the dependency between this parameter and the burst type, set for the RX frame trigger in the signaling application.

See [TRIGger:WLAN:SIGN<i>:RX:MACFrame:BTYPe](#)

Selecting a standard that is not compatible with the current scenario restores the "Standalone" scenario.

#### Parameters:

<Standard>      DSSS | LOFDm | HTOFdm | VHTofdm | HEOFdm | POFDm  
**DSSS:** 802.11b/g (DSSS)  
**LOFDm:** 802.11a/g (OFDM)  
**HTOFdm:** 802.11n  
**VHTofdm:** 802.11ac  
**HEOFdm:** 802.11ax  
**POFDm:** 802.11p  
**\*RST:**      LOFD

**Example:**      See [Specifying general measurement settings](#)

**Firmware/software:** V3.5.30  
V3.7.20: added HEOFdm

**Options:**      802.11n requires R&S CMW-KM651  
802.11p requires R&S CMW-KM655  
802.11ac requires R&S CMW-KM651 and -KM656  
802.11ax requires R&S CMW-KM651, -KM656, and -KM657

**Manual operation:**      See ["Standard"](#) on page 287

### CONFigure:WLAN:MEAS<i>:ISIGnal:RMODE <ReceiveMode>

Sets the receive mode.

Not all standards support MIMO. If you set a standard that is incompatible with the current receive mode, the receive mode automatically reverts to SISO.

**Parameters:**

<ReceiveMode> SISO | TMIMo | CMIMo  
**SISO:** SISO signal  
**CMIMo:** Composite MIMO  
**TMIMo:** True MIMO  
 \*RST: SISO

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V3.5.30  
 V3.7.31: added TMIMo

**Options:** R&S CMW-KM652 for CMIMo  
 R&S CMW-KM654 per R&S CMW for TMIMo

**Manual operation:** See ["Receive Mode"](#) on page 287

**CONFigure:WLAN:MEAS<i>:ISIGnal:BWIDth <BandWidth>**

Selects the channel bandwidth.

In the combined signal path (CSP), consider the dependency between this parameter and the trigger bandwidth, set for the RX frame trigger in the signaling application.

See [TRIGger:WLAN:SIGN<i>:RX:MACFrame:BW](#)

For 802.11ax and trigger source set to "HE\_TB Trigger", the setting depends on the trigger configuration:

[CONFigure:WLAN:SIGN<i>:CONNEction:HETF:CHBW](#)

**Parameters:**

<BandWidth> BW05mhz | BW10mhz | BW20mhz | BW40mhz | BW80mhz |  
 BW88mhz | BW16mhz  
**BW05mhz:** 5 MHz  
**BW10mhz:** 10 MHz  
**BW20mhz:** 20 MHz  
**BW40mhz:** 40 MHz  
**BW80mhz:** 80 MHz  
**BW88mhz:** 80+80 MHz  
**BW16mhz:** 160 MHz  
 \*RST: BW20

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V2.0.10, V3.2.10: added 80 MHz, V3.2.20: added 160 MHz,  
 V3.2.40: added 5 MHz and 10 MHz, V3.2.71: added 80+80 MHz  
 V3.7.60: 80 + 80 MHz supported also for 802.11ax

**Manual operation:** See ["Band Width"](#) on page 287

| Bandwidths  | Supported standards |
|-------------|---------------------|
| 5 MHz       | 802.11p, n, ac      |
| 10 MHz      | 802.11p, n, ac      |
| 20 MHz      | all standards       |
| 40 MHz      | 802.11n, ac, ax     |
| 80 MHz      | 802.11ac, ax        |
| 80 + 80 MHz | 802.11ac, ax        |
| 160 MHz     | 802.11ac, ax        |

---

**CONFigure:WLAN:MEAS<i>:ISIGnal:CDIStance** <ChannelDistance>

Configures the distance between the center frequencies of the two 80-MHz segments for the bandwidth of 80+80 MHz.

**Parameters:**

<ChannelDistance>    numeric  
                             Range:      80 MHz to 940 MHz  
                             Increment: 5 MHz  
                             \*RST:       80 MHz  
                             Default unit: MHz

**Firmware/software:** V3.5.121, V3.7.10 maximum reduced

**Options:**               R&S CMW-KM656

**Manual operation:**   See ["Ch. Distance"](#) on page 288

---

**CONFigure:WLAN:MEAS<i>:ISIGnal:BTYPe** <BurstType>

Sets the burst type for standard 802.11n. Do not use the command for other standards.

**Parameters:**

<BurstType>            MIXed | GREenfield  
                             **MIXed:** Compatibility mode, for coexistence with older standards  
                             **GREenfield:** Greenfield mode, incompatible with older standards  
                             \*RST:       MIX

**Example:**               See [Specifying general measurement settings](#)

**Firmware/software:** V1.0.10.50

**Manual operation:**   See ["Burst Type"](#) on page 294

---

**CONFigure:WLAN:MEAS<i>:ISIGnal:DSSS:ELENgth** <EvaluationLengthChips>[, <SkipPH>]

Specifies the evaluation length of the burst for DSSS signals.

**Parameters:**

<EvaluationLength> numeric  
 Number of payload chips  
 Range: 1000 chips to 270000 chips  
 \*RST: 1000 chips  
 Default unit: chip

<SkipPH> OFF | ON  
**OFF:** measure also preamble and header  
**ON:** skip preamble and header  
 \*RST: ON

**Firmware/software:** V3.5.121  
 V3.8.10: changed maximum for <EvaluationLength>

**Manual operation:** See ["Evaluation Length, Skip Preamble and Header"](#) on page 294

**CONFigure:WLAN:MEAS<i>:ISIGnal:OFDM:ELENgth** <EvaluationLengthSymbols>

Specifies the evaluation length of the burst for OFDM signals.

**Parameters:**

<EvaluationLength> numeric  
 Number of payload symbols  
 Range: 16 symbols to 1377 symbols  
 \*RST: 1377 symbols  
 Default unit: symbols

**Firmware/software:** V3.5.121

**Manual operation:** See ["Evaluation Length, Skip Preamble and Header"](#) on page 294

**CONFigure:WLAN:MEAS<i>:ISIGnal:PCLass** <PowerClass>

Sets the STA transmit power class for 802.11p and selects the transmit spectrum mask to be applied.

**Parameters:**

<PowerClass> CLA | CLB | CLCD | USERdefined  
**CLA:** class A transmit spectrum mask  
**CLB:** class B transmit spectrum mask  
**CLCD:** class C or D, no transmit spectrum limit check  
**USERdefined:** user-defined transmit spectrum mask  
 \*RST: CLA

**Firmware/software:** V3.2.40  
 V3.7.10 added USERdefined

**Options:** R&S CMW-KM655

**Manual operation:** See ["Power Class"](#) on page 295

---

### CONFigure:WLAN:MEAS<i>:ISIGnal:TDAa <FileName>

Specifies the training data source to be used as the reference signal for CMIMO measurements.

#### Parameters:

<FileName> "no data" | "Internal" | "<file name>"  
**"no data"**: Use no training data.  
**"Internal"**: Use in-memory data, acquired during preceding training mode runs. In-memory data must be available.  
**"<file name>"**: Use the training data file with the specified name, in the directory @USERDATA\MIMOData.  
 \*RST: "no data"

**Example:** See [CMIMO training mode](#)

**Firmware/software:** V2.1.10

**Options:** R&S CMW-KM652

**Manual operation:** See ["Training Data"](#) on page 296

---

### CONFigure:WLAN:MEAS<i>:ISIGnal:TDAa:FILE:DATE?

Returns the last modified date of the currently loaded training data file, if any.

#### Return values:

<FileDate> string  
 String with a formatted date or NAV

**Usage:** Query only

**Firmware/software:** V2.1.10

**Options:** R&S CMW-KM652

**Manual operation:** See ["Training Data"](#) on page 296

---

### CONFigure:WLAN:MEAS<i>:ISIGnal:MODFilter <ModulationFilter>

This command allows you to limit the evaluation to bursts of a particular modulation type. If the received burst has a different modulation, the reliability "Wrong Modulation" is displayed.

In the combined signal path (CSP), consider the dependency between this parameter and the data rate, set for the RX frame trigger in the signaling application.

See [TRIGger:WLAN:SIGN<i>:RX:MACFrame:RATE](#)

For 802.11ax and trigger source set to "HE\_TB Trigger", the setting depends on the trigger configuration:

CONFigure:WLAN:SIGN<i>:STA<s>:CONNection:HETF:MCS

**Parameters:**

<ModulationFilter> For OFDM: ALL | BPSK | QPSK | QAM16 | QAM64 | QAM256 | QAM1024  
For DSSS: ALL | DBPSk | DQPSk | CCK5\_5 | CCK11  
\*RST: ALL

**Firmware/software:** V3.5.10, V3.7.30: added QAM256, QAM1024

**Manual operation:** See ["Modulation Filter"](#) on page 295

CONFigure:WLAN:MEAS<i>:ISIGnal:IQSWap <IQSwap>

Swaps the role of the I and Q axes in the baseband.

**Parameters:**

<IQSwap> ON | OFF  
\*RST: OFF

**Example:** See [Specifying general measurement settings](#)

**Firmware/software:** V3.2.60

**Manual operation:** See ["IQ Swap"](#) on page 296

### 3.5.3 Multi-evaluation measurement control commands

The commands for the WLAN multi-evaluation measurement are divided into the groups listed below. The general measurement settings also affect the measurement, see [Chapter 3.5.2, "General measurement settings"](#), on page 355.

- [Measurement control and states](#).....370
- [Enabling results and views](#).....372
- [Measurement parameters](#).....377
- [List mode settings](#).....386
- [Trigger settings](#).....399

#### 3.5.3.1 Measurement control and states

The following commands control the measurement and return the current measurement state.

INITiate:WLAN:MEAS<i>:MEValuation.....371  
STOP:WLAN:MEAS<i>:MEValuation.....371  
ABORT:WLAN:MEAS<i>:MEValuation.....371  
FETCh:WLAN:MEAS<i>:MEValuation:STATe?.....371  
FETCh:WLAN:MEAS<i>:MEValuation:STATe:ALL?.....371

**INITiate:WLAN:MEAS<i>:MEValuation****STOP:WLAN:MEAS<i>:MEValuation****ABORt:WLAN:MEAS<i>:MEValuation**

Starts, stops, or aborts the measurement:

- **INITiate...** starts or restarts the measurement. The measurement enters the "RUN" state.
- **STOP...** halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- **ABORt...** halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

Use **FETCH...STATe?** to query the current measurement state.

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Event

**Firmware/software:** V1.0.10.1

**Manual operation:** See ["Multi Evaluation \(Softkey\)"](#) on page 256

**FETCH:WLAN:MEAS<i>:MEValuation:STATe?**

Queries the main measurement state. Use **FETCH:...:STATe:ALL?** to query the measurement state including the substates. Use **INITiate...**, **STOP...**, **ABORt...** to change the measurement state.

**Return values:**

<MultiEvalState> OFF | RUN | RDY

**OFF:** measurement off, no resources allocated, no results

**RUN:** measurement running, synchronization pending or adjusted, resources active or queued

**RDY:** measurement terminated, valid results can be available

\*RST: OFF

**Usage:** Query only

**Firmware/software:** V1.0.10.1

**Manual operation:** See ["Multi Evaluation \(Softkey\)"](#) on page 256

**FETCH:WLAN:MEAS<i>:MEValuation:STATe:ALL?**

Queries the main measurement state and the measurement substates. Both measurement substates are relevant for running measurements only. Use **FETCH:...:STATe?** to query the main measurement state only. Use **INITiate...**, **STOP...**, **ABORt...** to change the measurement state.

**Return values:**

|                           |                                                                                                                                                                                                                                                                                  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <MainState>               | OFF   RUN   RDY<br><b>OFF:</b> measurement off, no resources allocated, no results<br><b>RUN:</b> measurement running, synchronization pending or adjusted, resources active or queued<br><b>RDY:</b> measurement terminated, valid results can be available<br><b>*RST:</b> OFF |
| <SyncState>               | PEND   ADJ   INV<br><b>PEND:</b> waiting for resource allocation, adjustment, hardware switching ("pending")<br><b>ADJ:</b> adjustments finished, measurement running ("adjusted")<br><b>INV:</b> not applicable, <MainState> OFF or RDY ("invalid")                             |
| <ResourceState>           | QUE   ACT   INV<br><b>QUE:</b> measurement without resources, no results available ("queued")<br><b>ACT:</b> resources allocated, acquisition of results in progress but not complete ("active")<br><b>INV:</b> not applicable, <MainState> OFF or RDY ("invalid")               |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                                                                                       |
| <b>Firmware/software:</b> | V1.0.10.1                                                                                                                                                                                                                                                                        |
| <b>Manual operation:</b>  | See <a href="#">"Multi Evaluation (Softkey)"</a> on page 256                                                                                                                                                                                                                     |

**3.5.3.2 Enabling results and views**

The following commands select the evaluated results and the displayed views.

|                                                          |     |
|----------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult[:ALL]      | 373 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:MSCalar    | 374 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:EVM        | 374 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:EVMSymbol  | 374 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:EVMCarrier | 375 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:IQConst    | 375 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:UTERror    | 375 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:SFLatness  | 375 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:TSMask     | 376 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:RESult:PVTime     | 376 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:PVTime:BURSt      | 376 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:PVTime:REDGe      | 377 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:PVTime:FEDGE      | 377 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:SFLatness:DMODE   | 377 |
| CONFigure:WLAN:MEAS<i></i>:MEValuation:TSMask:DMODE      | 377 |



**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult[:ALL]** <ModScalar>, <PvT>, <EVMvsChip>, <EVMvsSym>, <EVMvsCarr>, <IQConst>, <SpecFlatness>, <TranSpecMask>[, <UnusedToneErr>]

Enables or disables the evaluation of results and shows or hides the views. This command combines all other **CONFigure:WLAN:MEAS<i>:MEvaluation:RESult...** commands.

Views can only be hidden for receive mode SISO.

To check which views are relevant for which standard, see [Measurement results](#).

**Parameters:**

|                 |                                                                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ModScalar>     | OFF   ON<br>Modulation scalar overview<br><b>OFF:</b> Do not evaluate results, hide the view<br><b>ON:</b> Evaluate results and show the view<br>*RST: ON |
| <PvT>           | OFF   ON<br>Power vs. time<br>*RST: OFF                                                                                                                   |
| <EVMvsChip>     | OFF   ON<br>EVM vs chip<br>*RST: ON                                                                                                                       |
| <EVMvsSym>      | OFF   ON<br>EVM vs symbol<br>*RST: ON                                                                                                                     |
| <EVMvsCarr>     | OFF   ON<br>EVM vs carrier<br>*RST: ON                                                                                                                    |
| <IQConst>       | OFF   ON<br>I/Q constellation diagram<br>*RST: OFF                                                                                                        |
| <SpecFlatness>  | OFF   ON<br>Spectrum flatness<br>*RST: OFF                                                                                                                |
| <TranSpecMask>  | OFF   ON<br>Transmit spectrum mask<br>*RST: OFF                                                                                                           |
| <UnusedToneErr> | OFF   ON<br>Unused tone error<br>*RST: OFF                                                                                                                |

- Example:** See [Specifying measurement-specific settings](#)
- Firmware/software:** V1.0.10.50, V1.0.15.0: <TranSpecMask>, V1.0.15.20: <SpecFlatness>, V2.1.10: <PvT>, V3.7.21: <UnusedToneErr>
- Manual operation:** See ["Assign Views"](#) on page 302

#### **CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:MSCalar <ModEnable>**

Enables or disables the evaluation of results and shows or hides the modulation scalar view.

**Parameters:**

<ModEnable> OFF | ON  
**OFF:** Do not evaluate results, hide the view  
**ON:** Evaluate results and show the view  
 \*RST: ON

**Firmware/software:** V1.0.10.50

**Manual operation:** See ["Assign Views"](#) on page 302

#### **CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:EVM <EVMEEnable>**

Enables or disables the evaluation of results and shows or hides the EVM vs. chip view.

**Parameters:**

<EVMEEnable> OFF | ON  
**OFF:** Do not evaluate results, hide the view  
**ON:** Evaluate results and show the view  
 \*RST: ON

**Firmware/software:** V1.0.10.50

**Manual operation:** See ["Assign Views"](#) on page 302

#### **CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:EVMSymbol <EVMEEnable>**

Enables or disables the evaluation of results and shows or hides the EVM vs. symbol view.

**Parameters:**

<EVMEEnable> OFF | ON  
**OFF:** Do not evaluate results, hide the view  
**ON:** Evaluate results and show the view  
 \*RST: ON

**Firmware/software:** V1.0.10.50

**Manual operation:** See ["Assign Views"](#) on page 302

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:EVMCarrier <EVMEEnable>**

Enables or disables the evaluation of results and shows or hides the EVM vs. carrier view.

**Parameters:**

<EVMEEnable>      OFF | ON  
                    **OFF:** Do not evaluate results, hide the view  
                    **ON:** Evaluate results and show the view  
                    \*RST:      ON

**Firmware/software:** V1.0.10.50

**Manual operation:** See ["Assign Views"](#) on page 302

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:IQConst <IQEnable>**

Enables or disables the evaluation of results and shows or hides the I/Q constellation diagram view.

**Parameters:**

<IQEnable>      OFF | ON  
                    **OFF:** Do not evaluate results, hide the view  
                    **ON:** Evaluate results and show the view  
                    \*RST:      OFF

**Firmware/software:** V1.0.10.50

**Manual operation:** See ["Assign Views"](#) on page 302

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:UTERror <UTE\_Enable>**

Enables or disables the evaluation of results and shows or hides the unused tone error view.

**Parameters:**

<UTE\_Enable>      OFF | ON  
                    **OFF:** Do not evaluate results, hide the view  
                    **ON:** Evaluate results and show the view  
                    \*RST:      OFF

**Firmware/software:** V3.7.21

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:SFlatness <SpecFlatness>**

Enables or disables the evaluation of results and shows or hides the spectrum flatness view.

**Parameters:**

<SpecFlatness> OFF | ON  
**OFF:** Do not evaluate results, hide the view  
**ON:** Evaluate results and show the view  
\*RST: OFF

**Firmware/software:** V1.0.15.20

**Manual operation:** See ["Assign Views"](#) on page 302

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:TSMask <SpecEnable>**

Enables or disables the evaluation of results and shows or hides the transmit spectrum mask view.

**Parameters:**

<SpecEnable> OFF | ON  
**OFF:** Do not evaluate results, hide the view  
**ON:** Evaluate results and show the view  
\*RST: OFF

**Firmware/software:** V1.0.15.0

**Manual operation:** See ["Assign Views"](#) on page 302

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:RESult:PVTime <PvTEnable>**

Enables or disables the evaluation of results and shows or hides the power vs. time view.

**Parameters:**

<PvTEnable> OFF | ON  
**OFF:** Do not evaluate results, hide the view  
**ON:** Evaluate results and show the view  
\*RST: OFF

**Firmware/software:** V2.1.10

**Manual operation:** See ["Assign Views"](#) on page 302

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:PVTime:BURSt <Burst>**

Enables or disables the evaluation of results and shows or hides the burst subview of the power vs. time view.

**Parameters:**

<Burst> OFF | ON  
**OFF:** Do not evaluate results, hide the subview  
**ON:** Evaluate results and show the subview  
\*RST: ON

**Firmware/software:** V2.1.10

---

#### **CONFigure:WLAN:MEAS<i>:MEValuation:PVTime:REDGe <Rising>**

Enables or disables the evaluation of results and shows or hides the rising edge sub-view (transmit power-on ramp) of the power vs. time view for DSSS signals.

##### **Parameters:**

<Rising>                      OFF | ON  
**OFF:** Do not evaluate results, hide the subview  
**ON:** Evaluate results and show the subview  
 \*RST:                      ON

**Firmware/software:** V2.1.10

---

#### **CONFigure:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE <Fall>**

Enables or disables the evaluation of results and shows or hides the falling edge sub-view (transmit power-down ramp) of the power vs. time view for DSSS signals.

##### **Parameters:**

<Fall>                      OFF | ON  
**OFF:** Do not evaluate results, hide the subview  
**ON:** Evaluate results and show the subview  
 \*RST:                      ON

**Firmware/software:** V2.1.10

---

#### **CONFigure:WLAN:MEAS<i>:MEValuation:SFlatness:DMODE <DispMode>**

#### **CONFigure:WLAN:MEAS<i>:MEValuation:TSMask:DMODE <DispMode>**

Selects the display mode of "Spectrum Flatness" and "Transmit Spectrum Mask" results to switch between relative and absolute result values (dB vs. dBm).

##### **Parameters:**

<DispMode>                      RELative | ABSolute  
 \*RST:                      REL

**Firmware/software:** V3.7.30, V3.7.60: added :TSMask: command

**Manual operation:** See ["Display Mode"](#) on page 326

### **3.5.3.3 Measurement parameters**

The following commands specify the scope of the measurement.

|                                                                            |     |
|----------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MEValuation:TOUT</a> .....        | 378 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MEValuation:REPetition</a> .....  | 378 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MEValuation:SCONdition</a> .....  | 379 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MEValuation:SMODE</a> .....       | 379 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MEValuation:MOEXception</a> ..... | 379 |

|                                                                      |     |
|----------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:SCOut:MODulation.....             | 380 |
| CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:PHASe.....  | 380 |
| CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:TIMing..... | 381 |
| CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:LEVel.....  | 381 |
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---

#### CONFigure:WLAN:MEAS<i>:MEValuation:TOUT <TCDTimeout>

Defines a timeout for the measurement. The timer is started when the measurement is initiated via a `READ` or `INIT` command. It is not started if the measurement is initiated manually ([`ON` | `OFF`] key or [`RESTART` | `STOP`] key).

When the measurement has completed the first measurement cycle (first single shot), the statistical depth is reached and the timer is reset.

If the first measurement cycle has not been completed when the timer expires, the measurement is stopped. The measurement state changes to `RDY`. The reliability indicator is set to 1, indicating that a measurement timeout occurred. Still running `READ`, `FETCh` or `CALCulate` commands are completed, returning the available results. At least for some results, there are no values at all or the statistical depth has not been reached.

A timeout of 0 s corresponds to an infinite measurement timeout.

The measurement of a DSSS signal with low data rate and large payload sizes can take up to 40 s. Set the measurement timeout to an adequate value, e.g. to 60 s.

#### Parameters:

<TCDTimeout>            numeric  
                               Default unit: s

**Firmware/software:** V2.0.10

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:REPetition <Repetition>

Specifies the repetition mode of the measurement. The repetition mode specifies whether the measurement is stopped after a single shot or repeated continuously. Use `CONFigure:::MEAS<i>:::SCOut` to determine the number of measurement intervals per single shot.

**Parameters:**

<Repetition> SINGleshot | CONTInuous  
**SINGleshot:** Single-shot measurement  
**CONTInuous:** Continuous measurement  
 \*RST: SING

**Firmware/software:** V1.0.10.1

**Manual operation:** See ["Repetition"](#) on page 297

**CONFigure:WLAN:MEAS<i>:MEValuation:SCONdition** <StopCondition>

Qualifies whether the measurement is stopped after a failed limit check or continued. **SLFail** means that the measurement is stopped and reaches the **RDY** state when one of the results exceeds the limits.

**Parameters:**

<StopCondition> NONE | SLFail  
**NONE:** Continue measurement irrespective of the limit check  
**SLFail:** Stop measurement on limit failure  
 \*RST: NONE

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.10.1

**Manual operation:** See ["Stop Condition"](#) on page 298

**CONFigure:WLAN:MEAS<i>:MEValuation:SMODE** <SynchronizationMode>

Sets the synchronization mode:

- Normal: synchronization according to preamble detection
- Tolerant: synchronization with the second part of preamble when the first part cannot be detected

**Parameters:**

<SyncMode> NORMAl | TOLerant  
 \*RST: NORM

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.20

**Manual operation:** See ["Synchronization Mode"](#) on page 298

**CONFigure:WLAN:MEAS<i>:MEValuation:MOEXception** <MeasOnException>

Specifies whether measurement results that the R&S CMW identifies as faulty or inaccurate are rejected.

**Parameters:**

&lt;MeasOnException&gt; OFF | ON

**OFF:** Faulty results are rejected**ON:** Results are never rejected

\*RST: OFF

**Example:**See [Specifying measurement-specific settings](#)**Firmware/software:** V1.0.10.1**Manual operation:** See ["Measure on Exception"](#) on page 298**CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:MODulation <StatisticCount>**

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

**Parameters:**

&lt;StatisticCount&gt; numeric

Number of measurement intervals for modulation measurements

Range: 1 to 2000

\*RST: 10

**Example:**See [Specifying measurement-specific settings](#)**Firmware/software:** V1.0.10.1**Manual operation:** See ["Statistic Count"](#) on page 298**CONFigure:WLAN:MEAS<i>:MEvaluation:COMPensation:TRACking:PHASe <Phase>**

Activate or deactivate phase tracking. With enabled tracking, fluctuations are compensated.

For composite MIMO and 802.11ac input signals, phase tracking is always enabled.

**Parameters:**

&lt;Phase&gt; OFF | ON

**OFF:** Tracking disabled**ON:** Tracking enabled

\*RST: ON

**Example:**See [Specifying measurement-specific settings](#)**Firmware/software:** V1.0.10.50**Manual operation:** See ["Tracking"](#) on page 299



---

**CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:TIMing**  
<Timing>

Activate or deactivate timing tracking. With enabled tracking, fluctuations are compensated.

**Parameters:**

<Timing>                    OFF | ON  
                              **OFF:** Tracking disabled  
                              **ON:** Tracking enabled  
                              \*RST:        ON

**Example:**                See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.10.50

**Manual operation:**    See ["Tracking"](#) on page 299

---

**CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:TRACking:LEVeI**  
<Level>

Activate or deactivate level tracking. With enabled tracking, fluctuations are compensated.

**Parameters:**

<Level>                    OFF | ON  
                              **OFF:** Tracking disabled  
                              **ON:** Tracking enabled  
                              \*RST:        OFF

**Example:**                See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.10.50

**Manual operation:**    See ["Tracking"](#) on page 299

---

**CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:CESTimation**  
<ChannelEst>

Specifies whether channel estimation is done in payload or preamble.

**Parameters:**

<ChannelEst>              PAYLoad | PREamble  
                              **PAYLoad:** Channel estimation in payload and preamble  
                              **PREamble:** Channel estimation in preamble only  
                              \*RST:        PRE

**Example:**                See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.10.50

**Manual operation:**    See ["Channel Estimation"](#) on page 300

**CONFigure:WLAN:MEAS<i>:MEValuation:DEMod:FFT:OFFSet** <Offset>

Sets the FFT start offset for OFDM signals.

**Parameters:**

<Offset>                      CENT | PEAK | AUTO  
**CENT:** Guard interval center used as start offset  
**PEAK:** Peak of fine-timing metric used to determine start offset  
**AUTO:** Automatic selection of optimal start offset  
 \*RST:                      AUTO

**Example:**                      See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.0.10

**Manual operation:**    See ["FFT Demodulation Offset"](#) on page 300

**CONFigure:WLAN:MEAS<i>:MEValuation:COMPensation:EFTaps**

<EqualizerFilterTaps\_enable>[, <EqualizerFilterTaps\_value>]

This command is relevant for DSSS signals only. It determines if and how accurate the transmit filter is estimated.

**Parameters:**

Enable                      ON | OFF  
 Enables or disables TX filter estimation  
 \*RST:                      ON  
 <EqualizerFilterTaps> decimal | 3 | 5 | 7 | 9 | 11 | 13 | 15  
 Number of equalizer filter taps  
 \*RST:                      11  
 Additional value: OFF (indicates that the TX filter estimation is disabled)

**Example:**                      See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.5.20

**Manual operation:**    See ["DSSS Tx Filter Estimation"](#) on page 300

**CONFigure:WLAN:MEAS<i>:MEValuation:EMETHod** <EVM\_Method11b>

This parameter is relevant for 802.11b signals only. It selects the EVM measurement method - according to standard IEEE Std 802.11-2007, IEEE Std 802.11b-1999, or according to standard IEEE Std 802.11-2016.

**Parameters:**

<EVM\_Method11b>    ST2007 | ST1999 | ST2016  
 \*RST:                      ST2007

**Example:**                      See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.5.120

V4.0.20: added ST2016

**Manual operation:** See ["DSSS EVM Method"](#) on page 300

**CONFigure:WLAN:MEAS<i>:MEvaluation:COMPensation:SKIPsymbols**  
 <SkipSymbols\_Head>, <SkipSymbols\_Tail>

Defines how many head and tail symbols are excluded from OFDM modulation measurements.

**Parameters:**

<SkipSymbols\_Head> decimal

Number of heading symbols to be skipped

Range: 0 Sym to 100 Sym

\*RST: 0 Sym

Default unit: symbol

<SkipSymbols\_Tail> decimal

Number of tailing symbols to be skipped

Range: 0 Sym to 100 Sym

\*RST: 0 Sym

Default unit: symbol

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.5.30

**Manual operation:** See ["Skip OFDM Symbols"](#) on page 300

**CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:TSMask** <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

**Parameters:**

<StatisticCount> numeric

Number of measurement intervals for the transmit spectrum mask measurement

Range: 1 to 1000

\*RST: 10

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.15.0

**Manual operation:** See ["Statistic Count"](#) on page 301

**CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:TROTime** <TriggerOffTime>

Specifies the trigger offset between trigger event and FFT operation.

**Parameters:**

<TriggerOffTime>    numeric  
                          Range:      0 s to 0.1E-3 s  
                          \*RST:      5E-6 s

**Example:**            See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.15.0

**Manual operation:** See "[Spectrum Trigger Time](#)" on page 301

**CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:AFFTnum <AverFFTNum>**

Specifies the number of FFT operations per burst.

**Parameters:**

<AverFFTNum>        numeric  
                          Range:      1 to 16  
                          \*RST:      8

**Example:**            See [Specifying measurement-specific settings](#)

**Firmware/software:** V1.0.15.0

**Manual operation:** See "[Average FFT Num](#)" on page 301

**CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:OBWPercent <OBWPower>**

Enables/disables OBW measurement and sets the OBW percentage.

**Parameters:**

<OBWPower>          numeric | ON | OFF  
                          Range:      50 % to 100 %  
                          \*RST:      99 %, OFF  
                          Additional values: ON | OFF (disables | enables OBW measurement)

**Firmware/software:** V3.2.11

**Manual operation:** See "[Occupied Bandwidth \(OBW\)](#)" on page 301

**CONFigure:WLAN:MEAS<i>:MEvaluation:TSMask:MSElection <MaskSelection>**

Selects the spectrum limit mask to be applied to 802.11p signals.

See "[Transmit spectrum mask OFDM, by regulation](#)" on page 320

**Parameters:**

<MaskSelection> IEEE | ETSI | ARIB

**IEEE:** Relative spectral density limits, IEEE 802.11

**ETSI:** Absolute emission limits, ETSI EN 302 571 V1.1.1 (2008-09)

**ARIB:** Absolute emission limits, ARIB STD-T109, only for 10 MHz bandwidth

\*RST: IEEE

**Firmware/software:** V3.2.40, V3.5.121 added ARIB

**Manual operation:** See "[Mask Selection](#)" on page 301

**CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:PVTime <StatisticCount>**

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

**Parameters:**

<StatisticCount> numeric

Number of measurement intervals for the power vs. time measurement

Range: 1 to 2000

\*RST: 10

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V2.1.10

**Manual operation:** See "[Statistic Count](#)" on page 302

**CONFigure:WLAN:MEAS<i>:MEvaluation:PVTime:ALENth <AvgLenth>**

Sets the length of the moving average filter, which smoothes the power trace and thus eliminates the modulation.

**Parameters:**

<AvgLenth> numeric

Range: 1 to 199

\*RST: 1

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See "[Average Length](#)" on page 302

**CONFigure:WLAN:MEAS<i>:MEvaluation:PVTime:RPOwer <RefPower>**

Sets the reference power to the maximum power or to the mean power of the burst.

In DSSS, the thresholds for rising and falling edge results are defined as percentage values of the reference power.

**Parameters:**

<RefPower>            MAXimum | MEAN  
                          \*RST:        MAX

**Example:**            See [Specifying measurement-specific settings](#)

**Firmware/software:** V2.1.30

**Manual operation:** See "[Reference Power](#)" on page 302

**3.5.3.4 List mode settings**

The following commands configure the list mode.

For retrieving list mode results, see [Chapter 3.5.5.24, "List mode results \(one segment\)"](#), on page 580 and [Chapter 3.5.5.25, "List mode results \(all segments\)"](#), on page 589.

For a description of the list mode, see [Chapter 3.2.7, "Multi-evaluation list mode"](#), on page 252.

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---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIST <ListModeEnable>**

Enables or disables the list mode.

**Parameters:**

<ListModeEnable> OFF | ON  
**OFF:** Disable list mode  
**ON:** Enable list mode  
 \*RST: OFF

**Example:** See [Performing single-shot measurements](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**Manual operation:** See "[List Mode > Enable](#)" on page 303

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:BTYPe <BurstTypes>...**

Specifies the burst types for standard 802.11n for all segments in list mode. Do not use the command for other standards.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Parameters:**

{<BurstType>}1... MIXed | GREenfield  
 {<BurstType>}n **MIXed:** for coexistence with other standards  
**GREenfield:** incompatible with other standards

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:BWIDth <BandWidths>...**

Specifies the channel bandwidths for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Parameters:**

{<BandWidth>}1... BW05mhz | BW10mhz | BW20mhz | BW40mhz | BW80mhz |  
 {<BandWidth>}n BW88mhz | BW16mhz  
**BW05mhz:** 5 MHz  
**BW10mhz:** 10 MHz  
**BW20mhz:** 20 MHz  
**BW40mhz:** 40 MHz  
**BW80mhz:** 80 MHz  
**BW88mhz:** 80+80 MHz  
**BW16mhz:** 160 MHz

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

| Bandwidths  | Supported standards |
|-------------|---------------------|
| 5 MHz       | 802.11p, n, ac      |
| 10 MHz      | 802.11p, n, ac      |
| 20 MHz      | all standards       |
| 40 MHz      | 802.11n, ac, ax     |
| 80 MHz      | 802.11ac, ax        |
| 80 + 80 MHz | 802.11ac, ax        |
| 160 MHz     | 802.11ac, ax        |

---

#### **CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT** <NoOfSegments>

Defines the number of segments in the entire measurement interval.

##### **Parameters:**

<NoOfSegments>    numeric  
                          Range:    1 to 100  
                          \*RST:     1

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

#### **CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:ENPower** <Levels>...

Specifies the expected nominal power of the measured RF signal for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

##### **Parameters:**

{<Level>}1...    numeric  
 {<Level>}n    The range of the expected nominal power can be calculated as follows: *Range (Expected Nominal Power) = Range (Input Power) + External Attenuation - User Margin*  
                          Range:    -47 dBm to 34 dBm for the input power at RF 1 COM and RF 2 COM (please notice also the ranges quoted in the data sheet).

**Example:** See [Specifying measurement-specific settings](#)



**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:FREQuency** <Frequencies>...

Specifies the measurement frequencies for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

#### **Parameters:**

{<Frequency>}1... numeric  
{<Frequency>}n Range: 70E+6 Hz to 6E+9 Hz

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:MOFFset** <MeasOffsets>...

Specifies the measurement offsets for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

#### **Parameters:**

{<MeasOffset>}1... numeric  
{<MeasOffset>}n Range: 1E-5 s to 1 s  
Increment: 1E-5 s

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:MTIME** <MeasTimes>...

Specifies the measurement times for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

#### **Parameters:**

{<MeasureTime>}1... numeric  
{<MeasureTime>}n Range: 1E-5 s to 1 s  
Increment: 1E-5 s

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

#### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:RESult:MODulation**

<EnableMod>...

#### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:RESult:TSMask <EnableSem>...**

Enables or disables the evaluation of results for modulation ( . . :MODulation) and transmit spectrum mask ( . . :TSMask) measurements in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

#### **Parameters:**

{<EnableResult>}1... OFF | ON

{<EnableResult>}n

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

#### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:RTRigger <ReTriggers>...**

Specifies, whether the measurement in list mode waits for a trigger event before measuring the segment, or not. For the first segment, the value OFF is always interpreted as ON.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

#### **Parameters:**

{<ReTrigger>}1... OFF | ON

{<ReTrigger>}n

**OFF:** measure the segment without retrigger

**ON:** wait for a trigger event from the trigger source configured via [TRIGger:WLAN:MEAS<i>:MEValuation:SOURce](#)

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

#### **CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SCount:MODulation**

<StatCountsMod>...

Specifies the statistical length for modulation measurements for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Parameters:**

{<StatCount>}1... numeric  
 {<StatCount>}n No. of bursts to be measured during modulation measurements, one result per segment  
 Range: 1 to 2000

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

### **CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:SCount:TSMask** <StatCountsSem>...

Specifies the spectrum statistical length for transmit spectrum mask measurements for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Parameters:**

{<StatCount>}1... numeric  
 {<StatCount>}n No. of bursts to be measured during spectrum measurements  
 Range: 1 to 1000

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

### **CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:STANdard** <Standards>...

Specifies the standard for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Parameters:**

{<Standard>}1... DSSS | LOFDm | HTOFDm | VHTofdm | HEOFdm | POFDm  
 {<Standard>}n  
**DSSS:** 802.11b/g (DSSS)  
**LOFDm:** 802.11a/g (OFDM)  
**HTOFdm:** 802.11n  
**VHTofdm:** 802.11ac  
**HEOFdm:** 802.11ax

**POFDm:** 802.11p

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012  
 R&S CMW-KM655 for 802.11p  
 R&S CMW-KM651 for 802.11n  
 R&S CMW-KM656 for 802.11ac  
 R&S CMW-KM657 for 802.11ax

### **CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:STIME <SegmentTimes>...**

Specifies the segment times for all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

#### **Parameters:**

{<SegmentTime>}1... numeric  
 {<SegmentTime>}n Range: 4E-5 s to 1 s  
 Increment: 1E-5 s

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

### **CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:SEGMENT<no>:BTYPE <BurstType>**

Specifies the burst types for standard 802.11n for segment <no> in list mode. Do not use the command for other standards.

#### **Suffix:**

<no> Segment

#### **Parameters:**

<BurstType> MIXed | GREenfield  
**MIXed:** for coexistence with other standards  
**GREenfield:** incompatible with other standards

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:BWIDth**  
 <BandWidth>

Specifies the channel bandwidth for segment <no> in list mode.

**Suffix:**

<no> 1..n  
Segment

**Parameters:**

<BandWidth> BW05mhz | BW10mhz | BW20mhz | BW40mhz | BW80mhz |  
BW88mhz | BW16mhz

**BW05mhz:** 5 MHz

**BW10mhz:** 10 MHz

**BW20mhz:** 20 MHz

**BW40mhz:** 40 MHz

**BW80mhz:** 80 MHz

**BW88mhz:** 80+80 MHz

**BW16mhz:** 160 MHz

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

| Bandwidths  | Supported standards |
|-------------|---------------------|
| 5 MHz       | 802.11p, n, ac      |
| 10 MHz      | 802.11p, n, ac      |
| 20 MHz      | all standards       |
| 40 MHz      | 802.11n, ac, ax     |
| 80 MHz      | 802.11ac, ax        |
| 80 + 80 MHz | 802.11ac, ax        |
| 160 MHz     | 802.11ac, ax        |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ENPower <Level>**

Specifies the expected nominal power of the measured RF signal for segment <no> in list mode.

**Suffix:**

<no> 1..n  
Segment

**Parameters:****<Level>** numeric

The range of the expected nominal power can be calculated as follows: *Range (Expected Nominal Power) = Range (Input Power) + External Attenuation - User Margin*

Range: -47 dBm to 34 dBm for the input power at RF 1 COM and RF 2 COM (please notice also the ranges quoted in the data sheet).

**Example:** See [Specifying measurement-specific settings](#)**Firmware/software:** V3.8.11**Options:** R&S CMW-KM012

**CONFigure:WLAN:MEAS< i>:MEvaluation:LIST:SEGment<no>:FREQuency**  
**<Frequency>**

Specifies the center frequency of the RF analyzer for segment <no> in list mode.

**Suffix:****<no>** 1..n**Parameters:****<Frequency>** numeric

Range: 70E+6 Hz to 6E+9 Hz

**Example:** See [Specifying measurement-specific settings](#)**Firmware/software:** V3.8.11**Options:** R&S CMW-KM012

**CONFigure:WLAN:MEAS< i>:MEvaluation:LIST:SEGment<no>:MOFFset**  
**<MeasOffset>**

Specifies the measurement offset for segment <no> in list mode.

**Suffix:****<no>** 1..n  
Segment**Parameters:****<MeasOffset>** numeric

Range: 0 s to 1 s

Increment: 1E-4 s

**Example:** See [Specifying measurement-specific settings](#)**Firmware/software:** V3.8.11**Options:** R&S CMW-KM012

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:SEGMENT<no>:MTIME**  
 <MeasTime>

Specifies the measurement time for segment <no> in list mode.

**Suffix:**

<no>                      1..n  
                              Segment

**Parameters:**

<MeasTime>              numeric  
                              Duration of measurement for the segment  
                              Range:        4E-4 s to 1 s  
                              Increment: 1E-4 s

**Example:**                See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:**                R&S CMW-KM012

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:SEGMENT<no>:RESULT**  
 <EnableMod>, <EnableSem>

Enables or disables the evaluation of results for modulation and transmit spectrum mask measurements for segment <no> in list mode.

**Suffix:**

<no>                      1..n  
                              Segment

**Parameters:**

<EnableMod>              OFF | ON  
 <EnableSem>              OFF | ON

**Example:**                See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:**                R&S CMW-KM012

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:SEGMENT<no>:RTRigger**  
 <ReTrigger>

Specifies for segment <no> in list mode, whether the measurement waits for a trigger event before measuring the segment, or not. For the first segment, the value OFF is always interpreted as ON.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Suffix:**

<no> 1..n  
Segment

**Parameters:**

<ReTrigger> OFF | ON  
**OFF:** measure the segment without retrigger  
**ON:** wait for a trigger event from the trigger source configured via [TRIGger:WLAN:MEAS<i>:MEvaluation:SOURce](#)

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIST:SEGment<no>:SCount**  
<StatCountMod>, <StatCountSem>

Specifies the modulation and spectrum statistical length for segment <no> in list mode.

**Suffix:**

<no> 1..n  
Segment

**Parameters:**

<StatCountMod> numeric  
No. of burst to be measured during modulation measurements  
Range: 1 to 2000  
\*RST: 10

<StatCountSem> numeric  
No. of bursts to be measured during spectrum measurements  
Range: 1 to 1000  
\*RST: 10

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIST:SEGment<no>:SETup**  
<SegmentTime>, <MeasTime>, <MeasOffset>, <Level>, <Frequency>,  
<Standard>, <BandWidth>

Specifies burst parameter settings for segment <no> in list mode. Send this command for all segments to be measured.



**Suffix:**

<no> 1..n  
 The segment number must not exceed the total number of segments measured (see [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#) on page 388).

**Parameters:**

<SegmentTime> numeric  
 Duration of the segment  
 Range: 4E-5 s to 1 s  
 Increment: 1E-5 s

<MeasTime> numeric  
 Duration of measurement for the segment  
 Range: 1E-5 s to 1 s  
 Increment: 1E-5 s

<MeasOffset> numeric  
 Measurement offset for the segment  
 Range: 1E-5 s to 1 s  
 Increment: 1E-5 s

<Level> numeric  
 Expected nominal power of the measured RF signal within the segment  
 The range of the expected nominal power can be calculated as follows: *Range (Expected Nominal Power) = Range (Input Power) + External Attenuation - User Margin*  
 Range: -47 dBm to 34 dBm for the input power at RF 1 COM and RF 2 COM (please notice also the ranges quoted in the data sheet).  
 Default unit: dBm

<Frequency> numeric  
 Configures the center frequency of the RF analyzer. Set it to the center frequency of the received WLAN channel.  
 Range: 70E+6 Hz to 6E+9 Hz

<Standard> DSSS | LOFDm | HTOFDm | VHTofdm | HEOFdm | POFDm  
**DSSS:** 802.11b/g (DSSS)  
**LOFDm:** 802.11a/g (OFDM)  
**HTOFdm:** 802.11n  
**VHTofdm:** 802.11ac  
**HEOFdm:** 802.11ax  
**POFDm:** 802.11p

<BandWidth> BW05mhz | BW10mhz | BW20mhz | BW40mhz | BW80mhz | BW88mhz | BW16mhz  
**BW05mhz:** 5 MHz  
**BW10mhz:** 10 MHz

**BW20mhz:** 20 MHz  
**BW40mhz:** 40 MHz  
**BW80mhz:** 80 MHz  
**BW88mhz:** 80+80 MHz  
**BW16mhz:** 160 MHz

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

| Bandwidths  | Supported standards |
|-------------|---------------------|
| 5 MHz       | 802.11p, n, ac      |
| 10 MHz      | 802.11p, n, ac      |
| 20 MHz      | all standards       |
| 40 MHz      | 802.11n, ac, ax     |
| 80 MHz      | 802.11ac, ax        |
| 80 + 80 MHz | 802.11ac, ax        |
| 160 MHz     | 802.11ac, ax        |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGMent<no>:STANdard**  
 <Standard>

Specifies the standard for segment <no> in list mode.

**Suffix:**

<no> 1..n  
 Segment

**Parameters:**

<Standard> DSSS | LOFDm | HTOFDm | VHTofdM | HEOFdm | POFDm  
**DSSS:** 802.11b/g (DSSS)  
**LOFDm:** 802.11a/g (OFDM)  
**HTOFdm:** 802.11n  
**VHTofdM:** 802.11ac  
**HEOFdm:** 802.11ax  
**POFDm:** 802.11p

**Example:** See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012  
 R&S CMW-KM655 for 802.11p  
 R&S CMW-KM651 for 802.11n  
 R&S CMW-KM656 for 802.11ac  
 R&S CMW-KM657 for 802.11ax

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:STIME**  
 <SegmentTime>

Specifies the duration of the segment for segment <no> in list mode.

**Suffix:**

<no>                      1..n  
                              Segment

**Parameters:**

<SegmentTime>        numeric  
                              Range:        4E-4 s to 1 s  
                              Increment: 1E-4 s

**Example:**                See [Specifying measurement-specific settings](#)

**Firmware/software:** V3.8.11

**Options:**                R&S CMW-KM012

### 3.5.3.5 Trigger settings

The following commands define the trigger parameters.

|                                                  |     |
|--------------------------------------------------|-----|
| TRIGger:WLAN:MEAS<i>:MEValuation:CATalog:SOURce? | 399 |
| TRIGger:WLAN:MEAS<i>:MEValuation:SOURce          | 400 |
| TRIGger:WLAN:MEAS<i>:MEValuation:SLOPe           | 400 |
| TRIGger:WLAN:MEAS<i>:MEValuation:THReshold       | 400 |
| TRIGger:WLAN:MEAS<i>:MEValuation:TOUT            | 401 |
| TRIGger:WLAN:MEAS<i>:MEValuation:MGAP            | 401 |

---

**TRIGger:WLAN:MEAS<i>:MEValuation:CATalog:SOURce? [<fullList>]**

Lists all trigger source values that can be set using [TRIGger:WLAN:MEAS<i>:MEValuation:SOURce](#) on page 400.

**Query parameters:**

<fullList>                OFF | ON  
                              Disables/ enables full list including also invalid trigger sources.

**Return values:**

<TrigSource>            string  
                              Comma-separated list of all supported values. Each value is represented as a string.

**Usage:**                Query only

**Firmware/software:** V1.0.10.50

**Manual operation:**    See ["Trigger Source"](#) on page 303

**TRIGger:WLAN:MEAS<i>:MEvaluation:SOURce** <TrigSource>

Selects the source of the trigger events. Some values are always available. They are listed below. Depending on the installed options, additional values are available. You can query a list of all supported values via `TRIGger: . . . :CATalog:SOURce?`.

**Parameters:**

<TrigSource>                      string  
                                       **'IF Power'**: Power trigger (received RF power)  
                                       \*RST:            'IF Power'

**Example:**                        See [Configuring the trigger system](#)

**Firmware/software:** V1.0.10.50, V3.7.60: removed 'Free Run'

**Manual operation:**    See ["Trigger Source"](#) on page 303

**TRIGger:WLAN:MEAS<i>:MEvaluation:SLOPe** <TrigSlope>

Qualifies whether the trigger event is generated at the rising or at the falling edge of the trigger pulse (valid for external and power trigger sources).

**Parameters:**

<TrigSlope>                      REDGe | FEDGe  
                                       **REDGe**: Rising edge  
                                       **FEDGe**: Falling edge  
                                       \*RST:            REDG

**Example:**                        See [Configuring the trigger system](#)

**Firmware/software:** V1.0.10.50

**Manual operation:**    See ["Trigger Slope"](#) on page 304

**TRIGger:WLAN:MEAS<i>:MEvaluation:THReshold** <TrigThreshold>

Defines the trigger threshold for power trigger sources.

**Parameters:**

<TrigThreshold>                numeric | ON | OFF  
                                       Range:        -50 dB to 0 dB  
                                       \*RST:        -30 dB  
                                       Default unit: dB (full scale, i.e. relative to reference level minus external attenuation)

**Example:**                        See [Configuring the trigger system](#)

**Firmware/software:** V1.0.10.50, V3.7.10 changed \*RST value

**Manual operation:**    See ["Trigger Threshold"](#) on page 304

**TRIGger:WLAN:MEAS<i>:MEValuation:TOUT <TrigTimeOut>**

Selects the maximum time that the measurement waits for a trigger event before it stops in remote control mode or indicates a trigger timeout in manual operation mode.

**Parameters:**

<TrigTimeOut>      numeric | ON | OFF  
                          Range:      0.01 s to 300 s  
                          \*RST:      1 s  
                          Default unit: s  
                          Additional values: OFF | ON (disables | enables the timeout)

**Example:**      See [Configuring the trigger system](#)

**Firmware/software:** V1.0.10.50  
                          V2.1.30: reset value changed

**Manual operation:** See ["Trigger Timeout"](#) on page 304

**TRIGger:WLAN:MEAS<i>:MEValuation:MGAP <MinTrigGap>**

Sets a minimum time during which the IF signal must be below the trigger threshold before the trigger is armed so that an IF power trigger event can be generated.

**Parameters:**

<MinTrigGap>      numeric  
                          Range:      5 µs to 10 ms  
                          \*RST:      50 µs  
                          Default unit: s

**Example:**      See [Configuring the trigger system](#)

**Firmware/software:** V1.0.10.50

**Manual operation:** See ["Trigger IF Min Gap"](#) on page 304

### 3.5.4 Measurement limit settings

The following commands configure the measurement limits.

|                                                                  |     |
|------------------------------------------------------------------|-----|
| • <a href="#">Power vs. time limits</a> .....                    | 402 |
| • <a href="#">Modulation limits: 802.11b/g DSSS</a> .....        | 404 |
| • <a href="#">Modulation limits: 802.11a/g OFDM</a> .....        | 406 |
| • <a href="#">Modulation limits: 802.11p OFDM</a> .....          | 409 |
| • <a href="#">Modulation limits: 802.11n SISO</a> .....          | 412 |
| • <a href="#">Modulation limits: 802.11ac</a> .....              | 415 |
| • <a href="#">Modulation limits: 802.11ax</a> .....              | 420 |
| • <a href="#">Spectrum flatness limits: 802.11a/g OFDM</a> ..... | 429 |
| • <a href="#">Spectrum flatness limits: 802.11p OFDM</a> .....   | 430 |
| • <a href="#">Spectrum flatness limits: 802.11n</a> .....        | 432 |
| • <a href="#">Spectrum flatness limits: 802.11ac</a> .....       | 433 |
| • <a href="#">Spectrum flatness limits: 802.11ax</a> .....       | 435 |

|                                                      |     |
|------------------------------------------------------|-----|
| • Transmit spectrum mask limits: 802.11b/g DSSS..... | 437 |
| • Transmit spectrum mask limits: 802.11a/g OFDM..... | 438 |
| • Transmit spectrum mask limits: 802.11p OFDM.....   | 439 |
| • Transmit spectrum mask limits: 802.11n.....        | 449 |
| • Transmit spectrum mask limits: 802.11ac.....       | 452 |
| • Transmit spectrum mask limits: 802.11ax.....       | 455 |

### 3.5.4.1 Power vs. time limits

The following commands define limits for results characterizing the pulse shape of DSSS signals and the timing error for OFDM standards.

|                                                                   |     |
|-------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:REDGe:ENABLE..... | 402 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:REDGe.....        | 402 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:FEDGE:ENABLE..... | 403 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:FEDGE.....        | 403 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:TERRor.....       | 403 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:TEDistrib.....    | 403 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:REDGe:ENABLE**  
<RisingEnable>

Enables/disables the upper limit for the rise time (transmit power-on ramp) of a DSSS signal.

**Parameters:**

<RisingEnable>      OFF | ON  
                             \*RST:      ON

**Example:**              See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V2.1.10

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:REDGe** <RisingLimit>

Sets the upper limit for the rise time (transmit power-on ramp) of a DSSS signal.

**Parameters:**

<RisingLimit>      numeric | ON | OFF  
                             Range:      0 s to 5E-6 s  
                             Increment: 1E-8 s  
                             \*RST:      2E-6 s  
                             Additional parameters: OFF | ON (disables the limit check | enables the limit check using the previous limit values)

**Example:**              See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V2.1.10, V3.0.11: finer granularity

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:FEDGE:ENABLE**  
 <FallEnable>

Enables/disables the upper limit for the fall time (transmit power-down ramp) of a DSSS signal.

**Parameters:**

<FallEnable>            OFF | ON  
                              \*RST:        ON

**Example:**                See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V2.1.10

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:FEDGE <FallingLimit>**

Sets the upper limit for the fall time (transmit power-down ramp) of a DSSS signal.

**Parameters:**

<FallingLimit>          numeric | ON | OFF  
                              Range:        0 s to 5E-6 s  
                              Increment: 1E-8 s  
                              \*RST:        2E-6 s  
                              Additional parameters: OFF | ON (disables the limit check | enables the limit check using the previous limit values)

**Example:**                See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V2.1.10, V3.0.11: finer granularity

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:TERRor <TimingError>**

Sets the upper limit for timing error for OFDM standards.

**Parameters:**

<TimingError>          numeric | ON | OFF  
                              Range:        0 s to 100E-6 s  
                              Additional parameters: OFF | ON (disables the limit check | enables the limit check using the previous limit values)

**Example:**                See [Specifying power vs. time limits](#)

**Firmware/software:** V3.7.20

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:PVTime:TEDistrib**  
 <TE\_Percentage>

Configure the limit of timing error distribution for PvT measurements for OFDM standards.

Exceeding the limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

**Parameters:**

<TE\_Percentage> numeric | ON | OFF  
 Default unit: %  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying power vs. time limits](#)

**Firmware/software:** V3.7.50

**3.5.4.2 Modulation limits: 802.11b/g DSSS**

The following commands define limits for results characterizing the modulation accuracy of DSSS signals.

CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:EVMPeak..... 404  
 CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:EVMRms..... 404  
 CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:IQOffset..... 405  
 CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:CFERror..... 405  
 CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:CCERror..... 405

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:EVMPeak**  
 <EVMPeak>

Defines and activates upper limits for the error vector magnitude (EVM) peak values of the data carriers (transmission scheme DSSS).

**Parameters:**

<EVMPeak> numeric | ON | OFF  
 Range: 0 % to 100 %  
 \*RST: 35 %  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V1.0.10.1

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:DSSS:EVMRms**  
 <EVMRMS>

Defines and activates upper limits for the error vector magnitude (EVM) RMS values of the data carriers (transmission scheme DSSS).

**Parameters:**

<EVMRMS> numeric | ON | OFF  
 Range: 0 % to 100 %  
 \*RST: 35 %  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying limits \(DSSS\)](#)



**Firmware/software:** V1.0.10.1

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:DSSS:IQOffset**  
 <IQOffset>

Defines and activates an upper limit for the I/Q origin offset (transmission scheme DSSS).

**Parameters:**

<IQOffset>            numeric | ON | OFF  
                          Range:        -100 dB to 0 dB  
                          \*RST:        -15 dB  
                          Default unit: dB  
                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**            See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V1.0.10.1

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:DSSS:CFError**  
 <FreqError>

Defines and activates an upper limit for the center frequency error (transmission scheme DSSS).

**Parameters:**

<FreqError>            numeric | ON | OFF  
                          Range:        0 Hz to 100E+6 Hz  
                          \*RST:        60000 Hz  
                          Default unit: Hz  
                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**            See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V1.0.10.1, V3.7.20: range extended

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:DSSS:CCError**  
 <ClockError>

Defines and activates an upper limit for the chip clock error (transmission scheme DSSS).

**Parameters:**

<ClockError>            numeric | ON | OFF  
                          Range:        0 ppm to 100 ppm  
                          \*RST:        25 ppm  
                          Default unit: ppm  
                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V1.0.10.1, V3.7.20: range extended

### 3.5.4.3 Modulation limits: 802.11a/g OFDM

The following commands define limits for results characterizing the modulation accuracy. The limits apply to 802.11a/g OFDM signals.

|                                                                         |     |
|-------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:LOFDm:EVM.....      | 406 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:LOFDm:EVMPilot..... | 407 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:LOFDm:IQOFset.....  | 408 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:LOFDm:CFERror.....  | 408 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:LOFDm:SCERror.....  | 408 |

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:LOFDm:EVM

<EVM6M>, <EVM9M>, <EVM12M>, <EVM18M>, <EVM24M>, <EVM36M>,  
<EVM48M>, <EVM54M>

Defines and activates upper limits for the error vector magnitude (EVM) of the data carriers (802.11a/g, OFDM).

#### Parameters:

|          |                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVM6M>  | numeric   ON   OFF<br>Limit for data rate 6 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -5 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM9M>  | numeric   ON   OFF<br>Limit for data rate 9 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -8 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM12M> | numeric   ON   OFF<br>Limit for data rate 12 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -10 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |

|          |                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVM18M> | numeric   ON   OFF<br>Limit for data rate 18 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -13 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVM24M> | numeric   ON   OFF<br>Limit for data rate 24 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -16 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVM36M> | numeric   ON   OFF<br>Limit for data rate 36 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -19 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVM48M> | numeric   ON   OFF<br>Limit for data rate 48 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -22 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVM54M> | numeric   ON   OFF<br>Limit for data rate 54 Mbps<br>Range: -100 dB to 0 dB<br>*RST: -25 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

---

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:LOFDm:EVMPilot**  
 <EVMPilot>

Defines and activates an upper limit for the error vector magnitude (EVM) of the pilot carriers (802.11a/g, OFDM).

**Parameters:**

<EVMPilot> numeric | ON | OFF  
 Range: -100 dB to 0 dB  
 \*RST: -8 dB  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:LOFDm:IQOffset**  
 <IQOffset>

Defines and activates an upper limit for the I/Q origin offset (802.11a/g, OFDM).

**Parameters:**

<IQOffset> numeric | ON | OFF  
 Range: -100 dB to 0 dB  
 \*RST: -15 dB  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:LOFDm:CFERror**  
 <CenterFreqError>

Defines and activates an upper limit for the center frequency error (802.11a/g, OFDM).

**Parameters:**

<CenterFreqError> numeric | ON | OFF  
 Range: 0 Hz to 100E+6 Hz  
 \*RST: 60E+3 Hz  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.20: range extended

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:LOFDm:SCERror**  
 <ClockError>

Defines and activates an upper limit for the symbol clock error (802.11a/g, OFDM).

**Parameters:**

<ClockError>            numeric | ON | OFF  
                              Range:        0 ppm to 100 ppm  
                              \*RST:        25 ppm  
                              Default unit: ppm  
                              Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**

See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.20: range extended

**3.5.4.4 Modulation limits: 802.11p OFDM**

The following commands define limits for results characterizing the modulation accuracy. The limits apply to 802.11p OFDM signals.

[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:EVM](#).....409  
[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:EVMPilot](#)..... 410  
[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:IQOffset](#).....411  
[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:CFError](#)..... 411  
[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:SCError](#).....411

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:EVM**

<BPSK12>, <BPSK34>, <QPSK12>, <QPSK34>, <Q16AM12>, <Q16AM34>,  
 <Q64AM23>, <Q64AM34>

Defines and activates upper limits for the error vector magnitude (EVM) of the data carriers in 802.11p signals.

**Parameters:**

<BPSK12>            numeric | ON | OFF  
                              Limit for data rate BPSK modulation and coding rate 1/2  
                              Range:        -100 dB to 0 dB  
                              \*RST:        -5 dB  
                              Additional parameters: OFF | ON (disables | enables the limit check)

<BPSK34>            numeric | ON | OFF  
                              Limit for data rate BPSK modulation and coding rate 3/4  
                              Range:        -100 dB to 0 dB  
                              \*RST:        -8 dB  
                              Additional parameters: OFF | ON (disables | enables the limit check)

<QPSK12>            numeric | ON | OFF  
                              Limit for data rate QPSK modulation and coding rate 1/2  
                              Range:        -100 dB to 0 dB  
                              \*RST:        -10 dB  
                              Additional parameters: OFF | ON (disables | enables the limit check)

|           |                                                                                                                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <QPSK34>  | numeric   ON   OFF<br>Limit for data rate QPSK modulation and coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -13 dB<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <Q16AM12> | numeric   ON   OFF<br>Limit for data rate 16-QAM modulation and coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -16 dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <Q16AM34> | numeric   ON   OFF<br>Limit for data rate 16-QAM modulation and coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -19 dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <Q64AM23> | numeric   ON   OFF<br>Limit for data rate 64-QAM modulation and coding rate 2/3<br>Additional parameters: OFF   ON (disables   enables the limit check)                                           |
| <Q64AM34> | numeric   ON   OFF<br>Limit for data rate 64-QAM modulation and coding rate 3/4<br>Additional parameters: OFF   ON (disables   enables the limit check)                                           |

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.2.40

**Options:** R&S CMW-KM655

---

#### **CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:POFDm:EVMPilot** <EVMPilot>

Defines and activates an upper limit for the error vector magnitude (EVM) of 802.11p pilot carriers.

#### **Parameters:**

|            |                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <EVMPilot> | numeric   ON   OFF<br>Range: -100 dB to 0 dB<br>*RST: -8 dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.2.40

**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:POFDm:IQOffset**  
<IQOffset>

Defines and activates an upper limit for the I/Q origin offset of 802.11p signals.

**Parameters:**

<IQOffset>            numeric | ON | OFF  
                          Range:        -100 dB to 0 dB  
                          \*RST:        -15 dB  
                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.2.40

**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:POFDm:CFERror**  
<CenterFreqError>

Defines and activates an upper limit for the center frequency error (802.11p).

**Parameters:**

<CenterFreqError>   numeric | ON | OFF  
                          Range:        0 Hz to 100E+6 Hz  
                          \*RST:        60E+3 Hz  
                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.2.40, V3.7.20: range extended

**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:POFDm:SCERror**  
<ClockError>

Defines and activates an upper limit for the symbol clock error (802.11p).

**Parameters:**

<ClockError>            numeric | ON | OFF  
                          Range:        0 ppm to 100 ppm  
                          \*RST:        25 ppm  
                          Default unit: ppm  
                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.2.40, V3.7.20: range extended

**Options:** R&S CMW-KM655

### 3.5.4.5 Modulation limits: 802.11n SISO

The following commands define limits for results characterizing the modulation accuracy. The limits apply to 802.11n SISO measurements.

|                                                                              |     |
|------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:EVM.....          | 412 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:EVMPilot.....     | 414 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:IQOFset:BW<BW>... | 414 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:CFERror.....      | 415 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:SCERror.....      | 415 |

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:EVM

<EVMBR12>, <EVMQR12>, <EVMQR34>, <EVMQ1M12>, <EVMQ1M34>,  
<EVMQ6M12>, <EVMQ6M34>, <EVMQ6M56>

Defines and activates upper limits for the error vector magnitude (EVM) of the data carriers (802.11n).

#### Parameters:

|           |                                                                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVMBR12> | numeric   ON   OFF<br>Limits for BPSK, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -5 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVMQR12> | numeric   ON   OFF<br>Limits for QPSK, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -10 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVMQR34> | numeric   ON   OFF<br>Limits for QPSK, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -13 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |



|                           |                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVMQ1M12>                | numeric   ON   OFF<br>Limits for 16-QAM, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -16 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVMQ1M34>                | numeric   ON   OFF<br>Limits for 16-QAM, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -19 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVMQ6M12>                | numeric   ON   OFF<br>Limits for 64-QAM, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -22 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVMQ6M34>                | numeric   ON   OFF<br>Limits for 64-QAM, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -25 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVMQ6M56>                | numeric   ON   OFF<br>Limits for 64-QAM, coding rate 5/6<br>Range: -100 dB to 0 dB<br>*RST: -27 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <b>Example:</b>           | See <a href="#">Specifying modulation limits (OFDM)</a>                                                                                                                                        |
| <b>Firmware/software:</b> | V3.5.121                                                                                                                                                                                       |
| <b>Options:</b>           | R&S CMW-KM651                                                                                                                                                                                  |

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:EVMPilot**  
**<EVMPilot>**

Defines and activates an upper limit for the error vector magnitude (EVM) of the pilot carriers (802.11n).

**Parameters:**

**<EVMPilot>** numeric | ON | OFF  
 Range: -100 dB to 0 dB  
 \*RST: -8 dB  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

**Options:** R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:IQOffset:**  
**BW<BW> <OffsetValue>**

Defines and activates an upper limit for the I/Q origin offset, for 802.11n and channel bandwidth <BW>.

**Suffix:**

**<BW>** 5, 10, 20, 40  
 Channel bandwidth in MHz

**Parameters:**

**<OffsetValue>** numeric | ON | OFF  
 Range: -100 dB to 0 dB  
 \*RST: see table below  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

| <BW> | *RST <OffsetValue> |
|------|--------------------|
| 5    | -15 dB             |
| 10   | -15 dB             |
| 20   | -15 dB             |
| 40   | -20 dB             |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:CFERror**  
**<CenterFreqError>**

Defines and activates an upper limit for the center frequency error (802.11n).

**Parameters:**

<CenterFreqError>    numeric | ON | OFF  
                           Range:        0 Hz to 100E+6 Hz  
                           \*RST:        60E+3 Hz  
                           Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.20: range extended

**Options:**                R&S CMW-KM651

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HTOFdm:SCERror**  
**<ClockError>**

Defines and activates an upper limit for the symbol clock error (802.11n).

**Parameters:**

<ClockError>            numeric | ON | OFF  
                           Range:        0 ppm to 100 ppm  
                           \*RST:        25 ppm  
                           Default unit: ppm  
                           Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.20: range extended

**Options:**                R&S CMW-KM651

### 3.5.4.6 Modulation limits: 802.11ac

The following commands define limits for results characterizing the modulation accuracy. The limits apply to 802.11ac measurements.

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:VHTofdm:EVMAIrr.....      | 416 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:VHTofdm:EVMPilot.....     | 418 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:VHTofdm:IQOFset:BW<BW>... | 418 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:VHTofdm:CFERror.....      | 419 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:VHTofdm:SCERror.....      | 419 |

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:VHTofdm:EVMall**

<EVMBR12>, <EVMQR12>, <EVMQR34>, <EVM16QAM12>,  
 <EVM16QAM34>, <EVM64QAM12>, <EVM64QAM34>, <EVM64QAM56>,  
 <EVM256QAM34>, <EVM256QAM56>[, <EVM1024QAM34>,  
 <EVM1024QAM56>]

Defines and activates upper limits for the error vector magnitude (EVM) of 802.11ac data carriers.

**Parameters:**

|              |                                                                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVMBR12>    | numeric   ON   OFF<br>Limits for BPSK, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -5 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)    |
| <EVMQR12>    | numeric   ON   OFF<br>Limits for QPSK, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -10 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVMQR34>    | numeric   ON   OFF<br>Limits for QPSK, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -13 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM16QAM12> | numeric   ON   OFF<br>Limits for 16-QAM, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -16 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |
| <EVM16QAM34> | numeric   ON   OFF<br>Limits for 16-QAM, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -19 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |

|                |                                                                                                                                                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVM64QAM12>   | <p>numeric   ON   OFF</p> <p>Limits for 64-QAM, coding rate 1/2</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -22 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>   |
| <EVM64QAM34>   | <p>numeric   ON   OFF</p> <p>Limits for 64-QAM, coding rate 3/4</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -25 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>   |
| <EVM64QAM56>   | <p>numeric   ON   OFF</p> <p>Limits for 64-QAM, coding rate 5/6</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -27 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>   |
| <EVM256QAM34>  | <p>numeric   ON   OFF</p> <p>Limits for 256-QAM, coding rate 3/4</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -30 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>  |
| <EVM256QAM56>  | <p>numeric   ON   OFF</p> <p>Limits for 256-QAM, coding rate 5/6</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -32 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>  |
| <EVM1024QAM34> | <p>numeric   ON   OFF</p> <p>Limits for 1024-QAM, coding rate 3/4</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -35 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p> |

<EVM1024QAM56> numeric | ON | OFF  
 Limits for 1024-QAM, coding rate 5/6  
 Range: -100 dB to 0 dB  
 \*RST: -35 dB  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

**Options:** R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:VHTofdm:EVMPilot**  
 <EVMPilot>

Defines and activates an upper limit for the error vector magnitude (EVM) of the pilot carriers in 802.11ac signals.

**Parameters:**

<EVMPilot> numeric | ON | OFF  
 Range: -100 dB to 0 dB  
 \*RST: -8 dB  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

**Options:** R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:VHTofdm:IQOffset:**  
**BW<BW>** <OffsetValue\_rel>[, <OffsetValue\_abs>]

Defines and activates upper limits for the I/Q origin offset, for 802.11ac and channel bandwidth <BW>.

**Suffix:**

<BW> 5, 10, 20, 40, 80, 160, 8080  
 Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<OffsetValue\_rel> numeric | ON | OFF  
 Relative limit  
 Range: -100 dB to 0 dB  
 \*RST: see table below  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

**<OffsetValue\_abs>** numeric | ON | OFF  
 Absolute limit, only for <BW>=8080  
 Range: -100 dBm to 0 dBm  
 \*RST: -20 dBm, OFF  
 Default unit: dBm  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.31: added bandwidth 5 and 10, V3.7.60: reset values changed, absolute limit can be enabled or disabled  
 V3.7.70: absolute limit implemented

**Options:** R&S CMW-KM656

| <BW> | *RST <OffsetValue_rel> |
|------|------------------------|
| 5    | -17.48 dB              |
| 10   | -17.48 dB              |
| 20   | -17.48 dB              |
| 40   | -20.57 dB              |
| 80   | -23.84 dB              |
| 8080 | -23.84 dB              |
| 160  | -26.85 dB              |

---

#### CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:VHTofdm:CFERror <CenterFreqError>

Defines and activates an upper limit for the center frequency error in 802.11ac signals.

##### Parameters:

**<CenterFreqError>** numeric | ON | OFF  
 Note that the reset value is identical for all standards  
 Range: 0 Hz to 100E+6 Hz  
 \*RST: 100E+3 Hz  
 Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121, V3.7.20: range extended

**Options:** R&S CMW-KM656

---

#### CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:VHTofdm:SCERror <ClockError>

Defines and activates an upper limit for the symbol clock error in 802.11ac signals.

**Parameters:**

<ClockError>            numeric | ON | OFF  
                              Range:        0 ppm to 100 ppm  
                              \*RST:        20 ppm  
                              Default unit: ppm  
                              Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.5.121

**Options:**                R&S CMW-KM656

**3.5.4.7 Modulation limits: 802.11ax**

The following commands define limits for results characterizing the modulation accuracy. The limits apply to 802.11ax measurements.

|                                                                                 |     |
|---------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:CFERror.....         | 420 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:CFDistrib.....       | 421 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMall.....          | 421 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMall:TBCodeRate..  | 424 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMall:TBHigh.....   | 425 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMall:TBLow.....    | 426 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMPilot.....        | 427 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMPilot:TBHigh..... | 427 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMPilot:TBLow.....  | 427 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:IQOffset:BW<BW>...   | 428 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:SCERror.....         | 428 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:UTEPower.....                          | 429 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:UTERror.....                           | 429 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:CFERror**  
                              <CenterFreqError>

Defines and activates an upper limit for the center frequency error in 802.11ax signals.

**Parameters:**

<CenterFreqError>    numeric | ON | OFF  
                              Range:        0 Hz to 100E+6 Hz  
                              \*RST:        100E+3 Hz  
                              Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.20

**Options:**                R&S CMW-KM657



---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:CFDistrib**  
 <CFO\_Percentage>, <CFO\_Frequency>

Configure the limit of carrier frequency offset (CFO) error distribution for HE modulation measurements.

Exceeding the limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

**Parameters:**

<CFO\_Percentage> numeric | ON | OFF  
 Upper limit for the tolerated CFO errors (CFO exceeding the specified <CFO\_Frequency>)  
 Default unit: %  
 Additional parameters: OFF | ON (disables | enables the limit check)

<CFO\_Frequency> numeric  
 Border value defining CFO error  
 Default unit: Hz

**Example:** See [Specifying power vs. time limits](#)

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMall**  
 <EVMBR14>, <EVMBR12>, <EVMQR14>, <EVMQR12>, <EVMQR34>,  
 <EVM16QAM14>, <EVM16QAM38>, <EVM16QAM12>, <EVM16QAM34>,  
 <EVM64QAM23>, <EVM64QAM34>, <EVM64QAM56>, <EVM256QAM34>,  
 <EVM256QAM56>[, <EVM1024QAM34>, <EVM1024QAM56>]

Defines and activates upper limits for the error vector magnitude (EVM) of 802.11ax data carriers.

**Parameters:**

<EVMBR14> numeric | ON | OFF  
 Limits for BPSK, coding rate 1/4, dual carrier modulation (DCM)  
 Range: -100 dB to 0 dB  
 \*RST: -5 dB  
 Default unit: dB  
 Additional parameters: OFF | ON (disables | enables the limit check)

|              |                                                                                                                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVMBR12>    | <p>numeric   ON   OFF</p> <p>Limits for BPSK, coding rate 1/2</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -5 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>        |
| <EVMQR14>    | <p>numeric   ON   OFF</p> <p>Limits for QPSK, coding rate 1/4 DCM</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -5 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>    |
| <EVMQR12>    | <p>numeric   ON   OFF</p> <p>Limits for QPSK, coding rate 1/2</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -10 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>       |
| <EVMQR34>    | <p>numeric   ON   OFF</p> <p>Limits for QPSK, coding rate 3/4</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -13 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p>       |
| <EVM16QAM14> | <p>numeric   ON   OFF</p> <p>Limits for 16-QAM, coding rate 1/4 DCM</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -10 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p> |
| <EVM16QAM38> | <p>numeric   ON   OFF</p> <p>Limits for 16-QAM, coding rate 3/8 DCM</p> <p>Range: -100 dB to 0 dB</p> <p>*RST: -13 dB</p> <p>Default unit: dB</p> <p>Additional parameters: OFF   ON (disables   enables the limit check)</p> |

|               |                                                                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <EVM16QAM12>  | numeric   ON   OFF<br>Limits for 16-QAM, coding rate 1/2<br>Range: -100 dB to 0 dB<br>*RST: -16 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM16QAM34>  | numeric   ON   OFF<br>Limits for 16-QAM, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -19 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM64QAM23>  | numeric   ON   OFF<br>Limits for 64-QAM, coding rate 2/3<br>Range: -100 dB to 0 dB<br>*RST: -22 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM64QAM34>  | numeric   ON   OFF<br>Limits for 64-QAM, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -25 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM64QAM56>  | numeric   ON   OFF<br>Limits for 64-QAM, coding rate 5/6<br>Range: -100 dB to 0 dB<br>*RST: -28 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM256QAM34> | numeric   ON   OFF<br>Limits for 256-QAM, coding rate 3/4<br>Range: -100 dB to 0 dB<br>*RST: -30 dB<br>Default unit: dB<br>Additional parameters: OFF   ON (disables   enables the limit check) |

- <EVM256QAM56> numeric | ON | OFF  
Limits for 256-QAM, coding rate 5/6  
Range: -100 dB to 0 dB  
\*RST: -32 dB  
Default unit: dB  
Additional parameters: OFF | ON (disables | enables the limit check)
- <EVM1024QAM34> numeric | ON | OFF  
Limits for 1024-QAM, coding rate 3/4  
Range: -100 dB to 0 dB  
\*RST: -35 dB  
Default unit: dB  
Additional parameters: OFF | ON (disables | enables the limit check)
- <EVM1024QAM56> numeric | ON | OFF  
Limits for 1024-QAM, coding rate 5/6  
Range: -100 dB to 0 dB  
\*RST: -35 dB  
Default unit: dB  
Additional parameters: OFF | ON (disables | enables the limit check)
- Example:** See [Specifying modulation limits \(OFDM\)](#)
- Firmware/software:** V3.7.20
- Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMall:  
TBCoderate** <CR\_BPSK>, <CR\_QPSK>, <CR\_16QAM>, <CR\_64QAM>,  
<CR\_256QAM>, <CR\_1024QAM>

Specifies the coding rate of HE TB PPDU per modulation type, used for the calculation of unused tone error limit line.

**Parameters:**

- <CR\_BPSK> CR14dcm | CR38dcm | CR12 | CR23 | CR34 | CR56  
Coding rate for BPSK modulation type  
**CR14dcm:** 1/4 DCM (coding rate 1/2 with DCM)  
**CR38dcm:** 3/8 DCM (coding rate 3/4 with DCM)  
**CR12:** 1/2 (coding rate 1/2 without DCM)  
**CR23:** 2/3 (coding rate 2/3 without DCM)  
**CR34:** 3/4 (coding rate 3/4 without DCM)  
**CR56:** 5/6 (coding rate 5/6 without DCM)
- <CR\_QPSK> CR14dcm | CR38dcm | CR12 | CR23 | CR34 | CR56
- <CR\_16QAM> CR14dcm | CR38dcm | CR12 | CR23 | CR34 | CR56
- <CR\_64QAM> CR14dcm | CR38dcm | CR12 | CR23 | CR34 | CR56

<CR\_256QAM> CR14dcm | CR38dcm | CR12 | CR23 | CR34 | CR56

<CR\_1024QAM> CR14dcm | CR38dcm | CR12 | CR23 | CR34 | CR56

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KM657

**Manual operation:** See ["Coding Rate"](#) on page 296

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMail:  
TBHigh** <EVM\_BPSK>, <EVM\_QPSK>, <EVM\_16QAM>, <EVM\_64QAM>,  
<EVM\_256QAM>, <EVM\_1024QAM>

Sets EVM limits for HE TB PPDU when transmit power is larger than the maximum power of MCS 7. The default values are in line with standard IEEE Std 802.11ax-2021, table 27-49 Allowed relative constellation error versus constellation size and coding rate.

**Parameters:**

|               |                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| <EVM_BPSK>    | numeric   ON   OFF<br>EVM limit for BPSK<br>Additional parameters: OFF   ON (disables   enables the limit check)     |
| <EVM_QPSK>    | numeric   ON   OFF<br>EVM limit for QPSK<br>Additional parameters: OFF   ON (disables   enables the limit check)     |
| <EVM_16QAM>   | numeric   ON   OFF<br>EVM limit for 16-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM_64QAM>   | numeric   ON   OFF<br>EVM limit for 64-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM_256QAM>  | numeric   ON   OFF<br>EVM limit for 256-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM_1024QAM> | numeric   ON   OFF<br>EVM limit for 1024-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check) |

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KM657

---

**CONFIGure:WLAN:MEAS<i>:MEvaluation:LIMit:MODulation:HEOFdm:EVMall:TBLow** <EVM\_BPSK>, <EVM\_QPSK>, <EVM\_16QAM>, <EVM\_64QAM>, <EVM\_256QAM>, <EVM\_1024QAM>

Sets EVM limits for HE TB PPDU when transmit power is less than or equal to the maximum power of MCS 7. The default values are in line with standard IEEE Std 802.11ax-2021, table 27-49 Allowed relative constellation error versus constellation size and coding rate.

**Parameters:**

|               |                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| <EVM_BPSK>    | numeric   ON   OFF<br>EVM limit for BPSK<br>Additional parameters: OFF   ON (disables   enables the limit check)     |
| <EVM_QPSK>    | numeric   ON   OFF<br>EVM limit for QPSK<br>Additional parameters: OFF   ON (disables   enables the limit check)     |
| <EVM_16QAM>   | numeric   ON   OFF<br>EVM limit for 16-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM_64QAM>   | numeric   ON   OFF<br>EVM limit for 64-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check)   |
| <EVM_256QAM>  | numeric   ON   OFF<br>EVM limit for 256-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check)  |
| <EVM_1024QAM> | numeric   ON   OFF<br>EVM limit for 1024-QAM<br>Additional parameters: OFF   ON (disables   enables the limit check) |

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMPilot**  
**<EVMPilot>**

Defines and activates an upper limit for the error vector magnitude (EVM) of the pilot carriers in 802.11ax signals.

**Parameters:**

**<EVMPilot>**                      numeric | ON | OFF  
                                          Range:        -100 dB to 0 dB  
                                          \*RST:        -8 dB  
                                          Default unit: dB  
                                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                      See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.20

**Options:**                      R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMPilot:**  
**TBHigh <EVMPilot>**

Sets EVM limits for a pilot subcarrier in 802.11ax signals, when transmit power is larger than the maximum power of MCS 7.

**Parameters:**

**<EVMPilot>**                      numeric | ON | OFF  
                                          \*RST:        -13 dB  
                                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                      See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:**                      R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:EVMPilot:**  
**TBLow <EVMPilot>**

Sets EVM limits for a pilot subcarrier in 802.11ax signals, when transmit power is less than or equal to the maximum power of MCS 7.

**Parameters:**

**<EVMPilot>**                      numeric | ON | OFF  
                                          Range:        -100 dB to 0 dB  
                                          \*RST:        -27 dB  
                                          Additional parameters: OFF | ON (disables | enables the limit check)

**Example:**                      See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:IQOffset:  
BW<BW> <OffsetValue\_rel>[, <OffsetValue\_abs>]**

Defines and activates upper limits for the I/Q origin offset for 802.11ax and channel bandwidth <BW>.

**Suffix:**

<BW> 5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<OffsetValue\_rel> numeric | ON | OFF  
Relative limit  
Range: -100 dB to 0 dB  
\*RST: -32 dB  
Default unit: dB  
Additional parameters: OFF | ON (disables | enables the limit check)

<OffsetValue\_abs> numeric | ON | OFF  
Absolute limit  
Range: -100 dBm to 0 dBm  
\*RST: -20 dBm, OFF  
Default unit: dBm  
Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.20, V3.7.60: added <BW> 8080, reset values changed, absolute values for all <BW> added  
V3.7.70: absolute limit implemented

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:MODulation:HEOFdm:SCERror  
<ClockError>**

Defines and activates an upper limit for the symbol clock error in 802.11ax signals.

**Parameters:**

<ClockError> numeric | ON | OFF  
Range: 0 ppm to 100 ppm  
\*RST: 20 ppm  
Default unit: ppm  
Additional parameters: OFF | ON (disables | enables the limit check)

**Example:** See [Specifying modulation limits \(OFDM\)](#)



**Firmware/software:** V3.7.20

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:UTEPower** <UTEPower>

Sets the type of limits to be applied for the unused tone error of HE TB PPDU.

**Parameters:**

<UTEPower> HIGH | LOW

**HIGH:** used limits are according to the transmit power larger than the maximum power of MCS 7

**LOW:** used limits are according to the transmit power less than or equal to the maximum power of MCS 7

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:UTERror** <UTELimits>

Enables/ disables an upper limit for the unused tone error of HE TB PPDU.

**Parameters:**

<UTELimits> ON | OFF

**Example:** See [Specifying modulation limits \(OFDM\)](#)

**Firmware/software:** V3.7.30

**Options:** R&S CMW-KM657

### 3.5.4.8 Spectrum flatness limits: 802.11a/g OFDM

The following commands define limits for results characterizing the spectrum flatness. The limits apply to 802.11a/g OFDM signals.

[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:LOFDm:ENABLE](#)..... 429

[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:LOFDm:UPPer](#)..... 430

[CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:LOFDm:LOWer](#)..... 430

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:LOFDm:ENABLE**  
<Enable>

Enables or disables the spectrum flatness limit check for 802.11a/g OFDM signals.

**Parameters:**

<Enable> ON | OFF

\*RST: ON

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:LOFDm:UPPer**  
 <Upper>

Defines an upper limit for the spectrum flatness of 802.11a/g OFDM signals.

The upper limit must be larger than the lower limits.

**Parameters:**

<Upper>                      numeric  
                                  Range:        -4 dB to 20 dB  
                                  \*RST:        4 dB

**Example:**                      See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:LOFDm:LOWer**  
 <Center>, <Side>

Defines lower limits for the spectrum flatness of the center subcarriers and the side subcarriers of 802.11a/g OFDM signals.

The lower limits must be smaller than the upper limit.

**Parameters:**

<Center>                      numeric  
                                  Range:        -20 dB to 4 dB  
                                  \*RST:        -4 dB

<Side>                        numeric  
                                  Range:        -20 dB to 4 dB  
                                  \*RST:        -6 dB

**Example:**                      See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10

### 3.5.4.9 Spectrum flatness limits: 802.11p OFDM

The following commands define limits for results characterizing the spectrum flatness. The limits apply to 802.11p OFDM signals.

CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:POFDm:BW<bandwidth>:  
     ENABLE.....431  
 CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:POFDm:BW<bandwidth>:UPPer 431  
 CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:POFDm:BW<bandwidth>:  
     LOWer.....431

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:POFDm:  
BW<bandwidth>:ENABle <Enable>**

Enables or disables the spectrum flatness limit check for 802.11p OFDM signals with the specified <bandwidth>.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<Enable>                ON | OFF  
\*RST:                    ON

**Example:**              See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10

**Options:**                R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:POFDm:  
BW<bandwidth>:UPPer <Upper>**

Defines an upper limit for the spectrum flatness of 802.11p OFDM signals with the specified <bandwidth>.

The upper limit must be larger than the lower limits.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<Upper>                 numeric  
Range:                 -4 dB to 20 dB  
\*RST:                   4 dB

**Example:**              See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10

**Options:**                R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:POFDm:  
BW<bandwidth>:LOWer <Center>, <Side>**

Defines lower limits for the spectrum flatness of the center subcarriers and the side subcarriers of 802.11p OFDM signals with the specified <bandwidth>.

The lower limits must be smaller than the upper limit.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

|          |                       |
|----------|-----------------------|
| <Center> | numeric               |
|          | Range: -20 dB to 4 dB |
|          | *RST: -4 dB           |
| <Side>   | numeric               |
|          | Range: -20 dB to 4 dB |
|          | *RST: -6 dB           |

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM656

### 3.5.4.10 Spectrum flatness limits: 802.11n

The following commands define limits for results characterizing the spectrum flatness. The limits apply to 802.11n measurements.

|                                                                                     |     |
|-------------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HTOFdm:BW<bandwidth>:ENABLE..... | 432 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HTOFdm:BW<bandwidth>:UPPer.....  | 432 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HTOFdm:BW<bandwidth>:LOWer.....  | 433 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HTOFdm:BW<bandwidth>:ENABLE <Enable>**

Enables or disables the spectrum flatness limit check for 802.11n signals with the specified <bandwidth>.

**Suffix:**

|             |                          |
|-------------|--------------------------|
| <bandwidth> | 5, 10, 20, 40            |
|             | Channel bandwidth in MHz |

**Parameters:**

|          |          |
|----------|----------|
| <Enable> | ON   OFF |
| *RST:    | ON       |

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HTOFdm:BW<bandwidth>:UPPer <Upper>**

Defines an upper limit for the spectrum flatness of 802.11n signals with the specified <bandwidth>.

The upper limit must be larger than the lower limits.

**Suffix:**

<bandwidth> 5, 10, 20, 40  
Channel bandwidth in MHz

**Parameters:**

<Upper> numeric  
Range: -4 dB to 20 dB  
\*RST: 4 dB  
Default unit: dB

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:MEvaluation:LIMit:SFLatness:HTOFdm:  
BW<bandwidth>:LOWer <Center>, <Side>**

Defines lower limits for the spectrum flatness of the center subcarriers and the side subcarriers for 802.11n signals with the specified <bandwidth>.

The lower limits must be smaller than the upper limit.

**Suffix:**

<bandwidth> 5, 10, 20, 40  
Channel bandwidth in MHz

**Parameters:**

<Center> numeric  
Range: -20 dB to 4 dB  
\*RST: -4 dB  
Default unit: dB

<Side> numeric  
Range: -20 dB to 4 dB  
\*RST: -6 dB  
Default unit: dB

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

### 3.5.4.11 Spectrum flatness limits: 802.11ac

The following commands define limits for results characterizing the spectrum flatness. The limits apply to 802.11ac measurements.

|                                                                                          |     |
|------------------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:VHTofdm:BW<bandwidth>:<br>ENABLE..... | 434 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:VHTofdm:BW<bandwidth>:<br>UPPer.....  | 434 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:VHTofdm:BW<bandwidth>:<br>LOWer.....  | 435 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:VHTofdm:  
BW<bandwidth>:ENABLE <Enable>**

Enables or disables the spectrum flatness limit check for 802.11ac signals with the specified <bandwidth>.

**Suffix:**

<bandwidth>            5, 10, 20, 40, 80, 160, 8080  
                          Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<Enable>                ON | OFF  
                          \*RST:        ON

**Example:**             See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:**              R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:VHTofdm:  
BW<bandwidth>:UPPer <Upper>**

Defines an upper limit for the spectrum flatness of 802.11ac signals with the specified <bandwidth>.

The upper limit must be larger than the lower limits.

**Suffix:**

<bandwidth>            5, 10, 20, 40, 80, 160, 8080  
                          Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<Upper>                numeric  
                          Range:        -4 dB to 20 dB  
                          \*RST:        4 dB

**Example:**             See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:**              R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:VHTofdm:  
BW<bandwidth>:LOWer <Center>, <Side>**

Defines lower limits for the spectrum flatness of the center subcarriers and the side subcarriers for 802.11ac signals with the specified <bandwidth>.

The lower limits must be smaller than the upper limit.

**Suffix:**

<bandwidth> 5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<Center> numeric  
Range: -20 dB to 4 dB  
\*RST: -4 dB

<Side> numeric  
Range: -20 dB to 4 dB  
\*RST: -6 dB

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM656

### 3.5.4.12 Spectrum flatness limits: 802.11ax

The following commands define limits for results characterizing the spectrum flatness. The limits apply to 802.11ax measurements.

|                                                                                         |     |
|-----------------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HEOFdm:BW<bandwidth>:<br>ENABLE..... | 435 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HEOFdm:BW<bandwidth>:<br>UPPer.....  | 436 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HEOFdm:BW<bandwidth>:<br>LOWer.....  | 436 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HEOFdm:  
BW<bandwidth>:ENABle <Enable>**

Enables or disables the spectrum flatness limit check for 802.11ax signals with the specified <bandwidth>.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<Enable> ON | OFF  
\*RST: ON

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HEOFdm:  
BW<bandwidth>:UPPer <Upper>**

Defines the upper limit for the spectrum flatness of 802.11ax signals with the specified <bandwidth>.

The upper limit must be larger than the lower limits.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<Upper> numeric  
Range: -4 dB to 20 dB  
\*RST: 4 dB  
Default unit: dB

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080

**Options:** R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:SFLatness:HEOFdm:  
BW<bandwidth>:LOWer <Center>, <Side>**

Defines the lower limits for the spectrum flatness of the center subcarriers and the side subcarriers for 802.11ax signals with the specified <bandwidth>.

The lower limits must be smaller than the upper limit.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<Center> numeric  
Range: -20 dB to 4 dB  
\*RST: -4 dB  
Default unit: dB

<Side> numeric  
Range: -20 dB to 4 dB  
\*RST: -6 dB  
Default unit: dB

**Example:** See [Specifying spectrum flatness limits](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080



Options: R&S CMW-KM657

### 3.5.4.13 Transmit spectrum mask limits: 802.11b/g DSSS

The following commands define the transmit spectrum mask. The limits apply to 802.11b/g DSSS signals.

|                                                                  |     |
|------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:DSSS:ENABLE..... | 437 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:DSSS:Y:AB.....   | 437 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:DSSS:Y:CD.....   | 437 |

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:DSSS:ENABLE <SpeLimEnable>

Activates or deactivates the transmit spectrum mask (transmission scheme DSSS), i.e. the limit check.

##### Parameters:

<SpeLimEnable> ON | OFF  
\*RST: ON

**Example:** See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V1.0.15.0

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:DSSS:Y:AB <YRelLevelAB>

Defines the power level of the horizontal spectrum mask line connecting point A and B, see "[Transmit spectrum mask DSSS](#)" on page 316.

##### Parameters:

<YRelLevelAB> numeric  
Range: -90 dB to 10 dB  
\*RST: -50 dB

**Example:** See [Specifying limits \(DSSS\)](#)

**Firmware/software:** V1.0.15.0

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:DSSS:Y:CD <YRelLevelCD>

Defines the power level of the horizontal spectrum mask line connecting point C and D, see "[Transmit spectrum mask DSSS](#)" on page 316.

##### Parameters:

<YRelLevelCD> numeric  
Range: -90 dB to 10 dB  
\*RST: -30 dB

**Example:** See [Specifying limits \(DSSS\)](#)

Firmware/software: V1.0.15.0

#### 3.5.4.14 Transmit spectrum mask limits: 802.11a/g OFDM

The following commands define and enable the transmit spectrum mask. The limits apply to 802.11a/g OFDM signals.

|                                                              |     |
|--------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:ENABLE | 438 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:A    | 438 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:B    | 438 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:C    | 439 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:D    | 439 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:ENABLE**  
<TSMLimEnable>

Activates or deactivates the transmit spectrum mask, i.e. the limit check (802.11a/g, OFDM).

**Parameters:**

<TSMLimEnable> ON | OFF  
\*RST: ON

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

Firmware/software: V3.7.10

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:A**  
<TSMLimYRelLevA>

Defines the Y-value for point A of the spectrum mask (802.11a/g, OFDM). See also ["Transmit spectrum mask OFDM, default masks"](#) on page 317.

**Parameters:**

<TSMLimYRelLevA> numeric  
Range: -90 dB to 10 dB  
\*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

Firmware/software: V3.7.10

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:B**  
<TSMLimYRelLevB>

Defines the Y-value for point B of the spectrum mask (802.11a/g, OFDM). See also ["Transmit spectrum mask OFDM, default masks"](#) on page 317.

**Parameters:**

<TSMLimYRelLevB> numeric  
Range: -90 dB to 10 dB  
\*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:C**  
**<TSMLimYRelLevC>**

Defines the Y-value for point C of the spectrum mask (802.11a/g, OFDM). See also ["Transmit spectrum mask OFDM, default masks"](#) on page 317.

**Parameters:**

**<TSMLimYRelLevC>** numeric  
 Range: -90 dB to 10 dB  
 \*RST: -28 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:LOFDm:Y:D**  
**<TSMLimYRelLevD>**

Defines the Y-value for point D of the spectrum mask (802.11a/g, OFDM). See also ["Transmit spectrum mask OFDM, default masks"](#) on page 317.

**Parameters:**

**<TSMLimYRelLevD>** numeric  
 Range: -90 dB to 10 dB  
 \*RST: -20 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

### 3.5.4.15 Transmit spectrum mask limits: 802.11p OFDM

The following commands define and enable the relative and absolute transmit spectrum masks for 802.11p OFDM signals.

The `CONFigure:WLAN:MEAS<i>:MEValuation:TSMask:MSElection` switch determines whether relative or absolute limits are used. The applicable relative limits are determined by the DUT's power class (`CONFigure:WLAN:MEAS<i>:ISIGnal:PClass`).

```
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:ENABle..440
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CA:Y:A...440
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CA:Y:B...441
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CA:Y:C...441
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CA:Y:D...442
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CA:Y:E...442
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CB:Y:A...442
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CB:Y:B...443
```

|                                                                                           |     |
|-------------------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CB:Y:C...             | 443 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CB:Y:D...             | 444 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:CB:Y:E...             | 444 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>UDEFined:Y:A..... | 444 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>UDEFined:Y:B..... | 445 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>UDEFined:Y:C..... | 445 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>UDEFined:Y:D..... | 446 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>UDEFined:Y:E..... | 446 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>ABSolute:Y:A..... | 446 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>ABSolute:Y:B..... | 447 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>ABSolute:Y:C..... | 447 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>ABSolute:Y:D..... | 448 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>ABSolute:Y:E..... | 448 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:BW<bandwidth>:<br>ABSolute:Y:F..... | 448 |

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:     BW<bandwidth>:ENABle <TSMLimEnable>

Activates or deactivates the transmit spectrum mask limit check for 802.11p signals with the specified <bandwidth>.

##### Suffix:

<bandwidth>            5, 10, 20

##### Parameters:

<TSMLimEnable>        ON | OFF  
                          \*RST:        ON

**Example:**                See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**                R&S CMW-KM655

---

#### CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:     BW<bandwidth>:CA:Y:A <TSMLimYRelLevA>

Defines the Y-value of point A ( $\Delta f = 2$  <bandwidth>) on the 802.11p spectrum mask for power class A and the specified <bandwidth>. For background information, see "[Transmit spectrum mask OFDM, by regulation](#)" on page 320.

**Suffix:**

&lt;bandwidth&gt; 5, 10, 20

**Parameters:**

&lt;TSMLimYRelLevA&gt; numeric

Range: -90 dB to 10 dB

\*RST: -40 dB

**Example:**See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10**Options:** R&S CMW-KM655**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CA:Y:B <TSMLimYRelLevB>**

Defines the Y-value of point B ( $\Delta f = 1.5 \cdot \text{<bandwidth>}$ ) on the 802.11p spectrum mask for power class A and the specified <bandwidth>. For background information, see "[Transmit spectrum mask OFDM, by regulation](#)" on page 320.

**Suffix:**

&lt;bandwidth&gt; 5, 10, 20

**Parameters:**

&lt;TSMLimYRelLevB&gt; numeric

Range: -90 dB to 10 dB

\*RST: -40 dB

**Example:**See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10**Options:** R&S CMW-KM655**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CA:Y:C <TSMLimYRelLevC>**

Defines the Y-value of point C ( $\Delta f = \text{<bandwidth>}$ ) on the 802.11p spectrum mask for power class A and the specified <bandwidth>. For background information, see "[Transmit spectrum mask OFDM, by regulation](#)" on page 320.

**Suffix:**

&lt;bandwidth&gt; 5, 10, 20

**Parameters:**

&lt;TSMLimYRelLevC&gt; numeric

Range: -90 dB to 10 dB

\*RST: -28 dB

**Example:**See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CA:Y:D <TSMLimYRelLevD>**

Defines the Y-value of point D ( $\Delta f = 0.55$  <bandwidth>) on the 802.11p spectrum mask for power class A and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevD> numeric

Range:            -90 dB to 10 dB

\*RST:            -20 dB

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CA:Y:E <TSMLimYRelLevE>**

Defines the Y-value of point E ( $\Delta f = 0.5$  <bandwidth>) on the 802.11p spectrum mask for power class A and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevE> numeric

Range:            -90 dB to 10 dB

\*RST:            -10 dB

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CB:Y:A <TSMLimYRelLevA>**

Defines the Y-value of point A ( $\Delta f = 2$  <bandwidth>) on the 802.11p spectrum mask for power class B and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevA> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
 BW<bandwidth>:CB:Y:B <TSMLimYRelLevB>**

Defines the Y-value of point B ( $\Delta f = 1.5 \cdot \text{bandwidth}$ ) on the 802.11p spectrum mask for power class B and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth> 5, 10, 20

**Parameters:**

<TSMLimYRelLevB> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
 BW<bandwidth>:CB:Y:C <TSMLimYRelLevC>**

Defines the Y-value of point C ( $\Delta f = \text{bandwidth}$ ) on the 802.11p spectrum mask for power class B and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth> 5, 10, 20

**Parameters:**

<TSMLimYRelLevC> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -28 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CB:Y:D <TSMLimYRelLevD>**

Defines the Y-value of point D ( $\Delta f = 0.55 \cdot \text{bandwidth}$ ) on the 802.11p spectrum mask for power class B and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevD>    numeric

Range:            -90 dB to 10 dB

\*RST:            -20 dB

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:CB:Y:E <TSMLimYRelLevE>**

Defines the Y-value of point E ( $\Delta f = 0.5 \cdot \text{bandwidth}$ ) on the 802.11p spectrum mask for power class B and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevE>    numeric

Range:            -90 dB to 10 dB

\*RST:            -16 dB

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:UDEFined:Y:A <TSMLimYRelLevA>**

Defines the Y-value of point A ( $\Delta f = 2 \cdot \text{bandwidth}$ ) on the 802.11p spectrum mask for power class "user defined" and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20



**Parameters:**

<TSMLimYRelLevA> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -50 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
 BW<bandwidth>:UDEFined:Y:B <TSMLimYRelLevB>**

Defines the Y-value of point B ( $\Delta f = 1.5 \cdot \text{bandwidth}$ ) on the 802.11p spectrum mask for power class "user defined" and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth> 5, 10, 20

**Parameters:**

<TSMLimYRelLevB> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -50 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
 BW<bandwidth>:UDEFined:Y:C <TSMLimYRelLevC>**

Defines the Y-value of point C ( $\Delta f = \text{bandwidth}$ ) on the 802.11p spectrum mask for power class "user defined" and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth> 5, 10, 20

**Parameters:**

<TSMLimYRelLevC> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:UDEFined:Y:D <TSMLimYRelLevD>**

Defines the Y-value of point D ( $\Delta f = 0.55$  <bandwidth>) on the 802.11p spectrum mask for power class "user defined" and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevD>    numeric

Range:            -90 dB to 10 dB

\*RST:            -32 dB

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:UDEFined:Y:E <TSMLimYRelLevE>**

Defines the Y-value of point E ( $\Delta f = 0.5$  <bandwidth>) on the 802.11p spectrum mask for power class "user defined" and the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, by regulation"](#) on page 320.

**Suffix:**

<bandwidth>            5, 10, 20

**Parameters:**

<TSMLimYRelLevE>    numeric

Range:            -90 dB to 10 dB

\*RST:            -26 dB

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:ABSolute:Y:A <TSMLimYAbsLevA>**

Defines the Y-value of point A ( $\Delta f = 2$  <BW>) on the ETSI ITS absolute emission mask for the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322.

**Suffix:**

<bandwidth>            10

**Parameters:**

&lt;TSMLimYAbsLevA&gt; numeric

Range: -150 dBm to 30 dBm

\*RST: -27 dBm

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:ABSolute:Y:B <TSMLimYAbsLevB>**

Sets/queries the Y-value of point B ( $\Delta f = 1.5 \cdot \text{BW}$ ) on the ETSI ITS absolute emission mask for the specified <bandwidth>. For background information, see "[Transmit spectrum mask OFDM, absolute limits](#)" on page 322.

**Suffix:**

&lt;bandwidth&gt; 10

**Parameters:**

&lt;TSMLimYAbsLevB&gt; numeric

Range: -150 dBm to 30 dBm

\*RST: -27 dBm

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:ABSolute:Y:C <TSMLimYAbsLevC>**

Defines the Y-value of point C ( $\Delta f = \text{BW}$ ) on the ETSI ITS absolute emission mask for the specified <bandwidth>. For background information, see "[Transmit spectrum mask OFDM, absolute limits](#)" on page 322.

**Suffix:**

&lt;bandwidth&gt; 10

**Parameters:**

&lt;TSMLimYAbsLevC&gt; numeric

Range: -150 dBm to 30 dBm

\*RST: -17 dBm

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10**Options:** R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:ABSolute:Y:D <TSMLimYAbsLevD>**

Defines the Y-value of point D ( $\Delta f = 0.55 \cdot \text{BW}$ ) on the ETSI ITS absolute emission mask for the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322.

**Suffix:**

<bandwidth>            10, 20

**Parameters:**

<TSMLimYAbsLevD> numeric

Range:            -150 dBm to 30 dBm

\*RST:            -9 dBm

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:ABSolute:Y:E <TSMLimYAbsLevE>**

Defines the Y-value of point E ( $\Delta f = 0.5 \cdot \text{BW}$ ) on the ETSI ITS absolute emission mask for the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322.

**Suffix:**

<bandwidth>            10, 20

**Parameters:**

<TSMLimYAbsLevE> numeric

Range:            -150 dBm to 30 dBm

\*RST:            -3 dBm

**Example:**            See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:**            R&S CMW-KM655

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:POFDm:  
BW<bandwidth>:ABSolute:Y:F <TSMLimYAbsLevF>**

Defines the Y-value of point F ( $\Delta f = 0.45 \cdot \text{BW}$ ) on the ETSI ITS absolute emission mask for the specified <bandwidth>. For background information, see ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322.

**Suffix:**

<bandwidth>            10, 20

**Parameters:**

<TSMLimYAbsLevF> numeric  
 Range: -150 dBm to 30 dBm  
 \*RST: 23 dBm

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10

**Options:** R&S CMW-KM655

**3.5.4.16 Transmit spectrum mask limits: 802.11n**

The following commands define and enable the transmit spectrum mask. The limits apply to 802.11n measurements.

|                                                                                              |     |
|----------------------------------------------------------------------------------------------|-----|
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:BW<bandwidth>:ENABLE                  | 449 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:BW<bandwidth>:<br>BAND<band>:Y:A..... | 449 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:BW<bandwidth>:<br>BAND<band>:Y:B..... | 450 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:BW<bandwidth>:<br>BAND<band>:Y:C..... | 450 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:BW<bandwidth>:<br>BAND<band>:Y:D..... | 451 |
| CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:BW<bandwidth>:<br>ABSLimit.....       | 451 |

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:  
 BW<bandwidth>:ENABLE <TSMLimEnable>**

Enables or disables the transmit spectrum mask for 802.11n signals with the specified <bandwidth>, i.e. activates or deactivates the corresponding limit check.

**Suffix:**

<bandwidth> 5, 10, 20, 40

**Parameters:**

<TSMLimEnable> ON | OFF  
 \*RST: ON

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:  
 BW<bandwidth>:BAND<band>:Y:A <TSMLimYRelLevA>**

Defines the relative spectral density limit for point A (frequency offset: 2\*bandwidth) on the transmit spectrum mask for 802.11n signals with the specified <bandwidth> and the selected <band>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40  
 <band> 2, 5  
 2.4-GHz band or 5-GHz band

**Parameters:**

<TSMLimYRelLevA> numeric  
 Range: -90 dB to 10 dB  
 \*RST: depends on <band>

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:  
 BW<bandwidth>:BAND<band>:Y:B <TSMLimYRelLevB>**

Defines the relative spectral density limit for point B (frequency offset:  $3/2 \cdot \text{bandwidth}$ ) on the transmit spectrum mask for 802.11n signals with the specified <bandwidth> and the selected <band>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40  
 <band> 2, 5  
 2.4-GHz band or 5-GHz band

**Parameters:**

<TSMLimYRelLevB> numeric  
 Range: -90 dB to 10 dB  
 \*RST: depends on <band>

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:  
 BW<bandwidth>:BAND<band>:Y:C <TSMLimYRelLevC>**

Defines the relative spectral density limit for point C (frequency offset:  $1 \cdot \text{bandwidth}$ ) on the transmit spectrum mask for 802.11n signals with the specified <bandwidth> and the selected <band>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40  
 <band> 2, 5  
 2.4-GHz band or 5-GHz band

**Parameters:**

<TSMLimYRelLevC> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -28 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:  
 BW<bandwidth>:BAND<band>:Y:D <TSMLimYRelLevD>**

Defines the relative spectral density limit for point D (frequency offset:  $1/2 \times \text{bandwidth} + 1$  MHz) on the transmit spectrum mask for 802.11n signals with the specified <bandwidth> and the selected <band>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40  
 <band> 2, 5  
 2.4-GHz band or 5-GHz band

**Parameters:**

<TSMLimYRelLevD> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -20 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM651

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HTOFdm:  
 BW<bandwidth>:ABSLimit <TSMLimAbs>**

Defines an absolute power limit for 802.11n signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322 for background information.

**Suffix:**

&lt;bandwidth&gt; 5, 10, 20, 40

**Parameters:**

&lt;TSMLimAbs&gt; numeric

Limit value, applies to frequency offsets greater than  $3/2 \times \text{bandwidth}$ , measured at 100 kHz RBW

Range: -90 dBm to 10 dBm

\*RST: depends on &lt;bandwidth&gt;

**Example:**See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10**Options:** R&S CMW-KM651**3.5.4.17 Transmit spectrum mask limits: 802.11ac**

The following commands define and enable the transmit spectrum mask. The limits apply to 802.11ac SISO measurements.

```

CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:BW<bandwidth>:ENABLE 452
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:BW<bandwidth>:Y:A..... 453
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:BW<bandwidth>:Y:B..... 453
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:BW<bandwidth>:Y:C..... 453
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:BW<bandwidth>:Y:D..... 454
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:BW<bandwidth>:
  ABSLimit.....454

```

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:  
BW<bandwidth>:ENABLE <TSMLimEnable>**

Enables or disables the transmit spectrum mask for 802.11ac signals with the specified <bandwidth>, i.e. activates or deactivates the corresponding limit checks.

**Suffix:**<bandwidth> 5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)**Parameters:**

&lt;TSMLimEnable&gt; ON | OFF

\*RST: ON

**Example:**See [Specifying transmit spectrum masks \(OFDM\)](#)**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10**Options:** R&S CMW-KM656



---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:  
BW<bandwidth>:Y:A <TSMLimYRelLevA>**

Defines the relative spectral density limit for point A (frequency offset:  $2 \times \text{bandwidth}$ ) on the transmit spectrum mask for 802.11ac signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth>                    5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevA> numeric  
Range:                    -90 dB to 10 dB  
\*RST:                    -40 dB

**Example:**                    See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:**                    R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:  
BW<bandwidth>:Y:B <TSMLimYRelLevB>**

Defines the relative spectral density limit for point B (frequency offset:  $3/2 \times \text{bandwidth}$ ) on the transmit spectrum mask for 802.11ac signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth>                    5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevB> numeric  
Range:                    -90 dB to 10 dB  
\*RST:                    -40 dB

**Example:**                    See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:**                    R&S CMW-KM656

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:  
BW<bandwidth>:Y:C <TSMLimYRelLevC>**

Defines the relative spectral density limit for point C (frequency offset:  $1 \times \text{bandwidth}$ ) on the transmit spectrum mask for 802.11ac signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevC> numeric  
Range: -90 dB to 10 dB  
\*RST: -28 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM656

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:  
BW<bandwidth>:Y:D <TSMLimYRelLevD>**

Defines the relative spectral density limit for point D (center frequency offset:  $1/2 \times \text{bandwidth} + 1$  MHz) on the transmit spectrum mask for 802.11ac signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevD> numeric  
Range: -90 dB to 10 dB  
\*RST: -20 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:** R&S CMW-KM656

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:VHTofdm:  
BW<bandwidth>:ABSLimit <TSMLimAbs>**

Defines an absolute power limit for 802.11ac signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322 for background information.

**Suffix:**

<bandwidth> 5, 10, 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimAbs>      numeric  
                          Limit value, applies to frequency offsets greater than  $3/2 \cdot \text{<bandwidth>}$ , measured at 100 kHz RBW  
                          Range:      -90 dBm to 10 dBm  
                          \*RST:      depends on <bandwidth>

**Example:**      See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.10, V3.7.31: added bandwidth 5 and 10

**Options:**      R&S CMW-KM656

**3.5.4.18 Transmit spectrum mask limits: 802.11ax**

The following commands define and enable the transmit spectrum mask. The limits apply to 802.11ax SISO measurements.

```
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:BW<bandwidth>:ENABle 455
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:BW<bandwidth>:Y:A..... 455
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:BW<bandwidth>:Y:B..... 456
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:BW<bandwidth>:Y:C..... 456
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:BW<bandwidth>:Y:D..... 457
CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:BW<bandwidth>:
  ABSLimit.....457
```

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:  
 BW<bandwidth>:ENABle <TSMLimEnable>**

Enables or disables the transmit spectrum mask for 802.11ax signals with the specified <bandwidth>, i.e. activates or deactivates the corresponding limit checks.

**Suffix:**

<bandwidth>      20, 40, 80, 160, 8080  
                          Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimEnable>    ON | OFF  
                          \*RST:      ON

**Example:**      See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080

**Options:**      R&S CMW-KM657

---

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:  
 BW<bandwidth>:Y:A <TSMLimYRelLevA>**

Defines the relative spectral density limit for point A (frequency offset:  $2 \cdot \text{bandwidth}$ ) on the transmit spectrum mask for 802.11ax signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevA> numeric  
Range: -90 dB to 10 dB  
\*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080

**Options:** R&S CMW-KM657

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:  
BW<bandwidth>:Y:B <TSMLimYRelLevB>**

Defines the relative spectral density limit for point B (frequency offset:  $3/2 \cdot \text{bandwidth}$ ) on the transmit spectrum mask for 802.11ax signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevB> numeric  
Range: -90 dB to 10 dB  
\*RST: -40 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.20

**Options:** R&S CMW-KM657

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:  
BW<bandwidth>:Y:C <TSMLimYRelLevC>**

Defines the relative spectral density limit for point C (frequency offset:  $1 \cdot \text{bandwidth}$ ) on the transmit spectrum mask for 802.11ax signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevC> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -28 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.20

**Options:** R&S CMW-KM657

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:  
 BW<bandwidth>:Y:D <TSMLimYRelLevD>**

Defines the relative spectral density limit for point D (center frequency offset:  $1/2 \times \text{bandwidth} + 1$  MHz) on the transmit spectrum mask for 802.11ax signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, default masks"](#) on page 317 for background information.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
 Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimYRelLevD> numeric  
 Range: -90 dB to 10 dB  
 \*RST: -20 dB

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080

**Options:** R&S CMW-KM657

**CONFigure:WLAN:MEAS<i>:MEValuation:LIMit:TSMask:HEOFdm:  
 BW<bandwidth>:ABSLimit <TSMLimAbs>**

Defines the absolute power limit for 802.11ax signals with the specified <bandwidth>.

See ["Transmit spectrum mask OFDM, absolute limits"](#) on page 322 for background information.

**Suffix:**

<bandwidth> 20, 40, 80, 160, 8080  
 Channel bandwidth in MHz (8080 means 80+80 MHz)

**Parameters:**

<TSMLimAbs> numeric  
 Limit value, applies to frequency offsets greater than  $3/2 \times \text{bandwidth}$ , measured at 25 kHz RBW  
 Range: -90 dBm to 10 dBm  
 \*RST: depends on <bandwidth>

**Example:** See [Specifying transmit spectrum masks \(OFDM\)](#)

**Firmware/software:** V3.7.20, V3.7.60: added <bandwidth> 8080

**Options:** R&S CMW-KM657

### 3.5.5 Measurement result commands

The following commands queries results.

|                                                                 |     |
|-----------------------------------------------------------------|-----|
| • <a href="#">Timing error traces</a> .....                     | 458 |
| • <a href="#">EVM versus chip traces</a> .....                  | 459 |
| • <a href="#">EVM versus symbol traces</a> .....                | 460 |
| • <a href="#">EVM versus carrier traces</a> .....               | 460 |
| • <a href="#">I/Q constellation traces</a> .....                | 465 |
| • <a href="#">Unused tone error traces</a> .....                | 466 |
| • <a href="#">Unused tone error margin values</a> .....         | 467 |
| • <a href="#">Carrier Frequency offset error traces</a> .....   | 468 |
| • <a href="#">Spectrum flatness traces</a> .....                | 469 |
| • <a href="#">Spectrum flatness margin values</a> .....         | 475 |
| • <a href="#">Spectrum flatness margin positions</a> .....      | 477 |
| • <a href="#">Transmit spectrum mask traces</a> .....           | 479 |
| • <a href="#">Transmit spectrum mask limit lines</a> .....      | 484 |
| • <a href="#">Transmit spectrum mask margin values</a> .....    | 487 |
| • <a href="#">Transmit spectrum mask margin positions</a> ..... | 492 |
| • <a href="#">Transmit spectrum OBW results</a> .....           | 498 |
| • <a href="#">Power vs. time burst traces</a> .....             | 500 |
| • <a href="#">Power vs. time ramp traces</a> .....              | 507 |
| • <a href="#">Power vs. time single values</a> .....            | 517 |
| • <a href="#">Modulation single values, DSSS</a> .....          | 522 |
| • <a href="#">Modulation single values, OFDM</a> .....          | 524 |
| • <a href="#">Modulation single values, OFDMA</a> .....         | 537 |
| • <a href="#">Signal fields info</a> .....                      | 541 |
| • <a href="#">List mode results (one segment)</a> .....         | 580 |
| • <a href="#">List mode results (all segments)</a> .....        | 589 |

#### 3.5.5.1 Timing error traces

The following commands return the timing error trace results (only for OFDM standards).

---

**FETCH:WLAN:MEAS<i>:MEvaluation:TRACe:TERRor?** [<start>, <count>[,  
<decimation>]]

**READ:WLAN:MEAS<i>:MEvaluation:TRACe:TERRor?** [<start>, <count>[,  
<decimation>]]

Return timing error traces. The number of results corresponds to the statistic count, see [CONFigure:WLAN:MEAS<i>:MEvaluation:SCount:PVTime](#).

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<TE\_Times> float  
 Comma-separated list of timing error values  
 Default unit: s

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.50

### 3.5.5.2 EVM versus chip traces

The following commands return the EVM vs chip trace results.

See also [Chapter 3.3.2.2, "Views error vector magnitude \(EVM\)"](#), on page 258

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:DSSS:CURRent?**  
 [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:DSSS:AVERage?**  
 [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:DSSS:MAXimum?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:DSSS:CURRent?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:DSSS:AVERage?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:DSSS:MAXimum?**  
 [<start>, <count>[, <decimation>]]

Return the values of the EVM vs chip traces. The results of the current, average and maximum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<EVM> float  
 Comma-separated list of EVM values, each value related to one chip. The maximum number of values is 1000.  
 Range: 0 % to 100 %  
 Default unit: %

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V1.0.10.50, V3.2.11 subarrays

### 3.5.5.3 EVM versus symbol traces

The following commands return the EVM vs symbol trace results.

See also [Chapter 3.3.2.2, "Views error vector magnitude \(EVM\)"](#), on page 258

---

```

FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:CURRent?
    [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:AVERage?
    [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:MINimum?
    [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:MAXimum?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:CURRent?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:AVERage?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:MINimum?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:SYMBOL:MAXimum?
    [<start>, <count>[, <decimation>]]

```

Return the values of the EVM vs. symbol traces for OFDM signals according to standard 802.11a, g, n, ac, ax, or p. The results of the current, average and maximum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

#### Return values:

|               |                                                                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                                         |
| <EVM>         | float<br>Comma-separated list of EVM values, one per OFDM symbol.<br>The maximum number of values is 1366.<br>Range: -100 dB to 0 dB<br>Default unit: dB |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

### 3.5.5.4 EVM versus carrier traces

The following commands return the EVM vs carrier trace results.

See also [Chapter 3.3.2.2, "Views error vector magnitude \(EVM\)"](#), on page 258



|                                                                           |     |
|---------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:CURRent?         | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:AVERage?         | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MINimum?         | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MAXimum?         | 462 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:CURRent?          | 462 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:AVERage?          | 462 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MINimum?          | 462 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MAXimum?          | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:    |     |
| CURRent?                                                                  | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:    |     |
| AVERage?                                                                  | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:    |     |
| MINimum?                                                                  | 462 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:    |     |
| MAXimum?                                                                  | 462 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:     |     |
| CURRent?                                                                  | 463 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:     |     |
| AVERage?                                                                  | 463 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:     |     |
| MINimum?                                                                  | 463 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:SEGMENT<seg>:     |     |
| MAXimum?                                                                  | 463 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:CURRent? | 463 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:AVERage? | 463 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:MINimum? | 463 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:MAXimum? | 463 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:CURRent?  | 463 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:AVERage?  | 464 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:MINimum?  | 464 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:MAXimum?  | 464 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:         |     |
| SEGMENT<seg>:CURRent?                                                     | 464 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:         |     |
| SEGMENT<seg>:AVERage?                                                     | 464 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:         |     |
| SEGMENT<seg>:MINimum?                                                     | 464 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:         |     |
| SEGMENT<seg>:MAXimum?                                                     | 464 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:          |     |
| SEGMENT<seg>:CURRent?                                                     | 464 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:          |     |
| SEGMENT<seg>:AVERage?                                                     | 464 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:          |     |
| SEGMENT<seg>:MINimum?                                                     | 465 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:          |     |
| SEGMENT<seg>:MAXimum?                                                     | 465 |

---

```

FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:CURRent?
    [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:AVERage?
    [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MINimum?
    [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MAXimum?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:CURRent?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:AVERage?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MINimum?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MAXimum?
    [<start>, <count>[, <decimation>]]

```

Return the values of the EVM vs. subcarrier traces for SISO connections. The results of the current, average, minimum and maximum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|               |                                                                                                                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                                                                                                            |
| <EVM_Trace>   | float<br>Comma-separated list of 2n+1 values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers / for 802.11ax, excluding subcarrier -3 to subcarrier 3)<br>n depends on the WLAN standard, channel bandwidth and mode, see <a href="#">OFDM subcarriers</a> .<br>Range: -100 dB to 0 dB<br>Default unit: dB |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

---

```

FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
    SEGMent<seg>:CURRent? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
    SEGMent<seg>:AVERage? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
    SEGMent<seg>:MINimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
    SEGMent<seg>:MAXimum? [<start>, <count>[, <decimation>]]

```

```

READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
  SEGMENT<seg>:CURRent? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
  SEGMENT<seg>:AVERAge? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
  SEGMENT<seg>:MINimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:
  SEGMENT<seg>:MAXimum? [<start>, <count>[, <decimation>]]

```

Return the values of the EVM vs. subcarrier traces for 80+80 MHz, SISO connections. The results of the current, average, minimum and maximum traces can be retrieved.

**Suffix:**

<seg>                      1..2  
Segment

**Query parameters:**

<start>, <count>,        numeric  
<decimation>              For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability>              decimal  
Reliability indicator

<EVMvsCarr>              float  
Comma-separated list of 2n+1 values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers / for 802.11ax, excluding subcarrier -3 to subcarrier 3)  
n depends on the WLAN standard, channel bandwidth and mode, see [OFDM subcarriers](#).  
Range:                      -100 dB to 0 dB  
Default unit: dB

**Example:**                See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:**                      Query only

**Firmware/software:** V3.7.70

---

```

FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:
  CURRent? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:
  AVERAge? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:
  MINimum? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:
  MAXimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:
  CURRent? [<start>, <count>[, <decimation>]]

```

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
AVERAge?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
MINimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
MAXimum?** [<start>, <count>[, <decimation>]]

Return the values of the EVM vs. subcarrier traces for MIMO connections. The results of the current, average, minimum and maximum traces can be retrieved.

**Suffix:**

<n> 1..8  
Antenna

**Query parameters:**

<start>, <count>, numeric  
<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<EVMvsCarr> float  
Comma-separated list of 2n+1 values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers / for 802.11ax, excluding subcarrier -3 to subcarrier 3)  
n depends on the WLAN standard, channel bandwidth and mode, see [OFDM subcarriers](#).  
Range: -100 dB to 0 dB  
Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.70

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMent<seg>:CURRent?** [<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMent<seg>:AVERAge?** [<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMent<seg>:MINimum?** [<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMent<seg>:MAXimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMent<seg>:CURRent?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMent<seg>:AVERAge?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMENT<seg>:MINimum? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:EVMagnitude:CARRier:MIMO<n>:  
SEGMENT<seg>:MAXimum? [<start>, <count>[, <decimation>]]**

Return the values of the EVM vs. subcarrier traces for 80+80 MHz, MIMO connections. The results of the current, average, minimum and maximum traces can be retrieved.

**Suffix:**

<n> 1..8  
Antenna

<seg> 1..2  
Segment

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<EVMvsCarr> float  
Comma-separated list of 2n+1 values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers / for 802.11ax, excluding subcarrier -3 to subcarrier 3)  
n depends on the WLAN standard, channel bandwidth and mode, see [OFDM subcarriers](#).  
Range: -100 dB to 0 dB  
Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.70

### 3.5.5.5 I/Q constellation traces

The following commands return the results in the I/Q constellation diagram.

See also [Chapter 3.3.2.3, "View IQ constellation"](#), on page 260

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:IQConst:INPHase?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:IQConst:QUADrature?**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:IQConst:INPHase?**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:IQConst:QUADrature?**

Return the results in the I/Q constellation diagram. The I (in phase) and Q (quadrature) components are retrieved via separate commands.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

|                           |                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <I_Amplitude_1>           | float                                                                                                                       |
| <Q_Amplitude_1>           | Comma-separated list of I or Q amplitudes. The maximum number of values is 1000 for DSSS signals and 2080 for OFDM signals. |
| <I_Amplitude_2>           |                                                                                                                             |
| <Q_Amplitude_2>...        |                                                                                                                             |
|                           | Range: -2 to 2                                                                                                              |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements (DSSS)</a>                                                              |
| <b>Usage:</b>             | Query only                                                                                                                  |
| <b>Firmware/software:</b> | V2.0.10                                                                                                                     |

### 3.5.5.6 Unused tone error traces

The following commands return the unused tone error trace results.

---

```

CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:CURRent?
CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:AVERage?
CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MINimum?
CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MAXimum?
FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:AVERage?
FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MINimum?
FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MAXimum?
READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:CURRent?
READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:AVERage?
READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MINimum?
READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MAXimum?

```

Return the values of the unused tone error traces according to standard 802.11ax. The results of the current, average, minimum and maximum traces can be retrieved.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8  
Antenna

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<UTE\_Yvals> float  
Comma-separated list of unused tone error results, one value per each 26-tone RU. The total number of RUs depends on the bandwidth, see table below.  
Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#).

**Usage:** Query only

**Firmware/software:** V3.7.21

|                          |    |    |    |     |
|--------------------------|----|----|----|-----|
| Channel bandwidth in MHz | 20 | 40 | 80 | 160 |
| Total No. of RUs         | 9  | 18 | 37 | 74  |

### 3.5.5.7 Unused tone error margin values

The following commands return the limit line and margin Y-values for the unused tone error results.

|                                                               |     |
|---------------------------------------------------------------|-----|
| CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:LIMit?          | 467 |
| FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:LIMit?              | 467 |
| READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:LIMit?               | 467 |
| CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:CURRent? | 468 |
| CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:AVErAge? | 468 |
| CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MINimum? | 468 |
| CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MAXimum? | 468 |
| FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:CURRent?     | 468 |
| FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:AVErAge?     | 468 |
| FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MINimum?     | 468 |
| FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MAXimum?     | 468 |
| READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:CURRent?      | 468 |
| READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:AVErAge?      | 468 |
| READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MINimum?      | 468 |
| READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MAXimum?      | 468 |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:LIMit?**

**FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:LIMit?**

**READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:LIMit?**

Displays unused tone error limit line.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n>                      1..8  
                            Antenna

**Return values:**

<Reliability>           decimal  
                            [Reliability indicator](#)

<UTE\_LimitLine>       float  
                            Comma-separated list of unused tone error limits, one value per  
                            26-tone RU (from left to right)  
                            Default unit: dB

**Example:**              See [Performing single-shot measurements \(OFDM, SISO\)](#).

**Usage:**                Query only

**Firmware/software:** V3.7.21



CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:CURRent?  
 CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:AVERage?  
 CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MINimum?  
 CALCulate:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MAXimum?  
 FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:CURRent?  
 FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:AVERage?  
 FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MINimum?  
 FETCh:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MAXimum?  
 READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:CURRent?  
 READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:AVERage?  
 READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MINimum?  
 READ:WLAN:MEAS<i>:MEValuation:UTERror<n>:MARGin:MAXimum?

Returns the margin values of the unused tone error measurement for the current, average, minimum and maximum traces.

A positive margin indicates a violation of the unused tone error limit line. The respective trace value is located above the upper limit line.

The values described below are returned by FETCh and READ commands. CALCulate commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8  
Antenna

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<UTE\_Margin> float  
Comma-separated list of margins to the unused tone error limits, one value per each 26-tone RU. The total number of RUs depends on the bandwidth, see table below.  
Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#).

**Usage:** Query only

**Firmware/software:** V3.7.21

|                          |    |    |    |     |
|--------------------------|----|----|----|-----|
| Channel bandwidth in MHz | 20 | 40 | 80 | 160 |
| Total No. of RUs         | 9  | 18 | 37 | 74  |

### 3.5.5.8 Carrier Frequency offset error traces

The following commands return the carrier frequency offset (CFO) error trace results.



**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:CFERror?** [<start>, <count>[,  
<decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:CFERror?** [<start>, <count>[,  
<decimation>]]

Return carrier frequency offset (CFO) error traces for HE. The number of results corresponds to the statistic count, see [CONFigure:WLAN:MEAS<i>:MEValuation:SCount:MODulation](#).

**Query parameters:**

<start>, <count>, numeric  
<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
Reliability indicator  
<CFO\_Frequencies> float  
Comma-separated list of CFO error values  
Default unit: Hz

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KM657

### 3.5.5.9 Spectrum flatness traces

The following commands return the spectrum flatness trace results.

For MIMO, use the commands with **MIMO** mnemonic.

For bandwidths with two segments, use the commands with **SEGment<seg>** mnemonic.

See also [Chapter 3.3.2.5, "View spectrum flatness"](#), on page 263

|                                                                                                                           |     |
|---------------------------------------------------------------------------------------------------------------------------|-----|
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:CURRent?</a>                                      | 471 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:AVERAge?</a>                                      | 471 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:MINimum?</a>                                      | 471 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:MAXimum?</a>                                      | 471 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:CURRent?</a>                                       | 471 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:AVERAge?</a>                                       | 471 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:MINimum?</a>                                       | 471 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness[:OFDM]:MAXimum?</a>                                       | 471 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness:MIMO:RXANtenna&lt;n&gt;:STReam&lt;s&gt;:CURRent?</a> | 472 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness:MIMO:RXANtenna&lt;n&gt;:STReam&lt;s&gt;:AVERAge?</a> | 472 |

|                                                                                                          |     |
|----------------------------------------------------------------------------------------------------------|-----|
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:MINimum?              | 472 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:MAXimum?              | 472 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:CURRent?                  | 472 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:AVERage?                  | 472 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:MINimum?                  | 472 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:MAXimum?                  | 472 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:CURRent?                   | 472 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:AVERage?                   | 472 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:MINimum?                   | 472 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:MAXimum?                   | 472 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:CURRent?                                 | 473 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:AVERage?                                 | 473 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:MINimum?                                 | 473 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:MAXimum?                                 | 473 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:CURRent?                                     | 473 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:AVERage?                                     | 473 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:MINimum?                                     | 473 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:MAXimum?                                     | 473 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:CURRent?                                      | 473 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:AVERage?                                      | 473 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:MINimum?                                      | 473 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:MAXimum?                                      | 473 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:CURRent? | 474 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:AVERage? | 474 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:MINimum? | 474 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:MAXimum? | 474 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:CURRent?     | 474 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:AVERage?     | 474 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:MINimum?     | 474 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:MAXimum?     | 474 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXANtenna<n>:<br>STReam<s>:SEGMENT<seg>:CURRent?      | 474 |

|                                                                                                                                      |     |
|--------------------------------------------------------------------------------------------------------------------------------------|-----|
| <code>READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness:MIMO:RXANtenna&lt;n&gt;:STReam&lt;s&gt;:SEGMENT&lt;seg&gt;:AVERAge?</code> | 474 |
| <code>READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness:MIMO:RXANtenna&lt;n&gt;:STReam&lt;s&gt;:SEGMENT&lt;seg&gt;:MINimum?</code> | 474 |
| <code>READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:SFLatness:MIMO:RXANtenna&lt;n&gt;:STReam&lt;s&gt;:SEGMENT&lt;seg&gt;:MAXimum?</code> | 474 |

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:CURRENT?**  
 [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:AVERAge?**  
 [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:MINimum?**  
 [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:MAXimum?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:CURRENT?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:AVERAge?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:MINimum?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness[:OFDM]:MAXimum?**  
 [<start>, <count>[, <decimation>]]

Return the values of the spectrum flatness traces for OFDM SISO signals according to standard 802.11a, g, n, ac, ax, or p. The results of the current, average, minimum and maximum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|               |                                                                                                                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                               |
| <Power>       | float<br>Comma-separated list of 2n+1 power values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers)<br>n depends on the WLAN standard, channel bandwidth and mode, see <a href="#">Table 3-2</a> .<br>Range: -20 dB to 20 dB |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V1.0.15.20, V3.2.11 subarrays

---

```

CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:
    RXAntenna<n>:STReam<s>:CURRent? [<start>, <count>[, <decimation>]]
CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:
    RXAntenna<n>:STReam<s>:AVERAge? [<start>, <count>[, <decimation>]]
CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:
    RXAntenna<n>:STReam<s>:MINimum? [<start>, <count>[, <decimation>]]
CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:
    RXAntenna<n>:STReam<s>:MAXimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:CURRent? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:AVERAge? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:MINimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:MAXimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:CURRent? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:AVERAge? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:MINimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:
    STReam<s>:MAXimum? [<start>, <count>[, <decimation>]]

```

Return the spectrum flatness traces for Rx antenna <n> and stream <s>, for switched and true MIMO measurements.

The results of the current, average, minimum and maximum traces can be retrieved.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

#### Suffix:

|     |      |
|-----|------|
| <n> | 1..8 |
| <s> | 1..8 |

#### Query parameters:

|                 |                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <start> <count> | numeric                                                                                                                                     |
| <decimation>    | For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |

#### Return values:

|                 |                                                                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                                |
| <SpecFlatTrace> | float<br>Comma-separated list of 2n+1 power values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers)<br>n depends on the WLAN standard, channel bandwidth and mode, see <a href="#">OFDM subcarriers</a> .<br>Default unit: dB |

**Usage:** Query only

**Firmware/software:** V3.7.31

---

```

CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  CURRENT? [<start>, <count>[, <decimation>]]
CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  CURRENT? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  CURRENT? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:SEGMENT<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]

```

Return the spectrum flatness traces for segment <seg>, for SISO measurements and bandwidths with two segments (80+80 MHz).

The results of the current, average, minimum and maximum traces can be retrieved.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

|       |                 |
|-------|-----------------|
| <seg> | 1..2<br>Segment |
|-------|-----------------|

**Query parameters:**

|                 |                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <start> <count> | numeric                                                                                                                                     |
| <decimation>    | For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a> |
|---------------|--------------------------------------------------|

|                    |                                                                                                                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <SpecFlatTrace>    | float                                                                                                                                                                                                                       |
|                    | Comma-separated list of 2n+1 power values, for subcarrier -n to subcarrier +n (including data, pilot and unused subcarriers)<br>n depends on the WLAN standard, channel bandwidth and mode, see <a href="#">Table 3-2</a> . |
|                    | Default unit: dB                                                                                                                                                                                                            |
| Usage:             | Query only                                                                                                                                                                                                                  |
| Firmware/software: | V3.7.30                                                                                                                                                                                                                     |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]**

**CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:AVERAge? [<start>, <count>[, <decimation>]]**

**CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:MINimum? [<start>, <count>[, <decimation>]]**

**CALCulate:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:MAXimum? [<start>, <count>[, <decimation>]]**

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]**

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:AVERAge? [<start>, <count>[, <decimation>]]**

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:MINimum? [<start>, <count>[, <decimation>]]**

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:MAXimum? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:AVERAge? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:MINimum? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:SFLatness:MIMO:RXAntenna<n>:STReam<s>:SEGMENT<seg>:MAXimum? [<start>, <count>[, <decimation>]]**

Return the spectrum flatness traces for Rx antenna <n>, stream <s>, and segment <seg> for switched MIMO measurements and bandwidths with two segments (80+80 MHz).

The results of the current, average, minimum and maximum traces can be retrieved.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

|     |      |
|-----|------|
| <n> | 1..8 |
|-----|------|

<s> 1..8

<seg> 1..2

**Query parameters:**

<start> <count> numeric

<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<SpecFlatTrace> float

Comma-separated list of  $2n+1$  power values, for subcarrier  $-n$  to subcarrier  $+n$  (including data, pilot and unused subcarriers)  
 $n$  depends on the WLAN standard, channel bandwidth and mode, see [OFDM subcarriers](#).

Default unit: dB

**Usage:** Query only

**Firmware/software:** V3.7.31

### 3.5.5.10 Spectrum flatness margin values

The following commands return the limit line margin Y-values for the spectrum flatness results.

For MIMO, use the commands with MIMO<n> mnemonic.

See also [Chapter 3.3.2.5, "View spectrum flatness"](#), on page 263

|                                                               |     |
|---------------------------------------------------------------|-----|
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:CURRent?         | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:AVERage?         | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MINimum?         | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MAXimum?         | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:CURRent?             | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:AVERage?             | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MINimum?             | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MAXimum?             | 476 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:CURRent?              | 476 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:AVERage?              | 476 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:MINimum?              | 476 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:MAXimum?              | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:CURRent? | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:AVERage? | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MINimum? | 476 |
| CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MAXimum? | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:CURRent?     | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:AVERage?     | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MINimum?     | 476 |
| FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MAXimum?     | 477 |



|                                                                                      |     |
|--------------------------------------------------------------------------------------|-----|
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:SFLatness:MIMO&lt;n&gt;:CURRent?</a> | 477 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:SFLatness:MIMO&lt;n&gt;:AVERAge?</a> | 477 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:SFLatness:MIMO&lt;n&gt;:MINimum?</a> | 477 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:SFLatness:MIMO&lt;n&gt;:MAXimum?</a> | 477 |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:AVERAge?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:AVERAge?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:AVERAge?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MAXimum?**

Returns the margin values of the spectrum flatness measurement for the current, average, minimum and maximum traces.

A positive margin indicates a violation of the spectrum flatness limit. The respective trace value is located above the upper or below the lower limit line.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

|                   |                                                                                                                                                                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>     | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                                                                                             |
| <UpperMargin>     | float                                                                                                                                                                                                                                                                                                                        |
| <LowerMargins>... | <b>Value 1:</b> trace margin to the upper spectrum flatness limit<br><b>Values 2 ...:</b> the trace margins to the lower spectrum flatness limit as comma-separated list of margins to the lower spectrum flatness limits, one value per subcarrier range (from left to right)<br>Range: -22 dB to 40 dB<br>Default unit: dB |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V2.1.10

---

**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:AVERAge?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:AVERAge?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MINimum?**



**FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:CURRENT?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:MAXimum?**

Return the single value margins for switched MIMO measurements, antenna/stream number <n>. There are current, average, minimum, and maximum results.

For the queries of subcarrier indices for spectrum flatness margins, see:

[FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:CURRENT?](#) etc.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8  
 Identical number of the antenna and stream, e.g.:  
**2** means antenna No. 2, stream No. 2

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 <SpecFlatMargins> float  
 Up to 10 comma-separated list of margins (one value per sub-carrier range from left to right)  
**Value 1:** trace margin to the upper spectrum flatness limit  
**Value 2 to 5:** for the trace margins to the lower spectrum flatness limit  
**Value 6:** trace margin to the upper spectrum flatness limit for segment 2 (80+80 MHz)  
**Value 7 to 10:** trace margins to the lower spectrum flatness limit for segment 2 (80+80 MHz)  
 Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.31

### 3.5.5.11 Spectrum flatness margin positions

The following commands return the limit line margin X-values (subcarrier indices) for the spectrum flatness results.

For MIMO, use the commands with **MIMO<n>** mnemonic.

See also [Chapter 3.3.2.5, "View spectrum flatness"](#), on page 263

[FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:X:CURRENT?](#)..... 478  
[FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:X:AVERage?](#)..... 478  
[FETCH:WLAN:MEAS<i>:MEValuation:SFLatness:X:MINimum?](#)..... 478

|                                                             |     |
|-------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:X:MAXimum?         | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:CURRent?          | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:AVERage?          | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:MINimum?          | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:MAXimum?          | 478 |
| FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:CURRent? | 478 |
| FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:AVERage? | 478 |
| FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MINimum? | 478 |
| FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MAXimum? | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:CURRent?  | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:AVERage?  | 478 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MINimum?  | 479 |
| READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MAXimum?  | 479 |

---

**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:X:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:X:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:X:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:X:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:X:MAXimum?**

Return the subcarrier indices for the current, average, minimum and maximum margin values.

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<UpperMargin> decimal

<LowerMargins>... Subcarrier index for:

**Value 1:** trace margin to the upper spectrum flatness limit

**Values 2 ...:** the trace margins to the lower spectrum flatness limit as comma-separated list of subcarrier indices for the trace margins, one value per subcarrier range (from left to right)

Range: Depends on used standard and channel bandwidth

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V2.1.10

---

**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:AVERage?**

**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MINimum?**

**READ:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:X:MAXimum?**

Return the subcarrier indices for the current, average, minimum and maximum margin values for switched MIMO, antenna/stream number <n>.

For the queries of spectrum flatness margins, see:

[FETCh:WLAN:MEAS<i>:MEValuation:SFLatness:MIMO<n>:CURRent?](#) etc.

**Suffix:**

<n> 1..8  
Identical number of the antenna and stream, e.g.:  
**2** means antenna No. 2, stream No. 2

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<MarginPositions> decimal

Up to 10 comma-separated list of subcarrier indices (one index per subcarrier range from left to right)

**Value 1:** subcarrier index of the trace margin to the upper spectrum flatness limit

**Value 2 to 5:** subcarrier indices, for the trace margin to the lower spectrum flatness limit

**Value 6:** subcarrier index of the trace margin to the upper spectrum flatness limit for segment 2 (80+80 MHz)

**Value 7 to 10:** subcarrier indices, for the trace margin to the lower spectrum flatness limit for segment 2 (80+80 MHz)

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.31

### 3.5.5.12 Transmit spectrum mask traces

The following commands return the transmit spectrum mask traces.

For MIMO, use the commands with MIMO<n> mnemonic.

For bandwidths with two segments, use the commands with SEGment<seg> mnemonic.

See also [Chapter 3.3.2.6, "View transmit spectrum mask"](#), on page 264

|                                                                            |     |
|----------------------------------------------------------------------------|-----|
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:CURRent?</a> | 480 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:AVERage?</a> | 480 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MINimum?</a> | 480 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MAXimum?</a> | 480 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:CURRent?</a>  | 480 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:AVERage?</a>  | 480 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MINimum?</a>  | 481 |

|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MAXimum?                           | 481 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:CURRent?                  | 481 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:AVERage?                  | 481 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MINimum?                  | 481 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MAXimum?                  | 481 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:CURRent?                   | 481 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:AVERage?                   | 481 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MINimum?                   | 481 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MAXimum?                   | 481 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:CURRent?             | 482 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:AVERage?             | 482 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:MINimum?             | 482 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:MAXimum?             | 482 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:CURRent?              | 482 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:AVERage?              | 482 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:MINimum?              | 482 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMENT<seg>:MAXimum?              | 482 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>CURRent? | 483 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>AVERage? | 483 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>MINimum? | 483 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>MAXimum? | 483 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>CURRent?  | 483 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>AVERage?  | 483 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>MINimum?  | 483 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:<br>MAXimum?  | 483 |

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:CURRent?** [<start>,  
<count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:AVERage?** [<start>,  
<count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MINimum?** [<start>,  
<count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MAXimum?** [<start>,  
<count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:CURRent?** [<start>,  
<count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:AVERage?** [<start>,  
<count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MINimum?** [<start>,  
<count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MAXimum?** [<start>,  
<count>[, <decimation>]]

Return the values of the transmit spectrum mask traces for SISO measurements and bandwidths with one segment.

The results of the current, average, minimum and maximum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

For 802.11p signals, use [CONFigure:WLAN:MEAS<i>:MEValuation:TSMask:MSElection](#) to switch between relative and absolute values.

**Return values:**

|                 |                                                                                                                                                                                                |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                               |
| <SpectrumTrace> | float<br>Comma-separated list of power values, trace from left to right<br>Range: -90 dB to 10 dB (relative) or -120 dBm to 25 dBm (absolute)<br>Default unit: dB (relative) or dBm (absolute) |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V1.0.15.0, V3.2.11 subarrays  
V3.2.40 absolute traces for 802.11p  
V3.5.121 MINimum commands

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:CURRent?**  
[<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:AVERAge?**  
[<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MINimum?**  
[<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MAXimum?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:CURRent?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:AVERAge?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MINimum?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:MAXimum?**  
[<start>, <count>[, <decimation>]]

Return the values of the transmit spectrum mask traces for MIMO measurements, antenna/stream number <n>, bandwidths with one segment.

The results of the current, average, minimum and maximum traces can be retrieved.

**Suffix:**

<n> 1..8

**Query parameters:**

<start> <count> numeric

<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<SpectrumTrace> float

Comma-separated list of power values, trace from left to right

Range: -90 dB to 10 dB

Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:CURRent?**  
[<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:AVERAge?**  
[<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:MINimum?**  
[<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:MAXimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:CURRent?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:AVERAge?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:MINimum?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:SEGMent<seg>:MAXimum?**  
[<start>, <count>[, <decimation>]]

Return the values of the transmit spectrum mask traces for segment <seg>, for SISO measurements and bandwidths with two segments.

The results of the current, average, minimum and maximum traces can be retrieved.

**Suffix:**

<seg> 1..2

**Query parameters:**

<start> <count> numeric  
 <decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<SpectrumTrace> float  
 Comma-separated list of power values, trace from left to right  
 Range: -90 dB to 10 dB  
 Default unit: dB

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 CURRent? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 AVERAge? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 MINimum? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 MAXimum? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 CURRent? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 AVERAge? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 MINimum? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MIMO<n>:SEGMENT<seg>:  
 MAXimum? [<start>, <count>[, <decimation>]]**

Return the values of the transmit spectrum mask traces for segment <seg> and antenna/stream number <n>, for MIMO measurements and bandwidths with two segments.

The results of the current, average, minimum and maximum traces can be retrieved.

**Suffix:**

<n> 1..8  
 <seg> 1..2

**Query parameters:**

<start> <count> numeric  
 <decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.



**Return values:**

|                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| <Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                                                      |
| <SpectrumTrace> | float<br>Comma-separated list of power values, trace from left to right<br>Range: -90 dB to 10 dB<br>Default unit: dB |

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

**3.5.5.13 Transmit spectrum mask limit lines**

The following commands return the transmit spectrum mask limit lines.

See also [Chapter 3.3.2.6, "View transmit spectrum mask"](#), on page 264

|                                                                                                          |     |
|----------------------------------------------------------------------------------------------------------|-----|
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:FREQuency?</a>                             | 484 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:FREQuency?</a>                              | 484 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK?</a>                                  | 485 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK?</a>                                   | 485 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK:MIMO&lt;n&gt;?</a>                    | 485 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK:MIMO&lt;n&gt;?</a>                     | 485 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK:SEGMENT&lt;seg&gt;?</a>               | 486 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK:SEGMENT&lt;seg&gt;?</a>                | 486 |
| <a href="#">FETCH:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK:MIMO&lt;n&gt;:SEGMENT&lt;seg&gt;?</a> | 486 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TRACe:TSMask:MASK:MIMO&lt;n&gt;:SEGMENT&lt;seg&gt;?</a>  | 486 |

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:FREQuency?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:FREQuency?** [<start>, <count>[, <decimation>]]

Return the frequency values (X-values) of the transmit spectrum mask limit line trace, for SISO and MIMO, bandwidths with one or two segments.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|                 |                                                                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                 |
| <SpecTraceFreq> | float<br>Comma-separated list of values, trace from left to right<br>0 Hz corresponds to the center of the channel, for 80+80 MHz signals to the center of the left segment.<br>Default unit: Hz |



**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.20

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK?** [<start>, <count>[, <decimation>]]

Return the power values (Y-values) of the transmit spectrum mask limit line trace, for SISO measurements and bandwidths with one segment.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<SpecTraceMask> float  
Comma-separated list of power values, trace from left to right  
Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.20

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK:MIMO<n>?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK:MIMO<n>?** [<start>, <count>[, <decimation>]]

Return the power values (Y-values) of the transmit spectrum mask limit line trace, for MIMO measurements, antenna/stream number <n>, bandwidths with one segment.

**Suffix:**

<n> 1..8

**Query parameters:**

<start> <count> numeric

<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<SpecTraceMaskTX> float  
Comma-separated list of power values, trace from left to right  
Default unit: dB

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK:SEGMENT<seg>?**  
[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK:SEGMENT<seg>?**  
[<start>, <count>[, <decimation>]]

Return the power values (Y-values) of the transmit spectrum mask limit line trace for segment <seg>, for SISO measurements and bandwidths with two segments.

**Suffix:**

<seg> 1..2

**Query parameters:**

<start> <count> numeric

<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<SpecTraceMask> float  
Comma-separated list of power values, trace from left to right  
Default unit: dB

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK:MIMO<n>:**  
**SEGMENT<seg>?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:TSMask:MASK:MIMO<n>:**  
**SEGMENT<seg>?** [<start>, <count>[, <decimation>]]

Return the power values (Y-values) of the transmit spectrum mask limit line trace for segment <seg> and antenna/stream number <n>, for MIMO measurements and bandwidths with two segments.

**Suffix:**

<n> 1..8

<seg> 1..2

**Query parameters:**

<start> <count> numeric

<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|                 |                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------|
| <Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                            |
| <SpecTraceMask> | float<br>Comma-separated list of power values, trace from left to right<br>Default unit: dB |

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

### 3.5.5.14 Transmit spectrum mask margin values

The commands in this section return the limit line margin Y-values of the transmit spectrum mask measurement.

For MIMO, use the commands with `MIMO<n>` mnemonic.

For bandwidths with two segments, use the commands with `SEGments` mnemonic.

See also "[Transmit spectrum mask DSSS](#)" on page 316 and subsequent sections.

|                                                                                        |     |
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| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:AVERage?</a>               | 488 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MINimum?</a>               | 488 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MAXimum?</a>               | 488 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:CURRent?</a>                   | 488 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:AVERage?</a>                   | 488 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MINimum?</a>                   | 488 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MAXimum?</a>                   | 488 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:CURRent?</a>                    | 488 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:AVERage?</a>                    | 488 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MINimum?</a>                    | 488 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MAXimum?</a>                    | 488 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:CURRent?</a> | 489 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:AVERage?</a> | 489 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:MINimum?</a> | 489 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:MAXimum?</a> | 489 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:CURRent?</a>     | 489 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:AVERage?</a>     | 489 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:MINimum?</a>     | 489 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:MAXimum?</a>     | 489 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:CURRent?</a>      | 489 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:AVERage?</a>      | 489 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:MINimum?</a>      | 489 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:MAXimum?</a>      | 489 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:SEGments:CURRent?</a>      | 490 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:SEGments:AVERage?</a>      | 490 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:SEGments:MAXimum?</a>      | 490 |

|                                                                     |     |
|---------------------------------------------------------------------|-----|
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MINimum?         | 490 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:CURRent?             | 490 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:AVERage?             | 490 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MAXimum?             | 490 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MINimum?             | 490 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:CURRent?              | 490 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:AVERage?              | 490 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MINimum?              | 490 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MAXimum?              | 490 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:CURRent? | 491 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:AVERage? | 491 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MINimum? | 491 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MAXimum? | 491 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:CURRent?     | 491 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:AVERage?     | 491 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MINimum?     | 491 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MAXimum?     | 491 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:CURRent?      | 491 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:AVERage?      | 491 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MINimum?      | 491 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MAXimum?      | 491 |

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**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MAXimum?**

Return the limit line margin values of the transmit spectrum mask for SISO measurements and bandwidths with one segment.

Margins for the current, average, minimum and maximum traces are returned. A positive result indicates that the trace is located above the limit line. The limit is exceeded.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<OutOfTol> float

Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.

|                           |                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Margin_AB> ...           | float                                                                                                                                                                                                                    |
| <Margin_BA>               | Comma-separated list of margin values, one value per spectrum mask area<br>The number of margin values depends on the selected standard, see <a href="#">Table B-1</a> .<br>Range: -100 dB to 100 dB<br>Default unit: dB |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements (OFDM, SISO)</a>                                                                                                                                                     |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                               |
| <b>Firmware/software:</b> | V3.5.121                                                                                                                                                                                                                 |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:MAXimum?**

Return the limit line margin values of the transmit spectrum mask for MIMO measurements, antenna/stream number <n>, bandwidths with one segment.

Margins for the current, average, minimum and maximum traces are returned. A positive result indicates that the trace is located above the limit line. The limit is exceeded.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<OutOfTol> float  
Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.  
Range: 0 % to 100 %  
Default unit: %

|                           |                                                                                                                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Margin_AB> ...           | float                                                                                                                                                                                                                    |
| <Margin_BA>               | Comma-separated list of margin values, one value per spectrum mask area<br>The number of margin values depends on the selected standard, see <a href="#">Table B-1</a> .<br>Range: -100 dB to 100 dB<br>Default unit: dB |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements (OFDM, CMIMO)</a>                                                                                                                                                    |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                               |
| <b>Firmware/software:</b> | V3.5.121                                                                                                                                                                                                                 |

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**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:MAXimum?**

Return the limit line margin values of the transmit spectrum mask, for SISO measurements and bandwidths with two segments.

Margins for the current, average, minimum and maximum traces are returned. A positive result indicates that the trace is located above the limit line. The limit is exceeded.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

|                 |                                                                                       |
|-----------------|---------------------------------------------------------------------------------------|
| <Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                      |
| <OutOfTol_Seg1> | float<br>Out of tolerance result, segment 1<br>Range: 0 % to 100 %<br>Default unit: % |
| <OutOfTol_Seg2> | float<br>Out of tolerance result, segment 2<br>Range: 0 % to 100 %<br>Default unit: % |

<Margin\_AB\_S1> float  
 <Margin\_AB\_S2> ... Comma-separated list of margin values, one value per spectrum  
 <Margin\_BA\_S1> mask area and segment  
 <Margin\_BA\_S2> The number of spectrum mask areas depends on the selected  
 standard, see [Table B-1](#).  
 Range: -100 dB to 100 dB  
 Default unit: dB  
**Usage:** Query only  
**Firmware/software:** V3.5.121

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:CURRENT?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:AVERage?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MINimum?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:AVERage?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MAXimum?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:CURRENT?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:AVERage?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MINimum?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:MAXimum?**

Return the limit line margin values of the transmit spectrum mask for MIMO measurements, antenna/stream number <n>, bandwidths with two segments.

Margins for the current, average, minimum and maximum traces are returned. A positive result indicates that the trace is located above the limit line. The limit is exceeded.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<OutOfTol\_Seg1> float  
 Out of tolerance result, segment 1  
 Range: 0 % to 100 %  
 Default unit: %



|                                                                                                                |                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;OutOfTol_Seg2&gt;</b>                                                                                   | float<br>Out of tolerance result, segment 2<br>Range: 0 % to 100 %<br>Default unit: %                                                                                                                                                               |
| <b>&lt;Margin_AB_S1&gt;,<br/>&lt;Margin_AB_S2&gt;, ...,<br/>&lt;Margin_BA_S1&gt;,<br/>&lt;Margin_BA_S2&gt;</b> | float<br>Comma-separated list of margin values, one value per spectrum mask area and segment<br>The number of spectrum mask areas depends on the selected standard, see <a href="#">Table B-1</a> .<br>Range: -100 dB to 100 dB<br>Default unit: dB |
| <b>Example:</b>                                                                                                | See <a href="#">Performing single-shot measurements (OFDM, MIMO, 80+80 MHz)</a>                                                                                                                                                                     |
| <b>Usage:</b>                                                                                                  | Query only                                                                                                                                                                                                                                          |
| <b>Firmware/software:</b>                                                                                      | V3.5.121                                                                                                                                                                                                                                            |

### 3.5.5.15 Transmit spectrum mask margin positions

The commands in this section return the limit line margin positions (X-values) of the transmit spectrum mask measurement.

For MIMO, use the commands with `MIMO<n>` mnemonic.

For bandwidths with two segments, use the commands with `SEGMENTS` mnemonic.

See also "[Transmit spectrum mask DSSS](#)" on page 316 and subsequent sections.

|                                                                                                  |     |
|--------------------------------------------------------------------------------------------------|-----|
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| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:AVERage?</a>               | 493 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:MINimum?</a>               | 493 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:MAXimum?</a>               | 493 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:CURRent?</a>                   | 493 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:AVERage?</a>                   | 493 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:MINimum?</a>                   | 493 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:MAXimum?</a>                   | 493 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:CURRent?</a>                    | 493 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:AVERage?</a>                    | 493 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:MINimum?</a>                    | 494 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:FREQuency:MAXimum?</a>                    | 494 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:CURRent?</a> | 494 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:AVERage?</a> | 494 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:MINimum?</a> | 494 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:MAXimum?</a> | 494 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:CURRent?</a>     | 494 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:AVERage?</a>     | 494 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:MINimum?</a>     | 494 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:MAXimum?</a>     | 494 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:MIMO&lt;n&gt;:FREQuency:CURRent?</a>      | 494 |



|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:AVERage?               | 494 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MINimum?               | 495 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MAXimum?               | 495 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:CURRent?         | 495 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:AVERage?         | 495 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MINimum?         | 495 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MAXimum?         | 495 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:CURRent?             | 495 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:AVERage?             | 495 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MINimum?             | 495 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MAXimum?             | 495 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:CURRent?              | 495 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:AVERage?              | 495 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MINimum?              | 496 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MAXimum?              | 496 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:CURRent? | 496 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:AVERage? | 496 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MINimum? | 496 |
| CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MAXimum? | 496 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:CURRent?     | 496 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:AVERage?     | 496 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MINimum?     | 496 |
| FETCh:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MAXimum?     | 497 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:CURRent?      | 497 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:AVERage?      | 497 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MINimum?      | 497 |
| READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MAXimum?      | 497 |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:AVERage?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:FREQuency:MAXimum?**

Return the X-positions of the limit line margins of the transmit spectrum mask.

For MIMO measurements, antenna/stream number <n>, bandwidths with one segment.

Positions for the current, average, minimum and maximum traces are returned.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

|                                |                                                                                                                                                                                                                                                                |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>                  | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                               |
| <OutOfTol>                     | float<br>Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.<br>Range: 0 % to 100 %<br>Default unit: %                          |
| <Margin_AB> ...<br><Margin_BA> | float<br>Comma-separated list of frequencies, one value per margin<br>The number of margins equals the number of spectrum mask areas and depends on the selected standard, see <a href="#">Table B-1</a> .<br>Range: -40E+6 Hz to 40E+6 Hz<br>Default unit: Hz |

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:CURRENT?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:AVERAge?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MINimum?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:AVERAge?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MAXimum?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:CURRENT?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:AVERAge?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:FREQuency:MAXimum?**

Return the X-positions of the limit line margins of the transmit spectrum mask, for MIMO measurements, antenna/stream number <n>, bandwidths with one segment.

Positions for the current, average, minimum and maximum traces are returned.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<OutOfTol> float

Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.

Range: 0 % to 100 %

<Margin\_AB> ... float

<Margin\_BA>

Comma-separated list of frequencies, one value per margin  
 The number of margins equals the number of spectrum mask areas and depends on the selected standard, see [Table B-1](#).

Range: -40E+6 Hz to 40E+6 Hz

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:CURRENT?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:AVERAge?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MINimum?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:AVERAge?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MAXimum?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:CURRENT?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:AVERAge?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:SEGMents:FREQuency:MAXimum?**

Return the X-positions of the limit line margins of the transmit spectrum mask, for SISO measurements, for bandwidths with two segments.

Positions for the current, average, minimum and maximum traces are returned.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

|                    |                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>      | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                         |
| <OutOfTol_Seg1>    | float<br>Out of tolerance result, segment 1<br>Range: 0 % to 100 %<br>Default unit: %                                                                                                    |
| <OutOfTol_Seg2>    | float<br>Out of tolerance result, segment 2<br>Range: 0 % to 100 %<br>Default unit: %                                                                                                    |
| <Margin_AB_S1>     | float                                                                                                                                                                                    |
| <Margin_AB_S2> ... | Comma-separated list of frequencies, one value per margin and segment                                                                                                                    |
| <Margin_BA_S1>     |                                                                                                                                                                                          |
| <Margin_BA_S2>     | The number of margins equals the number of spectrum mask areas and depends on the selected standard, see <a href="#">Table B-1</a> .<br>Range: -40E+6 Hz to 40E+6 Hz<br>Default unit: Hz |

**Usage:** Query only

**Firmware/software:** V3.5.121

---

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:CURRENT?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:AVERAge?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MINimum?**

**CALCulate:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:AVERAge?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MAXimum?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:CURRENT?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:AVERAge?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MINimum?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:MIMO<n>:SEGMents:FREQuency:MAXimum?**

Return the X-positions of the limit line margins of the transmit spectrum mask, for MIMO measurements, antenna/stream number <n>, bandwidths with two segments.

Positions for the current, average, minimum and maximum traces are returned.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<OutOfTol\_Seg1> float

Out of tolerance result for segment 1, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.

Range: 0 % to 100 %

Default unit: %

<OutOfTol\_Seg2> float

Out of tolerance result for segment 2, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.

Range: 0 % to 100 %

Default unit: %

<Margin\_AB\_S1> float

<Margin\_AB\_S2> ... Comma-separated list of frequencies, one value per margin and segment

<Margin\_BA\_S1> The number of margins equals the number of spectrum mask areas and depends on the selected standard, see [Table B-1](#).

<Margin\_BA\_S2> Range: -40E+6 Hz to 40E+6 Hz

Default unit: Hz

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

### 3.5.5.16 Transmit spectrum OBW results

The following commands return the occupied bandwidth (OBW) results.

See also ["Occupied bandwidth"](#) on page 267

|                                                                                         |     |
|-----------------------------------------------------------------------------------------|-----|
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW?</a>                        | 498 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW?</a>                         | 498 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW:MIMO&lt;n&gt;?</a>          | 498 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW:MIMO&lt;n&gt;?</a>           | 498 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW:SEGMents?</a>               | 499 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW:SEGMents?</a>                | 499 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW:MIMO&lt;n&gt;:SEGMents?</a> | 500 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:TSMask:OBW:MIMO&lt;n&gt;:SEGMents?</a>  | 500 |

---

**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:OBW?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:OBW?**

Return the OBW results for SISO measurements and bandwidths with one segment.

**Return values:**

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| <1_Reliability> | decimal<br><a href="#">Reliability indicator</a>          |
| <2_OBW_cur>     | float                                                     |
| <3_OBW_avg>     | Current, average, max and standard deviation OBW values   |
| <4_OBW_max>     | Default unit: Hz                                          |
| <5_OBW_sdev>    |                                                           |
| <6_OBW_left>    | float                                                     |
| <7_OBW_right>   | Lower and upper edge frequency of OBW<br>Default unit: Hz |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.2.11

---

**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:OBW:MIMO<n>?**

**READ:WLAN:MEAS<i>:MEValuation:TSMask:OBW:MIMO<n>?**

Return the OBW results for MIMO measurements, antenna/stream number <n>, bandwidths with one segment.

**Suffix:**

|     |      |
|-----|------|
| <n> | 1..8 |
|-----|------|

**Return values:**

|                 |                                                           |
|-----------------|-----------------------------------------------------------|
| <1_Reliability> | decimal<br><a href="#">Reliability indicator</a>          |
| <2_OBW_cur>     | float                                                     |
| <3_OBW_avg>     | Current, average, max and standard deviation OBW values   |
| <4_OBW_max>     | Default unit: Hz                                          |
| <5_OBW_sdev>    |                                                           |
| <6_OBW_left>    | float                                                     |
| <7_OBW_right>   | Lower and upper edge frequency of OBW<br>Default unit: Hz |

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

**FETCh:WLAN:MEAS<i>:MEValuation:TSMask:OBW:SEGMents?****READ:WLAN:MEAS<i>:MEValuation:TSMask:OBW:SEGMents?**

Return the OBW results for SISO measurements and bandwidths with two segments.

**Return values:**

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| <1_Reliability> | decimal<br><a href="#">Reliability indicator</a>                            |
| <2_OBWCurS1>    | float                                                                       |
| <3_OBWCurS2>    | Current, average, max and standard deviation OBW values for segment 1 and 2 |
| <4_OBWAvgS1>    | Default unit: Hz                                                            |
| <5_OBWAvgS2>    |                                                                             |
| <6_OBWMaxS1>    |                                                                             |
| <7_OBWMaxS2>    |                                                                             |
| <8_OBWStdDevS1> |                                                                             |
| <9_OBWStdDevS2> |                                                                             |
| <10_OBWleftS1>  | float                                                                       |
| <11_OBWleftS2>  | Lower edge frequency of OBW, segment 1 and 2<br>Default unit: Hz            |
| <12_OBWrightS1> | float                                                                       |
| <13_OBWrightS2> | Upper edge frequency of OBW, segment 1 and 2<br>Default unit: Hz            |

**Usage:** Query only

**Firmware/software:** V3.5.121

**FETCH:WLAN:MEAS<i>:MEValuation:TSMask:OBW:MIMO<n>:SEGMents?**  
**READ:WLAN:MEAS<i>:MEValuation:TSMask:OBW:MIMO<n>:SEGMents?**

Return the OBW results for MIMO measurements, antenna/stream number <n>, bandwidths with two segments.

**Suffix:**

<n> 1..8

**Return values:**

<1\_Reliability> decimal  
[Reliability indicator](#)

<2\_OBWCurS1> float  
 <3\_OBWCurS2> Current, average, max and standard deviation OBW values,  
 <4\_OBWAvgS1> segment 1 and 2  
 <5\_OBWAvgS2> Default unit: Hz  
 <6\_OBWMaxS1>  
 <7\_OBWMaxS2>  
 <8\_OBWStdDevS1>  
 <9\_OBWStdDevS2>

<10\_OBWleftS1> float  
 <11\_OBWleftS2> Lower edge frequency of OBW, segment 1 and 2  
 Default unit: Hz

<12\_OBWrightS1> float  
 <13\_OBWrightS2> Upper edge frequency of OBW, segment 1 and 2  
 Default unit: Hz

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121

### 3.5.5.17 Power vs. time burst traces

The following commands return the power vs. time burst trace results.

See also [Chapter 3.3.2.7, "View power vs. time"](#), on page 268

FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:CURRent?.....501  
 FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:AVERage?.....501  
 FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MINimum?.....501  
 FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MAXimum?.....501  
 READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:CURRent?.....501  
 READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:AVERage?.....502  
 READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MINimum?.....502  
 READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MAXimum?.....502  
 FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:CURRent?.....502  
 FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:AVERage?.....502  
 FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MINimum?.....502



|                                                                               |     |
|-------------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MAXimum?                  | 502 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:CURRent?                   | 502 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:AVERage?                   | 502 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MINimum?                   | 502 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MAXimum?                   | 502 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:CURRent?             | 503 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:AVERage?             | 503 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MINimum?             | 503 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MAXimum?             | 503 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:CURRent?              | 503 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:AVERage?              | 503 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MINimum?              | 503 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MAXimum?              | 503 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>CURRent? | 504 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>AVERage? | 504 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>MINimum? | 504 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>MAXimum? | 504 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>CURRent?  | 504 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>AVERage?  | 504 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>MINimum?  | 504 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:<br>MAXimum?  | 504 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:TIME?                             | 505 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:TIME?                              | 505 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:TIME?                     | 505 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:TIME?                      | 505 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:TIME?                | 506 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:TIME?                 | 506 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:TIME?        | 506 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMENT<seg>:TIME?         | 506 |

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:CURRent?** [<start>,  
<count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:AVERage?** [<start>,  
<count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MINimum?** [<start>,  
<count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MAXimum?** [<start>,  
<count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:CURRent?** [<start>, <count>[,  
<decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:AVERage?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MINimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MAXimum?** [<start>, <count>[, <decimation>]]

Return the values of the power vs. time traces for SISO connections. The results of the current, average, maximum and minimum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                        |
| <Power>       | float<br>Comma-separated list of max 1024 time values (1024 values without subarrays)<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm |

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V2.1.10, V3.2.11 subarrays

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:CURRent?** [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:AVERage?** [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MINimum?** [<start>, <count>[, <decimation>]]

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MAXimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:CURRent?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:AVERage?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MINimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:MAXimum?** [<start>, <count>[, <decimation>]]

Return the values of the power vs. time traces for MIMO. The results of the current, average, maximum and minimum traces can be retrieved.

**Suffix:**

|     |                 |
|-----|-----------------|
| <n> | 1..8<br>Antenna |
|-----|-----------------|

**Query parameters:**

<start>, <count>,      numeric  
 <decimation>      For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability>      decimal  
                      [Reliability indicator](#)

<Power>      float  
                      Comma-separated list of power values  
                      Range:      -100 dBm to 30 dBm  
                      Default unit: dBm

**Usage:**      Query only

**Firmware/software:** V3.7.40

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:CURRent?**  
 [<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:AVERAge?**  
 [<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MINimum?**  
 [<start>, <count>[, <decimation>]]

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MAXimum?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:CURRent?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:AVERAge?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MINimum?**  
 [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMENT<seg>:MAXimum?**  
 [<start>, <count>[, <decimation>]]

Return the values of the power vs. time traces for 80+80 MHz SISO connections. The results of the current, average, maximum and minimum traces can be retrieved.

**Suffix:**

<seg>      1..2  
                      Segment

**Query parameters:**

<start>, <count>,      numeric  
 <decimation>      For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability>      decimal  
                      [Reliability indicator](#)

|                           |                                                                            |
|---------------------------|----------------------------------------------------------------------------|
| <Power>                   | float<br>Comma-separated list of power values<br>Range: -100 dBm to 30 dBm |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements (DSSS)</a>             |
| <b>Usage:</b>             | Query only                                                                 |
| <b>Firmware/software:</b> | V3.7.40                                                                    |

---

```

FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  CURRent? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  CURRent? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]

```

Return the values of the power vs. time traces for 80+80 MHz MIMO. The results of the current, average, maximum and minimum traces can be retrieved.

**Suffix:**

|       |                 |
|-------|-----------------|
| <n>   | 1..8<br>Antenna |
| <seg> | 1..2<br>Segment |

**Query parameters:**

|                                   |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <start>, <count>,<br><decimation> | numeric<br>For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

**Return values:**

|               |                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                           |
| <Power>       | float<br>Comma-separated list of max 1024 time values (1024 values without subarrays)<br>Default unit: dBm |
| <b>Usage:</b> | Query only                                                                                                 |

**Firmware/software:** V3.7.40

---

**FETCH:WLAN:MEAS<i>:MEvaluation:TRACe:PVTime:TIME?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEvaluation:TRACe:PVTime:TIME?** [<start>, <count>[, <decimation>]]

Return the time indices for the current, average, minimum and maximum power vs. time traces, see [FETCH:WLAN:MEAS<i>:MEvaluation:TRACe:PVTime:CURRent?](#) etc.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                          |
| <TimeValues>  | float<br>Comma-separated list of max 1024 time values (1024 values without subarrays)<br>Default unit: µs |

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.20

---

**FETCH:WLAN:MEAS<i>:MEvaluation:TRACe:PVTime:MIMO<n>:TIME?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEvaluation:TRACe:PVTime:MIMO<n>:TIME?** [<start>, <count>[, <decimation>]]

Return the time indices for the current, average, minimum and maximum power vs. time traces (for MIMO), see [FETCH:WLAN:MEAS<i>:MEvaluation:TRACe:PVTime:MIMO<n>:CURRent?](#) etc.

**Suffix:**

|     |                 |
|-----|-----------------|
| <n> | 1..8<br>Antenna |
|-----|-----------------|

**Query parameters:**

|                                   |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <start>, <count>,<br><decimation> | numeric<br>For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a> |
|---------------|--------------------------------------------------|

<TimeValues> float  
 Comma-separated list of max 1024 time values (1024 values without subarrays)  
 Default unit: µs

**Usage:** Query only

**Firmware/software:** V3.7.40

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMent<seg>:TIME?**

[<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMent<seg>:TIME?**

[<start>, <count>[, <decimation>]]

Return the time indices for the current, average, minimum and maximum power vs. time traces (for 80+80 MHz), see [FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:SEGMent<seg>:CURRent?](#) etc.

**Suffix:**

<seg> 1..2  
 Segment

**Query parameters:**

<start>, <count>, numeric  
 <decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<TimeValues> float  
 Comma-separated list of max 1024 time values (1024 values without subarrays)  
 Default unit: µs

**Usage:** Query only

**Firmware/software:** V3.7.40

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:TIME?** [<start>, <count>[, <decimation>]]

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:TIME?** [<start>, <count>[, <decimation>]]

Return the time indices for the current, average, minimum and maximum power vs. time traces (for 80+80 MHz, MIMO), see [FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:MIMO<n>:SEGMent<seg>:CURRent?](#) etc.

**Suffix:**

<n> 1..8  
 Antenna

|                                   |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <seg>                             | 1..2<br>Segment                                                                                                                                        |
| <b>Query parameters:</b>          |                                                                                                                                                        |
| <start>, <count>,<br><decimation> | numeric<br>For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |
| <b>Return values:</b>             |                                                                                                                                                        |
| <Reliability>                     | decimal<br><a href="#">Reliability indicator</a>                                                                                                       |
| <TimeValues>                      | float<br>Comma-separated list of max 1024 time values (1024 values without subarrays)<br>Default unit: µs                                              |
| <b>Usage:</b>                     | Query only                                                                                                                                             |
| <b>Firmware/software:</b>         | V3.7.40                                                                                                                                                |

### 3.5.5.18 Power vs. time ramp traces

The following commands return the power vs. time ramp trace results.

See also [Chapter 3.3.2.7, "View power vs. time"](#), on page 268

|                                                                         |     |
|-------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:CURRent?              | 509 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:AVERage?              | 509 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MINimum?              | 509 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MAXimum?              | 509 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:CURRent?               | 509 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:AVERage?               | 509 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MINimum?               | 509 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MAXimum?               | 509 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:CURRent?              | 509 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:AVERage?              | 509 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MINimum?              | 510 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MAXimum?              | 510 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:CURRent?               | 510 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:AVERage?               | 510 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MINimum?               | 510 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MAXimum?               | 510 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:TIME?                 | 510 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:TIME?                 | 510 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:TIME?                  | 510 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:TIME?                  | 510 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:CURRent? | 511 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:AVERage? | 511 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:MINimum? | 511 |
| FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:MAXimum? | 511 |



|                                                                                     |     |
|-------------------------------------------------------------------------------------|-----|
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:CURRENT?              | 511 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:AVERAGE?              | 511 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:MINIMUM?              | 511 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:MAXIMUM?              | 511 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:CURRENT?             | 511 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:AVERAGE?             | 511 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:MINIMUM?             | 511 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:<br>MAXIMUM?         | 511 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:CURRENT?              | 511 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:AVERAGE?              | 511 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:MINIMUM?              | 512 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:MAXIMUM?              | 512 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:TIME?                | 512 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:TIME?                | 512 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:TIME?                 | 512 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:SEGMENT<seg>:TIME?                 | 512 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:CURRENT?                  | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:AVERAGE?                  | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:MINIMUM?                  | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:MAXIMUM?                  | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:CURRENT?                   | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:AVERAGE?                   | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:MINIMUM?                   | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:MAXIMUM?                   | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:CURRENT?                  | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:AVERAGE?                  | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:MINIMUM?                  | 513 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:MAXIMUM?                  | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:CURRENT?                   | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:AVERAGE?                   | 513 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:MINIMUM?                   | 514 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:MAXIMUM?                   | 514 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:TIME?                     | 514 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:TIME?                     | 514 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:TIME?                      | 514 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:TIME?                      | 514 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:CURRENT? | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:AVERAGE? | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:MINIMUM? | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:MAXIMUM? | 515 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:CURRENT?  | 515 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:AVERAGE?  | 515 |



|                                                                                     |     |
|-------------------------------------------------------------------------------------|-----|
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:MINimum?  | 515 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:MAXimum?  | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:CURRent? | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:AVERage? | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:MINimum? | 515 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:MAXimum? | 515 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:CURRent?  | 515 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:AVERage?  | 515 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:MINimum?  | 516 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:MAXimum?  | 516 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:TIME?    | 516 |
| FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:TIME?    | 516 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:<br>SEGMENT<seg>:TIME?     | 516 |
| READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:<br>SEGMENT<seg>:TIME?     | 516 |

---

```

FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:CURRent? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:AVERage? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MINimum? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MAXimum?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:CURRent? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:AVERage? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MINimum? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MAXimum? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:CURRent? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:AVERage? [<start>,
    <count>[, <decimation>]]

```

```

FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MINimum? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MAXimum?
    [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:CURREnt? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:AVERAge? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MINimum? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MAXimum? [<start>,
    <count>[, <decimation>]]

```

Return the values of the power vs. time ramp traces, falling edge (FEDGE) and rising edge (REDGe). The results of the current, average, maximum and minimum traces can be retrieved.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|               |                                                                                                                                                                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                                                  |
| <Power>       | float<br>Comma-separated list of power values<br>For DSSS signals, the values are returned in percent relative to the configured reference power.<br>For OFDM signals, the values are in dBm.<br>Range: 0 % to 200 % (DSSS) / -100 dBm to 30 dBm (OFDM)<br>Default unit: % or dBm |

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V2.1.10, V3.2.11 subarrays

---

```

FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:TIME? [<start>,
    <count>[, <decimation>]]
FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:TIME? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:TIME? [<start>,
    <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:TIME? [<start>,
    <count>[, <decimation>]]

```

Return the time indices corresponding to the power vs. time ramp traces, falling edge (FEDGE) and rising edge (REDGe). Refer to

[FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:CURREnt?](#) etc.

For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                          |
| <TimeValues>  | float<br>Comma-separated list of time indices corresponding to the ramp power results.<br>Default unit: s |

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.60

---

```

FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  CURRent? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  CURRent? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:
  CURRent? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:
  MINimum? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:
  MAXimum? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:
  CURRent? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:
  AVERAge? [<start>, <count>[, <decimation>]]

```

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:  
MINimum? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:  
MAXimum? [<start>, <count>[, <decimation>]]**

Return the values of the power vs. time ramp traces (for 80+80 MHz), rising edge (REDGe) and falling edge (FEDGE). The results of the current, average, maximum and minimum traces can be retrieved.

**Suffix:**

<seg> 1..2  
Segment

**Query parameters:**

<start>, <count>, numeric  
<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<Power> float  
Comma-separated list of power values  
Range: -100 dBm to 30 dBm  
Default unit: dBm

**Usage:** Query only

**Firmware/software:** V3.7.40

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:  
TIME? [<start>, <count>[, <decimation>]]**

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:  
TIME? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:  
TIME? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:SEGMENT<seg>:  
TIME? [<start>, <count>[, <decimation>]]**

Return the time indices for the power vs. time ramp traces (for 80+80 MHz), rising edge (REDGe) and falling edge (FEDGE). Refer to

[FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:SEGMENT<seg>:  
CURRent? etc.](#)

**Suffix:**

<seg> 1..2  
Segment

**Query parameters:**

<start> <count> numeric  
 <decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 <TimeValues> float  
 Comma-separated list of time indices corresponding to the ramp power results.  
 Default unit: s

**Usage:** Query only

**Firmware/software:** V3.7.60

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 CURRent? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 AVERAge? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 MINimum? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 MAXimum? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 CURRent? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 AVERAge? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 MINimum? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
 MAXimum? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
 CURRent? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
 AVERAge? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
 MINimum? [<start>, <count>[, <decimation>]]**  
**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
 MAXimum? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
 CURRent? [<start>, <count>[, <decimation>]]**  
**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
 AVERAge? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
MINimum? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
MAXimum? [<start>, <count>[, <decimation>]]**

Return the values of the power vs. time ramp traces for MIMO, rising edge (REDGe) and falling edge (FEDGE). The results of the current, average, maximum and minimum traces can be retrieved.

**Suffix:**

<n> 1..8  
Antenna

**Query parameters:**

<start>, <count>, numeric  
<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<Power> float  
Comma-separated list of power values  
For DSSS signals, the values are returned in percent relative to the configured reference power.  
For OFDM signals, the values are in dBm.  
Range: 0 % to 200 % (DSSS) / -100 dBm to 30 dBm (OFDM)  
Default unit: % or dBm

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.40

---

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:TIME?  
[<start>, <count>[, <decimation>]]**

**FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:TIME?  
[<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:TIME?  
[<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:TIME?  
[<start>, <count>[, <decimation>]]**

Return the time indices for the power vs. time ramp traces for MIMO, rising edge (REDGe) and falling edge (FEDGE). Refer to

[FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
CURRent?](#) etc.

**Suffix:**

<n> 1..8  
Antenna

**Query parameters:**

<start> <count> numeric  
<decimation> For the optional query parameters <start>, <count> and <decimation>, see [Chapter 3.5.1.4, "Trace Sub-Arrays"](#), on page 349.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<TimeValues> float  
Comma-separated list of time indices corresponding to the ramp power results.  
Default unit: s

**Usage:** Query only

**Firmware/software:** V3.7.60

---

```

FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:AVERAGE? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:MINIMUM? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:MAXIMUM? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:AVERAGE? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:MINIMUM? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:
    SEGMENT<seg>:MAXIMUM? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:
    SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:
    SEGMENT<seg>:AVERAGE? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:
    SEGMENT<seg>:MINIMUM? [<start>, <count>[, <decimation>]]
FETCh:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:
    SEGMENT<seg>:MAXIMUM? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:
    SEGMENT<seg>:CURRENT? [<start>, <count>[, <decimation>]]
READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGE:MIMO<n>:
    SEGMENT<seg>:AVERAGE? [<start>, <count>[, <decimation>]]

```



**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
SEGMENT<seg>:MINimum? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
SEGMENT<seg>:MAXimum? [<start>, <count>[, <decimation>]]**

Return the values of the power vs. time ramp traces (for 80+80 MHz, MIMO), rising edge (REDGe) and falling edge (FEDGE). The results of the current, average, maximum and minimum traces can be retrieved.

**Suffix:**

|       |                 |
|-------|-----------------|
| <n>   | 1..8<br>Antenna |
| <seg> | 1..2<br>Segment |

**Query parameters:**

|                                   |                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <start>, <count>,<br><decimation> | numeric<br>For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

**Return values:**

|               |                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                |
| <Power>       | float<br>Comma-separated list of power values<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm |

**Usage:** Query only

**Firmware/software:** V3.7.40

---

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
SEGMENT<seg>:TIME? [<start>, <count>[, <decimation>]]**

**FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
SEGMENT<seg>:TIME? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:FEDGE:MIMO<n>:  
SEGMENT<seg>:TIME? [<start>, <count>[, <decimation>]]**

**READ:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
SEGMENT<seg>:TIME? [<start>, <count>[, <decimation>]]**

Return the time indices for the power vs. time ramp traces (for 80+80 MHz, MIMO), rising edge (REDGe) and falling edge (FEDGE). Refer to

[FETCH:WLAN:MEAS<i>:MEValuation:TRACe:PVTime:REDGe:MIMO<n>:  
SEGMENT<seg>:CURRENT?](#)

**Suffix:**

|     |                 |
|-----|-----------------|
| <n> | 1..8<br>Antenna |
|-----|-----------------|



|                           |                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <seg>                     | 1..2<br>Segment                                                                                                                             |
| <b>Query parameters:</b>  |                                                                                                                                             |
| <start> <count>           | numeric                                                                                                                                     |
| <decimation>              | For the optional query parameters <start>, <count> and <decimation>, see <a href="#">Chapter 3.5.1.4, "Trace Sub-Arrays"</a> , on page 349. |
| <b>Return values:</b>     |                                                                                                                                             |
| <Reliability>             | decimal<br><a href="#">Reliability indicator</a>                                                                                            |
| <TimeValues>              | float<br>Comma-separated list of time indices corresponding to the ramp power results.<br>Default unit: s                                   |
| <b>Usage:</b>             | Query only                                                                                                                                  |
| <b>Firmware/software:</b> | V3.7.60                                                                                                                                     |

### 3.5.5.19 Power vs. time single values

The following commands return the statistical power vs. time results.

See also [Chapter 3.3.2.7, "View power vs. time"](#), on page 268

|                                                            |     |
|------------------------------------------------------------|-----|
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:CURRENT?   | 518 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:AVERage?   | 518 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:MAXimum?   | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:CURRENT?       | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:AVERage?       | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:MAXimum?       | 518 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:CURRENT?        | 518 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:AVERage?        | 518 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:MAXimum?        | 518 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:REDge:CURRENT?   | 518 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:REDge:AVERage?   | 518 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:REDge:MAXimum?   | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:PVTime:REDge:CURRENT?       | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:PVTime:REDge:AVERage?       | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:PVTime:REDge:MAXimum?       | 518 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:REDge:CURRENT?        | 518 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:REDge:AVERage?        | 518 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:REDge:MAXimum?        | 518 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:CURRENT?    | 519 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:AVERage?    | 519 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:MAXimum?    | 519 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:SDEviation? | 519 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RXAntenna<n>:CURRENT? | 519 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RXAntenna<n>:AVERage? | 519 |
| FETCH:WLAN:MEAS<i>:MEValuation:POWer:RXAntenna<n>:MAXimum? | 519 |

|                                                                         |     |
|-------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:POWer:RXANtenna<n>:SDEViation?           | 519 |
| FETCh:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:RXANtenna<n>:CURRent?    | 520 |
| FETCh:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:RXANtenna<n>:AVERage?    | 520 |
| FETCh:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:RXANtenna<n>:MAXimum?    | 520 |
| FETCh:WLAN:MEAS<i>:MEValuation:POWer:RUNit<ru>:RXANtenna<n>:SDEViation? | 520 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:CURRent?               | 520 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:AVERage?               | 520 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MINimum?               | 520 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MAXimum?               | 521 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:SDEViation?            | 521 |
| FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:CURRent?                   | 521 |
| FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:AVERage?                   | 521 |
| FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MINimum?                   | 521 |
| FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MAXimum?                   | 521 |
| FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:SDEViation?                | 521 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:CURRent?                    | 521 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:AVERage?                    | 521 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MINimum?                    | 521 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MAXimum?                    | 521 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:SDEViation?                 | 521 |
| CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TEDistrib?                    | 521 |
| FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TEDistrib?                        | 521 |
| READ:WLAN:MEAS<i>:MEValuation:PVTime:TEDistrib?                         | 521 |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:FEDGE:MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:REDGe:MAXimum?**

Returns the current, average and maximum ramp durations of the power vs. time measurement, for the falling edge (FEDGE) and rising edge (REDGe).

The values described below are returned by FETCh and READ commands. CALCulate commands return limit check results instead, one value for each result listed below.

**Return values:**

|               |                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                     |
| <Time>        | float<br>Range: 0 µs to 10 µs<br>Default unit: µs                                                                                                                                                    |
| <OutOfTol>    | float<br>Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for PVT measurements exceeding the specified power limit<br>Range: 0 % to 100 %<br>Default unit: % |

**Example:** See [Performing single-shot measurements \(DSSS\)](#)

**Usage:** Query only

**Firmware/software:** V2.1.10, V2.1.30 out of tolerance percentage added

**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RUNit<ru>:CURRENT?**  
**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RUNit<ru>:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RUNit<ru>:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RUNit<ru>:SDEViation?**

Returns single power value measured for RU at all antennas (OFDMA).

**Suffix:**

|      |                        |
|------|------------------------|
| <ru> | 1..74<br>Resource unit |
|------|------------------------|

**Return values:**

|                   |                                                            |
|-------------------|------------------------------------------------------------|
| <Reliability>     | decimal<br><a href="#">Reliability indicator</a>           |
| <PowerVsRUallAnt> | float<br>Power vs RU for all antennas<br>Default unit: dBm |

**Usage:** Query only

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KM657

**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RXAntenna<n>:CURRENT?**  
**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RXAntenna<n>:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RXAntenna<n>:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:POWER:RXAntenna<n>:SDEViation?**

Returns single power value measured at the specified antenna for all RUs (OFDMA).

**Suffix:**

<n> 1..8  
Antenna

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<PowVsAntAllRUs> float  
Power vs antenna for all RUs  
Default unit: dBm

**Usage:** Query only

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KM657

---

**FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUit<ru>:RXAntenna<n>:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUit<ru>:RXAntenna<n>:AVERage?**

**FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUit<ru>:RXAntenna<n>:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:POWer:RUit<ru>:RXAntenna<n>:SDEViation?**

Returns single power value measured for RU at the specified antenna (OFDMA).

**Suffix:**

<ru> 1..74  
Resource unit

<n> 1..8  
Antenna

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<PowVsAntVsRU> float  
Power vs RU vs antenna  
Default unit: dBm

**Usage:** Query only

**Firmware/software:** V3.7.40

**Options:** R&S CMW-KM657

---

**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:CURRENT?**

**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:AVERage?**

**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MINimum?**

**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:SDEVIation?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:SDEVIation?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:TERRor:SDEVIation?**

Return the current, average, minimum, maximum and standard deviation timing error single value results of the power vs. time measurement.

The commands are only supported for OFDM standards.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<TimingError> float  
 Default unit: s

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.20

**CALCulate:WLAN:MEAS<i>:MEValuation:PVTime:TEDistrib?**  
**FETCh:WLAN:MEAS<i>:MEValuation:PVTime:TEDistrib?**  
**READ:WLAN:MEAS<i>:MEValuation:PVTime:TEDistrib?**

Return the scalar results for timing error (TE) distribution.

The commands are only supported for OFDM standards.

Exceeding the limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<TE\_Percentage> float  
 Percentage of TEs  
 Default unit: %

<TE\_Outside> decimal  
 Number of detected TEs

|                           |                                                                      |
|---------------------------|----------------------------------------------------------------------|
| <TE_Total>                | decimal<br>Number of measured values                                 |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements (OFDM, SISO)</a> |
| <b>Usage:</b>             | Query only                                                           |
| <b>Firmware/software:</b> | V3.7.50                                                              |

### 3.5.5.20 Modulation single values, DSSS

The following commands return statistical modulation results for DSSS signals.

See also [Chapter 3.3.2.8, "View TX measurement \(scalar\) for DSSS"](#), on page 271

---

```

CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:CURRent?
CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:AVERAge?
CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:MINimum?
CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:MAXimum?
CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:SDEViation?
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:AVERAge?
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:MINimum?
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:MAXimum?
FETCh:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:SDEViation?
READ:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:CURRent?
READ:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:AVERAge?
READ:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:MINimum?
READ:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:MAXimum?
READ:WLAN:MEAS<i>:MEValuation:MODulation:DSSS:SDEViation?

```

Return the current, average, minimum, maximum and standard deviation single value results for DSSS signals.

The values described below are returned by `FETCh` and `READ` commands. `CALCulate` commands return limit check results instead, one value for each result listed below.

**Return values:**

|                   |                                                                                                                                                        |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <1_Reliability>   | decimal<br><a href="#">Reliability indicator</a>                                                                                                       |
| <2_ModType>       | DBPSk1   DQPSk2   CCK5   CCK11<br><b>DBPSk1:</b> 1 Mbps DBPSK<br><b>DQPSk2:</b> 2 Mbps DQPSK<br><b>CCK5:</b> 5.5 Mbps CCK<br><b>CCK11:</b> 11 Mbps CCK |
| <3_PLCPType>      | SHORTplcp   LONGplcp<br>Short or long PLCP                                                                                                             |
| <4_PayloadLength> | decimal<br>Range: 1 byte to 4095 bytes<br>Default unit: byte                                                                                           |

|                |                                                                                                                                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <5_BurstPower> | float<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                                                            |
| <6_EVMPeak>    | float<br>Error vector magnitude peak value<br>Range: 0 % to 100 %<br>Default unit: %                                                                                                                               |
| <7_EVMRMS>     | float<br>Error vector magnitude RMS value<br>Range: 0 % to 100 %<br>Default unit: %                                                                                                                                |
| <8_FreqError>  | float<br>Center frequency error<br>Range: -150E+6 Hz to 150E+6 Hz<br>Hz                                                                                                                                            |
| <9_ClockError> | float<br>Chip clock error<br>Range: -125 ppm to 125 ppm<br>Default unit: ppm                                                                                                                                       |
| <10_IQOffset>  | float<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                                                                                                                |
| <11_GainImbal> | float<br>Gain imbalance<br>Range: -100 dB to 100 dB<br>Default unit: dB                                                                                                                                            |
| <12_QuadError> | float<br>Quadrature error<br>Range: -180 deg to 180 deg<br>Default unit: deg                                                                                                                                       |
| <13_OutOfTol>  | float<br>Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits.<br>Range: 0 % to 100 %<br>Default unit: % |

|                             |                                                                                                                                                               |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;14_BurstRate&gt;</b> | float                                                                                                                                                         |
|                             | If a modulation filter is used, the burst rate indicates the share of bursts of the selected modulation type in the bursts received. Otherwise, it returns 1. |
|                             | See <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:ISIGnal:MODFilter</a> .                                                                                          |
|                             | Default unit: %                                                                                                                                               |
| <b>Example:</b>             | See <a href="#">Performing single-shot measurements (DSSS)</a>                                                                                                |
| <b>Usage:</b>               | Query only                                                                                                                                                    |
| <b>Firmware/software:</b>   | V1.0.10.1, V3.5.10 burst rate, V3.5.121 MINimum commands                                                                                                      |

### 3.5.5.21 Modulation single values, OFDM

The commands in this section return the statistical modulation results for OFDM signals.

The commands `...:MODulation:CURRent/...` query results that are antenna-independent, stream-independent and segment-independent.

For MIMO results per antenna/stream, use the commands with `MIMO<n>` mnemonic.

For results per segment (80+80 MHz signals), use the commands with `SEGments` mnemonic.

See also:

- [Chapter 3.3.2.9, "View TX measurement \(scalar\) for OFDM"](#), on page 272
- [Chapter 3.3.2.10, "View TX measurement \(scalar\) for CMIMO \(802.11n, ac, ax\)"](#), on page 275
- [Chapter 3.3.2.11, "View TX measurement \(scalar\) for SMIMO, TMIMO \(802.11n, ac, ax\)"](#), on page 276

|                                                                                     |     |
|-------------------------------------------------------------------------------------|-----|
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CURRent?</a>        | 526 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:AVERage?</a>        | 526 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MINimum?</a>        | 526 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MAXimum?</a>        | 526 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:SDEViation?</a>     | 526 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CURRent?</a>            | 526 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:AVERage?</a>            | 526 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MINimum?</a>            | 526 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MAXimum?</a>            | 526 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:SDEViation?</a>         | 526 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CURRent?</a>             | 526 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:AVERage?</a>             | 526 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MINimum?</a>             | 526 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MAXimum?</a>             | 526 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:SDEViation?</a>          | 526 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CMIMo:PSTS:CURRent?</a> | 529 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CMIMo:PSTS:AVERage?</a> | 529 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CMIMo:PSTS:MINimum?</a> | 529 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CMIMo:PSTS:MAXimum?</a> | 530 |



|                                                                            |     |
|----------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:SDEViation?           | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:CURREnt?               | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:AVERage?               | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:MINimum?               | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:MAXimum?               | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:SDEViation?            | 530 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:CURREnt?             | 530 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:AVERage?             | 530 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MINimum?             | 530 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MAXimum?             | 530 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SDEViation?          | 530 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:CURREnt?                 | 530 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:AVERage?                 | 530 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MINimum?                 | 530 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MAXimum?                 | 530 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SDEViation?              | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:CURREnt?                  | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:AVERage?                  | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MINimum?                  | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MAXimum?                  | 530 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SDEViation?               | 530 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:CURREnt?            | 533 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:AVERage?            | 533 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MINimum?            | 533 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MAXimum?            | 533 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:SDEViation?         | 533 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:CURREnt?                | 533 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:AVERage?                | 533 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MINimum?                | 533 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MAXimum?                | 533 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:SDEViation?             | 533 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:CURREnt?                 | 533 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:AVERage?                 | 533 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MINimum?                 | 533 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MAXimum?                 | 533 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:SDEViation?              | 533 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:CURREnt?    | 534 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:AVERage?    | 534 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:MINimum?    | 534 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:MAXimum?    | 534 |
| CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:SDEViation? | 534 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:CURREnt?        | 535 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:AVERage?        | 535 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:MINimum?        | 535 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:MAXimum?        | 535 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:SDEViation?     | 535 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:CURREnt?         | 535 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:AVERage?         | 535 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:MINimum?         | 535 |
| READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMENTS:MAXimum?         | 535 |

|                                                                                                         |     |
|---------------------------------------------------------------------------------------------------------|-----|
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:MIMO&lt;n&gt;:SEGments:SDEViation?</a> ..... | 535 |
| <a href="#">CALCulate:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CFDistrib?</a> .....                    | 536 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CFDistrib?</a> .....                        | 536 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:MEValuation:MODulation:CFDistrib?</a> .....                         | 536 |

---

**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SDEViation?**  
**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:SDEViation?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:SDEViation?**

Return the single value results for OFDM SISO measurements.

For MIMO measurements, the stream/antenna-independent values are returned.

For 80+80 MHz signals, the segment-independent values are returned. There are current, average, minimum, maximum and standard deviation results.

The values described below are returned by **FETCh** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

#### Return values:

|                 |                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <1_Reliability> | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                                                                                                                      |
| <2_OutOfTol>    | float<br>Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits.<br>Range: 0 % to 100 %<br>Default unit: %                                                                                    |
| <3_MCSIndex>    | decimal<br>Modulation and coding scheme index<br>Range: 0 to 76                                                                                                                                                                                                                                       |
| <4_Modulation>  | UNSPecified   BPSK14   BPSK12   BPSK34   QPSK14   QPSK12   QPSK34   16Q14   16Q38   16Q12   16Q34   64Q12   64Q23   64Q34   64Q56   256Q34   256Q56   1KQ34   1KQ56   BPSK   QPSK   16Q   64Q   256Q   1KQ<br>Modulation scheme and coding rate, stream <n><br><b>UNSPecified:</b> modulation unknown |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | <p><b>BPSK</b>: BPSK, coding rate unknown</p> <p><b>BPSK12, BPSK34</b> (BPSKab): BPSK, coding rate a/b</p> <p><b>BPSK14</b>: BPSK, coding rate 1/2 DCM</p> <p><b>QPSK</b>: QPSK, coding rate unknown</p> <p><b>QPSK12, QPSK34</b> (QPSKab): QPSK, coding rate a/b</p> <p><b>QPSK14</b>: QPSK, coding rate 1/2 DCM</p> <p><b>16Q</b>: 16-QAM, coding rate unknown</p> <p><b>16Q12, 16Q34</b> (16Qab): 16-QAM, coding rate a/b</p> <p><b>16Q14</b>: 16-QAM, coding rate 1/2 DCM</p> <p><b>16Q38</b>: 16-QAM, coding rate 3/4 DCM</p> <p><b>64Q</b>: 64-QAM, coding rate unknown</p> <p><b>64Q12, 64Q23, 64Q34, 64Q56</b> (64Qab): 64-QAM, coding rate a/b</p> <p><b>256Q</b>: 256-QAM, coding rate unknown</p> <p><b>256Q34, 256Q56</b> (256Qab): 256-QAM, coding rate a/b</p> <p><b>1KQ</b>: 1024-QAM, coding rate unknown</p> <p><b>1KQ34, 1KQ56</b> (1KQab): 1024-QAM, coding rate a/b</p> |
| <5_PayloadSym>    | <p>decimal</p> <p>Number of OFDM symbols in the payload of the measured burst</p> <p>Range: 1 symbol to 1366 symbols</p> <p>Default unit: symbol</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <6_MeasuredSym>   | <p>decimal</p> <p>Number of measured payload OFDM symbols</p> <p>Range: 1 symbol to 1366 symbols</p> <p>Default unit: symbol</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <7_PayloadBytes>  | <p>decimal</p> <p>Number of bytes in the payload of the measured burst</p> <p>Range: 1 byte to 4095 bytes</p> <p>Default unit: byte</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <8_GuardInterval> | <p>SHORT   LONG   GI08   GI16   GI32</p> <p><b>SHORT, LONG</b>: short or long guard interval (up to 802.11ac)</p> <p><b>GI08, GI16, GI32</b>: 0.8 <math>\mu</math>s, 1.6 <math>\mu</math>s, and 3.2 <math>\mu</math>s guard interval durations (for 802.11ax)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <9_NoSS>          | <p>decimal</p> <p>Number of spatial streams</p> <p>Range: 1 to 8</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <10_NoSTS>        | <p>decimal</p> <p>Number of space-time streams</p> <p>Range: 1 to 8</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                   |       |                                                                                                                                                                                                                                                               |
|-------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <11_BurstRate>    | float | If a modulation filter is used, the burst rate indicates the share of bursts of the selected modulation type in the bursts received. Otherwise, it returns 1.<br>See also <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:ISignal:MODFilter</a> .<br>Default unit: % |
| <12_PowerBackoff> | float | Minimum distance of signal power to reference level since the start of the measurement<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                                                                          |
| <13_BurstPower>   | float | RMS power of the measured burst<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                                                                             |
| <14_PeakPower>    | float | Peak power of the measured burst<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                                                                            |
| <15_CrestFactor>  | float | Range: 0 dB to 60 dB<br>Default unit: dB                                                                                                                                                                                                                      |
| <16_EVMAIICarr>   | float | EVM for all carriers<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                                                                                                                                            |
| <17_EVMDDataCarr> | float | EVM for data carriers<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                                                                                                                                           |
| <18_EVMPilotCarr> | float | EVM for pilot carriers<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                                                                                                                                          |
| <19_FreqError>    | float | Center frequency error<br>Range: -150E+6 Hz to 150E+6 Hz<br>Default unit: Hz                                                                                                                                                                                  |

|                           |                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <20_ClockError>           | float<br>Symbol clock error<br>Range: -125 ppm to 125 ppm<br>Default unit: ppm                                                                                                                                                                                              |
| <21_IQOffset>             | float<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                                                                                                                                                                         |
| <22_DCPower>              | float<br>Power of the DC subcarriers<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                                                                                      |
| <23_GainImbalance>        | float<br>Gain imbalance cannot be calculated if the spectrum is not symmetrical, e.g. for HT_TB and HE_MU.<br>Range: -100 dB to 100 dB<br>Default unit: dB                                                                                                                  |
| <24_QuadError>            | float<br>Quadrature error cannot be calculated if the spectrum is not symmetrical, e.g. for HT_TB and HE_MU.<br>Range: -180 deg to 180 deg<br>Default unit: deg                                                                                                             |
| <25_LTFPower>             | float<br>Power of long training fields (LTF) portion<br>Default unit: dBm                                                                                                                                                                                                   |
| <26_DataPower>            | float<br>Power of data portion<br>Default unit: dBm                                                                                                                                                                                                                         |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements (OFDM, SISO)</a>                                                                                                                                                                                                        |
| <b>Usage:</b>             | Query only                                                                                                                                                                                                                                                                  |
| <b>Firmware/software:</b> | V3.5.121<br>V3.7.20: added modulation UNSpecified, BPSK14, QPSK14, 16Q14, 16Q38 and guard interval GI08, GI16, GI32<br>V3.7.21: added modulation BPSK, QPSK, 16Q, 64Q, 256Q, 1KQ<br>V3.7.31: commands support also composite MIMO<br>V3.7.50: added <LTFPower>, <DataPower> |

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**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:SDEViation?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:CMIMo:PSTS:SDEViation?**

Return the single value RMS power results for the individual space-time streams. The current, average, minimum, maximum, and standard deviation results can be retrieved.

For a meaningful result, set the spatial mapping matrix in the DUT to direct mapping. It causes a one-to-one mapping of space time streams to TX antennas. Thus a broken TX chain (no power) is detected and a damaged chain is identified by its bad EVM.

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

<PowerSTS\_Tx> float

Eight values, one value per space-time stream

Default unit: dBm

**Example:** See [Performing single-shot measurements \(OFDM, CMIMO\)](#).

**Usage:** Query only

**Firmware/software:** V3.7.31

---

**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SDEViation?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SDEViation?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:CURRent?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SDEViation?**

Return the single value results for MIMO measurements, antenna/stream number <n>. For 80+80 MHz signals, the segment-independent values are returned. There are current, average, minimum, maximum and standard deviation results.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Suffix:**

<n> 1..8  
 Identical number of the antenna and stream, e.g.:  
**2** means antenna No. 2, stream No. 2

**Return values:**

<1\_Reliability> decimal  
[Reliability indicator](#)

<2\_Modulation> UNSPecified | BPSK14 | BPSK12 | BPSK34 | QPSK14 |  
 QPSK12 | QPSK34 | 16Q14 | 16Q38 | 16Q12 | 16Q34 | 64Q12 |  
 64Q23 | 64Q34 | 64Q56 | 256Q34 | 256Q56 | 1KQ34 | 1KQ56 |  
 BPSK | QPSK | 16Q | 64Q | 256Q | 1KQ  
 Modulation scheme and coding rate, stream <n>  
**UNSPecified**: modulation unknown  
**BPSK**: BPSK, coding rate unknown  
**BPSK12, BPSK34** (BPSKab): BPSK, coding rate a/b  
**BPSK14**: BPSK, coding rate 1/2 DCM  
**QPSK**: QPSK, coding rate unknown  
**QPSK12, QPSK34** (QPSKab): QPSK, coding rate a/b  
**QPSK14**: QPSK, coding rate 1/2 DCM  
**16Q**: 16-QAM, coding rate unknown  
**16Q12, 16Q34** (16Qab): 16-QAM, coding rate a/b  
**16Q14**: 16-QAM, coding rate 1/2 DCM  
**16Q38**: 16-QAM, coding rate 3/4 DCM  
**64Q**: 64-QAM, coding rate unknown  
**64Q12, 64Q23, 64Q34, 64Q56** (64Qab): 64-QAM, coding rate  
 a/b  
**256Q**: 256-QAM, coding rate unknown  
**256Q34, 256Q56** (256Qab): 256-QAM, coding rate a/b  
**1KQ**: 1024-QAM, coding rate unknown  
**1KQ34, 1KQ56** (1KQab): 1024-QAM, coding rate a/b

<3\_PowerBackoff> float  
 Power backoff, antenna <n>  
 Minimum distance of signal power to reference level since the  
 start of the measurement  
 Range: -100 dB to 0 dB  
 Default unit: dB

<4\_BurstPower> float  
 Burst power, antenna <n>  
 Range: -100 dBm to 30 dBm  
 Default unit: dBm

<5\_PeakPower> float  
 Peak power, antenna <n>  
 Range: -100 dBm to 30 dBm  
 Default unit: dBm

|                    |                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------|
| <6_CrestFactor>    | float<br>Crest factor, antenna <n><br>Range: 0 dB to 60 dB<br>Default unit: dB                      |
| <7_EVMAIICarr>     | float<br>EVM for all carriers, stream <n><br>Range: -100 dB to 0 dB<br>Default unit: dB             |
| <8_EVMDDataCarr>   | float<br>EVM for data carriers, stream <n><br>Range: -100 dB to 0 dB<br>Default unit: dB            |
| <9_EVMPilotCarr>   | float<br>EVM for pilot carriers, stream <n><br>Range: -100 dB to 0 dB<br>Default unit: dB           |
| <10_IQOffset>      | float<br>I/Q offset, antenna <n><br>Range: -100 dB to 0 dB<br>Default unit: dB                      |
| <11_DCPower>       | float<br>Power of the DC subcarriers, antenna <n><br>Range: -100 dBm to 30 dBm<br>Default unit: dBm |
| <12_GainImbalance> | float<br>Gain imbalance, antenna <n><br>Range: -100 dB to 100 dB<br>Default unit: dB                |
| <13_QuadError>     | float<br>Quadrature error, antenna <n><br>Range: -180 deg to 180 deg<br>Default unit: dB            |
| <14_LTFPower>      | float<br>Power of long training fields portion, antenna <n><br>Default unit: dBm                    |
| <15_DataPower>     | float<br>Power of data portion, antenna <n><br>Default unit: dBm                                    |

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)



**Usage:** Query only

**Firmware/software:** V3.5.121  
 V3.7.20: added modulation UNSpecified, BPSK14, QPSK14, 16Q14, 16Q38  
 V3.7.21: added modulation BPSK, QPSK, 16Q, 64Q, 256Q, 1KQ  
 V3.7.31: commands support also composite MIMO  
 V3.7.50: added <LTFPower>, <DataPower>

---

**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:CURRENT?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:AVERage?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:SDEVIation?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:CURRENT?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:SDEVIation?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:CURRENT?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:AVERage?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MINimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:MAXimum?**  
**READ:WLAN:MEAS<i>:MEValuation:MODulation:SEGMENTS:SDEVIation?**

Return the segment-specific single value results for 80+80 MHz SISO measurements.

There are current, average, minimum, maximum and standard deviation results.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

**Return values:**

<1\_Reliability> decimal  
[Reliability indicator](#)

<2\_EVMAAllCarrS1> float

<3\_EVMAAllCarrS2> EVM for all carriers, segment 1, 2  
 Range: -100 dB to 0 dB

<4\_EVMDDataCarrS1> float

<5\_EVMDDataCarrS2> EVM for data carriers, segment 1, 2  
 Range: -100 dB to 0 dB

<6\_EVMPilotCarrS1> float

<7\_EVMPilotCarrS2> EVM for pilot carriers, segment 1, 2  
 Range: -100 dB to 0 dB

|                    |                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <8_PowerBackoffS1> | float                                                                                                                                           |
| <9_PowerBackoffS2> | Power backoff, segment 1, 2<br>Minimum distance of signal power to reference level since the start of the measurement<br>Range: -100 dB to 0 dB |
| <10_BurstPowerS1>  | float                                                                                                                                           |
| <11_BurstPowerS2>  | RMS power of the measured burst, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                      |
| <12_PeakPowerS1>   | float                                                                                                                                           |
| <13_PeakPowerS2>   | Peak power of the measured burst, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                     |
| <14_CrestFactorS1> | float                                                                                                                                           |
| <15_CrestFactorS2> | Crest factor, segment 1, 2<br>Range: 0 dB to 60 dB                                                                                              |
| <16_IQOffsetS1>    | float                                                                                                                                           |
| <17_IQOffsetS2>    | I/Q origin offset, segment 1, 2<br>Range: -100 dB to 0 dB                                                                                       |
| <18_DCPowerS1>     | float                                                                                                                                           |
| <19_DCPowerS2>     | Power of the DC subcarriers, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                          |
| <20_LTFPowerS1>    | float                                                                                                                                           |
| <21_LTFPowerS2>    | Power of long training fields portion, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                |
| <22_DataPowerS1>   | float                                                                                                                                           |
| <23_DataPowerS2>   | Power of data portion, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                                |

**Usage:** Query only

**Firmware/software:** V3.5.121, V3.7.50: added <LTFPowerSx>, <DataPowerSx>

---

**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:**  
**CURRent?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:**  
**AVERAge?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:**  
**MINimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:**  
**MAXimum?**  
**CALCulate:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:**  
**SDEViation?**

```

FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    CURRent?
FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    AVERAge?
FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    MINimum?
FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    MAXimum?
FETCH:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    SDEViation?
READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    CURRent?
READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    AVERAge?
READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    MINimum?
READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    MAXimum?
READ:WLAN:MEAS<i>:MEValuation:MODulation:MIMO<n>:SEGMents:
    SDEViation?

```

Return the segment-specific single value results for switched MIMO measurements, antenna/stream number <n>. There are current, average, minimum, maximum and standard deviation results.

The values described below are returned by **FETCH** and **READ** commands. **CALCulate** commands return limit check results instead, one value for each result listed below.

#### Suffix:

<n> 1..8  
 Identical number of the antenna and stream, e.g.:  
 2 means antenna No. 2, stream No. 2

#### Return values:

<1\_Reliability> decimal  
[Reliability indicator](#)

<2\_EVMAIICarrS1> float

<3\_EVMAIICarrS2> EVM for all carriers, stream <n>, segment 1, 2  
 Range: -100 dB to 0 dB

<4\_EVMDDataCarrS1> float

<5\_EVMDDataCarrS2> EVM for data carriers, stream <n>, segment 1, 2  
 Range: -100 dB to 0 dB

<6\_EVMPilotCarrS1> float

<7\_EVMPilotCarrS2> EVM for pilot carriers, stream <n>, segment 1, 2  
 Range: -100 dB to 0 dB

|                    |                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <8_PowerBackoffS1> | float                                                                                                                                                        |
| <9_PowerBackoffS2> | Power backoff, antenna <n>, segment 1, 2<br>Minimum distance of signal power to reference level since the start of the measurement<br>Range: -100 dB to 0 dB |
| <10_BurstPowerS1>  | float                                                                                                                                                        |
| <11_BurstPowerS2>  | Burst power, antenna <n>, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                                          |
| <12_PeakPowerS1>   | float                                                                                                                                                        |
| <13_PeakPowerS2>   | Peak power, antenna <n>, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                                           |
| <14_CrestFactorS1> | float                                                                                                                                                        |
| <15_CrestFactorS2> | Crest factor, antenna <n>, segment 1, 2<br>Range: 0 dB to 60 dB                                                                                              |
| <16_IQOffsetS1>    | float                                                                                                                                                        |
| <17_IQOffsetS2>    | I/Q offset, antenna <n>, segment 1, 2<br>Range: -100 dB to 0 dB                                                                                              |
| <18_DCPowerS1>     | float                                                                                                                                                        |
| <19_DCPowerS2>     | Power of the DC subcarriers, antenna <n>, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                          |
| <20_LTFPowerS1>    | float                                                                                                                                                        |
| <21_LTFPowerS2>    | Power of long training fields portion, antenna <n>, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                |
| <22_DataPowerS1>   | float                                                                                                                                                        |
| <23_DataPowerS2>   | Power of data portion, antenna <n>, segment 1, 2<br>Range: -100 dBm to 30 dBm                                                                                |

**Example:** See [Performing single-shot measurements \(OFDM, MIMO, 80+80 MHz\)](#)

**Usage:** Query only

**Firmware/software:** V3.5.121, V3.7.50: added <LTFPowerSx>, <DataPowerSx>

---

**CALCulate:WLAN:MEAS<i>:MEvaluation:MODulation:CFDistrib?**

**FETCh:WLAN:MEAS<i>:MEvaluation:MODulation:CFDistrib?**

**READ:WLAN:MEAS<i>:MEvaluation:MODulation:CFDistrib?**

Return the scalar results for carrier frequency offset (CFO) error distribution.

The results are only supported for 802.11ax.

Exceeding the limit has no impact on the stop "On Limit Failure" condition or out-of-tolerance counter.

**Return values:**

|                  |                                                      |
|------------------|------------------------------------------------------|
| <Reliability>    | decimal<br><a href="#">Reliability indicator</a>     |
| <CFO_Percentage> | float<br>Percentage of CFO errors<br>Default unit: % |
| <CFO_Outside>    | decimal<br>Number of detected CFO errors             |
| <CFO_Total>      | decimal<br>Number of measured CFOs                   |

**Example:** See [Performing single-shot measurements \(OFDM, SISO\)](#)

**Usage:** Query only

**Firmware/software:** V3.7.50

**Options:** R&S CMW-KM657

### 3.5.5.22 Modulation single values, OFDMA

The commands in this section return the statistical modulation results for OFDMA signals.

See also [Chapter 3.3.2.13, "View TX measurement \(scalar\) for OFDMA"](#), on page 282.

|                                                                                              |     |
|----------------------------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:CURRent?                               | 537 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:AVERage?                               | 537 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:MAXimum?                               | 538 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:SDEViation?                            | 538 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:CURRent?                    | 538 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:AVERage?                    | 538 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:MAXimum?                    | 538 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:<br>SDEViation?             | 538 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:<br>STReam<str>:CURRent?    | 539 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:<br>STReam<str>:AVERage?    | 539 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:<br>STReam<str>:MAXimum?    | 539 |
| FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:<br>STReam<str>:SDEViation? | 539 |
| FETCh:WLAN:MEAS<i>:MEValuation:OFDMa:INFO?                                                   | 539 |
| FETCh:WLAN:MEAS<i>:MEValuation:OFDMa:UINFo<user>?                                            | 540 |

---

FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:CURRent?

FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:AVERage?

**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:SDEViation?**

Return the single value results per user for OFDMA SISO measurements. For MIMO measurements, the stream/antenna-independent values are returned.

There are current, average, minimum, maximum and standard deviation results.

**Return values:**

<1\_Reliability> decimal  
 See [Reliability indicator](#)

<2\_EVMvsU1\_All> float  
 <3\_EVMvsU1\_Data> EVM for all carriers, for data carriers, and for pilot carriers ver-  
 <4\_EVMvsU1\_Pilot> sus user 1 to 74

<5\_EVMvsU2\_All>  
 <6\_EVMvsU2\_Data>  
 <7\_EVMvsU2\_Pilot>...

<221\_EVMvsU74\_All>  
 <222\_EVMvsU74\_Data>  
 <223\_EVMvsU74\_Pilot>

**Usage:** Query only

**Firmware/software:** V3.7.40

---

**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:**  
**CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:**  
**AVERAge?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:**  
**MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:**  
**SDEViation?**

Return the single value results for OFDMA SISO measurements for the specified user. For MIMO measurements, the stream/antenna-independent values are returned.

There are current, average, minimum, maximum and standard deviation results.

**Suffix:**

<user> 1..74  
 User

**Return values:**

<reliability> decimal  
 See [Reliability indicator](#)

<EVMallCarr> float  
 EVM for all carriers for the specified user

|                           |                                               |
|---------------------------|-----------------------------------------------|
| <EVMdataCarr>             | float                                         |
|                           | EVM for data carriers for the specified user  |
| <EVMpilotCarr>            | float                                         |
|                           | EVM for pilot carriers for the specified user |
| <b>Usage:</b>             | Query only                                    |
| <b>Firmware/software:</b> | V3.7.40                                       |

---

**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:STReam<str>:CURRent?**

**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:STReam<str>:AVERAge?**

**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:STReam<str>:MAXimum?**

**FETCh:WLAN:MEAS<i>:MEValuation:MODulation:EVMagnitude:USER<user>:STReam<str>:SDEViation?**

Return the single value results for OFDMA MIMO measurements for the specified user and stream.

There are current, average, minimum, maximum and standard deviation results.

**Suffix:**

|        |        |
|--------|--------|
| <user> | 1..74  |
|        | User   |
| <str>  | 1..8   |
|        | Stream |

**Return values:**

|                           |                                                          |
|---------------------------|----------------------------------------------------------|
| <reliability>             | decimal                                                  |
|                           | See <a href="#">Reliability indicator</a>                |
| <EVMallCarr>              | float                                                    |
|                           | EVM for all carriers for the specified user and stream   |
| <EVMdataCarr>             | float                                                    |
|                           | EVM for data carriers for the specified user and stream  |
| <EVMpilotCarr>            | float                                                    |
|                           | EVM for pilot carriers for the specified user and stream |
| <b>Usage:</b>             | Query only                                               |
| <b>Firmware/software:</b> | V3.7.40                                                  |

---

**FETCh:WLAN:MEAS<i>:MEValuation:OFDMa:INFO?**

Queries OFDMA common information.

**Return values:**

|                           |                                                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <reliability>             | decimal<br><a href="#">Reliability indicator</a>                                                                                                                                                      |
| <NoOfUsers>               | decimal<br>No. of users                                                                                                                                                                               |
| <NoOfRUs>                 | decimal<br>No. of resource units                                                                                                                                                                      |
| <GuardInterval>           | SHORT   LONG   GI08   GI16   GI32<br><b>SHORT, LONG:</b> short or long guard interval (up to 802.11ac)<br><b>GI08, GI16, GI32:</b> 0.8 µs, 1.6 µs, and 3.2 µs guard interval durations (for 802.11ax) |
| <LTFSize>                 | LTF1   LTF2   LTF4<br>1x LTF (3.2 µs), 2x LTF (6.4 µs), 4x LTF (12.8 µs)                                                                                                                              |
| <b>Usage:</b>             | Query only                                                                                                                                                                                            |
| <b>Firmware/software:</b> | V3.7.40, 3.8.10: added <GuardInterval> and <LTFSize>                                                                                                                                                  |
| <b>Options:</b>           | R&S CMW-KM657                                                                                                                                                                                         |

**FETCh:WLAN:MEAS<i>:MEvaluation:OFDMa:UINFo<user>?**

Queries OFDMA user-specific information signaled in a HE signal field.

**Suffix:**

|        |               |
|--------|---------------|
| <user> | 1..74<br>User |
|--------|---------------|

**Return values:**

|               |                                                                            |
|---------------|----------------------------------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a>                           |
| <RUCount>     | decimal<br>Index of RUs of all sizes (users with STA-ID 2046 are included) |
| <RUIndex>     | decimal<br>Index of the RUs only with the used size <RUSize>               |
| <RU26Index>   | decimal<br>Index based on RU26                                             |
| <RUSize>      | decimal<br>RU size allocated by the user                                   |
| <MCS>         | decimal<br>Modulation and coding scheme                                    |
| <DCM>         | decimal<br>The value of DCM field                                          |



|                           |                                      |
|---------------------------|--------------------------------------|
| <STAID>                   | decimal<br>The value of STA-ID field |
| <NSTS>                    | decimal<br>The value of NSTS field   |
| <TxBf>                    | decimal<br>The value of TxBF field   |
| <Coding>                  | BCC   LDPC<br>Coding type            |
| <b>Usage:</b>             | Query only                           |
| <b>Firmware/software:</b> | V3.7.40                              |
| <b>Options:</b>           | R&S CMW-KM657                        |

### 3.5.5.23 Signal fields info

The commands in this section return the reported values of signal fields according to the used standard. See also [Chapter 3.3.2.12, "View signal fields info"](#), on page 278.

|                                                                                              |     |
|----------------------------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:CFIeld:CRC?                       | 543 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:CFIeld:CRU?                       | 544 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:CFIeld:<br>RUAllocation?          | 544 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:CFIeld:TAIL?                      | 544 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:CODing?     | 545 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:CRC?        | 545 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:DCM?        | 546 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:MCS?        | 546 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:NSTS?       | 547 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:REServed?   | 547 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:SPAConfig?  | 548 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:STAid?      | 548 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:TAIL?       | 549 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch_index>:<br>UField<usr_index>:TXBeamform? | 549 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BDCM?                                              | 549 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BMCS?                                              | 550 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BSSColor?                                          | 550 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BW?                                                | 551 |

|                                                                |     |
|----------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:CRC?                 | 551 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:DOPPler?             | 551 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:GILTfsize?           | 552 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:LDPC?                | 552 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:NLTFsymbols?         | 552 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:NSBSymbols?          | 553 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:PEDisambig?          | 553 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:PFECpadding?         | 553 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:REServed?            | 554 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:SBCompress?          | 554 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:SPATialreuse?        | 555 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:STBC?                | 555 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:TAIL?                | 555 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:TXOP?                | 556 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:ULDL?                | 556 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:BEAMchange?          | 556 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:BSSColor?            | 557 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:BW?                  | 557 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:CODing?              | 558 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:CRC?                 | 558 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:DCM?                 | 558 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:DOPPler?             | 559 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:FORMat?              | 559 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:GILTfsize?           | 559 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:LDPC?                | 560 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:MCS?                 | 560 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:NSTS?                | 561 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:PEDisambig?          | 561 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:PFECpadding?         | 561 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:REServed<index>?     | 562 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:SPATialreuse?        | 562 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:STBC?                | 562 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:TAIL?                | 563 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:TXBF?                | 563 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:TXOP?                | 564 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:ULDL?                | 564 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:BSSColor?            | 564 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:BW?                  | 565 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:CRC?                 | 565 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:FORMat?              | 566 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:REServed<index>?     | 566 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:SPATialreuse<index>? | 566 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:TAIL?                | 567 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HETB:TXOP?                | 567 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:AGGRegation?        | 568 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:CBW?                | 568 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:CRC?                | 568 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:FECCoding?          | 569 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:HTLength?           | 569 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:MCS?                | 570 |

|                                                              |     |
|--------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:NESS?             | 570 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:NSounding?        | 570 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:REServed?         | 571 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:SHORTgi?          | 571 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:SMOothing?        | 571 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:STBCoding?        | 572 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:TAIL?             | 572 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:LENGth?            | 573 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:PARity?            | 573 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:RATE?              | 573 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:REServed?          | 574 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:TAIL?              | 574 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:BEAMformed?      | 575 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:BW?              | 575 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:CRC?             | 575 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:FECCoding?       | 576 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:GID?             | 576 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:LDPC?            | 576 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:PAID?            | 577 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:REServed<index>? | 577 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SDISambig?       | 578 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SGI?             | 578 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SMCS?            | 578 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:STBC?            | 579 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SUNSts?          | 579 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:TAIL?            | 580 |
| FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:TXOP?            | 580 |

#### FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:CField: CRC?

Queries the value of CRC field signaled for the channel in HE-SIG-B common field for multi user MIMO.

##### Suffix:

<ch\_index> 1-2

##### Return values:

<reliability> decimal  
Reliability indicator

<value\_bin> string

<value\_dec> decimal

Range: 0 to 4.294967295E+9

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

**Manual operation:** See "HE-SIG-B" on page 282

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**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:CField:CRU?**

Queries the value of Center 26-tone RU field signaled for the channel in HE-SIG-B common field for multi user MIMO.

**Suffix:**

<ch\_index> 1-2

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4.294967295E+9

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:CField:RUAllocation?**

Queries the value of RU Allocation field signaled for the channel in HE-SIG-B common field for multi user MIMO.

**Suffix:**

<ch\_index> 1-2

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4.294967295E+9

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:CField:TAIL?**

Queries the value of Tail field signaled for the channel in HE-SIG-B common field for multi user MIMO.

**Suffix:**

&lt;ch\_index&gt; 1-2

**Return values:**

&lt;reliability&gt; decimal

[Reliability indicator](#)

&lt;value\_bin&gt; string

&lt;value\_dec&gt; decimal

Range: 0 to 4.294967295E+9

**Example:**See [Retrieving signal fields reports](#)**Usage:**

Query only

**Firmware/software:** V3.7.30**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UFieId<usr\_index>:CODing?**

Queries the value of Coding field signaled for the channel and user in HE-SIG-B user-specific field for multi user MIMO.

**Suffix:**

&lt;ch\_index&gt; 1-2

&lt;usr\_index&gt; 1..74

**Return values:**

&lt;reliability&gt; decimal

[Reliability indicator](#)

&lt;value\_bin&gt; string

&lt;value\_dec&gt; decimal

Range: 0 to 2047

**Example:**See [Retrieving signal fields reports](#)**Usage:**

Query only

**Firmware/software:** V3.7.30**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UFieId<usr\_index>:CRC?**

Queries the value of CRC field signaled for the channel and user in HE-SIG-B user-specific field for multi user MIMO.

**Suffix:**

&lt;ch\_index&gt; 1-2

&lt;usr\_index&gt; 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
 UField<usr\_index>:DCM?**

Queries the value of DCM field signaled for the channel and user in HE-SIG-B user-specific field for SU-MIMO.

**Suffix:**

<ch\_index> 1-2

<usr\_index> 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
 UField<usr\_index>:MCS?**

Queries the value of MCS field signaled for the channel and user in HE-SIG-B user-specific field for MIMO.

**Suffix:**

<ch\_index> 1-2

<usr\_index> 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

**<value\_bin>** string  
**<value\_dec>** decimal  
 Range: 0 to 2047  
**Example:** See [Retrieving signal fields reports](#)  
**Usage:** Query only  
**Firmware/software:** V3.7.30

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UField<usr\_index>:NSTS?**

Queries the value of NSTS field signaled for the channel and user in HE-SIG-B user-specific field for SU-MIMO.

**Suffix:**

**<ch\_index>** 1-2  
**<usr\_index>** 1..74

**Return values:**

**<reliability>** decimal  
[Reliability indicator](#)

**<value\_bin>** string  
**<value\_dec>** decimal  
 Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)  
**Usage:** Query only  
**Firmware/software:** V3.7.30

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UField<usr\_index>:REServed?**

Queries the value of Reserved field signaled for the channel and user in HE-SIG-B user-specific field for multi user MIMO.

**Suffix:**

**<ch\_index>** 1-2  
**<usr\_index>** 1..74

**Return values:**

**<reliability>** decimal  
[Reliability indicator](#)

**<value\_bin>** string  
**<value\_dec>** decimal  
 Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UField<usr\_index>:SPAConfig?**

Queries the value of Spatial Configuration field signaled for the channel and user in HE-SIG-B user-specific field for multi user MIMO.

**Suffix:**

<ch\_index> 1-2

<usr\_index> 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UField<usr\_index>:STAid?**

Queries the value of STA-ID field signaled for the channel and user in HE-SIG-B user-specific field for MIMO.

**Suffix:**

<ch\_index> 1-2

<usr\_index> 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30



---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UField<usr\_index>:TAIL?**

Queries the value of Tail field signaled for the channel and user in HE-SIG-B user-specific field for multi user MIMO.

**Suffix:**

<ch\_index> 1-2

<usr\_index> 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEB:CHANnel<ch\_index>:  
UField<usr\_index>:TXBeamform?**

Queries the value of Tx Beamforming field signaled for the channel and user in HE-SIG-B user-specific field for SU-MIMO.

**Suffix:**

<ch\_index> 1-2

<usr\_index> 1..74

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 2047

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BDCM?**

Queries the value of SIGB DCM field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**Manual operation:** See "HE-SIG-A (HE\_MU)" on page 281

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BMCS?**

Queries the value of SIGB MCS field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:BSSColor?**

Queries the value of BSS Color field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HEMU:BW?**

Queries the value of Bandwidth field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 63                        |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HEMU:CRC?**

Queries the value of CRC field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 127                       |
| <check>       | ON   OFF   1   0                                 |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HEMU:DOPPler?**

Queries the value of Doppler field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 63                        |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:GILTfsize?

Queries the value of GI+LTF Size field signaled in HE signal field for multi user MIMO (HE-SIG-A).

##### Return values:

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:LDPC?

Queries the value of LDPC Extra Symbol Segment field signaled in HE signal field for multi user MIMO (HE-SIG-A).

##### Return values:

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:NLTFsymbols?

Queries the value of Number of HE-LTF Symbols And Midamble Periodicity field signaled in HE signal field for multi user MIMO (HE-SIG-A).

##### Return values:

<reliability> decimal  
[Reliability indicator](#)

**<value\_bin>** string  
**<value\_dec>** decimal  
 Range: 0 to 127  
**Example:** See [Retrieving signal fields reports](#)  
**Usage:** Query only  
**Firmware/software:** V3.7.21

#### FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:NSBSymbols?

Queries the value of Number Of HE-SIG-B Symbols Or MU-MIMO Users field signaled in HE signal field for multi user MIMO (HE-SIG-A).

##### Return values:

**<reliability>** decimal  
[Reliability indicator](#)  
**<value\_bin>** string  
**<value\_dec>** decimal  
 Range: 0 to 63  
**Example:** See [Retrieving signal fields reports](#)  
**Usage:** Query only  
**Firmware/software:** V3.7.21

#### FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:PEDisambig?

Queries the value of PE Disambiguity field signaled in HE signal field for multi user MIMO (HE-SIG-A).

##### Return values:

**<reliability>** decimal  
[Reliability indicator](#)  
**<value\_bin>** string  
**<value\_dec>** decimal  
 Range: 0 to 127  
**Example:** See [Retrieving signal fields reports](#)  
**Usage:** Query only  
**Firmware/software:** V3.7.21

#### FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:PFECpadding?

Queries the value of Pre-FEC Padding Factor field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:REServed?**

Queries the value of Reserved field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:SBCompress?**

Queries the value of SIGB Compression field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:SPATialreuse?**

Queries the value of Spatial Reuse field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

|               |                                       |
|---------------|---------------------------------------|
| <reliability> | decimal                               |
|               | <a href="#">Reliability indicator</a> |
| <value_bin>   | string                                |
| <value_dec>   | decimal                               |
|               | Range: 0 to 63                        |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:STBC?**

Queries the value of STBC field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

|               |                                       |
|---------------|---------------------------------------|
| <reliability> | decimal                               |
|               | <a href="#">Reliability indicator</a> |
| <value_bin>   | string                                |
| <value_dec>   | decimal                               |
|               | Range: 0 to 127                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:TAIL?**

Queries the value of Tail field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

|               |                                       |
|---------------|---------------------------------------|
| <reliability> | decimal                               |
|               | <a href="#">Reliability indicator</a> |
| <value_bin>   | string                                |
| <value_dec>   | decimal                               |
|               | Range: 0 to 127                       |

<check> ON | OFF | 1 | 0  
Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.30

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:TXOP?**

Queries the value of TXOP field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HEMU:ULDL?**

Queries the value of UL/DL field signaled in HE signal field for multi user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:BEAMchange?**

Queries the value of Beam Change field signaled in HE signal field for single user MIMO (HE-SIG-A).



**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**Manual operation:** See "HE-SIG-A (HE\_SU)" on page 280

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:BSSColor?**

Queries the value of BSS Color field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:BW?**

Queries the value of Bandwidth field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:CODing?**

Queries the value of Coding field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 127                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:CRC?**

Queries the value of CRC field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

|               |                                                              |
|---------------|--------------------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a>             |
| <value_bin>   | string                                                       |
| <value_dec>   | decimal<br>Range: 0 to 127                                   |
| <check>       | ON   OFF   1   0<br>Indicates passed or failed check verdict |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:DCM?**

Queries the value of DCM field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:DOPPler?**

Queries the value of Doppler field signaled in HE signal field for single user MIMO (HE-SIG-A).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:FORMat?**

Queries the value of Format field signaled in HE signal field for single user MIMO (HE-SIG-A).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:GILTfsize?**

Queries the value of GI+LTF Size field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:LDPC?**

Queries the value of LDPC Extra Symbol Segment field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:MCS?**

Queries the value of MCS field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:NSTS?**

Queries the value of Nsts And Midamble Periodicity field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 63                        |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:PEDisambig?**

Queries the value of PE Disambiguity field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 127                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:PFECpadding?**

Queries the value of Pre-FEC Padding Factor field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 127                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:REServed<index>?**

Queries the value of Reserved field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Suffix:**

<index> 1-2  
 1: Reserved field in HE-SIG-A1  
 2: Reserved field in HE-SIG-A2

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:SPATialreuse?**

Queries the value of Spatial Reuse field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HESU:STBC?**

Queries the value of STBC field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:TAIL?**

Queries the value of Tail field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

<check> ON | OFF | 1 | 0  
 Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HESU:TXBF?**

Queries the value of TxBF field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HESU:TXOP?**

Queries the value of TXOP field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 127

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HESU:ULDL?**

Queries the value of UL/DL field signaled in HE signal field for single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HETB:BSSColor?**

Queries the value of BSS Color field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string



<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**Manual operation:** See ["HE-SIG-A \(HE\\_TRIG\)"](#) on page 281

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:BW?**

Queries the value of Bandwidth field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:CRC?**

Queries the value of CRC field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 511

<check> ON | OFF | 1 | 0  
 Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:FORMat?**

Queries the value of Format field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:REServed<index>?**

Queries the value of Reserved field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Suffix:**

<index> 1-2  
 1: Reserved field in HE-SIG-A1  
 2: Reserved field in HE-SIG-A2

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:SPATialreuse<index>?**

Queries the value of Spatial Reuse field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Suffix:**

<index> 1-4  
 Number of spatial reuse

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 63

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:TAIL?**

Queries the value of Tail field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 511

<check> ON | OFF | 1 | 0  
Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HETB:TXOP?**

Queries the value of TXOP field signaled in HE signal field for trigger based uplink single user MIMO (HE-SIG-A).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 511

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HTSig:AGGRegation?**

Queries the value of Aggregation field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**Manual operation:** See ["HT-SIG"](#) on page 279

---

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HTSig:CBW?**

Queries the value of CBW 20/40 field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 65.535E+3

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HTSig:CRC?**

Queries the value of CRC field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

<check> ON | OFF | 1 | 0  
 Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HTSig:FECCoding?**

Queries the value of FEC Coding field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:HTSig:HTLength?**

Queries the value of HT Length field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 65.535E+3

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:MCS?**

Queries the value of Modulation and Coding Scheme field signaled in high throughput signal field for 802.11n signal (HT-SIG).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 65.535E+3                 |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:NESS?**

Queries the value of Number of Extension Spatial Streams field signaled in high throughput signal field for 802.11n signal (HT-SIG).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 255                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:NSOunding?**

Queries the value of Not Sounding field signaled in high throughput signal field for 802.11n signal (HT-SIG).

**Return values:**

|               |                                                  |
|---------------|--------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a> |
| <value_bin>   | string                                           |
| <value_dec>   | decimal<br>Range: 0 to 255                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:REServed?**

Queries the value of Reserved field signaled in high throughput signal field for 802.11n signal (HT-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:SHORTgi?**

Queries the value of Short GI field signaled in high throughput signal field for 802.11n signal (HT-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

**FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:SMOothing?**

Queries the value of Smoothing field signaled in high throughput signal field for 802.11n signal (HT-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:STBCoding?**

Queries the value of STBC field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:HTSig:TAIL?**

Queries the value of Tail Bits field signaled in high throughput signal field for 802.11n signal (HT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

<check> ON | OFF | 1 | 0  
 Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21



**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:LSIG:LENGth?**

Queries the value of Length field signaled in legacy signal field for NON\_HT signal (L-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4095

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**Manual operation:** See "[L- SIG](#)" on page 279

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:LSIG:PARity?**

Queries the value of Parity field signaled in legacy signal field for NON\_HT signal (L-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4095

<check> ON | OFF | 1 | 0  
 Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:LSIG:RATE?**

Queries the value of Rate field signaled in legacy signal field for NON\_HT signal (L-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4095

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:REServed?**

Queries the value of Reserved field signaled in legacy signal field for NON\_HT signal (L-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4095

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:LSIG:TAIL?**

Queries the value of Tail field signaled in legacy signal field for NON\_HT signal (L-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 4095

<check> ON | OFF | 1 | 0  
 Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:BEAMformed?**

Queries the value of Beamformed field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Return values:**

|               |                                       |
|---------------|---------------------------------------|
| <reliability> | decimal                               |
|               | <a href="#">Reliability indicator</a> |
| <value_bin>   | string                                |
| <value_dec>   | decimal                               |
|               | Range: 0 to 255                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**Manual operation:** See "[VHT-SIG-A](#)" on page 279

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:BW?**

Queries the value of BW field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Return values:**

|               |                                       |
|---------------|---------------------------------------|
| <reliability> | decimal                               |
|               | <a href="#">Reliability indicator</a> |
| <value_bin>   | string                                |
| <value_dec>   | decimal                               |
|               | Range: 0 to 511                       |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:CRC?**

Queries the value of CRC field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Return values:**

|               |                                       |
|---------------|---------------------------------------|
| <reliability> | decimal                               |
|               | <a href="#">Reliability indicator</a> |
| <value_bin>   | string                                |
| <value_dec>   | decimal                               |
|               | Range: 0 to 255                       |

<check> ON | OFF | 1 | 0  
Indicates passed or failed check verdict

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:FECCoding?**

Queries the value of SU/MU[0] Coding field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:GID?**

Queries the value of Group ID field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
Range: 0 to 511

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:LDPC?**

Queries the value of LDPC Extra OFDM Symbol field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:VHTSig:PAID?**

Queries the value of Partial AID field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

For the filed NSTS refer to:

[FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:VHTSig:SUNSts?](#)

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 511

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEvaluation:SINFo:VHTSig:REServed<index>?**

Queries the value of Reserved field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Suffix:**

<index> 1-3  
**1:** Reserved field in VHT-SIG-A1 bit 2  
**2:** Reserved field in VHT-SIG-A1 bit 23  
**3:** Reserved field in VHT-SIG-A2 bit 9

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 511

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SDISambig?**

Queries the value of "Short GI  $N_{SYM}$  Disambiguation" field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SGI?**

Queries the value of Short GI field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

##### **Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

#### **FETCH:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SMCS?**

Queries the value of SU VHT-MCS/ MU[1- 3] Coding field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 255

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:STBC?**

Queries the value of STBC field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

**Return values:**

<reliability> decimal  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 511

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

**FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:SUNSts?**

Queries the value of NSTS field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

For the filed Partial AID refer to:

[FETCh:WLAN:MEAS<i>:MEValuation:SINFo:VHTSig:PAID?](#)

**Return values:**

<reliability> float  
[Reliability indicator](#)

<value\_bin> string

<value\_dec> decimal  
 Range: 0 to 511

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:VHTSig:TAIL?**

Queries the value of Tail field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

##### **Return values:**

|               |                                                              |
|---------------|--------------------------------------------------------------|
| <reliability> | decimal<br><a href="#">Reliability indicator</a>             |
| <value_bin>   | string                                                       |
| <value_dec>   | decimal<br>Range: 0 to 255                                   |
| <check>       | ON   OFF   1   0<br>Indicates passed or failed check verdict |

**Example:** See [Retrieving signal fields reports](#)

**Usage:** Query only

**Firmware/software:** V3.7.21

---

#### **FETCH:WLAN:MEAS<i>:MEvaluation:SINFo:VHTSig:TXOP?**

Queries the value of TXOP\_PS\_NOT\_ALLOWED field signaled in very high throughput signal field for 802.11ac signal (VHT-SIG).

##### **Return values:**

|                 |                                                      |
|-----------------|------------------------------------------------------|
| <reliability>   | decimal<br><a href="#">Reliability indicator</a>     |
| <value_bin>     | string                                               |
| <value_dec>     | decimal<br>Range: 0 to 511                           |
| <b>Example:</b> | See <a href="#">Retrieving signal fields reports</a> |
| <b>Usage:</b>   | Query only                                           |

**Firmware/software:** V3.7.21

### **3.5.5.24 List mode results (one segment)**

The following commands return the list mode results for a selected segment.

The segment number <no> in the following commands refers to the range of measured segments, see [CONFIGure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#) on page 388.

The commands return two reliability indicators. The first indicator covers all measured segments. The second indicator considers only the selected segment.



The indicated ranges apply to all statistical results except standard deviation results. The minimum for standard deviation results equals 0. The maximum equals the width of the indicated range divided by two. Exceptions are explicitly stated.

To configure the list mode, use the commands described in [Chapter 3.5.3.4, "List mode settings"](#), on page 386.

For a description of the list mode, see [Chapter 3.2.7, "Multi-evaluation list mode"](#), on page 252.

|                                                                                          |     |
|------------------------------------------------------------------------------------------|-----|
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:CURRent?              | 581 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:AVERage?              | 581 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:MINimum?              | 581 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:MAXimum?              | 581 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:SDEViation?           | 581 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:CURRent?         | 585 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:AVERage?         | 585 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:MINimum?         | 585 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:MAXimum?         | 585 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:<br>SDEViation?  | 585 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:CURRent?                  | 586 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:AVERage?                  | 586 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:MINimum?                  | 586 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:MAXimum?                  | 587 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:SDEViation?               | 587 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQuency:<br>CURRent?    | 587 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQuency:<br>AVERage?    | 587 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQuency:<br>MINimum?    | 588 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQuency:<br>MAXimum?    | 588 |
| FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQuency:<br>SDEViation? | 588 |

---

**FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:CURRent?**  
**FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:AVERage?**  
**FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:MINimum?**  
**FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:**  
**MAXimum?**  
**FETCh:WLAN:MEAS<i><no></i>:MEValuation:LIST:SEGMENT<no>:MODulation:**  
**SDEViation?**

Return OFDM/OFDMA modulation single value results for segment <no> in list mode.

**Suffix:**

<no>                      1..n  
                              Segment

**Return values:**

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <1_Reliability>    | decimal<br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <2_SegReliability> | decimal<br>Reliability indicator for the segment. The meaning of the returned values is the same as for the common reliability indicator, see previous parameter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <3_OutOfTol>       | float<br>Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits.<br>Range: 0 % to 100 %<br>Default unit: %                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <4_MCSIndex>       | decimal<br>Modulation and coding scheme index (see IEEE Std 802.11-2016, section 19.5)<br>Range: 0 to 76                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <5_ModType>        | UNSPecified   BPSK14   BPSK12   BPSK34   QPSK14   QPSK12   QPSK34   16Q14   16Q38   16Q12   16Q34   64Q12   64Q23   64Q34   64Q56   256Q34   256Q56   1KQ34   1KQ56   BPSK   QPSK   16Q   64Q   256Q   1KQ<br>Modulation scheme and coding rate<br><b>UNSPecified</b> : modulation unknown<br><b>BPSK</b> : BPSK, coding rate unknown<br><b>BPSK12, BPSK34</b> (BPSKab): BPSK, coding rate a/b<br><b>BPSK14</b> : BPSK, coding rate 1/2 DCM<br><b>QPSK</b> : QPSK, coding rate unknown<br><b>QPSK12, QPSK34</b> (QPSKab): QPSK, coding rate a/b<br><b>QPSK14</b> : QPSK, coding rate 1/2 DCM<br><b>16Q</b> : 16-QAM, coding rate unknown<br><b>16Q12, 16Q34</b> (16Qab): 16-QAM, coding rate a/b<br><b>16Q14</b> : 16-QAM, coding rate 1/2 DCM<br><b>16Q38</b> : 16-QAM, coding rate 3/4 DCM<br><b>64Q</b> : 64-QAM, coding rate unknown<br><b>64Q12, 64Q23, 64Q34, 64Q56</b> (64Qab): 64-QAM, coding rate a/b<br><b>256Q</b> : 256-QAM, coding rate unknown<br><b>256Q34, 256Q56</b> (256Qab): 256-QAM, coding rate a/b<br><b>1KQ</b> : 1024-QAM, coding rate unknown<br><b>1KQ34, 1KQ56</b> (1KQab): 1024-QAM, coding rate a/b |

|                   |                                                                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <6_PayloadSym>    | decimal<br>Number of OFDM symbols in the payload of the measured burst<br>Range: 1 symbol to 1366 symbols<br>Default unit: symbol                                                                                    |
| <7_MeasuredSym>   | decimal<br>Number of OFDM symbols in the payload to be measured<br>Range: 1 symbol to 1366 symbols<br>Default unit: symbol                                                                                           |
| <8_PayloadBytes>  | decimal<br>Number of bytes in the payload of the measured burst<br>Range: 1 byte to 4095 bytes<br>Default unit: byte                                                                                                 |
| <9_GuardInterval> | SHORT   LONG   GI08   GI16   GI32<br><b>SHORT, LONG:</b> short or long guard interval (up to 802.11ac)<br><b>GI08, GI16, GI32:</b> 0.8 $\mu$ s, 1.6 $\mu$ s, and 3.2 $\mu$ s guard interval durations (for 802.11ax) |
| <10_NoSS>         | decimal<br>Number of spatial streams<br>Range: 1 to 8                                                                                                                                                                |
| <11_NoSTS>        | decimal<br>Number of space-time streams<br>Range: 1 to 8                                                                                                                                                             |
| <12_BurstRate>    | float<br>Indicates the share of bursts of the selected modulation type <5_ModType> in the bursts received.<br>Default unit: %                                                                                        |
| <13_PowerBackoff> | float<br>Minimum distance of signal power to reference level since the start of the measurement<br>Range: -100 dB to 0 dB<br>Default unit: dB                                                                        |
| <14_BurstPower>   | float<br>RMS power of the measured burst<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                           |
| <15_PeakPower>    | float<br>Peak power of the measured burst<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                          |

|                           |                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------|
| <16_CrestFactor>          | float<br>Range: 0 dB to 60 dB<br>Default unit: dB                                     |
| <17_EVMAAllCarr>          | float                                                                                 |
| <18_EVMDDataCarr>         | EVM for all, data, and pilot carriers                                                 |
| <19_EVMPilotCarr>         | Range: -100 dB to 0 dB<br>Default unit: dB                                            |
| <20_FreqError>            | float<br>Center frequency error<br>Range: -150E+6 Hz to 150E+6 Hz<br>Default unit: Hz |
| <21_ClockError>           | float<br>Symbol clock error<br>Range: -125 ppm to 125 ppm<br>Default unit: ppm        |
| <22_IQOffset>             | float<br>Range: -100 dB to 0 dB<br>Default unit: dB                                   |
| <23_DCPower>              | float<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm                               |
| <24_GainImbalance>        | float<br>Range: -100 dB to 100 dB<br>Default unit: dB                                 |
| <25_QuadError>            | float<br>Quadrature error<br>Range: -180 deg to 180 deg<br>Default unit: deg          |
| <26_LTFPower>             | float<br>Power of long training fields (LTF) portion<br>Default unit: dBm             |
| <27_DataPower>            | float<br>Power of data portion<br>Default unit: dBm                                   |
| <b>Example:</b>           | See <a href="#">Using WLAN list mode</a>                                              |
| <b>Usage:</b>             | Query only                                                                            |
| <b>Firmware/software:</b> | V3.8.11,                                                                              |
| <b>Options:</b>           | R&S CMW-KM012                                                                         |

```

FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:
CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:
AVERAge?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:
MINimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:
MAXimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation:DSSS:
SDEViation?

```

Return the current, average, minimum, maximum and standard deviation single value results for DSSS signals for segment <no> in list mode.

**Suffix:**

<no>                      1..n  
Segment

**Return values:**

|                    |                                                                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <1_Reliability>    | decimal<br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| <2_SegReliability> | decimal<br>Reliability indicator for the segment. The meaning of the returned values is the same as for the common reliability indicator, see previous parameter.                                                                                         |
| <3_OutOfTol>       | float<br>Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits.<br>Range:      0 % to 100 %<br>Default unit: %                                   |
| <4_ModType>        | DBPSk1   DQPSk2   CCK5   CCK11<br>Modulation scheme and coding rate<br><b>DBPSk1:</b> 1 Mbps DBPSK<br><b>DQPSk2:</b> 2 Mbps DQPSK<br><b>CCK5:</b> 5.5 Mbps CCK<br><b>CCK11:</b> 11 Mbps CCK                                                               |
| <5_PLCPTType>      | SHORTplcp   LONGplcp<br>Short or long PLCP                                                                                                                                                                                                                |
| <6_PayloadLength>  | decimal<br>Number of bytes in the payload of the measured burst<br>Range:      1 byte to 4095 bytes<br>Default unit: byte                                                                                                                                 |

|                 |                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------|
| <7_BurstPower>  | float<br>RMS power of the measured burst<br>Range: -100 dBm to 30 dBm<br>Default unit: dBm |
| <8_EVMPeak>     | float<br>Error vector magnitude peak value<br>Range: 0 % to 100 %<br>Default unit: %       |
| <9_EVMRMS>      | float<br>Error vector magnitude RMS value<br>Range: 0 % to 100 %<br>Default unit: %        |
| <10_FreqError>  | float<br>Center frequency error<br>Range: -150E+6 Hz to 150E+6 Hz<br>Hz                    |
| <11_ClockError> | float<br>Chip clock error<br>Range: -125 ppm to 125 ppm<br>Default unit: ppm               |
| <12_IQOffset>   | float<br>Range: -100 dB to 0 dB<br>Default unit: dB                                        |
| <13_GainImbal>  | float<br>Gain imbalance<br>Range: -100 dB to 100 dB<br>Default unit: dB                    |
| <14_QuadError>  | float<br>Quadrature error<br>Range: -180 deg to 180 deg<br>Default unit: deg               |

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012

---

**FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:CURRent?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:SDEVIation?**

Return limit line margin results for segment <no> in list mode.

A positive result indicates that the trace is located above the limit line, i.e. the limit is exceeded.

Margins for the current, average, minimum, maximum and standard deviation results are returned.

**Suffix:**

<no> 1..n  
 Segment

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

<SegReliability> decimal  
 Reliability indicator for the segment. The meaning of the returned values is the same as for the common reliability indicator, see previous parameter.

<OutOfTol> float  
 Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.  
 Range: 0 % to 100 %  
 Default unit: %

<Margin\_AB> ...  
 <Margin\_BA> Comma-separated list of margin values, one value per spectrum mask area  
 The number of margin values depends on the selected standard, see [Table B-1](#).  
 Range: -100 dB to 100 dB  
 Default unit: dB

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQUENCY:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQUENCY:AVERAGE?**

**FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQUENCY:MINimum?**

**FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQUENCY:MAXimum?**

**FETCh:WLAN:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TSMask:FREQUENCY:SDEVIation?**

Return the X-positions of the limit line margins of the transmit spectrum mask for list mode, segment <no>.

Positions for the current, average, minimum, maximum and standard deviation results are returned.

**Suffix:**

<no> 1..n  
Segment

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

<SegReliability> decimal  
Reliability indicator for the segment. The meaning of the returned values is the same as for the common reliability indicator, see previous parameter.

<OutOfTol> float  
Out of tolerance result, i.e. percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified transmit spectrum mask limits.  
Range: 0 % to 100 %  
Default unit: %

<Frequency\_AB> ... float  
<Frequency\_BA> Comma-separated list of frequencies, one value per margin  
The number of margins equals the number of spectrum mask areas and depends on the selected standard, see [Table B-1](#).  
Range: -40E+6 Hz to 40E+6 Hz  
Default unit: Hz

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012



### 3.5.5.25 List mode results (all segments)

The following commands return a selected list mode result for all measured segments. The number of returned results depends on the number of measured segments, see [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#) on page 388.

To configure the list mode, use the commands described in [Chapter 3.5.3.4, "List mode settings"](#), on page 386.

The indicated ranges apply to all statistical results except standard deviation results. The minimum for standard deviation results equals 0. The maximum equals the width of the indicated range divided by two. Exceptions are explicitly stated.

For a description of the list mode, see [Chapter 3.2.7, "Multi-evaluation list mode"](#), on page 252.

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|                                                                                 |     |
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### FETCh:WLAN:MEAS<i>:MEValuation:LIST:SRELIability?

Returns the segment reliability for all measured list mode segments.

The number of active segments n is determined by `CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT`.

#### Return values:

<Reliability> decimal

#### Reliability indicator

A common reliability indicator of zero indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. If you get a non-zero common reliability indicator, you can use this command to retrieve the individual reliability values of all measured segments for further analysis.

<SegReliabilities> decimal

Comma-separated list of n values, one per measured segment  
The meaning of the returned values is the same as for the common reliability indicator, see previous parameter.

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

```

FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:BPOwer:CURRent?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:BPOwer:AVERage?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:BPOwer:MINimum?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:BPOwer:MAXimum?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:BPOwer:SDEViation?

```

Return the burst power results for OFDM/OFDMA signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<BurstPower>}1... float

{<BurstPower>}n RMS power value

Range: -100 dBm to 30 dBm

Default unit: dBm

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

```

FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DSSS:BPOwer:CURRent?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DSSS:BPOwer:AVERage?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DSSS:BPOwer:MINimum?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DSSS:BPOwer:MAXimum?
FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DSSS:BPOwer:SDEViation?

```

Return the burst power results for DSSS signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<BurstPower>}1... float  
 {<BurstPower>}n RMS power value  
 Range: -100 dBm to 30 dBm

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012

**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DCPower:CURRENT?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DCPower:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DCPower:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DCPower:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DCPower:SDEViation?**

Return the power results of DC subcarriers in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<DCPower>}1... float  
 {<DCPower>}n Range: -100 dBm to 30 dBm  
 Default unit: dBm

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DPOwer:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DPOwer:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DPOwer:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DPOwer:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:DPOwer:SDEViation?**

Return the power results of data portion of the burst in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

|                                |                                                                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;Reliability&gt;</b>     | decimal<br><br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| <b>{&lt;DataPower&gt;}1...</b> | float                                                                                                                                                                                                                                                         |
| <b>{&lt;DataPower&gt;}n</b>    | Range: -100 dBm to 30 dBm<br>Default unit: dBm                                                                                                                                                                                                                |

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:LTFPower:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:LTFPower:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:LTFPower:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:LTFPower:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:LTFPower:SDEViation?**

Return the power results of LTF portion of the burst in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

|                               |                                                                                                                                                                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;Reliability&gt;</b>    | decimal<br><br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| <b>{&lt;LTFPower&gt;}1...</b> | float                                                                                                                                                                                                                                                         |
| <b>{&lt;LTFPower&gt;}n</b>    | Range: -100 dBm to 30 dBm                                                                                                                                                                                                                                     |

**Example:** See [Using WLAN list mode](#)



**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:PPOwer:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:PPOwer:AVERAge?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:PPOwer:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:PPOwer:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:PPOwer:SDEViation?**

Return the peak power results of the burst in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<PeakPower>}1... float  
 {<PeakPower>}n Range: -100 dBm to 30 dBm

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:PBACkoff?**

Return the power backoff results in list mode.

The power backoff displays the minimum distance of signal power to reference level over all segments since the start of the measurement.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<PowerBackoff>}1... float  
 {<PowerBackoff>}n Range: -100 dB to 0 dB

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CCERror:CURRENT?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CCERror:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CCERror:MINimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CCERror:MAXimum?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CCERror:SDEViation?**

Return the chip clock error results for DSSS signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<ClockError>}1... float  
 {<ClockError>}n Chip clock error  
 Range: -125 ppm to 125 ppm  
 Default unit: ppm

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012

---

**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:CFActor:CURRENT?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:CFActor:AVERage?**  
**FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:CFActor:MINimum?**



**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFACtor:MAXimum?**  
**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFACtor:SDEVIation?**

Return the crest factor results in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS< i >:MEValuation:LIST:COUNT](#).

**Return values:**

|                     |                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>       | decimal<br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| {<CrestFactor>}1... | float                                                                                                                                                                                                                                                     |
| {<CrestFactor>}n    | Range: 0 dB to 60 dB<br>Default unit: dB                                                                                                                                                                                                                  |

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFERror:CURRent?**  
**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFERror:AVERage?**  
**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFERror:MINimum?**  
**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFERror:MAXimum?**  
**FETCH:WLAN:MEAS< i >:MEValuation:LIST:MODulation:CFERror:SDEVIation?**

Return the center frequency error results for OFDM/OFDMA signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS< i >:MEValuation:LIST:COUNT](#).

**Return values:**

|                   |                                                                                                                                                                                                                                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability>     | decimal<br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| {<FreqError>}1... | float                                                                                                                                                                                                                                                     |
| {<FreqError>}n    | Range: -150E+6 Hz to 150E+6 Hz<br>Default unit: Hz                                                                                                                                                                                                        |

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CFERror:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CFERror:AVERAge?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CFERror:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CFERror:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:CFERror:SDEViation?**

Return the center frequency error results for DSSS signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal

[Reliability indicator](#)

In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<FreqError>}1... float

{<FreqError>}n Range: -150E+6 Hz to 150E+6 Hz  
Default unit: Hz

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMall:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMall:AVERAge?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMall:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMall:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMall:SDEViation?**

Return the EVM results for OFDM/OFDMA signals for all carriers in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

&lt;Reliability&gt; decimal

[Reliability indicator](#)

In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{&lt;EVM\_All&gt;}1...

float

{&lt;EVM\_All&gt;}n

Range: -100 dB to 0 dB

Default unit: dB

**Example:**See [Using WLAN list mode](#)**Usage:**

Query only

**Firmware/software:** V3.8.11**Options:** R&S CMW-KM012

---

```

FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMDData:CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMDData:AVERage?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMDData:MINimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMDData:MAXimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMDData:SDEViation?

```

Return the EVM results for OFDM/OFDMA signals for data carriers in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>,

{...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:](#)

[WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

&lt;Reliability&gt; decimal

[Reliability indicator](#)

In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{&lt;EVM\_Data&gt;}1...

float

{&lt;EVM\_Data&gt;}n

Range: -100 dB to 0 dB

Default unit: dB

**Example:**See [Using WLAN list mode](#)**Usage:**

Query only

**Firmware/software:** V3.8.11**Options:** R&S CMW-KM012

---

```

FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMPilot:CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMPilot:AVERage?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMPilot:MINimum?

```

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMPilot:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:EVMPilot:SDEViation?**

Return the EVM results for OFDM/OFDMA signals for pilot carrier in list mode.

The values in curly brackets {} are specified for each active segment: {...}\_seg 1, {...}\_seg 2, ..., {...}\_seg n. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<EVM\_Pilot>}1... float  
 {<EVM\_Pilot>}n Range: -100 dB to 0 dB  
 Default unit: dB

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMPeak:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMPeak:AVERage?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMPeak:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMPeak:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMPeak:SDEViation?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMRms:CURRENT?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMRms:AVERage?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMRms:MINimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMRms:MAXimum?**

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:EVMRms:SDEViation?**

Return the current, average, minimum, maximum and standard deviation EVM results for DSSS signals in list mode.

Commands for EVM peak and EVM RMS values are available.

The values in curly brackets {} are specified for each active segment: {...}\_seg 1, {...}\_seg 2, ..., {...}\_seg n. The number of active segments n is determined by [CONFIGure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<EVM>}1... float  
 {<EVM>}n Range: 0 % to 100 %  
 Default unit: %

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:GIMBalance:CURRENT?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:GIMBalance:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:GIMBalance:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:GIMBalance:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEvaluation:LIST:MODulation:GIMBalance:SDEVIation?**

Return the gain imbalance results for OFDM/OFDMA signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}\_seg 1, {...}\_seg 2, ..., {...}\_seg n. The number of active segments n is determined by [CONFIGure: WLAN:MEAS<i>:MEvaluation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<GainImbal>}1... float  
 {<GainImbal>}n Range: -100 dB to 100 dB

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

```

FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:GIMBalance:
    CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:GIMBalance:
    AVERage?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:GIMBalance:
    MINimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:GIMBalance:
    MAXimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:GIMBalance:
    SDEViation?

```

Return the gain imbalance results for DSSS signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

|               |                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal                                                                                                                                                                                               |
|               | <a href="#">Reliability indicator</a>                                                                                                                                                                 |
|               | In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |

|                   |                          |
|-------------------|--------------------------|
| {<GainImbal>}1... | float                    |
| {<GainImbal>}n    | Range: -100 dB to 100 dB |

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012

---

```

FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:CURRent?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:AVERage?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:MINimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:MAXimum?
FETCh:WLAN:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:SDEViation?

```

Return the IQ offset results for OFDM/OFDMA signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure: WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

|               |                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal                                                                                                                                                                                               |
|               | <a href="#">Reliability indicator</a>                                                                                                                                                                 |
|               | In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |

|                           |                                          |
|---------------------------|------------------------------------------|
| {<IQOffset>}1...          | float                                    |
| {<IQOffset>}n             | Range: -100 dB to 0 dB                   |
| <b>Example:</b>           | See <a href="#">Using WLAN list mode</a> |
| <b>Usage:</b>             | Query only                               |
| <b>Firmware/software:</b> | V3.8.11                                  |
| <b>Options:</b>           | R&S CMW-KM012                            |

---

```

FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:IQOffset:
  CURRent?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:IQOffset:
  AVERage?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:IQOffset:
  MINimum?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:IQOffset:
  MAXimum?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:IQOffset:
  SDEViation?

```

Return the IQ offset results for DSSS signals in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

|               |                                                                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <Reliability> | decimal                                                                                                                                                                                               |
|               | <a href="#">Reliability indicator</a>                                                                                                                                                                 |
|               | In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |

|                           |                                                         |
|---------------------------|---------------------------------------------------------|
| {<IQOffset>}1...          | float                                                   |
| {<IQOffset>}n             | Range: -100 dB to 0 dB                                  |
| <b>Example:</b>           | See <a href="#">Performing single-shot measurements</a> |
| <b>Usage:</b>             | Query only                                              |
| <b>Firmware/software:</b> | V3.8.20                                                 |
| <b>Options:</b>           | R&S CMW-KM012                                           |

---

```

FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:QERRor:CURRent?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:QERRor:AVERage?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:QERRor:MINimum?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:QERRor:MAXimum?
FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:QERRor:SDEViation?

```

Return the quadrature error results for OFDM/OFDMA in list mode.

The values in curly brackets {} are specified for each active segment: {...}\_seg 1, {...}\_seg 2, ..., {...}\_seg n. The number of active segments n is determined by [CONFIGure: WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<QuadError>}1... float  
 {<QuadError>}n Range: -180 deg to 180 deg

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:QERRor:CURRENT?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:QERRor:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:QERRor:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:QERRor:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:DSSS:QERRor:SDEViation?**

Return the quadrature error results for DSSS in list mode.

The values in curly brackets {} are specified for each active segment: {...}\_seg 1, {...}\_seg 2, ..., {...}\_seg n. The number of active segments n is determined by [CONFIGure: WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<QuadError>}1... float  
 {<QuadError>}n Range: -180 deg to 180 deg

**Example:** See [Performing single-shot measurements](#)

**Usage:** Query only

**Firmware/software:** V3.8.20

**Options:** R&S CMW-KM012



**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:SCERror:CURRent?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:SCERror:AVERage?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:SCERror:MINimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:SCERror:MAXimum?**  
**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:SCERror:SDEViation?**

Return the symbol clock error results in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

|                               |                                                                                                                                                                                                                                                               |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;Reliability&gt;</b>    | decimal<br><br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| <b>{&lt;ClockErr&gt;}1...</b> | float                                                                                                                                                                                                                                                         |
| <b>{&lt;ClockErr&gt;}n</b>    | Range: -125 ppm to 125 ppm<br>Default unit: ppm                                                                                                                                                                                                               |

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:MODulation:SCOut?**

Returns the expired statistic counts for modulation results over all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFIGure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;Reliability&gt;</b>        | decimal<br><br><a href="#">Reliability indicator</a><br>In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. |
| <b>{&lt;ExpStatCount&gt;}1...</b> | decimal                                                                                                                                                                                                                                                       |
| <b>{&lt;ExpStatCount&gt;}n</b>    | No. of bursts measured during modulation measurements, one result per segment<br>Range: 1 to 2000                                                                                                                                                             |

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

---

**FETCH:WLAN:MEAS<i>:MEValuation:LIST:TSMask:SCount?**

Returns the expired statistic counts for transmit spectrum mask results over all segments in list mode.

The values in curly brackets {} are specified for each active segment: {...}<sub>seg 1</sub>, {...}<sub>seg 2</sub>, ..., {...}<sub>seg n</sub>. The number of active segments n is determined by [CONFigure:WLAN:MEAS<i>:MEValuation:LIST:COUNT](#).

**Return values:**

<Reliability> decimal  
[Reliability indicator](#)  
 In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

{<ExpStatCount>}1... decimal  
 {<ExpStatCount>}n No. of bursts measured during spectrum measurements, one result per segment  
 Range: 1 to 1000

**Example:** See [Using WLAN list mode](#)

**Usage:** Query only

**Firmware/software:** V3.8.11

**Options:** R&S CMW-KM012

### 3.5.6 CMIMO training mode

The following commands are related to the CMIMO training mode.

|                                                                                 |     |
|---------------------------------------------------------------------------------|-----|
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:MODE</a> .....                         | 606 |
| <a href="#">CLEar:WLAN:MEAS&lt;i&gt;:TMODE:DATA</a> .....                       | 607 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:TMODE:NOAntennas</a> .....             | 607 |
| <a href="#">INITiate:WLAN:MEAS&lt;i&gt;:TMODE:ANTenna&lt;Antennas&gt;</a> ..... | 607 |
| <a href="#">ABORt:WLAN:MEAS&lt;i&gt;:TMODE</a> .....                            | 608 |
| <a href="#">FETCh:WLAN:MEAS&lt;i&gt;:TMODE:ANTenna&lt;Antennas&gt;?</a> .....   | 608 |
| <a href="#">READ:WLAN:MEAS&lt;i&gt;:TMODE:ANTenna&lt;Antennas&gt;?</a> .....    | 608 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:TMODE:FILE:SAVE</a> .....              | 609 |
| <a href="#">CONFigure:WLAN:MEAS&lt;i&gt;:TMODE:FILE:DATE?</a> .....             | 609 |

---

**CONFigure:WLAN:MEAS<i>:MODE <TrainingMode>**

Switches between the measurement mode and the training mode.

**Parameters:**

<TrainingMode> TMODE | MMODE  
**TMODE:** Activate the training mode  
**MMODE:** Activate the measurement mode  
 \*RST: MMODE

**Example:** See [CMIMO training mode](#)

**Firmware/software:** V2.1.10

**Options:** R&S CMW-KM652

**Manual operation:** See ["Training Mode"](#) on page 290

**CLEar:WLAN:MEAS<i>:TMODE:DATA**

Clears the current training results and resets the internal data of the training mode.

**Example:** See [CMIMO training mode](#)

**Usage:** Event

**Firmware/software:** V2.1.10

**Options:** R&S CMW-KM652

**Manual operation:** See ["Clear Data \(Hotkey\)"](#) on page 328

**CONFigure:WLAN:MEAS<i>:TMODE:NOAntennas <NoOfAntennas>**

Gets/sets the number of TX antennas contributing to the MIMO training signal.

**Parameters:**

<NoOfAntennas> 1 to 8  
 \*RST: 4

**Example:** See [CMIMO training mode](#)

**Firmware/software:** V3.7.31

**Options:** R&S CMW-KM652

**Manual operation:** See ["No of Antennas \(Hotkey\)"](#) on page 329

**INITiate:WLAN:MEAS<i>:TMODE:ANTenna<Antennas>**

Starts the training data acquisition for the designated antenna.

**Suffix:**

<Antennas> 1..8  
 Antenna number; must not exceed the configured number of antennas

**Usage:** Event

**Firmware/software:** V2.1.10, V3.7.31: increased number of antennas

**Options:** R&S CMW-KM652

**Manual operation:** See ["Antenna <i> Data Acqui. \(Softkey\)"](#) on page 327

---

#### **ABORt:WLAN:MEAS<i>:TMODe**

Aborts the current MIMO training data acquisition.

**Usage:** Event

**Firmware/software:** V2.1.10

**Options:** R&S CMW-KM652

**Manual operation:** See ["Antenna <i> Data Acqui. \(Softkey\)"](#) on page 327

---

#### **FETCh:WLAN:MEAS<i>:TMODe:ANTenna<Antennas>?**

#### **READ:WLAN:MEAS<i>:TMODe:ANTenna<Antennas>?**

Return information about the training data acquired on the designated antenna and the decode status of the received signal.

##### **Suffix:**

<Antennas> 1..8  
Antenna number; must not exceed the configured number of antennas

##### **Return values:**

<Reliability> decimal  
[Reliability indicator](#)

<DecodeStatus> NAV | INV | OK  
The decode status of the received signal is not available (NAV) until all involved TX antenna signals were recorded. Then it changes to OK if the HT-SIG (high throughput signaling) fields of individual antenna signals are consistent, or to INValid otherwise.

<MCS> decimal  
Modulation and coding scheme of the recorded antenna signal, obtained from the HT-SIG field

<Power> float  
Absolute power of the measured antenna signal  
Default unit: dBm

<PilotEvm> float  
Error vector magnitude of the pilot subcarriers  
Default unit: dB

**Example:** See [CMIMO training mode](#)

**Usage:** Query only

**Firmware/software:** V2.1.10, V3.7.31: increased number of antennas

**Options:** R&S CMW-KM652

**Manual operation:** See ["Results"](#) on page 328

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#### **CONFigure:WLAN:MEAS<i>:TMODe:FILE:SAVE <FileName>**

Saves the current training data to a file or queries the file name of the last saved training data file.

##### **Parameters:**

<FileName> string

The name of the training data file - without path but including the file extension.

Training data files are saved to the directory  
@USERDATA\MIMOData.

**Example:** See [CMIMO training mode](#)

**Firmware/software:** V2.1.10

**Options:** R&S CMW-KM652

**Manual operation:** See ["Save Data to File \(Hotkey\)"](#) on page 328

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#### **CONFigure:WLAN:MEAS<i>:TMODe:FILE:DATE?**

Returns the last modified date of the last saved training data file, if any.

##### **Return values:**

<FileDate> string  
String with a formatted date or NAV

**Usage:** Query only

**Firmware/software:** V2.1.30

**Options:** R&S CMW-KM652

**Manual operation:** See ["Save Data to File \(Hotkey\)"](#) on page 328

### **3.5.7 Combined signal path commands**

For some settings, the command to be used depends on the active scenario. While the combined signal path (CSP) scenario is active, these settings are configured via commands of the signaling application. While the standalone (SA) scenario is active, they are configured via measurement commands.

The following mapping table provides an overview.

Table 3-18: Mapping of SA commands to CSP commands

| Setting                | Commands for SA scenario                                                      | Commands for CSP scenario                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Connector, converter   | <code>ROUTe:WLAN:MEAS&lt;i&gt;:SCENario:SA lone</code>                        | <code>ROUTe:WLAN:MEAS&lt;i&gt;:SCENario:CSPath</code><br><code>ROUTe:WLAN:SIGN&lt;i&gt;:SCENario:SCEl1:FLEXible</code><br><code>ROUTe:WLAN:SIGN&lt;i&gt;:SCENario:SCFading:FLEXible</code><br><code>ROUTe:WLAN:SIGN&lt;i&gt;:SCENario:MIMO:FLEXible</code><br><code>ROUTe:WLAN:SIGN&lt;i&gt;:SCENario:MIMFading:FLEXible</code>                                                                           |
| External attenuation   | <code>CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:EATTenuation</code>             | <code>CONFigure:WLAN:SIGN&lt;i&gt;:RFSettings:ANTenna&lt;n&gt;:EATTenuation:INPut</code>                                                                                                                                                                                                                                                                                                                  |
| Frequency              | <code>CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FREQuency</code>                | <code>CONFigure:WLAN:SIGN&lt;i&gt;:RFSettings:FREQuency</code><br><code>CONFigure:WLAN:SIGN&lt;i&gt;:RFSettings:CHANnel</code>                                                                                                                                                                                                                                                                            |
| Frequency off-set      | <code>CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:FOFFset</code>                  | <code>CONFigure:WLAN:SIGN&lt;i&gt;:RFSettings:FOFFset</code>                                                                                                                                                                                                                                                                                                                                              |
| Expected nominal power | <code>CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:ENPower&lt;connector&gt;</code> | <code>CONFigure:WLAN:SIGN&lt;i&gt;:RFSettings:ANTenna&lt;n&gt;:EPEPower</code>                                                                                                                                                                                                                                                                                                                            |
| Mixer level off-set    | <code>CONFigure:WLAN:MEAS&lt;i&gt;:RFSettings:MLOFFset</code>                 | <code>CONFigure:WLAN:SIGN&lt;i&gt;:RFSettings:ANTenna&lt;n&gt;:MLOFFset</code>                                                                                                                                                                                                                                                                                                                            |
| Standard               | <code>CONFigure:WLAN:MEAS&lt;i&gt;:ISignal:STANdard</code>                    | In CSP scenario, the standard is selectable by the measurement command and must comply the "RX Frame Trigger" configuration.<br><br>See <code>TRIGger:WLAN:SIGN&lt;i&gt;:RX:MACFrame:BTYPe</code><br><br>Selecting a standard that is not supported by the signaling application restores the standalone scenario.                                                                                        |
| Bandwidth              | <code>CONFigure:WLAN:MEAS&lt;i&gt;:ISignal:BWIDth</code>                      | In CSP scenario, the bandwidth is selectable by the measurement command and must comply the "RX Frame Trigger" configuration:<br><br><code>TRIGger:WLAN:SIGN&lt;i&gt;:RX:MACFrame:BW</code><br><br>For 802.11ax and trigger source set to "HE_TB Trigger", the setting must comply the trigger configuration:<br><br><code>CONFigure:WLAN:SIGN&lt;i&gt;:CONNection:HETF:CHBW</code>                       |
| Modulation filter      | <code>CONFigure:WLAN:MEAS&lt;i&gt;:ISignal:MODFilter</code>                   | In CSP scenario, the modulation filter is selectable by the measurement command and must comply the "RX Frame Trigger" configuration:<br><br><code>TRIGger:WLAN:SIGN&lt;i&gt;:RX:MACFrame:RATE</code><br><br>For 802.11ax and trigger source set to "HE_TB Trigger", the setting must comply the trigger configuration:<br><br><code>CONFigure:WLAN:SIGN&lt;i&gt;:STA&lt;s&gt;:CONNection:HETF:MCS</code> |

### 3.6 List of commands

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# Annex

## A Predefined fading settings

The predefined fading settings correspond to the test scenarios defined in the common mobile radio standards. The following tables provide a listing of the predefined standards along with the underlying test scenarios and the enabled settings.

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- [802.11n - SISO channel models](#)..... 643
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### A.1 802.11n - MIMO channel models

According to IEEE 802.11-03/940r4 TGn Channel Models.

Rx antenna distance = 1

Tx antenna distance = 0.5

Distribution = Laplacian

Profile = bell shape high throughput task group (TGn) indoor, exception model F, path 3 where the profile = bell shape TGn moving vehicle

Speed = 1.2 km/h, exception model F, path 3 where speed = 40 km/h

#### A.1.1 Model A

| Tap                                  | Path 1                |
|--------------------------------------|-----------------------|
| Cluster                              |                       |
| Profile                              | bell shape TGn indoor |
| Loss [dB]                            | 0                     |
| Delay [ns]                           | 0                     |
| Angle of arrival (AoA)               | 45                    |
| Angular spread (AS) at receiver (Rx) | 40                    |
| Angle of departure (AoD)             | 45                    |
| AS at transmitter (Tx)               | 40                    |
| Speed [km/h]                         | 1.2                   |

### A.1.2 Model B

| Tap          | Path 1                   | Path 2                   | Path 3                   |                          | Path 4                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 0                        | 5.4                      | 10.8                     | 3.2                      | 16.2                     | 6.3                      |
| Delay [ns]   | 0                        | 10                       | 20                       | 20                       | 30                       | 30                       |
| AoA          | 4.3                      | 4.3                      | 4.3                      | 118.4                    | 4.3                      | 118.4                    |
| AS (Rx)      | 14.4                     | 14.4                     | 14.4                     | 25.2                     | 14.4                     | 25.2                     |
| AoD          | 225.1                    | 225.1                    | 225.1                    | 106.5                    | 225.1                    | 106.5                    |
| AS (Tx)      | 14.4                     | 14.4                     | 14.4                     | 25.4                     | 14.4                     | 25.4                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 5                   |                          | Path 6                   | Path 7                   | Path 8                   | Path 9                   |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        |                          |                          |                          |                          |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 21.7                     | 9.4                      | 12.5                     | 15.6                     | 18.7                     | 21.8                     |
| Delay [ns]   | 40                       | 40                       | 50                       | 60                       | 70                       | 80                       |
| AoA          | 4.3                      | 118.4                    | 118.4                    | 118.4                    | 118.4                    | 118.4                    |
| AS (Rx)      | 14.4                     | 25.2                     | 25.2                     | 25.2                     | 25.2                     | 25.2                     |
| AoD          | 225.1                    | 106.5                    | 106.5                    | 106.5                    | 106.5                    | 106.5                    |
| AS (Tx)      | 14.4                     | 25.4                     | 25.4                     | 25.4                     | 25.4                     | 25.4                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

### A.1.3 Model C

| Tap        | Path 1                   | Path 2                   | Path 3                   | Path 4                   | Path 5                   | Path 6                   |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    |                          |                          |                          |                          |                          |                          |
| Profile    | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]  | 0                        | 2.1                      | 4.3                      | 6.5                      | 8.6                      | 10.8                     |
| Delay [ns] | 0                        | 10                       | 20                       | 30                       | 40                       | 50                       |

| Tap          | Path 1    | Path 2    | Path 3    | Path 4    | Path 5    | Path 6    |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| AoA          | 290.3     | 290.3     | 290.3     | 290.3     | 290.3     | 290.3     |
| AS (Rx)      | 24.6      | 24.6      | 24.6      | 24.6      | 24.6      | 24.6      |
| AoD          | 13.5      | 13.5      | 13.5      | 13.5      | 13.5      | 13.5      |
| AS (Tx)      | 24.7      | 24.7      | 24.7      | 24.7      | 24.7      | 24.7      |
| Speed [km/h] | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

| Tap          | Path 7                   |                          | Path 8                   |                          | Path 9                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 13                       | 5                        | 15.2                     | 7.2                      | 17.3                     | 9.3                      |
| Delay [ns]   | 60                       | 60                       | 70                       | 70                       | 80                       | 80                       |
| AoA          | 290.3                    | 332.3                    | 290.3                    | 332.3                    | 290.3                    | 332.3                    |
| AS (Rx)      | 24.6                     | 22.4                     | 24.6                     | 22.4                     | 24.6                     | 22.4                     |
| AoD          | 13.5                     | 56.4                     | 13.5                     | 56.4                     | 13.5                     | 56.4                     |
| AS (Tx)      | 24.7                     | 22.5                     | 24.7                     | 22.5                     | 24.7                     | 22.5                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 10                  |                          | Path 11                  | Path 12                  | Path 13                  | Path 14                  |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        |                          |                          |                          |                          |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 19.5                     | 11.5                     | 13.7                     | 15.8                     | 18                       | 20.2                     |
| Delay [ns]   | 90                       | 90                       | 110                      | 140                      | 170                      | 200                      |
| AoA          | 290.3                    | 332.3                    | 332.3                    | 332.3                    | 332.3                    | 332.3                    |
| AS (Rx)      | 24.6                     | 22.4                     | 22.4                     | 22.4                     | 22.4                     | 22.4                     |
| AoD          | 13.5                     | 56.4                     | 56.4                     | 56.4                     | 56.4                     | 56.4                     |
| AS (Tx)      | 24.7                     | 22.5                     | 22.5                     | 22.5                     | 22.5                     | 22.5                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

### A.1.4 Model D

| Tap          | Path 1                   | Path 2                   | Path 3                   | Path 4                   | Path 5                   | Path 6                   |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          |                          |                          |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 0                        | 0.9                      | 1.7                      | 2.6                      | 3.5                      | 4.3                      |
| Delay [ns]   | 0                        | 10                       | 20                       | 30                       | 40                       | 50                       |
| AoA          | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 158.9                    |
| AS (Rx)      | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 27.7                     |
| AoD          | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 332.1                    |
| AS (Tx)      | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 27.4                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 7                   | Path 8                   | Path 9                   | Path 10                  | Path 11                  |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 5.2                      | 6.1                      | 6.9                      | 7.8                      | 9                        | 6.6                      |
| Delay [ns]   | 60                       | 70                       | 80                       | 90                       | 110                      | 110                      |
| AoA          | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 320.2                    |
| AS (Rx)      | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 31.4                     |
| AoD          | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 49.3                     |
| AS (Tx)      | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 32.1                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 12                  |                          | Path 13                  |                          | Path 14                  |                          |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile    | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]  | 11.1                     | 9.5                      | 13.7                     | 12.1                     | 16.3                     | 14.7                     |
| Delay [ns] | 140                      | 140                      | 170                      | 170                      | 200                      | 200                      |
| AoA        | 158.9                    | 320.2                    | 158.9                    | 320.2                    | 158.9                    | 320.2                    |
| AS (Rx)    | 27.7                     | 31.4                     | 27.7                     | 31.4                     | 27.7                     | 31.4                     |



| Tap          | Path 12   |           | Path 13   |           | Path 14   |           |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| AoD          | 332.1     | 49.3      | 332.1     | 49.3      | 332.1     | 49.3      |
| AS (Tx)      | 27.4      | 32.1      | 27.4      | 32.1      | 27.4      | 32.1      |
| Speed [km/h] | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

| Tap          | Path 15                  |                          |                          | Path 16                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 19.3                     | 17.4                     | 18.8                     | 23.2                     | 21.9                     | 23.2                     |
| Delay [ns]   | 240                      | 240                      | 240                      | 290                      | 290                      | 290                      |
| AoA          | 158.9                    | 320.2                    | 276.1                    | 158.9                    | 320.2                    | 276.1                    |
| AS (Rx)      | 27.7                     | 31.4                     | 37.4                     | 27.7                     | 31.4                     | 37.4                     |
| AoD          | 332.1                    | 49.3                     | 275.9                    | 332.1                    | 49.3                     | 275.9                    |
| AS (Tx)      | 27.4                     | 32.1                     | 36.8                     | 27.4                     | 32.1                     | 36.8                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 17               |                       | Path 18               |
|--------------|-----------------------|-----------------------|-----------------------|
| Cluster      | 2                     | 3                     |                       |
| Profile      | bell shape TGn indoor | bell shape TGn indoor | bell shape TGn indoor |
| Loss [dB]    | 25.5                  | 25.2                  | 26.7                  |
| Delay [ns]   | 340                   | 340                   | 390                   |
| AoA          | 320.2                 | 276.1                 | 276.1                 |
| AS (Rx)      | 31.4                  | 37.4                  | 37.4                  |
| AoD          | 49.3                  | 275.9                 | 275.9                 |
| AS (Tx)      | 32.1                  | 36.8                  | 36.8                  |
| Speed [km/h] | 1.2                   | 1.2                   | 1.2                   |
| Distribution | Laplacian             | Laplacian             | Laplacian             |

### A.1.5 Model E

| Tap          | Path 1                   | Path 2                   | Path 3                              | Path 4                   | Path 5                   |                          |
|--------------|--------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                                     |                          | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn moving<br>vehicle | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 2.6                      | 3                        | 3.5                                 | 3.9                      | 4.5                      | 1.8                      |
| Delay [ns]   | 0                        | 10                       | 20                                  | 30                       | 50                       | 50                       |
| AoA          | 163.7                    | 163.7                    | 163.7                               | 163.7                    | 163.7                    | 251.8                    |
| AS (Rx)      | 35.8                     | 35.8                     | 35.8                                | 35.8                     | 35.8                     | 41.6                     |
| AoD          | 105.6                    | 105.6                    | 105.6                               | 105.6                    | 105.6                    | 293.1                    |
| AS (Tx)      | 36.1                     | 36.1                     | 36.1                                | 36.1                     | 36.1                     | 42.5                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                                 | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                           | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 6                   |                          | Path 7                   |                          | Path 8                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 5.6                      | 3.2                      | 6.9                      | 4.5                      | 8.2                      | 5.8                      |
| Delay [ns]   | 80                       | 80                       | 110                      | 110                      | 140                      | 140                      |
| AoA          | 163.7                    | 251.8                    | 163.7                    | 251.8                    | 163.7                    | 251.8                    |
| AS (Rx)      | 35.8                     | 41.6                     | 35.8                     | 41.6                     | 35.8                     | 41.6                     |
| AoD          | 105.6                    | 293.1                    | 105.6                    | 293.1                    | 105.6                    | 293.1                    |
| AS (Tx)      | 36.1                     | 42.5                     | 36.1                     | 42.5                     | 36.1                     | 42.5                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 9                   |                          |                          | Path 10                  |                          |                          |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile    | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]  | 9.8                      | 7.1                      | 7.9                      | 11.7                     | 9.9                      | 9.6                      |
| Delay [ns] | 180                      | 180                      | 180                      | 230                      | 230                      | 230                      |
| AoA        | 163.7                    | 251.8                    | 80                       | 163.7                    | 251.8                    | 80                       |
| AS (Rx)    | 35.8                     | 41.6                     | 37.4                     | 35.8                     | 41.6                     | 37.4                     |

| Tap          | Path 9    |           |           | Path 10   |           |           |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| AoD          | 105.6     | 293.1     | 61.9      | 105.6     | 293.1     | 61.9      |
| AS (Tx)      | 36.1      | 42.5      | 38        | 36.1      | 42.5      | 38        |
| Speed [km/h] | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

| Tap          | Path 11                  |                          |                          | Path 12                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 13.9                     | 10.3                     | 14.2                     | 16.1                     | 14.3                     | 13.8                     |
| Delay [ns]   | 280                      | 280                      | 280                      | 330                      | 330                      | 330                      |
| AoA          | 163.7                    | 251.8                    | 80                       | 163.7                    | 251.8                    | 80                       |
| AS (Rx)      | 35.8                     | 41.6                     | 37.4                     | 35.8                     | 41.6                     | 37.4                     |
| AoD          | 105.6                    | 293.1                    | 61.9                     | 105.6                    | 293.1                    | 61.9                     |
| AS (Tx)      | 36.1                     | 42.5                     | 38                       | 36.1                     | 42.5                     | 38                       |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 13                  |                          |                          | Path 14                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 18.3                     | 14.7                     | 18.6                     | 20.5                     | 18.7                     | 18.1                     |
| Delay [ns]   | 380                      | 380                      | 380                      | 430                      | 430                      | 430                      |
| AoA          | 163.7                    | 251.8                    | 80                       | 163.7                    | 251.8                    | 80                       |
| AS (Rx)      | 35.8                     | 41.6                     | 37.4                     | 35.8                     | 41.6                     | 37.4                     |
| AoD          | 105.6                    | 293.1                    | 61.9                     | 105.6                    | 293.1                    | 61.9                     |
| AS (Tx)      | 36.1                     | 42.5                     | 38                       | 36.1                     | 42.5                     | 38                       |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 15                  |                          |                          |                          | Path 16                  |                          | Path 17                  | Path 18                  |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 4                        | 2                        | 4                        | 4                        | 4                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 22.9                     | 19.9                     | 22.8                     | 20.6                     | 22.4                     | 20.5                     | 20.7                     | 24.6                     |
| Delay [ns]   | 490                      | 490                      | 490                      | 490                      | 560                      | 560                      | 640                      | 730                      |
| AoA          | 163.7                    | 251.8                    | 80                       | 182                      | 251.8                    | 182                      | 182                      | 182                      |
| AS (Rx)      | 35.8                     | 41.6                     | 37.4                     | 40.3                     | 41.6                     | 40.3                     | 40.3                     | 40.3                     |
| AoD          | 105.6                    | 293.1                    | 61.9                     | 275.7                    | 293.1                    | 275.7                    | 275.7                    | 275.7                    |
| AS (Tx)      | 36.1                     | 42.5                     | 38                       | 38.7                     | 42.5                     | 38.7                     | 38.7                     | 38.7                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

### A.1.6 Model F

| Tap          | Path 1                   | Path 2                   | Path 3                   | Path 4                   | Path 5                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 3.3                      | 3.6                      | 3.9                      | 4.2                      | 4.6                      | 1.8                      |
| Delay [ns]   | 0                        | 10                       | 20                       | 30                       | 50                       | 50                       |
| AoA          | 315.1                    | 315.1                    | 315.1                    | 315.1                    | 315.1                    | 180.4                    |
| AS (Rx)      | 48                       | 48                       | 48                       | 48                       | 48                       | 55                       |
| AoD          | 56.2                     | 56.2                     | 56.2                     | 56.2                     | 56.2                     | 183.7                    |
| AS (Tx)      | 41.6                     | 41.6                     | 41.6                     | 41.6                     | 41.6                     | 55.2                     |
| Speed [km/h] | 1.2                      | 1.2                      | 40                       | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 6                   |                          | Path 7                   |                          | Path 8                   |                          |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile    | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]  | 5.3                      | 2.8                      | 6.2                      | 3.5                      | 7.1                      | 4.4                      |
| Delay [ns] | 80                       | 80                       | 110                      | 110                      | 140                      | 140                      |

| Tap          | Path 6    |           | Path 7    |           | Path 8    |           |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| AoA          | 315.1     | 180.4     | 315.1     | 180.4     | 315.1     | 180.4     |
| AS (Rx)      | 48        | 55        | 48        | 55        | 48        | 55        |
| AoD          | 56.2      | 183.7     | 56.2      | 183.7     | 56.2      | 183.7     |
| AS (Tx)      | 41.6      | 55.2      | 41.6      | 55.2      | 41.6      | 55.2      |
| Speed [km/h] | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       | 1.2       |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

| Tap          | Path 9                   |                          |                          | Path 10                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 8.2                      | 5.3                      | 5.7                      | 9.5                      | 7.4                      | 6.7                      |
| Delay [ns]   | 180                      | 180                      | 180                      | 230                      | 230                      | 230                      |
| AoA          | 315.1                    | 180.4                    | 74.7                     | 315.1                    | 180.4                    | 74.7                     |
| AS (Rx)      | 48                       | 55                       | 42                       | 48                       | 55                       | 42                       |
| AoD          | 56.2                     | 183.7                    | 153                      | 56.2                     | 183.7                    | 153                      |
| AS (Tx)      | 41.6                     | 55.2                     | 47.4                     | 41.6                     | 55.2                     | 47.4                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 11                  |                          |                          | Path 12                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 11                       | 7                        | 10.4                     | 12.5                     | 10.3                     | 9.6                      |
| Delay [ns]   | 280                      | 280                      | 280                      | 330                      | 330                      | 330                      |
| AoA          | 315.1                    | 180.4                    | 74.7                     | 315.1                    | 180.4                    | 74.7                     |
| AS (Rx)      | 48                       | 55                       | 42                       | 48                       | 55                       | 42                       |
| AoD          | 56.2                     | 183.7                    | 153                      | 56.2                     | 183.7                    | 153                      |
| AS (Tx)      | 41.6                     | 55.2                     | 47.4                     | 41.6                     | 55.2                     | 47.4                     |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 13                     |                             |                             |                             | Path 14                     |                             |                             |                             |
|--------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Cluster      | 1                           | 2                           | 3                           | 4                           | 1                           | 2                           | 3                           | 4                           |
| Profile      | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor |
| Loss [dB]    | 14.3                        | 10.4                        | 14.1                        | 8.8                         | 16.7                        | 13.8                        | 12.7                        | 13.3                        |
| Delay [ns]   | 400                         | 400                         | 400                         | 400                         | 490                         | 490                         | 490                         | 490                         |
| AoA          | 315.1                       | 180.4                       | 74.7                        | 251.5                       | 315.1                       | 180.4                       | 74.7                        | 251.5                       |
| AS (Rx)      | 48                          | 55                          | 42                          | 28.6                        | 48                          | 55                          | 42                          | 28.6                        |
| AoD          | 56.2                        | 183.7                       | 153                         | 112.5                       | 56.2                        | 183.7                       | 153                         | 112.5                       |
| AS (Tx)      | 41.6                        | 55.2                        | 47.4                        | 27.2                        | 41.6                        | 55.2                        | 47.4                        | 27.2                        |
| Speed [km/h] | 1.2                         | 1.2                         | 1.2                         | 1.2                         | 1.2                         | 1.2                         | 1.2                         | 1.2                         |
| Distribution | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   |

| Tap          | Path 15                  |                          |                          |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 4                        | 5                        |
| Profile      | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor |
| Loss [dB]    | 19.9                     | 15.7                     | 18.5                     | 18.7                     | 12.9                     |
| Delay [ns]   | 600                      | 600                      | 600                      | 600                      | 600                      |
| AoA          | 315.1                    | 180.4                    | 74.7                     | 251.5                    | 68.5                     |
| AS (Rx)      | 48                       | 55                       | 42                       | 28.6                     | 30.7                     |
| AoD          | 56.2                     | 183.7                    | 153                      | 112.5                    | 291                      |
| AS (Tx)      | 41.6                     | 55.2                     | 47.4                     | 27.2                     | 33                       |
| Speed [km/h] | 1.2                      | 1.2                      | 1.2                      | 1.2                      | 1.2                      |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 16                  |                          | Path 17                  | Path 18                  |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    | 2                        | 5                        | 6                        | 6                        |
| Profile    | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor |
| Loss [dB]  | 19.9                     | 14.2                     | 16.3                     | 21.2                     |
| Delay [ns] | 730                      | 730                      | 880                      | 1050                     |
| AoA        | 180.4                    | 68.5                     | 246.2                    | 246.2                    |
| AS (Rx)    | 55                       | 30.7                     | 38.2                     | 38.2                     |

| Tap          | Path 16   |           | Path 17   | Path 18   |
|--------------|-----------|-----------|-----------|-----------|
| AoD          | 183.7     | 291       | 62.3      | 62.3      |
| AS (Tx)      | 55.2      | 33        | 38        | 38        |
| Speed [km/h] | 1.2       | 1.2       | 1.2       | 1.2       |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian |

## A.2 802.11n - SISO channel models

These fading profiles are implemented as the 802.11n - MIMO channel models, expect that:

- Correlation Path = off
- Coefficient, % = 100
- Phase, deg = 0

See [Chapter A.1, "802.11n - MIMO channel models"](#), on page 633.

## A.3 802.11ac, ax - MIMO channel models

According to IEEE 802.11-03/940r4 TGn Channel Models.

Rx antenna distance = 1

Tx antenna distance = 0.5

Distribution = Laplacian

Profile = bell shape high throughput task group (TGn) indoor, exception model F, path 3 where the profile = bell shape TGn moving vehicle.

Speed = 0.089 km/h, exception model F, Path 3 where speed = 40 km/h



The 802.11ac, ax - MIMO channel models are applicable to channel bandwidth  $\leq 40$  MHz.

### A.3.1 Model A ( $\leq 40$ MHz)

| Tap        | Path 1                |
|------------|-----------------------|
| Cluster    |                       |
| Profile    | bell shape TGn indoor |
| Loss [dB]  | 0                     |
| Delay [ns] | 0                     |

| Tap                                  | Path 1 |
|--------------------------------------|--------|
| Angle of arrival (AoA)               | 45     |
| Angular spread (AS) at receiver (Rx) | 40     |
| Angle of departure (AoD)             | 45     |
| AS at transmitter (Tx)               | 40     |
| Speed [km/h]                         | 0.089  |

### A.3.2 Model B ( $\leq 40$ MHz)

| Tap          | Path 1                   | Path 2                   | Path 3                   |                          | Path 4                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 0                        | 5.4                      | 10.8                     | 3.2                      | 16.2                     | 6.3                      |
| Delay [ns]   | 0                        | 10                       | 20                       | 20                       | 30                       | 30                       |
| AoA          | 4.3                      | 4.3                      | 4.3                      | 118.4                    | 4.3                      | 118.4                    |
| AS (Rx)      | 14.4                     | 14.4                     | 14.4                     | 25.2                     | 14.4                     | 25.2                     |
| AoD          | 225.1                    | 225.1                    | 225.1                    | 106.5                    | 225.1                    | 106.5                    |
| AS (Tx)      | 14.4                     | 14.4                     | 14.4                     | 25.4                     | 14.4                     | 25.4                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 5                   |                          | Path 6                   | Path 7                   | Path 8                   | Path 9                   |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        |                          |                          |                          |                          |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 21.7                     | 9.4                      | 12.5                     | 15.6                     | 18.7                     | 21.8                     |
| Delay [ns]   | 40                       | 40                       | 50                       | 60                       | 70                       | 80                       |
| AoA          | 4.3                      | 118.4                    | 118.4                    | 118.4                    | 118.4                    | 118.4                    |
| AS (Rx)      | 14.4                     | 25.2                     | 25.2                     | 25.2                     | 25.2                     | 25.2                     |
| AoD          | 225.1                    | 106.5                    | 106.5                    | 106.5                    | 106.5                    | 106.5                    |
| AS (Tx)      | 14.4                     | 25.4                     | 25.4                     | 25.4                     | 25.4                     | 25.4                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |



### A.3.3 Model C ( $\leq 40$ MHz)

| Tap          | Path 1                   | Path 2                   | Path 3                   | Path 4                   | Path 5                   | Path 6                   |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          |                          |                          |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 0                        | 2.1                      | 4.3                      | 6.5                      | 8.6                      | 10.8                     |
| Delay [ns]   | 0                        | 10                       | 20                       | 30                       | 40                       | 50                       |
| AoA          | 290.3                    | 290.3                    | 290.3                    | 290.3                    | 290.3                    | 290.3                    |
| AS (Rx)      | 24.6                     | 24.6                     | 24.6                     | 24.6                     | 24.6                     | 24.6                     |
| AoD          | 13.5                     | 13.5                     | 13.5                     | 13.5                     | 13.5                     | 13.5                     |
| AS (Tx)      | 24.7                     | 24.7                     | 24.7                     | 24.7                     | 24.7                     | 24.7                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 7                   |                          | Path 8                   |                          | Path 9                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 13                       | 5                        | 15.2                     | 7.2                      | 17.3                     | 9.3                      |
| Delay [ns]   | 60                       | 60                       | 70                       | 70                       | 80                       | 80                       |
| AoA          | 290.3                    | 332.3                    | 290.3                    | 332.3                    | 290.3                    | 332.3                    |
| AS (Rx)      | 24.6                     | 22.4                     | 24.6                     | 22.4                     | 24.6                     | 22.4                     |
| AoD          | 13.5                     | 56.4                     | 13.5                     | 56.4                     | 13.5                     | 56.4                     |
| AS (Tx)      | 24.7                     | 22.5                     | 24.7                     | 22.5                     | 24.7                     | 22.5                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 10                  |                          | Path 11                  | Path 12                  | Path 13                  | Path 14                  |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    | 1                        | 2                        |                          |                          |                          |                          |
| Profile    | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]  | 19.5                     | 11.5                     | 13.7                     | 15.8                     | 18                       | 20.2                     |
| Delay [ns] | 90                       | 90                       | 110                      | 140                      | 170                      | 200                      |
| AoA        | 290.3                    | 332.3                    | 332.3                    | 332.3                    | 332.3                    | 332.3                    |
| AS (Rx)    | 24.6                     | 22.4                     | 22.4                     | 22.4                     | 22.4                     | 22.4                     |

| Tap          | Path 10   |           | Path 11   | Path 12   | Path 13   | Path 14   |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| AoD          | 13.5      | 56.4      | 56.4      | 56.4      | 56.4      | 56.4      |
| AS (Tx)      | 24.7      | 22.5      | 22.5      | 22.5      | 22.5      | 22.5      |
| Speed [km/h] | 0.089     | 0.089     | 0.089     | 0.089     | 0.089     | 0.089     |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

### A.3.4 Model D (≤ 40 MHz)

| Tap          | Path 1                   | Path 2                   | Path 3                   | Path 4                   | Path 5                   | Path 6                   |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          |                          |                          |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 0                        | 0.9                      | 1.7                      | 2.6                      | 3.5                      | 4.3                      |
| Delay [ns]   | 0                        | 10                       | 20                       | 30                       | 40                       | 50                       |
| AoA          | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 158.9                    |
| AS (Rx)      | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 27.7                     |
| AoD          | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 332.1                    |
| AS (Tx)      | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 27.4                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 7                   | Path 8                   | Path 9                   | Path 10                  | Path 11                  |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 5.2                      | 6.1                      | 6.9                      | 7.8                      | 9                        | 6.6                      |
| Delay [ns]   | 60                       | 70                       | 80                       | 90                       | 110                      | 110                      |
| AoA          | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 158.9                    | 320.2                    |
| AS (Rx)      | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 27.7                     | 31.4                     |
| AoD          | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 332.1                    | 49.3                     |
| AS (Tx)      | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 27.4                     | 32.1                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 12                  |                          | Path 13                  |                          | Path 14                  |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 11.1                     | 9.5                      | 13.7                     | 12.1                     | 16.3                     | 14.7                     |
| Delay [ns]   | 140                      | 140                      | 170                      | 170                      | 200                      | 200                      |
| AoA          | 158.9                    | 320.2                    | 158.9                    | 320.2                    | 158.9                    | 320.2                    |
| AS (Rx)      | 27.7                     | 31.4                     | 27.7                     | 31.4                     | 27.7                     | 31.4                     |
| AoD          | 332.1                    | 49.3                     | 332.1                    | 49.3                     | 332.1                    | 49.3                     |
| AS (Tx)      | 27.4                     | 32.1                     | 27.4                     | 32.1                     | 27.4                     | 32.1                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 15                  |                          |                          | Path 16                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 19.3                     | 17.4                     | 18.8                     | 23.2                     | 21.9                     | 23.2                     |
| Delay [ns]   | 240                      | 240                      | 240                      | 290                      | 290                      | 290                      |
| AoA          | 158.9                    | 320.2                    | 276.1                    | 158.9                    | 320.2                    | 276.1                    |
| AS (Rx)      | 27.7                     | 31.4                     | 37.4                     | 27.7                     | 31.4                     | 37.4                     |
| AoD          | 332.1                    | 49.3                     | 275.9                    | 332.1                    | 49.3                     | 275.9                    |
| AS (Tx)      | 27.4                     | 32.1                     | 36.8                     | 27.4                     | 32.1                     | 36.8                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 17               |                       | Path 18               |
|------------|-----------------------|-----------------------|-----------------------|
| Cluster    | 2                     | 3                     |                       |
| Profile    | bell shape TGn indoor | bell shape TGn indoor | bell shape TGn indoor |
| Loss [dB]  | 25.5                  | 25.2                  | 26.7                  |
| Delay [ns] | 340                   | 340                   | 390                   |
| AoA        | 320.2                 | 276.1                 | 276.1                 |
| AS (Rx)    | 31.4                  | 37.4                  | 37.4                  |
| AoD        | 49.3                  | 275.9                 | 275.9                 |
| AS (Tx)    | 32.1                  | 36.8                  | 36.8                  |

| Tap          | Path 17   |           | Path 18   |
|--------------|-----------|-----------|-----------|
| Speed [km/h] | 0.089     | 0.089     | 0.089     |
| Distribution | Laplacian | Laplacian | Laplacian |

### A.3.5 Model E ( $\leq 40$ MHz)

| Tap          | Path 1                   | Path 2                   | Path 3                   | Path 4                   | Path 5                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                          |                          | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 2.6                      | 3                        | 3.5                      | 3.9                      | 4.5                      | 1.8                      |
| Delay [ns]   | 0                        | 10                       | 20                       | 30                       | 50                       | 50                       |
| AoA          | 163.7                    | 163.7                    | 163.7                    | 163.7                    | 163.7                    | 251.8                    |
| AS (Rx)      | 35.8                     | 35.8                     | 35.8                     | 35.8                     | 35.8                     | 41.6                     |
| AoD          | 105.6                    | 105.6                    | 105.6                    | 105.6                    | 105.6                    | 293.1                    |
| AS (Tx)      | 36.1                     | 36.1                     | 36.1                     | 36.1                     | 36.1                     | 42.5                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 6                   |                          | Path 7                   |                          | Path 8                   |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 5.6                      | 3.2                      | 6.9                      | 4.5                      | 8.2                      | 5.8                      |
| Delay [ns]   | 80                       | 80                       | 110                      | 110                      | 140                      | 140                      |
| AoA          | 163.7                    | 251.8                    | 163.7                    | 251.8                    | 163.7                    | 251.8                    |
| AS (Rx)      | 35.8                     | 41.6                     | 35.8                     | 41.6                     | 35.8                     | 41.6                     |
| AoD          | 105.6                    | 293.1                    | 105.6                    | 293.1                    | 105.6                    | 293.1                    |
| AS (Tx)      | 36.1                     | 42.5                     | 36.1                     | 42.5                     | 36.1                     | 42.5                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap     | Path 9                   |                          |                          | Path 10                  |                          |                          |
|---------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |

| Tap          | Path 9    |           |           | Path 10   |           |           |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Loss [dB]    | 9.8       | 7.1       | 7.9       | 11.7      | 9.9       | 9.6       |
| Delay [ns]   | 180       | 180       | 180       | 230       | 230       | 230       |
| AoA          | 163.7     | 251.8     | 80        | 163.7     | 251.8     | 80        |
| AS (Rx)      | 35.8      | 41.6      | 37.4      | 35.8      | 41.6      | 37.4      |
| AoD          | 105.6     | 293.1     | 61.9      | 105.6     | 293.1     | 61.9      |
| AS (Tx)      | 36.1      | 42.5      | 38        | 36.1      | 42.5      | 38        |
| Speed [km/h] | 0.089     | 0.089     | 0.089     | 0.089     | 0.089     | 0.089     |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

| Tap          | Path 11                  |                          |                          | Path 12                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 13.9                     | 10.3                     | 14.2                     | 16.1                     | 14.3                     | 13.8                     |
| Delay [ns]   | 280                      | 280                      | 280                      | 330                      | 330                      | 330                      |
| AoA          | 163.7                    | 251.8                    | 80                       | 163.7                    | 251.8                    | 80                       |
| AS (Rx)      | 35.8                     | 41.6                     | 37.4                     | 35.8                     | 41.6                     | 37.4                     |
| AoD          | 105.6                    | 293.1                    | 61.9                     | 105.6                    | 293.1                    | 61.9                     |
| AS (Tx)      | 36.1                     | 42.5                     | 38                       | 36.1                     | 42.5                     | 38                       |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 13                  |                          |                          | Path 14                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 18.3                     | 14.7                     | 18.6                     | 20.5                     | 18.7                     | 18.1                     |
| Delay [ns]   | 380                      | 380                      | 380                      | 430                      | 430                      | 430                      |
| AoA          | 163.7                    | 251.8                    | 80                       | 163.7                    | 251.8                    | 80                       |
| AS (Rx)      | 35.8                     | 41.6                     | 37.4                     | 35.8                     | 41.6                     | 37.4                     |
| AoD          | 105.6                    | 293.1                    | 61.9                     | 105.6                    | 293.1                    | 61.9                     |
| AS (Tx)      | 36.1                     | 42.5                     | 38                       | 36.1                     | 42.5                     | 38                       |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 15                     |                             |                             |                             | Path 16                     |                             | Path 17                     | Path 18                     |
|--------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Cluster      | 1                           | 2                           | 3                           | 4                           | 2                           | 4                           | 4                           | 4                           |
| Profile      | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor |
| Loss [dB]    | 22.9                        | 19.9                        | 22.8                        | 20.6                        | 22.4                        | 20.5                        | 20.7                        | 24.6                        |
| Delay [ns]   | 490                         | 490                         | 490                         | 490                         | 560                         | 560                         | 640                         | 730                         |
| AoA          | 163.7                       | 251.8                       | 80                          | 182                         | 251.8                       | 182                         | 182                         | 182                         |
| AS (Rx)      | 35.8                        | 41.6                        | 37.4                        | 40.3                        | 41.6                        | 40.3                        | 40.3                        | 40.3                        |
| AoD          | 105.6                       | 293.1                       | 61.9                        | 275.7                       | 293.1                       | 275.7                       | 275.7                       | 275.7                       |
| AS (Tx)      | 36.1                        | 42.5                        | 38                          | 38.7                        | 42.5                        | 38.7                        | 38.7                        | 38.7                        |
| Speed [km/h] | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       |
| Distribution | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   |

### A.3.6 Model F ( $\leq 40$ MHz)

| Tap          | Path 1                   | Path 2                   | Path 3                              | Path 4                   | Path 5                   |                          |
|--------------|--------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      |                          |                          |                                     |                          | 1                        | 2                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn moving<br>vehicle | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 3.3                      | 3.6                      | 3.9                                 | 4.2                      | 4.6                      | 1.8                      |
| Delay [ns]   | 0                        | 10                       | 20                                  | 30                       | 50                       | 50                       |
| AoA          | 315.1                    | 315.1                    | 315.1                               | 315.1                    | 315.1                    | 180.4                    |
| AS (Rx)      | 48                       | 48                       | 48                                  | 48                       | 48                       | 55                       |
| AoD          | 56.2                     | 56.2                     | 56.2                                | 56.2                     | 56.2                     | 183.7                    |
| AS (Tx)      | 41.6                     | 41.6                     | 41.6                                | 41.6                     | 41.6                     | 55.2                     |
| Speed [km/h] | 0.089                    | 0.089                    | 40                                  | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                           | Laplacian                | Laplacian                | Laplacian                |

| Tap       | Path 6                   |                          | Path 7                   |                          | Path 8                   |                          |
|-----------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster   | 1                        | 2                        | 1                        | 2                        | 1                        | 2                        |
| Profile   | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB] | 5.3                      | 2.8                      | 6.2                      | 3.5                      | 7.1                      | 4.4                      |

| Tap          | Path 6    |           | Path 7    |           | Path 8    |           |
|--------------|-----------|-----------|-----------|-----------|-----------|-----------|
| Delay [ns]   | 80        | 80        | 110       | 110       | 140       | 140       |
| AoA          | 315.1     | 180.4     | 315.1     | 180.4     | 315.1     | 180.4     |
| AS (Rx)      | 48        | 55        | 48        | 55        | 48        | 55        |
| AoD          | 56.2      | 183.7     | 56.2      | 183.7     | 56.2      | 183.7     |
| AS (Tx)      | 41.6      | 55.2      | 41.6      | 55.2      | 41.6      | 55.2      |
| Speed [km/h] | 0.089     | 0.089     | 0.089     | 0.089     | 0.089     | 0.089     |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian | Laplacian |

| Tap          | Path 9                   |                          |                          | Path 10                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 8.2                      | 5.3                      | 5.7                      | 9.5                      | 7.4                      | 6.7                      |
| Delay [ns]   | 180                      | 180                      | 180                      | 230                      | 230                      | 230                      |
| AoA          | 315.1                    | 180.4                    | 74.7                     | 315.1                    | 180.4                    | 74.7                     |
| AS (Rx)      | 48                       | 55                       | 42                       | 48                       | 55                       | 42                       |
| AoD          | 56.2                     | 183.7                    | 153                      | 56.2                     | 183.7                    | 153                      |
| AS (Tx)      | 41.6                     | 55.2                     | 47.4                     | 41.6                     | 55.2                     | 47.4                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 11                  |                          |                          | Path 12                  |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 1                        | 2                        | 3                        |
| Profile      | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor | bell shape<br>TGn indoor |
| Loss [dB]    | 11                       | 7                        | 10.4                     | 12.5                     | 10.3                     | 9.6                      |
| Delay [ns]   | 280                      | 280                      | 280                      | 330                      | 330                      | 330                      |
| AoA          | 315.1                    | 180.4                    | 74.7                     | 315.1                    | 180.4                    | 74.7                     |
| AS (Rx)      | 48                       | 55                       | 42                       | 48                       | 55                       | 42                       |
| AoD          | 56.2                     | 183.7                    | 153                      | 56.2                     | 183.7                    | 153                      |
| AS (Tx)      | 41.6                     | 55.2                     | 47.4                     | 41.6                     | 55.2                     | 47.4                     |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap          | Path 13                        |                             |                             |                             | Path 14                     |                             |                             |                             |
|--------------|--------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| Cluster      | 1                              | 2                           | 3                           | 4                           | 1                           | 2                           | 3                           | 4                           |
| Profile      | bell shape<br>TGnTGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor | bell shape<br>TGn<br>indoor |
| Loss [dB]    | 14.3                           | 10.4                        | 14.1                        | 8.8                         | 16.7                        | 13.8                        | 12.7                        | 13.3                        |
| Delay [ns]   | 400                            | 400                         | 400                         | 400                         | 490                         | 490                         | 490                         | 490                         |
| AoA          | 315.1                          | 180.4                       | 74.7                        | 251.5                       | 315.1                       | 180.4                       | 74.7                        | 251.5                       |
| AS (Rx)      | 48                             | 55                          | 42                          | 28.6                        | 48                          | 55                          | 42                          | 28.6                        |
| AoD          | 56.2                           | 183.7                       | 153                         | 112.5                       | 56.2                        | 183.7                       | 153                         | 112.5                       |
| AS (Tx)      | 41.6                           | 55.2                        | 47.4                        | 27.2                        | 41.6                        | 55.2                        | 47.4                        | 27.2                        |
| Speed [km/h] | 0.089                          | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       | 0.089                       |
| Distribution | Laplacian                      | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   | Laplacian                   |

| Tap          | Path 15                  |                          |                          |                          |                          |
|--------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster      | 1                        | 2                        | 3                        | 4                        | 5                        |
| Profile      | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor |
| Loss [dB]    | 19.9                     | 15.7                     | 18.5                     | 18.7                     | 12.9                     |
| Delay [ns]   | 600                      | 600                      | 600                      | 600                      | 600                      |
| AoA          | 315.1                    | 180.4                    | 74.7                     | 251.5                    | 68.5                     |
| AS (Rx)      | 48                       | 55                       | 42                       | 28.6                     | 30.7                     |
| AoD          | 56.2                     | 183.7                    | 153                      | 112.5                    | 291                      |
| AS (Tx)      | 41.6                     | 55.2                     | 47.4                     | 27.2                     | 33                       |
| Speed [km/h] | 0.089                    | 0.089                    | 0.089                    | 0.089                    | 0.089                    |
| Distribution | Laplacian                | Laplacian                | Laplacian                | Laplacian                | Laplacian                |

| Tap        | Path 16                  |                          | Path 17                  | Path 18                  |
|------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Cluster    | 2                        | 5                        | 6                        | 6                        |
| Profile    | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor | bell shape TGn<br>indoor |
| Loss [dB]  | 19.9                     | 14.2                     | 16.3                     | 21.2                     |
| Delay [ns] | 730                      | 730                      | 880                      | 1050                     |
| AoA        | 180.4                    | 68.5                     | 246.2                    | 246.2                    |
| AS (Rx)    | 55                       | 30.7                     | 38.2                     | 38.2                     |



| Tap          | Path 16   |           | Path 17   | Path 18   |
|--------------|-----------|-----------|-----------|-----------|
| AoD          | 183.7     | 291       | 62.3      | 62.3      |
| AS (Tx)      | 55.2      | 33        | 38        | 38        |
| Speed [km/h] | 0.089     | 0.089     | 0.089     | 0.089     |
| Distribution | Laplacian | Laplacian | Laplacian | Laplacian |

## A.4 802.11ac, ax - SISO channel models

These fading profiles are implemented as the 802.11ac, ax - MIMO channel models, expect that:

- Correlation Path = off
- Coefficient, % = 100
- Phase, deg = 0

See [Chapter A.3, "802.11ac, ax - MIMO channel models"](#), on page 643.

## B Spectrum mask areas

This section lists information that is referenced by several command descriptions.

**Table B-1: Spectrum mask areas**

| Standard (mask)                                   | Spectrum mask areas with margins                                                                         |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| 802.11b/g DSSS                                    | 4 areas (AB, CD, DC, BA)                                                                                 |
| 802.11a/g OFDM, n, ac, ax<br>Exception: 80+80 MHz | 8 areas (AB, BC, CD, DE, ED, DC, CB, BA)<br>up to 16 areas (number of areas depends on channel distance) |
| 802.11p (ETSI or IEEE)                            | 10 areas (AB, BC, CD, DE, EF, FE, ED, DC, CB, BA)                                                        |
| 802.11p (ARIB)                                    | 4 areas (AB, BC, CB, BA)                                                                                 |

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